A Dichotomy for Complex Boolean Holant with Binary Disequality

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Abstract

We classify Boolean Holant problems with a supplied binary disequality signature and an arbitrary fixed finite set of algebraic complex-valued signatures. Each problem is polynomial-time computable or #P-hard, and the tractability criterion is decidable. The proof combines arity reduction with the growth of the available binary transfer group and uses stable pairs to recover the structure of higher-arity signatures. We also prove a dichotomy for the disequality-edge model when the signature set contains an endpoint-nondegenerate irreducible eight-vertex signature, resolving the corresponding arity-four cases in the main proof.

1 Introduction

Many counting problems ask for a global sum determined by local constraints: each assignment receives the product of its local weights, and these products are summed over all assignments. A complexity dichotomy identifies the choices of local signatures that admit polynomial-time evaluation and proves #P-hardness for every remaining choice. On the Boolean domain, this program has a long history in counting constraint satisfaction problems (#CSP). Creignou and Hermann classified the unweighted case [CH96]. Later dichotomies treated nonnegative rational weights [DGJ09], rational weights of mixed signs [BDG$^+$09], and complex weights [CLX14]. Complex weights permit cancellations that yield tractable cases absent from the nonnegative setting [CLX14], making the algebra of the weights part of the classification. Over arbitrary finite domains, counting-CSP dichotomies have likewise been established for unweighted constraints [Bul13, DR13], nonnegative rational weights [BDG$^+$12], nonnegative algebraic weights [CCL16], and algebraic complex weights [CC17].

The Holant framework brings the availability of local constraints into this classification problem [CLX09]. A counting-CSP instance can be represented by a bipartite graph whose variable vertices carry equality constraints, enforcing agreement among all occurrences of a variable. Holant allows the available vertex signatures to be prescribed without supplying these equalities. This accommodates problems such as counting perfect matchings, where each vertex requires exactly one incident edge to be selected. It also makes the construction of useful local signatures a part of the classification proof. Holographic transformations, originating in Valiant's algorithms [Val08], give another source of tractability: a simultaneous change of basis can expose structure in the local signatures while preserving the value of the total sum.

Boolean Holant dichotomies have addressed both restrictions on the signatures and assumptions about auxiliary signatures. With no prescribed auxiliary signatures, classifications were obtained for symmetric real-valued signatures [HL16], symmetric complex-valued signatures [CGW16], arbitrary nonnegative signatures [LW18], and arbitrary real-valued signatures [SC20]. Here symmetry means that a signature depends only on the number of ones in its input. Another line of work supplies unary signatures: Holant* supplies all of them [CLX11], Holant$^+$ supplies four distinguished ones [Bac17], and Holant^c supplies the two pinnings $\Delta_0 = [1, 0]^{\mathsf{T}}$ and $\Delta_1 = [0, 1]^{\mathsf{T}}$. The last model was classified for symmetric complex signatures [CHL12], arbitrary real signatures [CLX18], and algebraic complex signatures [Bac21, Bac25]. The real-valued classification introduced the local-affine tractable family $\mathscr{L}$, which also occurs here.

Meng, Wang, Xia, and Zheng proved a tensor-factor extraction theorem and classified sets containing a nonzero odd-arity signature. Their condition (PC) lists seven cases [MWXZ25, conference version, Theorem 30]. For two cases involving weighted Eulerian orientations [MWX25], the original bound was $\mathsf{FP}^{\mathsf{NP}}$. Guan, Shao, and Shi gave polynomial-time algorithms for these cases [GSS26]. We write Tract for the resulting seven-case criterion, the most general tractability condition currently known for algebraic complex-valued Boolean Holant. Even-arity signatures present a different resource constraint. Every gadget built entirely from them has an even number of external inputs, because each internal edge consumes two inputs. Such gadgets cannot supply a unary signature, so binary signatures are the smallest positive-arity objects available for further constructions. We study the case in which binary disequality, requiring two bits to differ, is supplied.

There is also a precedent for prescribing disequality in counting CSP. In #CSP$_d$, each variable occurs a number of times divisible by d. Cai, Fu, and Shao classified #CSP$_d(\neq_2, \mathcal{F})$ for complex signatures [CFS20b, full version, Section 5.1 and Theorem 5.3]. Shao and Cai used its hardness conclusion in their real Holant dichotomy [SC20, full version, Theorem 2.24].

Prescribing binary disequality is useful for three further reasons. First, ordinary edges enforce

equality, and the supplied binary signature enforces disequality. Together they restrict the binary groups in Xia's framework [Xia26, Section 4 and Table 2] to the marked forms analyzed here. Second, inserting binary vertices lets us use disequality edges with equality supplied as a vertex signature. We can thus choose the edge model suited to a construction while keeping the original signatures in their original coordinates. Third, the classification applies to any Holant problem for which a reduction obtains disequality. For the complementary problems, the inability to obtain it supplies a structural restriction for further classification.

The problem and main results. A Boolean signature of arity n is a function $f : \{0,1\}^n \to \mathbb{C}$. In a Holant instance, a graph vertex of degree n carries such a signature, with its incident edges ordered. For a fixed finite set $\mathcal{F}$ of signatures, the problem Holant($\mathcal{F}$) evaluates

$$\sum_{\sigma:E\to\{0,1\}} \prod_{v\in V} f_v(\sigma|_{E(v)}), \qquad f_v \in \mathcal{F}.$$

Here $E(v)$ is the ordered list of edge incidences at v. Section F.2.3 gives the conventions for loops and multiple edges. The graph and its vertex labels form the input. The set $\mathcal{F}$ is fixed. Write

$$X = (\neq_2) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad I = (=_2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Our problem is Holant($\mathcal{F}, X$) := Holant($\mathcal{F} \cup \{X\}$). Thus any number of degree-two vertices may enforce opposite values on their two incident edges.

With X supplied, three of the seven cases of Tract are excluded (Lemma F.2.7). Write $\mathsf{ReTract}$ for the restriction to the other four. In one case every signature is a tensor product of signatures of arity at most two. In each of the other three, a common change of basis puts the edge and all vertex signatures in the affine class $\mathscr{A}$, the product class $\mathscr{P}$, or the local-affine class $\mathscr{L}$ (Section 2). The prefix *Re* records this restriction and the correspondence with the tractable cases of the real-valued Holant dichotomy of Shao and Cai [SC20].

Theorem 1.1 (Binary-disequality Holant dichotomy)**.** *For every finite set of algebraic complex-valued Boolean signatures $\mathcal{F}$, the problem* Holant($\mathcal{F}, X$) *is polynomial-time computable when* $\mathsf{ReTract}(\mathcal{F} \cup \{X\})$ *holds and is* #P*-hard otherwise. The tractability criterion is decidable from exact algebraic input.*

The proof also classifies sets containing a particular kind of arity-four signature, without a supplied equality signature. Following Meng, Wang, Xia, and Zheng [MWXZ25], write K-Holant($\mathcal{F}$) := Holant($\neq_2 | \mathcal{F}$) for the bipartite Holant problem with disequality on the left and signatures from $\mathcal{F}$ on the right. Equivalently, each connection between signature inputs enforces disequality.

A quaternary signature q is *endpoint-nondegenerate* when

$$q(\mathbf{x}) = 0 \quad \text{for odd } \mathrm{wt}(\mathbf{x}), \qquad q(0000)q(1111) \neq 0.$$

Here $\mathrm{wt}(\mathbf{x})$ counts the ones in $\mathbf{x}$. Only the eight even-weight inputs can have nonzero values, giving the eight-vertex form used below. A nonzero positive-arity signature is *irreducible* if it has no factorization $a \otimes b$ on two nonempty blocks of inputs, allowing a permutation of inputs and a nonzero scalar.

Theorem 1.2 (Quaternary K-Holant dichotomy)**.** *Let $\mathcal{F}$ be a finite set of algebraic complex-valued Boolean signatures. Suppose that $\mathcal{F}$ contains an endpoint-nondegenerate irreducible quaternary signature. Then* K-Holant($\mathcal{F}$) *is polynomial-time computable if* $\mathsf{ReTract}(\mathcal{F})$ *holds and is* #P*-hard otherwise. The tractability criterion is decidable from exact algebraic input.*

The dichotomy for the eight-vertex model [CF23, arXiv version 2, Theorem 3.1] classifies a single eight-vertex signature q with nonzero endpoints. Even if q alone is tractable, using it with other signatures can make the problem hard. Theorem 1.2 settles the complexity of the whole set whenever a construction yields an endpoint-nondegenerate irreducible quaternary signature.

From existing tools to stable pairs. The odd-arity classification and tensor-factor extraction of [MWXZ25] provide two starting points. Obtaining a nonzero odd-arity signature invokes a known dichotomy. In the cases considered here, their factor-extraction theorem allows us to simulate irreducible factors and retain them for subsequent constructions. Two existing approaches suggest how to use these tools. Shao and Cai's real-valued proof uses arity descent [SC20, Sections 3–4], but its orthogonality and norm arguments do not directly control cancellation for arbitrary complex signatures. Xia's finite-group approach [Xia26, Sections 4–5] organizes the available binary constructions. The remaining task is to determine how these constructions constrain higher-arity signatures. We couple these approaches by tracking a nonbinary irreducible signature f of minimum arity together with a group G. After the preliminary reductions, G consists of the transfer classes $[BX]$ of all ordered nonzero binaries B directly realizable by gadgets over the current set. Brackets identify matrices up to nonzero scalar.

Close a pair of inputs of f with an available binary and retain the irreducible factors of the output by reduction. An unresolved nonbinary factor has smaller arity. A binary factor whose transfer class lies outside G gives a different form of progress. Once retained, it can be used directly in gadgets over the enlarged set. The complete group strictly enlarges, so we can test the same f with new contractions even when its arity has not decreased. Figure 1 shows the possible strict inclusions. Bounded arity descent, a uniform bound on group orders, and finite processing at each stage ensure termination.

If neither form of progress occurs and the complexity remains unresolved, every prescribed one-copy contraction, including those obtained by attaching realizable binaries to selected inputs, gives zero or a product of retained binaries on a perfect matching. This is the stable condition. We call the outputs cards. Fix the pair of inputs being closed and compare its nonzero cards. To recover a common factor, we first seek a matching compatible with the cards. Even when the pairings agree, the binary factors on an edge from which we want to extract a common factor must be proportional across the cards.

Gadgets test these requirements by preserving a proposed disagreement down to bounded arity, where exhaustive classifications apply. When the contractions recover the slices of f, a common factor of the cards lifts to f, contradicting irreducibility. Other outcomes determine the complexity or enlarge the binary group. In the cyclic case the surviving arity-eight core, the indicator of the Reed–Muller code $RM(1,3)$ up to the equivalences in Section 7, permits group enlargement. In the standard Klein-four and internal tetrahedral cases the whole set can admit a common real or affine basis, respectively.

The short account below gives the classification argument and selected proof mechanisms. The full paper follows the shared references. Numbers prefixed by F refer to its statements and proofs. Recognition takes the signature set as input, whereas the signature set is fixed in each evaluation problem.

2 Classification results and reductions

In the disequality-edge model, the equality constraint of an ordinary edge is realized by two X-edges joined through an I-vertex, since $XIX = I$. Conversely, an X-edge becomes an X-vertex in the

ordinary model, and an I-vertex can be suppressed. Inserting and suppressing these binary vertices, using $X^2 = I$, gives

$$\text{Holant}(\mathcal{F}, X) \equiv_{\text{T}} \text{K-Holant}(\mathcal{F} \cup \{I\}).$$

Thus proving Theorem 1.1 amounts to classifying disequality-edge problems whose signature sets contain I. For counting problems A, B, write $A \leq_{\text{T}} B$ if A has a polynomial-time Turing reduction to B, and $A \equiv_{\text{T}} B$ if both reductions hold. A finite set $\mathcal{S}$ is *simulated* by $\mathcal{F}$ if $\text{K-Holant}(\mathcal{F} \cup \mathcal{S}) \leq_{\text{T}} \text{K-Holant}(\mathcal{F})$. We use *realized* for a signature obtained by a gadget.

The second model equivalence is a holographic transformation [Val08]. Set

$$K = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \qquad K^{\mathsf{T}} K = X, \qquad K\mathcal{F} = \{K^{\otimes \operatorname{arity}(f)} f : f \in \mathcal{F}\}.$$

Transforming every signature by K makes contraction along an ordinary edge equal to the original contraction against X. The partition function is unchanged, so

$$\text{K-Holant}(\mathcal{F}) \equiv_{\text{T}} \text{Holant}(K\mathcal{F}).$$

The tractable classes used here are the affine class $\mathscr{A}$ [CLX09, CLX14], the product class $\mathscr{P}$ [DGJ09, CLX14], and the local-affine class $\mathscr{L}$ [CLX18]. We use their standard definitions from these papers. The fourth case is the tensor closure $\langle \mathscr{T} \rangle$, where $\mathscr{T}$ consists of signatures of arity at most two. Splitting these tensor products reduces evaluation to paths and cycles.

Holographic transformations explain the common-transformability condition. For edge signature $E \in \{I, X\}$ and $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$, it requires a single invertible matrix M such that

$$M^{\mathsf{T}} E M \in \mathscr{C}, \qquad M^{-1}\mathcal{F} \subseteq \mathscr{C}.$$

Here M^{-1} acts on every input of each signature. Transforming all vertices by M^{-1} and all edges by $M^{\mathsf{T}} E M$ preserves the partition function. Both conditions put the entire network in the same tractable class, using one common basis. Write $\mathsf{ReTract}_E$ for these three possibilities together with $\mathcal{F} \subseteq \langle \mathscr{T} \rangle$. Theorem 1.1 uses $E = I$ and Theorem 1.2 uses $E = X$. Elsewhere $\mathsf{ReTract} = \mathsf{ReTract}_I$. The corresponding criteria in the two models satisfy

$$\begin{aligned} \mathsf{ReTract}_X(\mathcal{F}) &\iff \mathsf{ReTract}(K\mathcal{F}), \\ \mathsf{ReTract}(\mathcal{F} \cup \{X\}) &\iff \mathsf{ReTract}(K(\mathcal{F} \cup \{I\})). \end{aligned}$$

The proofs are in Section F.2.10.3 and the discussion following Lemma F.5.1.

Both predicates use the zero and nullary preprocessing of Section F.2.10.3. In Theorem 1.1, $X \in \mathcal{F} \cup \{X\}$ excludes the three additional cases of Tract. In the equivalent disequality-edge model the set contains I, so its image under K contains $K^{\otimes 2} I = \operatorname{diag}(1, -1)$, which excludes the same cases (Lemma F.2.7).

A nonzero odd-arity signature, including a unary, invokes the known classification and algorithms [MWXZ25, GSS26]. A nonzero singular binary has unary tensor factors. For a nonsingular binary B, the transfer BX has infinite projective order if no positive power is a scalar matrix. Such a binary yields a rank-one binary by interpolation. For the signature sets containing I, factor extraction and the unary reductions therefore resolve these cases (Lemma F.3.8 and Lemma F.3.11).

The factor-extraction theorem of [MWXZ25] lets us simulate the factors of an available tensor product unless all supports lie on the same side of the middle Hamming layer (including that layer). Weighted Eulerian-orientation results classify this exception [MWX25, GSS26]. Since $I(00) = I(11) = 1$, a set containing I has support on both sides. We retain all previous signatures

when passing to the set containing the simulated factors. The problems remain Turing-equivalent, hardness transfers to the preceding problem, and the same common basis establishes the tractable condition for the preceding set. We call the set factor-closed once all its nonzero positive-arity irreducible tensor factors have been retained.

Two further reductions will be used in the proof of Theorem 1.2. A generalized equality is supported exactly on the all-zero and all-one strings, with both values nonzero. At arity at least four it permits the unary reduction. An equality anchor is a realized quaternary for which one change of basis turns the edge into binary equality and the quaternary into four-ary equality, up to nonzero scalars. It likewise permits a nonzero unary to be simulated. These reductions classify the whole set (Lemma F.3.16 and Lemma F.3.15).

3 From an eight-vertex signature to a signature-set dichotomy

The dichotomy for the eight-vertex model classifies Holant$(X \mid q)$ for a single endpoint-nondegenerate eight-vertex signature q [CF23, arXiv version 2, Theorem 3.1]. Its hard case gives hardness for any set realizing q. In the tractable cases we must determine how q interacts with the rest of the set. We allow q to be reducible here because later gadgets can produce such a signature and supply its factors.

Proposition 3.1 (Eight-vertex reduction)**.** *Let $\mathcal{F}$ be a finite factor-closed retained set directly realizing an endpoint-nondegenerate eight-vertex signature q. Then* K-Holant$(\mathcal{F})$ *is* $\#\mathsf{P}$*-hard, or* $\mathsf{Tract}(K\mathcal{F})$ *holds, or the original q factors across two pairs of inputs. In the last case both binary factors can be adjoined by reduction, and one is a nonsingular weighted equality* $\mathrm{diag}(p, q')$*, with $pq' \neq 0$.*

To prove the proposition, we examine the three exceptions of the single-signature theorem. Product transformability leaves three structural possibilities: a generalized equality, a product of two weighted equalities, or a dense even-parity form that realizes an equality anchor. The generalized equality and the anchor yield a simulated unary as explained in Section 2, and therefore classify the whole set. The product of two binaries is the factorization outcome of the proposition. The classification of product forms uses the support and the ranks of the three pair flattenings, so it identifies factors of q itself.

The endpoint conditions reduce local-affine transformability to affine transformability. In the affine case, a common diagonal change of basis preserves the disequality edge up to scalar and puts q into a normal form with support of size two, four, or eight. The diagonal change preserves the two nonzero endpoints of q, so factor extraction remains available throughout this analysis.

Support two gives a generalized equality, so the same unary reduction applies. For support four, constant assignments on the two input pairs define logical bits with coefficient matrix $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, where $AD = \pm BC$. If $AD = -BC$, a two-copy gadget gives a generalized equality. If $AD = BC$, the normal form factors into two weighted equalities. Undoing the diagonal change gives the required factorization of the original q.

On full even-parity support, the constructions give an equality anchor, a nonzero rank-one binary, or a tensor product of nonsingular weighted equalities. In the last case, extracting a factor and normalizing it to equality allows a single-input flip, producing a quaternary with full odd-parity support. A finite analysis of its contractions gives an equality anchor, a nonzero rank-one binary, a binary of infinite projective transfer order, or an endpoint-nondegenerate eight-vertex signature with support of size six. The anchor invokes the equality-anchor reduction. A rank-one binary supplies a unary factor, while an infinite-order binary first yields a rank-one binary by interpolation. The

support-six output lies outside the three transformable classes and is hard by the dichotomy for the eight-vertex model [CF23, arXiv version 2, Theorem 3.1]. Thus each full-support case classifies the entire retained set.

The reductions transfer these conclusions back to the original set. The normal forms and finite gadget analysis, including the exact computations and coverage arguments, are given in Theorem F.A.4, Theorem F.A.5, and Theorem F.A.6.

In Theorem 1.2, irreducibility excludes the factorization of q. Its two nonzero endpoints give support on both sides of the middle layer. Lemma F.2.7 therefore reduces $\mathsf{Tract}(K\mathcal{F})$ to $\mathsf{ReTract}(K\mathcal{F})$. An irreducible q of arity four also excludes the tensor-product case. By the criterion correspondence in Section 2, the three common-transformability cases give the tractable side of the theorem (Corollary F.4.8).

4 Stabilizing a signature and its binary group

We return to Theorem 1.1. By Section 2 we may work with disequality edges and supplied I. After factor extraction and the binary and odd-arity reductions, choose a nonbinary irreducible f of minimum arity $n = \nu(\mathcal{F})$ in the retained factor-closed set $\mathcal{F}$. Its arity is even. We follow the pair $(f, G(\mathcal{F}))$, recording $\mathcal{F}$ as part of the state because it determines the available gadgets.

For an ordered binary B, define its transfer by $T_B = BX$. Series composition realizes BXC, with $T_{BXC} = T_B T_C$. After the binary reductions, every nonzero realizable binary is nonsingular with finite projective transfer order. We call such a binary *safe*. Put

$$G(\mathcal{F}) = \{[BX] : B \text{ is a safe ordered binary directly realizable over } \mathcal{F}\}.$$

Here $[BX]$ is the transfer class of B, identifying matrices up to nonzero scalar. This *complete transfer group* includes the classes of all realizable binary gadgets, of arbitrary size. It is finite. In a fixed coefficient field L, an eigenvalue ratio of order r satisfies $\varphi(r) \le 2[L : \mathbb{Q}]$, bounding the possible orders. Powers supply inverses. The faithful adjoint representation and Burnside's finite-exponent theorem give finiteness [Bur05]. The finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$ then give a uniform order bound in this field (Lemma F.5.3).

Fix realizable ordered representatives B_g for $g \in G$. A G-matching signature is

$$c \bigotimes_{e \in M} B_{g_e}^{(e)}, \qquad c \neq 0,$$

where M is an oriented perfect matching of the inputs and the superscript places the binary on that ordered pair. The scalar absorbs the choice of representatives. To test f, attach a representative, in either orientation, to any subset of its inputs through X-edges. Write f^δ for the resulting dressed signature. Close one ordered pair (i, j) with another representative B_h^ω, where $B_h^{\mathrm{id}} = B_h$ and $B_h^{\mathrm{rev}} = B_h^{\mathsf{T}}$. The *matching deck* is the finite family

$$\mathfrak{D}_G(f) = \{0\} \ \cup \ \left\{ \rho\Big(\partial_{(i,j)}^{XB_h^\omega X} f^\delta \Big) : \delta, i \neq j, h, \omega, \rho \right\},$$

which collects the outputs obtained by applying the allowed binary dressings to a single copy of f and then closing one self-loop. Here $\partial_{(i,j)}^{Q} u$ sums the two inputs against the kernel $Q = XB_h^\omega X$, with one X-edge at each end of the closing binary. The permutation ρ ranges over residual input orders. Exactly one pair is closed and at least two inputs remain. Every nonzero card is a direct gadget output.

Definition 4.1 (Stable state)**.** Let $\mathcal{F}$ be finite, factor-closed, and contain I. Let $f \in \mathcal{F}$ be a minimum-arity nonbinary irreducible signature, and let $G = G(\mathcal{F})$ be finite with fixed realizable binary representatives. The state $(\mathcal{F}, G, f)$ is *deck-stable* if every nonzero card of $\mathfrak{D}_G(f)$ is a G-matching signature and all its factors are retained in $\mathcal{F}$.

Thus stability says that every allowed self-loop, including the specified input dressings, gives zero or a tensor product of already available binaries. Including input dressings makes this condition apply to the changes of binary action used in later comparisons.

Lemma 4.2 (Stabilization, Lemma F.5.9)**.** *Let $\mathcal{F}_0 \ni I$ be a retained factor-closed finite set, and let $f \in \mathcal{F}_0$ be an even nonbinary irreducible signature of minimum arity. The reductions either establish that* K-Holant$(\mathcal{F}_0)$ *is polynomial-time computable or* $\#\mathsf{P}$*-hard, or yield a finite retained factor-closed extension $\mathcal{F}' \supseteq \mathcal{F}_0$ with* K-Holant$(\mathcal{F}') \equiv_{\mathrm{T}}$ K-Holant$(\mathcal{F}_0)$ *such that either its minimum nonbinary arity is smaller or $(\mathcal{F}', G(\mathcal{F}'), f)$ is deck-stable. The procedure can strictly enlarge the complete binary group before reaching either extension.*

Process each card and retain its irreducible factors using factor extraction. A nonbinary factor has arity at most $n - 2$. Odd-arity factors and unsafe binaries invoke the established complexity classifications. Otherwise there are two distinct forms of progress:

(i) *Lowering the arity.* An even nonbinary factor gives an extension with smaller minimum nonbinary arity. Choose a minimum-arity irreducible in this extension as the new core signature and restart stabilization.

(ii) *Enlarging the binary group.* Retain the new binary factors. If a safe binary B has $[BX] \notin G$, the complete transfer group strictly enlarges. Keep the same core f and regenerate its deck using the larger group, which supplies additional contraction tests.

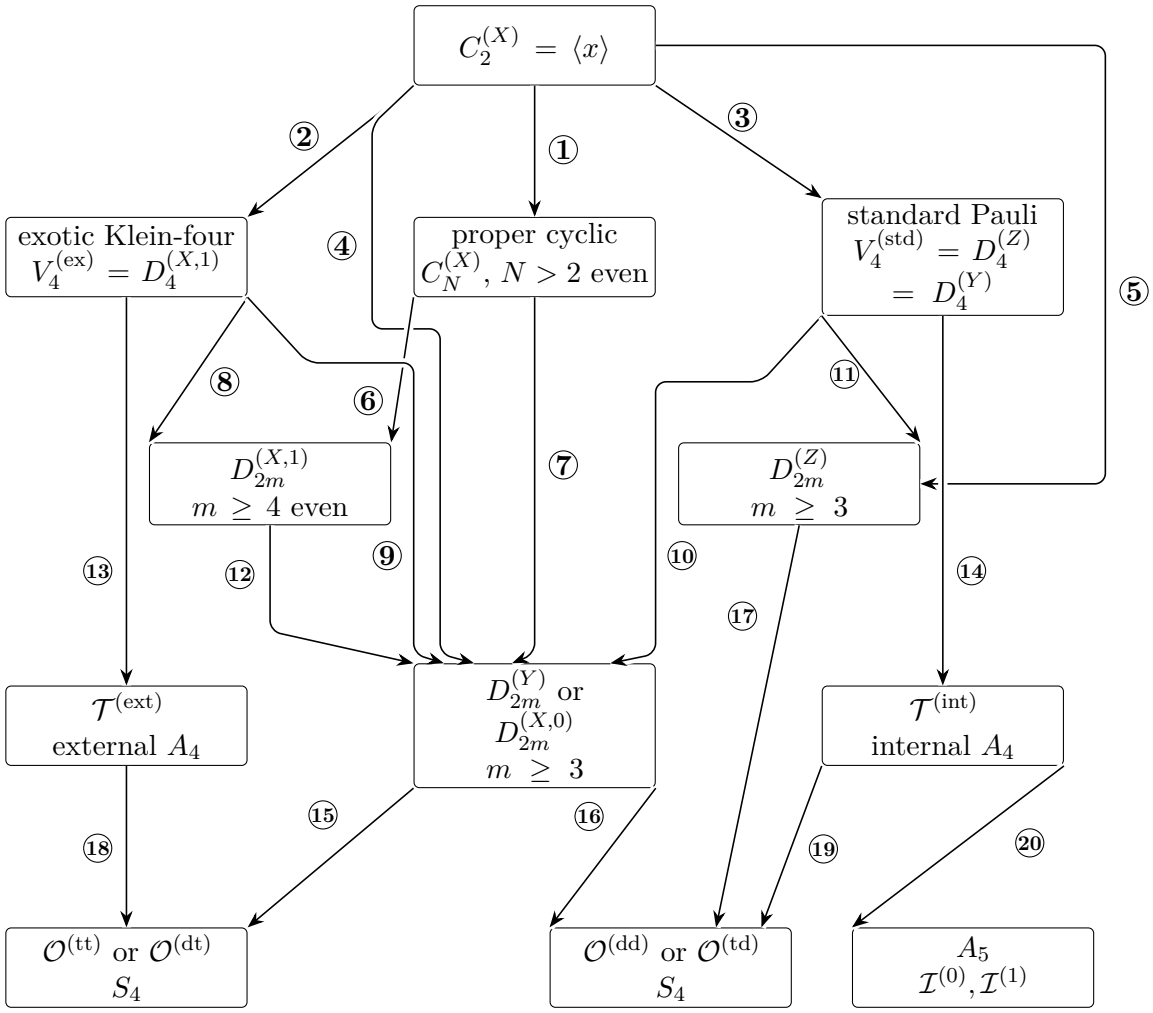
Each extension retains the previous signatures, including I, and preserves Turing equivalence. Binary factors whose transfer classes already lie in G do not by themselves give strict group growth.

Iterating these steps must eventually reach a stable pair unless the complexity is determined first. The two forms of progress are finite for different reasons. Each replacement of the core strictly decreases its arity. Since arity is an integer bounded below, only finitely many such replacements occur. Each group enlargement is strict, and all previous binary resources remain available even when the core changes. In the original coordinates, gadget evaluation and normalized factor extraction stay in one fixed number field. The complete groups therefore form an increasing chain with the uniform order bound established above, so only finitely many strict enlargements occur.

For a fixed core and group, the deck and its factorization trees are finite, with each distinct card of this deck processed once. After the last core replacement and group enlargement, this finite processing either determines the complexity or leaves every nonzero card as a tensor product of retained binaries from the current group. This is the stable condition, relative to the final retained set and its complete group.

5 Marked groups and their possible growth

The complete binary group determines the contractions available in the matching argument. Its abstract type is cyclic, dihedral, or one of the Platonic groups A_4, S_4, A_5, by the classification of finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$. The supplied I puts $x = [X]$ in G. Set $Z = \mathrm{diag}(1, -1)$, $Y = ZX$, and $j = [Z]$. Figure 1 shows the possible strict inclusions $G < G^+$. Its superscripts distinguish groups of the same abstract type that supply different contractions in our coordinates.



① $C_2^{(X)} \to C_N^{(X)}$
② $C_2^{(X)} \to V_4^{(\mathrm{ex})}$
③ $C_2^{(X)} \to V_4^{(\mathrm{std})}$
④ $C_2^{(X)} \to D_{2m}^{(Y)}$, m odd.
⑤ $C_2^{(X)} \to D_{2m}^{(Z)}$, m odd.
⑥ $C_N^{(X)} \to D_{2m}^{(X,1)}$, $N \mid m$.
⑦ $C_N^{(X)} \to D_{2m}^{(X,0)}$, $N \mid m$, j enters G^+.
⑧ $V_4^{(\mathrm{ex})} \to D_{2m}^{(X,1)}$, $m \equiv 2 \pmod 4$.
⑨ $V_4^{(\mathrm{ex})} \to D_{2m}^{(X,0)}$, $4 \mid m$, j enters G^+.
⑩ $V_4^{(\mathrm{std})} \to D_{2m}^{(Y)}$ or $D_{2m}^{(X,0)}$, $2 \mid m$.
⑪ $V_4^{(\mathrm{std})} \to D_{2m}^{(Z)}$, $2 \mid m$.
⑫ $D_{2m}^{(X,1)} \to D_{4m}^{(X,0)}$, j enters G^+.
⑬ $V_4^{(\mathrm{ex})} \to \mathcal{T}^{(\mathrm{ext})}$.
⑭ $V_4^{(\mathrm{std})} \to \mathcal{T}^{(\mathrm{int})}$.
⑮ $D_8^{(Y)} \to \mathcal{O}^{(\mathrm{tt})}$ or $D_8^{(X,0)} \to \mathcal{O}^{(\mathrm{dt})}$.
⑯ $D_8^{(Y)} \to \mathcal{O}^{(\mathrm{dd})}$ or $D_8^{(X,0)} \to \mathcal{O}^{(\mathrm{dd})}$.
⑰ $D_8^{(Z)} \to \mathcal{O}^{(\mathrm{dd})}$ or $\mathcal{O}^{(\mathrm{td})}$.
⑱ $\mathcal{T}^{(\mathrm{ext})} \to \mathcal{O}^{(\mathrm{dt})}$, j enters G^+.
⑲ $\mathcal{T}^{(\mathrm{int})} \to \mathcal{O}^{(\mathrm{dd})}$.
⑳ $\mathcal{T}^{(\mathrm{int})} \to \mathcal{I}^{(0)}$ or $\mathcal{I}^{(1)}$.

Here j enters G^+ when $j \notin G$ and $j \in G^+$. The parameters satisfy strict inclusion, with $m \geq 3$ for dihedral targets and even $m \geq 4$ for $D_{2m}^{(X,1)}, D_{2m}^{(X,0)}$.

Figure 1: Possible strict growth of the complete transfer group with the supplied equality fixed. Numbered arrows indicate subgroup inclusions between marked forms, with their parameter conditions listed below the graph. Divisibility steps within a family are suppressed. Grouped nodes collect distinct marked forms at the same level of the diagram.

Reversing the inputs of a binary replaces its transfer T by $XT^{\mathsf{T}}X$, with $[XT^{\mathsf{T}}X] = j[T]^{-1}j$. Since reversal preserves realizability and G is closed under inverses, $jGj = G$. Thus

$$\widehat{G} = \langle G, j\rangle, \qquad [\widehat{G} : G] \le 2, \qquad \langle x, j\rangle \cong V_4.$$

The positions of x and j in $\widehat{G}$, including whether $j \in G$, distinguish the marked forms.

Cyclic and dihedral groups. Write C_N for the abstract cyclic group of order N and D_{2m} for the abstract dihedral group of order $2m$. Let μ_m be the m-th roots of unity and put $D(t) = \mathrm{diag}(1, t)$, $C(t) = (1 + t)I + i(1 - t)Y$, and $A(t) = (1 + t)I + (1 - t)X$. In the original coordinates the cyclic transfer group is $C_N^{(X)} = \{[A(t)] : t \in \mu_N\}$, with N even. Its smallest form is $C_2^{(X)} = \langle x\rangle$. For $N > 2$, $j \notin G$ and $\widehat{G} \cong D_{2N}$. The dihedral forms are

$$D_{2m}^{(Z)} = \{[D(t)], [XD(t)] : t \in \mu_m\}, \quad D_{2m}^{(Y)} = \{[C(t)], [XC(t)] : t \in \mu_m\},$$
$$D_{2m}^{(X,0)} = \{[A(t)], [ZA(t)] : t \in \mu_m\}, \quad D_{2m}^{(X,1)} = \{[A(t)], [ZA(\eta t)] : t \in \mu_m\}.$$

Here $m \ge 3$, with even m in the last two forms and $\eta \in \mu_{2m} \setminus \mu_m$. The subscript gives the group order. The superscripts Z, Y, X specify the rotation axis, with rotation matrices in $\mathrm{span}\{I, Z\}$, $\mathrm{span}\{I, Y\}$, or $\mathrm{span}\{I, X\}$. The labels $(X, 0)$ and $(X, 1)$ distinguish the unshifted and shifted reflection cosets. In the former $j \in G$, whereas in the latter $j \notin G$. At the Klein-four boundary the two groups are isomorphic to V_4. In our coordinates the standard Pauli subgroup is $V_4^{(\mathrm{std})} = D_4^{(Z)} = D_4^{(Y)}$ and contains j. The exotic subgroup is $V_4^{(\mathrm{ex})} = D_4^{(X,1)}$ and does not contain j.

Platonic groups. The tetrahedral forms are $\mathcal{T}^{(\mathrm{int})}$, with $G = \widehat{G} \cong A_4$, and $\mathcal{T}^{(\mathrm{ext})}$, with $G \cong A_4$ normal in $\widehat{G} \cong S_4$. Their superscripts record whether j is internal or external to G. For the four octahedral forms $\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$, one has $G = \widehat{G} \cong S_4$. Each superscript gives the cycle types of (x, j), where d denotes a double transposition and t a transposition. Finally $\mathcal{I}^{(0)}, \mathcal{I}^{(1)}$ are two coordinate forms of the same abstract marked icosahedral type, with $G = \widehat{G} \cong A_5$. Explicit matrices and the complete marked classification appear in Theorem F.5.16.

If a retained extension gives $G < G^+$, a path in Figure 1 leads from the marked type of G to that of G^+, with divisibility steps within a family suppressed (Proposition F.5.18). The binaries realizable over the enlarged set determine which inclusions occur and which additional contractions can be used.

6 Reconstruction from stable contractions

Fix a stable pair with core f and retained set $\mathcal{F}$. Every allowed one-copy self-loop on f gives zero or a tensor product of retained binaries. We use these contractions to recover enough of f to determine the complexity of K-Holant($\mathcal{F}$) or to contradict the irreducibility of f. The difficulty is that each nonzero card comes with its own pairing of the remaining inputs and its own binary factors on those pairs.

Comparing contractions at a fixed pair. Fix a pair p of inputs of f. Closing p with different binary kernels gives cards on the same remaining inputs. We can therefore compare their values at each assignment. Write $\Phi_p^f(Q) = \partial_p^Q f$ for the card obtained with kernel Q. For example, suppose three available kernels satisfy $Q_3 = aQ_1 + bQ_2$, and write $C_i = \Phi_p^f(Q_i)$. Linearity of contraction gives

$$C_3(z) = aC_1(z) + bC_2(z)$$

for every assignment z to the remaining inputs. Wherever C_1 and C_2 both vanish, C_3 must vanish as well. A proposed choice of matchings and factors that violates this identity at even one assignment is ruled out.

Testing a disagreement at low arity. Suppose two cards pair one input with different partners. To test whether these pairings are compatible, we keep the inputs that exhibit the difference and close pairs elsewhere. On disjoint pairs these closures commute with the contraction at p. The two smaller cards can thus be compared as contractions of the same smaller signature. Each closure sums coefficients, so a choice of kernel can erase the very entries that distinguish the pairings. The group-specific argument must exhibit available kernels that keep the relevant entries nonzero. If the difference survives down to bounded arity, the low-arity classification can rule it out or yield a further reduction.

Once the matching is fixed, extracting a common binary on a chosen edge requires the factors there to be proportional across the nonzero cards. To test this, fix the inputs outside that edge and list the four values obtained by varying its two inputs. If two such vectors, from different cards or different assignments, were not proportional, no single binary could factor out on that edge. A gadget testing this possibility must preserve entries that witness the failure of proportionality.

Lifting a common factor to f. Suppose the available kernels span $\mathrm{Mat}_2(\mathbb{C})$. Choose four that form a basis and close the same ordered pair p with each. Write $f_{ab}(z) = f(a, b, z)$, ordering the inputs with p first and letting the remaining assignment z vary. The coordinate kernel selecting ab is a linear combination of the four basis kernels, so f_{ab} is the same linear combination of their cards. If the nonzero cards all share a factor P on the same nonempty block of remaining inputs, each slice has the form $P \otimes r_{ab}$, allowing $r_{ab} = 0$. Putting the slices back together gives $f = P \otimes r$ up to input order, where the four r_{ab} are the slices of r obtained by fixing p to ab. Since r contains the inputs of p, this factorization contradicts irreducibility. If the kernels recover only combinations of slices, the group analysis must determine what those combinations imply.

Some comparisons require additional gadgets, including gadgets built from two copies of f. Stability applies to the prescribed one-copy cards. For each additional gadget we first extract the factors of its output and apply the reductions of Section 4 (Lemma F.5.12 and Lemma F.10.1).

7 How the group cases use the common mechanism

Dihedral groups. For $G = D_{2m}^{(Z)}$, $m \geq 3$, the allowed kernels at a fixed pair give two families of cards, up to nonzero scalars,

$$E(t) = f_{00} + t f_{11}, \qquad O(t) = f_{10} + t f_{01}, \qquad t \in \mu_m.$$

Within each family, the support constraints force one matching. On that matching the coefficient minors have degree at most two in t. They vanish at three distinct phases, forcing all but at most one binary factor to be fixed up to scalar (Lemma F.6.8). For arity at least eight, suppose two cards have different matchings or differ on two edges of the same matching. Monomial closures outside the deleted pair preserve this disagreement in an arity-six contraction of f. Its two inherited cards are incompatible with a matching product. Yet its first closure is a stable card, and further monomial closures preserve the matching product whenever the result is nonzero. This rules out the disagreement (Lemma F.6.6 and Lemma F.6.7). If two cards differ on edge e, a third differing from the first on another edge would differ from the second on both. Thus the nonzero cards share

a product on all but one common edge (any edge suffices if all cards are proportional). Two phase evaluations in each family recover the four slices, so this product factors out of f by Section 6.

At arity six a direct support and pencil analysis forces factorization (Lemma F.6.11). At arity four invertible dressings give an eight-vertex signature to which Section 3 applies. The forms $D_{2m}^{(Y)}$ and $D_{2m}^{(X,0)}$ reduce to the standard analysis by transformations of the whole network, including its edges. For $D_{2m}^{(X,1)}$, once the support matching is fixed, weighted two-copy gadgets compare the ratios of coefficients obtained by changing the assignment on one edge. After the lower-arity outputs have been processed, the resulting identities force these ratios to be independent of the other edge assignments. This yields the required factorization. Section F.6 gives the transformations and the low-arity classifications.

Cyclic groups. In the original coordinates $G = C_N^{(X)}$ and its transfer span is $\mathrm{span}\{I, X\}$. Write $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. Since $P^{-1}A(t)P = 2D(t)$, the group in Walsh coordinates is $P^{-1}GP = C_N^{\mathrm{std}} = \{[D(t)] : t \in \mu_N\}$. For $\widetilde{f} = (P^{-1})^{\otimes n} f$, the closures test the sum and difference of two slices. The nonzero cards are matching products whose overlaps constrain both support and coefficients. After the binary and arity-four reductions, a surviving core forces $N = 2$, excluding the $N > 2$ cyclic node of Figure 1. In the original coordinates

$$f \sim \mathrm{RM}_8 := \mathbf{1}[RM(1,3)], \qquad G = C_2^{(X)} = \{[I], [X]\}.$$

Its Walsh-coordinate group is $C_2^{\mathrm{std}} = \{[I], [Z]\}$. The code $RM(1,3)$ consists of the eight-bit evaluation vectors of affine linear functions on $\mathbb{F}_2^3$. Here $\sim$ allows a nonzero scalar, input permutation, and realizable input X-dressings. To identify it, the matching slices reconstruct a doubly-even self-dual support, with a unique weight-four word through each triple of coordinates. This forces the Reed–Muller support. Comparisons on its affine planes then fix the coefficients up to a sign character, removed by the allowed dressings. Applying the coefficient constraint after every available single-input dressing also forces the complete group to be $C_2^{(X)}$ in the original coordinates (Lemma F.7.10 and Lemma F.7.14).

This core supplies a reduction that strictly enlarges the binary group. Once RM_8 is available and the binary and arity-four cases have been resolved, the limited-appearance reduction of Shao and Cai [SC20, full version, Section 8.2, Lemmas 8.9–8.10] permits adjoining the Bell signatures. In the original coordinates the binaries Z, Y supply transfer classes $[ZX] = [Y]$ and $[YX] = [Z]$. Their matrix representatives lie outside $\mathrm{span}\{I, X\}$. The complete group therefore strictly enlarges and stabilization resumes. This is the shortcut stated in Remark F.7.13, which also describes the stronger internal structural classification retained in the full proof.

Klein-four groups. The standard Pauli subgroup $V_4^{(\mathrm{std})}$ supplies the four Bell kernels I, X, Z, Y. For two signatures of arity at least five, suppose their corresponding cards are linearly dependent for every input pair and each of the four kernels. The relation may depend on the pair and the kernel, and zero cards are allowed. Then the signatures themselves are linearly dependent (Lemma F.8.2). Apply this to a retained signature and its complex conjugate. If the signature is not proportional to a real signature, some pair card must have the same obstruction. Descent reaches arity four, where classification resolves the set or enlarges the group, unless an earlier reduction already gives progress. If all retained signatures are proportional to real signatures in the same coordinates, the real Holant dichotomy [SC20] applies.

For $V_4^{(\mathrm{ex})}$, the common matching method forces a proper factor. The arity-six classification supplies the base case. At higher arity the four cards at a fixed deleted pair share one residual

matching. If an edge has a fixed binary factor across these cards, reconstruction extracts it from f. Otherwise the four labels on each edge vary by a permutation. Crossing two edges compares these permutations and creates a support incompatible with a matching product. The arity-eight boundary needs two such contraction choices. From arity ten one suffices. This reduces the branch to the arity-four analysis. Theorem F.8.3 and Theorem F.8.10 give the two conclusions.

Platonic groups. Fix the deleted pair p and regard each available kernel as a projective point. For three collinear points, a linear relation among their kernel representatives gives the same relation among their cards under Φ_p^f. This constrains the possible matching products as in Section 6. A common factor can lift to f. In the octahedral and icosahedral cases a surviving disagreement can be localized to arity six. Finite classifications then give a quaternary signature covered by Section 3 or a binary whose transfer class enlarges the group (Section F.9). For $\mathcal{T}^{(\mathrm{ext})}$, the gadgets determine the complexity, strictly enlarge the group, or expose a proper factor of f, contradicting its irreducibility in the last case (Theorem F.9.28).

For $\mathcal{T}^{(\mathrm{int})}$ the whole retained set may admit a common affine basis. Work in the coordinates of the internal tetrahedral form. If a remaining irreducible is nonaffine, the shortening argument exposes a factor or produces a smaller signature that is still nonaffine in those coordinates (Lemma F.9.23). Iteration would reach the already excluded cases of arity at most eight. Applying this argument in the same basis to every remaining irreducible gives the common affine presentation.

8 Completing the dichotomy

For a set containing I, stabilization brings each unresolved case to a stable pair. The group arguments compare its contractions to reconstruct a proper factor, contradicting irreducibility, or reduce a disagreement to low arity. Together with the structural exceptions and the reductions that enlarge the binary group, these arguments classify stable pairs (Theorem F.5.14). Factor extraction and stabilization then give the dichotomy for sets containing I (Theorem F.5.2).

The model equivalence and criterion correspondence in Section 2 transfer this dichotomy to $\mathrm{Holant}(\mathcal{F}, X)$, giving Theorem 1.1. The recognition procedure of Theorem F.3.3 tests the four cases of $\mathsf{ReTract}(\mathcal{F} \cup \{X\})$ on the exact algebraic input, so the criterion is decidable.

AI methodology statement

We used OpenAI's GPT-5.6 Sol to assist with mathematical exploration and proof development for the arguments in this paper. In Sections F.6–F.9, we used AI to carry out most of the analysis within each group case, including developing proofs and writing code for the computations used in those proofs. All companion verification code was generated by OpenAI Codex and subsequently reviewed by the authors.

We used OpenAI's GPT-6 Astra to improve the writing and presentation, and support the Lean formalization. We formalized the proof in Lean and incorporated two corrections to the original arXiv version (arXiv:2609.00219v1). We corrected the signs in the two mixed-mark identities in Equation (132) and restored a common nonzero scalar factor in Lemma 9.19 and its application in the proof of Lemma 9.20. The corrected statements were verified in Lean. The authors independently checked the mathematical claims, proofs, and computational results and take full responsibility for the correctness and originality of the paper.

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Full paper

F.1 Introduction

Many counting problems ask for a global sum determined by local constraints: each assignment receives the product of its local weights, and these products are summed over all assignments. A complexity dichotomy identifies the choices of local signatures that admit polynomial-time evaluation and proves #P-hardness for every remaining choice. On the Boolean domain, this program has a long history in counting constraint satisfaction problems (#CSP). Creignou and Hermann classified the unweighted case [CH96]. Later dichotomies treated nonnegative rational weights [DGJ09], rational weights of mixed signs [BDG$^+$09], and complex weights [CLX14]. Complex weights permit cancellations that yield tractable cases absent from the nonnegative setting [CLX14], making the algebra of the weights part of the classification. Over arbitrary finite domains, counting-CSP dichotomies have likewise been established for unweighted constraints [Bul13, DR13], nonnegative rational weights [BDG$^+$12], nonnegative algebraic weights [CCL16], and algebraic complex weights [CC17].

The Holant framework brings the availability of local constraints into this classification problem [CLX09]. A counting-CSP instance can be represented by a bipartite graph whose variable vertices carry equality constraints, enforcing agreement among all occurrences of a variable. Holant allows the available vertex signatures to be prescribed without supplying these equalities. This accommodates problems such as counting perfect matchings, where each vertex requires exactly one incident edge to be selected. It also makes the construction of useful local signatures a part of the classification proof. Holographic transformations, originating in Valiant's algorithms [Val08], give another source of tractability: a simultaneous change of basis can expose structure in the local signatures while preserving the value of the total sum.

Boolean Holant dichotomies have addressed both restrictions on the signatures and assumptions about auxiliary signatures. With no prescribed auxiliary signatures, classifications were obtained for symmetric real-valued signatures [HL16], symmetric complex-valued signatures [CGW16], arbitrary nonnegative signatures [LW18], and arbitrary real-valued signatures [SC20]. Here symmetry means that a signature depends only on the number of ones in its input. Another line of work supplies unary signatures: Holant* supplies all of them [CLX11], Holant$^+$ supplies four distinguished ones [Bac17], and Holant^c supplies the two pinnings $\Delta_0 = [1, 0]^{\mathsf{T}}$ and $\Delta_1 = [0, 1]^{\mathsf{T}}$. The last model was classified for symmetric complex signatures [CHL12], arbitrary real signatures [CLX18], and algebraic complex signatures [Bac21, Bac25]. The real-valued classification introduced the local-affine tractable family $\mathscr{L}$, which also occurs here.

Meng, Wang, Xia, and Zheng proved a tensor-factor extraction theorem and classified complex-valued Boolean Holant when a nonzero odd-arity signature is present [MWXZ25, Theorems 29 and 30]. They collected seven cases in condition (PC). For two cases defined by support, the weighted Eulerian-orientation results of [MWX25] gave an $\mathrm{FP}^{\mathrm{NP}}$ upper bound. Guan, Shao, and Shi subsequently gave polynomial-time algorithms for these cases [GSS26]. Taken together, the seven cases give the most general known sufficient condition for tractability of algebraic complex-valued Boolean Holant. We denote their disjunction by Tract. Even-arity signatures present a different resource constraint. Every gadget built entirely from them has an even number of external inputs, because each internal edge consumes two inputs. Such gadgets cannot supply a unary signature, so binary signatures are the smallest positive-arity objects available for further constructions. We study the case in which binary disequality, requiring two bits to differ, is supplied.

There is also a precedent for prescribing disequality in counting CSP. In $\#\mathrm{CSP}_d$, each variable occurs a number of times divisible by d. Cai, Fu, and Shao classified $\#\mathrm{CSP}_d(\neq_2, \mathcal{F})$ for complex

signatures [CFS20b, full version, Section 5.1 and Theorem 5.3]. Shao and Cai used its hardness conclusion in their real Holant dichotomy [SC20, full version, Theorem 2.24].

Prescribing binary disequality is useful for three further reasons. First, ordinary edges enforce equality and the supplied binary signature enforces disequality. Equivalently, we may use disequality edges and supply binary equality as a vertex signature (Lemma F.5.1). Together these constraints restrict the realizable binaries in Xia's finite-group framework [Xia26, Section 4 and Table 2] to the cyclic, dihedral, Klein-four, and Platonic forms in Theorem F.5.16. Second, switching between the two formulations only requires inserting or removing binary vertices. Every signature in the original set keeps its entries, so we can use whichever edge model suits a construction or tractable case. Third, the resulting classification can be applied to other Holant problems whenever a reduction makes binary disequality available. When disequality cannot be obtained, that failure gives a structural restriction on the remaining problems.

The problem and main results. A Boolean signature of arity n is a function $f : \{0,1\}^n \to \mathbb{C}$. In a Holant instance, a graph vertex of degree n carries such a signature, with its incident edges ordered. For a fixed finite set $\mathcal{F}$ of signatures, the problem Holant($\mathcal{F}$) evaluates

$$\sum_{\sigma:E\to\{0,1\}} \prod_{v\in V} f_v(\sigma|_{E(v)}), \qquad f_v \in \mathcal{F}.$$

Here $E(v)$ is the ordered list of edge incidences at v. Section F.2.3 gives the conventions for loops and multiple edges. The graph and its vertex labels form the input. The set $\mathcal{F}$ is fixed. Write

$$X = (\neq_2) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad I = (=_2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Our problem is Holant($\mathcal{F}, X$) := Holant($\mathcal{F} \cup \{X\}$). Thus any number of degree-two vertices may enforce opposite values on their two incident edges.

With X supplied, Lemma F.2.7 rules out three of the seven cases of Tract. We write ReTract for the disjunction of the remaining four. The signature set lies in the tensor closure $\langle \mathscr{T} \rangle$ of signatures of arity at most two, or it is transformable into the affine $\mathscr{A}$, product $\mathscr{P}$, or local-affine $\mathscr{L}$ class. In each transformability case, one change of basis must place the edge signature of the model and all the vertex signatures in that class. The classes and common transformability are defined in Section F.2.10.1 and Definition F.2.3, and the four cases are collected in Section F.2.10.3. Here "Re" refers both to the restriction of the general tractability criterion Tract to these four tractable cases and to their correspondence with the tractable cases in the real-valued Holant dichotomy of Shao and Cai [SC20].

Theorem F.1.1 (Binary-disequality Holant dichotomy)**.** *For every finite set of algebraic complex-valued Boolean signatures $\mathcal{F}$, the problem* Holant($\mathcal{F}, X$) *is polynomial-time computable when* ReTract($\mathcal{F} \cup \{X\}$) *holds and is* #P*-hard otherwise. The tractability criterion is decidable from exact algebraic input.*

With disequality edges, the proof also classifies sets containing an endpoint-nondegenerate irreducible signature of arity four, without requiring a supplied equality signature. Following Meng, Wang, Xia, and Zheng [MWXZ25], write K-Holant($\mathcal{F}$) := Holant($\neq_2 | \ \mathcal{F}$) for the bipartite Holant problem with disequality on the left and signatures from $\mathcal{F}$ on the right. Each connection between signature inputs thus enforces disequality. In the following theorem, transformability in ReTract uses this edge X.

A quaternary signature q is *endpoint-nondegenerate* when

$$q(\mathbf{x}) = 0 \quad \text{for odd } \mathrm{wt}(\mathbf{x}), \qquad q(0000)q(1111) \neq 0.$$

Here $\mathrm{wt}(\mathbf{x})$ counts the ones in $\mathbf{x}$. The eight even-weight inputs are the allowed configurations of an eight-vertex signature. The two endpoint entries are required to be nonzero. A nonzero positive-arity signature is *irreducible* if it has no factorization $a \otimes b$ on two nonempty blocks of inputs, allowing a permutation of inputs and a nonzero scalar.

Theorem F.1.2 (Quaternary K-Holant dichotomy)**.** *Let $\mathcal{F}$ be a finite set of algebraic complex-valued Boolean signatures. Suppose that $\mathcal{F}$ contains an endpoint-nondegenerate irreducible quaternary signature. Then* K-Holant($\mathcal{F}$) *is polynomial-time computable if* $\mathsf{ReTract}(\mathcal{F})$ *holds and is* #P*-hard otherwise. The tractability criterion is decidable from exact algebraic input.*

Cai and Fu's eight-vertex classification determines the complexity for a single eight-vertex signature q with nonzero endpoints [CF23, arXiv version 2, Theorem 3.1]. If q alone is tractable, its interaction with the remaining signatures can still make the problem hard. Section F.4 treats these interactions. Consequently, whenever a later reduction realizes an endpoint-nondegenerate irreducible eight-vertex signature, Theorem F.1.2 classifies the entire available set.

F.1.1 Proof strategy

Two approaches to Boolean Holant dichotomies motivate our proof. Shao and Cai use arity descent in the real-valued setting, reducing a signature outside the relevant tensor-product class to a smaller such signature or a low-arity case [SC20, Sections 3–4]. Their orthogonality and norm arguments rely on real-valuedness, or the corresponding conjugation symmetry after a holographic transformation. Arbitrary complex signatures lack this guarantee, so the descent needs another way to control cancellations. Xia's approach organizes the remaining problems by their finite binary groups [Xia26, Sections 4–5]. This specifies the binary constructions available in each case, but leaves open what they force about higher-arity signatures.

We combine the two approaches in the disequality-edge formulation, where binary equality is available by Lemma F.5.1. After the preliminary reductions in Section F.3 (see also [CFX18, CFS20a]), choose a nonbinary irreducible signature f of minimum arity in the current set $\mathcal{F}$. The complete transfer group G of $\mathcal{F}$ consists of the projective classes $[BX]$ as B ranges over all nonsingular ordered binary signatures directly realizable over $\mathcal{F}$, with transfer matrices identified up to nonzero scalar. Close a pair of inputs of f using such a binary. By Lemma F.3.4, we can retain the resulting signature's irreducible factors by reduction. An unresolved nonbinary factor has smaller arity than f and continues the descent.

A contraction that produces only binary factors can also advance the proof. If a retained factor has transfer class outside G, the complete group of the enlarged set strictly contains G. The factor is available by reduction, so gadgets over the enlarged set can use binaries that were not directly realizable over $\mathcal{F}$. We can now test the same f with additional contractions. Figure F.1 shows the possible strict inclusions. Which one occurs depends on the binaries realizable over the enlarged set.

If neither the minimum arity nor the group changes and the reductions have not settled the complexity, processing the prescribed contractions leaves every nonzero output as a product of retained binaries on a perfect matching of the remaining inputs. We call the contraction outputs *cards*. Together with zero, the cards obtained from one copy of f by attaching available binaries to selected inputs and closing one pair form its *matching deck*. When every nonzero card has such a decomposition with its factors retained, the state is *deck-stable* (Definitions F.5.7 and F.5.8). The

group-order bound, arity descent, and finite processing within each stage give the descent measure of Lemma F.5.12. Lemma F.10.1 verifies descent in all four group branches. Since each extension retains the preceding set, hardness transfers back along the reductions, and a common tractable basis restricts to that set (Lemma F.3.5).

F.1.2 Structural arguments in the group cases

The availability of both $=_2$ and $\neq_2$ restricts the forms of the finite binary groups in the given basis. In the disequality-edge formulation, equality supplies a fixed involution in the group, while reversal of binary inputs imposes an additional symmetry. Together these constraints refine the abstract group classification into the specific forms of Theorem F.5.16. These forms give explicit restrictions on the supports and coefficients of the cards.

Fix a pair of inputs of f. Each nonzero card obtained by closing this pair has a matching decomposition, but two cards may pair the same remaining input with different partners. Even when their matchings agree, extracting a common binary factor on an edge requires the factors there to be proportional across the cards being compared. The proportionality scalar can be absorbed into the rest of each product. The group arguments use further contractions to test a disagreement at lower arity or use compatible factors to recover a factor of f, as explained in Section F.5.5.

Dihedral groups. The group binaries constrain the supports and coefficient ratios of the cards, but these constraints must agree across contractions. Superposing two matchings gives alternating cycles. Section F.6 uses these cycles to localize matching disagreements to six inputs. In the standard case, excluding these disagreements forces a common factor, which two phase evaluations recover in the original signature. This contradicts irreducibility. The weighted dihedral case requires additional support and coefficient reconstruction, including Laurent interpolation, to obtain the same reduction to arity four.

Cyclic groups. Fewer independent contractions are available in this case. After the arity-four cases are resolved or yield group enlargement, Section F.7 identifies the surviving minimum-arity signature, up to input permutations, bit flips and a nonzero scalar, with the indicator of the length-eight Reed–Muller code $RM(1,3)$. The proof reconstructs its support and coefficients by comparing contractions across input pairs. This signature then realizes sums over affine fibres (Lemma F.7.16). These sums test the supports and coefficients of the rest of the set. When the tests produce no further reduction, they place the whole set and the edge in the affine class in one basis.

Klein-four groups. In the standard Pauli form, compare two signatures of the same arity above four by contracting each input pair with each of the four binary Bell signatures. If the corresponding cards are linearly dependent in every comparison, the signatures are linearly dependent. Section F.8 applies this test to each signature and its complex conjugate to detect whether the signature can be made real by rescaling. A failure descends to a quaternary signature with the same obstruction, whose analysis resolves the complexity or produces further progress. If every signature can be made real, the real Holant dichotomy applies to the whole set. The other Klein-four form requires a direct factorization argument. Its six-input classification forces agreement among residual factors, which the crossed contractions then use to obtain factors at higher arities. These factors exclude a surviving nonbinary irreducible signature.

Platonic groups. Fix a pair of inputs and regard the available contraction kernels as projective points. Linear relations among their representatives pass to the corresponding cards and constrain

their matchings. Section F.9 uses these relations to obtain a common factor or localize differing matchings to six inputs. A common factor contradicts irreducibility, while localized disagreements give small signatures to which the results for arity-four signatures or binary-group analysis can be applied. The octahedral and icosahedral analyses then produce quaternary signatures or new binaries. The internal tetrahedral case requires testing whether the whole set is affine in one basis. Stabilizer-code constructions shorten a non-affine signature while preserving its failure of affine membership, unless a complexity conclusion, factor, or larger group is obtained first. Once the descent reaches at most eight inputs, the low-arity classifications rule out a surviving non-affine obstruction. This completes the induction to a common affine basis for the remaining set and its edge.

The low-arity classifications that require exact finite enumeration, together with their completeness arguments, are stated in Appendix F.A. Section F.10.1 assembles the four branches, and Section F.10 transfers their conclusions to the original problem. The recognition procedure tests the criterion on the input set itself (Lemma F.2.7 and Theorem F.3.3).

F.2 Preliminaries

We fix the common algebraic, tensor, and reduction conventions, define the signature classes, and state the imported dichotomies used across the proof. The constructive recognition and retained-reduction arguments are in Section F.3. Specialized tools are introduced where their applications begin.

F.2.1 Foundational algebraic conventions

F.2.1.1 Sets, maps, and exact fields

We use $[n] = \{1, \dots, n\}$, $[0] = \varnothing$, the indicator $\mathbf{1}[\mathsf{P}]$, cardinality $|S|$, complement S^c in the stated ambient set, disjoint union $\sqcup$, restriction $f|_S$, and fibres $\pi^{-1}(y) = \{x : \pi(x) = y\}$. Tuples and binary strings are ordered lexicographically.

For a field L, write $L^\times = L \setminus \{0\}$, $\mathrm{Mat}_{m\times n}(L)$ for the m-by-n matrices over L, and $\mathrm{Mat}_n(L) = \mathrm{Mat}_{n\times n}(L)$. We use $\mathbb{Z}_m = \mathbb{Z}/m\mathbb{Z}$ and the binary field $\mathbb{F}_2 = \mathbb{Z}_2$. The function $\varphi(m)$ counts the residue classes modulo m coprime to m.

We write $\overline{\mathbb{Q}} \subseteq \mathbb{C}$ for the field of algebraic numbers. We write $\overline{L}$ for an algebraic closure, $[L' : L]$ for a finite extension degree, and $L(\alpha)$ for the generated field. An L-conjugate of α is its image under an embedding $L(\alpha) \hookrightarrow \overline{L}$ fixing L. $\bar{z}$ denotes complex conjugation. Exact algebraic values are represented by a minimal polynomial and a rational isolating region selecting the intended complex root [Coh93].

F.2.1.2 Linear, polynomial, and tensor conventions

All vector spaces are finite-dimensional unless stated otherwise. We use $\ker T$, $\operatorname{im} T$, $\operatorname{rank} T$, $\operatorname{span}_L S$, $U \oplus V$, $W^\vee = \mathrm{Hom}_L(W, L)$, and W/U with their usual meanings. An affine subspace is a coset $\mathbf{a} + U$, with translation space U. For matrices, A^{T} denotes transpose, $\operatorname{tr} A$ trace, I_n the identity, and $\mathrm{GL}_n(L)$ the invertible matrices. $\mathrm{GL}(V)$ denotes the invertible linear maps on V. A pivot is a selected nonzero entry used in elimination or normalization. An L-matrix algebra is an L-subspace of $\mathrm{Mat}_n(L)$ closed under multiplication. It is unital if it contains I_n and transpose-stable if it is closed under transpose. Matrix units E_{ij} use the stated index order. Boolean indices give $E_{00}, E_{01}, E_{10}, E_{11}$. We use the polynomial ring $L[x_1, \dots, x_n]$ and monomials $\mathbf{x}^{\mathbf{u}} = \prod_j x_j^{u_j}$, $\mathbf{u} \in \mathbb{Z}^n_{\geq 0}$.

With fixed bases, tensor products have coordinates

$$(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_r)_{i_1,\ldots,i_r} = (\mathbf{v}_1)_{i_1} \cdots (\mathbf{v}_r)_{i_r}.$$

Such a product is a simple tensor. The map $A^{\otimes r}$ applies A to every factor. A bipartite flattening groups selected tensor coordinates into a row index and the complementary coordinates into a column index. Its flattening rank is the resulting matrix rank.

F.2.1.3 Groups and projective spaces

We use $H \le G$ for subgroups, $H \lhd G$ for normal subgroups, $[G : H]$ for index, G/N for a quotient, and $\langle S \rangle$ for the subgroup generated by S. A group presentation is written $\langle S : \mathcal{R} \rangle$. The order of an element is its least positive identity power, or infinity. An involution has order two. The exponent of a finite group is the least common multiple of its element orders. Bounded torsion means that one integer $m \ge 1$ satisfies $g^m = 1$ for every element. We use the center $Z(G)$ and automorphism group $\mathrm{Aut}(G)$. For a group action, Gx and G_x denote the orbit and stabilizer. A linear representation is faithful when its homomorphism to $\mathrm{GL}(V)$ is injective. An action of H by automorphisms of N gives $N \rtimes H$, with product $(n, h)(n', h') = (n(h \cdot n'), hh')$. The normalizer and centralizer are

$$N_G(H) = \{g : gHg^{-1} = H\}, \qquad C_G(H) = \{g : gh = hg \text{ for every } h \in H\}.$$

Write S_m, A_m, C_m, D_{2m}, and V_4 for the symmetric, alternating, cyclic, dihedral, and Klein four groups. Our convention is $D_{2m} = \langle r, s : r^m = s^2 = 1,\ srs = r^{-1} \rangle$, of order $2m$: $\langle r \rangle$ consists of rotations and its other coset of reflections. For a finite set V, $\mathrm{Sym}(V)$ is its permutation group. A double transposition is a product of two disjoint transpositions.

For a finite-dimensional L-space V, its projective space is $\mathbb{P}_L(V) := (V \setminus \{0\})/L^\times$. When $L = \mathbb{C}$ or is clear we write $\mathbb{P}(V)$. The class of $\mathbf{v} \neq 0$ is $[\mathbf{v}]$, and the line through distinct $[\mathbf{u}], [\mathbf{v}]$ is $\mathbb{P}(\mathrm{span}_L\{\mathbf{u}, \mathbf{v}\})$. A projective subspace $\mathbb{P}(U)$ has dimension $\dim U - 1$. An invertible linear map induces a projective automorphism, and scalar multiples induce the same one. We put

$$\mathrm{PGL}(V) := \mathrm{GL}(V)/(L^\times I_V), \qquad \mathrm{PGL}_n(L) := \mathrm{GL}_n(L)/(L^\times I_n).$$

Here I_V is the identity linear map on V. The diagonal torus in $\mathrm{PGL}_n(L)$ is the subgroup of projective classes of nonsingular diagonal matrices. For a nonsingular 2-by-2 matrix T, the *projective order* of $[T] \in \mathrm{PGL}_2(\overline{\mathbb{Q}})$ is the least $m \ge 1$ for which $T^m = \lambda I$ for some $\lambda \in \overline{\mathbb{Q}}^\times$, and is infinite if no such m exists.

F.2.2 Complexity conventions

We use polynomial-time Turing reductions $\le_{\mathrm{T}}$, allowing adaptive exact oracle queries. $\equiv_{\mathrm{T}}$ means reduction in both directions. The classes FP, $\#\mathsf{P}$, and $\mathsf{FP}^{\mathsf{NP}}$ have their usual meanings. For algebraic-valued problems, $\#\mathsf{P}$-hardness means that every integer-valued $\#\mathsf{P}$ function reduces to the exact algebraic oracle, including polynomial-time postprocessing of its answers.

Signature sets, their coefficient fields, and fixed gadgets are fixed independently of the input network. Thus polynomial time measures the network's encoding length. Exact answers use the chosen effective number field. Recognition of a tractable signature set is a separate task whose input consists of finite exact signature tables. We prove decidability for that task without an encoding-size complexity bound.

F.2.3 Signatures, tensors, and Holant instances

We use the fields $\overline{\mathbb{Q}}$ and $\mathbb{F}_2$ fixed above. For $r \geq 0$, an arity-r Boolean signature is a function $f : \{0,1\}^r \to \overline{\mathbb{Q}}$, identified with its lexicographically ordered column tensor. Its r input positions are ordered, and this order is part of the data. A *relation* is a $\{0,1\}$-valued signature, identified with its support. Arity zero, one, two, three, and four are called nullary, unary, binary, ternary, and quaternary. We write $\mathrm{arity}(f) = r$ and $\mathrm{supp}(f) = \{\mathbf{x} : f(\mathbf{x}) \neq 0\}$. For a Boolean string $\mathbf{x} \in \{0,1\}^r$, we also write $\mathrm{supp}(\mathbf{x}) := \{j \in [r] : x_j = 1\}$. The argument type distinguishes string support from signature support. Whole vectors are written in bold, while their scalar coordinates and input-position labels are not. All binary-vector arithmetic is over $\mathbb{F}_2$, with $\oplus$ used when bitwise addition should be explicit. The symbols $\mathbf{0}^r$ and $\mathbf{1}^r$ denote the all-zero and all-one strings, respectively. At $r = 0$, both denote the unique empty string.

For $T \subseteq [r]$, $\mathbf{1}_T \in \{0,1\}^r$ denotes the incidence vector of T. In particular $\mathbf{1}_\varnothing = \mathbf{0}^r$. We write $\mathbf{e}_j = \mathbf{1}_{\{j\}}$ for the j-th standard basis vector whenever the ambient arity is clear.

We write $\mathrm{diag}(a, b)$ for the diagonal matrix with entries a, b. For $s \in \mathbb{C}$, put $D(s) := \mathrm{diag}(1, s)$. This matrix is invertible exactly when $s \neq 0$, and has algebraic entries when s is algebraic.

Following [MWXZ25, Section 2.3.1], the signature matrix $M_f = M_{x_1 \cdots x_\ell, x_{\ell+1} \cdots x_r}(f)$ has rows indexed by the first ℓ bits and columns by the remaining bits, both in lexicographic order. Its entries are the corresponding values of f. The subscripts may also specify any other ordered partition of the input positions. For a binary signature, $M_f = (f(a, b))_{a,b \in \{0,1\}}$. We also use f for this matrix when the input order is fixed.

For $\tau \in S_r$, define the reordering of inputs $(\tau f)(\mathbf{x}) = f(x_{\tau^{-1}(1)}, \ldots, x_{\tau^{-1}(r)})$. With this convention, $\tau(\sigma f) = (\sigma \circ \tau) f$. On disjoint ordered inputs, $(f \otimes g)(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) g(\mathbf{y})$. A nonzero positive-arity signature is *irreducible* if, up to permutation of inputs and nonzero scalar, it cannot be written as $a \otimes b$ with both a and b of positive arity. A nonzero positive-arity signature that is not irreducible is *reducible*. We use *irreducible* as in [MWXZ25, Section 2.3.4]. The positive-arity convention excludes scalar factors. Prime factorization into irreducible signatures is unique up to these operations and factor order [MWXZ25, Lemma 16]. For any class $\mathscr{C}$ of signatures, its tensor closure $\langle \mathscr{C} \rangle$ consists of tensor products of members of $\mathscr{C}$, up to permutation of inputs and nonzero scalar. By convention, it also contains zero and the arity-zero empty product. Thus $\langle \mathscr{C} \rangle$ denotes tensor closure, whereas $\langle S \rangle$ denotes the subgroup generated by S. A *tensor factorization* partitions the ordered inputs into blocks and writes the tensor, up to a nonzero scalar and permutation of inputs, as a tensor product on those blocks. For example, a $2 + 2$ factorization of a quaternary has two binary factors.

For $\mathbf{a} \in \{0,1\}^r$, define $\delta_{\mathbf{a}} : \{0,1\}^r \to \overline{\mathbb{Q}}$ by

$$\delta_{\mathbf{a}}(\mathbf{x}) = \mathbf{1}[\mathbf{x} = \mathbf{a}].$$

We use the unary pinnings

$$\Delta_0 = [1, 0]^{\mathsf{T}}, \qquad \Delta_1 = [0, 1]^{\mathsf{T}},$$

which are point masses forcing the incident bit to 0 and 1, respectively.

For $r \geq 1$, a *complementary-pair signature* (CP) has the form

$$c_0 \delta_{\mathbf{a}} + c_1 \delta_{\bar{\mathbf{a}}}, \qquad c_0, c_1 \in \overline{\mathbb{Q}}, \quad c_0 c_1 \neq 0,$$

where $\bar{\mathbf{a}}$ is the bitwise complement. It is a *generalized equality* (GE) when $\mathbf{a} = \mathbf{0}^r$: its support is $\{\mathbf{0}^r, \mathbf{1}^r\}$, with the two nonzero weights still arbitrary. Replacing each argument bit x_j by $x_j \oplus a_j$ puts a CP signature in GE form as a tensor identity.

Define the nullary equality/unit by $(=_0)(\varnothing) := 1$. More generally, for $r \geq 1$, define the arity-r equality signature by

$$(=_r)(x_1, \ldots, x_r) = \mathbf{1}[x_1 = \cdots = x_r].$$

The binary equality and disequality tensors are

$$I = (=_2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad X = (\neq_2) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

A *weighted equality* is a binary signature with matrix $\mathrm{diag}(p, q)$, where $p, q \in \overline{\mathbb{Q}}$. It is nonsingular when $pq \neq 0$. Binary equality is the case $p = q = 1$.

Graphs are finite multigraphs with ordered incidences at each vertex. Loops and parallel edges are allowed, and a loop contributes two incidences to the degree. A matching is a family of disjoint two-element subsets, and is perfect when it covers its ground set.

A closed signature network Ω has ordinary tensor vertices v, each assigned a signature f_v and a linearly ordered set H_v of $\mathrm{arity}(f_v)$ half-edges. The sets H_v are disjoint. Edges pair these half-edges in a perfect matching. Loops join two half-edges of one vertex, and parallel edges are allowed. Each half-edge has its own Boolean variable. An edge $e = \{h, h'\}$ contributes $W_e(\tau_h, \tau_{h'})$, and a vertex contributes $f_v(\tau|_{H_v})$. The partition function sums their product over all half-edge assignments. The ordinary model uses $W_e = I$. The disequality-edge model uses $W_e = X$.

Write $Z_{\mathbf{W}}(\Omega)$ for this partition function with edge kernels $\mathbf{W} = (W_e)_e$, and abbreviate it to $Z_W(\Omega)$ when every edge has kernel W. An instance uses either the ordinary model Z_I or the disequality-edge model Z_X, unless a displayed I or X is supplied as an explicit degree-two signature/gadget. The problems $\mathrm{Holant}(\mathcal{F})$ and $\text{K-Holant}(\mathcal{F})$ ask for $Z_I(\Omega)$ and $Z_X(\Omega)$, respectively, on closed networks over $\mathcal{F}$. When all edge kernels are I, summing the half-edge variables forces the two endpoints of each edge to agree and recovers the usual one-bit-per-edge multigraph definition (including loops). For signature sets $\mathcal{G}$ and $\mathcal{F}$, the bipartite problem $\mathrm{Holant}(\mathcal{G} \mid \mathcal{F})$ is the ordinary (I-edge) half-edge problem on a bipartite network: vertices in the left part are assigned signatures from $\mathcal{G}$, and vertices in the right part are assigned signatures from $\mathcal{F}$. For X-edges no one-bit compression is made. Subdivision inserts degree-two vertices on an edge. Subdividing every X-edge by a degree-two X-vertex, with ordinary equality edges on its two sides, gives the equivalent bipartite formulation

$$\text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \mathrm{Holant}(X \mid \mathcal{F}).$$

Throughout, comma-separated signature or signature-set arguments mean adding their union. For example, $\mathrm{Holant}(\mathcal{F}, a, b)$ abbreviates $\mathrm{Holant}(\mathcal{F} \cup \{a, b\})$.

We write

$$\mathrm{Holant}(\mathcal{F}, X) = \mathrm{Holant}(\mathcal{F} \cup \{X\})$$

for the same ordinary equality-edge problem with the additional binary signature X available as a vertex. Subdividing every ordinary edge by an $=_2$-vertex gives

$$\mathrm{Holant}(\mathcal{F}) = \mathrm{Holant}(=_2 \mid \mathcal{F}).$$

An *open gadget* is a finite signature network with ordered, labelled dangling inputs. Fixing their bits and summing over the internal variables gives its signature T_Ω. Let $\partial\Omega$ be the ordered list of unpaired vertex half-edges and $E_{\mathrm{int}}(\Omega)$ the internal edges. For this boundary convention,

$$T_\Omega(\mathbf{z}) = \sum_{\substack{\tau : \bigsqcup_v H_v \to \{0,1\} \\ \tau|_{\partial\Omega} = \mathbf{z}}} \prod_v f_v(\tau|_{H_v}) \prod_{\{h,h'\} \in E_{\mathrm{int}}(\Omega)} W_{\{h,h'\}}(\tau_h, \tau_{h'}).$$

We also allow an isolated edge E_W as a binary gadget with signature W. A supplied I- or X-signature instead labels a tensor vertex. A closed component contributes its partition function, and the empty

network contributes the nullary unit 1. Joining inputs contracts their bits against the edge kernel, as specified in Section F.2.6. Appendix F.B gives the equivalent general representation with free edge endpoints and its boundary-coordinate correction. Both representations use the same replacement signature T_Ω.

F.2.4 Zero and nullary preprocessing

An instance containing an identically zero signature has value zero. A nullary signature contributes a scalar: delete its vertex and multiply that scalar back into the final value. The empty residual instance has value 1. After this preprocessing, we continue to write $\mathcal{F}$ for the input signature set and assume it contains neither zero nor nullary signatures. The tractable predicates are evaluated after this preprocessing. Intermediate constructions may still produce zero or nullary signatures, which are handled as specified in each argument.

F.2.5 Holographic transformations and the K-basis

Let $i \in \mathbb{C}$ satisfy $i^2 = -1$, and fix the K-coordinate matrix

$$K = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \qquad K^{-1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}.$$

The transformations below rewrite every tensor and its incident edge consistently, giving exact identities between partition functions. Their fixed algebraic coefficients lie in a finite extension of the input coefficient field. We follow the exact evaluation model of Section F.2.2.

For tensors or matrices, write $A \doteq B$ when $A = \lambda B$ for some $\lambda \in \overline{\mathbb{Q}}^\times$. Throughout the complexity arguments, we ignore fixed nonzero scalar factors of signatures. Rescaling a signature multiplies an instance value by a computable nonzero factor, so it does not change the complexity. This convention does not discard relative coefficients in sums or assignment-dependent weights.

In Holant($\mathcal{G} \mid \mathcal{F}$), write each left-side signature g of arity r as a row vector and each right-side signature f of arity s as a column vector, both in lexicographic order. For $M \in \mathrm{GL}_2(\overline{\mathbb{Q}})$, replace them by

$$gM^{\otimes r}, \qquad (M^{-1})^{\otimes s} f,$$

respectively. On each edge the two matrices cancel by $MM^{-1} = I$, so the transformed instance has the same value. Write $gM := gM^{\otimes r}$, $Mf := M^{\otimes s} f$, and

$$\mathcal{G}M := \{gM : g \in \mathcal{G}\}, \qquad M\mathcal{F} := \{Mf : f \in \mathcal{F}\}.$$

Applying the transformation in both directions gives

$$\mathrm{Holant}(\mathcal{G} \mid \mathcal{F}) \equiv_{\mathrm{T}} \mathrm{Holant}(\mathcal{G}M \mid M^{-1}\mathcal{F}).$$

For binary f, the right tensor action has signature matrix $M_{Mf} = MM_f M^{\mathsf{T}}$. For Theorem F.1.1, Lemma F.5.1 gives an equivalent disequality-edge problem with I supplied. For a signature in ordinary coordinates, write $\widehat{f} = K^{-1} f$ and $\widehat{\mathcal{S}} = K^{-1}\mathcal{S}$.

For ordinary Holant, the left signature $(=_2)$ has row vector $(1, 0, 0, 1)$. After right multiplication by $M^{\otimes 2}$, its (a, b)-entry is $\sum_{x \in \mathbb{F}_2} M_{xa} M_{xb} = (M^{\mathsf{T}} M)_{ab}$. Consequently,

$$\mathrm{Holant}(=_2 \mid \mathcal{F}) = \mathrm{Holant}((=_2)M \mid M^{-1}\mathcal{F}), \qquad (=_2)M = M^{\mathsf{T}} M.$$

Write

$$GO(X) = \{D \in \mathrm{GL}_2(\overline{\mathbb{Q}}) : D^{\mathsf{T}} X D = \lambda X \text{ for some } \lambda \in \overline{\mathbb{Q}}^\times\}$$

for legal global transformations preserving the disequality edge X projectively. For a closed network, if $D\Omega$ applies D at every ordinary vertex and $D^{\mathsf{T}}XD = \lambda X$, then

$$Z_X(D\Omega) = \lambda^{|E_{\mathrm{int}}(\Omega)|} Z_X(\Omega). \tag{F.1}$$

Thus the two problems differ only by the known edge-count scalar. For $m \geq 1$, define the group of m-th roots of unity by

$$\mu_m = \{z \in \overline{\mathbb{Q}} : z^m = 1\}.$$

An element $\omega \in \mu_m$ is a *primitive* m-th root of unity when m is the least positive integer satisfying $\omega^m = 1$. Since $K^{\mathsf{T}}K = X$, putting $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$ gives

$$\mathrm{Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \mathrm{Holant}(X \mid \widehat{\mathcal{F}}) \equiv_{\mathrm{T}} \text{K-Holant}(\widehat{\mathcal{F}}). \tag{F.2}$$

For original-coordinate signatures and signature sets, the hats in Equation (F.2) record passage to K-coordinates. Once we work with a signature set already in K-coordinates, we suppress these hats and denote its member signatures simply by f. Unless stated otherwise, signatures and signature sets appearing in a K-Holant problem are understood to be in K-coordinates, with edge signature X.

F.2.6 Gadgets, interpolation, and factor reductions

A *direct gadget* over $\mathcal{F}$ is a finite X-edge network of $\mathcal{F}$-vertices, possibly using the edge E_X, with an ordered exposed boundary. If its signature, defined in Section F.2.3, is $T_\Omega = \rho h$, where $\rho \in \overline{\mathbb{Q}}^{\times}$ is a nonzero scalar, then

$$\text{K-Holant}(\mathcal{F}, h) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}). \tag{F.3}$$

A fixed gadget has constant size. A gadget realizing ρh, $\rho \neq 0$, implements the same Turing reduction after an instance with N occurrences is rescaled by ρ^{-N}. We call such an h *realizable* over $\mathcal{F}$. More generally, a finite set S is *available by reduction* over $\mathcal{F}$ if

$$\text{K-Holant}(\mathcal{F}, S) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}).$$

Joining two boundary inputs contracts their signature coordinates against the default edge kernel W, in the inherited input order. On an ordinary boundary, this inserts a W-edge. An isolated E_W splices into one edge, and closing its two ends contributes $\sum_{a,b} W(a,b)^2 = 2$. Vertex replacement gives Equation (F.3). The same rules for general free endpoints, and their compatibility with $GO(X)$-transport, are proved in Appendix F.B. Every realizable signature is available by reduction. Tensor factorization and interpolation may also give reductions without realizing the resulting signature by a gadget. A gadget using such a signature therefore gives a reduction to the preceding set, but need not be realizable over that set. Algebraic operations on tensors may establish identities or select contractions. Making their outputs available still requires a gadget or a reduction. A *slice* fixes selected argument values. Implementing it by pinnings requires the corresponding signatures. In particular, bit flips at individual inputs are used only when available or inside a stated global transform. Interpolation recovers finitely many unknown coefficients from oracle evaluations of a parameterized gadget family by solving an invertible linear system. It supplies an oracle reduction.

Let f have arity r. For an ordered pair of distinct inputs $p = (i, j)$, a binary contraction kernel $C : \mathbb{F}_2^2 \to \overline{\mathbb{Q}}$, and an assignment to the remaining inputs $\mathbf{z}$ indexed by $[r] \setminus \{i, j\}$ in increasing input order, let $\mathbf{z}^{i \leftarrow a, j \leftarrow b}$ be the length-r string obtained by inserting a, b at inputs i, j. Define the pair contraction by

$$(\partial_p^C f)(\mathbf{z}) = \sum_{a,b \in \mathbb{F}_2} C(a,b) f(\mathbf{z}^{i \leftarrow a, j \leftarrow b}).$$

In the notation of Shao–Cai [SC20, full version, Section 2.4], their $\widehat{\partial}_{ij}\widehat{f}$ and $\partial_{ij}f$ are, respectively, our $\partial^X_{(i,j)}\widehat{f}$ and $\partial^I_{(i,j)}f$. Input order fixes the ordered pair (i,j). A *pair card* of f is one such contraction $\partial^C_p f$. We shorten this to *card* when the contraction is specified. A card is *proper* if its arity is strictly smaller than its parent's. Iterated Pauli and exotic cards are defined where contractions on lists of disjoint pairs are used.

For binary matrices U, V, the Frobenius pairing is bilinear:

$$\langle U, V\rangle_{\mathrm{Frob}} = \mathrm{tr}(U^{\mathsf{T}}V) = \sum_{a,b\in\mathbb{F}_2} U(a,b)V(a,b),$$

without conjugation. A *binary chain* is a serial composition of ordered binary gadgets joined by X-edges. A *dressed pair card* $\partial^C_p f$ uses a closing binary chain over the given signature set, attached through the two edge kernels. Suppose the closing subnetwork has ordered boundary tensor B. Its two attachment X-edges complement both boundary bits, so

$$C(a,b) = \sum_{u,v\in\mathbb{F}_2} X(a,u)B(u,v)X(v,b) = (XBX)(a,b) = B(1-a,1-b).$$

Exchanging the two ordered boundary inputs replaces B by B^{T}, and the same attachment therefore gives $XB^{\mathsf{T}}X$. These formulas fix the input order used below. Every dressed pair card so constructed is a gadget. Linear combinations of cards are only analytic comparisons.

For every ordered binary signature B, its transfer matrix is $T_B := BX$. If B is nonsingular, its transfer class is $[T_B] = [BX] \in \mathrm{PGL}_2(\overline{\mathbb{Q}})$. A nonzero singular matrix has a class only in the projective matrix space, and the zero matrix has no projective class.

F.2.7 Charge and Hamming layers

All charge terminology in this subsection is in K-coordinates. For $\mathbf{x} \in \{0,1\}^r$, define its Hamming weight by $\mathrm{wt}(\mathbf{x}) := |\{j : x_j = 1\}|$ and its doubled charge by

$$q_r(\mathbf{x}) = 2\,\mathrm{wt}(\mathbf{x}) - r.$$

A *Hamming layer* is the set of strings of one fixed Hamming weight. A string has even or odd parity according as its weight is even or odd, and a signature has *even-parity support* when it vanishes on every odd-parity string. We distinguish $\mathrm{supp}\, f \subseteq M$ from $\mathrm{supp}\, f = M$. Coefficients in a prescribed support set may vanish unless equality is required. Define the Hamming half-spaces

$$HW^{\geq}_r = \{\mathbf{x} : q_r(\mathbf{x}) \geq 0\}, \quad HW^{\leq}_r = \{\mathbf{x} : q_r(\mathbf{x}) \leq 0\}, \quad HW^{=}_r = \{\mathbf{x} : q_r(\mathbf{x}) = 0\},$$

and define $HW^{>}_r, HW^{<}_r$ analogously. A signature is $HW^{\geq}$ (respectively, $HW^{\leq}$) when its support lies in $HW^{\geq}_r$ (respectively, $HW^{\leq}_r$). A signature set is *common one-sided* when all members are $HW^{\geq}$ or all are $HW^{\leq}$. Otherwise it is *two-sided*, meaning that both charge signs occur across the supports of its members. A signature is *two-sided* when its own support contains strings of both charge signs. Common one-sided support is the exception to tensor-factor extraction in Theorem F.2.10. An arity-$2d$ *EO signature* is supported in $HW^{=}_{2d}$.

In a closed X-edge instance, each contributing assignment has one 0-bit and one 1-bit on every structural edge. Hence $\sum_v q_{\mathrm{arity}(f_v)}(\tau|_{H_v}) = 0$. For an open gadget, the same cancellation equates the total vertex charge with the boundary charge. The general-boundary calculation, including the free-endpoint correction, is given in Appendix F.B.2.

Remark F.2.1 (Notation relative to prior work). We follow [MWXZ25, Sections 2.1–2.3] in writing $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$, $HW^{\geq}$, $HW^{\leq}$, $HW^{=}$, and *EO signature*. We also use their equality and disequality notation $=_r$ and $\neq_2$, with matrix aliases $I = (=_2)$ and $X = (\neq_2)$. Subscripts in $HW^{\bullet}_r$ record arity.

F.2.8 Binary codes and affine geometry

The support reconstructions in Sections F.6 to F.8 use binary affine spaces and coefficient ratios to describe supports and phases. We fix their shared Boolean and affine conventions here. The symmetric-space tools are introduced with the flat-Lagrangian networks.

For a finite coordinate set S, put $\mathbb{F}_2^S := \{\mathbf{x} : S \to \mathbb{F}_2\}$. A *binary code* of length n is a subset of $\mathbb{F}_2^n$. Its elements are *codewords*. It is linear when it is a subspace and affine when it is a coset of a linear code. The truth table of a Boolean function lists its values in the fixed lexicographic input order. For a commutative ring R, a Boolean polynomial over R is an element of the quotient ring of $R[x_1, \dots, x_n]$ by the ideal generated by the polynomials $x_j^2 - x_j$, $j \in [n]$, and hence has a unique multilinear representative. Its degree is the largest number of distinct variables in a nonzero monomial, and a mixed term contains at least two distinct variables. An affine-linear form is a constant plus a linear form. A *quadratic phase* is $i^{Q(\mathbf{x})}$, where $Q : \mathbb{F}_2^n \to \mathbb{Z}_4$ has multilinear degree at most two and even coefficients on mixed terms. We call Q its quadratic phase polynomial. A character of an additive group is a group homomorphism into the multiplicative group of the coefficient field. These Boolean normal forms are instances of Boolean-lattice Möbius inversion [Car21, Chapters 1–2].

For an affine subspace $\mathbf{a} + U$, U is its *translation space*. An affine map $\mathbf{a} + U \to \mathbf{b} + V$ has the form $\mathbf{a} + \mathbf{u} \mapsto \mathbf{b} + T(\mathbf{u})$ for a linear map $T : U \to V$. If an affine surjection $\pi : \mathbf{a} + U \twoheadrightarrow D$ onto an affine space D is fixed, an *affine section* is an affine map $\sigma : D \to \mathbf{a} + U$ with $\pi \circ \sigma = \mathrm{id}_D$. For $S \subseteq [n]$, in increasing coordinate order, define coordinate projection and zero extension by

$$\pi_S : \mathbb{F}_2^n \to \mathbb{F}_2^S, \quad \pi_S(\mathbf{x}) = (x_i)_{i \in S}, \qquad \iota_S : \mathbb{F}_2^S \to \mathbb{F}_2^n, \quad (\iota_S \mathbf{y})_i = \begin{cases} y_i, & i \in S, \\ 0, & i \notin S. \end{cases}$$

For $Q : \mathbb{F}_2^d \to R$, with R a commutative ring, its *Boolean Möbius expansion* is the unique multilinear expression

$$Q(\mathbf{x}) = \sum_{S \subseteq [d]} \widehat{Q}_S \prod_{j \in S} x_j, \qquad \widehat{Q}_S = \sum_{T \subseteq S} (-1)^{|S|-|T|} Q(\mathbf{1}_T).$$

The coefficient table $(\widehat{Q}_S)_S$ is the Boolean Möbius transform. The formula is interpreted in the stated coefficient ring, including $\mathbb{F}_2$ and $\mathbb{Z}_4$. Direct substitution is the Möbius-inversion proof of uniqueness. This additive transform is distinct from the multiplicative Möbius inversion used later.

F.2.9 Bell signatures

The Bell tensors form a common basis for binary contractions. We fix their ordered signs.

We also fix the sign and skew matrices

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Y = ZX = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For $d, \ell, u, v \in \mathbb{F}_2$, define the ordered Bell signature

$$B_{d,\ell}(u, v) = \mathbf{1}[u \oplus v = d](-1)^{\ell u}.$$

Write $B_{\boldsymbol{\gamma}} = B_{d,\ell}$ for $\boldsymbol{\gamma} = (d, \ell)^{\mathsf{T}}$. Input order matters for $B_{1,1}$, which is negated under reversal.

With truth-table order $00, 01, 10, 11$, these four signatures are

$$\begin{aligned} B_{0,0} &= (1, 0, 0, 1) = I, \quad B_{0,1} = (1, 0, 0, -1) = Z, \\ B_{1,0} &= (0, 1, 1, 0) = X, \quad B_{1,1} = (0, 1, -1, 0) = Y. \end{aligned} \tag{F.4}$$

A *Pauli binary* is a nonzero scalar multiple of one of the four ordered Bell tensors $B_{\boldsymbol{\gamma}}$, retaining its input order. It is *realizable* over a signature set when a direct gadget over that set realizes it up to a nonzero scalar, as in Section F.2.6.

Remark F.2.2 (The Bell-signature dictionary of Shao–Cai). The tuple in Equation (F.4) is Shao–Cai's $(=_2^+, =_2^-, \neq_2^+, \neq_2^-)$ [SC20, Section 2]. $Y^{\mathsf{T}} = -Y$. Their holographic matrix named Z is our K, whereas our Z is only the diagonal sign matrix.

The Bell tables satisfy the exact bilinear orthogonality identity

$$\langle B_{\boldsymbol{\gamma}}, B_{\boldsymbol{\gamma}'} \rangle_{\mathrm{Frob}} = 2\,\mathbf{1}[\boldsymbol{\gamma} = \boldsymbol{\gamma}'].$$

For an ordered binary tensor A and an ordered pair of inputs (p, q), the notation $A^{(p,q)}$ places the first and second inputs of A on inputs p and q, respectively. For an ordered pair $e = (p, q)$, we also write $A^{(e)}$. For an even finite input set R, a *Bell product* on R is a nonzero scalar multiple of

$$\bigotimes_{e=(p_e,q_e)\in M} B_{\gamma_e}^{(p_e,q_e)}, \qquad \boldsymbol{\gamma}_e \in \mathbb{F}_2^2,$$

where M is a perfect matching of R and each pair of M has been given the displayed orientation. Reversing a pair carrying $B_{1,1}$ only changes the overall scalar sign, so this class is independent of the chosen orientations. For $R = \varnothing$, a Bell product is a nonzero nullary signature.

F.2.10 Tractable signature sets and imported dichotomies

F.2.10.1 The standard signature classes

We include the zero signature in every class below. The affine class $\mathscr{A}$ consists of signatures

$$f(\mathbf{x}) = \lambda\,\mathbf{1}[A\mathbf{x} = \mathbf{b}]\, i^{Q(\mathbf{x})}, \tag{F.5}$$

where $\lambda \in \overline{\mathbb{Q}} \setminus \{0\}$ and, for $r = \mathrm{arity}(f)$, there is an $m \geq 0$ with $A \in \mathbb{F}_2^{m\times r}$ and $\mathbf{b} \in \mathbb{F}_2^m$. The function $Q : \{0,1\}^r \to \mathbb{Z}_4$, where $\mathbb{Z}_4 = \mathbb{Z}/4\mathbb{Z}$, has a multilinear representative

$$Q(\mathbf{x}) = q_0 + \sum_{j=1}^{r} q_j x_j + \sum_{1\leq j<\ell\leq r} q_{j\ell} x_j x_\ell \pmod 4, \qquad q_{j\ell} \in \{0, 2\},$$

so it has degree at most two and even mixed coefficients. Equivalently, the phase is a product of fourth-root phases of affine linear forms interpreted as $\{0,1\}$-valued integers. Membership in $\mathscr{A}$ always refers to the displayed computational coordinates.

The product class $\mathscr{P}$ consists of zero and all signatures of the form $\lambda \prod_j h_j$, where $0 \neq \lambda \in \overline{\mathbb{Q}}$ and every factor is either a nonzero unary signature $u(x_i)$, a binary equality $I(x_i, x_j)$, or a binary disequality $X(x_i, x_j)$ on designated signature variables. Empty products and repeated variables are allowed. An inconsistent product denotes the zero signature.

Define the primitive eighth root and the phase matrix by

$$\zeta_8 := e^{\pi i/4} = \frac{1+i}{\sqrt{2}}, \qquad T_2 := D(\zeta_8).$$

The local-affine class $\mathscr{L}$ consists of the signatures f such that

$$\left(\bigotimes_{j=1}^{\mathrm{arity}(f)} T_2^{\alpha_j}\right) f \in \mathscr{A} \qquad \text{for every } \boldsymbol{\alpha} \in \mathrm{supp}(f).$$

Let $\mathscr{T}$ be the set of all unary and binary signatures, and define the signature class $\mathscr{M}$ by

$$\mathscr{M} = \{f : f(\mathbf{x}) = 0 \text{ whenever } \mathrm{wt}(\mathbf{x}) > 1\}.$$

The Reed–Muller arguments also use the following subclass of $\mathscr{A}$, whose supports are affine translates of self-dual binary codes. Let L be a linear subspace of $\mathbb{F}_2^{2k}$. Its orthogonal complement under the standard dot product $\mathbf{x} \cdot \mathbf{y} := \sum_{j=1}^{2k} x_j y_j \in \mathbb{F}_2$ is

$$L^{\perp} = \{\mathbf{y} \in \mathbb{F}_2^{2k} : \mathbf{x} \cdot \mathbf{y} = 0 \text{ for every } \mathbf{x} \in L\}.$$

Call L *self-dual for the dot product* when $L = L^{\perp}$, and an arity-$2k$ nonzero signature *flat-Lagrangian* when it has the form

$$f(\mathbf{x}) = c\,\mathbf{1}[\mathbf{x} \in \mathbf{a} + L], \qquad 0 \neq c \in \overline{\mathbb{Q}}, \quad \mathbf{a} \in \mathbb{F}_2^{2k}, \quad L = L^{\perp}.$$

Let FLag denote the zero signature together with all flat-Lagrangian signatures of every even arity.

Definition F.2.3 (Common transformability and K-presentation)**.** Let $\mathscr{C}$ be a signature class closed under nonzero scalars and permutations of inputs. A finite signature set $\mathcal{F}$ is $\mathscr{C}$-transformable if one $M \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ satisfies

$$M^{\mathsf{T}} M \in \mathscr{C}, \qquad M^{-1}\mathcal{F} \subseteq \mathscr{C}.$$

Equivalently, for $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$, one matrix $U = K^{-1}M$ gives the common K-presentation

$$U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1}\widehat{\mathcal{F}} \subseteq \mathscr{C}.$$

We abbreviate this condition by

$$\mathsf{Pres}_K(\mathcal{F}; \mathscr{C}) \iff \exists U \in \mathrm{GL}_2(\overline{\mathbb{Q}}) : \quad U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1}(K^{-1}\mathcal{F}) \subseteq \mathscr{C}. \tag{F.6}$$

If $\mathcal{F}$ is already in K-coordinates, its common $\mathscr{C}$ K-presentation is precisely

$$\mathsf{Pres}_K(K\mathcal{F}; \mathscr{C}) \iff \exists U \in \mathrm{GL}_2(\overline{\mathbb{Q}}) : \quad U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1}\mathcal{F} \subseteq \mathscr{C}.$$

These conditions include the edge and require one basis for the entire signature set, not one basis per signature. We use this definition for $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$ and for the flat-Lagrangian subclass. The character-flat subclass is introduced where it is used in Appendix F.D.

Since $\mathsf{FLag} \subseteq \mathscr{A}$, a common flat-Lagrangian K-presentation is a common affine K-presentation.

F.2.10.2 The Eulerian-orientation predicate

Two cases of Tract use the same Eulerian-orientation criterion. In the common one-sided case, we restrict each signature to its central Hamming layer. In the single-layer case with both charge signs represented across the set, we append pins to each signature to obtain central support. For an arity-r signature g, its central restriction is

$$g|_{\mathrm{EO}}(\mathbf{x}) = g(\mathbf{x})\mathbf{1}[q_r(\mathbf{x}) = 0].$$

It is zero for odd r. We omit zero restrictions from the signature set and define

$$\mathcal{F}|_{\mathrm{EO}} = \{g|_{\mathrm{EO}} : g \in \mathcal{F},\ g|_{\mathrm{EO}} \not\equiv 0\}.$$

This omission is a convention for the set. In a closed instance over a common-one-sided set, an occurrence whose central restriction is zero makes the instance value zero.

Let $g \neq 0$ be a signature of arity k supported on the single Hamming layer of weight d. Its *EO completion* [MWX25, Section 6.2] is the ordered tensor

$$g_{\to\mathrm{EO}} = \begin{cases} g \otimes \Delta_0^{\otimes(2d-k)}, & 2d \geq k, \\ g \otimes \Delta_1^{\otimes(k-2d)}, & 2d < k. \end{cases} \tag{F.7}$$

The original inputs precede the added unaries. A zeroth power is the nullary unit. The completed signature has even arity and central support, hence is EO. For a signature set of nonzero single-layer signatures, define its EO-completion signature set by

$$\mathcal{F}_{\to\mathrm{EO}} = \{g_{\to\mathrm{EO}} : g \in \mathcal{F}\}.$$

For an arity-$2d$ EO signature g, an oriented perfect pairing $\pi = ((a_1, b_1), \ldots, (a_d, b_d))$ has unordered pairs partitioning $[2d]$. For $\mathbf{z} \in \mathbb{F}_2^d$, let $\mathbf{y} \in \mathbb{F}_2^{2d}$ be the unique string with $y_{a_j} = z_j$ and $y_{b_j} = 1 - z_j$. The pairing induces

$$g^\pi(z_1, \ldots, z_d) = g(\mathbf{y}).$$

Definition F.2.4 (The predicate $\mathsf{EOTract}$)**.** For a finite EO signature set $\mathcal{F}$, $\mathsf{EOTract}(\mathcal{F})$ holds if there exist a single sign $\varepsilon \in \{+1, -1\}$ and a single class $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}\}$, both shared by the entire signature set, such that:

1. for every $g \in \mathcal{F}$ and all $\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma} \in \mathrm{supp}(g)$, if $\boldsymbol{\delta} = \boldsymbol{\alpha} \oplus \boldsymbol{\beta} \oplus \boldsymbol{\gamma}$, then $\boldsymbol{\delta} \in \mathrm{supp}(g)$ or $\varepsilon q_{\mathrm{arity}(g)}(\boldsymbol{\delta}) > 0$;
2. for every $g \in \mathcal{F}$ and every oriented perfect pairing π of its inputs, $g^\pi \in \mathscr{C}$.

In particular, $\mathsf{EOTract}(\varnothing)$ holds.

The common-sign condition follows [MWX25, Theorem 26]. The opposite-pair restrictions defining the affine and product EO classes appear in [MWX24, Definition 22]. Our compressed form substitutes $y_{b_j} = 1 - y_{a_j}$. After ordering the inputs as $(a_1, \ldots, a_d, b_1, \ldots, b_d)$, the full zero-extended restriction is

$$R_\pi g(\mathbf{z}, \mathbf{w}) = g(\mathbf{z}, \mathbf{w}) \prod_{j=1}^{d} X(z_j, w_j) = g^\pi(\mathbf{z}) \prod_{j=1}^{d} X(z_j, w_j).$$

For $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}\}$, adjoining these disequality factors preserves $\mathscr{C}$, so $g^\pi \in \mathscr{C}$ implies $R_\pi g \in \mathscr{C}$. Conversely, each z_j determines exactly one w_j, and

$$\sum_{\mathbf{w} \in \mathbb{F}_2^d} R_\pi g(\mathbf{z}, \mathbf{w}) = g^\pi(\mathbf{z}).$$

This variable elimination preserves affine and product normal forms: in the affine case it is affine-variable summation, and in the product case substituting $w_j = 1 - z_j$ preserves unary and equality/disequality factors. Hence $R_\pi g \in \mathscr{C}$ if and only if $g^\pi \in \mathscr{C}$. These are identities of normal forms. Gadget availability follows the conventions of Section F.2.6.

The source papers use unoriented pairs. Reversing one pair applies the bit permutation $z_j \mapsto 1 - z_j$, and reordering pairs is a permutation of inputs. Both preserve $\mathscr{A}$ and $\mathscr{P}$. Hence the oriented predicate and the source predicate are equivalent. For a fixed signature set, the number of pairings and orientations is a constant independent of the network input. If the signature set itself is input, these are finite exact checks within the decidability procedure of Section F.2.2. No polynomial bound in that input's encoding length is asserted.

F.2.10.3 The tractability predicates

The reductions use all seven cases of Tract for intermediate signature sets. Lemma F.2.7 identifies when these reduce to the four cases of $\mathsf{ReTract}$.

When either predicate below is applied to an arbitrary finite input set, its argument is first reduced to the nonzero positive-arity part according to Section F.2.4. The definitions may therefore be stated for preprocessed signature sets without carrying a separate superscript in the public criteria.

For a preprocessed signature set $\mathcal{F}$, define its K-coordinate image $\widehat{\mathcal{F}} = K^{-1}\mathcal{F}$. Define $\mathsf{Tract}(\mathcal{F})$ to be the disjunction of the following seven cases.

(1) The signature set $\widehat{\mathcal{F}}$ is common one-sided, in the explicit sense that

$$[\forall \widehat{f} \in \widehat{\mathcal{F}},\ \mathrm{supp}(\widehat{f}) \subseteq HW^{\geq}_{\mathrm{arity}(\widehat{f})}] \quad \text{or} \quad [\forall \widehat{f} \in \widehat{\mathcal{F}},\ \mathrm{supp}(\widehat{f}) \subseteq HW^{\leq}_{\mathrm{arity}(\widehat{f})}],$$

and $\mathsf{EOTract}(\widehat{\mathcal{F}}|_{\mathrm{EO}})$ holds.

(2) Every $\widehat{f} \in \widehat{\mathcal{F}}$ is supported on one Hamming weight d_f. Across the signature set, some string in a signature support has negative charge and some (possibly other) string in a signature support has positive charge, and $\mathsf{EOTract}(\widehat{\mathcal{F}}_{\to \mathrm{EO}})$ holds, with the completion operation defined in Equation (F.7).

(3) $\mathcal{F} \subseteq \langle \mathscr{T} \rangle$.

(4) $\mathcal{F} \subseteq \langle K\mathscr{M} \rangle$, or $\mathcal{F} \subseteq \langle KX\mathscr{M} \rangle$.

(5) $\mathcal{F}$ is $\mathscr{A}$-transformable.

(6) $\mathcal{F}$ is $\mathscr{P}$-transformable.

(7) $\mathcal{F}$ is $\mathscr{L}$-transformable.

Remark F.2.5. The matrix symbols in Item (4) act tensorwise at the arity of each signature. In particular,

$$f \in \langle K\mathscr{M} \rangle \iff \text{every irreducible factor of } K^{-1}f \text{ lies in } \mathscr{M},$$
$$f \in \langle KX\mathscr{M} \rangle \iff \text{every irreducible factor of } XK^{-1}f \text{ lies in } \mathscr{M}.$$

The four cases indexed by $\mathscr{T}, \mathscr{A}, \mathscr{P}, \mathscr{L}$ form the reduced tractability predicate used in the main theorems.

Definition F.2.6 (The reduced tractability predicate $\mathsf{ReTract}$)**.** For a preprocessed signature set $\mathcal{F}$, define

$$\begin{aligned} \mathsf{ReTract}(\mathcal{F}) \iff\ & \mathcal{F} \subseteq \langle \mathscr{T} \rangle, \\ & \text{or } \mathcal{F} \text{ is } \mathscr{A}\text{-transformable}, \\ & \text{or } \mathcal{F} \text{ is } \mathscr{P}\text{-transformable}, \\ & \text{or } \mathcal{F} \text{ is } \mathscr{L}\text{-transformable}. \end{aligned}$$

Thus $\mathsf{ReTract}$ is exactly the disjunction of Items (3) and (5) to (7).

To make the edge condition explicit, for $E \in \{I, X\}$ write $\mathsf{ReTract}_E(\mathcal{F})$ for the same four cases, with transformability requiring $M^{\mathsf{T}}EM \in \mathscr{C}$ and $M^{-1}\mathcal{F} \subseteq \mathscr{C}$ for one common M. The introduction

suppresses E, using I in Theorem F.1.1 and X in Theorem F.1.2. Elsewhere the unadorned $\mathsf{ReTract}$ is $\mathsf{ReTract}_I$, as in Definition F.2.6. Since $K^{\mathsf{T}}K = X$,

$$\mathsf{ReTract}_X(\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{ReTract}(K\mathcal{F}).$$

Indeed, substituting $M = KU$ identifies the edge condition $M^{\mathsf{T}}M \in \mathscr{C}$ with $U^{\mathsf{T}}XU \in \mathscr{C}$ and gives $M^{-1}K\mathcal{F} = U^{-1}\mathcal{F}$. Invertible single-input maps also preserve $\langle \mathscr{T} \rangle$. Thus the K-change expresses the same four tractable cases in ordinary-Holant coordinates.

Lemma F.2.7 (Reduction to $\mathsf{ReTract}$)**.** *Let $\mathcal{F}$ be a preprocessed signature set. If $K^{-1}\mathcal{F}$ contains a two-sided irreducible signature of arity at least two, then*

$$\mathsf{Tract}(\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{ReTract}(\mathcal{F}).$$

In particular, this equivalence holds if $X \in \mathcal{F}$ or $Z = \mathrm{diag}(1, -1) \in \mathcal{F}$.

Proof. Let $g \in K^{-1}\mathcal{F}$ be two-sided and irreducible, with $n := \mathrm{arity}(g) \geq 2$. Its support contains strings of both charge signs and hence of different Hamming weights. Thus Items (1) and (2) are impossible.

By irreducibility and the characterization preceding Definition F.2.6, Item (4) would require $g \in \mathscr{M}$ or $X^{\otimes n}g \in \mathscr{M}$. A signature in $\mathscr{M}$ is supported at weights at most one, whose doubled charges are at most $2 - n \leq 0$. Complementing all inputs reverses the charge sign. Either possibility therefore makes g one-sided, a contradiction. Only the four cases of $\mathsf{ReTract}$ remain. The reverse implication holds by definition.

For the binary special cases, $K^{-1}X = -iZ$ and $K^{-1}Z = I$. Both images have matrix rank two and nonzero values at 00 and 11, so they are irreducible and two-sided. □

In Theorem F.1.1, $\mathcal{F} \cup \{X\}$ contains X, so the first binary special case applies directly. In Theorem F.5.2, $I \in \mathcal{F}$ gives $Z = K^{\otimes 2}I \in K\mathcal{F}$, so the second applies. For Theorem F.1.2, the nonzero values at 0000 and 1111 make the irreducible quaternary two-sided. Its arity also excludes Item (3), since invertible maps at individual inputs preserve tensor irreducibility. Thus only the three transformability cases of $\mathsf{ReTract}$ remain under that hypothesis.

F.2.10.4 Imported dichotomies and decomposition

Unless an ordinary Holant symbol or an explicit source edge model $\mathrm{Holant}(X \mid \cdot)$ is displayed, the imported K-Holant statements below use K-coordinates (so the edge tensor is X). Applying the fixed K-change gives the corresponding ordinary statement. All signature sets in these statements are fixed and preprocessed. In particular, every later use of Theorem F.2.9 on a signature set $\mathcal{F}$ means the holographic equivalence

$$\text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \mathrm{Holant}(K\mathcal{F}), \qquad K\mathcal{F} = \{K^{\otimes \mathrm{arity}(g)}g : g \in \mathcal{F}\},$$

followed by the ordinary theorem on $K\mathcal{F}$ and the inverse holographic equivalence. The known nonzero edge and gadget scalars are restored by the record.

Theorem F.2.8 (Weighted Eulerian orientations [MWX25, Theorem 26]; [GSS26, Theorem 1.1])**.** *For each fixed finite preprocessed algebraic EO signature set $\mathcal{F}$, if $\mathsf{EOTract}(\mathcal{F})$ holds, then* K-Holant$(\mathcal{F}) \in \mathsf{FP}$. *Otherwise* K-Holant$(\mathcal{F})$ *is* $\#\mathsf{P}$*-hard.*

Theorem F.2.9 (Odd-arity Holant [MWXZ25, full version, Theorem 33; conference version, Theorem 30])**.** *Let $\mathcal{F}$ be a finite preprocessed algebraic signature set containing a nonzero odd-arity signature. Then* $\mathrm{Holant}(\mathcal{F}) \in \mathsf{FP}$ *if* $\mathsf{Tract}(\mathcal{F})$ *holds, and is* $\#\mathsf{P}$*-hard otherwise.*

Its first two cases were originally in $\mathsf{FP}^{\mathsf{NP}}$. The stated FP conclusion follows from [GSS26, Theorem 1.1 and Corollary 1.2]. Thus a set containing a nonzero odd-arity signature is covered by this classification. The remaining structural analysis concerns even-arity signatures. The next theorem allows us to separate their tensor factors.

Theorem F.2.10 (Tensor-factor extraction [MWXZ25, full version, Theorem 32 and Remark 39; conference version, Theorem 29])**.** *Let $\mathcal{F}$ be a finite preprocessed algebraic signature set in K-coordinates, and let $h = a \otimes b$ be a nonzero tensor product with algebraic factors a, b of positive arity (equivalently, apply the source theorem to the ordinary set $K(\mathcal{F} \cup \{h\})$). Any nullary scalar factor is absorbed during preprocessing. Unless $\mathcal{F} \cup \{h\}$ is common one-sided,*

$$\text{K-Holant}(\mathcal{F}, a, b) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}, h).$$

In the common-one-sided exception, K-Holant$(\mathcal{F}, h)$ is in FP when $\mathsf{EOTract}((\mathcal{F} \cup \{h\})|_{\mathrm{EO}})$ holds and is #P-hard otherwise, by the one-sided clause of the cited theorem and Theorem F.2.8.

In the remaining case, the displayed equivalence permits factor extraction. We next justify iterating this reduction and handle any odd-arity factors that it produces.

F.3 Shared Reductions and Effective Recognition

The structural classification needs both algorithms for its tractable cases and reductions that apply across several branches. We first establish recognition and evaluation, then show how adding factors and changing basis preserve the connection to the original problem. The remaining tools use binary interpolation, single-occurrence replication, and higher equalities to obtain nonzero unaries. The odd-arity dichotomy then classifies the whole signature set. Direct realization and Turing access follow Section F.2.6.

F.3.1 Recognition and evaluation of tractable cases

Recognition of the predicates in Section F.2.10 combines finite support and factor tests with common-basis tests for the transformable cases. Polynomial constraints decide the latter and produce an exact witness for the edge and all signatures. For each fixed signature set, the resulting representations give polynomial-time instance evaluation.

F.3.1.1 Effective algebraic geometry

The transformability test in Lemma F.3.1 asks for one matrix satisfying polynomial conditions for the edge and every signature. Work over the exact coefficient field L. For an ideal $I \subseteq L[x_1, \dots, x_n]$, write $V_{\overline{L}}(I)$ for its common zero set. Appendix F.C records the algebraic tools used to decide whether this set is empty and to recover an exact witness when it is nonempty.

Lemma F.3.1 (Effective algebraic transformability witnesses)**.** *Let $E \in \mathrm{Mat}_2(\overline{\mathbb{Q}})$, let $\mathcal{F}$ be a finite algebraic signature set, and let $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$. Suppose there is a matrix $U \in \mathrm{GL}_2(\mathbb{C})$ such that*

$$U^{\mathsf{T}} E U \in \mathscr{C}, \qquad U^{-1}\mathcal{F} \subseteq \mathscr{C},$$

where membership in $\mathscr{C}$ allows complex values for the free scalar and unary parameters in its normal forms. Then such a matrix exists in $\mathrm{GL}_2(\overline{\mathbb{Q}})$. Moreover, for exact algebraic input, the existence of a matrix is decidable and an algebraic matrix can be produced.

Proof. Let L be a number field containing the entries of E and $\mathcal{F}$, as well as i and $\zeta_8 = (1+i)/\sqrt{2}$ (equivalently i and $\sqrt{2}$). There are only finitely many combinatorial normal-form types compatible with the arities: affine supports and phase coefficients, product partitions with equality or disequality signs and unary weights, and, for $\mathscr{L}$, the finitely many prescribed T_2-twisted affine conditions. Enumerate one shared type choice for the transformed edge and all transformed signatures.

For $\mathscr{P}$, repeated factors cause no infinitude: at fixed arity we canonically enumerate the support set, tensor partition, factor multiplicities, equality/disequality signs, unary phase type, and permutation of inputs. On a fixed pair of variables, repeated copies of the same I- or X-factor are idempotent and are retained as a single factor. The simultaneous presence of an I- and an X-factor makes the whole product zero. A loop $I(x,x)$ is the scalar one and a loop $X(x,x)$ is zero. Unary factors on one variable are multiplied and combined into one unary (or zero). These simplifications give a finite canonical enumeration. For $\mathscr{L}$, enumerate the prospective transformed support set and affine phase data. Requiring entries to be zero outside this support and nonzero on it, together with the twist conditions, gives finitely many polynomial equalities and inequations. Thus every candidate, including one whose transformed support is not known in advance, is one of finitely many constructible systems.

Fix one such branch, and let $\mathbf{x}$ be the tuple of all its scalar variables: the four entries of U, the scalar and unary parameters in the chosen normal forms, and the auxiliary variables introduced below. Introduce an auxiliary scalar variable z and impose

$$z \det U = 1.$$

Here $\operatorname{adj}(U)$ denotes the classical adjugate, characterized by $U \operatorname{adj}(U) = (\det U) I$. Hence on this solution set $U^{-1} = z \operatorname{adj}(U)$. After this substitution, equality of every transformed tensor entry with its proposed normal form is a polynomial equation over L. If the branch also has the nonvanishing polynomial constraints $p_1, \dots, p_s \neq 0$, introduce one more auxiliary scalar variable t and replace all these inequations by the single equation

$$t \prod_{\ell=1}^{s} p_\ell = 1.$$

Thus the branch is exactly the zero set of a finitely generated ideal $I \subseteq L[\mathbf{x}]$.

Compute a Gröbner basis of I over the exact number field L. For the algebraic closure $\overline{L} \subseteq \mathbb{C}$ of this embedded field, ideal membership and the weak Nullstellensatz give

$$V_{\overline{L}}(I) = \varnothing \quad \Longleftrightarrow \quad 1 \in I,$$

and the latter is equivalent to the reduced Gröbner basis being $\{1\}$. Hence emptiness of each branch is decidable. Moreover, a complex solution makes I proper, so the weak Nullstellensatz supplies a point over $\overline{L} = \overline{\mathbb{Q}}$. Conversely every such point is a complex solution. In a zero-dimensional branch, standard elimination followed by triangular decomposition produces exact minimal polynomials and isolating data for one point. A nonempty positive-dimensional branch is also effective: enumerate algebraic tuples fairly, first by total exact encoding height and then lexicographically, and test every displayed equation exactly. The weak Nullstellensatz guarantees that this search eventually reaches a point of the branch. Reading the U-coordinates gives the required algebraic witness. These effective procedures follow from Gröbner-basis theory [Coh93, CLO15], without an encoding-size or running-time bound. □

F.3.1.2 Evaluation for signatures in $\mathscr{M}$

For signatures in $\mathscr{M}$, each vertex accepts at most one incident bit equal to one. This reduces evaluation to trees and unicyclic components, as the following lemma shows. Here a factor graph records signature vertices and their incidences with edge kernels. A unicyclic component has exactly one cycle, including a loop or a two-cycle formed by parallel edges. Cycles are identified under reversal and cyclic changes of starting point. Suppression removes subdivision vertices.

Lemma F.3.2 (Evaluation for $\mathscr{M}$)**.** *Fix a finite algebraic signature set, and let Γ be a closed factor-expanded finite X-edge network over it (each occurrence has first been replaced by its fixed irreducible factors, with the induced input connections and scalar). If a vertex signature is zero, return zero. Otherwise, for every vertex w, let h_w denote its nonzero signature, define its arity by $d_w := \mathrm{arity}(h_w)$, and suppose $\mathrm{supp}(h_w) \subseteq \{\mathbf{0}^{d_w}, \mathbf{e}_1, \dots, \mathbf{e}_{d_w}\}$. Define its weight at the all-zero string $a_w := h_w(\mathbf{0}^{d_w})$ and weight-one input coefficients $b_{w,p} := h_w(\mathbf{e}_p)$. Then $Z_X(\Gamma)$ is computable using $O(|V(\Gamma)| + |E_{\mathrm{int}}(\Gamma)|)$ arithmetic operations in the fixed coefficient field, and hence in polynomial bit time. $V(\Gamma)$ is the vertex set.*

Proof. For an X-edge exactly one of its two half-edges receives bit 1. Consequently assignments are in bijection with choices of one endpoint of each edge, subject to at most one chosen endpoint at each vertex. For such a feasible endpoint-choice assignment ω, write $\mathrm{occ}_\omega(w) \in \{0,1\}$ for the number of selected endpoints at w, and let $s_w(\omega)$ be the selected input when it exists. The weight of ω is

$$\prod_{\mathrm{occ}_\omega(w)=0} a_w \prod_{\mathrm{occ}_\omega(w)=1} b_{w,s_w(\omega)}.$$

The product of the fixed nonzero scalar factors recorded during the factor expansion is multiplied into the final component product (and a zero expansion scalar was already handled by returning zero), so it does not change the recurrence below. For a connected component, let e and v denote its edge and vertex counts. A component with $e > v$ has value zero. Every surviving component has $e = v - 1$ (a tree) or $e = v$ (unicyclic). On a rooted tree, let p_w be the parent-edge input at w, and let p_{wc} be the input at w of the child edge wc. If $F_w(1)$ (resp. $F_w(0)$) denotes the subtree sum when the parent edge selects w (resp. its parent), then

$$F_w(1) = b_{w,p_w} \prod_c F_c(1),$$
$$F_w(0) = a_w \prod_c F_c(1) + \sum_c b_{w,p_{wc}} F_c(0) \prod_{d \neq c} F_d(1).$$

At the root use the second line without a parent term. Order the children as $c_1, \dots, c_m$. Products and sums are evaluated without division by the prefix products $P_j := \prod_{\ell<j} F_{c_\ell}(1)$ and suffix products $S_j := \prod_{\ell>j} F_{c_\ell}(1)$, for which $P_j S_j = \prod_{\ell \neq j} F_{c_\ell}(1)$ even when some factors vanish. For a unicyclic component delete one cycle edge e, with endpoint inputs p_u at u and p_v at v. Enumerate the two legal choices $p \in \{p_u, p_v\}$, and let $v(p)$ be the vertex incident to the chosen input. The inputs are distinct even when e is a loop. For each choice, root the resulting tree at $v(p)$ and impose that p is already occupied through e: at that root use

$$b_{v(p),p} \prod_c F_c(1).$$

This forces every remaining incident edge at that root to select its child. Use the ordinary tree recurrence on all proper subtrees and perform this same forced-root calculation separately for

each p. Sum the two values. The prefix/suffix evaluations use linearly many field operations. Every intermediate entry is a sum of at most exponentially many products of $O(|V| + |E_{\text{int}}|)$ fixed coefficients. In a fixed rational basis of their number field, a common denominator and the numerator sizes therefore have $O(|V| + |E_{\text{int}}|)$ bits, giving polynomial bit time. □

F.3.1.3 Decidability and polynomial-time evaluation

Theorem F.3.3 (Recognition and evaluation)**.** *For every finite algebraic signature set $\mathcal{F}$, the predicate* $\mathsf{Tract}(\mathcal{F})$*, with the preprocessing convention of Section F.2.4, is decidable. If it holds, then* $\text{Holant}(\mathcal{F}) \in \mathsf{FP}$.

Proof. The two EO cases are decided by finitely many exact tests of support, layers, affine subspaces, and phases. Enumerate all oriented pairings and the four shared $(\varepsilon, \mathscr{C})$-choices. A signature's tensor decomposition can be found by enumerating all nontrivial bipartitions $S \mid S^c$, where $\varnothing \neq S \subsetneq [\text{arity}(f)]$ and $S^c = [\text{arity}(f)] \setminus S$, and testing whether the corresponding matrix flattening has rank one. Recursively applying this test to the factors decides Items (3) and (4). Meanwhile, Lemma F.3.1 with $E = I$ decides Items (5) to (7) and returns the common witness.

For the two EO cases, evaluation uses value-preserving transformations of the input instance. Charge conservation reduces Item (1) to Theorem F.2.8: in a closed contributing assignment all local charges have the common sign and sum to zero, so every local charge is zero and only the central restrictions contribute. Deleting the off-central entries therefore preserves the value. In Item (2), an occurrence of arity k_f and weight d_f has charge $c_f = 2d_f - k_f$. Each connected component C must satisfy

$$\sum_{v \in V(C)} c_{f_v} = 0.$$

Failure gives zero. Otherwise construct an auxiliary EO instance: replace each occurrence f_v by its fixed completion $(f_v)_{\to \text{EO}}$, and in each component pair the newly appended Δ_0-inputs (from $c_{f_v} > 0$) with the appended Δ_1-inputs (from $c_{f_v} < 0$) by X-edges. The charge balance makes the two numbers equal, and every original assignment has a unique extension to this auxiliary instance (and conversely), so its partition function is unchanged [MWX25, Section 6.2, proof of Corollary 68]. The finite completed signature set is EO, so Theorem F.2.8 supplies the polynomial-time evaluator. After expansion into irreducible factors, the arity-two branch is a graph of maximum degree two and is evaluated by transfer matrices on paths and cycles. For the matching cases, first apply the common K-coordinate equivalence, the possible global X-toggle, and expansion into irreducible factors. The toggle is a change preserving the disequality edge because $X^{\mathsf{T}} X X = X$. After this change, every counted irreducible factor h satisfies $h(\mathbf{x}) = 0$ for $\text{wt}(\mathbf{x}) > 1$. Only structural equality kernels, or degree-two equality representations introduced solely to state the source theorem, are suppressed before the factor graph is built. An explicit arity-two signature occurrence remains a factor vertex unless it has first been eliminated by the path/cycle preprocessing of Lemma F.5.1, with its scalar recorded. Thus Lemma F.3.2 gives a polynomial-time evaluator using linearly many field operations. The expansion has constant size per occurrence because the signature set is fixed. Finally, a common change-of-basis matrix M gives

$$\text{Holant}((=_2)M \mid M^{-1}\mathcal{F}), \qquad (=_2)M = M^{\mathsf{T}} M,$$

with both sides in the same class $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$. Forgetting the bipartition yields an ordinary $\text{Holant}(\mathscr{C})$ instance, computable in FP: affine instances are quadratic Gauss sums, product instances split into unary/equality/disequality components, and local-affine instances are a restriction of the corrected complex-valued Holant^c evaluator of [Bac21, Bac25] (the real-valued presentation [CLX18, Corollary 3.1] gives the same local routine). □

F.3.2 Retained reductions and normalization

Fix the preprocessed signature set $\mathcal{F}_0$ at the beginning of a structural branch. We form finite signature sets from $\mathcal{F}_0$ by adding realizable signatures, adding factors through the decomposition theorem, or, within the eight-vertex reduction, applying a global $GO(X)$ change of basis. We call a set obtained in this way a *retained set*. Each addition is justified by its gadget or factor reduction, and each change of basis acts on the whole set. Interpolation is used only to resolve the current case.

We record the current coordinates and signature set as $(A, \mathcal{F})$, with

$$A \in GO(X), \qquad A\mathcal{F}_0 \subseteq \mathcal{F}, \qquad \text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}_0).$$

Adding signatures in the current basis leaves A unchanged. The normalization data in Lemma F.3.7 records the coefficient field and pulled-back group. The eight-vertex interface returns its outputs in the original coordinates, as specified in Corollary F.4.8 and Section F.5.

For each signature, specify its gadget or reduction, input order, and changes of basis. Inverse transformations express it in any earlier basis. Nonzero scalar factors follow the convention in Section F.2.5.

F.3.2.1 Finite induction and termination

The reductions in Section F.5.2 retain tensor factors and sometimes enlarge the finite group of binary transfers. The orders below distinguish finite processing of a fixed family from a restart with a smaller arity or larger group.

A strict order is irreflexive and transitive. It is *well founded* when it has no infinite descending chain. A finite multiset over a set E records elements of E with multiplicity. The multiset sum $\uplus$ adds multiplicities. Given a strict order $<$ on E and finite multisets A, B over E, its strict multiset extension is defined by $A <_{\mathrm{mul}} B$ when

$$B = Z \uplus X, \qquad A = Z \uplus Y, \qquad X \neq \varnothing,$$

for finite multisets X, Y, Z over E, and every $y \in Y$ satisfies $y < x$ for some $x \in X$. The multiset extension of a well-founded strict order is well founded [DM79]. Lexicographic order compares the first coordinate at which two tuples differ. Finite lexicographic products of well-founded orders are well founded by induction on the number of coordinates. Well-founded induction proves a property at x assuming it at every strict predecessor of x.

We index unfinished signature occurrences by a finite multiset. Each occurrence retains its realizing gadget or reduction and its ordered inputs. To identify duplicate signatures, divide by the first nonzero coefficient in the fixed input order. Repeated occurrences are still counted with multiplicity in the induction measures. The reductions test their listed cases in order. The first applicable case determines the next step.

F.3.2.2 Factor closure

Record each recursive tensor factorization as a tree, retaining the ordered blocks of inputs, permutation of inputs, and nonzero scalar at every split. The leaves are the irreducible factors.

For a preprocessed finite signature set $\mathcal{F}$, choose an exact irreducible tensor factorization of every member and let $\Pi(\mathcal{F})$ be the joint signature set of all irreducible factors, with ordered blocks of inputs retained. We call $\mathcal{F}$ *factor-closed* when $\Pi(\mathcal{F}) \subseteq \mathcal{F}$.

Lemma F.3.4 (Factor closure)**.** *If $\mathcal{F}$ is not common one-sided, then*

$$\text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}, \Pi(\mathcal{F})). \tag{F.8}$$

Moreover, every invocation of factor closure is finite. The factorization procedure and its strictly decreasing potential are given in the proof.

Proof. Start with one root occurrence for each member of the current finite retained set and add a factor only through the decomposition theorem. Let h_u denote the signature carried by a nonzero node u. Such a node, carrying its ordered block of inputs, is either proved irreducible, or is replaced by one split

$$h_u \doteq a_u \otimes b_u, \qquad 1 \le \text{arity}(a_u), \text{arity}(b_u) < \text{arity}(h_u),$$

with the nonzero scalar and the permutation of inputs recorded with the factorization. Record and discard nullary scalars, and remove irreducible unary nodes before forming the unresolved queue. Normalize each node by its first nonzero entry and retain its ordered input block. Identical normalized signatures may share one retained signature, while their occurrences remain distinct work items. Let $\mathcal{W}$ be the multiset of unresolved occurrences, all of arity at least two. The potential

$$\Phi(\mathcal{W}) = \sum_{u \in \mathcal{W}} (\text{arity}(h_u) - 1) \tag{F.9}$$

is a nonnegative integer. Splitting an item of arity $r + s$ removes $r + s - 1$ and adds at most $(r - 1) + (s - 1) = r + s - 2$. Already pending or processed normalized children only lower this amount further, and identifying an irreducible signature removes its summand. Thus every step strictly decreases Φ, and $\Pi(\mathcal{F})$ is the finite set of canonical irreducible leaves. Its independence of processing order follows from uniqueness of irreducible tensor factorization in Section F.2.3.

Apply the decomposition theorem at each queued split $h_u = a_u \otimes b_u$ to the current signature set [MWXZ25, full version, Theorem 32 and Remark 39]. The original non-one-sided signature set remains a subset, so the exceptional case never reappears. Iteration adds every factor, while set inclusion gives the reverse reduction. These two directions give Equation (F.8). All reduction scalars and ordered blocks are available for later gadgets. □

For the initial reduction, each original signature can also be reconstructed as the tensor product of its irreducible factors, with its input permutation and nonzero scalar restored. Consequently, outside the common-one-sided case,

$$\text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{K-Holant}(\Pi(\mathcal{F})).$$

We may therefore take the initial signature set to consist of irreducible signatures and continue to denote it by $\mathcal{F}$. If a nonzero odd-arity factor occurs, Theorem F.2.9 applies after the fixed K-change. If all factors have arity at most two, the evaluation in Theorem F.3.3 applies. The remaining analysis starts with even-arity irreducible signatures and at least one of arity at least four. Later constructions can have reducible outputs. At those steps we use the extension $\mathcal{F} \cup \Pi(\mathcal{F})$ in Equation (F.8), retaining the outputs as well as their factors for the subsequent group comparison.

For a factor-closed signature set $\mathcal{F}$ containing an irreducible signature of arity at least three, define its minimum arity of a nonbinary irreducible signature by

$$\nu(\mathcal{F}) = \min\{\text{arity}(h) : h \in \mathcal{F} \text{ is irreducible and } \text{arity}(h) \ge 3\}.$$

This parameter is used only after the branch in which every irreducible factor has arity at most two has been removed.

F.3.2.3 Preservation under taking subsets and global normalization

To recover the tractability criterion for the original set, we must both restrict to a subset and undo any global normalization. The next two lemmas justify these steps.

Lemma F.3.5 (Preservation under taking subsets)**.** *If $\mathcal{F} \subseteq \mathcal{F}'$ are preprocessed signature sets and* $\mathsf{Tract}(K\mathcal{F}')$ *holds, then* $\mathsf{Tract}(K\mathcal{F})$ *holds.*

Proof. All cases except the single-weight one are hereditary by definition. In the latter case a signature subset either retains both charge signs and the same EO-completion witness, or becomes one-sided. In the latter event, its central members are unchanged by both restriction and EO completion, whereas all off-central members restrict to zero. Hence its nonzero central restrictions form a subset of the original completion signature set, and the same shared EO witness applies. □

Lemma F.3.6 ($GO(X)$-invariance of transported tractability)**.** *Let $\mathcal{F}$ be a finite preprocessed signature set and let $D \in GO(X)$. Then*

$$\mathsf{Tract}(K\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{Tract}\big(K(D\mathcal{F})\big).$$

Proof. Write the matrix $D = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, and let $\lambda \in \overline{\mathbb{Q}}^{\times}$ be its conformal multiplier, defined by $D^{\mathsf{T}} X D = \lambda X$. Comparing entries gives $ac = bd = 0$ and $ad + bc = \lambda \neq 0$. Thus D is diagonal or anti-diagonal. The former preserves every string in the support. The latter complements it. On a fixed layer its multiplier is constant. Hence the two EO cases are preserved (with the support side and shared sign possibly exchanged), irreducible factor arities are unchanged, and $\langle \mathscr{M} \rangle$ and $\langle X\mathscr{M} \rangle$ are preserved or exchanged.

Define the conjugate matrix $A = KDK^{-1}$, so $A^{\mathsf{T}}A = \lambda I$. If the matrix W is a common $\mathscr{C}$-transformability witness, define the matrix $W' := AW$. It satisfies

$$(W')^{-1} A(K\mathcal{F}) = W^{-1}(K\mathcal{F}) \subseteq \mathscr{C}, \qquad (W')^{\mathsf{T}} W' = \lambda W^{\mathsf{T}} W \in \mathscr{C}$$

for $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$. Applying the same argument to D^{-1} proves the converse. □

The following lemma records the data needed to transport reductions, group enlargements, and tractable presentations back to the original signature set. Its continuing-state use is confined to the eight-vertex reduction.

Lemma F.3.7 (Normalization data)**.** *Let $\mathcal{F}_0$ be the signature set at the beginning of a structural branch in which coordinate transport is permitted. Let L_0 be a fixed number field containing all entries of $\mathcal{F}_0$, their complex conjugates, and all algebraic constants already fixed by the coordinate conventions. Thus L_0 is the fixed initial coefficient field, not a field enlarged after each normalization. Every continuing normalization inside that branch is an explicitly recorded $GO(X)$ change of coordinates, and tensor splits are performed after pulling back by the current A, using a nonzero pivot and row–column ratios. Thus the pulled-back data (rather than the displayed current coordinates) stay over L_0, and the pulled-back group remains inside* $\mathrm{PGL}_2(L_0)$*. The structural reductions maintain a current signature set $\mathcal{F}$, a matrix $A \in GO(X)$, and its nonzero conformal factor $\mu_A \in \overline{\mathbb{Q}}^{\times}$, satisfying*

$$A\mathcal{F}_0 \subseteq \mathcal{F}, \qquad A^{\mathsf{T}} X A = \mu_A X, \qquad \text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}_0). \tag{F.10}$$

Every positive-arity $h \in \mathcal{F}$, up to a nonzero scalar, also has the field-compatible form

$$h = A^{\otimes \operatorname{arity}(h)} h_0 \quad \textit{for a tensor } h_0 \textit{ over } L_0.$$

At a residual node where every directly realizable nonsingular transfer has finite projective order, these transfers form a group. Denote by $G(\mathcal{F})$ the group of their classes $[BX]$, where B ranges over directly realizable nonsingular ordered binaries over $\mathcal{F}$. Then the pulled-back group

$$\overline{G}(A,\mathcal{F}) := [A]^{-1}G(\mathcal{F})[A] \le \mathrm{PGL}_2(L_0)$$

is unchanged by a global normalization and is monotone when signatures are added. Adding realizable signatures and factor closure preserve A. If $D \in GO(X)$ is applied globally to the current signature set, the record updates by

$$(A,\mathcal{F}) \longmapsto (DA, D\mathcal{F}).$$

The new reduction state again satisfies Equation (F.10)*, up to the known edge scalar from Equation* (F.1)*. Before a continuing branch leaves Section F.4, apply the inverse of the accumulated matrix A. The returned reduction state therefore has $A = I$ in the fixed original coordinates and retains any supplied equality signature. Generic interpolation is used only as a terminal transition and is not a record-preserving update. If an interpolation output were ever allowed to re-enter a continuing finite-group state, the effective coefficient field and the pulled-back group would first have to be enlarged explicitly.*

If $\mathsf{Tract}(K\mathcal{F})$ *holds, then* $\mathsf{Tract}(K\mathcal{F}_0)$ *holds. Separately, if V is a common $\mathscr{C}$ K-presentation of $\mathcal{F}$, where $\mathscr{C}$ is closed under nonzero scalar multiplication, then the pullback matrix*

$$W := A^{-1}V$$

is a common $\mathscr{C}$ K-presentation of $\mathcal{F}_0$. In particular, this applies to $\mathscr{A}, \mathscr{P}, \mathscr{L}$ and to the flat-Lagrangian subclass.

Proof. The assertions hold initially for $(I, \mathcal{F}_0)$. Adding realizable signatures and factor closure preserve them by Equations (F.3) and (F.8). A global $D \in GO(X)$, with nonzero conformal factor $\mu_D \in \overline{\mathbb{Q}}^{\times}$ defined by $D^{\mathsf{T}}XD = \mu_D X$, replaces A by DA. Then $(DA)^{\mathsf{T}}X(DA) = \mu_D\mu_A X$, and Equation (F.1) supplies the stated equivalence. Conversely, the transported inclusion $A\mathcal{F}_0 \subseteq \mathcal{F}$, followed by the inverse global transformation, reduces the initial problem to the current one.

Inductively pull every construction through A. Gadget contraction stays over L_0. At a tensor split, normalizing one nonzero entry and taking row–column ratios chooses the factors over L_0. Moreover $A^{\mathsf{T}}X \doteq XA^{-1}$. Thus, if a current binary B has the record form $B = A^{\otimes 2}B_0$ for a binary tensor B_0 over L_0, its projective transfer pulls back by conjugation:

$$[BX] = [A(B_0X)A^{-1}].$$

The identity transfer is realized by the disequality edge, and chaining realizes products. At such a finite-projective-order residual node, the inverse of each transfer is a positive power of that transfer, so the realizable classes form a group. The displayed conjugation shows that global normalization leaves $\overline{G}(A,\mathcal{F})$ unchanged, while adding signatures can only add realizable transfers. This proves the field and group assertions. Finally, if $\mathsf{Tract}(K\mathcal{F})$ holds, preservation under taking subsets first gives $\mathsf{Tract}(K(A\mathcal{F}_0))$, and Lemma F.3.6 then gives $\mathsf{Tract}(K\mathcal{F}_0)$. If V is an explicit common K-presentation, then $W = A^{-1}V$ satisfies

$$W^{-1}\mathcal{F}_0 \subseteq V^{-1}\mathcal{F} \subseteq \mathscr{C}, \qquad W^{\mathsf{T}}XW = \mu_A^{-1}V^{\mathsf{T}}XV \in \mathscr{C},$$

which proves the last assertion. □

F.3.3 Binary Interpolation

A singular nonzero binary gives unary factors. For an infinite-order transfer, interpolation first gives a rank-one binary. The following lemma establishes both reductions, including the size and query bounds needed for polynomial-time interpolation.

Lemma F.3.8 (Binary interpolation)**.** *Let $\mathcal{F}$ be a finite algebraic, retained, two-sided signature set, let $B \neq 0$ be an ordered binary directly realizable over $\mathcal{F}$, and define its transfer matrix $A = BX$.*

1. *If B has the rank-one factorization $B = u \otimes v$, where u and v are nonzero algebraic unary signatures, then*
$$\text{K-Holant}(\mathcal{F}, u, v) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$
2. *If A is nonsingular and has infinite projective order, there is a nonzero rank-one binary H such that*
$$\text{K-Holant}(\mathcal{F}, H) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$
For $m \geq 1$ designated occurrences of H, the interpolation uses $m+1$ oracle queries, $O(m^2)$ added vertices per query, and hence $O(m^3)$ added vertices over all queries.

Proof. For the singular case, choose a nonzero algebraic pivot of the rank-one matrix and take the corresponding algebraic row and column factors (absorbing the pivot into one factor). Then $B = u \otimes v$ with u, v algebraic, and Theorem F.2.10 applies. Two-sidedness excludes its one-sided exception. For the other case, a length-t chain realizes the ordered binary signature $B_t := A^t X$. If A is diagonalizable, choose an eigenbasis matrix $V \in \text{GL}_2(\overline{\mathbb{Q}})$ and nonzero eigenvalues $\lambda_0, \lambda_1 \in \overline{\mathbb{Q}}^{\times}$ such that $A = V \,\text{diag}(\lambda_0, \lambda_1) V^{-1}$. For $j \in \{0, 1\}$, let E_{jj} be the 2×2 matrix with (a, b)-entry $\mathbf{1}[a = b = j]$, for $a, b \in \{0, 1\}$, and define the rank-one binary matrix $H_j = V E_{jj} V^{-1} X$. Then

$$B_t = \lambda_0^t H_0 + \lambda_1^t H_1,$$

with H_0, H_1 rank one. Replacing the m designated H_0-vertices by B_t and dividing by λ_1^{tm} yields evaluations of a polynomial $P(z)$ of degree at most m at the distinct points $z = (\lambda_0/\lambda_1)^t$, $1 \leq t \leq m + 1$. The coefficient of z^m is the all-H_0 value. These points are distinct because $[A]$ has infinite projective order, so λ_0/λ_1 is not a root of unity.

In the remaining infinite-order Jordan case, write $A = \lambda I + N$, where $\lambda \in \overline{\mathbb{Q}}^{\times}$ is the eigenvalue and N is a nonzero nilpotent 2×2 matrix satisfying $N^2 = 0$. Then $B_t = \lambda^t (X + t\lambda^{-1} N X)$. After normalization the oracle value is a polynomial in t of degree at most m, whose coefficient of t^m is the value with the rank-one binary $\lambda^{-1} N X$ at every designated vertex. Indeed, $N \neq 0$ and $N^2 = 0$ in dimension two imply $\text{rank}(N) = 1$, and right multiplication by X preserves rank. Both cases use $m + 1$ queries. $t \leq m + 1$ gives $O(m^2)$ replacement size for each query and $O(m^3)$ in total. The eigenvalues and basis matrices are fixed algebraic numbers. Their powers and the interpolation systems have polynomial encoding length, so exact interpolation is polynomial-time postprocessing. □

Both reductions give access to a nonzero unary after factor extraction. The dichotomy in Lemma F.3.11 below then applies to the whole current set. Thus the later unresolved binary analysis concerns nonsingular transfers of finite projective order.

F.3.4 Single-occurrence reductions and nonzero unaries

The next reductions start with one occurrence of an auxiliary signature. A fixed gadget passes that signature to the next copy and supplies one additional output. Repeating the construction supplies as many outputs as needed while using the auxiliary signature only once. The single-occurrence replication result in Lemma F.3.10 originates in an unpublished expanded version of [MWXZ25], which is not publicly available. We include the statement and proof here, with the ordered connections explicit.

Define the ordered carrier signature

$$\Delta = \Delta_0 \otimes \Delta_1,$$

with the displayed order of its two inputs. For a signature g, the superscripts $g^{=1}$ and $g^{\le 1}$ mean that an oracle instance contains, respectively, exactly one or at most one g-vertex.

Lemma F.3.9 (Universal one-pair reduction)**.** *For every fixed finite algebraic signature set $\mathcal{F}$,*

$$\text{K-Holant}(\mathcal{F}, \Delta^{\le 1}) \le_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

Proof. The zero-occurrence case is already an instance over $\mathcal{F}$. Consider an instance with one Δ-vertex. Delete that vertex, retaining its two incident X-attachments as edges to ordered free boundary inputs. With the boundary bits read at these free endpoints, the raw signature is

$$H = \begin{pmatrix} h_{00} & h_{01} \\ h_{10} & h_{11} \end{pmatrix}, \qquad \text{the desired one-}\Delta\text{ value is } h_{01}.$$

Each free endpoint carries an X-edge, so the signature for vertex replacement is XHX by Equation (F.154). Closing the two exposed inputs by an X-edge gives an oracle value

$$S = h_{01} + h_{10}.$$

Let $\mathcal{B}_{\mathcal{F}}$ be the set of ordered binary tensors realized by finite $\mathcal{F}$-networks. If every $B \in \mathcal{B}_{\mathcal{F}}$ satisfies $B_{01} = B_{10}$, then $XHX \in \mathcal{B}_{\mathcal{F}}$ is symmetric, so H is symmetric and the desired value is $S/2$.

Otherwise fix once and for all a realized ordered binary

$$B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad b \ne c.$$

Insert its fixed network at the deleted location in the two boundary orientations. The two oracle values are

$$T_+ = ah_{00} + bh_{01} + ch_{10} + dh_{11}, \qquad T_- = ah_{00} + ch_{01} + bh_{10} + dh_{11}.$$

Thus

$$h_{01} = \frac{1}{2}\left(S + \frac{T_+ - T_-}{b - c}\right).$$

The ordered inputs are essential in the subtraction. The witness network for B is fixed with $\mathcal{F}$, hence has constant size, and the reduction uses at most three oracle queries. □

The preceding reduction allows one carrier occurrence. To obtain unrestricted access to a new signature, we use a gadget that preserves the carrier while producing another output. The next lemma states the required replication and nonzero-closure conditions for ordinary Holant and K-Holant, with equality and disequality edges, respectively.

Lemma F.3.10 (Single-occurrence replication)**.** *Let $\mathcal{F}$ be a fixed signature set and let g, h be nonzero signatures. For ordinary Holant, all connections use I. Suppose that*

1. $\text{Holant}(\mathcal{F}, g^{=1}) \leq_{\text{T}} \text{Holant}(\mathcal{F})$*;*
2. *a fixed gadget C over $\mathcal{F}$ has three disjoint ordered boundary blocks A, P, H, where $|A| = |P| = \text{arity}(g)$ and $|H| = \text{arity}(h)$. Connecting one g-vertex coordinatewise to A realizes $\rho\, g(P) \otimes h(H)$, for a nonzero scalar $\rho \in \overline{\mathbb{Q}}^{\times}$. Between consecutive copies $C_\ell, C_{\ell+1}$, connect P_ℓ coordinatewise to $A_{\ell+1}$, with no other input identifications. For every $k \geq 1$, this chain begins with exactly one g-vertex and realizes $\rho^k g(P_k) \otimes \bigotimes_{\ell=1}^{k} h(H_\ell)$, with pairwise disjoint output blocks $P_k, H_1, \ldots, H_k$; and*
3. *some fixed $\text{Holant}(\mathcal{F}, g^{=1})$-instance J has $Z_I(J) \neq 0$.*

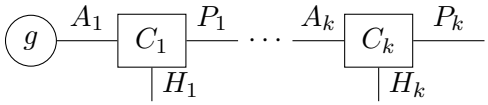
Then

$$\text{Holant}(\mathcal{F}, h) \leq_{\text{T}} \text{Holant}(\mathcal{F}).$$

For K-Holant, all connections instead use X, assumption 1 is $\text{K-Holant}(\mathcal{F}, g^{=1}) \leq_{\text{T}} \text{K-Holant}(\mathcal{F})$, and assumption 3 requires a $\text{K-Holant}(\mathcal{F}, g^{=1})$-instance J with $Z_X(J) \neq 0$. With item 2 imposed using these X-edges, the conclusion is

$$\text{K-Holant}(\mathcal{F}, h) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

Proof. By the ordered chain composition in item 2, iterating the gadget produces $g \otimes h^{\otimes k}$ with one g-vertex, size $O(k)$, and a nonzero scalar. Hence an instance containing one g and k designated h-vertices reduces to an instance containing exactly one g. Divide out the scalar. For an instance Ω over $\mathcal{F} \cup \{h\}$, take the disjoint union $\Omega \sqcup J$. Its value is $Z_I(\Omega) Z_I(J)$, so division by the fixed nonzero number $Z_I(J)$ inserts the one carrier required by the first reduction. The context and J are fixed, so the construction has size linear in the number of h-occurrences. Composing with item 1 proves the ordinary-Holant claim. For K-Holant, the same construction uses X-edges and the disjoint-union value is $Z_X(\Omega) Z_X(J)$. Division by the stipulated nonzero $Z_X(J)$ and the K-Holant version of item 1 give its stated reduction. □



The chain in item 2 of Lemma F.3.10. Each connection represents the corresponding ordered block of edges.

Once these reductions provide a nonzero unary, the imported odd-arity dichotomy applies to the extended set. Its conclusion transfers to the original set as follows.

Lemma F.3.11 (Dichotomy with a nonzero unary)**.** *Let $\mathcal{F}$ be a signature set and let $u \neq 0$ be unary. In ordinary Holant, if $\text{Holant}(\mathcal{F}, u) \leq_{\text{T}} \text{Holant}(\mathcal{F})$, then $\text{Holant}(\mathcal{F})$ is $\#\mathsf{P}$-hard unless $\mathsf{Tract}(\mathcal{F})$ holds. In the disequality-edge model, if $\text{K-Holant}(\mathcal{F}, u) \leq_{\text{T}} \text{K-Holant}(\mathcal{F})$, then $\text{K-Holant}(\mathcal{F})$ is $\#\mathsf{P}$-hard unless $\mathsf{Tract}(K\mathcal{F})$ holds.*

Proof. For the first assertion, apply Theorem F.2.9 to $\mathcal{F} \cup \{u\}$. A tractable presentation of that set restricts to one for $\mathcal{F}$. For the second, transform to ordinary coordinates:

$$\text{K-Holant}(\mathcal{F}, u) \equiv_{\text{T}} \text{Holant}(K\mathcal{F}, Ku).$$

The signature Ku is a nonzero unary, so the same argument applies. □

For unary columns g, u, their contraction is the bilinear scalar

$$g^{\mathsf{T}} u := \sum_{x \in \{0,1\}} g(x)u(x),$$

with no complex conjugation. For the higher-equality construction below, the carrier must have a nonzero closure. The next lemma supplies a companion unary with this property and a reduction allowing one occurrence of the pair.

Lemma F.3.12 (Unary companion)**.** *Let $\mathcal{F}$ be a fixed finite algebraic signature set and let $g \neq 0$ be unary. There is an algebraic unary u such that*

$$g^{\mathsf{T}} u \neq 0, \qquad \operatorname{Holant}(\mathcal{F}, (g \otimes u)^{=1}) \leq_{\mathrm{T}} \operatorname{Holant}(\mathcal{F}).$$

Proof. We first prove the claim for $g = \Delta_0$. Let $\mathcal{B}_{\mathcal{F}}$ be the set of ordered binary tensors realized by finite $\mathcal{F}$-networks with ordinary equality edges. Deleting one occurrence of $\Delta_0 \otimes u$, where $u = [s, t]^{\mathsf{T}}$, leaves a binary context $H = (h_{ij})$, and the desired value is $sh_{00} + th_{01}$. Equality-closing the exposed inputs gives the oracle value

$$S := h_{00} + h_{11}.$$

We choose u by the following exhaustive alternatives.

If every $H \in \mathcal{B}_{\mathcal{F}}$ satisfies $h_{00} = h_{11}$, take $u = [2, 0]^{\mathsf{T}}$. Then the desired value is S. Otherwise fix

$$G_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \in \mathcal{B}_{\mathcal{F}}, \qquad a_1 \neq d_1.$$

If every realized binary has equal off-diagonal entries, then $b_1 = c_1 = b$. For $b = 0$, take $u = [1, 0]^{\mathsf{T}}$. Inserting G_1 gives the oracle value $T := a_1 h_{00} + d_1 h_{11}$, whence

$$h_{00} = \frac{T - d_1 S}{a_1 - d_1}.$$

For $b \neq 0$, take $u = [a_1 - d_1, 2b]^{\mathsf{T}}$. The same insertion, now using $h_{01} = h_{10}$, gives

$$T - d_1 S = (a_1 - d_1)h_{00} + 2bh_{01}.$$

It remains that some

$$G_2 = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \in \mathcal{B}_{\mathcal{F}} \quad \text{satisfies} \quad b_2 \neq c_2.$$

Take $u = [a_1 - d_1, b_1 + c_1]^{\mathsf{T}}$. Let T_1 be the value obtained by inserting G_1, and let T_2, T_3 use the two orientations of G_2. Then

$$\frac{T_2 - T_3}{b_2 - c_2} = h_{01} - h_{10},$$

and therefore

$$T_1 - d_1 S + \frac{c_1}{b_2 - c_2}(T_2 - T_3) = (a_1 - d_1)h_{00} + (b_1 + c_1)h_{01}.$$

In every case the first entry of u is nonzero, so $\Delta_0^{\mathsf{T}} u \neq 0$.

Now write the unary column vector $g = [a, b]^{\mathsf{T}}$. If $a^2 + b^2 = 0$, then projectively $g \doteq [1, i]^{\mathsf{T}}$ or $g \doteq [1, -i]^{\mathsf{T}}$. In the first case take $u = [1, -i]^{\mathsf{T}}$, and in the second take $u = [1, i]^{\mathsf{T}}$. Under the

ordinary-to-disequality-edge K^{-1}-transformation the pair is, up to a nonzero scalar and input order, $\Delta_0 \otimes \Delta_1$. The reduction follows from Lemma F.3.9, and $g^{\mathsf{T}}u \neq 0$.

If $a^2 + b^2 \neq 0$, choose the scalar $r \in \overline{\mathbb{Q}}$ with $r^2 = a^2 + b^2$ and define the matrix

$$Q = \frac{1}{r}\begin{pmatrix} a & b \\ b & -a \end{pmatrix}.$$

Then $Q^{\mathsf{T}}Q = I$ and $Qg = r\Delta_0$. Apply the point-mass case to $Q\mathcal{F}$, obtaining $u' = [s, t]^{\mathsf{T}}$ with $s \neq 0$, and define the unary vector $u = Q^{-1}u'$. Because $Q^{\mathsf{T}}Q = I$, the holographic identity leaves every ordinary equality edge unchanged and pulls the reduction back to $\mathcal{F}$, while

$$g^{\mathsf{T}}u = (Qg)^{\mathsf{T}}u' = rs \neq 0. \qquad \square$$

Corollary F.3.13 (Higher equality exposes unaries)**.** *Let $d \geq 4$ and let $\mathcal{F}$ be a finite algebraic signature set. In ordinary Holant, the reduction* $\text{Holant}(\mathcal{F} \cup \{=_d\}) \leq_{\text{T}} \text{Holant}(\mathcal{F})$ *implies*

$$\text{Holant}(\mathcal{F} \cup \{\Delta_0, \Delta_1\}) \leq_{\text{T}} \text{Holant}(\mathcal{F}).$$

In the disequality-edge model, the reduction $\text{K-Holant}(\mathcal{F} \cup \{=_d\}) \leq_{\text{T}} \text{K-Holant}(\mathcal{F})$ *implies*

$$\text{K-Holant}(\mathcal{F} \cup \{\Delta_0, \Delta_1\}) \leq_{\text{T}} \text{K-Holant}(\mathcal{F}).$$

Proof. First work with ordinary equality edges and add the accessible $=_d$. For $j \in \{0, 1\}$, apply Lemma F.3.12 to $\mathbf{e}_j = \delta_j$, obtaining u_j with $u_j(j) \neq 0$ and the one-occurrence carrier $v_j = \mathbf{e}_j \otimes u_j$. Attach the $\mathbf{e}_j$-input of that carrier to one input of $=_d$. Up to input order the output is

$$v_j \otimes \mathbf{e}_j^{\otimes(d-2)}.$$

Closing the carrier through equality has value $u_j(j) \neq 0$, so Lemma F.3.10 gives access to $\mathbf{e}_j^{\otimes(d-2)}$. Passing temporarily to K-coordinates, the unary $K^{-1}\mathbf{e}_j$ has two nonzero entries. Hence this tensor power is not in the common-one-sided exception of Theorem F.2.10. Factor extraction therefore exposes $\mathbf{e}_j$ after transforming back.

Now work in K-Holant and use the carrier Δ. Attach its two factors, through X-edges and in the two orientations, to two copies of $=_d$. The output is

$$\Delta^{\otimes(d-1)} = \Delta \otimes \Delta^{\otimes(d-2)}$$

up to input order. The X-contraction of Δ has value one, so Lemmas F.3.9 and F.3.10 give access to $\Delta^{\otimes(d-2)}$. The available $=_d$ contains both endpoint strings $\mathbf{0}^d, \mathbf{1}^d$, of doubled charges $-d, +d$, and hence excludes the common-one-sided exception. Applying Theorem F.2.10 exposes Δ_0 and Δ_1. Finally compose with the assumed reduction that removes $=_d$. $\square$

Either point mass is a nonzero unary signature, so Lemma F.3.11 determines the complexity of the original problem. Thus a reduction obtaining $=_d$, for $d \geq 4$, ends the current structural case.

F.3.5 Equality and generalized-equality anchors

We now apply the higher-equality reduction after a common change of basis, and then adapt the carrier construction to arbitrary nonzero endpoint weights. The first application requires the same basis change to turn the edge into equality and the quaternary signature into a multiple of four-ary equality. We call this configuration an equality anchor.

Definition F.3.14 (Equality anchor)**.** A signature set $\mathcal{F}$ has an *equality anchor* if it directly realizes an ordered quaternary signature g for which there exist a matrix $U \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ and nonzero algebraic scalars $\kappa, \lambda \in \overline{\mathbb{Q}}^{\times}$ satisfying

$$(U^{-1})^{\mathsf{T}} X U^{-1} = \kappa\,(=_2), \qquad U^{\otimes 4} g = \lambda\,(=_4).$$

Lemma F.3.15 (Dichotomy from a transformed quaternary equality)**.** *Let $\mathcal{F}$ be a finite algebraic signature set with an equality anchor g, and fix witnesses U, κ, λ as in Definition F.3.14. Then either* K-Holant$(\mathcal{F})$ *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{F})$ *holds.*

Proof. Apply the disequality-edge holographic identity with $M = U^{-1}$ to the whole problem. It sends each right signature f to Uf and the disequality edge to $(U^{-1})^{\mathsf{T}} X U^{-1} = \kappa(=_2)$. For a queried instance Ω, the edge factor $\kappa^{|E_{\mathrm{int}}(\Omega)|}$ is known. Thus the direct anchor becomes $Ug = \lambda(=_4)$, so $=_4$ is accessible in the ordinary-Holant problem over $U\mathcal{F}$. By Corollary F.3.13, a nonzero unary u is accessible there. Pulling the reduction back exposes the nonzero unary $U^{-1}u$ over $\mathcal{F}$. The conclusion is now Lemma F.3.11. □

Lemma F.3.16 (Dichotomy from a generalized equality)**.** *Suppose a retained finite algebraic signature set $\mathcal{F}$ directly realizes the arity-q generalized-equality signature*

$$G_q := c_0 \delta_{\mathbf{0}^q} + c_1 \delta_{\mathbf{1}^q}, \qquad c_0, c_1 \in \overline{\mathbb{Q}}^{\times}, \qquad q \geq 4.$$

Then either K-Holant$(\mathcal{F})$ *is* #P*-hard or* $\mathsf{Tract}(K\mathcal{F})$ *holds.*

Proof. Direct realizability gives K-Holant$(\mathcal{F} \cup \{G_q\}) \leq_{\mathrm{T}}$ K-Holant$(\mathcal{F})$. We use this extension in the construction below. Use the ordered carrier Δ from Section F.3.4. Attach its two factors through X-edges, in opposite orientations, to one input of each of two copies of G_q. The two copies are forced onto opposite endpoint strings, and the external tensor is

$$c_0 c_1\, \Delta^{\otimes(q-1)} = c_0 c_1\, \Delta \otimes \Delta^{\otimes(q-2)}$$

up to input order. The X-contraction of the carrier has value one. Thus Lemmas F.3.9 and F.3.10 give access to $\Delta^{\otimes(q-2)}$. Since G_q contains both endpoint strings, of doubled charges $-q, +q$, Theorem F.2.10 has no common-one-sided exception and exposes Δ_0 and Δ_1. Apply Lemma F.3.11 over $\mathcal{F} \cup \{G_q\}$. A hard conclusion composes back to $\mathcal{F}$, while $\mathsf{Tract}(K(\mathcal{F} \cup \{G_q\}))$ implies $\mathsf{Tract}(K\mathcal{F})$ by Lemma F.3.5. □

F.4 Eight-Vertex Signatures with Nonzero Endpoints

The eight-vertex dichotomy classifies the problem with a single quaternary signature. When that signature is realized over a larger set, its hard case transfers immediately, but its tractable cases do not classify the other signatures. We analyze those product, affine, and local-affine cases using the unary and equality reductions of Section F.3. They either classify the whole retained set or factor the original quaternary into weighted equalities. Its irreducibility in Theorem F.1.2 rules out this last outcome. The group analysis also calls this reduction on possibly reducible quaternary signatures and can retain their factors.

Theorem F.4.1 (Quaternary eight-vertex boundary [CF23, arXiv version 2, Theorem 3.1])**.** *Let M be an algebraic quaternary signature satisfying*

$$\mathrm{supp}(M) \subseteq \{\mathbf{x} \in \{0,1\}^4 : \mathrm{wt}(\mathbf{x}) = 2\} \cup \{0000, 1111\}, \qquad M(0000)M(1111) \neq 0.$$

Then Holant$(X \mid M)$ *is* #P*-hard unless there exist* $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$ *and* $U \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ *such that*

$$U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1} M \in \mathscr{C}.$$

Every exception is in FP*. The source permits* U *over* $\mathbb{C}$*. Lemma F.3.1 makes it algebraic.*

Definition F.4.2 (Endpoint-nondegenerate eight-vertex signature)**.** An *endpoint-nondegenerate eight-vertex signature* is a quaternary signature q satisfying

$$q(\mathbf{x}) = 0 \quad \text{when } \mathrm{wt}(\mathbf{x}) \text{ is odd}, \qquad q(0000)q(1111) \neq 0.$$

Thus its support is contained in the eight even-parity strings, while both endpoint values are nonzero.

We use the two entries $q(0000), q(1111)$, the six weight-two entries, and the three flattenings with row–column splits $12 \mid 34$, $13 \mid 24$, and $14 \mid 23$, respectively. For example, $12 \mid 34$ uses (x_1, x_2) as row index and (x_3, x_4) as column index. We list their ranks in this order. A complementary pair of two-element input blocks is $J, J^c \subseteq [4]$ with $|J| = 2$.

We first handle product transformability and reduce local-affine transformability to affine transformability. The affine case then uses the finite interfaces in Theorems F.A.4 to F.A.6. For an irreducible quaternary, the continuing factorization is impossible, giving Theorem F.1.2.

Throughout this section, weighted equality and nonsingularity have the meaning fixed in Section F.2.3. For a signature class $\mathscr{C}$, the statement that one signature M is $\mathscr{C}$-transformable uses the singleton convention: the set $\{M\}$ and the transformed disequality edge have one common transformability witness, as in Definition F.2.3.

F.4.1 Product and local-affine exceptions

For a product-transformable signature, we distinguish the forms that yield an equality reduction from the form that splits into two binaries.

Lemma F.4.3 (Product-transformable eight-vertex signatures)**.**
Let M *be an endpoint-nondegenerate eight-vertex signature. If* M *is* $\mathscr{P}$*-transformable in the sense of Theorem F.4.1, then exactly one of the following structural outcomes applies (up to a permutation of inputs):*

1. M *is a generalized equality;*
2. M *is a tensor product of two nonsingular weighted equalities;*
3. M *has full even-parity support, has the dense form Equation* (F.11)*, and every retained set directly realizing* M *also directly realizes an equality anchor.*

In particular, every $\mathscr{P}$*-transformable endpoint-nondegenerate eight-vertex signature has support size* 2*,* 4*, or* 8*.*

Proof. Choose an algebraic witness matrix $T = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ by Lemma F.3.1, and define the transformed signature $H = (T^{-1})^{\otimes 4} M \in \mathscr{P}$. The transformed edge is

$$T^{\mathsf{T}} X T = \begin{pmatrix} 2ac & ad + bc \\ ad + bc & 2bd \end{pmatrix}.$$

Because this transformed edge is nonsingular and belongs to $\mathscr{P}$, its support is either equality or disequality. Disequality support makes T monomial. Equality support gives $ad + bc = 0$. Define the

nonzero scalar ratio $\lambda := b/a$. Then

$$T = \operatorname{diag}(a, c) \begin{pmatrix} 1 & \lambda \\ 1 & -\lambda \end{pmatrix}, \qquad ac\lambda \neq 0.$$

If H is irreducible, product type makes it a CP signature for some $\mathbf{s} \in \mathbb{F}_2^4$, with nonzero coefficients $A, B \in \overline{\mathbb{Q}}^\times$:

$$H = A\delta_{\mathbf{s}} + B\delta_{\bar{\mathbf{s}}}, \qquad AB \neq 0.$$

In the monomial case the two nonzero endpoint values force $\{\mathbf{s}, \bar{\mathbf{s}}\} = \{\mathbf{0}^4, \mathbf{1}^4\}$, giving outcome 1. Otherwise, for $k = \operatorname{wt}(\mathbf{s})$ and $w = \operatorname{wt}(\mathbf{x})$,

$$M(\mathbf{x}) = a^{4-w} c^w (-1)^{\mathbf{s}\cdot\mathbf{x}} (A\lambda^k + B\lambda^{4-k}(-1)^w).$$

The odd-weight zeros are equivalent to $B/A = \lambda^{2k-4}$. Hence all even entries are nonzero. Define the nonzero overall scale $\rho := 2A\lambda^k a^4$ and weight-ratio scalar $r := c/a$. The entries then have the form

$$M(\mathbf{x}) = \rho r^{\operatorname{wt}(\mathbf{x})} (-1)^{\mathbf{s}\cdot\mathbf{x}} \mathbf{1}[\operatorname{wt}(\mathbf{x}) \text{ is even}], \qquad \rho r \neq 0. \tag{F.11}$$

For this dense case choose nonzero algebraic scalars $q_0, q_1 \in \overline{\mathbb{Q}}^\times$ satisfying the first two equations below, define the diagonal matrix $Q := \operatorname{diag}(q_0, q_1)$, and define the normalization matrix $R := QT^{-1}$:

$$q_0^2 = 2ac, \qquad q_1/q_0 = i\lambda, \qquad R = QT^{-1}.$$

Then, since $\lambda^{2k-4}(i\lambda)^{4-2k} = (-1)^k$, there is a nonzero scalar $\eta \in \overline{\mathbb{Q}}^\times$ such that

$$(R^{-1})^{\mathsf{T}} X R^{-1} = I, \qquad R^{\otimes 4} M = \eta(\delta_{\mathbf{s}} + (-1)^k \delta_{\bar{\mathbf{s}}}). \tag{F.12}$$

Define the signature $F_{\mathbf{s}} = \delta_{\mathbf{s}} + (-1)^k \delta_{\bar{\mathbf{s}}}$. In these coordinates the disequality edge is equality by Equation (F.12). A nonzero multiple of $=_4$ uses at most two copies of $F_{\mathbf{s}}$: one for $k = 0, 4$. For $k = 2$, expose the two $s = 0$ inputs of the first and the two $s = 1$ inputs of the second, pairing the remaining inputs. For $k = 1$, expose the three majority-zero inputs of the first and the minority-one input of the second, join the first's minority input to one majority input of the second, and equality-close the second's last two majority inputs. The construction for $k = 3$ complements all bits. In each two-copy construction, an all-equal external string has exactly one compatible internal assignment, while every other external string has zero contribution. The resulting signature is $=_4$ for $k = 2$ and $-(=_4)$ for $k = 1, 3$. Pulling it back through R gives a direct gadget g over M with $R^{\otimes 4} g = \lambda'(=_4)$, $\lambda' \neq 0$. Together with Equation (F.12) this is outcome 3.

If H is not irreducible, neither is M. Its nonzero endpoints make every factor nonzero at both endpoints. Even parity excludes an odd-sized block, so the only proper tensor-factor partition is $[4] = S \sqcup S^c$ with $|S| = |S^c| = 2$, where $S^c = [4] \setminus S$. The odd-weight zeros then force both binary factors to be diagonal. Thus they are nonsingular weighted equalities and

$$M(\mathbf{0}^4) M(\mathbf{1}^4) = M(\mathbf{1}_S) M(\mathbf{1}_{S^c}).$$

This is outcome 2 and completes the classification. □

Corollary F.4.4 (Reductions from product-transformable eight-vertex signatures)**.** *Let $\mathcal{F}$ be a retained, factor-closed finite algebraic signature set that directly realizes a product-transformable endpoint-nondegenerate eight-vertex signature M. Then at least one of the following holds:*

(i) K-Holant($\mathcal{F}$) *is #*P*-hard;*

(ii) $\mathsf{Tract}(K\mathcal{F})$ *holds;*

(iii) *both factors in the two-binary factorization of* M*, including a nonsingular weighted-equality factor, can be added to* $\mathcal{F}$ *by the factor-closure reduction. The extension is finite, retained, and factor-closed, and its* K-Holant *problem reduces to* K-Holant$(\mathcal{F})$*.*

Proof. Apply Lemma F.4.3. Its generalized-equality form is closed by Lemma F.3.16, and its dense form is closed by Lemma F.3.15. These give alternatives (i)–(ii), with hardness composing to $\mathcal{F}$ and tractability descending to $\mathcal{F}$ by Lemma F.3.5.

In the remaining form, add the directly realized M and retain the complete factor batch from its decomposition across two pairs of inputs. The endpoint-nondegenerate eight-vertex signature is nonzero on both endpoint strings $\mathbf{0}^4$ and $\mathbf{1}^4$, whose doubled charges are -4 and $+4$, respectively, so the common-one-sided exception in Lemma F.3.4 is impossible. The resulting finite factor-closed signature set reduces to $\mathcal{F}$ and contains both nonsingular weighted-equality factors. This is alternative (iii). □

It remains to treat the affine and local-affine cases. The next lemma shows that even parity lets us replace a local-affine witness by an affine witness, so one affine analysis will cover both.

Lemma F.4.5 (Local-affine reduction to affine)**.** *Let* M *be a nonzero quaternary even-parity signature. If one matrix* $T \in \mathrm{GL}_2(\mathbb{C})$ *satisfies*

$$T^{\mathsf{T}} X T \in \mathscr{L}, \qquad (T^{-1})^{\otimes 4} M \in \mathscr{L},$$

then there is a matrix $T_0 \in \mathrm{GL}_2(\mathbb{C})$ *with*

$$T_0^{\mathsf{T}} X T_0 \in \mathscr{A}, \qquad (T_0^{-1})^{\otimes 4} M \in \mathscr{A}.$$

If $T \in \mathrm{GL}_2(\overline{\mathbb{Q}})$*, then* T_0 *is algebraic.*

Proof. For any nonempty affine set $S \subseteq \mathbb{F}_2^n$, choose a base point $\mathbf{s} \in S$ and write $S = \mathbf{s} + C$, where $C \leq \mathbb{F}_2^n$ is a linear subspace. The subspace

$$C = \{\mathbf{x} \oplus \mathbf{y} : \mathbf{x}, \mathbf{y} \in S\},$$

is independent of $\mathbf{s}$. It is self-orthogonal if $\mathbf{u} \cdot \mathbf{v} = 0$ over $\mathbb{F}_2$ for all $\mathbf{u}, \mathbf{v} \in C$.

Use the phase scalar ζ_8 from Section F.2.10.1. For an arity-n signature F and $\mathbf{z}, \mathbf{y} \in \mathbb{F}_2^n$, define the local twist selected by $\mathbf{z}$ as

$$\Theta_{\mathbf{z}} := \bigotimes_{j=1}^{n} T_2^{z_j}, \qquad (\Theta_{\mathbf{z}} F)(\mathbf{y}) = \zeta_8^{\langle \mathbf{z}, \mathbf{y} \rangle_{\mathbb{Z}}} F(\mathbf{y}), \qquad \langle \mathbf{z}, \mathbf{y} \rangle_{\mathbb{Z}} := \sum_{j=1}^{n} z_j y_j.$$

By the local-affine criterion [CLX18, Definition 3.1 and Theorem 3.2], the translation space of every support is self-orthogonal, and $\Theta_{\mathbf{z}} F \in \mathscr{A}$ for every $\mathbf{z} \in \mathrm{supp}(F)$ when $F \in \mathscr{L}$. Apply this first to the symmetric nonsingular binary $E = T^{\mathsf{T}} X T$. Self-orthogonality rules out full support, and nonsingularity rules out singleton support. Hence $\mathrm{supp}(E)$ is equality or disequality. Disequality support would give

$$E = pX \notin \mathscr{L},$$

since $01 \in \mathrm{supp}(E)$ and the twist required by this string in the support has relative phase $\zeta_8 \notin \mu_4$, so it is not affine. Thus $E = \mathrm{diag}(p, q)$, where $p, q \in \overline{\mathbb{Q}}^{\times}$ are its nonzero diagonal entries (their algebraicity follows from $E \in \mathscr{L}$). The 00-twist is the identity, so $q/p \in \mu_4$. Writing $T = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the

off-diagonal equation yields $q/p = -(b/a)^2$. Hence, with the scalar parameters $\rho := b/a \in \mu_8$ and $t := c/a \in \mathbb{C}^\times$, using the diagonal family defined in Section F.2.3,

$$T = aD(t)PD(\rho), \quad at \neq 0, \quad \rho^8 = 1.$$

Remove the right factor $D(\rho)$ by defining $T_0 := aD(t)P$. To prove that this is an affine witness, we examine the resulting signature through the sum

$$W_t(\mathbf{y}) = \sum_{\substack{\mathbf{x}\in\mathbb{F}_2^4 \\ \mathrm{wt}(\mathbf{x})\equiv 0 \pmod 2}} t^{-\mathrm{wt}(\mathbf{x})} M(\mathbf{x})(-1)^{\mathbf{x}\cdot\mathbf{y}}.$$

Let $N := (T^{-1})^{\otimes 4} M$ denote the transformed local-affine signature. Since $P^{-1} = P/2$, define

$$\kappa_0 := (2a)^{-4} \neq 0, \qquad W'_t := \kappa_0 W_t.$$

$$(T^{-1})^{\otimes 4} M(\mathbf{y}) = \rho^{-\mathrm{wt}(\mathbf{y})} W'_t(\mathbf{y}), \qquad (T_0^{-1})^{\otimes 4} M(\mathbf{y}) = W'_t(\mathbf{y}), \tag{F.13}$$

Since $N \in \mathscr{L}$ and $\rho \in \mu_8$, $W'_t(\mathbf{y}) = \rho^{\mathrm{wt}(\mathbf{y})} N(\mathbf{y})$ is algebraic for every $\mathbf{y}$. Moreover,

$$T_0^{\mathsf{T}} X T_0 = 2a^2 t \,\mathrm{diag}(1,-1) = p\,\mathrm{diag}(1,-1) \in \mathscr{A},$$

and parity gives $W'_t(\bar{\mathbf{y}}) = W'_t(\mathbf{y})$.

The support of N, which is also the support of W'_t, is an affine coset $\mathbf{s} + C$. Self-orthogonality and the identity $W'_t(\bar{\mathbf{y}}) = W'_t(\mathbf{y})$, which makes this support closed under bitwise complement, give $\mathbf{1}^4 \in C$ and $1 \leq \dim C \leq 2$. For dimension one, the support is a complementary pair on which W'_t has equal values, so $W'_t \in \mathscr{A}$. For dimension two, after permuting inputs,

$$C = \mathrm{span}_{\mathbb{F}_2}\{\mathbf{1}^4, \mathbf{w}\}, \qquad \mathrm{wt}(\mathbf{w}) = 2.$$

For $u, v \in \mathbb{F}_2$, index the four strings in the support by $\mathbf{x}_{uv} := \mathbf{s} \oplus u\mathbf{1}^4 \oplus v\mathbf{w}$. For a signature F that is nonzero on these four strings, define its cross-ratio functional by $\mathrm{cr}(F) := F(\mathbf{x}_{00})F(\mathbf{x}_{11})/(F(\mathbf{x}_{10})F(\mathbf{x}_{01}))$. Complement symmetry gives $\mathrm{cr}(W'_t) = 1$, while the nowhere-zero restriction of any member of $\mathscr{A}$ with this support has cross ratio ± 1. The quotient $\mathbb{F}_2^4/C$ has exactly three nonzero classes. Up to a permutation of inputs and $\mathbf{w} \leftrightarrow \bar{\mathbf{w}}$, each of the two odd cosets has representative $(\mathbf{s}, \mathbf{w}) = (0001, 0011)$, and the unique even coset has representative $(\mathbf{s}, \mathbf{w}) = (0101, 0011)$. Thus the displayed cases exhaust all three nonzero quotient classes. In both representatives the $\rho^{-\mathrm{wt}(\mathbf{y})}$ factor has cross ratio one and the direct four-point check gives

$$\langle \mathbf{s}, \mathbf{x}_{00}\rangle_{\mathbb{Z}} + \langle \mathbf{s}, \mathbf{x}_{11}\rangle_{\mathbb{Z}} - \langle \mathbf{s}, \mathbf{x}_{10}\rangle_{\mathbb{Z}} - \langle \mathbf{s}, \mathbf{x}_{01}\rangle_{\mathbb{Z}} = 2.$$

The local twist selected by $\mathbf{s}$ is $N_{\mathbf{s}} := \Theta_{\mathbf{s}} N$, so $N_{\mathbf{s}}(\mathbf{y}) = \zeta_8^{\langle \mathbf{s},\mathbf{y}\rangle_{\mathbb{Z}}} N(\mathbf{y})$. It would therefore have cross ratio $\zeta_8^2 = i$, contradicting $N_{\mathbf{s}} \in \mathscr{A}$. Hence $\mathbf{s} + C = C$.

If the complement-symmetric values of W'_t are A on the endpoints and B on the central pair, then N has values $(A, \rho^{-4}A, \rho^{-2}B, \rho^{-2}B)$. Since $0 \in \mathrm{supp}(N)$, its selected local twist is the identity, so $N \in \mathscr{A}$ and $\rho^{-2}B/A \in \mu_4$. As $\rho^2 \in \mu_4$, also $B/A \in \mu_4$, whence $W'_t \in \mathscr{A}$. Now Equation (F.13) proves the claim. If the original witness is algebraic, so are t and T_0. □

F.4.2 The finite affine quaternary certificate

The affine witness may initially use an arbitrary invertible matrix. We first reduce it to a diagonal change of basis, which preserves the disequality edge up to scalar. The proof derives a finite list of matrix forms and applies the classification in Appendix F.A.1, checked by `certificates/verify_affine_q4.py`. The corollary then uses the resulting signatures in reductions for the whole retained set.

Proposition F.4.6 (Affine quaternary certificate)**.** *Let M be an algebraic affine-transformable endpoint-nondegenerate eight-vertex signature. Then M admits an algebraic invertible diagonal matrix $D_{\mathrm{aff}} = \mathrm{diag}(d_0, d_1)$, with $d_0, d_1 \in \overline{\mathbb{Q}}^{\times}$, such that*

$$(D_{\mathrm{aff}}^{-1})^{\otimes 4} M \in \mathscr{A}, \qquad D_{\mathrm{aff}}^{\mathsf{T}} X D_{\mathrm{aff}} \doteq X.$$

For some $J \subseteq [4]$ with $|J| = 2$, write $J^c := [4] \setminus J$. In these coordinates its support is exactly one of

$$\{\mathbf{0}^4, \mathbf{1}^4\}, \qquad \{\mathbf{0}^4, \mathbf{1}^4, \mathbf{1}_J, \mathbf{1}_{J^c}\}, \qquad \{\mathbf{x} : \mathrm{wt}(\mathbf{x}) \text{ is even}\}.$$

The first support is a generalized equality.

Proof. By Lemma F.3.1, choose an algebraic affine change-of-basis matrix $U = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, and define the transformed binary edge $E := U^{\mathsf{T}} X U$. A nonsingular symmetric affine binary has disequality support, equality support, or full support. In the first case,

$$U = D_0 K_0, \qquad K_0 \in \{I, X\},$$

for an invertible diagonal D_0. In the second, $ad + bc = 0$, so

$$U = D_0 \begin{pmatrix} 1 & r \\ 1 & -r \end{pmatrix}, \qquad r \in \mu_8,$$

for an invertible diagonal D_0. In the full-support case, after normalizing its 00-entry, E has the affine form

$$\begin{pmatrix} 1 & u \\ u & -u^2 \end{pmatrix}, \qquad u \in \mu_4.$$

Writing $U = D_0 \left(\begin{smallmatrix} 1 & r \\ 1 & s \end{smallmatrix}\right)$ gives $r + s = 2u$ and $rs = -u^2$. Hence $\{r, s\} = \{u(1+\sqrt{2}), u(1-\sqrt{2})\}$. Denote the resulting eighteen-element set of representative matrices by $\mathscr{K}_{\mathrm{aff}}$. Its explicit list appears in Theorem F.A.4. These three calculations give precisely

$$U = D_0 K_0, \qquad K_0 \in \mathscr{K}_{\mathrm{aff}},$$

which proves completeness of the list.

Now define the affine signature $f = (U^{-1})^{\otimes 4} M \in \mathscr{A}$. Since M is an endpoint-nondegenerate eight-vertex signature and D_0 is diagonal, $K_0^{\otimes 4} f$ satisfies the hypotheses of Theorem F.A.4. The certificate is projective, so the normalization factor for f can be absorbed into a global scalar. The same classification theorem therefore yields

$$\gamma, t \in \overline{\mathbb{Q}}^{\times}$$

and a normalized affine signature $h \in \mathscr{A}$, with $h(\mathbf{0}^4) = 1$, such that

$$K_0^{\otimes 4} f = \gamma D(t)^{\otimes 4} h.$$

Moreover, $\mathrm{supp}(h)$ is one of the three supports displayed above. Consequently the single algebraic diagonal matrix $D_\star = D_0 D(t)$ satisfies $(D_\star^{-1})^{\otimes 4} M \doteq h$, and $D_\star^{\mathsf{T}} X D_\star \doteq X$. Taking $D_{\mathrm{aff}} = D_\star$ proves the asserted diagonal normalization. □

Corollary F.4.7 (Reductions from affine quaternaries)**.** *Let $\mathcal{F}$ be a retained, factor-closed finite algebraic signature set that directly realizes an affine-transformable endpoint-nondegenerate eight-vertex signature M. Then at least one of the following holds:*

(i) K-Holant($\mathcal{F}$) *is #*P*-hard;*

(ii) $\mathsf{Tract}(K\mathcal{F})$ *holds;*

(iii) *the complete factor batch from the gadget construction can be added to $\mathcal{F}$ by reduction, giving a finite retained factor-closed extension whose* K-Holant *problem reduces to* K-Holant($\mathcal{F}$)*. The original signature M has, up to a permutation of inputs and a nonzero scalar, a genuine factorization across two pairs of inputs. One of its proper binary factors is a nonsingular weighted equality $E := \mathrm{diag}(p, q)$, where $p, q \in \overline{\mathbb{Q}}^{\times}$, retained in this extension through the factor-closure reduction.*

Proof. Choose $D_\star = D_{\mathrm{aff}}$ from Proposition F.4.6. Apply the global change $D_\star^{-1}$, and normalize the resulting quaternary to an affine signature h with $h(\mathbf{0}^4) = 1$. Thus

$$M \doteq D_\star^{\otimes 4} h.$$

Keep every direct output over the current signature set, and add proper tensor factors only through factor closure. Record these constructions and the chosen normal form in the normalization data of Lemma F.3.7. The normalized signature h has nonzero entries at both endpoints, so every resulting set is two-sided and Lemma F.3.4 applies.

The normal form has three possible supports. The support-two form is a generalized equality. For support four, the distinction between an equality construction and a factorization is determined by the following coefficient matrix. Let $J \sqcup J^c = [4]$ be the two pairs of inputs. For $u, v \in \mathbb{F}_2$, put $\mathbf{x}_{uv} := u\mathbf{1}_J \oplus v\mathbf{1}_{J^c}$, and define the logical coefficient matrix by

$$F := \big(h(\mathbf{x}_{uv})\big)_{u,v\in\mathbb{F}_2} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad ABCD \neq 0.$$

Here u is the row bit and v is the column bit. For an ordered binary tensor E, write $E^{(J)}$ for its placement on the two inputs of J in their inherited order, and define $E^{(J^c)}$ analogously. If $AD = -BC$, the two-copy X-edge gadget in Theorem F.A.5(ii) directly realizes a nonzero pure generalized equality. If $AD = BC$, then, on these two blocks,

$$h = E_1^{(J)} \otimes E_2^{(J^c)}$$

for nonsingular weighted equalities E_1, E_2. Since $D_\star$ is diagonal,

$$M \doteq (D_\star E_1 D_\star^{\mathsf{T}})^{(J)} \otimes (D_\star E_2 D_\star^{\mathsf{T}})^{(J^c)}.$$

Thus the original endpoint-nondegenerate eight-vertex signature itself has a genuine factorization across two pairs of inputs into nonsingular weighted equalities. Add M and apply factor closure to obtain both factors by reduction.

On full even-parity support, Theorem F.A.5(iii), together with Lemma F.4.3, gives, over the current signature set, either a directly realized equality anchor, a directly realized nonzero rank-one binary, or a directly realized product

$$cD(r) \otimes D(r), \qquad c \in \overline{\mathbb{Q}}^{\times}, \qquad r \in \mu_4,$$

where $D(r)$ is a nonsingular weighted-equality matrix. In the last case, Lemma F.3.4 makes $D(r)$ available by reduction. We use it to construct an odd-parity signature and continue the full-support analysis. Choose the scalar $s \in \mu_8$ with $s^2 = r^{-1}$, and use $D(s)$ for the equality normalization. Then

$$(D(s))^{\mathsf{T}} X D(s) = sX, \qquad D(s)D(r)(D(s))^{\mathsf{T}} = I.$$

Apply this normalization and define the transformed signature $h_r = (D(s))^{\otimes 4} h$. Attach one input of h_r by a X-edge to one input of the retained equality I, leaving the other equality input external. This is a gadget over the current signature set, and it complements that input of h_r, producing, up to a nonzero scalar, an odd-parity quaternary p that is nonzero on all eight odd-weight strings and is used in Theorem F.A.6. Because $s \in \mu_8$, its normalized nonzero entries are fourth roots of unity.

The twelve disequality-loop and equality-mediated binary gadgets of Theorem F.A.6 are therefore realizable at this node. If a test gives a nonzero rank-one binary or a nonsingular binary of infinite projective order, the unary reductions below apply. Otherwise the same theorem classifies the remaining signatures, including the simultaneous-zero cases of its loop tests. Its exceptional family directly realizes the certified endpoint-nondegenerate eight-vertex signature with support of cardinality six. For every other remaining signature, choose a certified $\delta \in \mathbb{Z}_4$, choose $\tau_\delta \in \mu_8$ with $\tau_\delta^2 = i^\delta$, and apply the further legal global-normalization matrix $D(\tau_\delta)$. Let p_δ denote the certified phase-shifted quaternary signature with odd-parity support from Theorem F.A.6. Since p is supported only at Hamming weights one and three,

$$D(\tau_\delta)^{\otimes 4} p \doteq p_\delta.$$

Theorem F.A.6 now gives, through its disequality self-loop and two-copy constructions, either a nonzero rank-one binary or a product-transformable endpoint-nondegenerate eight-vertex quaternary in the dense form Equation (F.11). The latter directly yields an equality anchor by Lemma F.4.3.

For the eight-vertex signature with support of size six, Lemma F.4.3 rules out $\mathscr{P}$-transformability. The diagonal normalization in Proposition F.4.6 preserves support and allows only sizes $2, 4, 8$, ruling out $\mathscr{A}$-transformability. Lemma F.4.5 then rules out $\mathscr{L}$-transformability, so Theorem F.4.1 gives hardness.

The support-two case and the support-four case $AD = -BC$ are resolved by Lemma F.3.16. Equality anchors are resolved by Lemma F.3.15. A nonzero rank-one binary exposes a unary by Lemma F.3.8, and an infinite-order binary first interpolates such a binary. In both cases Lemma F.3.11 applies. An exposed odd-arity signature is resolved by Theorem F.2.9 after the holographic K-coordinate equivalence. Every case with full even-parity support above therefore gives hardness or tractability.

The sole continuing case, the support-four form $AD = BC$, gives the displayed factorization of the original M into two nonsingular weighted-equality factors after undoing the diagonal normalization. By Lemma F.3.7, hardness reductions compose back to $\mathcal{F}$, tractable conclusions descend to $K\mathcal{F}$, and this continuing state returns to the original coordinates. □

F.4.3 Consequences for the Full Signature Set

We now combine the three transformability cases. The conclusion either classifies the entire signature set or supplies proper binary factors of the original quaternary for the later group analysis.

Corollary F.4.8 (Reductions from eight-vertex signatures)**.** *Let $\mathcal{F}$ be a retained, factor-closed finite algebraic signature set containing a directly realizable endpoint-nondegenerate eight-vertex signature M. Then at least one of the following holds:*

(i) K-Holant($\mathcal{F}$) *is* #P*-hard;*

(ii) $\mathsf{Tract}(K\mathcal{F})$ *holds;*

(iii) *the complete factor batch from a genuine factorization of the original* M *across two pairs of inputs can be added to* $\mathcal{F}$ *by reduction, giving a finite retained factor-closed extension whose* K-Holant *problem reduces to* K-Holant($\mathcal{F}$)*. This extension contains a proper tensor factor of* M *of the form*

$$E := \mathrm{diag}(p, q), \qquad p, q \in \overline{\mathbb{Q}}^{\times}.$$

In item (iii), the factor E *is available by reduction from the original* M*.*

Proof. If all six weight-two entries vanish, the signature is a generalized equality and is closed by Lemma F.3.16. Otherwise apply the Cai–Fu quaternary boundary Theorem F.4.1. Its hard branch is item (i). Its product-transformable branch is handled by Corollary F.4.4. A local-affine branch first collapses to affine by Lemma F.4.5 and then uses Corollary F.4.7. An affine branch uses that corollary directly.

These two corollaries give the three stated outcomes. Composing the reductions gives hardness for $\mathcal{F}$, and restricting the tractability criterion gives the same conclusion for $K\mathcal{F}$. Apply Lemma F.3.7 to return to the original basis. The sole continuing output is the retained nonsingular weighted-equality factor in item (iii). □

The continuing outcome factors the original quaternary signature, so the irreducibility hypothesis in Theorem F.1.2 rules it out.

Proof of Theorem F.1.2. Choose an endpoint-nondegenerate irreducible quaternary $q \in \mathcal{F}$. Apply Section F.2.4, retaining the name $\mathcal{F}$. The signature q survives, and the tractability criterion is unchanged. By the coordinate identity in Section F.2.10.3, the criterion in the theorem is $\mathsf{ReTract}(K\mathcal{F})$ in ordinary-Holant coordinates. Since q is two-sided, Lemma F.2.7 gives

$$\mathsf{Tract}(K\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{ReTract}(K\mathcal{F}).$$

If $\mathsf{ReTract}(K\mathcal{F})$ holds, then Theorem F.3.3 and Equation (F.2) gives K-Holant($\mathcal{F}$) $\in \mathsf{FP}$, and the root evaluator restores the deleted nullary scalars. Now suppose that $\mathsf{ReTract}(K\mathcal{F})$ fails, so $\mathsf{Tract}(K\mathcal{F})$ fails as well. The two-sided signature q excludes the common-one-sided exception. By Lemma F.3.4, adding $\Pi(\mathcal{F})$ gives a finite factor-closed extension equivalent to the original problem. By Lemma F.3.5, its tractability criterion also fails. Continue to denote this extension by $\mathcal{F}$. It still contains the original irreducible signature q.

Apply Corollary F.4.8 to q over $\mathcal{F}$. Its tractable case is excluded by the preceding paragraph. Its continuing outcome gives a genuine $2 + 2$ factorization of the original q, contradicting irreducibility. Hence K-Holant($\mathcal{F}$) is #P-hard. The factor-closure equivalence and preprocessing transfer this hardness to the original input problem. Finally, the same predicate equivalence and Theorem F.3.3 give decidability. □

F.5 Equality-Accessible Matching Decks

This section organizes the structural proof of the binary-disequality dichotomy. By the equivalence below, we work with disequality edges and the supplied binary equality $I = (=_2)$. The central task is to combine the classification of binary signatures with reductions of higher-arity irreducible

signatures. Binary transfers determine the available contractions, while factors of their outputs can supply new transfers and change the group under analysis.

After the unary reductions have settled the singular and infinite-order binary cases, the remaining transfers form a finite group. The supplied I gives the transfer class $[X]$, and reversal makes $[Z]$ normalize the group. These two distinguished classes restrict its form. We then choose a minimum-arity nonbinary irreducible signature and contract it using the available binaries. Retaining the outputs and their factors can settle the complexity, lower the minimum arity, or enlarge the complete transfer group. When the remaining process stabilizes, the contraction outputs are products of binaries on perfect matchings. This is the structure analyzed in the four group branches.

The progress mechanism gives those branches a common task: determine the complexity of the current set or produce a reduction that advances the structural analysis. A fixed number field bounds strict group enlargement, and finite processing controls the steps that leave both the group and minimum arity unchanged. A further induction treats the unresolved irreducible occurrences in the whole set. These ingredients connect the local constructions in Sections F.6 to F.9 to the final dichotomy proof in Section F.10.

Throughout, the supplied I stays in the original coordinates. The eight-vertex reductions return to these coordinates, and any additional weighted equality is retained as a binary signature. Continuing changes of basis preserve both X and I projectively. Other basis changes are used within reductions that settle the complexity. A *resolved complexity leaf*, or *resolved leaf*, means hardness, Tract for the entire retained set, or a common tractable K-presentation as in Definition F.2.3.

Lemma F.5.1 (Equality-edge-model equivalence)**.** *For every signature set $\mathcal{F}$,*

$$\text{K-Holant}(\mathcal{F}, I) \equiv_{\mathrm{T}} \text{Holant}(\mathcal{F}, X), \tag{F.14}$$

where the right side uses ordinary equality edges and has X as an available binary signature.

Proof. Forward, subdivide each X-edge by an X-vertex and suppress degree-two I-vertices as equality edges. Conversely, splice the two incident inputs of the ordinary X-vertices after first preprocessing the subgraph induced by those vertices. Every such component is a path or a cycle, because X has arity two. A path with two incidences at non-X vertices collapses to I or X according to the parity of the number of X-vertices. A cycle with no external incidence contributes zero for odd length and the scalar 2 for even length. (A self-loop is the length-one case.) These are exact identities obtained by summing the internal equality-edge bits, and they also cover adjacent X-vertices and parallel edges. For each resulting X kernel use one structural E_X edge (no additional tensor vertex). For each resulting I kernel, and for each ordinary equality edge whose endpoints are not in an X-component, use

$$\text{input} - X - I - X - \text{input}.$$

Indeed, $\sum_{u,v} X(a,u)I(u,v)X(v,b) = I(a,b)$. Multiply the final answer by the known product of the even-cycle scalars (or return zero for an odd cycle). The preprocessing and gadget replacements are linear-size and preserve ordered incidences. □

The signatures in $\mathcal{F}$ keep their entries under this gadget replacement. The four tractable cases correspond as well:

$$\mathsf{ReTract}(\mathcal{F} \cup \{X\}) \iff \mathsf{ReTract}_X(\mathcal{F} \cup \{I\}) \iff \mathsf{ReTract}(K(\mathcal{F} \cup \{I\})).$$

The last equivalence is the coordinate identity in Section F.2.10.3. The arity-two case is unchanged by adjoining either binary. For $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$, nonsingular symmetric binaries in $\mathscr{C}$ are closed

under inversion and under EBE. The latter is Lemma F.6.14. Inversion preserves the binary affine and product normal forms, and the local-affine forms $a\,\mathrm{diag}(1, i^k)$ identified in its proof. For an ordinary-Holant witness M, put $E = M^{\mathsf{T}}M$ and $B = M^{-1}XM^{-\mathsf{T}}$. Then the disequality edge becomes $M^{\mathsf{T}}XM = EBE$, and the supplied I becomes E^{-1}, both in $\mathscr{C}$. Conversely, for a disequality-edge witness U, put $E = U^{\mathsf{T}}XU$ and $B = U^{-1}IU^{-\mathsf{T}}$. The ordinary edge becomes $U^{\mathsf{T}}U = EBE$, and the supplied X becomes $U^{-1}XU^{-\mathsf{T}} = BEB$, both in $\mathscr{C}$. The same witness transforms every member of $\mathcal{F}$ in each direction.

Consequently, the following statement is equivalent to Theorem F.1.1 and is the target of the structural analysis beginning here. Its proof is completed in Section F.10.

Theorem F.5.2 (Equality-accessible K-Holant dichotomy)**.** *For every finite set $\mathcal{F}$ of algebraic complex-valued Boolean signatures with $I = (=_2) \in \mathcal{F}$, the problem* K-Holant($\mathcal{F}$) *is polynomial-time computable when* $\mathsf{ReTract}(K\mathcal{F})$ *holds and is* #P*-hard otherwise. The tractability criterion is decidable from exact algebraic input.*

For a set containing I, the transformed set contains $Z = K^{\otimes 2}I = \mathrm{diag}(1, -1)$. Thus Lemma F.2.7 reduces the full criterion Tract to the four cases of $\mathsf{ReTract}$.

Since $I(00) = I(11) = 1$, the supplied I excludes the common one-sided support exception, so Lemma F.3.4 allows the initial reduction to irreducible signatures while retaining I. Odd-arity factors are covered by Theorem F.2.9, and a set of binary factors is covered by the arity-two case. We therefore begin the remaining analysis with a finite set $\mathcal{F}$ of nonzero even-arity irreducible signatures, containing I and at least one signature of arity at least four. Such a set is factor-closed. Later outputs and their factors are retained as they are added. The final proof in Section F.10 transfers the resulting classification to the original set. We first control the binary transfers.

F.5.1 Binary transfers and complete groups

Binary chains turn composition into matrix multiplication. Bounding the group they generate gives a finite set of contraction kernels for the matching deck.

For an ordered binary signature B, recall its transfer

$$T_B = BX.$$

Joining B to another ordered binary signature C in series through one X-edge realizes BXC, and therefore

$$T_{BXC} = T_B T_C.$$

For a transfer matrix T, define its reversal by the anti-involution

$$T^{\#} := XT^{\mathsf{T}}X; \qquad T^{\#} = (\det T)ZT^{-1}Z \quad \text{if } T \text{ is nonsingular.} \tag{F.15}$$

Here "anti-involution" means explicitly that

$$(ST)^{\#} = T^{\#}S^{\#}, \qquad (T^{\#})^{\#} = T$$

for all transfer matrices S, T.

Lemma F.5.3 (Bounded torsion)**.** *Fix a number field L, and let G be the set of projective transfers of all directly realizable nonzero nonsingular ordered binary gadgets over a fixed signature set with entries in L. If $G \neq \varnothing$ and every element of G has finite projective order, then G is a finite subgroup of* $\mathrm{PGL}_2(L)$.

Proof. Define the field degree $D = [L : \mathbb{Q}]$. Let $[T] \in G$ have projective order r. By definition,

$$T^r = \eta I$$

for some $\eta \in L^\times$, and r is the least positive integer with this property.

The minimal polynomial m_T divides $x^r - \eta$, which is square-free because $\gcd(x^r - \eta, rx^{r-1}) = 1$ in characteristic zero. Thus T is diagonalizable over the algebraic closure. Choose an eigenbasis matrix $V \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ and eigenvalues $\lambda_1, \lambda_2 \in \overline{\mathbb{Q}}^\times$ such that

$$V^{-1}TV = \mathrm{diag}(\lambda_1, \lambda_2).$$

Define the eigenvalue ratio

$$\zeta := \frac{\lambda_1}{\lambda_2}.$$

The equality $T^r = \eta I$ gives $\zeta^r = 1$, and the minimality of r shows that ζ is a primitive r-th root of unity. Moreover,

$$\zeta + \zeta^{-1} = \frac{(\mathrm{tr}\, T)^2}{\det T} - 2 \in L.$$

For $r > 2$,

$$[\mathbb{Q}(\zeta + \zeta^{-1}) : \mathbb{Q}] = \frac{\varphi(r)}{2},$$

so $\varphi(r) \le 2D$, also valid for $r = 1, 2$. The integers with this bound form an effectively finite set: each prime power $p^a \mid r$ satisfies $p^{a-1}(p-1) \le 2D$, bounding both p and a. Consequently all projective orders divide the computable exponent

$$e_D = \mathrm{lcm}\{r : \varphi(r) \le 2D\}.$$

Series composition gives product closure. A positive power of every element is the identity, so the identity and inverses also belong to G. Hence G is a subgroup. Put

$$\mathfrak{sl}_2(L) := \{A \in \mathrm{Mat}_2(L) : \mathrm{tr}\, A = 0\}.$$

The adjoint representation is

$$\mathrm{Ad} : \mathrm{PGL}_2(L) \hookrightarrow \mathrm{GL}(\mathfrak{sl}_2(L)) \cong \mathrm{GL}_3(L), \qquad \mathrm{Ad}([T])(A) = TAT^{-1}.$$

The conjugation homomorphism has trivial kernel: a matrix commuting with all trace-zero matrices is scalar. Burnside's finite-exponent theorem states that a subgroup $H \le \mathrm{GL}_n(\mathbb{C})$ satisfying $h^e = I$ for one fixed e and every $h \in H$ is finite [Bur05]. Apply it to $\mathrm{Ad}(G) \le \mathrm{GL}_3(\mathbb{C})$ with $e = e_D$. Thus all binary gadgets, despite their unbounded sizes, have only finitely many projective transfer classes. □

For a finite algebraic retained, factor-closed, two-sided signature set $\mathcal{F}$, call a nonzero directly realizable ordered binary *safe* if it is nonsingular and its transfer has finite projective order. Call it *unsafe* if it is singular or its transfer has infinite projective order. Call the set *binary-residual* when every nonzero realizable ordered binary is safe. We work in this case. Otherwise Lemmas F.3.8 and F.3.11 resolves the branch. Series composition and Lemma F.5.3 then give a finite group: take L to be the number field generated by the entries of $\mathcal{F}$, which also contains the entries of every direct gadget transfer. The group is

$$G(\mathcal{F}) = \{[BX] : B \text{ is a safe ordered binary directly realizable over } \mathcal{F}\}.$$

The definition uses all directly realizable binaries. Below we distinguish cases according to this group for each signature set. The decision algorithm for the resulting tractability cases is given by Theorem F.3.3. It does not enumerate gadgets of unbounded size. We choose realizable representatives as follows.

Realizable binary representatives. For each class g of a finite realized projective group $G \leq \mathrm{PGL}_2(\overline{\mathbb{Q}})$, choose an ordered binary gadget B_g with transfer $T_{B_g} = B_g X$ and $[T_{B_g}] = g$. For $1 = [I]$, take $B_1 := \mathsf{E}_X = X$, so $T_{B_1} = I$. The empty network is the nullary unit. Every realizable ordered binary with transfer class g has the form $B = \rho B_g$ for a unique $\rho \in \overline{\mathbb{Q}}^{\times}$. Retain the multiplication and reversal scalars for these representatives, with the input orders and reduction data of Section F.3.2. This fixes the scalar factors when projective identities are used in gadgets.

Whenever the proof says that an external input is X-translated, it means attaching the binary chain in the current set whose transfer class is $[X]$ and recording its nonzero scalar in the input data. The displayed chain realizes this action.

Lemma F.5.4 (Relative completeness test for realizable binaries)**.** *Let $\mathcal{F}_0 \subseteq \mathcal{F}$ be finite algebraic retained, factor-closed signature sets with $I \in \mathcal{F}_0$, and suppose that every nonzero binary directly realizable over $\mathcal{F}_0$ is safe. Put $H = G(\mathcal{F}_0)$. The following exhaustive ordered case distinction applies:*

1. *if $\mathcal{F}$ directly realizes a nonzero singular ordered binary, its nonzero unary factors are available by the factor reduction;*
2. *otherwise, if $\mathcal{F}$ directly realizes a nonsingular ordered binary of infinite projective transfer order, a nonzero rank-one binary is available by interpolation;*
3. *otherwise every nonzero binary directly realizable over $\mathcal{F}$ is safe, $H \leq G(\mathcal{F})$, and every nonsingular realizable binary B over $\mathcal{F}$ with $[BX] \notin H$ witnesses the strict inclusion $H < G(\mathcal{F})$.*

In the third case, if $G(\mathcal{F}) = H$, every nonzero realizable binary over $\mathcal{F}$ is projectively one of the fixed representatives B_h, $h \in H$.

Proof. The supplied I has both charge signs, so both signature sets are two-sided. A singular binary is handled by Lemmas F.3.8 and F.3.11, and an infinite-order binary by Lemma F.3.8. If neither exists, $\mathcal{F}$ is binary-residual, so Lemma F.5.3 defines $G(\mathcal{F})$. Every gadget over $\mathcal{F}_0$ remains a gadget over $\mathcal{F}$, hence $H \leq G(\mathcal{F})$. The remaining assertions follow directly from the definitions and the fixed representative choice. □

In the first two cases, Lemma F.3.11 determines the complexity after factor extraction. In the third, a new transfer class gives strict group enlargement, while equality of the groups permits reuse of the fixed binary representatives.

Remark F.5.5 (Connection with binary-signature classifications)*.* The singular, nonsingular infinite-projective-order, and nonsingular finite-projective-order split is the K-Holant form of the binary split used by Cai–Lu–Xia [CLX09, CLX18]. The four ordered Bell signatures $\mathcal{B} = \{B_{0,0}, B_{0,1}, B_{1,0}, B_{1,1}\}$ represent the projective Bell classes of Shao–Cai [SC20]. Beyond the binary classification, we need the complete transfer group and uniform control of its higher-arity cards.

Finite projective groups and marked matrices. The classification of finite projective groups gives the uniform order bound needed for stabilization. We record its concrete models here and analyze the distinguished classes $[X]$ and $[Z]$ in Section F.5.3.

We use the quaternion matrix basis

$$Q_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad Q_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad Q_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Q_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \tag{F.16}$$

It is a basis of $\mathrm{Mat}_2(\mathbb{C})$. Moreover, $Q_0 = I$, $Q_j^2 = -I$ for $1 \leq j \leq 3$, and $Q_j Q_k = -Q_k Q_j$ for distinct $j, k \in \{1, 2, 3\}$. These identities are what the adjective *quaternion* records.

For the concrete Platonic models below, int records that $[Z]$ belongs to the tetrahedral group, *dd* records that $[X], [Z]$ are double transpositions in the octahedral group, and 0 labels the base icosahedral display.

Proposition F.5.6 (Finite subgroups of $\mathrm{PGL}_2(\mathbb{C})$)**.** *Every finite subgroup of* $\mathrm{PGL}_2(\mathbb{C})$ *has exactly one of the following isomorphism types, and subgroups of the same type are projectively conjugate:*

$$C_m, \qquad D_{2m}, \qquad A_4, \qquad S_4, \qquad A_5,$$

where $m \geq 1$ *for* C_m*,* $m \geq 2$ *for* D_{2m}*,* $|C_m| = m$*, and* $|D_{2m}| = 2m$*. If* ω *is a primitive* m*-th root of unity, explicit cyclic and dihedral models are*

$$C_m^{\mathrm{std}} = \left\{ \left[\begin{pmatrix} 1 & 0 \\ 0 & \omega^k \end{pmatrix} \right] : 0 \leq k < m \right\},$$

and

$$D_{2m}^{\mathrm{std}} = C_m^{\mathrm{std}} \cup \left\{ \left[X \begin{pmatrix} 1 & 0 \\ 0 & \omega^k \end{pmatrix} \right] : 0 \leq k < m \right\}.$$

The diagonal classes are the rotations, and the anti-diagonal coset consists of the reflections.

In the basis Equation (F.16)*, take the tetrahedral model*

$$\begin{aligned} \mathcal{T}^{(\mathrm{int})} = {} & \{[Q_0], [Q_1], [Q_2], [Q_3]\} \\ & \cup \{[Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3] : \epsilon_j \in \{1, -1\}\}, \end{aligned} \tag{F.17}$$

and the octahedral model

$$\mathcal{O}^{(\mathrm{dd})} = \mathcal{T}^{(\mathrm{int})} \cup \{[Q_\mu + \epsilon Q_\nu] : 0 \leq \mu < \nu \leq 3,\ \epsilon \in \{1, -1\}\}. \tag{F.18}$$

Finally, define the golden-ratio scalar and its reciprocal by

$$\phi := \frac{1 + \sqrt{5}}{2}, \qquad \tau := \phi^{-1},$$

and define the icosahedral model by

$$\begin{aligned} \mathcal{I}^{(0)} = \mathcal{T}^{(\mathrm{int})} \ \cup \ \{[v_0 Q_0 + v_1 Q_1 + v_2 Q_2 + v_3 Q_3] : {} & (v_0, v_1, v_2, v_3) \\ & = \pi(0, \epsilon_1, \epsilon_2 \phi, \epsilon_3 \tau), \quad \pi \in A_4,\ \epsilon_j \in \{1, -1\}\}, \end{aligned} \tag{F.19}$$

where A_4 *acts by even coordinate permutations and projective duplicates are removed. The additional part of* $\mathcal{I}^{(0)}$ *has* 48 *projective classes. Thus*

$$\mathcal{T}^{(\mathrm{int})} \cong A_4, \qquad \mathcal{O}^{(\mathrm{dd})} \cong S_4, \qquad \mathcal{I}^{(0)} \cong A_5,$$

of orders $12, 24, 60$*, respectively.*

Proof. The conjugacy classification is classical [Bea83, Section 5.1]. The first two displays are the standard rotation and reflection models. Direct matrix multiplication verifies closure, cardinalities, and element-order distributions of the three remaining displayed sets, identifying them respectively with A_4, S_4, A_5. □

For the concrete models in Proposition F.5.6, the finite matrix checks in its proof consist of reducing products modulo nonzero scalar, testing closure of the displayed projective sets, and counting their distinct classes and element orders. The scalar factors remain separate from the realizable-gadget records above.

We now choose a minimum-arity nonbinary irreducible signature and retain the factors of all its dressed contractions.

F.5.2 Stable matching decks and restart control

Fix a retained, factor-closed set $\mathcal{F} \ni I$ with complete transfer group $G = G(\mathcal{F})$. Attach available binaries to a minimum-arity nonbinary irreducible signature and close one pair of inputs. A nonbinary factor of the output has smaller arity. If only binary factors remain, they either settle the complexity, enlarge the group, or form a product on a perfect matching. The *matching deck* collects these contractions, using the ordered representatives from Section F.5.1.

We first stabilize this deck for the selected signature. We then give the descent conditions for subsequent constructions in the group branches. The final proof also uses an outer induction over unresolved irreducible occurrences to pass from the selected signature to the whole set.

For an even finite input set R, a *perfect matching* is a partition of R into two-element subsets. An *oriented perfect matching* also chooses an order within each pair, and a matching product places one ordered binary factor on every oriented pair.

Definition F.5.7 (Matching deck)**.** Let $\mathcal{F} \ni I$ be factor-closed and let $G = G(\mathcal{F})$. On a fixed ordered input set, a *G-matching signature* of arity $2m$ is

$$c \bigotimes_{e \in M} B^{(e)}_{g_e}, \qquad c \neq 0, \quad g_e \in G,$$

where M is an oriented perfect matching in the preceding sense and $B^{(e)}_{g_e}$ places the chosen ordered representative B_{g_e} on the oriented edge e. We allow a simultaneous permutation of inputs. The scalar c absorbs all representative scalars, so the notion depends only on G.

For $\omega \in \{\mathrm{id}, \mathrm{rev}\}$, put $B^{\mathrm{id}}_g = B_g$ and $B^{\mathrm{rev}}_g = B^{\mathsf{T}}_g$. Let δ specify a subset of the inputs and, at each selected input, a representative B_g and one of these two orientations. Attach its recorded binary through one X-edge and retain the other binary input in the same position. Write f^δ for the resulting realizable dressed signature. Here the dressing acts on the specified inputs of f, before the pair contraction defining a deck card below.

For each δ, choose distinct ordered inputs (i, j), a closing representative B_h with orientation ω, and a residual input order ρ, leaving at least two inputs. The resulting card is

$$\rho\Big(\partial^{XB^{\omega}_h X}_{(i,j)} f^\delta\Big).$$

The *matching deck* $\mathfrak{D}_G(f)$ contains zero and all such one-copy proper pair cards. Thus one copy of f is used and exactly one ordered input pair is closed. The binary dressings and closing chain use the fixed representatives. Every nonzero card is a direct gadget output.

Definition F.5.8 (Deck-stable state)**.** A retained reduction state $(\mathcal{F}, G, f)$ is *deck-stable* if $\mathcal{F}$ is finite, factor-closed, and contains I; $f \in \mathcal{F}$ is a minimum-arity irreducible nonbinary signature; $G = G(\mathcal{F})$ is the complete transfer group, which is finite, with fixed safe binary representatives; and the deck for the resulting set, with all card factors retained, consists of zero and G-matching signatures. Realizability is always relative to $\mathcal{F}$. Added factors are only available by reduction over their predecessor.

Adding signatures available by reduction gives a Turing-equivalent problem: the defining reduction removes the added signatures, and inclusion gives the reverse reduction. Direct outputs use Equation (F.3). Their factors use Lemma F.3.4.

To prove stabilization, fix the finite family of contractions that can be processed while the group and minimum arity stay unchanged. We record this family and its factorization trees as follows. Fix $(A, \mathcal{F}, f)$ and $H := \overline{G}(A, \mathcal{F})$, the complete transfer group in the original coordinates. Throughout

this stabilization and descent argument, H has this meaning. While H and $\nu(\mathcal{F})$ remain unchanged, fix one representative of each group class. A one-copy proper card is specified by the input dressings, the ordered deleted pair, the closing representative B_h and its orientation ω (giving the kernel $XB_h^{\omega}X$), and the residual input order. Let $\mathscr{U}(A, f, H)$ consist of zero and the signatures obtained by pulling these cards back by A and dividing each nonzero result by its first nonzero entry, with the residual inputs kept in order.

Each nonzero member of $\mathscr{U}(A, f, H)$ indexes one *token* in the local queue, even if several constructions yield that member. Store one such construction, its discarded scalar, and the factorization tree of the selected card from Lemma F.3.4. Equal factors at different positions in the tree remain distinct occurrences. Write $\mathscr{U}^{\mathrm{fac}}(A, f, H)$ for the token set with this attached factor work, and write

$$Q_{\mathrm{done}} \subseteq \mathscr{U}^{\mathrm{fac}}(A, f, H), \qquad Q = \mathscr{U}^{\mathrm{fac}}(A, f, H) \setminus Q_{\mathrm{done}}$$

for the completed and unfinished tokens, respectively. Processing a factor occurrence takes place within its token's tree, so the queue stays within this finite set while H and ν are fixed. A lower ν, a strict enlargement of the pulled-back group, or a complexity conclusion stops the local construction. The binary equality I and the original coordinates remain fixed throughout.

The next lemma either produces a deck-stable state, resolves the complexity of the current set, or exposes a smaller nonbinary irreducible signature.

Lemma F.5.9 (Deck stabilization under factor access)**.** *Let $\mathcal{F}_0$ be a retained, factor-closed finite algebraic signature set containing the supplied equality I, and let $f \in \mathcal{F}_0$ be an even-arity irreducible nonbinary signature of arity $n = \nu(\mathcal{F}_0)$.*

Then an established reduction determines the complexity, or $\mathsf{Tract}(K\mathcal{F}_0)$ *holds, or one obtains one of the following retained extensions:*

1. *a finite retained factor-closed set $\mathcal{F}_< \supseteq \mathcal{F}_0$ such that*

$$\text{K-Holant}(\mathcal{F}_<) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}_0), \qquad \nu(\mathcal{F}_<) < n;$$

every nonbinary irreducible signature retained in this successor has even arity;

2. *a finite retained factor-closed set $\mathcal{F}_\star \supseteq \mathcal{F}_0$ such that*

$$\text{K-Holant}(\mathcal{F}_\star) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}_0),$$

and, for $G_\star = G(\mathcal{F}_\star)$, every member of $\mathfrak{D}_{G_\star}(f)$ is zero or a $G_\star$-matching signature.

The stable successor retains the chosen representatives, the regenerated complete deck, and every card factor. Such factors are available by reduction over, but need not be directly realizable from, their predecessor.

Proof. Construct states $(A_i, \mathcal{F}_i, H_i, Q_i)$, where $H_i = \overline{G}(A_i, \mathcal{F}_i) \leq \mathrm{PGL}_2(L_0)$ is in the fixed pulled-back coordinates and Q_i is the unfinished queue defined above. At stage i, apply the reductions for unsafe binaries, consider the complete group of the resulting set $G_i = G(\mathcal{F}_i)$, retain one chosen representative of every class, and generate each normalized card in $\mathscr{U}(A_i, f, H_i)$ at most once. Preprocess and jointly close the resulting finite batch under irreducible factors to obtain $\mathcal{F}_{i+1}$. The supplied I excludes the one-sided exception. Inclusion supplies the reverse reduction at every exposure step, so every $\mathcal{F}_i$ is Turing-equivalent to $\mathcal{F}_0$.

Every nonbinary card factor has arity at most $n-2$. Odd factors are handled by Theorem F.2.9. An even one gives successor 1. Otherwise only binaries remain. No global normalization is made in this fixed-I procedure. Completeness therefore gives $H_i \leq H_{i+1}$. Only a strict inclusion $H_i \subsetneq H_{i+1}$

is a successor with a strictly larger complete transfer group, and it alone triggers regeneration of the complete deck.

Use the fixed pulled-back field L_0 from Lemma F.3.7. Gadget evaluation and pivot-normalized rank-one splitting are performed in pulled-back coordinates, so $H_i \le \mathrm{PGL}_2(L_0)$. Let $d_{L_0} := [L_0 : \mathbb{Q}]$ be the field degree, and define the torsion exponent and group-order bound by

$$e_{L_0} = \mathrm{lcm}\{r \ge 1 : \varphi(r) \le 2d_{L_0}\}, \qquad M_{L_0} = \max\{2e_{L_0}, 60\}, \tag{F.20}$$

where φ denotes Euler's totient function. Proposition F.5.6 and Lemma F.5.3 give $|H_i| \le M_{L_0}$. Hence the strict subgroup chain has at most $M_{L_0} - 1$ changes. Between changes, stabilization processes only the tokens indexed by the nonzero cards in $\mathscr{U}(A_i, f, H_i)$. Each token is removed once, and its factorization tree is finite by the decreasing potential Φ in Equation (F.9). A lower-arity factor, new group class, or complexity conclusion stops the current stage. Consequently there is a first stage at which both $H_{i+1} = H_i$ and the queue is empty. Call this the stabilized stage. At that stage define the stable retained set and its complete group by

$$\mathcal{F}_\star = \mathcal{F}_{i+1}, \qquad G_\star = G_{i+1}.$$

Every $0 \ne h \in \mathfrak{D}_{G_\star}(f)$ then has retained irreducible binary factors $B_1, \dots, B_m$ satisfying

$$[B_j X] \in G(\mathcal{F}_\star) = G_\star.$$

Writing $B_j = \beta_j B_{g_j}$, their blocks form a perfect matching, so

$$h = c \bigotimes_{j=1}^{m} B_{g_j}$$

for $c \ne 0$, as required. □

Remark (The role of stabilization). The contraction and factor arguments interact through the complete transfer group. Direct gadgets use the current set, whereas a factor may become available only after a reduction. Once retained, that factor can participate in further gadgets and enlarge the group, changing both the available contractions and the applicable group case. The procedure therefore retains every factor, compares the complete groups, and regenerates the deck after strict enlargement. This comparison includes multi-copy binary gadgets beyond the finite batch that produced the factor.

The fixed number field bounds the number of strict enlargements. Between them, the finite card indices and their factorization trees control the remaining work. Thus stabilization supplies a common starting point for the group-specific arguments: each can establish rigidity of the selected signature or obtain progress that is handled by the same descent proof. The conditions below make this requirement precise for constructions beyond the one-copy deck.

Later constructions can lower ν or enlarge the complete group. We call either change an *outer deck successor*. Processing a fixed family while both remain unchanged is a *within-node closure*.

Definition F.5.10 (Processing a finite family of signatures)**.** For deck-stable $(\mathcal{F}, G, f)$, take a finite family of signatures retained in or directly realized over $\mathcal{F}$, with their input orders and realization scalars. Add them, preprocess, jointly close the enlarged set under irreducible factors, apply the reductions for odd and unsafe factors, and consider the group of the resulting set. After a strict enlargement, regenerate the complete deck. We refer to these steps as processing the family. Each added signature has its realization or reduction, as stipulated above. In the fixed partition

$\mathscr{U}^{\mathrm{fac}} = Q_{\mathrm{done}} \sqcup Q$, each token moves from Q to Q_{done} once its card and all factor occurrences in its tree have been processed. A completed token stays in Q_{done}. For families indexed by constructions, completion is recorded at the specified index even when two constructions give the same signature. The descent conditions for continuations with the same group and minimum arity are stated in Definition F.5.11 and verified in the group-specific arguments.

Applications of the equality-accessible arity-four boundary also satisfy the requirements stated after Theorem F.5.22.

The current case is complete once finitely many such families resolve every branch and $Q = \varnothing$. Under the conditions of Definition F.5.11, a transition leaving the pulled-back group and ν unchanged completes a pending index, decreases its stored factor potential, or certifies that none remains, and hence is a within-node closure. Otherwise, ν decreases or the pulled-back group enlarges.

Standing predicates for relative cases. To keep the group-specific statements short, write

$$\mathsf{Stable}(\mathcal{F}, G, f) \iff (\mathcal{F}, G, f) \text{ is deck-stable in the sense of Definition F.5.8,}$$
$$\mathsf{Even}(\mathcal{F}) \iff \forall g \in \Pi(\mathcal{F}), \quad \mathrm{arity}(g) \geq 3 \Longrightarrow \mathrm{arity}(g) \in 2\mathbb{Z}.$$

Each later group branch must show that its constructions satisfy the following conditions. In particular, a step that preserves both H and ν must consume work in the already fixed finite family.

Definition F.5.11 (Conditions for descent)**.** Fix a deck-stable tuple $(\mathcal{F}, G, f)$, a coordinate matrix A, the pulled-back group $H = \overline{G}(A, \mathcal{F})$, and a fixed partition

$$\mathscr{U}^{\mathrm{fac}}(A, f, H) = Q_{\mathrm{done}} \sqcup Q,$$

where Q is the unfinished queue. Let $\mathfrak{B}$ be a collection of admissible families. Processing any member must give, on each branch, one of:

1. a resolved complexity leaf;
2. a successor with smaller ν;
3. a strict enlargement of the pulled-back complete group;
4. unchanged (H, ν), in which case every continuing card or factor occurrence retained for processing is indexed by a token of Q, and its processing is confined to that token's stored card, reduction, and attached factorization tree. The processing procedure either moves at least one such token to Q_{done}, or, while processing one fixed unfinished token, strictly decreases the factor potential of its stored worklist.

A token is marked done only after its full factorization tree is exhausted.

For a family $\mathfrak{C}$ of tractable signature classes,

$$\mathsf{Close}_{\mathfrak{C}}(\mathcal{F}, G, f) \tag{F.21}$$

means that finitely many applications of Definition F.5.10 resolve all branches for $(\mathcal{F}, G, f)$, with $Q = \varnothing$, and every surviving tractable case with a common presentation satisfies $\mathsf{Pres}_K(K\mathcal{F}; \mathscr{C})$ for some $\mathscr{C} \in \mathfrak{C}$. Established complexity conclusions and outer deck successors are also part of completion. The class restriction applies to cases certified by a common presentation. $\mathfrak{C} = \varnothing$ means that no tractable case with a common presentation remains. Thus Close is a completion predicate for this reduction procedure, whereas tensor closure refers to the class $\langle \mathscr{C} \rangle$. The predicates retain all procedure hypotheses and transition requirements.

The verification is necessarily branch-by-branch and uses the reductions proved in Sections F.6 to F.9. It is therefore deferred to Section F.10, after those four arguments are available.

The following lexicographic measure strictly decreases on every continuing step satisfying these conditions.

Lemma F.5.12 (Deck-transition dichotomy and termination)**.** *Fix the initial number field L_0. At a continuing all-even, binary-residual deck-stable node, carry the reduction data $(A, \mathcal{F}, f)$, a finite token queue Q, and the supplied equality I. Let $\mathfrak{B}$ satisfy Definition F.5.11, let $\mathcal{F}^+$ be the result of processing one batch in $\mathfrak{B}$ over $\mathcal{F}$, and define the pulled-back group by*

$$\overline{G}(A, \mathcal{F}) = [A]^{-1} G(\mathcal{F})[A].$$

Then

$$\text{K-Holant}(\mathcal{F}^+) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}). \tag{F.22}$$

Batch outputs are direct over $\mathcal{F}$, while their factors need only be available by reduction there.

After considering odd-arity signatures, unsafe binaries, and unary factors over the whole set, the only continuing transitions are:

1. *$\overline{G}(A, \mathcal{F}^+) = \overline{G}(A, \mathcal{F})$ and $\nu(\mathcal{F}^+) = \nu(\mathcal{F})$: process a previously unfinished token of Q, complete it or strictly decrease its factor potential, or certify that Q is empty.*
2. *$\overline{G}(A, \mathcal{F}^+) = \overline{G}(A, \mathcal{F})$ and $\nu(\mathcal{F}^+) < \nu(\mathcal{F})$: choose a new minimum-arity nonbinary irreducible signature and reapply Lemma F.5.9.*
3. *$\overline{G}(A, \mathcal{F}) \subsetneq \overline{G}(A, \mathcal{F}^+)$: choose a minimum-arity nonbinary irreducible signature in the resulting set, regenerate its deck, and restart marked routing.*

At every continuing node define the pair by

$$\Xi(A, \mathcal{F}) = (M_{L_0} - |\overline{G}(A, \mathcal{F})|, \nu(\mathcal{F})).$$

Refine it to the tuple

$$\Xi_{\text{full}}(A, \mathcal{F}, f, Q) = (M_{L_0} - |\overline{G}(A, \mathcal{F})|, \nu(\mathcal{F}), |Q|, \Phi_Q). \tag{F.23}$$

For each $u \in Q$, let $\mathcal{W}_u$ be the multiset of still-open occurrences in its attached factor worklist, and define

$$\Phi_Q := \sum_{u \in Q} \Phi(\mathcal{W}_u),$$

using the potential in Equation (F.9)*. The displayed tuple is ordered lexicographically. Items 2–3 strictly decrease its first two coordinates. In item 1, completing a token strictly decreases the third coordinate until the queue is empty. If a token remains unfinished during a genuine factor split, the fourth coordinate Φ_Q strictly decreases. If an exposed factor changes the pulled-back group, the operation is instead item 3 and is charged to the strict subgroup chain.*

Proof. Gadget replacement and factor closure prove Equation (F.22). I excludes the one-sided exception, and inclusion reverses the reduction. Adding signatures cannot increase ν. The continuing state remains in fixed-I coordinates. New classes can arise only from an exposed factor. Such a class is precisely a strict successor in the pulled-back group. By Lemma F.3.7 and Equation (F.20), every continuing group lies in $\mathrm{PGL}_2(L_0)$ and has order at most M_{L_0}. Therefore a lower ν decreases the second coordinate of Ξ, strict subgroup inclusion decreases the first. A fixed-(H, ν) step either decreases the finite queue counter in Ξ_{full} or, within its fixed token, decreases Φ. By Definitions F.5.10 and F.5.11, item 1 processes only a previously fixed token of Q, so these decreases exhaust every fixed-(H, ν) continuation. □

Reading the transition measure. The table matches each outcome of Lemma F.5.12 with the coordinate that decreases. The field L_0, original coordinates and I remain fixed. Write $H^+ = \overline{G}(A, \mathcal{F}^+)$ and $\nu^+ = \nu(\mathcal{F}^+)$.

Outcome	Next operation	Descent or exit
Resolved complexity leaf	Return the established whole-set conclusion.	Terminal
$H \subsetneq H^+$	Choose a new minimum-arity nonbinary irreducible signature, regenerate its deck, and restart marked routing.	First coordinate of Ξ_{full}
$H^+ = H,\ \nu^+ < \nu$	Choose a new minimum-arity nonbinary irreducible signature and reapply Lemma F.5.9.	Second coordinate
Same (H, ν), completed token	Move that token from Q to Q_{done} after its full factor tree is processed.	Third coordinate
Same (H, ν), unfinished token	Continue inside its fixed factor tree. A genuine factor split decreases Φ_Q.	Fourth coordinate

When Q is empty, the local family has been processed.

Definition F.5.13 (The conclusion required of each construction)**.** Write

$$\Sigma = (A, \mathcal{F}, H, f, Q), \qquad H = \overline{G}(A, \mathcal{F}),$$

with the record, deck-stability, fixed queue, and fixed-I data of Definitions F.5.10 and F.5.11. Write

$$\Xi_{\text{full}}(\Sigma) := \Xi_{\text{full}}(A, \mathcal{F}, f, Q) = \big(M_{L_0} - |H|, \nu(\mathcal{F}), |Q|, \Phi_Q\big)$$

for the lexicographic measure in Equation (F.23).

For each concrete construction at Σ, we prove that processing its outputs by Definition F.5.10 gives either

1. a resolved complexity leaf: hardness, a tractable case satisfying Tract, or a tractable case with a common K-presentation; or
2. a deck-stable tuple Σ', with the same required realization and induction data, satisfying

$$\Xi_{\text{full}}(\Sigma') <_{\text{lex}} \Xi_{\text{full}}(\Sigma).$$

We call the tuple in the second outcome a *strict stable successor*. This outcome includes a decrease of ν, a strict enlargement of the pulled-back complete group, completion of a fixed token, or a strict decrease of that token's factor potential. Every same-(H, ν) continuation must be indexed by the already fixed queue and processes only its stored cards, reductions, and factorization trees. A continuing successor includes this queue data and the strict descent asserted in item 2.

Item 1 resolves the complexity question. Item 2 establishes strict progress and continues the induction at Σ'. A *resolved construction*, in the terminology of the relative branch statements, is one for which either item has been proved, including realizability, factor access, and descent. A *resolved complexity leaf* always means item 1. The assertion $\mathsf{Close}_{\mathfrak{C}}(\mathcal{F}, G, f)$ additionally covers every case for the current (H, ν) and restricts the surviving common presentations to $\mathfrak{C}$, as in Equation (F.21).

For a selected stable tuple Σ, the continuing outcome of Definition F.5.13 decreases Ξ_{full}. To classify the whole set, the final proof in Section F.10 also uses an outer worklist induction over unresolved irreducible occurrences. Completing the current occurrence and choosing another of the same arity advances that outer induction. This separates progress for the selected signature from completion of the entire set.

F.5.3 The main stable-state theorem and four-way routing

The preceding reductions leave a deck-stable tuple $(\mathcal{F}, G, f)$. The next theorem is proved in Section F.10 from the four branches routed below. Together with stabilization, it gives the equality-accessible main theorem Theorem F.5.2.

Theorem F.5.14 (Relative matching-deck classification)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, G, f), \qquad \mathrm{arity}(f) = \nu(\mathcal{F}).$$

In particular, the deck-stable state contains the supplied equality I. If $\mathsf{Tract}(K\mathcal{F})$ *holds, then* $\text{K-Holant}(\mathcal{F}) \in \mathsf{FP}$. *Otherwise* $\text{K-Holant}(\mathcal{F})$ *is* $\#\mathsf{P}$*-hard. Equivalently, by Lemma F.2.7, the tractable condition is* $\mathsf{ReTract}(K\mathcal{F})$.

In each tractable case reached after the hardness reductions, the proof supplies one common product, affine, or flat-Lagrangian K-presentation. By Equation (F.6), the same basis also places the transformed edge in that class.

The supplied equality gives $x = [X] \in G$. Reversal then makes $j = [Z]$ normalize G. Let $\widehat{G} := \langle G, j\rangle$ be the extension obtained by adjoining the mark. The reversal identity gives $G \triangleleft \widehat{G}$ and $[\widehat{G} : G] \le 2$, while the distinct commuting involutions x, j generate a Klein-four subgroup of $\widehat{G}$. Their positions refine the abstract finite-group classification into the four branches used below. We record the group facts and matrix notation needed to make this refinement explicit.

Lemma F.5.15 (Finite projective-group facts)**.** *The following properties will be used below.*

1. *A cyclic group has at most one involution, and it has one exactly when its order is even.*

2. *For*
$$D_{2m} = \langle r, s : r^m = s^2 = 1,\ srs = r^{-1}\rangle,$$
the rotations $\langle r\rangle \cong C_m$ form a normal index-two subgroup, and every element outside it is a reflection. If m is odd, the rotation group is the unique index-two subgroup and the center is trivial. If m is even and $m \ge 4$, the center is $\{1, r^{m/2}\}$. For $m = 2$, $D_4 \cong V_4$ and all three nonidentity elements are central. Every subgroup of a cyclic or dihedral group is cyclic or dihedral.

3. *The three involutions of A_4, together with the identity, form its unique normal V_4. The group has no index-two subgroup, and the centralizer of an involution is V_4.*

4. *The unique index-two subgroup of S_4 is A_4. Its involutions are six transpositions and three double transpositions. The centralizer of a transposition has order 4, whereas the centralizer of a double transposition is D_8. The three double transpositions are the nonidentity elements of the unique normal V_4. Conjugation is transitive both on ordered pairs of distinct double transpositions and on ordered pairs of disjoint transpositions.*

5. *The group A_5 is simple and has no index-two subgroup. Its fifteen involutions form one conjugacy class, and the centralizer of each is V_4.*

6. *The element-order distributions are*

	1	2	3	*order* 4 *or* 5
A_4	1	3	8	0
S_4	1	9	8	6 *of order* 4
A_5	1	15	20	24 *of order* 5.

7. For the displayed models,

$$N_{\mathrm{PGL}_2(\mathbb{C})}(\mathcal{T}^{(\mathrm{int})}) = \mathcal{O}^{(\mathrm{dd})}, \qquad N_{\mathrm{PGL}_2(\mathbb{C})}(\mathcal{O}^{(\mathrm{dd})}) = \mathcal{O}^{(\mathrm{dd})}, \qquad N_{\mathrm{PGL}_2(\mathbb{C})}(\mathcal{I}^{(0)}) = \mathcal{I}^{(0)}.$$

8. The projective centralizers of the distinguished involutions are

$$C_{\mathrm{PGL}_2(\mathbb{C})}([Z]) = \left\{ \left[\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \right] : ab \neq 0 \right\} \sqcup \left\{ \left[\begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix} \right] : ab \neq 0 \right\}.$$

Thus $[Z]$ is the only nonidentity diagonal involution, while every nonsingular anti-diagonal class is an involution. Moreover,

$$C_{\mathrm{PGL}_2(\mathbb{C})}(\langle [X], [Z] \rangle) = \langle [X], [Z] \rangle \cong V_4.$$

Every finite subgroup of the displayed diagonal torus is cyclic.

Proof. Items 1–6 follow directly from the cyclic and dihedral presentations and the usual permutation models of A_4, S_4, A_5 [DM96]. For item 7, conjugation embeds each normalizer modulo its projective centralizer into the corresponding automorphism group. The projective centralizers are trivial. $\mathrm{Aut}(A_4) \cong S_4$, the displayed $\mathcal{O}^{(\mathrm{dd})}$ normalizes $\mathcal{T}^{(\mathrm{int})}$, and every automorphism of S_4 is inner. Since $\mathrm{Aut}(A_5) \cong S_5$, an additional outer normalizer of $\mathcal{I}^{(0)}$ would give a finite projective group H of order 120 containing A_5. The classification forces $H \cong C_{120}$ or D_{120}. Every subgroup of a cyclic or dihedral group is cyclic or dihedral, so H cannot contain A_5. Item 8 follows by solving $UZ \doteq ZU$, and then imposing the same identity with X. The ratio map $[\mathrm{diag}(a, b)] \mapsto a/b$ embeds the diagonal torus into $\mathbb{C}^\times$. Every finite subgroup of $\mathbb{C}^\times$ is a cyclic group of roots of unity. □

The abstract classification is up to projective conjugacy. Our group cases also retain the ordered marks $[X], [Z]$ in the original coordinates. A global change of basis or a binary gadget is used only with its corresponding holographic identity or realization.

Matrix and marked-pair dictionary. Combining Equations (F.4) and (F.16) gives the matrix identities

$$I = Q_0 = B_{0,0}, \qquad iX = Q_1 = iB_{1,0}, \qquad Y = Q_2 = B_{1,1}, \qquad iZ = Q_3 = iB_{0,1}.$$

Consequently the marked projective notation used throughout is

$$1 = [I] = [Q_0] = [B_{0,0}], \quad x = [X] = [Q_1] = [B_{1,0}],$$
$$j = [Z] = [Q_3] = [B_{0,1}], \quad xj = [Q_2] = [B_{1,1}].$$

Here the last identity is projective: $XZ = -Y$, whereas $ZX = Y$. For the transfer convention $T_B = BX$, the four Bell binaries give

B	Bell name	$[B]$	$[T_B] = [BX]$
I	$B_{0,0}$	1	x
X	$B_{1,0}$	x	1
Z	$B_{0,1}$	j	xj
Y	$B_{1,1}$	xj	j

The subscripted $B_{d,\ell}$ is a fixed ordered Bell signature, whereas B_g denotes a chosen realizable binary representative of a general transfer class g. Reversal fixes I, X, Z and sends $Y = B_{1,1}$ to $-Y$, without changing its projective class.

Use the diagonal family $D(t)$ defined in Section F.2.3. For the dihedral forms, the subscript gives the group order and the superscript records the direction of the rotation subgroup: Z, Y, or X. The labels $(X,0)$ and $(X,1)$ distinguish the unshifted and shifted reflection cosets, respectively. For $m \geq 2$, define the matrix family $C(t)$ and the first two marked dihedral families

$$\begin{aligned} D_{2m}^{(Z)} &= D_{2m}^{\mathrm{std}} = \{[D(t)], [XD(t)] : t \in \mu_m\}, \\ C(t) &= (1+t)I + i(1-t)Y, \\ D_{2m}^{(Y)} &= \{[C(t)], [XC(t)] : t \in \mu_m\}. \end{aligned}$$

For even $N \geq 2$, define the cyclic group

$$A(t) = (1+t)I + (1-t)X, \qquad C_N^{(X)} := \{[A(t)] : t \in \mu_N\}.$$

Its unique involution is $x = [X] = [A(-1)]$. For even m, fix $\eta \in \mu_{2m} \setminus \mu_m$ and define

$$D_{2m}^{(X,1)} = C_m^{(X)} \cup \{[ZA(\eta t)] : t \in \mu_m\}.$$

In the same even parameter range, define the unshifted marked form

$$D_{2m}^{(X,0)} = \{[A(t)], [ZA(t)] : t \in \mu_m\}.$$

These are projective displays. Any suppressed factor of two belongs to the coordinate model and does not normalize a chosen representative.

The symbols $\mathcal{T}^{(\mathrm{int})}, \mathcal{T}^{(\mathrm{ext})}$, $\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$, and $\mathcal{I}^{(0)}, \mathcal{I}^{(1)}$ denote the marked tetrahedral, octahedral, and icosahedral forms specified in the theorem. Here int and ext record whether $j \in G$ or $j \notin G$. For octahedral forms, d denotes a double transposition and t a transposition, in the order (x, j). The labels $0, 1$ distinguish the two icosahedral coordinate displays of the same abstract marked type. Section F.9 gives the explicit fixed-equality representatives and coefficient correspondences.

Theorem F.5.16 (Marked finite-group routing)**.** *Let $\mathcal{F}$ be a retained factor-closed set containing the supplied equality I, and let $G = G(\mathcal{F})$ be its complete transfer group (finite, with safe binary representatives). Recall the ordered marked pair and its generated group:*

$$(x, j) := ([X], [Z]), \qquad x \in G, \qquad \widehat{G} := \langle G, j \rangle.$$

In the fixed coordinates containing the supplied I, exactly one of the following alternatives holds:

1. *$G \cong D_{2m}$, $m \geq 3$, and the marked configuration $(G; x, j)$ has one of the forms denoted by $D_{2m}^{(Z)}, D_{2m}^{(Y)}, D_{2m}^{(X,1)}, D_{2m}^{(X,0)}$. The last two families occur only for even m.*
2. *$G = C_N^{(X)} \cong C_N$, where N is even and $j \notin G$. This is the cyclic branch.*
3. *$G \cong V_4$, and G is either the standard Pauli group $V_4^{(\mathrm{std})} = D_4^{(Z)} = D_4^{(Y)}$ or the exotic group $V_4^{(\mathrm{ex})} = D_4^{(X,1)}$.*
4. *G is one of the two marked tetrahedral forms $\mathcal{T}^{(\mathrm{int})}, \mathcal{T}^{(\mathrm{ext})}$, the four fixed-equality marked octahedral forms $\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$, or one of the two icosahedral displays $\mathcal{I}^{(0)}, \mathcal{I}^{(1)}$.*

Positions of the marks. *All conjugacy statements in this paragraph are taken in $\widehat{G}$. In every form, x and j are distinct commuting involutions and $\langle x, j \rangle = \{1, x, j, xj\} \cong V_4$. In the four dihedral forms*

the marked positions are

$$D_{2m}^{(Z)}: \quad x=[XD(1)], \quad j=[D(-1)], \quad xj=[XD(-1)], \qquad \widehat{G}=\begin{cases} D_{2m}^{(Z)}, & 2 \mid m, \\ D_{4m}^{(Z)}, & 2 \nmid m; \end{cases}$$

$$D_{2m}^{(Y)}: \quad x=[XC(1)], \quad j=[XC(-1)], \quad xj=[C(-1)], \qquad \widehat{G}=\begin{cases} D_{2m}^{(Y)}, & 2 \mid m, \\ D_{4m}^{(Y)}, & 2 \nmid m; \end{cases}$$

$$\begin{aligned} D_{2m}^{(X,1)}: \quad & x=[A(-1)] \in G, \quad j=[ZA(1)] \in \widehat{G} \setminus G, \quad xj=[ZA(-1)] \in \widehat{G} \setminus G; \\ & \widehat{G}=\{[A(t)] : t \in \mu_{2m}\} \cup \{[ZA(t)] : t \in \mu_{2m}\} \cong D_{4m}; \\ D_{2m}^{(X,0)}: \quad & x=[A(-1)], \quad j=[ZA(1)], \quad xj=[ZA(-1)], \qquad \widehat{G}=G=D_{2m}^{(X,0)}. \end{aligned}$$

For $G=D_{2m}^{(Z)}$, the central half-turn of $\widehat{G}$ is j and the other two marks are reflections. For $G=D_{2m}^{(Y)}$, it is xj and the reflections are x, j. In both cases $j \in G$ exactly when m is even. The two marked reflections are conjugate in $\widehat{G}$ exactly when $4 \mid m$. In $D_{2m}^{(X,1)}$, where m is even, x is the central half-turn and j, xj are conjugate reflections in $\widehat{G} \setminus G$. In $D_{2m}^{(X,0)}$, x is again the central half-turn, while $j, xj \in G$ are reflections. They are conjugate in G exactly when $4 \mid m$. When m is odd in either $D_{2m}^{(Z)}$ or $D_{2m}^{(Y)}$, one has $j, xj \in \widehat{G} \setminus G$. When m is even, all three marks lie in G.

In the cyclic branch $G=C_N^{(X)}$, the order N is even, $j, xj \in \widehat{G} \setminus G$, $\widehat{G} \cong D_{2N}$, x is the central half-turn, and j, xj are reflections, conjugate exactly when $4 \mid N$.

At $m=2$, the standard Pauli form has $\widehat{G}=G$, with x, j, xj the three nonidentity elements (three singleton conjugacy classes) of $V_4^{(\mathrm{std})}$. The exotic form has $j, xj \in \widehat{G} \setminus G$, $\widehat{G} \cong D_8$, central half-turn x, and conjugate outside reflections j, xj. In $\mathcal{T}^{(\mathrm{int})}$, $\widehat{G}=G \cong A_4$ and $x=[Q_1], j=[Q_3], xj=[Q_2]$ are the three double transpositions, hence one conjugacy class. In $\mathcal{T}^{(\mathrm{ext})}$, $G \cong A_4$, $G \triangleleft \widehat{G}$, and $\widehat{G} \cong S_4$. $x=[Q_1] \in G$ is a double transposition, and $j=[Q_3], xj=[Q_2] \in \widehat{G} \setminus G$ are disjoint transpositions in one class. In $\mathcal{O}^{(\mathrm{dd})}$, $\widehat{G}=G \cong S_4$ and x, j, xj are the three double transpositions. In $\mathcal{O}^{(\mathrm{tt})}$, x, j are disjoint transpositions and xj is a double transposition. In $\mathcal{O}^{(\mathrm{dt})}$, x is a double transposition and j, xj are transpositions. In $\mathcal{O}^{(\mathrm{td})}$, j is a double transposition and x, xj are transpositions. In all four cases $\widehat{G}=G \cong S_4$. Finally, in each icosahedral coordinate display $\widehat{G}=G \cong A_5$, and $x=[Q_1], j=[Q_3], xj=[Q_2]$ lie in its unique involution class.

Proof. The reversal identity $[T] \mapsto j[T]^{-1}j=[ZT^{-1}Z]$ (with $j=[Z]$) and completeness give

$$jGj=G, \qquad G \triangleleft \widehat{G}, \qquad [\widehat{G}:G] \leq 2.$$

The two distinct projective involutions x, j commute, so $V_4^{(\mathrm{std})}=\langle x, j \rangle \leq \widehat{G}$. By Proposition F.5.6, this leaves an even dihedral group or one of A_4, S_4, A_5.

Suppose first that $G \cong D_{2m}$ with $m \geq 3$. Then $\widehat{G}$ is also dihedral. Indeed, if $\widehat{G}$ were Platonic and $[\widehat{G}:G]=2$, then G would be an index-two normal subgroup. By Lemma F.5.15, the only such subgroup among the Platonic groups is $A_4 \triangleleft S_4$, which is not dihedral. The case $\widehat{G}=G$ is immediate. Choose a rotation generator r and a reflection generator s so that $\widehat{G}=\langle r, s : r^n=s^2=1,\ srs=r^{-1} \rangle$. Since $\widehat{G}$ contains $\langle x, j \rangle \cong V_4$, n is even. Exactly one of x, j, xj is the central half-turn $r^{n/2}$, and the other two are reflections. If the central mark is j or xj, we obtain $D_{2m}^{(Z)}$ or $D_{2m}^{(Y)}$, respectively. If it is x, membership $j \in G$ gives $D_{2m}^{(X,0)}$, while $j \notin G$ gives $D_{2m}^{(X,1)}$. For $j \in G$, one has $\widehat{G}=G$, so $n=m$ is even. For $j \notin G$, one has $n=2m$ and $G=\langle r^2, r^k s \rangle$ for some $k \in \{0,1\}$. The central half-turn r^m of $\widehat{G}$ belongs to G exactly when m is even. Thus m is odd in the first two outside-mark cases and even in $D_{2m}^{(X,1)}$, where $x \in G$. This exhausts the four forms and their parameter ranges.

In $D_{2m}^{(X,0)}$, the mark $j = [Z]$ belongs to the complete group. Its fixed chosen representative B_j satisfies $[B_j X] = j$, hence $B_j \doteq Y$. This realizable binary and the already available I give the signature-preserving edge switch used in Theorem F.6.16. The distinguished equality is never replaced or re-anchored.

The marked eigenline equations give the displayed matrices, with

$$ZC(t)Z = tC(t^{-1}), \qquad x = [XC(1)],$$

while $x = [A(-1)]$ is central in both $D_{2m}^{(X,1)}$ and $D_{2m}^{(X,0)}$. Hence the four marked configurations are legally distinct.

Next suppose that G is cyclic. By Lemma F.5.15, its unique involution is x. A class commuting with $[X]$ lies in $\mathrm{span}\{I, X\}$ or $\mathrm{span}\{Z, Y\}$, and every nonsingular member of the latter is another involution. Thus all transfer classes have representatives in $\mathrm{span}\{I, X\}$. Their eigenvalue ratios form μ_N, where $N = |G|$. Thus $G = C_N^{(X)}$, N is even, and $j \notin G$. Conjugation by j inverts the eigenvalue ratio of every class in this algebra, so

$$\widehat{G} = G \rtimes \langle j \rangle \cong D_{2N}.$$

Consequently, x is the central half-turn and j, xj are the two marked reflections.

It remains to consider $G \cong V_4$, corresponding to $m = 2$ in the dihedral notation. Here

$$V_4^{(\mathrm{std})} = D_4^{(Z)} = D_4^{(Y)} = \{[I], [X], [Z], [Y]\}.$$

For $\eta = i$,

$$\frac{ZA(i)}{1-i} = Y + iZ, \qquad \frac{ZA(-i)}{1+i} = Y - iZ,$$

so $V_4^{(\mathrm{ex})} = \{[I], x, [Y + iZ], [Y - iZ]\}$. Conversely, every subgroup $G \cong V_4$ containing x has the form

$$\{[I], x, [aZ + bY], [bZ + aY]\}, \qquad a^2 \neq b^2,$$

and $jGj = G$ gives either $ab = 0$ (the Pauli form) or $a^2 = -b^2$ (the exotic form $V_4^{(\mathrm{ex})}$).

Thus the Pauli form has $\widehat{G} = G$, with x, j, xj its three nonidentity elements. In the exotic form, $j, xj \notin G$ and $\widehat{G} = \langle G, j \rangle \cong D_8$. Relative to this $\widehat{G}$, x is the central half-turn and j, xj are conjugate reflections in the outside coset.

It remains to route the Platonic groups while retaining the ordered marked pair. Let $G \cong A_4$. If $j \in G$, then $\widehat{G} = G$, and the three nonidentity elements x, j, xj of $V_4^{(\mathrm{std})}$ are precisely the double transpositions. This is $\mathcal{T}^{(\mathrm{int})}$. If $j \notin G$, then $|\widehat{G}| = 24$. By Proposition F.5.6 and Lemma F.5.15, the finite-PGL_2 classification and the subgroup structure of cyclic and dihedral groups force $\widehat{G} \cong S_4$ with $G \triangleleft \widehat{G}$ and $G \cong A_4$. Thus $x \in G$ is a double transposition, whereas the odd involutions $j, xj \in \widehat{G} \setminus G$ are transpositions. Since they are distinct and commute, they are disjoint. This is $\mathcal{T}^{(\mathrm{ext})}$.

Now let $G \cong S_4$. If $j \notin G$, then $\widehat{G} = \langle G, j \rangle$ has order 48 and is cyclic or dihedral by Proposition F.5.6, contrary to containing S_4 by Lemma F.5.15. Hence $j \in G$.

The two commuting involutions may independently lie in the transposition or double-transposition class, subject to the class of their product. The four resulting ordered class patterns are

$$(dd; d), \qquad (tt; d), \qquad (dt; t), \qquad (td; t),$$

where the entries record the classes of $(x, j; xj)$. Conjugation in S_4 is transitive on ordered pairs in each pattern by Lemma F.5.15. They are therefore exactly the four fixed-equality marked types

$$\mathcal{O}^{(\mathrm{dd})}, \qquad \mathcal{O}^{(\mathrm{tt})}, \qquad \mathcal{O}^{(\mathrm{dt})}, \qquad \mathcal{O}^{(\mathrm{td})}.$$

Their ordered marks distinguish the four types in the fixed coordinates. Concrete representatives and direct finite exits are recorded in Section F.9 and Appendix F.A.5.

For $G \cong A_5$, if $j \notin G$, then $\widehat{G}$ has order 120 and is cyclic or dihedral by Proposition F.5.6, contrary to containing A_5 by Lemma F.5.15. Thus $\widehat{G} = G$. Finally, Lemma F.5.15 shows that A_5 has one involution class, with x, j, xj in that class. Once an ordered pair (x, j) is fixed, two conjugators differ by

$$C_{\mathrm{PGL}_2(\mathbb{C})}(\langle x, j\rangle) = \langle x, j\rangle.$$

Each of these four projective classes has a representative in $GO(X)$. Hence the abstract marked type is unique. Fixed-equality coordinates give the two displays $\mathcal{I}^{(0)}, \mathcal{I}^{(1)}$. Section F.9 verifies their explicit representatives and a direct coefficient-and-card correspondence that leaves the supplied I untouched. This deferred coordinate calculation is not used to classify the abstract group, so the four-way list is exhaustive. □

Each of the four group-specific sections, Sections F.6 to F.9, proves a relative theorem for its corresponding stable tuple. It either determines the complexity at that tuple, or produces a reduction step whose processing yields a strictly smaller active tuple under Ξ_{full}. In the second case, the deck is stabilized again and the new tuple is routed anew. Movement between distinct same-arity irreducible signatures in a proof for the whole signature set remains governed by the separate outer worklist induction of Definition F.5.13.

Coefficient and open-network correspondences in the branch proofs leave the retained set and supplied I fixed. If factor access enlarges G, reapply Lemma F.5.9 and the marked classification to the new complete group.

Remark F.5.17 (Connection with Xia's binary-group taxonomy). Version 1 of Xia's framework [Xia26, Sections 1 and 4] organizes the finite projective binary residue into nine *unmarked* rotation-group types. Here the group consists of transfer classes $[T_B] = [BX]$, rather than projective classes $[B]$ of the underlying binary matrices, and the supplied binary equality fixes the marks

$$x = [X] \in G, \qquad j = [Z], \qquad jGj = G.$$

Consequently, our routing refines Xia's taxonomy by the ordered marked pair (x, j). The table compares the unmarked types with the cases compatible with the supplied I in this section.

Xia's unmarked type	Corresponding case in this proof
C_1	absent at a continuing node, where a nonidentity transfer is already retained
C_2	the order-two cyclic case $C_2^{(X)}$, including the Reed–Muller case of Section F.7
C_n, $n \geq 3$	the fixed-I cyclic case $C_N^{(X)} \ni [X]$, with even $N \geq 4$
D_{2m}, m odd	$D_{2m}^{(Z)}, D_{2m}^{(Y)}$, with odd $m \geq 3$
$V_4 = D_4$	$V_4^{(\mathrm{std})}, V_4^{(\mathrm{ex})}$
D_{2m}, even $m \geq 4$	$D_{2m}^{(Z)}, D_{2m}^{(Y)}, D_{2m}^{(X,0)}, D_{2m}^{(X,1)}$
A_4	$\mathcal{T}^{(\mathrm{int})}, \mathcal{T}^{(\mathrm{ext})}$
S_4	$\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$
A_5	the two fixed-equality coordinate displays

The marked classification and the corresponding signature-set reductions are proved here.

F.5.4 Strict growth of the complete transfer group

The preceding classification identifies the branch for a fixed complete group. A later contraction or factor reduction may enlarge that group and return the proof to a different branch. Figure F.1 shows how these branches fit together, retaining the marked distinctions that determine which later argument applies.

Work in the fixed coordinates containing the supplied I. Let $\mathcal{F} \subseteq \mathcal{F}^+$ be a retained factor-closed set, put

$$G = G(\mathcal{F}), \qquad G^+ = G(\mathcal{F}^+),$$

and suppose that the unsafe-binary case distinction has not already applied. If a newly exposed ordered binary B is safe and $[BX] \notin G$, then Lemma F.5.4 gives the strict inclusion $G < G^+$. If $[BX] \in G$, its transfer class is already present. We still consider the complete group of the resulting set, which may realize additional multi-copy binaries beyond the batch that exposed B. Accordingly, G^+ is the complete group of realizable transfer classes, and need not equal the subgroup generated by G and the first new class. Whenever $G < G^+$, regenerate the deck and reroute before invoking a group-specific theorem.

Recall the marked extension $\widehat{G} = \langle G, j\rangle$ from Theorem F.5.16. The diagram tracks inclusions of G. An outside mark j contributes to growth only when it enters the complete group of the enlarged set. For example, adjoining j to $D_{2m}^{(X,1)}$ gives $D_{4m}^{(X,0)}$. The standard Klein-four node is $V_4^{(\mathrm{std})} = D_4^{(Z)} = D_4^{(Y)}$, and the exotic node is $V_4^{(\mathrm{ex})} = D_4^{(X,1)}$.

Proposition F.5.18 (Finite marked-successor routing)**.** *Let $\mathcal{F} \subseteq \mathcal{F}^+$ be retained factor-closed sets containing the supplied equality I, both in the binary-residual regime, and suppose*

$$G = G(\mathcal{F}) < G^+ = G(\mathcal{F}^+).$$

After recomputing the complete group and applying the marked classification Theorem F.5.16, the marked type of G^+ is reachable from the marked type of G by a directed path in Figure F.1, allowing the suppressed same-family divisibility steps. The paths describe the group inclusions compatible with the fixed marks. Their occurrence for a particular signature set depends on its realizable binaries.

Proof. The fixed marks $x = [X]$ and $j = [Z]$, together with reversal closure, give

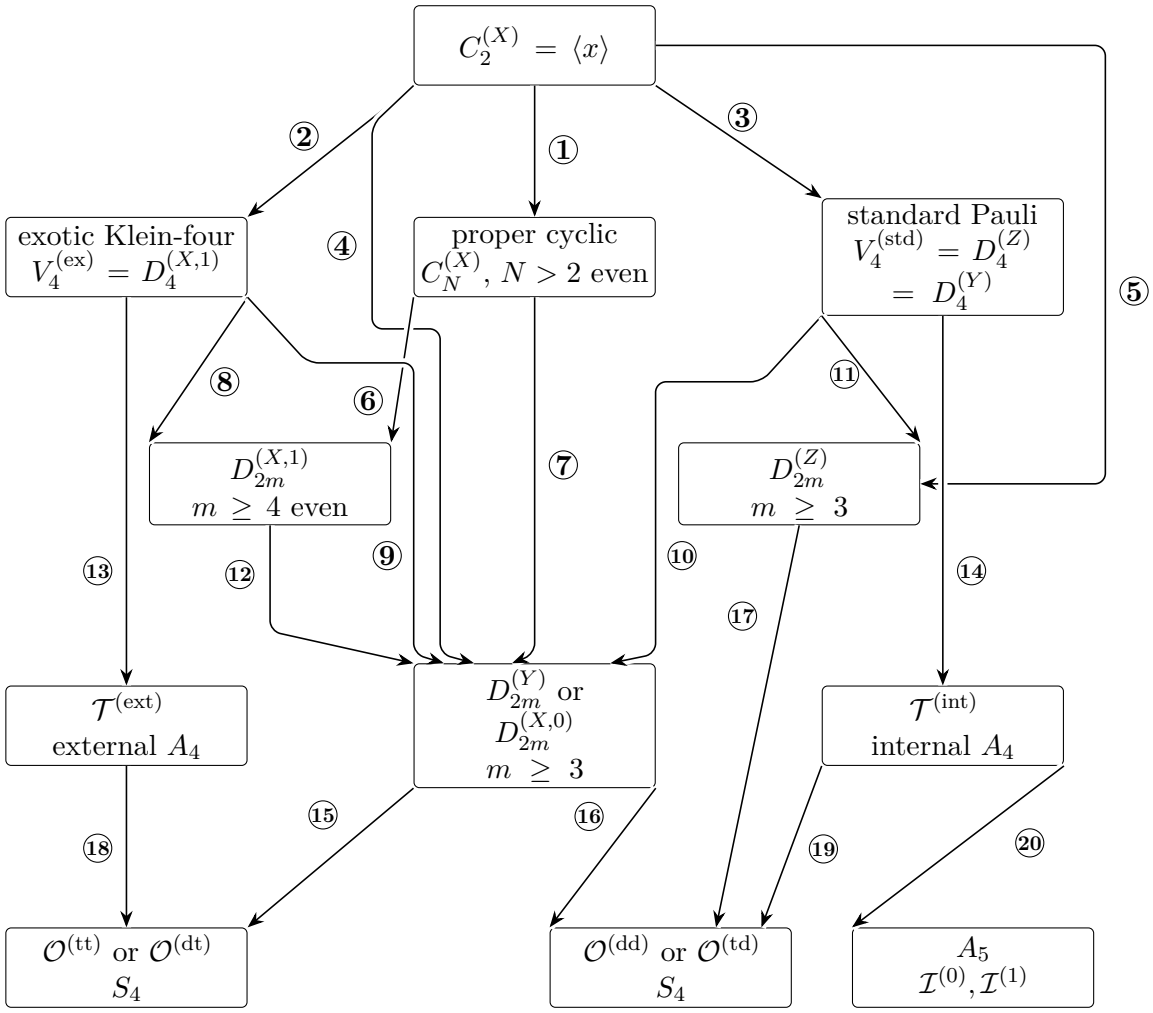
$$x \in G < G^+, \qquad jGj = G, \qquad jG^+j = G^+.$$

Hence $\widehat{G} \leq \langle G^+, j\rangle$, and if $j \in G^+$ then $\widehat{G} \leq G^+$. Apply Proposition F.5.6 and Lemma F.5.15 and retain this marked-pair constraint throughout.

Suppose first that $G = C_N^{(X)}$, where N is even. For $N > 2$, every embedding in a dihedral overgroup places $C_N^{(X)}$ in the rotation subgroup; therefore its unique involution x is the central half-turn. If $j \notin G^+$, the routed form is $D_{2m}^{(X,1)}$; if $j \in G^+$, the routed form is $D_{2m}^{(X,0)}$. Enlarging first inside the cyclic family accounts for the displayed condition $N \mid m$. The group $C_2^{(X)} = \langle x\rangle$ is exceptional because x may instead be a reflection, giving the odd-parameter entrances to $D_{2m}^{(Z)}$ and $D_{2m}^{(Y)}$; the even-parameter entrances factor through one of the displayed Klein forms.

For the Klein boundary, $V_4^{(\mathrm{std})}$ already contains j, so a dihedral overgroup has all three marked involutions and routes to $D_{2m}^{(Z)}$, $D_{2m}^{(Y)}$, or $D_{2m}^{(X,0)}$ with even m. These three cases are distinguished by which of j, xj, x, respectively, is the central half-turn of the dihedral overgroup. In the exotic form $V_4^{(\mathrm{ex})} = D_4^{(X,1)}$, the mark j lies outside. The reflection parameters in $D_{2m}^{(X,1)}$ form $\mu_{2m} \setminus \mu_m$. Since $\mu_4 \setminus \mu_2 = \{i, -i\}$, for even $m > 2$ we have

$$D_4^{(X,1)} < D_{2m}^{(X,1)} \quad \Longleftrightarrow \quad i^m = -1 \quad \Longleftrightarrow \quad m \equiv 2 \pmod 4.$$



① $C_2^{(X)} \to C_N^{(X)}$
② $C_2^{(X)} \to V_4^{(\mathrm{ex})}$
③ $C_2^{(X)} \to V_4^{(\mathrm{std})}$
④ $C_2^{(X)} \to D_{2m}^{(Y)}$, m odd.
⑤ $C_2^{(X)} \to D_{2m}^{(Z)}$, m odd.
⑥ $C_N^{(X)} \to D_{2m}^{(X,1)}$, $N \mid m$.
⑦ $C_N^{(X)} \to D_{2m}^{(X,0)}$; $N \mid m$, j enters G^+.
⑧ $V_4^{(\mathrm{ex})} \to D_{2m}^{(X,1)}$; $m \equiv 2 \pmod 4$.
⑨ $V_4^{(\mathrm{ex})} \to D_{2m}^{(X,0)}$; $4 \mid m$, j enters G^+.
⑩ $V_4^{(\mathrm{std})} \to D_{2m}^{(Y)}$ or $D_{2m}^{(X,0)}$, $2 \mid m$.
⑪ $V_4^{(\mathrm{std})} \to D_{2m}^{(Z)}$, $2 \mid m$.
⑫ $D_{2m}^{(X,1)} \to D_{4m}^{(X,0)}$, j enters G^+.
⑬ $V_4^{(\mathrm{ex})} \to \mathcal{T}^{(\mathrm{ext})}$.
⑭ $V_4^{(\mathrm{std})} \to \mathcal{T}^{(\mathrm{int})}$.
⑮ $D_8^{(Y)} \to \mathcal{O}^{(\mathrm{tt})}$ or $D_8^{(X,0)} \to \mathcal{O}^{(\mathrm{dt})}$.
⑯ $D_8^{(Y)} \to \mathcal{O}^{(\mathrm{dd})}$ or $D_8^{(X,0)} \to \mathcal{O}^{(\mathrm{dd})}$.
⑰ $D_8^{(Z)} \to \mathcal{O}^{(\mathrm{dd})}$ or $\mathcal{O}^{(\mathrm{td})}$.
⑱ $\mathcal{T}^{(\mathrm{ext})} \to \mathcal{O}^{(\mathrm{dt})}$, j enters G^+.
⑲ $\mathcal{T}^{(\mathrm{int})} \to \mathcal{O}^{(\mathrm{dd})}$.
⑳ $\mathcal{T}^{(\mathrm{int})} \to \mathcal{I}^{(0)}$ or $\mathcal{I}^{(1)}$.

Here j enters G^+ when $j \notin G$ and $j \in G^+$. The parameters satisfy strict inclusion, with $m \geq 3$ for dihedral targets and even $m \geq 4$ for $D_{2m}^{(X,1)}, D_{2m}^{(X,0)}$.

Figure F.1: Possible strict growth of the complete transfer group with the supplied equality fixed. Numbered arrows indicate subgroup inclusions between marked forms, with their parameter conditions listed below the graph. Divisibility steps within a family are suppressed. Grouped nodes collect distinct marked forms at the same level of the diagram.

When $4 \mid m$, the inclusion $V_4^{(\mathrm{ex})} < D_{2m}^{(X,0)}$ factors through the subgroup $\langle V_4^{(\mathrm{ex})}, j\rangle = D_8^{(X,0)}$. More generally,

$$\langle D_{2m}^{(X,1)}, j\rangle = D_{4m}^{(X,0)}.$$

The remaining dihedral-over-dihedral possibilities are precisely the routine divisibility inclusions suppressed from the drawing. In the $(X,1)$-family, an odd index preserves the shifted reflection coset; an even index enters an unshifted $D_{2n}^{(X,0)}$ form. Thus all cyclic, Klein, and dihedral successors follow the displayed paths.

It remains to attach the Platonic groups. The unique normal Klein four in A_4, together with whether j belongs to the current group, gives

$$V_4^{(\mathrm{std})} < \mathcal{T}^{(\mathrm{int})}, \qquad V_4^{(\mathrm{ex})} < \mathcal{T}^{(\mathrm{ext})}.$$

The displayed models in Equations (F.17) to (F.19) give

$$\mathcal{T}^{(\mathrm{int})} < \mathcal{O}^{(\mathrm{dd})}, \qquad \mathcal{T}^{(\mathrm{int})} < \mathcal{I}^{(0)} \cap \mathcal{I}^{(1)}.$$

For the external tetrahedral form, adjoining j gives the mixed fixed-I form $\mathcal{O}^{(\mathrm{dt})}$.

The only marked dihedral groups of parameter at least three that can enter a Platonic successor have parameter four, with fixed-mark inclusions

$$D_8^{(Z)} < \mathcal{O}^{(\mathrm{dd})} \text{ or } \mathcal{O}^{(\mathrm{td})}, \qquad D_8^{(Y)} < \mathcal{O}^{(\mathrm{dd})} \text{ or } \mathcal{O}^{(\mathrm{tt})}, \qquad D_8^{(X,0)} < \mathcal{O}^{(\mathrm{dd})} \text{ or } \mathcal{O}^{(\mathrm{dt})}.$$

Indeed, an S_4 dihedral subgroup of parameter at least three has parameter 3 or 4, by the element orders in Lemma F.5.15. At parameter 3, the marked extension $\widehat{G}$ has a rotation of order 6, which S_4 does not. At parameter 4, the $D_8^{(Z)}$, $D_8^{(Y)}$, and $D_8^{(X,0)}$ marked positions give the displayed inclusions, while $\widehat{D_8^{(X,1)}} \cong D_{16}$ cannot occur. The domain and second-form matrices in Equations (F.18) and (F.69), together with the conjugations and mark checks in Equations (F.70) to (F.72), verify the three inclusions. Here the second-form similarity identifies subgroup classes. The same argument for A_5 leaves only parameters 3 and 5, but their marked extensions contain rotations of order 6 and 10, absent from A_5. No parameter-at-least-three dihedral group lies in A_4: the only possible parameter is 3, and an order-six subgroup would have index two, contrary to Lemma F.5.15.

Finally, $\mathcal{T}^{(\mathrm{ext})}$ cannot grow to a group isomorphic to A_5, because the target contains j and would therefore contain $\langle \mathcal{T}^{(\mathrm{ext})}, j\rangle \cong S_4$, whereas $24 \nmid 60$. The groups S_4 and A_5 have no strict finite overgroup in $\mathrm{PGL}_2(\mathbb{C})$: the finite-group classification and orders exclude every remaining type. This proves exhaustiveness. □

Every strict inclusion increases $|G|$, and therefore decreases the group coordinate of Ξ_{full} in Lemma F.5.12. The bounded-field argument bounds the possible orders along a run, including the same-family divisibility steps suppressed in the figure. Thus the diagram complements the arity descent: when a construction produces new binary transfers instead of a smaller irreducible signature, the proof continues in a strictly larger group.

F.5.5 Shared methods for the four branch proofs

The group branches use two common tools. Binary contractions determine either a full two-input slice or only part of it, which separates the cyclic analysis from the other cases. At arity four, a shared gadget construction connects the stable deck to the eight-vertex reductions.

F.5.5.1 Two-input slices and contractions

For two binary chains L, R, order each matrix with the free input as its row and the joined input as its column. Joining their second inputs through X gives the external tensor LXR^{T}. This fixes the transpose convention for the contraction kernels below.

For an arity-r tensor f, ordered inputs $p = (i, j)$, and an assignment to the remaining inputs $\mathbf{z}$, the *two-input slice* is the binary matrix

$$F^f_{p,\mathbf{z}}(a, b) := f(\mathbf{z}^{i\leftarrow a, j\leftarrow b}), \qquad a, b \in \mathbb{F}_2.$$

If $\mathscr{K} \leq \mathrm{Mat}_2(\mathbb{C})$ is a space of contraction matrices, its Frobenius-orthogonal space is

$$\mathscr{K}^{\perp_{\mathrm{Frob}}} := \{F \in \mathrm{Mat}_2(\mathbb{C}) : \langle C, F\rangle_{\mathrm{Frob}} = 0 \text{ for every } C \in \mathscr{K}\}.$$

Thus contractions against $\mathscr{K}$ determine the slice modulo $\mathscr{K}^{\perp_{\mathrm{Frob}}}$.

We determine the transfer span and use its fixed identification with contraction kernels to recover two-input slices.

Definition F.5.19 (The transfer span $\mathscr{K}_G$)**.** For a retained set in the fixed-I, X coordinates, with complete transfer group G, let $\mathscr{K}_G$ be the complex span of the transfer matrices $T_B = BX$ of all finite ordered two-input chains B made from the fixed representatives B_g, $g \in G$, the explicit I- or X-edge links, and their input reversals. This is equivalently the span of independently dressed I/X kernels after the fixed right multiplication by X, which is invertible and therefore only relabels the corresponding slice coordinates. For a closing binary B, its contraction kernel is $C = XBX = XT_B$. Thus $\langle C, F\rangle_{\mathrm{Frob}} = \langle T_B, XF\rangle_{\mathrm{Frob}}$. We use this fixed identification when reconstructing slices from $\mathscr{K}_G$.

The information determined by binary contractions depends on $\mathscr{K}_G$. The following theorem gives its two possible values.

Define the unnormalized Walsh matrix P and, for a signature f already in K-coordinates, its Walsh-coordinate tensor $\widetilde{f}$ by

$$P := \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \widetilde{f} := (P^{-1})^{\otimes \operatorname{arity}(f)} f. \tag{F.24}$$

The normalized Walsh matrix used in the proof of Theorem F.5.20 is $S = P/\sqrt{2}$. Thus $SXS = Z$, whereas $P^{\mathsf{T}}XP = 2Z$. Since $P^{-1}A(t)P = 2D(t)$, the cyclic group in Walsh coordinates is $P^{-1}C_N^{(X)}P = C_N^{\mathrm{std}}$.

Theorem F.5.20 (The span of realizable binary transfers)**.** *Let $\mathcal{F}$ be a retained set with normalized I, X, and let $G = G(\mathcal{F})$. With $\mathscr{K}_G$ as in Definition F.5.19,*

$$\mathscr{K}_G \in \{\mathrm{span}\{I, X\}, \mathrm{Mat}_2(\mathbb{C})\}.$$

Moreover,

$$\mathrm{span}\{I, X\}^{\perp_{\mathrm{Frob}}} = \left\{ \begin{pmatrix} u & v \\ -v & -u \end{pmatrix} : u, v \in \mathbb{C} \right\}.$$

If $\mathscr{K}_G = \mathrm{Mat}_2(\mathbb{C})$, four realizable kernel evaluations determine each two-input slice. If $\mathscr{K}_G = \mathrm{span}\{I, X\}$, they determine it modulo the displayed orthogonal complement.

Proof. Because $B_1 = X$, the span contains I, and the explicit I-link gives X. Series composition, with the second chain taken in its inward (reversed) input order, has tensor in the original basis BXC and satisfies $T_{BXC} = T_B T_C$, so the span is an algebra. Reversal gives $T^{\#} = XT^{\mathsf{T}}X$. Since X is in the span, $T^{\mathsf{T}} = XT^{\#}X$ is also in it. Thus $\mathscr{K}_G$ is unital and transpose-stable.

Conjugate by $S = P/\sqrt{2}$, so $SXS = Z$. For A in the conjugated algebra, define its diagonal and off-diagonal parts by $D := (A + ZAZ)/2$ and $O := (A - ZAZ)/2$. If some $O \neq 0$, write $O = \left(\begin{smallmatrix} 0 & b \\ c & 0 \end{smallmatrix}\right)$. Since Z and O lie in the algebra, so does $ZO = \left(\begin{smallmatrix} 0 & b \\ -c & 0 \end{smallmatrix}\right)$. If $bc \neq 0$, $O + ZO = 2bE_{01}$ and $O - ZO = 2cE_{10}$, where E_{01} and E_{10} use Boolean row and column indices. If one entry is zero, transpose-stability supplies the missing unit. Products supply the diagonal units, hence the algebra is $\mathrm{Mat}_2(\mathbb{C})$. If every $O = 0$, it is diagonal and contains I, Z, so it is $\mathrm{span}\{I, Z\}$, which conjugates back to $\mathrm{span}\{I, X\}$.

In the full case, four realizable transfer matrices can be greedily chosen as a basis. Their corresponding kernel evaluations reconstruct every two-input slice under the fixed identification above. In the proper case, orthogonality to I, X is $a + d = b + c = 0$, giving exactly $\left(\begin{smallmatrix} u & v \\ -v & -u \end{smallmatrix}\right)$. □

For the finite groups routed above, observability identifies the routing split intrinsically:

$$G \text{ is cyclic} \quad \Longleftrightarrow \quad \mathscr{K}_G = \mathrm{span}\{I, X\},$$

$$G \text{ is dihedral, Klein four, or Platonic} \quad \Longleftrightarrow \quad \mathscr{K}_G = \mathrm{Mat}_2(\mathbb{C}).$$

Indeed, the cyclic transfer classes lie in $\mathrm{span}\{I, X\}$, whereas a finite projective subgroup of the units of this proper algebra is cyclic. Thus Section F.7 treats $\mathscr{K}_G = \mathrm{span}\{I, X\}$. The other three branches have $\mathscr{K}_G = \mathrm{Mat}_2(\mathbb{C})$.

To see what the kernel span tells us about a stable f, fix an ordered pair p of its inputs and vary the closing kernel, keeping the other inputs in the same order. If three available kernels satisfy $C_3 = \alpha C_1 + \beta C_2$, their cards satisfy

$$\partial_p^{C_3} f = \alpha\, \partial_p^{C_1} f + \beta\, \partial_p^{C_2} f.$$

At an assignment where the first two cards vanish, the third must vanish as well. The proposed matchings and binary factors must give values consistent with this identity at every assignment.

When the kernels span $\mathrm{Mat}_2(\mathbb{C})$, choose four that form a basis. Fixing p to $a, b \in \{0, 1\}$ gives a residual signature f_{ab} with $f_{ab}(\mathbf{z}) = F^f_{p,\mathbf{z}}(a, b)$. Thus each f_{ab} is a linear combination of the four cards. Suppose the nonzero cards among them all have a common factor h on the same nonempty block of remaining inputs. Then $f_{ab} = h \otimes r_{ab}$ for each a, b, allowing $r_{ab} = 0$. Define r by $r(a, b, \cdot) = r_{ab}$ in the input order with p first. Then $f = h \otimes r$ up to input order. Since r contains the two inputs in p, this contradicts irreducibility. In the cyclic case the contractions recover only combinations of the four residual signatures. Section F.7 compares these combinations across different input pairs.

F.5.5.2 The arity-four reductions

We first process the binary contractions of a quaternary signature. This local step checks which complete group applies before the common gadget construction is used.

For a quaternary tensor q with ordered inputs $1, 2, 3, 4$, let $M_{x_i x_j, x_k x_\ell}(q)$ be its 4×4 signature matrix, with row index (x_i, x_j) and column index (x_k, x_ℓ). For a finite complete transfer group G and its fixed representatives B_g, $g \in G$, define

$$\mathcal{M}_4(G) = \left\{ c\, B_g(x_i, x_j) B_h(x_k, x_\ell) : \begin{array}{l} c \in \overline{\mathbb{Q}}^{\times},\ ((i,j),(k,\ell)) \text{ is an oriented} \\ \text{perfect matching of } [4], \quad g, h \in G \end{array} \right\}.$$

Thus $\mathcal{M}_4(G)$ is the arity-four G-matching notion of Definition F.5.7. Define the complete oriented one-copy binary batch as follows. For an ordered binary card C, set $C^{\mathrm{id}} = C$ and $C^{\mathrm{rev}} = C^{\mathsf{T}}$, and put

$$\mathscr{D}_G^{(1)}(q) = \{0\} \cup \left\{ \left(\partial_{(i,j)}^{XB_gX} q\right)^{\epsilon} : i,j \in [4],\ i \neq j,\ g \in G,\ \epsilon \in \{\mathrm{id}, \mathrm{rev}\} \right\}.$$

The order (i,j) records the orientation of the deleted pair, and the choice of ϵ records either orientation of the residual pair. Every displayed member is a direct gadget: XB_gX is the realizable kernel obtained by attaching B_g through its two X-edges, and residual reversal is a permutation of inputs. In this local case analysis, a *nonbinary irreducible signature* means an irreducible signature of arity at least three.

Lemma F.5.21 (Binary contractions of a quaternary signature)**.** *Let $\mathcal{F}$ be a finite algebraic retained, factor-closed, binary-residual signature set containing the supplied equality I, suppose every retained nonbinary irreducible signature has even arity, and put $G = G(\mathcal{F})$. Let q be a realizable signature over $\mathcal{F}$, with its ordered inputs, realization, and nonzero scalar recorded. Assume that q is endpoint-nondegenerate, irreducible, and*

$$q \notin \mathcal{M}_4(G).$$

The three flattenings satisfy

$$\operatorname{rank} M_{x_1x_2,x_3x_4}(q) \geq 2, \qquad \operatorname{rank} M_{x_1x_3,x_2x_4}(q) \geq 2, \qquad \operatorname{rank} M_{x_1x_4,x_2x_3}(q) \geq 2.$$

These inequalities follow from irreducibility and give the rank conditions used in the later arity-four arguments.

Let $\mathcal{F}_q$ be the preprocessed signature set obtained by adding q and all nonzero tensor cards in $\mathscr{D}_G^{(1)}(q)$, followed by factor closure of this finite batch. Then

$$\text{K-Holant}(\mathcal{F}_q) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}),$$

and one of the following holds:

(i) *an established complexity conclusion applies, including one obtained from an odd-arity or unary signature, a nonzero singular binary, or interpolation from a realizable infinite-order binary. Alternatively, a retained lower-arity nonbinary irreducible signature gives an outer successor with smaller minimum arity;*

(ii) *$\mathcal{F}_q$ is binary-residual and*

$$G < G_q := G(\mathcal{F}_q),$$

with the strict inclusion witnessed by an ordered binary B realizable over $\mathcal{F}_q$ such that $[BX] \in G_q \setminus G$;

(iii) *$\mathcal{F}_q$ is binary-residual, $G_q = G$, q is a minimum-arity nonbinary irreducible signature in $\mathcal{F}_q$, and every member of $\mathscr{D}_G^{(1)}(q)$ is zero or projectively equal to one of the fixed B_g, $g \in G$. All nonzero card factors are retained. On this continuing branch every nonzero card has a trivial factorization tree.*

Moreover, on branch (iii), let r be an auxiliary realizable quaternary signature over $\mathcal{F}_q$, assume $r \notin \mathcal{M}_4(G)$, and suppose that r has a genuine $2+2$ factorization. Let $\mathcal{F}_{q,r}$ be the preprocessed retained, factor-closed signature set obtained by adding r and its nonzero factor batch. Then either an established reduction determines the complexity or

$$G < G(\mathcal{F}_{q,r}).$$

Thus there is no same-group continuing branch. The case distinction is finite.

Proof. The batch $\mathscr{D}_G^{(1)}(q)$ is finite: there are six unordered deleted pairs, two orders for the closing kernel, two residual orders, and finitely many closing-representative choices B_g. Its tensor cards are realizable over $\mathcal{F}$, since q and every B_g have direct realizations. Adding them therefore gives one Turing reduction by Equation (F.3), while inclusion gives the reverse one. Because every resulting set contains I and is two-sided, Lemma F.3.4 may retain the finite factorization tree without changing the problem up to Turing equivalence.

Dispatch every odd-arity or unary factor and every singular binary factor. If $\mathcal{F}_q$ directly realizes any nonsingular binary of infinite projective transfer order, apply Lemma F.3.8. This is item (i). Otherwise $\mathcal{F}_q$ is binary-residual, so Lemma F.5.3 defines the finite group $G_q = G(\mathcal{F}_q)$. Retained inclusion gives $G \le G_q$. If the inclusion is strict, the definition of G_q supplies a realizable binary representative of a class in $G_q \setminus G$, proving item (ii). The group-enlargement witness is a realizable binary gadget over $\mathcal{F}_q$, possibly using several copies of q.

It remains that $G_q = G$. Every nonzero card in $\mathscr{D}_G^{(1)}(q)$ is then nonsingular and has transfer class in $G_q = G$, hence is projectively one of the fixed B_g. Such a binary has no proper positive-arity tensor factorization. The tensor q remains irreducible under tensor factorization. Any nonbinary irreducible signature of arity below four would have arity three and would already have triggered the odd-arity reduction, so q has minimum arity of a nonbinary irreducible signature. This proves item (iii).

Finally, write $r \doteq C_1 \otimes C_2$ for the auxiliary factorization. Factor closure makes C_1, C_2 signatures of $\mathcal{F}_{q,r}$. They are not asserted to have been realizable over $\mathcal{F}_q$. If either is singular, or if the resulting set realizes an infinite-order binary, an established reduction determines the complexity. Otherwise $\mathcal{F}_{q,r}$ is binary-residual and $G \le G(\mathcal{F}_{q,r})$. If equality held, both factors would be nonsingular and would represent classes in G, whence $r \in \mathcal{M}_4(G)$, contradicting the stated hypothesis. The inclusion is therefore strict. □

Only item (iii) of Lemma F.5.21 is passed to the following arity-four construction. The preceding proof uses the finite local batch and the complete-group comparison, before any appeal to the global descent measure. The complete stabilized deck permits the following gadget construction. Its output is a realizable nonmatching eight-vertex signature to which Corollary F.4.8 applies.

Theorem F.5.22 (Quaternary signatures with stable matching decks)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, G, q), \qquad \operatorname{arity}(q) = \nu(\mathcal{F}) = 4,$$

and that G is one of

1. $G = \{1, [X]\} \cong C_2$;
2. $G = \{[D(t)], [XD(t)] : t \in \mu_m\}$, $m \ge 3$;
3. *the standard Pauli* $V_4^{(\mathrm{std})}$;
4. *the exotic Klein group*

$$V_4^{(\mathrm{ex})} = \left\{ [I], [X], \left[\begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix} \right], \left[\begin{pmatrix} -i & 1 \\ -1 & i \end{pmatrix} \right] \right\}.$$

Then a gadget using at most two copies of q produces an endpoint-nondegenerate eight-vertex signature that is not G-matching.

Proof. The gadget in Theorem F.A.8 uses at most two copies of the selected signature and gives the asserted nonmatching endpoint-nondegenerate eight-vertex output. Its directly realized signature r satisfies

$$\text{K-Holant}(\mathcal{F}, r) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}).$$ □

To apply Theorem F.5.22, first regenerate the current quaternary card and its complete oriented one-copy deck, close under factors, and process them as in Lemma F.5.21. Auxiliary cards used inside the finite certificate are local to that proof and are not appended to Q. Only an added factor batch returned by the eight-vertex reduction can be passed to an outer closure.

The quantifiers include zero cards, both input orientations for all six contracted pairs, every current-group representative assignment on the remaining matched edges, and the fixed representative scalars. Apply Corollary F.4.8 to the output over the whole retained set. In items 2–4 and the one-copy case of item 1, invertible dressings preserve irreducibility, so only a hard or tractable case remains.

In the exceptional $C_2^{(X)}$ subcase, write $a_S = \widetilde{q}(\mathbf{1}_S)$. Call it *mixed-sign* when $a_{\varnothing} = a_{[4]} = 0$ and two nonzero complementary weight-two blocks $\{S, \bar{S}\}$ have different ratios $a_{\bar{S}}/a_S \in \{1, -1\}$, as in Theorem F.A.8(i). Only this mixed-sign $C_2^{(X)}$ subcase can reach the continuing factor outcome after the two-copy construction. Retain its complete added factor batch. The last assertion of Lemma F.5.21 resolves the complexity or gives a strictly larger complete transfer group, since the output is nonmatching. Every added factor remains available by reduction over the predecessor, and only this outer successor is passed back to stabilization.

Every bounded enumeration appears as an exact proposition in Appendix F.A. The main text supplies the reduction for the whole signature set and its direction. The coverage map `certificates/theorem-to-script-map.md` records the manuscript anchors and finite claims checked by each entry point, and `certificates/README.md` gives the replay instructions and the package-wide computational scope. The next four sections use the available contraction kernels to constrain higher-arity signatures in their respective group cases. Each supplies a complexity conclusion or proves the strict progress required by Definition F.5.13. Section F.10 combines these arguments with stabilization and the outer induction over irreducible occurrences to classify the whole signature set. We begin with the marked dihedral case.

F.6 Marked Dihedral Matching Decks

This section proves the marked-dihedral case of the deck-stable classification. A deck-stable signature has only zero or G-matching proper cards. The difficulty is that different cards may use different matchings or different binary factors. We show how these local choices force a proper factor of a higher-arity signature, contradicting its irreducibility. At arity four, realizable eight-vertex signatures give the complexity conclusion for the entire retained set.

For the standard group $D_{2m}^{(Z)}$, monomial kernels let us shorten alternating cycles and preserve selected differences between two cards. This reduces a disagreement to six inputs, where the pencil classification and a finite support analysis apply. When the cards agree outside one binary factor, two phase evaluations recover the four residual signatures obtained by fixing the deleted pair, giving a common tensor factor of the original signature.

The other marked groups require control of the edge matrix as well as the binary transfers. Whole-network equivalences carry $D_{2m}^{(Y)}$ and $D_{2m}^{(X,0)}$ to the standard case. For $D_{2m}^{(X,1)}$, explicit weighted kernels preserve the local comparisons. The support and coefficient reconstruction developed below

supplies the weighted-kernel argument, including its six-input case. Each branch gives the conclusion required in Definition F.5.13; their combination proves the following theorem.

Theorem F.6.1 (Marked-dihedral deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, G, f), \qquad \mathsf{Even}(\mathcal{F}),$$

and suppose that the complete transfer group has one of the following marked forms:

$$G \in \{D_{2m}^{(Z)} : m \geq 3\} \cup \{D_{2m}^{(Y)} : m \geq 3\} \cup \{D_{2m}^{(X,0)} : m \geq 4,\ 2 \mid m\} \cup \{D_{2m}^{(X,1)} : m \geq 4,\ 2 \mid m\}.$$

Then $\mathsf{Close}_{\{\mathscr{P}\}}(\mathcal{F}, G, f)$.

The four cases are proved in Theorems F.6.12 and F.6.15 to F.6.17.

Contractions in the standard case. Until Section F.6.4, fix $G = D_{2m}^{(Z)}$, $m \geq 3$. Let $n := \mathrm{arity}(f)$. We use the conventions of Section F.2.6 and Definitions F.5.10 and F.5.13 throughout.

If $n = 4$, first regenerate the complete deck of f and close it under irreducible factors, apply Theorem F.5.22, and apply the eight-vertex reduction to its realizable endpoint-nondegenerate output using Corollary F.4.8. The added factor batch is closed by Lemmas F.3.4 and F.5.4. It gives a resolved complexity leaf or a strict stable successor. Thus the arity-four base is resolved, and throughout the remaining standard $D_{2m}^{(Z)}$ proof we may assume

$$n = 2k + 2 \geq 6.$$

For every $t \in \mu_m$, the representative convention of Section F.2.6 fixes realizable ordered binary outputs

$$B_{[D(t)]} = \sigma_t^{\mathrm{diag}} D(t) X, \qquad B_{[XD(t)]} = \sigma_t^{\mathrm{anti}} X D(t) X, \qquad \sigma_t^{\mathrm{diag}}, \sigma_t^{\mathrm{anti}} \neq 0.$$

For a deleted set $p = \{i, j\}$ equipped with the input order (i, j), write the four residual signatures obtained by fixing that pair as

$$A := f(00, \cdot), \quad B := f(01, \cdot), \quad C := f(10, \cdot), \quad D_{11} := f(11, \cdot),$$

and define the associated $(n-2)$-ary *sector pencils* by

$$E(t) := A + tD_{11}, \qquad O(t) := C + tB.$$

Here $E(t)$ is the equal/even sector associated with the deleted-pair values $00, 11$, while $O(t)$ is the unequal/odd sector associated with $10, 01$. A pencil means a one-parameter linear family $U + tV$. The corresponding contraction-kernel spaces

$$\mathrm{span}\{E_{00}, E_{11}\} \quad \text{and} \quad \mathrm{span}\{E_{01}, E_{10}\}$$

are the even and odd *parity planes*, with the displayed matrix units as coordinate axes. For $t \in \mu_m$, the X-edges give the exact realizable outputs $\sigma_t^{\mathrm{diag}} O(t)$ and $\sigma_t^{\mathrm{anti}} E(t)$, since

$$X B_{[D(t)]} X = \sigma_t^{\mathrm{diag}} X D(t), \qquad X B_{[XD(t)]} X = \sigma_t^{\mathrm{anti}} D(t).$$

For $t \in \mu_m$, these outputs are deck cards with the displayed nonzero scalars. The four slices fix the inputs of the deleted pair and leave all other inputs free. They are the coefficient tensors of the two sector pencils and will be called their *pencil endpoints*. Two phase evaluations recover these

tensors by linear inversion. Later $K_E(t), K_O(t) \in \mathrm{Mat}_2(\mathbb{C})$ denote the varying binary factors of the two sector pencils.

We use *deck card* for a member of the matching deck in Definition F.5.7. A sequence of contractions gives a *contraction output*; constructions using two copies give *two-copy outputs*.

Write a nonsingular monomial binary, up to scalar, as

$$M_\alpha(a)(x,y) = \mathbf{1}[x \oplus y = \alpha]a^x, \qquad \alpha \in \mathbb{F}_2, \quad a \neq 0.$$

Here α is its support parity and $M_\alpha(a) = D(a)X^\alpha$. In particular, $M_0(a) = D(a)$, while $XD(t) = tM_1(t^{-1})$ for $t \neq 0$. Reversal changes only the weight and a nonzero scalar.

F.6.1 Trace covers and alternating-cycle localization

Consider two nonzero sampled cards of the same signature. Their matchings and binary factor classes can disagree. We will contract common parts while preserving a disagreement, leaving only four or six inputs to analyze.

Projective tensor geometry. A matching product is a simple tensor after its inputs are grouped into pairs. Segre geometry expresses this factorization by rank equations.

We use the projective-space conventions of Section F.2.1.3. Whenever a projective object occurs in a tensor equation, a displayed or previously fixed nonzero representative is understood. We set $\mathbb{P}(0) := \varnothing$.

A nonzero tensor $\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_s$ is *simple*. For finite-dimensional spaces $V_1, \ldots, V_s$, the Segre map is

$$\mathbb{P}(V_1) \times \cdots \times \mathbb{P}(V_s) \longrightarrow \mathbb{P}(V_1 \otimes \cdots \otimes V_s), \qquad ([\mathbf{v}_1], \ldots, [\mathbf{v}_s]) \longmapsto [\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_s].$$

Its image is the *Segre variety*. Its affine cone consists of zero and the simple tensors. The Segre ideal is generated by the 2-by-2 minors of the bipartite flattenings. A *one-factor Segre ruling* is a projective linear family in which all factors but one are fixed, and every projective line contained in a Segre variety belongs to such a ruling [Lan12, Chapter 2].

For two perfect matchings M_0, M_1, an *alternating M_0/M_1-cycle* is an even cycle in $M_0 \triangle M_1$ whose edges alternate between the two matchings. We call a sequence of the contractions below a *trace cover*. It closes common edges and shortens alternating cycles while retaining a chosen discrepancy. Its ordered contraction graph records the connections, edge orientations, and external-input order.

Definition F.6.2 (Disagreements between matching products)**.** The *factor class* of an ordered nonzero binary B is its projective class $[B]$. Normalize the sampled deck cards by their nonzero realization scalars. Two nonzero cards *disagree* if their residual matchings differ, or if their ordered binary factor classes differ on at least two edges of the same matching. A parity change changes the factor class. Variation on one designated edge, including its parity, is allowed.

A one-factor Segre ruling, as defined in the projective tensor conventions, allows one binary factor to vary and fixes the other factors on the matching. Thus products on one such ruling differ in at most that factor class.

Consider two realizable nonzero sector cards

$$P_0 = \bigotimes_{e \in M_0} B_{0,e}, \qquad P_1 = \bigotimes_{e \in M_1} B_{1,e}, \tag{F.25}$$

where every $B_{\epsilon,e}$ is a nonsingular monomial binary from the normalized group.

We close common edges whose ordered binary factors agree projectively. Connecting one input on each of two residual edges by a realizable binary fuses those edges in each product. The following identities show how to choose these contractions while preserving selected parity differences or nonzero determinants.

Definition F.6.3 (Parity and coefficient discrepancies)**.** After aligning the common edges of two sector products, a residual matched factor is a *parity discrepancy* if its support parity changes between the two products. It is a *coefficient discrepancy* if the parities agree but the two ordered nonzero-entry vectors are not projectively proportional. In that case the associated nonzero 2×2 determinant records the coefficient discrepancy.

Lemma F.6.4 (Compatible contractions and cycle shortening)**.** *Common factors with the same projective class in Equation* (F.25) *can be closed by realizable group binaries while keeping both products nonzero. An alternating M_0/M_1-cycle can be shortened by two inputs without cancellation, and a selected coefficient determinant is only multiplied by a nonzero scalar. Two edges with discrepancies can be fused by a realizable connection while preserving any distinguished remaining discrepancy.*

Proof. For ordered monomial factors one has, up to the fixed orientation scalar,

$$\sum_{x,y} M_\alpha(a)(x,y)M_\alpha(t)(x,y) = 1 + at,$$
$$\sum_{z} M_\alpha(a)(u,z)M_\beta(b)(z,v) = b^\alpha M_{\alpha+\beta}(ab^{1-2\alpha})(u,v).$$

Thus on a common edge the two product traces are nonzero affine functions of $t \in \mu_m$. Together they exclude at most two phases. Since $m \geq 3$, one allowed phase keeps both nonzero. Closing an edge then splices the two incident edges in the other matching. The displayed fusion has a unique internal lift: each supported external assignment extends to exactly one supported internal assignment. Thus there is no cancellation, and every selected minor is multiplied by nonzero trace and fusion factors.

For an ordered edge with a discrepancy $q = (u_q, v_q)$, let $\alpha_{\xi,q} \in \mathbb{F}_2$ denote its support parity in product $\xi \in \mathbb{F}_2$. Put

$$\delta_q := \alpha_{0,q} + \alpha_{1,q} \in \mathbb{F}_2 \qquad (q \in \{r, s\}), \qquad \alpha_{\xi,q} = \alpha_{0,q} + \xi\delta_q.$$

One ordered connection $M_\gamma(t)$ fuses the two factors in each product. Applying the displayed fusion identity twice gives

$$M_\alpha(a)M_\gamma(t)M_\beta(b) = t^\alpha b^{\alpha\oplus\gamma} M_{\alpha\oplus\gamma\oplus\beta}\Big(a\, t^{1-2\alpha} b^{1-2(\alpha\oplus\gamma)}\Big).$$

For each supported exposed string the internal lift is unique, and the fusion scalar is nonzero. The fused parity difference is $\delta_r + \delta_s$. A mixed pair therefore leaves a parity discrepancy. Equal types leave at most one coefficient discrepancy, which may also cancel. A distinguished discrepancy outside the two spliced edges is preserved: its determinant is multiplied only by the nonzero orientation and fusion scalars of the two products. Each connection removes two inputs, whether or not the two spliced discrepancies cancel. All operations use only the X-edge, the explicit normalized I-link, and fixed realizable group representatives. □

Lemma F.6.5 (Binary factor classes and a quaternary contraction)**.** *For distinct $t_0, t_1 \in \mu_m$, suppose the four nonzero realizable cards represented by $E(t_0), E(t_1), O(t_0), O(t_1)$ at a deleted pair p*

have a common residual matching edge e, with the same support parity. If all four ordered factors on e are proportional to one binary B_e, the signature f has the exact tensor factorization

$$f = B_e \otimes h$$

with a nonsingular monomial binary B_e. Otherwise choose two cards whose ordered factors on e are nonproportional. If their other matching edges and support parities agree, a contraction sequence retaining $p \cup e$ has a nonzero 2×2 flattening minor. Without this parity compatibility no simultaneous trace is asserted. The factor identity is analytic. Its factors are exposed only by factor closure.

Proof. For the nonproportional pair with the stated compatibility, close every other common matching edge successively. At each step the two traces exclude at most two roots, so Lemma F.6.4 keeps both products nonzero. The two surviving e-vectors remain nonproportional. They are contractions of the resulting realizable quaternary by two kernels on p. Hence that quaternary has a nonzero flattening minor. The two kernels are linearly independent: two distinct phases are independent inside one parity plane, while an even and an odd kernel have different support parities. Thus nonproportional retained e-vectors really give flattening rank at least two.

For the factor alternative all four ordered e-factors are proportional. Choose their common representative

$$B_e(x_u, x_v) = \mathbf{1}[x_u \oplus x_v = \alpha] b_{x_u}, \qquad b_0 b_1 \neq 0,$$

and absorb each card's nonzero scalar into its complementary tensor. The two-card inversions

$$A = \frac{t_1 E(t_0) - t_0 E(t_1)}{t_1 - t_0}, \quad D_{11} = \frac{E(t_1) - E(t_0)}{t_1 - t_0}, \qquad C = \frac{t_1 O(t_0) - t_0 O(t_1)}{t_1 - t_0}, \quad B = \frac{O(t_1) - O(t_0)}{t_1 - t_0}$$

put every slice in $\mathbb{C}B_e \otimes V$, where V is the tensor space on the residual inputs outside e. Reassembling the four slices gives the displayed tensor identity. □

Lemma F.6.6 (Two-card localization to six inputs)**.** *Fix the deleted pair p, and let two nonzero realizable sampled p-cards of the same signature disagree. There is a realizable one-copy contraction h, retaining p and four further inputs, whose two inherited p-cards remain nonzero and either have different residual matchings or differ on both edges of their common residual matching. The contraction closes only inputs outside p and uses $O(\mathrm{arity}(f))$ fixed realizable group representatives. In particular, h is not a G-matching signature.*

Proof. For each selected pair of cards, choose contractions that keep both outputs nonzero and preserve their disagreement.

First suppose the two residual matchings differ. Choose one alternating cycle as distinguished. On every other alternating cycle, close an edge of the first matching by a kernel of the same support parity. The trace in the first product excludes at most one phase, whereas the other product has a unique monomial lift. Thus both products remain nonzero. Repeat until that nondistinguished cycle becomes one common edge.

Every common edge outside the distinguished cycle can now be absorbed into the cycle without any parity compatibility assumption. Write that edge as $\{u, w\}$, select a cycle vertex v, and connect u, v by any realizable monomial group kernel. If the two cycle neighbours of v are v_0, v_1, the new matching edges are respectively $\{w, v_0\}$ and $\{w, v_1\}$. Thus the alternating cycle is unchanged except that w replaces v. Both internal lifts are unique and have nonzero weights, including when the old common edge had different support parities in the two products. Absorb every remaining common edge in this way. Finally shorten the distinguished cycle, using the same trace-and-fusion operation,

until it is a four-cycle. Stop before closing an edge of that four-cycle. The two residual four-input matchings are therefore still different.

Now suppose the residual matching agrees and at least two factor classes differ. Select two such edges whose discrepancies will be retained. Every other edge whose two support parities agree can be traced directly: the two traces exclude at most two phases, so $m \geq 3$ leaves an allowed phase keeping both products nonzero. The two selected edges are untouched by these traces.

An edge outside the selected pair with different support parities is instead fused with one selected edge by a realizable binary connection. The other selected edge remains untouched. If the selected edge used in the fusion has equal parities but different factor classes, the fused edge has different support parities and retains a parity discrepancy. If both fused edges have different support parities, the new support parities agree, but the ratio of their normalized fused weights is

$$\kappa t^{\pm 2}, \qquad \kappa \neq 0.$$

This follows directly from the three-factor fusion identity in Lemma F.6.4. At most two phases make this ratio equal to one. Choose another allowed phase. The fused edge then carries a coefficient discrepancy. In both cases the operation reduces the number of edges by one and leaves two edges whose factor classes differ between the products. Continue until exactly these two edges remain.

Perform the selected contractions on the original signature, leaving the inputs of p open. Tensor contraction commutes with either of the two original p-kernels, so its inherited cards are precisely the two nonzero products just constructed, with their nonzero scalars. Each operation removes two inputs, giving the stated size bound.

If h were a product on three matching edges, deleting p would either trace out one of its matching edges, leaving the other two factors fixed, or fuse its two incident matching edges, leaving the third factor fixed. Whenever the two resulting cards are nonzero, their residual matching is therefore the same and at most the fused factor can vary. This contradicts either of the two constructed configurations. □

Lemma F.6.7 (Reduction to four inputs or a tensor factor)**.** *Every arity-six signature produced in Lemma F.6.6 has a complete reduction through its proper factor and deck batches. It cannot close as a G-matching signature. A quaternary child is processed by its own complete factor and deck batch. A crossed factorization into safe binaries completes that child only.*

Proof. Let h be the realizable six-input output, with its inherited nonzero cards, input order, realization scalar, and originating fixed token retained. By Lemma F.6.6, h is not a three-factor matching product. Add h, jointly close the enlarged set under irreducible factors, and close every proper factor and every proper oriented deck card. Odd or unary factors and unsafe binaries or safe binaries whose transfer classes lie outside the current group follow their established exits. Any proper nonbinary irreducible signature has arity at most four and is processed by the lower-arity induction, including the arity-four quaternary deck classification stated in the setup. Thus this processing does not use the six-input case being proved.

If no resolved leaf or strict successor occurs, all surviving proper factors are G-matching signatures. A proper factorization of h would therefore make h itself a matching product, a contradiction. Otherwise h is irreducible and every nonzero proper pair card is a G-matching signature. Lemma F.6.11 excludes this case by the sector-ruling and binary-pencil lemmas together with the Boolean support classification.

For a realizable quaternary child, first close its complete factor batch. If it factors into safe current-group binaries, this completes the child card index. Otherwise use the established binary exits or retain its irreducible quaternary occurrence, regenerate the complete oriented deck, and

apply Theorem F.5.22 and Corollary F.4.8. Every returned factor batch is closed by Lemmas F.3.4 and F.5.4. A factorization completes the corresponding child output. The selected signature remains under analysis.

In particular, for an active deck-stable signature of arity at least eight there is an immediate contradiction: the first proper pair contraction outside p is zero or a matching product, and further monomial contractions preserve that property. Yet localization keeps the two disagreeing inherited cards nonzero. The same argument applies after the proper lower-arity batches of a later occurrence have been closed. □

Common-factor consequence. Let $\mathscr{S}$ be a nonempty family of nonzero matching products on one fixed, nonempty oriented matching M. Suppose that, for every $S, T \in \mathscr{S}$, their binary factor classes differ on at most one edge of M. Then there are an edge $e \in M$ and a nonzero product P on $M \setminus \{e\}$ such that

$$\mathscr{S} \subseteq \mathbb{C}P \otimes V_e, \qquad V_e := (\mathbb{C}^2)^{\otimes e}.$$

Here $(\mathbb{C}^2)^{\otimes e} := \bigotimes_{j \in e} \mathbb{C}^2$, with factors in the fixed input order, and proportional factors are absorbed into the factor on e. If S, T differ on e, a third sample differing from S on another edge would differ from T on two edges. If all samples are proportional, choose any edge. This proves the assertion, including a matching with only one edge, where P is a nonzero scalar.

Apply the preceding localization and reduction to each pair of nonzero samples at a fixed deleted pair. The closures need preserve only that pair and may differ between comparisons. On the continuing branch they give the hypothesis above for the union of the two sectors, so the same P works in both. Each nonzero pencil has at least two nonzero samples at distinct phases because $m \geq 3$. Inverting those two evaluations puts both endpoint slices in $\mathbb{C}P \otimes V_e$. An identically zero pencil has two zero endpoint slices. Thus all four slices share P, which gives the factorization of f used below.

Lemma F.6.8 (Sector-ruling rigidity)**.** *Let $U + tV$, $t \in \mu_m$, $m \geq 3$, be a tensor pencil on an even number of inputs whose samples are zero or current $D_{2m}^{(Z)}$-matching signatures. Its nonzero samples have one fixed matching and all but at most one binary factor class fixed projectively. In particular this holds for each of $E(t)$ and $O(t)$.*

Proof. We use only the sampled matching hypothesis. A proportional pencil has one projective value. Otherwise its m samples are nonzero and projectively distinct. Write $N = 2^s$ for the support size of an s-edge matching product. Two different matching supports S_0, S_1 are affine cosets of the same dimension, so $|S_0 \cap S_1| \leq N/2$. Every nontrivial linear combination of the corresponding tensors has at least $2N - 2|S_0 \cap S_1| \geq N$ supported strings outside the intersection. For a third sample to have exactly N strings requires equality and cancellation on the whole intersection. A single common string fixes that linear combination projectively. There is therefore at most one such third matching point, contradicting $m \geq 4$.

For $m = 3$, every ratio of two nonzero coefficients within a matching sample lies in μ_3. Use a string of $S_0 \cap S_1$ to express the required cancellation, and choose one string in each of $S_0 \setminus S_1$ and $S_1 \setminus S_0$. Their coefficient ratio in the third sample lies in $-\mu_3$, disjoint from μ_3. This is again impossible. Thus all sample supports coincide.

The weight-two strings of the common support translation space determine its matching uniquely, including the parity of each support coset. In that fixed tensor factorization every 2×2 Segre flattening minor is a polynomial in t of degree at most two. It vanishes at all $m \geq 3$ sampled phases and hence identically. The whole projective line consists of simple tensors. The Segre-line criterion

in the projective tensor conventions therefore permits variation in only one factor, proving the assertion. □

F.6.2 Binary pencils and endpoint compatibility

By Lemma F.6.8, each sector has one fixed residual matching and at most one varying factor class. This subsection classifies those binary factor pencils and proves the six-input base for the standard case. Further endpoint comparisons will be used in the weighted reconstruction below.

F.6.2.1 Möbius geometry

These Möbius conventions control the ratio of the two nonzero entries in a binary factor. That ratio is a fractional-linear function of the contraction phase. Requiring it to remain in the same finite phase group forces the two endpoint matrices onto complementary coordinate axes.

The Riemann sphere is $\widehat{\mathbb{C}} := \mathbb{C} \cup \{\infty\} \cong \mathbb{P}(\mathbb{C}^2)$, with affine chart $z = [z : 1]$ and $\infty = [1 : 0]$. A Möbius transformation is a projective automorphism of this sphere. In the affine chart it is

$$z \longmapsto \frac{az + b}{cz + d}, \qquad ad - bc \neq 0,$$

where $cz + d = 0$ is sent to ∞ and ∞ is interpreted projectively. A *generalized circle* is a Euclidean circle $\{z : |z - z_0| = r\}$, $r > 0$, or a real affine line together with ∞. Such a line has the form $\{z : \mathrm{Re}(\overline{\alpha} z) = \beta\}$ for $\alpha \in \mathbb{C}^\times$ and $\beta \in \mathbb{R}$. Möbius transformations carry generalized circles to generalized circles [Bea83, Chapter 3]. The unit circle is $S^1 := \{z \in \mathbb{C} : |z| = 1\}$. Its counterclockwise cyclic order is the order of arguments modulo 2π.

Lemma F.6.9 (Möbius stabilizer of a root polygon)**.** *Let $m \geq 3$, and let the Möbius map*

$$\phi : \widehat{\mathbb{C}} \longrightarrow \widehat{\mathbb{C}}, \qquad \phi : z \longmapsto \frac{c + dz}{a + bz}, \qquad ad - bc \neq 0,$$

have no pole on μ_m and permute μ_m. Then

$$\phi(z) = \alpha z \qquad or \qquad \phi(z) = \alpha / z, \qquad \alpha \in \mu_m.$$

Proof. A Möbius map sends generalized circles to generalized circles. Three distinct roots and their images force ϕ to preserve the unit circle. Its action on the regular m-gon preserves or reverses cyclic order, so it is a rotation or a rotation followed by reflection. The corresponding Möbius maps agree with ϕ on at least three points and hence agree identically. □

Use the even and odd parity planes and their ordered coordinate axes defined above. For a nonsingular monomial matrix, its *ordered nonzero-entry ratio* is the coefficient of the second displayed axis divided by that of the first, in the fixed input orientation.

Lemma F.6.10 (Classification of binary pencils)**.** *Let $A, B \in \mathrm{Mat}_2(\mathbb{C})$ be binary signatures, and define the binary matrix pencil*

$$K(t) := A + tB, \qquad t \in \mu_m, \qquad m \geq 3.$$

Suppose every sampled binary signature $K(t)$ is zero or a nonsingular monomial binary whose ordered nonzero-entry ratio lies in μ_m. Then either A, B are proportional, or there are a fixed parity plane with coordinate axes E_0, E_1, $\lambda \neq 0$, and $\alpha \in \mu_m$ such that

$$K(t) = \lambda(E_0 + \alpha t E_1) \qquad or \qquad K(t) = \lambda(tE_0 + \alpha E_1).$$

Here $K(1) = A + B$ is a sampled value, whereas $A = K(0)$ is an analytic endpoint because $0 \notin \mu_m$. In the application below, $K(t)$ is the one varying binary factor of a higher-arity realizable matching card. Its coefficients are recovered from the sector pencil.

Proof. In the nonproportional case no sample is zero and the projective samples are distinct. Two of the at least three samples lie in one of the two monomial parity planes. They span the pencil, so all samples lie in that plane. In these axes, write the matrix pencil with coefficient scalars a, b, c, d and projective coordinate ratio ϕ as

$$K(t) = (a + tb)E_0 + (c + td)E_1, \qquad \phi(t) = \frac{c + td}{a + tb}.$$

Nonproportionality gives $ad - bc \neq 0$, and nonsingularity excludes zeros and poles on μ_m. Thus ϕ permutes μ_m. By Lemma F.6.9, it equals αt or α/t. Cross-multiplication gives a degree-two identity valid on at least three points. Its coefficients yield respectively $b = c = 0, d = \alpha a$, or $a = d = 0, c = \alpha b$, which are the two displayed forms. A zero sample already makes the endpoints proportional. □

Use the following names for the three binary-pencil outcomes of Lemma F.6.10:

Z	:	identically zero,
P	:	nonzero proportional,
C	:	nonproportional complementary-axis.

For a deleted pair p, factor the even-sector card into the common matching factor P_E and endpoint matrix pencil $K_E(t)$, after ordering the $2k = \mathrm{arity}(f) - 2$ residual inputs of a nonconstant even ruling:

$$E(t) = P_E \otimes K_E(t), \qquad K_E(t) := K_{E,0} + tK_{E,1}, \qquad K_E(t), K_{E,0}, K_{E,1} \in \mathrm{Mat}_2(\mathbb{C}),$$

with $k - 1$ fixed factors. Define $O(t)$ and its binary factor pencil $K_O(t)$ analogously. To obtain one tensor identity with a fixed scalar, take two distinct nonzero samples, use their linear inversion to reconstruct the two analytic endpoint tensors, normalize the common factor P_E by a fixed linear functional $\eta_E(P_E) = 1$, and define

$$K_E(t) = (\eta_E \otimes \mathrm{id})(E(t)).$$

The normalization fixes the common factor once for the whole pencil. The whole sector inherits the type of this exact residual binary pencil. An identically zero whole pencil is type Z, and a nonzero proportional whole pencil is type P before any varying factor is selected. In a C/C branch the variable endpoints are complementary rank-one matrices.

F.6.2.2 The six-input base

Lemma F.6.11 (Direct six-input base case)**.** *Let $h \neq 0$ be a retained arity-six signature in the standard $D^{(Z)}_{2m}$ sector, where $m \geq 3$. Suppose every realizable oriented pair card of h, in both kernel parities and at every phase in μ_m, is zero or a product of two nonsingular current-group monomial binaries. Then, after a permutation of inputs, h has an exact nontrivial factorization of type $1 + 5$ or $2 + 4$. In particular, h is not irreducible.*

Proof. Apply Lemmas F.6.8 and F.6.10 at each deleted pair p. A type-P sector is a scalar pencil times one fixed nonsingular matching product on the four residual inputs. A type-C sector is one fixed nonsingular binary factor tensored with a complementary-axis pencil. Its two endpoint supports are complementary singletons on the varying edge. A type-Z sector vanishes identically.

If a pair has type Z/P, then $h = U_p \otimes P$ for a nonzero binary U_p and a nonzero residual matching product P. If it has type Z/C, the fixed binary factor of the C ruling factors from h. The reversed cases are identical, and Z/Z would give $h = 0$. Each remaining case containing Z therefore proves the assertion.

Suppose a pair has type P/P. Write

$$h = U_p \otimes P_0 + V_p \otimes P_1,$$

where $U_p, V_p \neq 0$ are supported on the two opposite parities of p, and P_0, P_1 are nonsingular matching products on the four residual inputs. Choose a residual pair e belonging to neither matching. Such a pair exists because the two matchings contain at most four of the six residual pairs. Contract e by either realizable parity kernel at phase 1. This crosses the two factors of each P_j. For every supported p-string, the output has precisely two complementary residual strings, each with one internal lift and a nonzero coefficient. Different p-strings cannot cancel. The resulting realizable quaternary therefore has

$$2(|\operatorname{supp} U_p| + |\operatorname{supp} V_p|)$$

nonzero entries. Six or eight entries contradict the card hypothesis. Otherwise both supports are singletons. Their opposite-parity p-strings differ in exactly one bit, so the other input of p is fixed. Its point mass is a unary tensor factor of h.

For type P/C, choose a residual pair e belonging to neither the P matching nor the fixed/varying matching of the C ruling. Apply the same realizable parity contraction. Each supported P-sector p-string leaves two residual strings. Each of the two C-sector p-strings leaves exactly one: the singleton varying endpoint and the kernel parity force the bit on the fixed factor, and then force both remaining residual bits. These lifts are unique and all coefficients are nonzero. If the P sector has two p-strings, the card has six nonzero entries. If it has one, the card has four nonzero entries but its projection onto p has three strings. A product of two nonsingular monomial binaries projects onto any two inputs in exactly two strings when those inputs are matched, and in four strings otherwise. Both alternatives contradict the card hypothesis. Type C/P is identical.

Unless one of the stated factorizations has already occurred, every deleted pair must consequently have type C/C. Fix one such pair. Its fixed binary factors and complementary-axis endpoints give exactly the eight-string support in Equation (F.124). The support classification in Proposition F.A.9 either gives an injective I/X-contraction with six or eight projected strings, or identifies a three-edge matching coset. In the first case, injectivity gives one nonzero summand per output string, so this realizable quaternary contradicts the card hypothesis. In the second case, one parity sector at any edge of that matching is identically zero, contradicting the conclusion that every deleted pair is C/C. This completes the proof.

All contractions in this proof are applied directly to h. The factors are exact tensor factors, and are exposed only through Theorem F.2.10 and Lemma F.3.4 when the reduction procedure uses them. □

F.6.3 Completion of the standard dihedral case

Theorem F.6.12 (Standard $D_{2m}^{(Z)}$-sector deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, D_{2m}^{(Z)}, f), \qquad \mathsf{Even}(\mathcal{F}), \qquad \operatorname{arity}(f) = \nu(\mathcal{F}) = 2k+2, \qquad m \geq 3,$$

where

$$D_{2m}^{(Z)} = \{[D(t)], [XD(t)] : t \in \mu_m\}.$$

Then $\mathsf{Close}_{\{\mathscr{P}\}}(\mathcal{F}, D_{2m}^{(Z)}, f)$.

Completion of Theorem F.6.12. If $\mathrm{arity}(f) = 4$, regenerate and close its complete deck. In the standard $D_{2m}^{(Z)}$ case the output of Theorem F.5.22 is an invertible dressing of the irreducible quaternary. It is therefore a realizable endpoint-nondegenerate irreducible eight-vertex signature. Apply Theorem F.1.2 to its signature set. Direct realization and preservation under taking subsets give a hard or tractable case satisfying $\mathsf{ReTract}(K\mathcal{F})$ for the entire retained set $\mathcal{F}$.

For $\mathrm{arity}(f) = 6$, Lemma F.6.11 contradicts the irreducibility of the distinguished irreducible signature. Now let $n := \mathrm{arity}(f) \geq 8$. At any deleted pair p, two disagreeing nonzero sampled cards would give the six-input output in Lemma F.6.6. The first proper contraction of the distinguished irreducible signature is zero or a matching product, and further monomial contractions preserve that property. This contradicts the two inherited disagreeing nonzero cards. Hence all samples at p share one matching and all factor classes outside one designated edge e. The common-factor consequence following Lemma F.6.7, applied to both sectors, therefore gives $f = P \otimes q_{p,e}$. Here P is a nonzero product containing at least one binary factor, contradicting irreducibility.

Consequently no higher-arity deck-stable signature branch remains. The quaternary case already resolves the whole signature set, and completes this branch. These are the allowed outcomes of $\mathsf{Close}_{\{\mathscr{P}\}}(\mathcal{F}, D_{2m}^{(Z)}, f)$. □

The restriction $m \geq 3$ is essential here. For $m = 2$, two forbidden trace roots can exhaust μ_2, and evaluation on μ_2 does not determine a polynomial of coordinate degree two. That case belongs to the separate normalized $V_4^{(\mathrm{std})}$ analysis.

F.6.4 Marked coordinates and whole-network reductions

For the nonstandard marked groups, the same binary transfers acquire a different edge matrix. We first compute that matrix, then determine which parts of the standard proof survive. The matrices U_τ below change analytic coordinates. The $D_{2m}^{(Y)}$ and $D_{2m}^{(X,0)}$ proofs additionally give whole-network equivalences. The $D_{2m}^{(X,1)}$ proof instead keeps the original network and tracks the resulting weights.

Lemma F.6.13 (Marked-sector coordinates and edge matrices)**.**
Use the form labels $Z, Y, (X,0), (X,1)$ *as sector indices. For* $\tau = Z, Y$*, let* $m \geq 3$ *and set*

$$U_Z = I, \qquad U_Y = I - iX.$$

For $\tau = (X,1)$*, let* $m \geq 4$ *be even, fix* $\eta \in \mu_{2m} \setminus \mu_m$*, choose* $s_0 \in \overline{\mathbb{Q}}$ *with* $s_0^2 = \eta^{-1}$*, and set*

$$U_{X,1} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \mathrm{diag}(1, s_0), \quad s_0^2 = \eta^{-1}.$$

Conjugation by U_τ *sends* $D_{2m}^{(Z)}$ *projectively onto the corresponding marked group. Define the transformed edge matrix* S_τ *by*

$$S_\tau = U_\tau^{\mathsf{T}} X U_\tau.$$

Then, projectively,

$$S_Z = X, \qquad S_Y = I, \qquad S_{X,1} = \mathrm{diag}(1, -\eta^{-1}).$$

For $T \in D_{2m}^{(Z)}$, choose a fixed realizable marked-sector binary B_T whose sector-coordinate transfer is $[T]$, and set $B'_T := U_\tau^{-1} B_T U_\tau^{-\mathsf{T}}$. Then, for $T, U \in D_{2m}^{(Z)}$,

$$B'_T \doteq T S_\tau^{-1}, \qquad (T S_\tau^{-1}) S_\tau (U S_\tau^{-1}) = T U S_\tau^{-1}. \tag{F.26}$$

Thus multiplication and reversal are preserved, and every monomial fusion has the same unique internal lift. Consequently such a fusion introduces no cancellation.

Proof.

$$U_Y D(t) U_Y^{-1} \doteq C(t), \qquad U_{X,1} D(t) U_{X,1}^{-1} \doteq A(t), \qquad U_{X,1} X D(t) U_{X,1}^{-1} \doteq Z A(\eta t).$$

$$U_Y^{\mathsf{T}} X U_Y = -2iI, \qquad U_{X,1}^{\mathsf{T}} X U_{X,1} = 2\,\mathrm{diag}(1, -\eta^{-1}).$$

Congruence gives $B'_T \doteq T S_\tau^{-1}$, and the fusion identity is matrix multiplication. All omitted scalars are fixed nonzero. □

Sector coordinates and available kernels. For an r-ary current-coordinate signature h, write

$$h^{(\tau)} = (U_\tau^{-1})^{\otimes r} h$$

for its analytic sector-coordinate tensor. A *sector-X dressing* is input dressing by the fixed realizable binary B_X. By Equation (F.26), its transported transfer fuses as X. Put

$$\beta_Y := 1, \qquad \delta := -\eta^{-1}, \quad \beta_{X,1} := \delta^{-1}, \qquad \mathscr{R}_\tau := \beta_\tau \mu_m,$$

where the quantities involving η and δ apply only in the $(X,1)$-sector. For a quotient in $\mathscr{R}_\tau$, division by β_τ gives a point of μ_m. This division acts on coefficient ratios. In one monomial parity plane, a *complementary-axis pencil* has its two coefficient matrices on the two coordinate axes. A sample with exactly one vanishing coordinate is a singular binary.

Lemma F.6.14 (Symmetric binary pullback)**.** *Let $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$. If $E, F \in \mathscr{C}$ are nonsingular symmetric binaries, then $EFE \in \mathscr{C}$.*

Proof. Closure under contraction proves the affine case, and the equality/disequality normal forms prove the product case. For the local-affine case, write $E = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$. The diagonal twists in the definition of $\mathscr{L}$ preserve support, which must be affine. Nonsingularity excludes support size one, and affine support excludes size three. Full support is also impossible: the $\alpha = 00$ condition requires $b/a \in \mu_4$, whereas the $\alpha = 01$ twist requires $\zeta_8 b/a \in \mu_4$. On anti-diagonal support symmetry gives $E = bX$, and the same $\alpha = 01$ twist again has a nonzero-entry ratio outside μ_4. Consequently $E = a\,\mathrm{diag}(1, i^k)$, with $a \neq 0$ and $k \in \mathbb{Z}/4\mathbb{Z}$. Conversely these four diagonal forms satisfy both supported twists, $\alpha = 00, 11$, and belong to $\mathscr{L}$. Writing $F = d\,\mathrm{diag}(1, i^\ell)$ in the same way gives $EFE = a^2 d\,\mathrm{diag}(1, i^{2k+\ell}) \in \mathscr{L}$, as required. □

Theorem F.6.15 (Signature-set-wide reduction of the $D_{2m}^{(Y)}$ sector)**.** *If*

$$\mathsf{Stable}(\mathcal{F}, D_{2m}^{(Y)}, f), \qquad \mathsf{Even}(\mathcal{F}), \qquad m \geq 3,$$

then $\mathsf{Close}_{\{\mathscr{P}\}}(\mathcal{F}, D_{2m}^{(Y)}, f)$.

The tractable cases here include $\mathsf{Pres}_K(K\mathcal{F}; \mathscr{A})$ and $\mathsf{Pres}_K(K\mathcal{F}; \mathscr{L})$. The product case is the only remaining tractable case.

Proof. Define the transformed signature set $\Gamma := U_Y^{-1}\mathcal{F}$ and the transformed active irreducible signature

$$f^{(Y)} := (U_Y^{-1})^{\otimes \operatorname{arity}(f)} f \in \Gamma.$$

This whole-problem change gives

$$U_Y^{\mathsf{T}} X U_Y = -2iI, \qquad U_Y^{-1} I U_Y^{-\mathsf{T}} = \frac{i}{2} X, \qquad U_Y^{-1} X U_Y^{-\mathsf{T}} = \frac{i}{2} I.$$

Thus edge and equality exchange, and Lemma F.5.1 gives

$$\text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{Holant}(\Gamma, X) \equiv_{\mathrm{T}} \text{K-Holant}(\Gamma, I).$$

Define the signature set $\mathcal{F}^\sharp := \Gamma \cup \{I\}$. Here the *open-network bijection* is the vertexwise change by U_Y^{-1}, together with the displayed edge/equality exchange, applied also to the ordered open boundary legs. The inverse change restores the original network. It preserves arity, factorization, minimum arity of a nonbinary irreducible signature, and parity of the arity of every irreducible signature. Hence $\mathsf{Even}(\mathcal{F}^\sharp)$.

For a realizable nonsingular b over $\mathcal{F}^\sharp$, its pullback B satisfies

$$B \doteq U_Y b U_Y^{\mathsf{T}}, \qquad [BX] = [U_Y b U_Y^{-1}],$$

using $U_Y^{\mathsf{T}} X = -2iU_Y^{-1}$. Infinite order pulls back to the interpolation reduction of Lemma F.3.8. Otherwise Lemma F.5.4 applies to the pullback transfer $[BX]$. Conjugating by U_Y^{-1} first puts $[b]$ in $D_{2m}^{(Z)}$. Since $[X] \in D_{2m}^{(Z)}$, closure under right multiplication then puts the realizable transformed transfer $[bX]$ in $D_{2m}^{(Z)}$.

Conversely, for the chosen representative B_T, define its nonzero representative scalar β_T by $B_T X = \beta_T U_Y T U_Y^{-1}$. Then

$$B'_T = U_Y^{-1} B_T U_Y^{-\mathsf{T}} = \frac{i\beta_T}{2} T.$$

Since $[D(t)X] = [XD(t^{-1})]$ and $[XD(t)X] = [D(t^{-1})]$, every class occurs and

$$G(\mathcal{F}^\sharp) = D_{2m}^{(Z)}.$$

The open-network bijection also transports the complete deck: absorbing an equality edge replaces a label by XT, TX, or XTX, which permute $D_{2m}^{(Z)}$. Hence zeros, matching products, factor closure, and minimum arity among irreducible signatures correspond, so the transformed state is deck-stable. In particular,

$$\mathsf{Stable}(\mathcal{F}^\sharp, D_{2m}^{(Z)}, f^{(Y)}), \qquad \operatorname{arity}(f^{(Y)}) = \nu(\mathcal{F}^\sharp) = \operatorname{arity}(f).$$

Apply Theorem F.6.12. Complete closure of each transported concrete batch is resolved. On the remaining branch there are $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$ and one $V \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ such that

$$V^{-1}(\Gamma \cup \{I\}) \subseteq \mathscr{C}, \qquad V^{\mathsf{T}} X V \in \mathscr{C}.$$

Define the common change-of-basis matrix $W := U_Y V$. Then $W^{-1}\mathcal{F} \subseteq \mathscr{C}$. With

$$E = V^{\mathsf{T}} X V, \qquad F = V^{-1} I V^{-\mathsf{T}},$$

both E, F are nonsingular symmetric members of $\mathscr{C}$, and Lemma F.6.14 gives

$$W^{\mathsf{T}} X W = -2iV^{\mathsf{T}} I V = -2iEFE \in \mathscr{C}.$$

Thus W is common. The $\mathscr{A}, \mathscr{L}$ cases are resolved tractable cases satisfying $\mathsf{Tract}(K\mathcal{F})$. $\mathscr{P}$ is the remaining tractable case. The reductions above use U_Y for the whole network. $W = U_Y V$ records its contribution to the final witness. □

Theorem F.6.16 (Reduction of the $D_{2m}^{(X,0)}$ sector)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, D_{2m}^{(X,0)}, f), \qquad \mathsf{Even}(\mathcal{F}), \qquad m \geq 4, \quad 2 \mid m.$$

Then $\mathsf{Close}_{\{\mathscr{P}\}}(\mathcal{F}, D_{2m}^{(X,0)}, f)$.

Proof. Use the Walsh matrix P from Equation (F.24), with $P^{-1} = P/2$. Direct multiplication gives, projectively,

$$PD(t)P^{-1} \doteq A(t), \qquad PXD(t)P^{-1} \doteq ZA(t), \tag{F.27}$$

so $P^{-1}D_{2m}^{(X,0)}P = D_{2m}^{(Z)}$. More importantly,

$$P^{\mathsf{T}}XP = 2Z, \qquad P^{-1}IP^{-\mathsf{T}} = \frac{1}{2}I. \tag{F.28}$$

Thus this whole-problem change preserves the available binary equality.

Make available by reduction the fixed representative B_j of the mark $j = [Z]$, retaining its realization scalar and input orientation. Since $[B_jX] = j$, one has $B_j \doteq Y$. Put

$$\mathcal{F}^+ = \mathcal{F} \cup \{B_j\}, \qquad \Gamma = P^{-1}\mathcal{F}^+, \qquad f^{(X,0)} = (P^{-1})^{\otimes \operatorname{arity}(f)} f.$$

After recording the two nonzero representative scalars, use $I, Y \in \Gamma$. Indeed

$$P^{-1}IP^{-\mathsf{T}} = \frac{1}{2}I, \qquad P^{-1}YP^{-\mathsf{T}} = -\frac{1}{2}Y.$$

Write $\text{Holant}^Z(\Gamma)$ for the homogeneous Z-edge half-edge problem. The whole-problem change in Equation (F.28) gives

$$\text{K-Holant}(\mathcal{F}^+) \equiv_{\mathrm{T}} \text{Holant}^Z(\Gamma),$$

where the factor 2 per edge is known. The exact local identities

$$Z = XIXYX, \qquad X = -ZIZYZ \tag{F.29}$$

give a signature-preserving open-network equivalence

$$\text{Holant}^Z(\Gamma) \equiv_{\mathrm{T}} \text{K-Holant}(\Gamma). \tag{F.30}$$

For the forward reduction, replace a Z-edge by three X-edges with the available ordered vertices I, Y. For the reverse reduction, replace an X-edge by three Z-edges with the same two vertices. Reversing the ordered Y-vertex changes only its recorded nonzero sign. Define

$$\Gamma^\sharp := \Gamma \cup \{Z\}.$$

The first identity in Equation (F.29) directly realizes Z in the X-edge model, so adding it changes neither the problem nor the set of directly realizable signatures.

The two edge presentations therefore have the same projective set $\mathscr{B}$ of directly realizable non-singular ordered binary tensors. In the Z-edge model, open-network transport and Equation (F.27) give

$$\{[bZ] : b \in \mathscr{B}\} = P^{-1}D_{2m}^{(X,0)}P = D_{2m}^{(Z)}.$$

Since m is even, $[Y] = [XD(-1)] \in D_{2m}^{(Z)}$. Hence, in the X-edge presentation,

$$G(\Gamma^\sharp) = \{[bX] : b \in \mathscr{B}\} = \{[bZ][ZX] : b \in \mathscr{B}\} = D_{2m}^{(Z)}[Y] = D_{2m}^{(Z)}.$$

The complete deck transports as well. Under P^{-1}, the original X-edge deck is the corresponding Z-edge deck. Translating any target dressing or closing edge by the second identity in Equation (F.29), and then absorbing the inserted I, Y binary chain into the adjacent fixed chosen representative, merely permutes the $D_{2m}^{(Z)}$ labels. The same argument with the first identity gives the inverse correspondence. Thus zeros, matching products, factorization trees, minimum arity of a nonbinary irreducible signature, and parity of the irreducible factors' arities correspond. Therefore

$$\mathsf{Stable}(\Gamma^\sharp, D_{2m}^{(Z)}, f^{(X,0)}), \qquad \mathrm{arity}(f^{(X,0)}) = \nu(\Gamma^\sharp) = \mathrm{arity}(f), \qquad \mathsf{Even}(\Gamma^\sharp).$$

Apply Theorem F.6.12. Every resolved complexity leaf, concrete factor batch, and strict successor transports back through the two open-network equivalences. On a surviving tractable case with a common presentation, there are $\mathscr{C} \in \{\mathscr{A}, \mathscr{P}, \mathscr{L}\}$ and $V \in \mathrm{GL}_2(\overline{\mathbb{Q}})$ such that

$$V^{-1}\Gamma^\sharp \subseteq \mathscr{C}, \qquad E := V^{\mathsf{T}} X V \in \mathscr{C}.$$

Set $W := PV$. Then $W^{-1}\mathcal{F} \subseteq \mathscr{C}$. Because $Z \in \Gamma^\sharp$, also

$$F := V^{-1} Z V^{-\mathsf{T}} \in \mathscr{C}.$$

The binaries E, F are nonsingular and symmetric, so Lemma F.6.14 and $XZX = -Z$ give

$$EFE = V^{\mathsf{T}} X Z X V = -V^{\mathsf{T}} Z V, \qquad W^{\mathsf{T}} X W = 2V^{\mathsf{T}} Z V = -2EFE \in \mathscr{C}.$$

Thus W is a common change-of-basis matrix for the original problem. The $\mathscr{A}, \mathscr{L}$ cases are resolved tractable cases satisfying $\mathsf{Tract}(K\mathcal{F})$ and $\mathscr{P}$ is the remaining tractable case. The witness $W = PV$ combines the whole-network equivalence with the transformed common presentation. I remains available throughout. □

F.6.5 The weighted $D_{2m}^{(X,1)}$ case

In the shifted dihedral case, a matching describes the support, but factorization also requires the coefficients to split across its edges. We test this by joining two copies. The resulting identities force the ratio for changing one edge assignment to be independent of the others. We develop the test in standard coordinates and then show how to realize the required contractions with the weighted kernels of $D_{2m}^{(X,1)}$.

Theorem F.6.17 ($D_{2m}^{(X,1)}$-sector deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, D_{2m}^{(X,1)}, f), \qquad \mathsf{Even}(\mathcal{F}), \qquad m \geq 4, \quad 2 \mid m.$$

Then $\mathsf{Close}_{\{\mathscr{P}\}}(\mathcal{F}, D_{2m}^{(X,1)}, f)$.

F.6.5.1 Relative stability and endpoint compatibility

We work in the standard $D_{2m}^{(Z)}$ coordinates up to and including Lemma F.6.30, using the standard kernels and sector pencils with $m \geq 3$ and even arity $n = 2k + 2 \geq 6$. The support and coefficient reconstruction applies to the selected occurrences under the deck hypotheses stated below. Lemmas F.6.33 and F.6.34 then transfer these results to the weighted $D_{2m}^{(X,1)}$ kernels. We begin by specifying relative deck stability.

Definition F.6.18 (Occurrence-level relative deck stability)**.** For a retained occurrence h at current reduction data $(\mathcal{F}, G)$, write

$$\mathrm{RelStable}(\mathcal{F}, G; h)$$

when its complete oriented deck has been regenerated, every already obtained proper card and its complete factorization tree have been processed, the complete group remains G, and, on the continuing branch, every surviving nonzero proper card is a G-matching signature, while every unresolved lower-arity nonbinary occurrence has been placed earlier in the occurrence induction. An active deck-stable minimum occurrence is automatically relatively deck-stable.

Under this hypothesis, the proper outputs needed below have already been processed. Lemma F.6.30 checks the use of the reconstruction results with this hypothesis, and Section F.6.5.6 records their occurrence-level descent.

For a fixed deleted pair $p = \{i, j\}$, a *residual matching* of a nonzero card

$$P = \bigotimes_{\{u,v\} \in \mathcal{M}_p} B_{uv}$$

is the perfect matching $\mathcal{M}_p$ of the residual inputs $[n] \setminus p$. Its matching translation space is

$$M(\mathcal{M}_p) = \mathrm{span}\{\mathbf{e}_u + \mathbf{e}_v : \{u, v\} \in \mathcal{M}_p\}.$$

The even and odd families have the same residual matching at p when their unordered edge sets agree, equivalently when their matching translation spaces agree. Support-parity and ordered factor-class disagreements remain separate discrepancies. For example, on residual inputs $3, 4, 5, 6$, the pairings $\{\{3, 4\}, \{5, 6\}\}$ in both sectors give the same matching, whereas that pairing and $\{\{3, 5\}, \{4, 6\}\}$ form a matching disagreement whose union contains an alternating four-cycle. The support integration below relates these local matchings for different deleted pairs.

For the following support comparison suppose that both sector pencils have type C. All ordered pairs containing Z or P are handled separately in Lemma F.6.22 below.

Compare one realizable nonzero even card with one realizable nonzero odd card at the fixed deleted pair p. If their residual matchings differ, or their ordered factor classes differ on at least two edges (including parity changes), this is a cross-sector disagreement and is consumed by Lemmas F.6.6 and F.6.7. Hence the continuing branch has one common residual matching with all nondesignated factor classes aligned.

If the two sector rulings have the same residual matching but designate different varying edges, close only their common fixed factors. Up to affine translations, the remaining support, in the distinguished bits x, y and the two varying-edge bits u, v, is

$$\begin{aligned} x = y &\implies u = x + \alpha, \quad v \text{ free}, \\ x \neq y &\implies v = x + \beta, \quad u \text{ free}. \end{aligned}$$

A realizable parity kernel imposing $u = x + \alpha$ has a unique lift and leaves a six-string quaternary signature, hence not a two-monomial product. Its support contains a complementary pair. Realizable logical X-dressings may move that pair to the two endpoints. Add this quaternary signature and its irreducible factors. A retained nonbinary factor has arity at most four: for the active minimum-arity nonbinary irreducible signature it is a strict lower-ν successor, and for a later occurrence it is assigned to the lower-arity occurrence induction. Stabilize the state of that quaternary occurrence, regenerate its complete oriented deck, and only then invoke the equality-accessible quaternary boundary and eight-vertex reduction. Thus the different-edge branch is closed. We may therefore assume below that both sectors use one common varying edge.

To assemble the four slices into one affine support, their offsets must fit the same affine translation space. After aligning the common factors, write the four pencil-endpoint supports as

$$\operatorname{supp} h_\xi = \mathbf{a}_\xi + L, \qquad h_{00} = A,\ h_{01} = B,\ h_{10} = C,\ h_{11} = D_{11},$$

where

$$L = \operatorname{span}\{\mathbf{e}_{u_r} + \mathbf{e}_{v_r} : 1 \le r < k\}, \qquad \dim L = k - 1.$$

Define the endpoint displacement vectors

$$\mathbf{d}_E = \mathbf{a}_{00} + \mathbf{a}_{11}, \qquad \mathbf{d}_O = \mathbf{a}_{01} + \mathbf{a}_{10}.$$

Then the realizable even/odd card support translation spaces are

$$L_E = L + \operatorname{span}_{\mathbb{F}_2}\{\mathbf{d}_E\}, \qquad L_O = L + \operatorname{span}_{\mathbb{F}_2}\{\mathbf{d}_O\}, \qquad \mathbf{d}_E, \mathbf{d}_O \notin L.$$

The condition we need is

$$\mathbf{a}_{00} + \mathbf{a}_{01} + \mathbf{a}_{10} + \mathbf{a}_{11} \in L \tag{F.31}$$

Equivalently,

$$\mathbf{d}_E + \mathbf{d}_O \in L, \qquad [\mathbf{a}_{00}] + [\mathbf{a}_{11}] = [\mathbf{a}_{01}] + [\mathbf{a}_{10}] \quad \text{in } \mathbb{F}_2^{n-2}/L, \qquad L + \operatorname{span}_{\mathbb{F}_2}\{\mathbf{d}_E\} = L + \operatorname{span}_{\mathbb{F}_2}\{\mathbf{d}_O\}.$$

This condition is independent of the representatives $\mathbf{a}_\xi$, since changing one representative changes the four-term sum by an element of L.

Lemma F.6.19 (Realizable-card endpoint test)**.** *In the complementary-axis* C/C *branch with one common varying edge, the residual matchings of the realizable nonzero cards represented by $E(t)$ and $O(s)$ agree if and only if Equation* (F.31) *holds. Failure is localized using the realizable cards, not the endpoint slices.*

Proof. A monomial matching is determined by its weight-two strings, so the two matchings agree exactly when their translation spaces agree. Since $\mathbf{d}_E, \mathbf{d}_O \notin L$,

$$L_E = L_O \iff \mathbf{d}_E + \mathbf{d}_O \in L \iff \mathbf{a}_{00} + \mathbf{a}_{01} + \mathbf{a}_{10} + \mathbf{a}_{11} \in L.$$

Failure therefore invokes Lemmas F.6.6 and F.6.7 on the realizable cards and is resolved. □

Lemma F.6.20 (Zero and proportional endpoint pencils)**.** *Every endpoint pair in which at least one sector has type* Z *or* P *yields at least one of the following concrete outcomes:*

1. *the branch is impossible because it would give $f = 0$;*
2. *all four slices have one common residual tensor, giving an exact binary–residual factorization of f;*
3. *all but one residual matching factor are common, giving an exact factorization $f = P \otimes q$, where q is a nonzero quaternary factor on the deleted pair and the exceptional residual edge;*
4. *realizable nonzero cards have a disagreement, which is localized and consumed by Lemmas F.6.6 and F.6.7.*

In item 3 the quaternary has matching-flattening rank two whenever one of the two pencils has type C*. The factor q is available by the factor theorem and is a strict lower-ν successor.*

Proof. For a type-Z even pencil, two-card inversion gives $A = D_{11} = 0$, hence the support condition $x_i \oplus x_j = 1$. If the odd pencil also has type Z, then $f = 0$, contrary to the active nonzero irreducible signature. If it has type P, the same inversion gives

$$f = G_p \otimes H_O, \qquad G_p = \begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix},$$

where G_p is the displayed binary signature on p and H_O is the complementary $(n-2)$-ary signature. This is an exact factorization of f, and its factors are exposed only by Theorem F.2.10. If the odd pencil has type C, write its exact ruling as $O(t) = P_O \otimes K_O(t)$, where P_O is the product of the $k-1$ fixed residual factors and $K_O(t)$ is the nonproportional binary pencil on its varying edge e. Two-card inversion puts both B and C in $P_O \otimes \mathrm{Mat}_2(\mathbb{C})$. The zero slices A, D_{11} lie there as well. Hence

$$f = P_O \otimes q_{p,e}.$$

The complementary-axis endpoints of K_O are nonproportional, so the $p \mid e$ logical flattening of $q_{p,e}$ has rank two. Apply Theorem F.2.10 and Lemma F.3.4. The quaternary available by reduction is a deck-stable lower-ν successor, while every binary factor is sent through Lemma F.5.4. Exchanging the sectors covers every type-Z case. A single zero sample in a type-P pencil is included in the exact proportional inversion and is never used by itself to infer a factor.

For a proportional even pencil, inversion gives

$$A = aH_E, \qquad D_{11} = dH_E.$$

Choose a nonzero realizable even card as the representative of the line $\mathbb{C}H_E$, and compare it with two nonzero realizable odd cards. Any matching change or variation on at least two residual factor classes, including parity changes, is a disagreement and is consumed by Lemmas F.6.6 and F.6.7. If no bounded output occurs, the two sectors have one common residual matching and differ on at most one designated edge e. Let P be the product of all fixed factors outside e. Four-card inversion then puts $A, B, C, D_{11} \in P \otimes \mathrm{Mat}_2(\mathbb{C})$, and therefore proves the exact identity

$$f = P \otimes q_{p,e}.$$

If no edge differs, absorb the last common factor into P. This is the stronger identity $f = G_p \otimes H$. If one edge differs, the remaining quaternary is nonzero. In a P/C or C/P case its $p \mid e$ flattening has rank two because the C-endpoints are nonproportional. In the P/P case different projective factor classes give the same rank-two conclusion, while equal classes give the stronger binary–residual factorization. Only after this exact identity has been proved are P and $q_{p,e}$ exposed by Theorem F.2.10 and Lemma F.3.4. This covers P/P, P/C, C/P, and their possible zero samples. □

By Lemma F.6.20, it remains to analyze two nonproportional sector pencils.

We now describe the six-input contraction output obtained by retaining the deleted pair and two residual edges. Label these inputs by $p = \{1, 2\}$ and $R = \{3, 4, 5, 6\}$. For each sector $\epsilon = x_1 \oplus x_2$, let $M_\epsilon = \{e_\epsilon, e'_\epsilon\}$ be its residual matching, where $e_\epsilon = \{u_\epsilon, v_\epsilon\}$ is the fixed edge and $e'_\epsilon = (r_\epsilon, s_\epsilon)$ is the ordered varying edge. There are parameters $a_\epsilon, b_\epsilon, o_\epsilon \in \mathbb{F}_2$ such that its six-input support is parametrized by

$$\begin{aligned} (x_1, x_2) &= (z, z + \epsilon), \\ x_{u_\epsilon} \oplus x_{v_\epsilon} &= a_\epsilon, \qquad\qquad \epsilon, z \in \mathbb{F}_2. \\ (x_{r_\epsilon}, x_{s_\epsilon}) &= (o_\epsilon + z, o_\epsilon + z + b_\epsilon), \end{aligned} \tag{F.32}$$

For fixed (ϵ, z), the parity equation on the fixed edge has two solutions and the varying edge is determined. Hence each of the four endpoint slices indexed by (ϵ, z) is a two-string affine flat, each sector has four strings, and the two sectors have the eight-string support used in Equation (F.32). The varying-edge assignments for $z = 0, 1$ are bitwise complements. The coefficients on these slices need not be equal.

For a perfect matching M on $2r$ inputs and parities $(c_e)_{e\in M}$, the associated *matching coset* is

$$\{\mathbf{x} \in \mathbb{F}_2^{2r} : x_u + x_v = c_{\{u,v\}} \text{ for every } \{u, v\} \in M\}.$$

Lemma F.6.21 (A six-to-four contraction)**.** *Suppose a standard $D_{2m}^{(Z)}$ localization leaves exactly the distinguished pair p and two residual monomial pairs, with support as in Equation* (F.32)*. Then either the six-input support is a single three-edge matching coset, or a realizable I/X-contraction realizes a nonzero quaternary of support size six or eight. In the latter case, the output is not a product of current-group monomial binaries and the procedure does not re-enter a six-input state.*

Proof. The explicit support in Equation (F.32) is the Boolean family exhaustively covered by Theorem F.A.11. That classification either identifies the global matching coset or supplies a pair and parity γ with injective projection. Contracting it by the realizable parity-γ kernel—I for $\gamma = 0$ and X for $\gamma = 1$—gives one nonzero internal extension per string, so there is no cancellation and the output signature has the certified six- or eight-string support. Add this direct output and close its complete factor batch. A singular/unsafe factor, unary or odd factor, or a smaller irreducible signature is handled by the stabilized procedure. A safe binary whose transfer class lies outside the current group gives a successor with a strictly larger complete transfer group. If the output itself is irreducible, retain it as a lower-arity quaternary occurrence, stabilize its state, and regenerate its complete oriented deck, and only then invoke the equality-accessible quaternary boundary and eight-vertex reduction. Since a product of two nonsingular current-group monomial binaries has support size four, the support-six/eight output cannot close as an unrecorded matching product. This completes the reduction of the arity-six tensor through its four-input output and factor batch. □

Lemma F.6.22 (Reduction of the endpoint cases)**.** *For every deleted pair p, all endpoint combinations containing type* Z *or* P *are resolved. On the remaining branch, the endpoint pair is* C/C*, Equation* (F.31) *holds, and there is one common residual matching.*

Proof. The Z/Z entry gives $f = 0$. Every Z/P, Z/C, P/P, P/C, and transposed entry is covered by Lemma F.6.20.

It remains to take both types equal to C. A matching change or two different ordered factor classes invokes the trace cover Lemmas F.6.6 and F.6.7. After closing common factors with the same projective class, two different variable edges give the six-string quaternary constructed above. Apply the lower-arity stabilized-quaternary route given there. For one common variable edge, failure of Equation (F.31) is again a realizable-card disagreement. If the parallelogram holds, then, because $n \geq 6$, retain the varying edge and one fixed residual edge together with p. This gives the arity-six tensor of Equation (F.32). By Lemma F.6.21, any nonmatching quaternary output is processed through its full factor batch. The sole support-level remaining case is a three-edge matching coset. These alternatives exhaust C/C. □

F.6.5.2 Global support integration and the Reed–Muller alternative

The Z/P/C endpoint classification has resolved every case containing Z or P and every C/C family with a disagreement. The remaining case satisfies Equation (F.31), and the two sectors have the

same residual matching at every deleted pair. These local translation spaces determine a self-dual code. A weight-two codeword identifies a matched pair and then the global matching coset. If there is no such codeword, the weight-four supports determine $RM(1,3)$. A realizable contraction with a minor proportional to $1-t^2$ excludes that remaining case.

Lemma F.6.23 (Endpoint-support saturation)**.** *Assume the standard $D_{2m}^{(Z)}$ analysis is on the remaining endpoint- type branch. Then, for each deleted pair $p=\{i,j\}$, there is a fixed-edge matching translation space L_p on the residual coordinates and endpoint differences $\mathbf{d}_{E,p},\mathbf{d}_{O,p}\notin L_p$ such that the four endpoint slices have full support on affine cosets $\mathbf{a}_{ab}+L_p$, with every coefficient on each such coset nonzero. Moreover*

$$M_p := L_p + \operatorname{span}_{\mathbb{F}_2}\{\mathbf{d}_{E,p}\} = L_p + \operatorname{span}_{\mathbb{F}_2}\{\mathbf{d}_{O,p}\}$$

is the support translation space of the equal- and unequal-p realizable cards. It is the translation space $M(\mathcal{M}_p)$ of the residual matching coset, abbreviated M_p, and hence $M_p = M_p^{\perp}$.

Proof. On the remaining C/C branch, Lemma F.6.8 gives one fixed nonzero monomial factor on every residual matching edge except the designated variable edge. Two samples with nonzero output and Equation (F.31) reconstruct each slice by the four-card inversion used in Lemma F.6.5. Hence every string in the resulting affine matching coset has a nonzero coefficient. More explicitly, the two complementary-axis endpoint matrices have disjoint monomial supports, so each assignment to the remaining inputs is supplied by exactly one endpoint. Its coefficient is a product of nonzero fixed monomial entries and a nonzero endpoint entry, with no cancellation between endpoints. For $\mathbf{d}_{E,p}=\mathbf{a}_{00}+\mathbf{a}_{11}$ and $\mathbf{d}_{O,p}=\mathbf{a}_{01}+\mathbf{a}_{10}$, the parallelogram equation says $\mathbf{d}_{E,p}+\mathbf{d}_{O,p}\in L_p$, which identifies the two displayed descriptions of M_p and the four offsets. Matchings belonging to different deleted pairs are integrated into one common code by Lemma F.6.25, rather than compared directly here. □

For $d\geq 1$,

$$\mathrm{AGL}(d,2) := \mathbb{F}_2^d \rtimes \mathrm{GL}_d(\mathbb{F}_2)$$

acts by $\mathbf{x}\mapsto A\mathbf{x}+\mathbf{b}$. An *affine relabelling* of inputs indexed by $\mathbb{F}_2^d$ is the corresponding permutation of those inputs. Indexing the eight coordinates by $\mathbb{F}_2^3$ in lexicographic order, define the first-order Reed–Muller code by

$$RM(1,3) := \{(\ell(\mathbf{t}))_{\mathbf{t}\in\mathbb{F}_2^3} : \ell(\mathbf{t}) = a+u_1t_1+u_2t_2+u_3t_3,\ a,u_1,u_2,u_3\in\mathbb{F}_2\}.$$

Thus $RM(1,3)$ consists of the sixteen truth tables of affine-linear functions on $\mathbb{F}_2^3$. The natural action satisfies $\mathrm{AGL}(2,2)\cong S_4$. $\mathrm{AGL}(3,2)$ is two-transitive and preserves $RM(1,3)$, because affine precomposition preserves affine truth tables [Car21, Section 4.1].

We now regard the support as a set of binary strings, using the projections and affine operations of Section F.2.8. The local matching equations will determine a self-dual binary code. A weight-two codeword identifies a global matched pair. If no such codeword exists, the weight-four codewords identify the eight-coordinate Reed–Muller code. This reconstructs support only. Coefficients are treated afterward.

For a function $F:\mathbb{F}_2^d\to R$ into an abelian group, its additive derivative is

$$(\Delta_{\mathbf{a}}F)(\mathbf{x}) := F(\mathbf{x}+\mathbf{a})-F(\mathbf{x}).$$

The identities $\Delta_{\mathbf{a}}\Delta_{\mathbf{b}}F=0$ for all $\mathbf{a},\mathbf{b}$ hold exactly when $F(\mathbf{x})=F(0)+\ell(\mathbf{x})$ for a group homomorphism $\ell:(\mathbb{F}_2^d,+)\to(R,+)$: the identities make every $\Delta_{\mathbf{a}}F$ constant, and the resulting $\ell(\mathbf{x}):=F(\mathbf{x})-F(0)$ is additive.

Lemma F.6.24 (Self-dual Steiner reconstruction)**.** *Let $D \le \mathbb{F}_2^n$ be self-dual, where $n > 4$, and suppose that D has no codeword of weight two. If every three distinct coordinates lie in the support of a unique weight-four codeword of D, then $n = 8$. After identifying the coordinates with $\mathbb{F}_2^3$, one has $D = RM(1,3)$.*

Proof. The weight-four supports form a *Steiner quadruple system*: a family of four-element blocks in which every three coordinates belong to exactly one block. Self-orthogonality of D makes two blocks meet evenly. Uniqueness through triples then makes the intersection of two distinct blocks either empty or a pair.

Fix a coordinate 0. Define a binary operation $+$ on the coordinates by $x + x = 0$, $x + 0 = x$, and, for distinct nonzero x, y, let $x + y$ be the fourth point in the block through $0, x, y$. This operation is commutative and every element has order two. For the associativity check, define the auxiliary points $u := x + y$, $v := y + z$, and $w := u + z$. The symmetric difference of the blocks $\{0, x, y, u\}$ and $\{0, y, z, v\}$ is the weight-four codeword $\{x, u, z, v\}$. Symmetric difference with $\{0, u, z, w\}$ gives the block $\{0, x, v, w\}$. Uniqueness of the block through $0, x, v$ gives

$$(x + y) + z = w = x + v = x + (y + z).$$

If an auxiliary point coincides—for example $z = x + y$—the same identity reduces immediately to $x + x = 0$. Thus the displayed distinct-block calculation and these collision cases cover all inputs. The coordinates form an elementary abelian group $V \cong \mathbb{F}_2^r$, and $n = 2^r$. If $\{a, b, c, d\}$ is any block, symmetric difference with the block through $0, a, b$ shows $d = a + b + c$. Hence the blocks are exactly the affine planes of V.

Let B be the binary span of the block indicator vectors. An affine plane is a translate of a two-dimensional subspace of V. Orthogonality to every such plane is equivalent to vanishing of every second additive derivative $\Delta_{\mathbf{a}}\Delta_{\mathbf{b}}F(\mathbf{x})$. Hence $B^\perp$ is exactly the space of truth tables of affine-linear functions $F : V \to \mathbb{F}_2$. Thus $\dim B = 2^r - r - 1$. Since $B \subseteq D$ and $\dim D = 2^{r-1}$, the inequality $2^r - r - 1 \le 2^{r-1}$, with $r > 2$, forces $r = 3$. Equality gives $D = B = RM(1,3)$. □

Lemma F.6.25 (Support integration)**.** *Assume the remaining endpoint-type case, so that Equation* (F.31) *holds for every deleted pair. Then* $\mathrm{supp}(f) = \mathbf{a} + D$ *for a self-dual binary linear code $D \le \mathbb{F}_2^n$. Exactly one of the following support alternatives remains:*

1. *D is the matching translation space of one perfect matching on all n inputs, so $\mathbf{a} + D$ is one* global matching coset*;*
2. *$n = 8$ and, after relabelling the inputs, $D = RM(1,3)$.*

The second alternative is incompatible with the standard $D_{2m}^{(Z)}$ deck for $m \ge 3$. Consequently the sole remaining support is one global matching coset.

Proof. Apply Lemma F.6.23. Translation by the 00-slice gives, for any fixed deleted pair p,

$$\mathbf{a} + D, \qquad D = (\{00\} \times L_p) + \mathrm{span}\{(01, \mathbf{a}_{00} + \mathbf{a}_{01}), (10, \mathbf{a}_{00} + \mathbf{a}_{10})\}.$$

The two displayed generators are independent modulo $\{00\} \times L_p$. Hence $\dim D = n/2$. For an arbitrary deleted pair $p = \{i, j\}$, define the equal-coordinate slice $H_p := \{\mathbf{x} \in D : x_i = x_j\}$. The endpoint saturation and the parallelogram give

$$\pi_{\{i,j\}} : D \longrightarrow \mathbb{F}_2^2 \text{ surjective}, \qquad \pi_{[n]\setminus\{i,j\}}(H_{\{i,j\}}) = M_{\{i,j\}},$$

where $M_p = M_p^\perp$ is the residual matching translation space from Lemma F.6.23. To see self-orthogonality, take arbitrary $\mathbf{x}, \mathbf{y} \in D$. Among the $n > 4$ coordinate columns $(x_\ell, y_\ell) \in \mathbb{F}_2^2$, two

are equal. Relabel them i, j. Then $\mathbf{x}, \mathbf{y} \in H_p$, so their residual projections lie in M_p and have zero mutual dot product. Their contribution on the deleted pair is $x_i y_i + x_j y_j = 0$. Hence $\mathbf{x} \cdot \mathbf{y} = 0$, proving $D \subseteq D^\perp$. Since $\dim D = n/2$, equality follows: $D = D^\perp$.

Suppose first that $\mathbf{e}_u + \mathbf{e}_v \in D$. Self-duality implies $x_u = x_v$ for every $\mathbf{x} \in D$, so $H_{\{u,v\}} = D$. Put $E = [n] \setminus \{u, v\}$. Endpoint saturation gives

$$\pi_E(D) = M_{\{u,v\}},$$

the translation space of the residual perfect matching. For $\mathbf{y} \in M_{\{u,v\}}$, choose a lift $\mathbf{x} \in D$. Subtracting $x_u(\mathbf{e}_u + \mathbf{e}_v)$ gives the zero extension $\iota_E(\mathbf{y}) \in D$. Hence

$$D = \operatorname{span}_{\mathbb{F}_2}\{\mathbf{e}_u + \mathbf{e}_v\} \oplus \iota_E(M_{\{u,v\}}),$$

which is exactly the translation space of the global perfect matching $\{\{u, v\}\} \cup \mathcal{M}_{\{u,v\}}$. This is a support identity only: it does not yet assert a tensor factorization of the coefficients.

It remains that D has no weight-two codeword. Fix $i \neq j$ and put $E = [n] \setminus \{i, j\}$. The projection $H_{\{i,j\}} \to \mathbb{F}_2^E$ is injective, because its only possible nonzero nullspace is $\mathbf{e}_i + \mathbf{e}_j$. The functional $\mathbf{x} \mapsto x_i + x_j$ is nonzero on D, since otherwise $\mathbf{e}_i + \mathbf{e}_j \in D^\perp = D$. Thus $\dim H_{\{i,j\}} = n/2 - 1$, and endpoint saturation identifies its image with $M_{\{i,j\}}$. For a third coordinate ℓ, the residual matching pairs ℓ with a unique q. The unique lift of $\mathbf{e}_\ell + \mathbf{e}_q \in M_{\{i,j\}}$ cannot have 00 on i, j, since that would be a weight-two codeword. It is therefore $\mathbf{e}_i + \mathbf{e}_j + \mathbf{e}_\ell + \mathbf{e}_q \in D$. Hence every three coordinates lie in a unique weight-four support, and Lemma F.6.24 gives $n = 8$ and $D = RM(1, 3)$.

To exclude this residue, identify the eight coordinate positions with $\mathbb{F}_2^3$ and, after an affine relabelling, take the deleted pair to be $\{000, 001\}$. On its equal-coordinate slice the residual matching is

$$\{010, 011\}, \qquad \{100, 101\}, \qquad \{110, 111\}.$$

If y_1, y_2, y_3 are the three logical matching bits, the common bit on the deleted pair is

$$b = y_1 + y_2 + y_3.$$

Contract the deleted pair by the realizable sector phase t. The projection of the equal slice is injective, so every string in the residual support has one internal lift and its coefficient is multiplied by t^b. Fixing y_3 and inspecting the 2×2 face indexed by y_1, y_2, the phase exponents are $(0, 1, 1, 0)$, or their complement. Thus the face has one of the raw forms

$$\begin{pmatrix} A & tB \\ tC & D \end{pmatrix}, \qquad \begin{pmatrix} tA & B \\ C & tD \end{pmatrix}, \qquad A, B, C, D \neq 0.$$

The realizable $t = 1$ card lies in the fixed Segre ruling, so $AD = BC$. Nonzero row and column scalings therefore give, for some $\varepsilon \in \mathbb{F}_2$, the normalized block in the original basis

$$\begin{aligned} H_t &= \kappa \operatorname{diag}(r_0, r_1) C_\varepsilon(t) \operatorname{diag}(s_0, s_1), \\ C_0(t) &= \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}, \qquad C_1(t) = \begin{pmatrix} t & 1 \\ 1 & t \end{pmatrix}, \\ \det H_t &= (-1)^\varepsilon \kappa^2 r_0 r_1 s_0 s_1 (1 - t^2). \end{aligned}$$

with all scalar factors nonzero. If an orientation produces t^{-1}, clear its nonzero monomial factor. Since $m \geq 3$, some realizable $t \in \mu_m \setminus \{\pm 1\}$ exists, and the displayed nonzero minor contradicts the fixed Segre ruling. □

The support is therefore one global matching coset. To obtain a tensor factorization, we still need to separate its coefficients. We first handle the degenerate six-input coefficient tensors, then use two-copy outputs to prove the coefficient product formula.

F.6.5.3 The six-input coefficient calculation

We now return to the matching-supported six-input coefficient cases retained by the support analysis. Trace-cover six-input disagreements have already been reduced to quaternary and resolved. The only remaining case is the fixed three-edge-matching coefficient pattern. This part first handles all degenerate coefficient cases and then analyzes the coefficient tensor with all eight entries nonzero.

For a quaternary contraction whose support is contained in two fixed residual matching edges, write its logical coefficient matrix as

$$Q = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

We include the cases $0 < |\operatorname{supp} Q| < 4$. A *support-matching disagreement* means that its support in the original variables uses a residual pairing different from the fixed pairing. This is the combinatorial disagreement of the trace-cover reduction. By contrast, *not G-matching* means that the signature has no tensor factorization into current-group binaries. These two notions are not interchangeable.

For each directly realized quaternary, first add it with its input order, scalar, and reduction. Jointly add all its irreducible factors. Test all three $2 + 2$ flattenings and consider every exposed factor. Make a realizable endpoint dressing. Invoke Lemma F.5.21 only after tensor irreducibility, endpoint nondegeneracy, non-G-matching status, and both crossed rank hypotheses have been verified.

Lemma F.6.26 (Reductions of six-input coefficient tensors)**.** *Let h be a directly realized six-input signature whose support is exactly the eight-string coset of a fixed three-edge matching. Before a quotient of its logical coefficients is formed, close every already constructed four- or six-input batch as follows:*

1. *a zero quaternary is converted into an exact factorization of its parent arity-six tensor by the proportional-slice identity below;*
2. *a support-matching disagreement is sent to Lemmas F.6.6 and F.6.7;*
3. *for a quaternary on the two retained matching edges, test Q: if $\det Q = 0$, factor it exactly. If $\det Q \neq 0$, use the quaternary reduction just described;*
4. *jointly close all outputs under irreducible factors and consider every unary, singular, unsafe, infinite-order, or safe binary factor whose transfer class lies outside G;*
5. *if no preceding concrete output applies, retain h only when all eight logical coefficients are nonzero.*

A quaternary with $0 < |\operatorname{supp} Q| < 4$ enters item 3. Every noncurrent factor or nonmatching quaternary outcome gives a resolved leaf or strict successor. If a nonzero child quaternary factors into two current-group binaries, complete that output, record its rank-one equation, and continue the finite analysis of the same originating arity-six tensor.

Proof. Let u_0, u_1 be the two standard-basis binary signatures supported on the two allowed states of a selected matching edge. Write

$$h = u_0 \otimes H_0 + u_1 \otimes H_1.$$

An allowed phase kernel has two nonzero coefficients α_t, β_t, and its quaternary contraction is

$$q_t = \alpha_t H_0 + \beta_t H_1, \qquad \alpha_t \beta_t \neq 0.$$

If $q_t = 0$, first suppose $H_0 = 0$. Then $H_1 = 0$, so $h = 0$ and its token is completed directly. Otherwise $H_0 \neq 0$ and

$$H_1 = -\frac{\alpha_t}{\beta_t} H_0, \qquad h = \left(u_0 - \frac{\alpha_t}{\beta_t} u_1\right) \otimes H_0.$$

The zero quaternary output therefore gives an exact binary–quaternary factorization of h. Close the complete factor batch by Theorem F.2.10 and Lemma F.3.4. A nonbinary factor gives a quaternary or lower-ν successor. Unary factors and unsafe binaries enter their established reductions. A safe binary B with $[BX] \notin G$ strictly enlarges the complete group. If all irreducible factors are already safe current-group binaries, the parent is a G-matching signature and its fixed token is completed.

Now let $q_t \neq 0$ have logical coefficient matrix Q on the two retained matching edges. If $\det Q = 0$, then Q has rank one and the quaternary signature has an exact $2+2$ factorization. When $0 < |\operatorname{supp} Q| < 4$, at least one logical factor has a zero entry, so one exposed binary is singular. If both exposed factors are safe current-group binaries, complete this quaternary output, retain the equation $\det Q = 0$, and resume the coefficient analysis of the same parent arity-six tensor. The child factorization alone does not complete the parent token. If $\det Q \neq 0$, its fixed-matching flattening has rank two. Add the realizable card with its input order, scalar, and reduction. If Q is supported on exactly two entries, nonzero determinant forces them to be opposite logical entries. Realizable logical X-translations then make the quaternary a nonzero generalized equality, and Lemma F.3.16 applies. If exactly three entries are nonzero, realizable logical flips put a nonzero opposite pair at the two endpoints, so the card is endpoint-nondegenerate. After factor closure and removal of every crossed rank-one factorization, it is irreducible, non-G-matching, and has both crossed ranks at least two. The case with four nonzero logical coefficients is endpoint-nondegenerate after the same endpoint choice. Thus the quaternary reduction just described is applicable. Invoke Lemma F.5.21, Theorem F.5.22, and Corollary F.4.8 only after their hypotheses have been verified. A support-matching disagreement instead enters the typed trace-cover reduction. These routes also process every exposed factor by Lemma F.5.4.

For the matching-supported arity-six tensor retained from Lemma F.6.25, the support is already the complete eight-string global-matching coset, so its eight logical coefficients are nonzero. The case analysis processes each phase contraction produced below and each output of a later mixed two-copy batch within the originating fixed local queue. We may therefore write its coefficients as c_{abc}, $a, b, c \in \mathbb{F}_2$, and normalize $c_{000} = 1$. □

Lemma F.6.27 (One-coordinate six-input minor localization)**.** *Let $q = (q_{ab\xi})_{a,b,\xi\in\mathbb{F}_2}$ be the logical coefficient tensor of an arity-six tensor whose support is contained in a fixed three-edge matching. Close every quaternary and factor batch produced below by the quaternary reduction just described and the binary case analysis. On any branch with no resolved leaf or strict stable successor, fully close every factor batch. Each resulting arity-six tensor is then either zero, a G-matching signature, or has rank at most one in each of*

$$a \mid (b, \xi), \qquad b \mid (a, \xi), \qquad \xi \mid (a, b)$$

The first two alternatives also have all three ranks at most one. Their rank equations are recorded before their child tokens are completed, so the conclusion holds at every allowed phase assignment in any later two-copy batch to which this lemma is applied.

Proof. Consider $a \mid (b, \xi)$. Close the ξ-edge by every allowed phase, obtaining the logical matrix pencil

$$A + tB$$

in the a, b coordinates. If $\det(A + tB) \neq 0$ at an allowed phase, the resulting quaternary signature enters the preceding nonzero-minor reduction. If the determinant vanishes for every $t \in \mu_m$, it is a degree-two polynomial with at least three roots and is therefore identically zero.

The two-dimensional rank-one-line classification is elementary. Write nonzero rank-one endpoints as $A = \mathbf{u}\mathbf{v}^{\mathsf{T}}$ and $B = \mathbf{x}\mathbf{y}^{\mathsf{T}}$. The mixed coefficient of $\det(A + tB)$ is, up to sign, $\det[\mathbf{u}, \mathbf{x}] \det[\mathbf{v}, \mathbf{y}]$. Hence either $\mathbf{u}, \mathbf{x}$ are proportional or $\mathbf{v}, \mathbf{y}$ are proportional. Zero or proportional endpoints satisfy the same conclusion directly. In the first case A, B have one common a-factor, so the $a \mid (b, \xi)$ flattening of q has rank at most one. In the second case they have one common b-factor, giving an exact binary factor of q. Take factor closure and apply the factorization reduction. If closing this factorization gives a resolved leaf or strict stable successor, the parent branch ends. Otherwise all exposed factors are current-group matching factors, so q itself is a G-matching signature and all three logical flattenings have rank at most one. Record these equations before completing its child token. Thus every branch continuing the parent analysis has the stated a-rank equation. Repeating the same argument after permuting a, b, ξ proves the other two rank bounds. □

After Lemma F.6.26, all eight logical coefficients on the continuing branch are nonzero. Retain their notation and normalize the remaining global-matching six-input support to $X^{(1,2)} \otimes X^{(3,4)} \otimes X^{(5,6)}$, where the superscript records the ordered pair of inputs on which the binary disequality tensor X acts. For each selected pair of matching edges, use both realizable parity-zero and parity-one kernels. The parity-zero contraction on the first two edges gives

$$c_{000}c_{111} = c_{001}c_{110}.$$

The parity-one contractions on the three choices of edge pair give the remaining equalities among

$$c_{001}c_{110}, \qquad c_{010}c_{101}, \qquad c_{011}c_{100}.$$

Every nonzero determinant is handled by the quaternary reduction just described. Hence, on the continuing branch, all four products are equal:

$$c_{000}c_{111} = c_{001}c_{110} = c_{010}c_{101} = c_{011}c_{100}. \tag{F.33}$$

Define

$$\alpha = \frac{c_{100}}{c_{000}}, \qquad \beta = \frac{c_{010}}{c_{000}}, \qquad \gamma = \frac{c_{001}}{c_{000}}.$$

These three quantities are exposed by explicit factor cards. For α, join inputs 3 and 5, the first inputs of the second and third matching edges, by the realizable parity-zero kernel $M_0(1) = I$. The remaining logical coefficient matrix, with the first matching edge as its row edge, is

$$\begin{pmatrix} c_{000} & c_{011} \\ c_{100} & c_{111} \end{pmatrix}.$$

Its determinant vanishes by Equation (F.33), so exact factor extraction exposes on the first matching edge a binary with ordered ratio $\alpha^{\pm 1}$. Cycling the three matching edges exposes $\beta^{\pm 1}$ and $\gamma^{\pm 1}$ in the same way. Process every exposed binary by Lemma F.5.4. Hence on the continuing branch

$$\alpha, \beta, \gamma \in \mu_m.$$

Fuse the fixed representatives with transfers $D(\alpha^{-1})$, $D(\beta^{-1})$, and $D(\gamma^{-1})$ onto the three oriented matching edges, and divide by the recorded common scalar c_{000}. Equation Equation (F.33) then gives

$$N_r(\mathbf{z}) = \begin{cases} 1, & \mathrm{wt}(\mathbf{z}) \leq 1, \\ r, & \mathrm{wt}(\mathbf{z}) \geq 2, \end{cases} \qquad \mathbf{z} \in \mathbb{F}_2^3, \qquad r := \frac{c_{000}c_{110}}{c_{100}c_{010}}.$$

It remains to prove that r is also a group phase. On this normalized six-input contraction output, contract inputs 1 and 3, the first inputs of the first and second matching edges, by the realizable parity-one kernel $M_1(u) = M_{\alpha=1}(a = u)$, $u \in \mu_m$. In the displayed input orientation this is

$$M_1(u) = \begin{pmatrix} 0 & 1 \\ u & 0 \end{pmatrix}.$$

Write the bits on the six inputs as $x_1, \dots, x_6$. The support matching means $x_2 = 1 - x_1$, $x_4 = 1 - x_3$, and $x_6 = 1 - x_5$. The kernel joining inputs 1 and 3 therefore imposes $x_1 \oplus x_3 = 1$ and contributes the factor u^{x_1}. After this contraction, use x_2 (with $x_4 = 1 - x_2$) as the row bit and x_5 (with $x_6 = 1 - x_5$) as the column bit. Ordering the rows and columns changes only the nonzero representative scalar or reverses an input. The resulting quaternary signature on inputs $2, 4, 5, 6$, up to reversal and a nonzero representative scalar, is

$$\begin{pmatrix} 1 & r \\ u & ur \end{pmatrix} = \begin{pmatrix} 1 \\ u \end{pmatrix} \begin{pmatrix} 1 & r \end{pmatrix}.$$

This is an exact tensor factorization, and its two binary factors have ordered ratios $u^{\pm 1}$ and $r^{\pm 1}$. Expose them through Theorem F.2.10 and Lemma F.3.4 before testing the complete transfer group. Process both factors by Lemma F.5.4 within the selected occurrence's fixed local queue. Thus the coefficient calculation continues only when both factors are safe with transfer classes in $D_{2m}^{(Z)}$. On that branch, $r^{\pm 1} \in \mu_m$, and hence $r \in \mu_m$. Thus the parameter in the displayed coefficient tensor N_r belongs to μ_m. Closing the third matching pair gives the quaternary whose logical coefficient matrix is

$$Q_r = \begin{pmatrix} 2 & 1 + r \\ 1 + r & 2r \end{pmatrix}, \qquad \det Q_r = -(r - 1)^2.$$

Attach the supplied I through an X-edge at the second input of each residual matching pair. These two realizable bit flips give a quaternary supported on $\{0000, 0011, 1100, 1111\}$, with the same logical coefficient matrix Q_r. Its endpoint entries $2, 2r$ are nonzero. For $r \neq -1$, all four logical coefficients are nonzero, and their cross-ratio is

$$\mathrm{cr}(Q_r) = \frac{(Q_r)_{00}(Q_r)_{11}}{(Q_r)_{01}(Q_r)_{10}} = \frac{4r}{(1 + r)^2}.$$

If this support-four quaternary is affine- or local-affine-transformable, Lemma F.4.5 reduces the latter case to affine transformability. The diagonal normal form of Proposition F.4.6 then gives an affine reference signature on the same four-point support. On the two equal-input pairs, a diagonal change of basis rescales the rows and columns of Q_r, preserving its cross-ratio. Clause (ii) of Theorem F.A.5 therefore gives $\mathrm{cr}(Q_r) \in \{1, -1\}$. The value $+1$ is equivalent here to $\det Q_r = 0$, hence to $r = 1$, which is the product case removed below. Therefore a nonproduct affine signature has $\mathrm{cr}(Q_r) = -1$, giving $r + r^{-1} = -6$. Every other outcome is handled by the quaternary reduction. This is a direct quaternary signature with its inherited input order and nonzero representative scalar. Add it and its complete batch of irreducible factors before applying the quaternary reduction. If $r = 1$, then $N_1 \equiv 1$, so the parent arity-six tensor itself is the product of its three matching binaries. Complete the originating fixed token after the binary case distinction. If $r \neq 1$, the displayed fixed-matching minor is nonzero, and the quaternary reduction of Lemmas F.6.7 and F.6.26 applies. For the exceptional value $r = -1$, the equality exit is the following explicit two-copy construction. Define the arity-six signature

$$F_r(z_1, \dots, z_6) = \prod_{j=1}^{3} \mathbf{1}[z_{2j-1} \oplus z_{2j} = 1]\, N_r((z_1, z_3, z_5)),$$

where the three displayed logical bits are the first inputs of the three paired-disequality edges. Take two copies, leave inputs $1, 2$ of each copy external, and join corresponding inputs $3, 4, 5, 6$ by four realizable equality links. The resulting quaternary tensor is

$$W_r(a, b, c, d) = \sum_{u,v,w,z \in \mathbb{F}_2} F_r(a, b, u, v, w, z) F_r(c, d, u, v, w, z).$$

Direct substitution into N_{-1} gives

$$W_{-1}(a, b, c, d) = 4\,\mathbf{1}[(a, b, c, d) \in \{0101, 1010\}].$$

Applying a realizable X-translation on the second external input of each copy sends these two strings to $0000, 1111$, so the output is $4(=_4)$ in the displayed normalization. This terminates by Corollary F.3.13 and Lemma F.3.11. Otherwise the affine cross-ratio test would force $r + r^{-1} = -6$, impossible for a root of unity. Hence every nonproduct case not already terminated enters Corollary F.4.8.

F.6.5.4 Laurent polynomials

The minors below are polynomials in the phase variables. Reversals and weighted kernels can also introduce negative exponents, so we use the following Laurent-polynomial convention. For a commutative ring R, a Laurent polynomial is a finite sum

$$P(\boldsymbol{\lambda}) = \sum_{\mathbf{u} \in \mathbb{Z}^r} c_{\mathbf{u}} \boldsymbol{\lambda}^{\mathbf{u}}, \qquad \boldsymbol{\lambda}^{\mathbf{u}} := \prod_{j=1}^{r} \lambda_j^{u_j},$$

with $c_{\mathbf{u}} \in R$ and all but finitely many $c_{\mathbf{u}}$ zero. Equivalently, $P \in R[\lambda_1^{\pm 1}, \ldots, \lambda_r^{\pm 1}]$. Coefficient extraction is $[\boldsymbol{\lambda}^{\mathbf{u}}]P := c_{\mathbf{u}}$. When a Boolean vector is used as an exponent, its entries are lifted from $\{0, 1\}$ to $\mathbb{Z}$, so exponent arithmetic is never modulo two. Multiplication by a monomial with sufficiently large nonnegative exponents clears all negative exponents. The coordinate degree in λ_j of a nonzero P is

$$\deg_{\lambda_j} P := \max\{u_j : c_{\mathbf{u}} \neq 0\} - \min\{u_j : c_{\mathbf{u}} \neq 0\}.$$

By convention, $\deg_{\lambda_j} 0 := -\infty$. Consequently, every finite upper degree bound used below includes the zero Laurent polynomial.

Lemma F.6.28 (Grid interpolation for bounded Laurent degree)**.** *Let $m \geq 3$, let $c_1, \ldots, c_r \in \mathbb{C}^\times$, and let $P(\lambda_1, \ldots, \lambda_r)$ be a Laurent polynomial such that, after multiplying by a monomial, its degree in each variable is at most two. If P vanishes on the product of cosets $\prod_{j=1}^{r} c_j \mu_m$, then P is identically zero.*

Proof. After the monomial shift, substitute $\lambda_j = c_j t_j$. Fix all variables except the last one. The result is a univariate polynomial of degree at most two with at least three distinct roots $t_j \in \mu_m$, hence is zero. Induction on the number of variables proves that every coefficient polynomial vanishes. The nonzero monomial used to clear the Laurent exponents never affects a value on the stated cosets. □

F.6.5.5 Polarized coefficient integration

The remaining tensor has a fixed matching support and nonzero logical coefficients. We test their factorization by the following two-copy construction, called *polarization*. Each phase assignment

gives a realizable arity-six tensor. Its rank-one equations give polynomial identities in the phases. One set of minors makes the coefficient ratio constant within each of two layers. A second minor equates the ratios between the layers. Interpolation then gives a ratio independent of all other logical inputs, which yields the coefficient product formula.

Let $w : \mathbb{F}_2^s \to \mathbb{C}^\times$, $s \geq 3$, be the logical coefficient tensor on the global matching, with every matching edge in its fixed orientation. Fix distinct logical coordinates i, k and take two equally oriented copies. Leave the entire i-edge of each copy external. On the two k-edges, join one pair of corresponding inputs by a realizable equality link and leave the other two inputs external. After one recorded X-dressing they form the third external matching edge and carry one common logical bit ξ. For every $\ell \notin \{i, k\}$, close the two ℓ-edges by one equality link and one phase-dressed equality link. Unique monomial lifts give, after division by the nonzero representative scalar, the logical coefficient tensor of the direct six-input output

$$q^{i,k}_{ab\xi}(\boldsymbol{\lambda}) = \sum_{\mathbf{z}} \boldsymbol{\lambda}^{\mathbf{z}} w_i(a, \xi, \mathbf{z}) w_i(b, \xi, \mathbf{z}), \qquad \boldsymbol{\lambda}^{\mathbf{z}} = \prod_{\ell \notin \{i,k\}} \lambda_\ell^{z_\ell}. \tag{F.34}$$

Here $w_i(a, \xi, \mathbf{z})$ means that coordinates i, k of w are a, ξ, respectively, and the other coordinates are ordered as $\mathbf{z}$. The fixed realization scalars may replace μ_m in each variable by a nonzero coset $c_\ell \mu_m$. Lemma F.6.28 applies to these cosets. Their Cartesian product is the *polarization grid*. Every grid coordinate is nonzero. Lemma F.6.28 turns the allowed phase evaluations into identities among the original coefficients.

Lemma F.6.29 (Polarized determinant)**.** *Assume the global matching support has been fixed by Lemma F.6.25, and use Lemma F.6.27 to retain every one-coordinate rank equation across completed current-group factor batches. Form every realizable two-copy output $q^{i,k}(\boldsymbol{\lambda})$ in Equation* (F.34)*. Suppose that, after the concrete six-input case analysis has removed every violating batch, each retained two-copy output has rank at most one in each of its three one-logical-coordinate flattenings*

$$a \mid (b, \xi), \qquad b \mid (a, \xi), \qquad \xi \mid (a, b).$$

Any two-copy output violating this hypothesis is resolved before this lemma is invoked by Lemmas F.5.4, F.6.7 and F.6.26. Then

$$w(\mathbf{z}) = w(\mathbf{0}^s) \prod_{i=1}^{s} \rho_i^{z_i} \tag{F.35}$$

for nonzero scalars ρ_i. Consequently the signature is a tensor product of binary signatures on the global perfect matching. This is an exact tensor identity. The individual factors are made available only through Theorem F.2.10 and Lemma F.3.4 and are then sent through the binary case analysis.

Proof. Fix $i \neq k$ and $\xi \in \mathbb{F}_2$, and abbreviate

$$u_{\mathbf{z}} := w_i(0, \xi, \mathbf{z}), \qquad v_{\mathbf{z}} := w_i(1, \xi, \mathbf{z}).$$

Rank at most one of the $a \mid b$ face, including a zero output, gives the determinant polynomial

$$\Delta_\xi(\boldsymbol{\lambda}) := \left(\sum_{\mathbf{z}} \boldsymbol{\lambda}^{\mathbf{z}} u_{\mathbf{z}}^2 \right) \left(\sum_{\mathbf{z}} \boldsymbol{\lambda}^{\mathbf{z}} v_{\mathbf{z}}^2 \right) - \left(\sum_{\mathbf{z}} \boldsymbol{\lambda}^{\mathbf{z}} u_{\mathbf{z}} v_{\mathbf{z}} \right)^2.$$

This polynomial vanishes on the full product of phase cosets. Its coordinate degree is at most two, so Lemma F.6.28 makes it the zero polynomial. If $\mathbf{z}, \mathbf{z}'$ are adjacent in the remaining $(s-2)$-cube, the coefficient of $\boldsymbol{\lambda}^{\mathbf{z}+\mathbf{z}'}$ is

$$(u_{\mathbf{z}} v_{\mathbf{z}'} - u_{\mathbf{z}'} v_{\mathbf{z}})^2.$$

All values of w are nonzero. Hence $w_i(1,\xi,\mathbf{z})/w_i(0,\xi,\mathbf{z})$ is independent of $\mathbf{z}$ for fixed ξ.

To equate the ratios between the two ξ-layers, use an $a \mid \xi$ face of the same arity-six output. Rank at most one of $a \mid (b,\xi)$ gives the genuine face minor

$$\Gamma(\boldsymbol{\lambda}) = q^{i,k}_{000}(\boldsymbol{\lambda})q^{i,k}_{101}(\boldsymbol{\lambda}) - q^{i,k}_{001}(\boldsymbol{\lambda})q^{i,k}_{100}(\boldsymbol{\lambda}) = 0.$$

The polynomial Γ vanishes on the full product of phase cosets and has coordinate degree at most two. Hence Lemma F.6.28 makes it identically zero. At $\boldsymbol{\lambda}^{2\mathbf{z}}$, only the ordered pair $(\mathbf{z},\mathbf{z})$ contributes, and its coefficient is

$$w_i(0,0,\mathbf{z})w_i(0,1,\mathbf{z})\big(w_i(0,0,\mathbf{z})w_i(1,1,\mathbf{z}) - w_i(1,0,\mathbf{z})w_i(0,1,\mathbf{z})\big).$$

It vanishes, so the edge ratio in coordinate i is the same for $\xi = 0$ and $\xi = 1$. Combining this equality with the preceding constancy over the remaining $(s-2)$-coordinate cube shows that

$$\rho_i := \frac{w(1,\mathbf{x}_{-i})}{w(0,\mathbf{x}_{-i})}$$

is independent of every coordinate of $\mathbf{x}_{-i}$, where $\mathbf{x}_{-i}$ is the ordered tuple of all coordinates except i, and $w(b,\mathbf{x}_{-i})$ inserts b in coordinate i. Repeat for each i. Starting at $\mathbf{0}^s$ and flipping the nonzero coordinates of $\mathbf{z}$ in any order then multiplies by the corresponding constant edge ratios, giving Equation (F.35). The vanishing two-face minors make the result independent of the order of the flips. □

Together, the two minors make every logical-coordinate ratio constant on the coefficient cube. The interpolation uses the full phase grid specified above.

F.6.5.6 Composition of reductions and occurrence-level descent

The reconstruction results supply an exact factorization when all local comparisons continue. We now record how their gadget outputs and factor reductions compose, and how processing further occurrences decreases the occurrence measure.

Every contraction above uses fixed representatives. The only analytic objects are slices, pencil endpoints, inverse evaluation formulas, and coefficient comparisons. The two-root choices are legal for $m \geq 3$. Unique-lift monomial splices are cancellation-free. Phase traces and two-copy sums may cancel and are handled by their explicit zero-output analysis or by grid interpolation. Realizable four- or six-input outputs therefore have the direct-gadget reduction direction. In contrast, Lemmas F.6.5, F.6.25 and F.6.29 use analytic reconstruction only to prove tensor identities. Their proper factors are added through Theorem F.2.10 and Lemma F.3.4 with the Turing reduction supplied by the factor theorem recorded. All bounded outputs have size $O(\mathrm{arity}(f))$. Their complete concrete batches are resolved by Lemmas F.5.12, F.6.7 and F.6.26 and Definition F.5.10. In particular, no nonmatching arity-six tensor is recursively re-entered.

For relative worklist invocations, let $\mathcal{W}$ be the finite multiset of unresolved irreducible occurrences in the retained factorization trees. Distinct occurrences of the same normalized signature are counted separately. For each unresolved occurrence u, with underlying irreducible signature h_u, let A denote the current coordinate matrix and put $H := \overline{G}(A,\mathcal{F})$. Fix the occurrence's own finite card/factor universe

$$\mathscr{U}^{\mathrm{fac}}(A,h_u,H) = Q_{u,\mathrm{done}} \sqcup Q_u$$

when that occurrence first enters the worklist. Let Φ_u be its stored factor potential. Define

$$\Theta_{\mathrm{MD}}(\mathcal{W}) = \{\{(\mathrm{arity}(h_u), |Q_u|, \Phi_u) : u \in \mathcal{W}\}\},$$

using strict lexicographic order on each triple and its well-founded multiset extension, with the conventions of Section F.3.2.1.

Certifying an occurrence removes its triple. If the selected signature itself factors, its triple is replaced by triples having strictly smaller first coordinate. If processing one child-card index exposes lower-arity occurrences while the parent remains unresolved, that index is first completed or its factor potential strictly decreases. Hence the old parent triple is replaced by a smaller parent triple together with child triples of smaller arity. This also decreases the multiset measure. The bounded four- or six-input reductions create no fresh unresolved same-arity occurrence: their complete factorization trees are processed inside the originating Q_u.

Lemma F.6.30 (Relative further applications)**.** *Let h be an irreducible occurrence of even arity in the current reduction data $(\mathcal{F}, D_{2m}^{(Z)})$ in the standard $D_{2m}^{(Z)}$ coordinates, and suppose* $\mathrm{RelStable}(\mathcal{F}, D_{2m}^{(Z)}; h)$*, under the standing assumption $m \geq 3$. Every invocation of deck stability of a minimum-arity irreducible signature in the preceding active-signature argument remains valid with h as the selected occurrence and with the other local hypotheses unchanged. In particular, this applies to Lemmas F.6.8, F.6.22, F.6.25 to F.6.27 and F.6.29. Specifically, every surviving nonzero proper card is a $D_{2m}^{(Z)}$-matching signature, and every lower-arity nonbinary occurrence exposed during its processing is handled earlier in the occurrence induction. The chosen representatives, current group and fixed local queue are those of h. These are the replacements for the minimum-arity argument in the preceding active-signature proof. All its other hypotheses and reduction directions are retained.*

Proof. The sector-ruling, endpoint, support, six-input, and polarization arguments use minimum arity only to exclude a surviving proper nonmatching card or nonbinary factor. Under the present hypothesis each such object has already produced a resolved leaf or strict stable successor, or has been certified as a $D_{2m}^{(Z)}$-matching signature. All other inputs—the fixed representatives, normalized group, even arity, and tensor irreducibility of h—are unchanged. Thus every local construction and reduction used in the active-signature proof remains applicable. □

F.6.5.7 Weighted kernels and arity-four reductions

We now return to the marked coordinates of Lemma F.6.13. The following support calculation applies to both displayed marked forms. Its $D_{2m}^{(X,1)}$ case supplies the arity-four step for the weighted argument.

Lemma F.6.31 (Supports of marked quaternaries)**.** *Let $q \neq 0$ be a retained quaternary occurrence in a current $D_{2m}^{(Y)}$-state ($m \geq 3$) or $D_{2m}^{(X,1)}$-state (even $m \geq 4$). Suppose its complete oriented deck has been regenerated and every member of that deck is zero or a safe current-group binary. Let $\tau = Y$ or $(X, 1)$, respectively, and define the sector-coordinate signature*

$$q^{(\tau)} := (U_\tau^{-1})^{\otimes 4} q.$$

After realizable sector-X dressings, for some $J \subseteq \{1, 2, 3\}$, nonzero a, h, and nonzero b_ℓ, d_ℓ ($\ell \in J$),

$$q^{(\tau)} = a\delta_{\mathbf{0}^4} + h\delta_{\mathbf{1}^4} + \sum_{\ell \in J} (b_\ell \delta_{\mathbf{p}_\ell} + d_\ell \delta_{\overline{\mathbf{p}}_\ell}), \tag{F.36}$$

where $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$ represent the three complementary pairs of weight-two strings. The assertion includes all six contracted pairs, both orientations, all zero-card patterns, and the fixed representative scalars.

Proof. For an ordered input pair with residual slices (A, B, C, D_{11}), the complete card pencils, up to their nonzero representative scalars, are

	$E(t)$	$O(t)$	
$D_{2m}^{(Y)}$	$A + tD_{11}$	$tB + C$	$t \in \mu_m$.
$D_{2m}^{(X,1)}$	$A + \delta t D_{11}$	$tB + \delta C, \quad \delta = -\eta^{-1},$	

Their effective kernels are $D(t), XD(t)$ for $D_{2m}^{(Y)}$, and

$$D(\delta)D(t) = D(\delta t), \qquad D(\delta)XD(t) = \begin{pmatrix} 0 & t \\ \delta & 0 \end{pmatrix},$$

for $D_{2m}^{(X,1)}$. The diagonal kernels are symmetric. Define the off-diagonal kernel $K^{\mathrm{off}}(t) := \left(\begin{smallmatrix} 0 & t \\ \delta & 0 \end{smallmatrix}\right)$. It satisfies

$$K^{\mathrm{off}}(t)^{\mathsf{T}} = \frac{t}{\delta} K^{\mathrm{off}}(\delta^2/t), \qquad \delta^2/t \in \mu_m; \tag{F.37}$$

so reversal permutes the labels and restores a nonzero scalar. The $D_{2m}^{(Y)}$ formula is $\delta = 1$.

With β_τ fixed above, three phases imply that each two-coordinate pencil is zero, lies on a fixed line with ordered ratio in $\mathscr{R}_\tau$, or has its endpoints on the two complementary coordinate axes. Indeed, a zero makes the endpoints proportional. Otherwise the quotient $\phi(t)$ of the two independent affine coordinates is injective on μ_m and takes its values in the fixed coset $\beta_\tau \mu_m$. Hence $\psi(t) = \beta_\tau^{-1}\phi(t)$ permutes μ_m, and Lemma F.6.9 gives $\psi(t) = \lambda t$ or λ/t. Cross-multiplication leaves the opposite coordinate axes. Multiplication by β_τ^{-1} is only an analytic scaling of one endpoint coordinate. Formula Equation (F.37) covers the opposite orientation with its nonzero scalar.

Apply this three-type pencil classification to both card columns for each of the six deleted pairs. After analytic quotient normalization, the Boolean support constraints are exactly those enumerated in Theorem F.A.8(ii): every nonempty solution consists of one total parity together with a nonempty subset of its four complementary string pairs. Quotient normalization changes no zero pattern, so the same exact enumeration applies in marked coordinates. Choose a nonzero complementary pair and use realizable sector-X dressings, justified by Equation (F.26), to send it to $0000, 1111$. The other pairs of nonzero entries are then among the three complementary weight-two blocks, giving Equation (F.36). This includes both orientations and every zero-card pattern. The resulting support description is in sector coordinates. □

In Equation (F.36), an included complementary weight-two block has $b_\ell d_\ell \neq 0$. All other blocks are zero. Ratios below use the displayed order of the two nonzero binary entries.

Lemma F.6.32 (The $D_{2m}^{(X,1)}$ quaternary coset)**.** *Let $m \geq 4$ be even, let $G(\mathcal{F}) = D_{2m}^{(X,1)}$, and let $0 \neq q$ be a retained quaternary occurrence satisfying*

$$\mathrm{RelStable}(\mathcal{F}, D_{2m}^{(X,1)}; q).$$

Suppose every member of the complete oriented dressed deck of q is zero or a safe $D_{2m}^{(X,1)}$-binary. Then the quaternary analysis yields an established resolved complexity leaf or a successor with a strictly larger complete transfer group, a genuine factorization of q into two binary signatures whose complete factor batch is retained, or a realizable endpoint-nondegenerate eight-vertex signature in the current coordinates. In this last alternative the original sector-coordinate signature also satisfies

$$(U_{X,1}^{-1})^{\otimes 4} q \in \mathscr{P}.$$

Proof. Apply Lemma F.6.31 and fix the resulting realizable sector-X dressings. In sector coordinates those dressings are invertible monomial transformations, so they preserve membership in $\mathscr{P}$, reducibility, all flattening ranks, and exact factorization. We may therefore work with Equation (F.36) and undo the dressings afterward.

Recall the phase scalar δ and ratio coset $\mathscr{R}_{X,1}$:

$$\delta = -\eta^{-1}, \qquad \mathscr{R}_{X,1} = \delta^{-1}\mu_m = \eta\mu_m.$$

Since $\eta \notin \mu_m$ and m is even,

$$\mathscr{R}_{X,1} \cap \mu_m = \varnothing, \qquad \mathscr{R}_{X,1}^{-1} = \mathscr{R}_{X,1}.$$

By Lemma F.6.13, every safe sector binary has ratio in $\mathscr{R}_{X,1}$. In Equation (F.36), denote the endpoint pair by (a, h) and an active central pair by (b_i, d_i).

For one active block, the corresponding diagonal pencil is

$$\operatorname{diag}(a + \delta t d_i,\ b_i + \delta t h), \qquad t \in \mu_m.$$

The root-pencil type classification in Lemma F.6.31 excludes a sample with exactly one vanishing coordinate and forces

$$(b_i, d_i) = (a\rho_i, h/\rho_i), \qquad \rho_i \in \mathscr{R}_{X,1}, \qquad b_i d_i = ah,$$

whereas an inactive block gives the diagonal card pencil

$$\operatorname{diag}(a, \delta t h), \qquad t \in \mu_m.$$

Its ordered ratio $\delta t h/a$ belongs to $\mathscr{R}_{X,1} = \delta^{-1}\mu_m$. Since $\delta^2 = \eta^{-2} \in \mu_m$, this is equivalent to

$$h/a \in \delta^{-2}\mu_m = \mu_m.$$

If two blocks j, k are active, the crossing deleted pair and the realizable reflections

$$\begin{pmatrix} 0 & t \\ \delta & 0 \end{pmatrix}, \qquad t \in \mu_m$$

give, up to common endpoint factors,

$$ta\rho_j + \delta h/\rho_k, \qquad ta\rho_k + \delta h/\rho_j.$$

Writing

$$L(t) = ta + \frac{\delta h}{\rho_j \rho_k},$$

these entries are $\rho_j L(t)$ and $\rho_k L(t)$. Every nonzero sample therefore has ratio $\rho_j/\rho_k \in \mu_m$, because both ρ_j, ρ_k lie in the same coset $\mathscr{R}_{X,1}$. This ratio is outside the coset $\mathscr{R}_{X,1}$. The nonzero affine function $L(t)$ vanishes for at most one phase, so another allowed phase gives a nonsingular binary whose transfer class lies outside the current group. Complete binary closure by Lemma F.5.4 gives a resolved complexity leaf or a successor with a strictly larger complete transfer group, contradicting a remaining deck-stable branch.

Thus at most one block is active. If exactly one block i is active, then $ah - b_i d_i = 0$, giving a genuine $2 + 2$ factorization exposed by Theorem F.2.10 and Lemma F.3.4. Its two binaries are

then considered individually: a safe binary B with $[BX] \notin G$ gives a successor with a strictly larger complete transfer group, and unsafe binary factors reach their established complexity conclusions.

With no active block, let $\lambda = h/a \in \mu_m$. Before any further rotation, the sector-coordinate tensor is already

$$a\delta_{0000} + h\delta_{1111},$$

a generalized equality and hence a member of $\mathscr{P}$. Undoing the fixed sector-X dressings preserves $\mathscr{P}$, proving

$$(U_{X,1}^{-1})^{\otimes 4} q \in \mathscr{P}.$$

Moreover $\eta^2/\lambda \in \mu_m$, so it is the product of four available phase parameters. Realizable rotations with this product of input phases give the sector tensor

$$a(\delta_{0000} + \eta^2 \delta_{1111}).$$

Since $s_0^4 = \eta^{-2}$,

$$\begin{aligned} U_{X,1}^{\otimes 4} a(\delta_{0000} + \eta^2 \delta_{1111}) &= a((1,1)^{\otimes 4} + (1,-1)^{\otimes 4}) \\ &= 2a\, \mathbf{1}[\mathrm{wt}(\mathbf{x}) \text{ is even}]. \end{aligned}$$

This is a current-coordinate endpoint-nondegenerate eight-vertex signature. The rotations are realizable by Equation (F.26). $U_{X,1}$ only verifies the identity. □

To apply Lemma F.6.32 to the selected quaternary occurrence or a quaternary output of a higher-arity construction, first add it with its input order, scalar, and reduction, jointly add all its irreducible factors, consider the complete transfer group of the resulting set, and regenerate its complete oriented one-copy deck. Test all three $2 + 2$ flattenings. Every rank-one flattening is retained as an exact factor identity. Unary factors and unsafe binaries reach their established reductions, and a safe binary whose transfer class lies outside $D_{2m}^{(X,1)}$ gives a successor with a strictly larger complete transfer group. If all exposed binary factors are safe current-group binaries, the child quaternary is a $D_{2m}^{(X,1)}$-matching signature and its originating fixed token is completed. If the quaternary itself is the selected irreducible occurrence, such a factorization contradicts its irreducibility.

On the remaining branch the quaternary is irreducible, every member of its complete deck is zero or a safe binary whose transfer class lies in $D_{2m}^{(X,1)}$, and the closure just performed gives

$$\mathrm{RelStable}(\mathcal{F}, D_{2m}^{(X,1)}; q)$$

in the sense of Definition F.6.18. For a later occurrence, processing of its fixed local queue and prior processing of every lower-arity occurrence give this certificate directly. Apply Lemma F.6.32, which starts from the support in sector coordinates given by Lemma F.6.31 and constructs the required eight-vertex signature in current coordinates.

In the no-active-block branch the output signature is $2a\, \mathbf{1}[\mathrm{wt}(\mathbf{x}) \text{ is even}]$, with $a \neq 0$. It is irreducible: the GE signature used to construct it has rank two across every nontrivial partition, and the invertible transformations at individual inputs preserve these ranks. The output is realizable, has nonzero endpoints, and is an eight-vertex signature. Apply the already proved Theorem F.1.2 to the signature set. This gives a hard or tractable case satisfying $\mathsf{ReTract}(K\mathcal{F})$ for the entire current signature set, with the reduction supplied by realization. The procedure is the *marked-coordinate reduction.*

F.6.5.8 Gauges and product weights

An *input-wise gauge* is an invertible diagonal rescaling on specified tensor inputs. On a binary flattening it gives invertible diagonal row or column scaling. Its use as a gadget requires a separate realization. A genuine character of $(\mathbb{F}_2^r, +)$ has the form $\mathbf{z} \mapsto \prod_j \epsilon_j^{z_j}$ with $\epsilon_j \in \{1, -1\}$. More generally,

$$\mathbf{z} \longmapsto \prod_j \gamma_j^{z_j}, \qquad \gamma_j \in \mathbb{C}^\times,$$

is a *product weight*. It is a character exactly when every $\gamma_j^2 = 1$.

For the remainder of the $D_{2m}^{(X,1)}$-sector, fix

$$\delta := -\eta^{-1},$$

with the input-wise gauge defined above. In particular, $(D(\delta)^{(i)}h)(\mathbf{x}) = \delta^{x_i} h(\mathbf{x})$.

To compare coefficients across several outputs, we need the kernel weights to act on the same parent inputs. Call a finite family of marked-sector contraction outputs a *gauge-coherent comparison batch* if its members use the same ordered contraction multigraph, the same orientations on all contracted edges, and the same external-input order, and differ only in the chosen phase or product at designated kernels. For a set $\mathscr{S}$ of parent inputs write

$$D(\delta)^{(\mathscr{S})} := \prod_{i \in \mathscr{S}} D(\delta)^{(i)}, \qquad (D(\delta)^{(\mathscr{S})}h)(\mathbf{x}) = \delta^{\sum_{i \in \mathscr{S}} x_i} h(\mathbf{x}).$$

The embedded single-input operators commute. The set $\mathscr{S}$, and hence the gauged parent $D(\delta)^{(\mathscr{S})}h$, is fixed throughout one comparison batch. Different ordered contraction graphs may yield different sets.

Lemma F.6.33 (Explicit weighted-gauge transport)**.** *Let $m \geq 4$ be even and use the fixed $\delta = -\eta^{-1}$, where $\eta \in \mu_{2m} \setminus \mu_m$, and the diagonal gauge matrix $D(\delta)$ above. In the analytic $D_{2m}^{(X,1)}$-sector the effective diagonal and reflection kernels are*

$$K^0(t) = D(\delta)D(t) = \mathrm{diag}(1, \delta t), \qquad K^1(t) = D(\delta)XD(t) = \begin{pmatrix} 0 & t \\ \delta & 0 \end{pmatrix}, \qquad t \in \mu_m.$$

For every binary kernel C and every two-input slice g on ordered inputs (i, j), pointwise in all remaining variables, every occurrence of $D(\delta)$ can be moved from the kernel to the corresponding endpoint of the parent tensor:

$$\langle D(\delta)C, g\rangle_{\mathrm{Frob}} = \langle C, D(\delta)^{(i)}g\rangle_{\mathrm{Frob}}, \qquad \langle CD(\delta), g\rangle_{\mathrm{Frob}} = \langle C, D(\delta)^{(j)}g\rangle_{\mathrm{Frob}}, \tag{F.38}$$

where i and j are the first and second ordered inputs and $D(\delta)^{(i)}$ means multiplication by δ^{x_i}. Consequently for every gauge-coherent comparison batch there is one fixed set $\mathscr{S}$, independent of the phase and product choices, such that the entire batch is exactly the corresponding standard $D_{2m}^{(Z)}$ batch on the one locally gauged parent $D(\delta)^{(\mathscr{S})}h$, followed by invertible row/column gauges and the substitutions

$$\widehat{t} = c\, t^{\varepsilon}, \qquad c \in \{1, \delta, \delta^{-1}\}, \quad \varepsilon \in \{1, -1\}. \tag{F.39}$$

In particular:

1. *support, zero or incomplete-support status, matching translation spaces, all flattening ranks, and exact factorization are unchanged by the gauges;*

2. *whenever both product coefficients on a common edge are nonzero and their trace is not identically zero, its trace has the form $\kappa(1+\theta t^{\varepsilon})$ with $\kappa\theta \neq 0$ and $\varepsilon \in \{-1,1\}$. A one-term trace has no forbidden phase, while an identically-zero product is a case of the zero-output or incomplete-support analysis, according to the other product. Whenever the corresponding selected minor is defined, its transported value is a nonzero monomial in δ, t times the standard selected determinant;*

3. *reversal sends t to $\delta^2/t \in \mu_m$, so both orientations use the same finite phase set, and the ordered ratios of all safe sector binaries lie in the single coset $\delta^{-1}\mu_m$.*

4. *the four samples $E(t_0), E(t_1), O(t_0), O(t_1)$ at one fixed ordered deleted pair, the two products of one fixed trace-cover graph, the phase family producing a quaternary from an arity-six tensor, and the two-copy outputs on the complete polarization grid are gauge-coherent. Hence every joint multi-card comparison in the transported proof is made on one common locally gauged parent.*

Proof. The two displayed kernel identities are direct multiplication. For the first identity in Equation (F.38), expand the Frobenius pairing: $\sum_{a,b} \delta^a K(a,b) g(a,b)$. The second is identical with the factor δ^b. Transposing or reversing an ordered kernel merely switches i and j and contributes the nonzero scalar in Equation (F.37). Induction over the contraction edges therefore pushes all $D(\delta)$'s to vertex inputs. For a fixed ordered contraction graph, the endpoint at which each $D(\delta)$ lands is determined only by the ordered edge, not by its phase or by the choice of diagonal versus reflection product. Hence every member of a gauge-coherent batch has the same parent-input set $\mathscr{S}$. Reversed cards are first put in the common orientation using Equation (F.37), which only changes the scalar and replaces the phase by δ^2/t. On a matching product the remaining external gauges are row and column gauges of its binary factors, so they preserve support, nonsingularity, ranks, and exact tensor-factor identities. Unrelated ordered contraction graphs may induce different sets $\mathscr{S}$, but their raw coefficients are never compared: support assertions are gauge-invariant, and coefficient identities pull back with their explicit nonzero product-weight factors.

Pushing the entire gauge to the parent leaves the phase unchanged, so $c = 1$. Alternatively, keeping the raw parent uses $K^0(t) = D(\delta t)$ and $K^1(t) = \delta X D(t/\delta)$, giving $c = \delta$ and $c = \delta^{-1}$, respectively. These are the substitutions in Equation (F.39). A reversed reflection has δ^2/t, and $\delta^2 = \eta^{-2} \in \mu_m$. A trace of two monomial factors is a sum of two nonzero monomials. If the two terms have the same exponent and cancel, that product is identically zero (the full card may consequently be zero or partially supported). Otherwise, after extracting one term it is $\kappa(1+\theta t)$ with $\kappa\theta \neq 0$ (or a one-term trace). A unique-lift fusion multiplies a selected minor by the nonzero entry and orientation scalars. A row/column gauge multiplies a determinant by the product of its diagonal entries. This proves the second item and also shows that a zero or partially supported card is zero or partially supported before and after transport. Finally, $\delta^{-1}\mu_m = \eta\mu_m$ is disjoint from μ_m and is invariant under inversion because $\eta^{-2} \in \mu_m$. This proves the ratio and orientation assertions. □

The six-input argument with weighted kernels. The weighted kernels also preserve the support argument of Lemma F.6.11. We record this observation to explain the support part of the transport below. The proof of Theorem F.6.17 uses the full reconstruction in Lemma F.6.34. Work analytically with $h^{(X,1)} = (U_{X,1}^{-1})^{\otimes 6} h$. The exact sector-ruling argument uses only monomial supports and Segre factorization. In its individual binary pencils, the ratios of safe sector binaries lie in $\delta^{-1}\mu_m$. Dividing that one ratio by δ^{-1} gives the same $\mathsf{Z}/\mathsf{P}/\mathsf{C}$ support classification. This is a separate analytic calculation for each pencil, not a common input gauge or an available binary. In

the proof of Lemma F.6.11, replace the parity kernels I, X by the realizable effective kernels

$$D(\delta)D(1) = \begin{pmatrix} 1 & 0 \\ 0 & \delta \end{pmatrix}, \qquad D(\delta)XD(1) = \begin{pmatrix} 0 & 1 \\ \delta & 0 \end{pmatrix}.$$

They have the same respective supports and nonzero entries. Every cross-pair lift in the P/P and P/C arguments and every injective lift in the support classification of Proposition F.A.9 therefore acquires only a nonzero weight. All support counts and projection obstructions are unchanged. A matching coset still has an identically zero opposite-parity sector. The resulting exact factorization of $h^{(X,1)}$ pulls back by the tensor powers of $U_{X,1}$ to an exact factorization of the signature h in the original basis, with its nonzero scalar.

Lemma F.6.34 (Transport of contractions and reconstruction in the $D_{2m}^{(X,1)}$ sector)**.** *Let*

$$\mathsf{Stable}(\mathcal{F}, D_{2m}^{(X,1)}, f_0), \qquad \mathsf{Even}(\mathcal{F}), \qquad m \geq 4 \text{ even}.$$

Use the phase group μ_m and recall the fixed scalar, gauge matrix, and ratio coset

$$\delta = -\eta^{-1}, \qquad D(\delta) = \operatorname{diag}(1, \delta), \qquad \mathscr{R}_{X,1} = \delta^{-1}\mu_m.$$

Let $h \in \Pi(\mathcal{F})$ be a selected unresolved nonbinary irreducible signature occurrence of even arity at least six. Regenerate and close its complete realizable oriented deck over the current signature set. Suppose that this occurrence is relatively deck-stable and has produced neither a resolved complexity leaf nor a strict stable successor. In particular, all lower-arity irreducible occurrences exposed by its proper cards or factors have already been processed, and every surviving binary factor has its transfer class in the current complete group. Then every concrete local comparison batch in the standard $D_{2m}^{(Z)}$ proof has a gauge-coherent counterpart in the original basis. Its required invariants are the following:

1. *The realizable kernels are monomial, their traces remain affine in $t \in \mu_m$, reversal permutes μ_m, and the ordered ratio of every safe sector binary lies in the fixed coset $\mathscr{R}_{X,1}$. Analytically dividing that ratio by δ^{-1} gives the pencil normal form. Hence all support, ruling, trace-cover, coefficient-minor, endpoint-type, and bounded-reduction arguments transport with their zero cases. Unique-lift monomial fusions remain cancellation-free.*

2. *For a logical coefficient tensor $c = (c_{\mathbf{z}})_{\mathbf{z}\in\mathbb{F}_2^d}$ on an oriented matching and orientation signs $\epsilon_j \in \{1, -1\}$, the product-weight change $\widetilde{c}_{\mathbf{z}} = \delta^{\sum_j \epsilon_j z_j} c_{\mathbf{z}}$ sends the logical ratios of safe binary factors from $\mathscr{R}_{X,1}$ to μ_m while preserving Equation* (F.33) *and the cross ratio r. The final quaternary signature is handled by the marked-coordinate reduction.*

3. *Fixed δ-weights do not alter support integration. They multiply the Reed–Muller obstruction by a nonzero scalar. In the polarized determinant, they merely reparameterize the full grid and multiply each tested coefficient by a nonzero product-weight value.*

Consequently every branch either

1. *reaches a resolved complexity leaf or a strict stable successor;*

2. *returns an exact proper-factor or child-card factor batch to the lower-arity occurrence induction; or*

3. *certifies*

$$(U_{X,1}^{-1})^{\otimes \operatorname{arity}(h)} h \in \mathscr{P}.$$

When $h = f_0$, the first two alternatives are resolved in the sense of Definition F.5.13. For a later occurrence they are charged to the occurrence-level induction described in Section F.6.5.6. No decrease of the active tuple's Ξ_{full} is asserted. The bounded outputs are realizable quaternary or arity-six tensors in the current coordinates. Factor alternatives are exact tensor identities pulled back to those coordinates and exposed through Theorem F.2.10 and Lemma F.3.4. Their Turing reductions are retained in the worklist. The conclusion concerns the selected occurrence h.

Proof. By Lemma F.6.13, a realizable sector binary is a nonzero multiple of $TD(\delta)^{-1}$, and its effective kernel is $D(\delta)T$. The gauge identity Lemma F.6.33 moves each kernel weight to a specified parent input. It gives the effective kernels

$$D(\delta)D(t) = \mathrm{diag}(1, \delta t), \qquad D(\delta)XD(t) = \begin{pmatrix} 0 & t \\ \delta & 0 \end{pmatrix},$$

so the two sector pencils are $A + \delta t D_{11}$ and $tB + \delta C$. For a nonproportional pencil, multiply the ratio of a safe sector binary by δ and use the coset-normalized pencil argument. The resulting ratio is in μ_m, so Lemmas F.6.9 and F.6.10 applies. This is an analytic rescaling only. Reversal sends t to $\delta^2/t \in \mu_m$, and the gauge lemma shows that a common-edge trace has at most one forbidden phase in each product. Hence the two-product trace cover still has an admissible phase because $m \geq 4$, while every zero or partially supported card is preserved exactly by the invertible gauges. By item 4 of Lemma F.6.33, each comparison of samples, products, or a full phase grid uses one common locally gauged parent. Its determinant identity gives, with an explicitly nonzero scalar κ,

$$\det_{X,1}(t) = \kappa\, \delta^e t^u \det_{Z}(c\, t^{\varepsilon}), \qquad e, u \in \mathbb{Z}, \quad c \neq 0, \quad \varepsilon \in \{-1, 1\},$$

where $\det_Z$ denotes the same selected minor for the locally gauged tensor in the standard sector, so selected discrepancies, the six-to-four Laurent coefficient, and the endpoint minor are nonzero exactly in the corresponding standard calculation. Therefore Lemmas F.6.4 to F.6.8 and F.6.19 to F.6.22 transport with the same realizable closures. Every bounded quaternary output is handled by the marked-coordinate reduction, and every exact factor identity is pulled back by the corresponding tensor power of $U_{X,1}$ and then handled with a recorded Turing reduction. For the active minimum occurrence these transported concrete batches are resolved. For a later occurrence, the role of minimum arity is replaced by the closed proper batches and the earlier lower-arity occurrences, exactly as in the hypothesis check of Lemma F.6.30. The changes of kernel are justified separately by Lemma F.6.33. Each batch is completely closed inside that occurrence's fixed local queue and charged to the outer occurrence induction. On the branch surviving the corresponding closure, the standard support, ruling, and coefficient invariants remain available.

Use the product-weight convention of Section F.6.5.8. For this logical coefficient cube put

$$\chi(\mathbf{z}) := \prod_j \gamma_j^{z_j} \qquad (\gamma_j \in \mathbb{C}^{\times}).$$

Multiplication by χ preserves supports and all two-face parallelogram identities. It also preserves every complementary-pair product up to one common nonzero scalar. For oriented matched pairs, the logical ratio of a safe binary factor is $\delta^{-\epsilon_j} h_j$, $h_j \in \mu_m$, $\epsilon_j \in \{1, -1\}$. The product-weight change

$$\widetilde{c}_{\mathbf{z}} = \delta^{\sum_j \epsilon_j z_j} c_{\mathbf{z}}$$

puts all ratios in μ_m, multiplies both sides of every equality in Equation (F.33) equally, and leaves

$$r = \frac{c_{000} c_{110}}{c_{100} c_{010}}$$

unchanged. Here the product-weight change is analytic. Since $\delta^2 \in \mu_m$, absorb orientation changes into the phases and write the ratios of the three initial safe binary factors as $\delta^{-1}h_j$, $h_j \in \mu_m$. The crossing factor cards used in the standard six-input calculation expose these raw ratios: the extra δt multiplies a column of the card and does not change the ratio on its unclosed matching edge. Realizable rotations $D(h_j^{-1})$ therefore leave the signature in the current coordinates

$$c_{\mathbf{z}} = \delta^{-\operatorname{wt}(\mathbf{z})} N_r(\mathbf{z}), \qquad N_r(\mathbf{z}) = \begin{cases} 1, & \operatorname{wt}(\mathbf{z}) \le 1, \\ r, & \operatorname{wt}(\mathbf{z}) \ge 2, \end{cases}$$

after its nonzero realization scalar is recorded. To check $r \in \mu_m$ without realizing the analytic weight change, use the standard $X_{12}X_{34}X_{56}$ support and close inputs $(1,3)$ by $K^1(t)$. The resulting quaternary signature, with logical row x_2 and column x_5, has coefficient matrix

$$\begin{pmatrix} 1 & r/\delta \\ t/\delta & tr/\delta^2 \end{pmatrix} = \begin{pmatrix} 1 \\ t/\delta \end{pmatrix} (1, r/\delta).$$

Expose its exact binary factors through the factor theorem. Unless a binary exit occurs, $r/\delta \in \delta^{-1}\mu_m$, proving $r \in \mu_m$. Closing the third matching pair of the signature in the current coordinates by $K^1(t)$ gives

$$Q(t) = \operatorname{diag}(1, \delta^{-1}) \begin{pmatrix} 1+t & r+t \\ r+t & r(1+t) \end{pmatrix} \operatorname{diag}(1, \delta^{-1}), \qquad \det Q(t) = \delta^{-2}(r-1)(t^2 - r).$$

If $r = 1$, the displayed arity-six signature already factors. Otherwise at most two phases satisfy $t^2 = r$. Choose another phase in μ_m. The resulting nonzero quaternary signature enters the marked-coordinate reduction with the displayed nonzero minor.

Support integration is Boolean and unchanged. On its sole Reed–Muller residue the raw face, in one orientation, is

$$\begin{pmatrix} A & \delta t B \\ \delta t C & D \end{pmatrix}, \qquad A, B, C, D \neq 0.$$

The realizable $t = 1$ matching card gives $AD = \delta^2 BC$, so its determinant is $\delta^2 BC(1 - t^2)$. The complementary exponent pattern gives the same obstruction up to sign and a nonzero scalar. It is nonzero for some $t \in \mu_m \setminus \{\pm 1\}$. In polarization, each grid variable is merely reparameterized as $\lambda_j = c_j t_j$, $c_j \in \mathbb{C}^\times$, and the remaining weights form a nonzero product weight χ. The grid is a product of cosets $c_j \mu_m$, so Lemma F.6.28 applies without requiring $c_j \in \mu_m$. Hence the adjacent coefficient becomes

$$[\lambda^{\mathbf{z}+\mathbf{z}'}]\Delta_\xi = \chi(\mathbf{z})\chi(\mathbf{z}')(u_{\mathbf{z}} v_{\mathbf{z}'} - u_{\mathbf{z}'} v_{\mathbf{z}})^2 \qquad \text{for adjacent } \mathbf{z}, \mathbf{z}'.$$

The cross-slice coefficient is similarly multiplied by $\chi(\mathbf{z})^2$. Thus the full grid and all vanishing tests in Lemmas F.6.25, F.6.27 and F.6.29 remain valid.

The transported support integration therefore gives one global matching support for the selected occurrence, and the transported polarization identities integrate its nonzero coefficients into a product of binary factors on that matching. This is precisely $(U_{X,1}^{-1})^{\otimes \operatorname{arity}(h)} h \in \mathscr{P}$. If an intermediate identity factors the selected signature h itself, replace its occurrence by the irreducible occurrences of its proper factors. If instead only a constructed child card factors, process that child's complete factorization tree inside the selected occurrence's fixed local queue. The token for h remains unresolved until its analysis resumes. Together with the product certificate above, this proves the occurrence-level alternatives. □

F.6.5.9 Completion of the weighted dihedral case

Proof of Theorem F.6.17. Fix $U_{X,1}$ from Lemma F.6.13 and let $n = \mathrm{arity}(f)$ be the active minimum arity. If $n = 4$, the marked-coordinate reduction, using Lemma F.6.32 after its prescribed factor and deck batches, gives a resolved complexity leaf for the whole signature set or an established strict successor. For $n = 6$, apply Lemma F.6.34 to the active minimum occurrence. It is relatively deck-stable by Definition F.6.18. On a branch without a resolved leaf or strict successor, its lower-arity batches have already been processed. The weighted support and polarization calculation in that lemma gives a proper factorization, contradicting irreducibility.

Suppose $n \geq 8$, and work analytically with $f^{(X,1)} = (U_{X,1}^{-1})^{\otimes n} f$. The two-card localization proof transports directly to the realizable weighted kernels: common-edge traces exclude at most two phases. Joining two parity discrepancies gives a fused weight ratio $\kappa t^{\pm 2}$, $\kappa \neq 0$. Cycle shortening or vertex replacement has the same unique-lift matching support. These are the explicit kernel and fusion identities in Lemmas F.6.6 and F.6.33. As $m \geq 4$, each required allowed phase exists. A disagreeing pair of sampled cards would therefore localize to a nonmatching arity-six tensor. But the first proper gadget contraction of the distinguished irreducible signature is zero or a product of safe current-group binaries, and subsequent contractions preserve that property by the group fusion law. The inherited nonzero disagreeing cards give a contradiction.

Thus at a fixed deleted pair all nonzero samples have one matching and one common varying edge. The common-factor consequence fixes the other factor classes across both sectors. Its hypothesis concerns projective factor classes on the common matching and is unchanged by the coherent invertible gauges. For each weighted pencil $A + \delta t D_{11}$ and $tB + \delta C$, evaluation at two distinct nonzero phases gives an invertible linear system for its two slices, since $\delta \neq 0$. The four recovered slices have one common nonzero binary product factor, proving an exact proper factorization of $f^{(X,1)}$. Pulling it back by the invertible matrices at individual inputs $U_{X,1}$ gives a proper factorization of the signature f in the current coordinates, again impossible.

No higher-arity deck-stable signature branch survives, and the quaternary route already resolves the entire retained set. This proves $\mathsf{Close}_{\{\mathscr{P}\}}(\mathcal{F}, D_{2m}^{(X,1)}, f)$. □

F.6.6 Completion of the marked-dihedral classification

Proof of Theorem F.6.1. If $G = D_{2m}^{(Z)}$, then deck stability and $\mathsf{Even}(\mathcal{F})$ give $\mathrm{arity}(f) = \nu(\mathcal{F}) = 2k+2$ for some k, so apply Theorem F.6.12. If $G = D_{2m}^{(Y)}$, apply Theorem F.6.15. If $G = D_{2m}^{(X,0)}$, apply Theorem F.6.16. If $G = D_{2m}^{(X,1)}$, apply Theorem F.6.17. These are exactly the four cases in the statement. □

F.7 Cyclic Matching Decks and Reed–Muller Rigidity

In this case the transfer span is $\mathscr{K}_G = \mathrm{span}\{I, X\}$. In Walsh coordinates, the corresponding contractions impose equality of two bits with a possible phase on their common value. In the remaining higher-arity case, the matching cards determine the Reed–Muller signature RM_8 up to the equivalences in Theorem F.7.1. If the core remains in this case under every realizable single-input dressing, Lemma F.7.14 forces the current complete group to be $C_2^{(X)}$. This excludes every larger cyclic group in the surviving case. After the binary and arity-four cases have been resolved, the reduction in Remark F.7.13 can adjoin Bell signatures and continue the complexity classification with a strictly larger group. For the synthesis in Section F.10, we prove the stronger conclusion that the entire retained set and the edge admit one common flat-Lagrangian presentation in the surviving cyclic branch.

To obtain this presentation, we use RM_8 to test the other retained signatures. Networks built from the core realize flat-Lagrangian signatures. They also realize Radon contractions of a signature g. On a chosen hyperplane, the new signature takes the value $g(\mathbf{x}) + g(\mathbf{x}+\mathbf{v})$ on each fibre $\{\mathbf{x}, \mathbf{x}+\mathbf{v}\}$. It has two fewer inputs, allowing a failure of affine structure to be localized at lower arity. An induction, followed by affine patching and recognition, yields one common flat-Lagrangian presentation for the whole set. Appendix F.D extends the network results to all four Bell signatures.

Use the lexicographic indexing of the eight inputs by $\mathbb{F}_2^3$ fixed in Section F.6.5.2. Define the eight-ary Reed–Muller indicator signature $\mathrm{RM}_8 : \mathbb{F}_2^8 \to \{0,1\}$ by

$$\mathrm{RM}_8(\mathbf{x}) := \mathbf{1}[\mathbf{x} \in RM(1,3)]. \tag{F.40}$$

For $S \subseteq \mathbb{F}_2^r$, write $\mathbf{1}[S]$ for the indicator signature $\mathbf{x} \mapsto \mathbf{1}[\mathbf{x} \in S]$.

Theorem F.7.1 (Reed–Muller deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, G, f), \qquad \mathrm{arity}(f) = \nu(\mathcal{F}), \qquad \mathscr{K}_G = \mathrm{span}\{I, X\}.$$

Then $\mathsf{Close}_{\{\mathsf{FLag}\}}(\mathcal{F}, G, f)$*. In the remaining tractable case with a common flat-Lagrangian presentation,*

$$G = \{[I], [X]\}, \qquad f \sim \mathrm{RM}_8 = \mathbf{1}[RM(1,3)],$$

where $\sim$ *means equality up to a nonzero scalar, a permutation of inputs, and realizable single-input* X*-dressings.*

For $s \in \overline{\mathbb{Q}}$, the Walsh coordinates satisfy

$$P^{-1}\left[\frac{1}{2}\begin{pmatrix} 1+s & 1-s \\ 1-s & 1+s \end{pmatrix}\right] P = D(s), \qquad P^{\mathsf{T}} X P = 2Z$$

F.7.1 Cyclic phase pencils and binary attachments

The cyclic group supplies a finite family of phase contractions. The next three lemmas describe which binary pencils they permit and which changes to a quaternary signature their realizable input dressings can realize.

A *sampled phase pencil* evaluates a pencil $U + tV$ on a finite root-of-unity group, here μ_N, retaining zero members and the projective classes of nonzero members.

For each integer $N \geq 3$, use the diagonal phase matrices $D(s)$, $s \in \mu_N$, defined in Section F.2.3. Realizable networks use X-edges. Equation (F.24) gives their analytic matrix $P^{\mathsf{T}} X P = 2Z$ in the tables below.

For $N \geq 3$ and $a, b, c, d \in \overline{\mathbb{Q}}$, define the condition $\mathcal{R}_N(a,b,c,d)$ by declaring that, for some $\alpha \in \mu_N$, at least one of the following systems holds:

$$c = -\alpha a, \qquad d = -\alpha b, \qquad \text{(proportional)}, \tag{F.41}$$

$$b = c = 0, \qquad d = \alpha a, \qquad \text{(diagonal monomial)}, \tag{F.42}$$

$$a = d = 0, \qquad c = \alpha b, \qquad \text{(anti-diagonal monomial)}. \tag{F.43}$$

The alternatives may overlap and include the all-zero quadruple.

For a quaternary signature q, write $\widetilde{q}$ for its Walsh coordinates from Equation (F.24). When it is supported on even Hamming weight, define its endpoint and central Walsh coordinates by

$$p = \widetilde{q}_{0000}, \qquad t = \widetilde{q}_{1111}, \qquad x_{ab} = \widetilde{q}_{\mathbf{1}_{\{a,b\}}}, \qquad 1 \leq a < b \leq 4. \tag{F.44}$$

Here $\mathbf{1}_{\{a,b\}} \in \mathbb{F}_2^4$ has ones precisely at inputs a, b, so x_{ab} indexes one of the six weight-two strings.

Lemma F.7.2 (Cyclic phase-pencil rigidity, including zeros)**.** *For every integer $N \geq 3$, the condition $\mathcal{R}_N(a,b,c,d)$ is equivalent to the following condition: for every $s \in \mu_N$,*

$$\operatorname{diag}(a - sb, c - sd)Z \quad \textit{is zero or is projectively } D(\eta) \textit{ for some } \eta \in \mu_N.$$

Proof. If $a = b = c = d = 0$, both conditions hold, so assume otherwise. Set

$$u(s) = -c + sd, \qquad v(s) = a - sb.$$

The phase-card condition says that $u(s)$ and $v(s)$ are either both zero or both nonzero, and in the latter case $u(s)/v(s) \in \mu_N$. If the two affine linear forms are dependent, the possibility that exactly one is identically zero is excluded by this condition. Hence $u = \alpha v$ for some $\alpha \in \mu_N$, giving Equation (F.41). If they are independent, neither form vanishes on μ_N, and $\psi(s) := u(s)/v(s)$ is a nonconstant Möbius transformation. The card condition gives $\psi(\mu_N) \subseteq \mu_N$. Injectivity and equality of the finite cardinalities give $\psi(\mu_N) = \mu_N$. Thus Lemma F.6.9 applies and gives $\psi(s) = \alpha s$ or α/s, with $\alpha \in \mu_N$. Cross multiplication yields Equations (F.42) and (F.43). Direct substitution proves the converse, including the all-zero case. □

For the two cyclic lemmas below, fix a finite algebraic retained set $\mathcal{F}$ in K-coordinates, with edge signature X, and fix an integer $N \geq 3$. For every $s \in \mu_N$, fix a realizable ordered binary gadget C_s over $\mathcal{F}$, with a specified orientation, whose analytic Walsh matrix is

$$\widetilde{C}_s = \kappa_s D(s) Z, \qquad \kappa_s \in \overline{\mathbb{Q}}^{\times}.$$

The nonzero scalar κ_s is recorded with the gadget. Throughout these lemmas, "realizable" and "directly realizable" are relative to this same $\mathcal{F}$, and every displayed phase-card matrix is divided by its nonzero scalar. In the cyclic case, $P^{-1}G(\mathcal{F})P = C_N^{\mathrm{std}} = \{[D(s)] : s \in \mu_N\}$. The Walsh transfer class of C_s is $[\widetilde{C}_s Z] = [D(s)]$, so these binaries represent all classes in that form.

Lemma F.7.3 (Cyclic quaternary phase family)**.** *Let q be a realizable quaternary signature over $\mathcal{F}$, and suppose $\widetilde{q} = (P^{-1})^{\otimes 4} q$ has even-parity support. Define p, t, x_{ab} by Equation* (F.44)*. Assume that, for every complementary pair, both orders of its two residual inputs, and every $s \in \mu_N$, contraction with the fixed chosen representative C_s gives either the zero card or a nonsingular binary card whose Walsh transfer class belongs to $C_N^{\mathrm{std}} = \{[D(\eta)] : \eta \in \mu_N\}$. Equivalently, define*

$$\mathfrak{I} = \{\{1,2\},\{1,3\},\{1,4\}\}, \qquad I^c = [4] \setminus I,$$

and write $x_I := x_{ab}$ for $I = \{a,b\}$. For every $I \in \mathfrak{I}$, $s \in \mu_N$, and $\varepsilon \in \{+,-\}$, the corresponding normalized realizable Walsh card has matrix

$$B_{I,+}(s) = \operatorname{diag}(p - s x_I, x_{I^c} - st),$$
$$B_{I,-}(s) = \operatorname{diag}(p - s x_{I^c}, x_I - st).$$

Here $+$ records deletion of I and $-$ deletion of I^c. Both orders of the remaining two inputs are part of the assumption. Each displayed card is zero or satisfies

$$[B_{I,\varepsilon}(s) Z] \in C_N^{\mathrm{std}}.$$

The right factor Z comes from the Walsh transfer convention. Organize the eight coordinates into the endpoint block $\mathbf{v}_0 = (p,t)$ and the three central complement blocks $\mathbf{v}_I = (x_I, x_{I^c})$, $I \in \mathfrak{I}$. Then exactly one of the following holds:

1. If $p, t \neq 0$, then $\lambda = t/p \in \mu_N$, and each central block $\mathbf{v}_I$ is zero or

$$\mathbf{v}_I = (-p\lambda\alpha_I^{-1}, -p\alpha_I), \qquad \alpha_I \in \mu_N.$$

2. If $p = t = 0$, each central block $\mathbf{v}_I$ is zero or

$$\mathbf{v}_I = \rho_I(1, \alpha_I), \qquad \rho_I \in \overline{\mathbb{Q}}^\times, \quad \alpha_I \in \mu_N.$$

In particular, exactly one of p, t cannot be nonzero. Conversely, every even Walsh coefficient tensor in these alternatives satisfies the stated phase-card condition under every phase contraction.

Proof. For a complementary block I, I^c, the phase card is

$$\operatorname{diag}(p - sx_I, x_{I^c} - st), \qquad s \in \mu_N.$$

Apply Lemma F.7.2 to $(a, b, c, d) = (p, x_I, x_{I^c}, t)$, and then interchange I, I^c. If $p, t \neq 0$, the anti-diagonal system is impossible. If one of the two complementary-deletion systems is diagonal, then $x_I = x_{I^c} = 0$ and $t/p \in \mu_N$. The other system is then diagonal as well. In the remaining case both systems are proportional. Taking their parameters to be $\alpha, \beta \in \mu_N$ gives

$$x_{I^c} = -\alpha p, \qquad x_I = -\beta p, \qquad t = \alpha\beta p.$$

Thus $t/p = \alpha\beta \in \mu_N$, and choosing $\alpha_I = \alpha$ gives the displayed parametrization. If exactly one endpoint is nonzero, neither monomial system is possible and the two proportional systems force respectively $x_I = 0, x_{I^c} \neq 0$ and $x_{I^c} = 0, x_I \neq 0$, a contradiction. If $p = t = 0$, proportional and diagonal systems give only the zero block. Every nonzero block is therefore anti-diagonal, $\mathbf{v}_I = \rho_I(1, \alpha_I)$. Direct substitution checks both complementary deletions and every card arising from a zero complementary block, proving the converse. □

Lemma F.7.4 (Realizable quaternary cyclic dressing)**.** *Let q be a realizable quaternary signature over $\mathcal{F}$. For $r_1, r_2, r_3, r_4 \in \mu_N$, attach the fixed representative C_{r_i}, in its specified orientation, to input i of q through one X-edge, leaving the other input of C_{r_i} exposed. Denote the normalized output signature by h. After division of the direct-gadget boundary tensor by the nonzero scalar $2^4 \prod_i \kappa_{r_i}$, its Walsh coordinates are*

$$\widetilde{h}_{\mathbf{y}} = \widetilde{q}_{\mathbf{y}} \prod_{i=1}^{4} r_i^{y_i}.$$

Define the map $F_N : \mu_N^4 \to \mu_N^4$ by

$$F_N(r_1, r_2, r_3, r_4) = (R, r_{12}, r_{13}, r_{14}), \qquad R = \prod_{i=1}^{4} r_i, \quad r_I = \prod_{i \in I} r_i,$$

and put $(\mu_N)^2 := \{z^2 : z \in \mu_N\}$. A target $(R, u, v, w) \in \mu_N^4$ lies in the image of F_N exactly when

$$\frac{uvw}{R} \in (\mu_N)^2. \tag{F.45}$$

For odd N, F_N is bijective.

Proof. At input i, the fixed Walsh matrix $\kappa_{r_i} D(r_i) Z$, followed by the analytically transformed disequality edge $P^{\mathsf{T}} X P = 2Z$, acts as $2\kappa_{r_i} D(r_i)$. The four attachments therefore give the stated realizable dressing and scalar. Necessity is $uvw/R = r_1^2$. For sufficiency choose

$$r_1^2 = uvw/R, \qquad r_2 = u/r_1, \qquad r_3 = v/r_1, \qquad r_4 = w/r_1.$$

These choices satisfy $F_N(r_1, r_2, r_3, r_4) = (R, u, v, w)$. For odd N, squaring on μ_N is bijective. □

F.7.2 The arity-four case

The cyclic groups in Remark F.5.17 have transfer span $\mathscr{K}_G = \mathrm{span}\{I, X\}$. We treat arity four before reconstructing the higher-arity Reed–Muller signature.

Lemma F.7.5 (The cyclic arity-four case)**.** *Let $(\mathcal{F}, G, q)$ be a deck-stable retained state with the supplied equality I available,* $\mathrm{arity}(q) = \nu(\mathcal{F}) = 4$*, transfer span $\mathscr{K}_G = \mathrm{span}\{I, X\}$, and finite cyclic $G \ni [X]$. Then either an unsafe realizable binary is exposed, factor access strictly enlarges the complete transfer group of the resulting set, or a gadget using at most two copies of q realizes an irreducible endpoint-nondegenerate eight-vertex signature. Thus one obtains a resolved complexity leaf, an outer deck successor under Lemma F.5.12, or the tractable case* $\mathsf{Tract}(K\mathcal{F})$*.*

Proof. In Walsh coordinates the transfer span is diagonal. Therefore

$$P^{-1}GP = C_N^{\mathrm{std}} = \{[D(s)] : s \in \mu_N\}, \qquad 2 \mid N.$$

Here $[X]$ becomes $[Z] = [D(-1)]$, and chosen representatives are proportional to $D(s)Z$. To justify the parity reduction, fix an ordered deleted pair and an odd-weight string $\mathbf{y}$ on the two residual inputs. Up to a fixed nonzero representative scalar and a possible sign, the corresponding off-diagonal phase-card entry is $u + sv$, where u and v are the contributions of the two Walsh coefficients whose deleted-pair assignments are 00 and 11, respectively, and whose assignment to the remaining inputs is $\mathbf{y}$. A zero or cyclic card is diagonal, so this entry vanishes. At the two allowed phases $s = 1, -1$ we obtain $u + v = u - v = 0$, hence $u = v = 0$. Running through all six deleted pairs and both input orders (equivalently, the complete ordered deck) forces every odd-weight Walsh coefficient to vanish. Thus $\widetilde{q}$ is supported on even Walsh parity, as required by Lemma F.7.3. Item 1 of Theorem F.5.22 handles $N = 2$. Hence assume $N \geq 4$ and use the endpoint and complement-block coordinates of Equation (F.44), indexed by

$$I_1 = \{1, 2\}, \qquad I_2 = \{1, 3\}, \qquad I_3 = \{1, 4\}.$$

The cyclic quaternary-family result in Lemma F.7.3 leaves two endpoint regimes. We treat the zero-endpoint regime first. In the nonzero regime, the dressing-image criterion further separates successful dressing from the residual square obstruction. Define the block phase parameters $\alpha_i := \alpha_{I_i}$ and, for nonzero endpoints, the endpoint ratio $\lambda := t/p$. Call the complementary block indexed by I_i *active* when $(x_{I_i}, x_{I_i^c}) \neq (0, 0)$, and *inactive* otherwise.

For each two-copy output below, let $\mathcal{F}^+$ be the signature set after preprocessing, complete factor closure, and group recomputation. Zero, nullary, unary, odd, singular, and infinite-order factors are handled by the corresponding reductions. The remaining calculations concern the all-even finite-group branch. Proper factors become available over $\mathcal{F}^+$ through the factor theorem, with its Turing reduction and nonzero normalization scalars recorded.

Zero endpoints. Write each complementary block with scale ρ_i as $(x_{I_i}, x_{I_i^c}) = \rho_i(1, \alpha_i)$, taking $\rho_i = 0$ for an inactive block and then choosing any $\alpha_i \in \mu_N$, and define the auxiliary scalar $X_i := \alpha_i \rho_i^2$. For each i, choose the remaining indices $j < k$, so $\{i, j, k\} = \{1, 2, 3\}$, and order $I_i = (1, i + 1)$. Connect the first input of one copy to the first input of the other by the chosen representative $D(\alpha_j \alpha_k)Z$, and connect the second inputs by I (with the same orientation on both copies). If Q_i is the $I_i \mid I_i^c$ flattening, write the normalized output in Walsh coordinates as $\widetilde{g}_i$, where

$$\widetilde{g}_i = Q_i^{\mathsf{T}} K_i Q_i, \qquad K_i = (D(\alpha_j \alpha_k)Z) \otimes I = \mathrm{diag}(1, 1, -\alpha_j \alpha_k, -\alpha_j \alpha_k).$$

Order the residual inputs as the two inputs of I_i^c in the first copy followed by those in the second copy, each in inherited order. The only potentially nonzero complementary pairs of $\widetilde{g}_i$ are

$$E = (\widetilde{g}_i(0000), \widetilde{g}_i(1111)), \qquad D = (\widetilde{g}_i(0101), \widetilde{g}_i(1010)).$$

Their entries are

$$(E_0, E_1) = (-\alpha_j\alpha_k\rho_i^2,\ \alpha_i^2\rho_i^2),$$
$$(D_0, D_1) = (\alpha_j(X_j - X_k),\ -\alpha_k(X_j - X_k)),$$

and, whenever both displayed pairs E and D are nonzero,

$$\frac{D_1/D_0}{E_1/E_0} = \left(\frac{\alpha_k}{\alpha_i}\right)^2.$$

Set

$$d_i := (X_j - X_k)^2 - X_i^2.$$

We now verify endpoint nondegeneracy, including the case of an inactive i-block. If $\rho_i \neq 0$, the binary-gadget attachment criterion Lemma F.7.4 and Equation (F.45) is applied in the displayed residual-input order. Put $(u, v, w) = (r_{12}, r_{13}, r_{14})$ for these four output positions. The entries at 0101 and 1010 are multiplied by R/v and v, respectively. Take

$$R = (E_1/E_0)^{-1}, \qquad u = 1, \qquad v = -\alpha_j/\alpha_i, \qquad w = R/v.$$

All four parameters lie in μ_N, since N is even, and $uvw/R = 1$. Moreover

$$E_1R = E_0, \qquad D_0R/v = D_1v = \frac{\alpha_j\alpha_k}{\alpha_i}(X_j - X_k).$$

Thus the two entries in each displayed block are equal. Since the Walsh support is even, the two endpoint values in the original basis coincide. Up to one common nonzero scalar their common value is

$$(X_j - X_k) - X_i.$$

It is nonzero whenever $d_i \neq 0$. If instead $\rho_i = 0$, then $E_0 = E_1 = 0$, while $d_i \neq 0$ implies $D_0D_1 \neq 0$. Set

$$u = v = 1, \qquad R = D_1/D_0 = -\alpha_k/\alpha_j, \qquad w = R.$$

Here $uvw/R = 1$, so the same criterion applies and makes the two dressed entries of the D-block equal and nonzero. Inverse Walsh transformation again gives two nonzero endpoint values in the original basis. Thus every index with $d_i \neq 0$, active or inactive, yields an endpoint-nondegenerate eight-vertex signature. In the displayed four-string matching-code support, the two crossed $2 + 2$ flattenings have rank at least two. The remaining logical flattening has determinant, up to a nonzero scalar,

$$\alpha_j\alpha_k d_i.$$

If some $d_i \neq 0$, the corresponding realizable output has no binary factor. The dressing is a product of nonsingular local maps, so it preserves tensor factorizations. Even parity and its complementary endpoints exclude a unary factor, so the dressed endpoint-nondegenerate eight-vertex signature is irreducible.

It remains to exclude simultaneous vanishing of the three minors. If $d_1 = d_2 = d_3 = 0$, choose signs $\epsilon_i \in \{1, -1\}$ such that

$$X_2 - X_3 = \epsilon_1 X_1,$$
$$X_3 - X_1 = \epsilon_2 X_2,$$
$$X_1 - X_2 = \epsilon_3 X_3.$$

The coefficient matrix of this system is

$$M_\epsilon = \begin{pmatrix} -\epsilon_1 & 1 & -1 \\ -1 & -\epsilon_2 & 1 \\ 1 & -1 & -\epsilon_3 \end{pmatrix}, \qquad \det M_\epsilon = -(\epsilon_1\epsilon_2\epsilon_3 + \epsilon_1 + \epsilon_2 + \epsilon_3).$$

For the two constant sign triples this determinant is nonzero. For each of the other six, direct substitution gives, up to permutation, precisely the kernel lines $\mathrm{span}_{\mathbb{C}}\{(1,1,0)\}$, $\mathrm{span}_{\mathbb{C}}\{(1,0,1)\}$, and $\mathrm{span}_{\mathbb{C}}\{(0,1,1)\}$. Hence every nonzero solution is a permutation of $(\xi,\xi,0)$, with $\xi \neq 0$. Since $q \neq 0$, not all three X_i vanish.

Relabel so that $(X_1, X_2, X_3) = (\xi, \xi, 0)$. Then $\rho_1\rho_2 \neq 0$ and $\rho_3 = 0$. In the $I_3 \mid I_3^c$ flattening of $\widetilde{q}$, order the odd assignments on either side as $10, 01$. Its only nonzero block is

$$\begin{pmatrix} \rho_1 & \rho_2 \\ \alpha_2\rho_2 & \alpha_1\rho_1 \end{pmatrix}, \qquad \det = \alpha_1\rho_1^2 - \alpha_2\rho_2^2 = X_1 - X_2 = 0.$$

All four entries are nonzero, so the block has rank one and $\widetilde{q}$ has a genuine binary-by-binary tensor factorization across $I_3 \mid I_3^c$. The Walsh change is a product of invertible local maps, so q has the same factorization, contradicting its irreducibility. Therefore some $d_i \neq 0$, as required.

Nonzero endpoints and successful dressing. Define the ratio scalar $R := \lambda^{-1}$ and the dressing scalar $r_I := \epsilon_I\alpha_I\lambda^{-1}$ on each active block, where $\epsilon_I \in \{\pm 1\}$. Abbreviate $\epsilon_i := \epsilon_{I_i}$. An inactive block removes the square constraint: its otherwise irrelevant dressing parameter may be chosen so that the criterion's product equation $uvw/R = 1$ holds. With all three active, Lemma F.7.4 and Equation (F.45) applies exactly when $\epsilon_1\epsilon_2\epsilon_3\alpha_1\alpha_2\alpha_3$ is a square. If $N \equiv 2 \pmod 4$, signs remove the obstruction. When $4 \mid N$, this works precisely if $\alpha_1\alpha_2\alpha_3$ is a square. The dressed endpoint block is (p, p). Every active complementary block becomes $(-\epsilon_I p, -\epsilon_I p)$, while every inactive block remains $(0, 0)$. Let h be the resulting signature in the original basis. Equality of complementary Walsh blocks makes h even-parity, while its even Walsh support makes it complement-invariant, meaning $h(\mathbf{y}) = h(\mathbf{y} + \mathbf{1}^4)$ for every $\mathbf{y} \in \mathbb{F}_2^4$. It is nonzero because $p \neq 0$. Choose an even string $\mathbf{x}$ with $h(\mathbf{x}) \neq 0$, and attach realizable X-dressings on exactly the coordinates where $\mathbf{x}$ is one. This even translation preserves parity, and the translated signature h' satisfies

$$h'(0000) = h(\mathbf{x}) = h(\mathbf{x} + \mathbf{1}^4) = h'(1111) \neq 0.$$

Thus h' is an endpoint-nondegenerate eight-vertex signature. Every dressing used here is an invertible single-input map, so it also preserves the irreducibility of q.

Square obstruction. The only remaining case is

$$4 \mid N, \qquad p, t \neq 0, \qquad \text{all three blocks active}, \qquad \alpha_1\alpha_2\alpha_3 \notin (\mu_N)^2.$$

Normalize $p = 1$. For a permutation (i, j, k) of $(1, 2, 3)$, abbreviate

$$(a, b, c) := (\alpha_i, \alpha_j, \alpha_k),$$

and order the $I_i \mid I_i^c$ flattening as $00, 01, 10, 11$:

$$Q_i = \begin{pmatrix} 1 & 0 & 0 & -a \\ 0 & -b & -c & 0 \\ 0 & -\lambda c^{-1} & -\lambda b^{-1} & 0 \\ -\lambda a^{-1} & 0 & 0 & \lambda \end{pmatrix}.$$

Let $D := Z \otimes Z$ be the diagonal matrix on the four two-bit assignments, and let S be the permutation matrix that swaps $01, 10$. The realizable joins are

$$\widetilde{G}_i^{\parallel} = 4 Q_i D Q_i^{\mathsf{T}},$$
$$\widetilde{G}_i^{\times} = 4 Q_i D S Q_i^{\mathsf{T}}.$$

The crossed one is

$$\frac{1}{4}\widetilde{G}_i^{\times} = \begin{pmatrix} 1+a^2 & 0 & 0 & -\lambda(a+a^{-1}) \\ 0 & -2bc & -2\lambda & 0 \\ 0 & -2\lambda & -2\lambda^2/(bc) & 0 \\ -\lambda(a+a^{-1}) & 0 & 0 & \lambda^2(1+a^{-2}) \end{pmatrix}.$$

By Lemma F.5.4, an unsafe phase card is already a binary and resolves the complexity. Otherwise the cyclic pencil of the displayed central block forces, when $1 + a^2 \neq 0$,

$$\frac{2\lambda}{1+a^2} \in \mu_N.$$

Since a is a root of unity, $|1 + a^2| = 2$ gives $a^2 = 1$, whereas $1 + a^2 = 0$ gives $a^2 = -1$. Hence

$$\alpha_i^2 \in \{1, -1\} \qquad (i = 1, 2, 3).$$

If $8 \mid N$, all fourth roots are squares, contradicting the obstruction. Thus $N \equiv 4 \pmod 8$, and an odd number of the three squares equals -1. Choose the permutation (i, j, k) so that $\alpha_i^2 = a^2 = -1$, define the ratio scalar $r := \lambda/(bc)$, and evaluate the straight join. Its sole nonzero block has rank one and factors through the nonsingular anti-diagonal matrix $M_1(r) = D(r)X$, giving

$$\frac{1}{4}\widetilde{G}_i^{\parallel} = \pm 2 M_1(r) \otimes M_1(r), \qquad M_1(r) = \begin{pmatrix} 0 & 1 \\ r & 0 \end{pmatrix}, \qquad r \neq 0,$$

on the two residual copies of I_i. Lemma F.3.4 exposes the nonsingular anti-diagonal $M_1(r)$. Its transfer obeys

$$(M_1(r)Z)^2 = -rI, \qquad (M_1(r)Z)D(s)(M_1(r)Z)^{-1} = sD(s^{-1}) \quad (s \in \mu_N).$$

The class $[M_1(r)Z]$ is an involution outside C_N^{std} and acts on it by inversion. Together they generate a subgroup isomorphic to D_{2N}. After factor closure, the complete transfer group $G(\mathcal{F}^+)$ in the original coordinates contains $G(\mathcal{F})$ and the additional class $[PM_1(r)ZP^{-1}]$, so $G(\mathcal{F}) < G(\mathcal{F}^+)$. The first two cases pass a realizable endpoint-nondegenerate eight-vertex signature to Corollary F.4.8, while the third strictly enlarges the complete group. This proves the lemma. □

The arity-four case therefore gives a resolved complexity leaf or an outer deck successor. We next reconstruct the higher-arity core needed for Theorem F.7.1.

F.7.3 Reconstruction of the minimum Reed–Muller core

Let f be a minimum-arity nonbinary irreducible signature of the retained set, and let its arity be $n := \mathrm{arity}(f)$. If n is odd, apply Theorem F.2.9 to the ordinary-coordinate set $K\mathcal{F}$. Its hardness conclusion pulls back through the holographic K-coordinate equivalence, while its tractable conclusion is exactly $\mathsf{Tract}(K\mathcal{F})$. Thus, on the continuing branch, n is even. Every proper card is

zero or a nonsingular G-matching signature. In Walsh coordinates the latter is a weighted-equality product. Define its analytic Walsh coefficients by

$$a_S := \widetilde{f}(\mathbf{1}_S), \qquad S \subseteq [n].$$

For distinct i, j, label and normalize the two realizable cards obtained from the I/X-closures as C^+_{ij} and C^-_{ij} so that, for every $T \subseteq [n] \setminus \{i, j\}$,

$$\widetilde{C}^+_{ij}(\mathbf{1}_T) = 2(a_T + a_{Tij}), \qquad \widetilde{C}^-_{ij}(\mathbf{1}_T) = 2(a_T - a_{Tij}), \tag{F.46}$$

where $Tij = T \cup \{i, j\}$. A nonzero card has the form

$$c \prod_{\{p,q\}\in M} \mathbf{1}[z_p = z_q] r_{pq}^{z_p}, \qquad c \prod_{\{p,q\}\in M} r_{pq} \neq 0, \tag{F.47}$$

for a perfect matching M. It is nonzero at zero, and its supported weight-two strings are exactly the edges of M. Arity four is Lemma F.7.5, so assume $n \geq 6$.

Define the zero-card graph Z_f on $[n]$, writing ij for the unordered pair $\{i, j\}$, by

$$ij \in E(Z_f) \quad \Longleftrightarrow \quad C^+_{ij} = C^-_{ij} = 0.$$

By Equation (F.46),

$$ij \in E(Z_f) \quad \Longleftrightarrow \quad \operatorname{supp} \widetilde{f} \subseteq \{\mathbf{z} : z_i + z_j = 1\}.$$

Lemma F.7.6 (Elimination of the zero-card graph)**.** *Let $f \neq 0$ be the fixed minimum-arity nonbinary irreducible signature of arity n. Every connected component of Z_f is complete bipartite. If $n > 4$, then $E(Z_f) = \varnothing$.*

Proof. Zero-edge equations alternate along paths, so every nontrivial connected component is complete bipartite. Suppose that $uv \in E(Z_f)$. Since $n > 4$, the set $[n] \setminus \{u, v\}$ has at least three vertices. The graph induced there is a disjoint union of bipartite graphs and therefore is not complete. Choose distinct $i, j \notin \{u, v\}$ with $ij \notin E(Z_f)$. At least one of C^+_{ij}, C^-_{ij} is then a nonzero realizable card. Because u, v remain external and $\operatorname{supp} \widetilde{f} \subseteq \{\mathbf{z} : z_u + z_v = 1\}$, that card vanishes at the all-zero boundary string. This contradicts Equation (F.47), under which every nonzero card is nonzero at zero. Hence $E(Z_f) = \varnothing$. □

Thus, for the remaining arities, no coordinate pair is invisible to both realizable contractions: every pair supports at least one nonzero matching card. We may therefore compare the local card matchings through the common pencil Equation (F.46). The first global datum extracted from that comparison is the following weight-two graph.

The weight-two graph. Assume henceforth that $n \geq 6$ and $E(Z_f) = \varnothing$. Define the weight-two edge set

$$W_f := \{ij : 1 \leq i < j \leq n,\ a_{ij} \neq 0\}.$$

Let $\mathcal{W}_f := ([n], W_f)$ be the corresponding simple graph. For $S \subseteq [n]$, write

$$W_f[S] := \{pq \in W_f : p, q \in S\}, \qquad W_f[S] - k := W_f[S \setminus \{k\}]$$

for the restricted edge sets.

Lemma F.7.7 (Matching consistency)**.** *Let f be the fixed minimum-arity nonbinary irreducible signature of arity $n \geq 6$, and assume $E(Z_f) = \varnothing$. Then the following statements hold.*

1. *$a_\varnothing \neq 0$.*
2. *W_f is a matching.*
3. *If both C_{ij}^+ and C_{ij}^- are nonzero, their support matchings are equal.*
4. *Every edge $pq \in W_f$ is a global Walsh equality:*

$$\operatorname{supp} \widetilde{f} \subseteq \{\mathbf{z} : z_p = z_q\}.$$

Proof. If $a_\varnothing = 0$, then $\widetilde{C}_{ij}^{\pm}(\mathbf{0}) = \pm 2a_{ij}$. Since $E(Z_f) = \varnothing$ and matching cards are nonzero at zero, every $a_{ij} \neq 0$. The two ij-cards would then cover every edge on the remaining at least four inputs, impossible for two matchings. Hence $a_\varnothing \neq 0$.

For a deleted pair i, j, let P, Q be the support matchings when both cards survive, and write $P \triangle Q$ for their symmetric difference. Their weight-two entries imply

$$W_f[[n] \setminus \{i, j\}] \subseteq P \cup Q, \qquad P \triangle Q \subseteq W_f[[n] \setminus \{i, j\}].$$

Indeed, for $pq \in W_f[[n] \setminus \{i, j\}]$ the two pencil values cannot both vanish. If one whole card vanishes, the other has value $\pm 4a_{pq}$ at pq. Thus, with one nonzero card, $W_f[[n] \setminus \{i, j\}]$ lies in its perfect matching. We use the following elementary graph argument. If a vertex of $\mathcal{W}_f$ had degree at least three, delete two vertices outside it and its three neighbours (possible since $n \geq 6$). The induced graph has degree at least three, whereas the union of two perfect matchings has maximum degree two (and one matching has maximum degree one), a contradiction. Hence every component of $\mathcal{W}_f$ is a path or a cycle.

Suppose a path component has at least two edges. If two vertices lie outside that component, delete them. Otherwise (then the component has at least $n - 1 \geq 5$ vertices) delete two consecutive vertices at one end, leaving a path with at least two edges. For this deleted pair, one nonzero card is impossible. If both cards are nonzero, the two edges at the middle of the retained path must lie one in P and one in Q, hence in $P \triangle Q$. Alternating along the path forces an edge of $P \triangle Q$ at an endpoint, but $P \triangle Q$ is a disjoint union of cycles and $P \triangle Q \subseteq W_f[[n] \setminus \{i, j\}]$, whose endpoint has no second edge. This is impossible. A cycle is handled similarly: a triangle is retained by deleting vertices outside it and cannot alternate. For a 4-cycle delete one cycle vertex and one outside vertex, and for a cycle of length at least five delete two adjacent cycle vertices, leaving a path with at least two edges. Thus every component has at most one edge, so W_f is a matching. Finally, for any pair with both cards nonzero, $P \triangle Q$ is a union of cycles contained in the matching W_f. Therefore $P \triangle Q = \varnothing$ and $P = Q$.

For $pq \in W_f$, suppose $\mathbf{z} \in \operatorname{supp} \widetilde{f}$ and $z_p \neq z_q$. Choose distinct $i, j \in [n] \setminus \{p, q\}$ with $z_i = z_j$. Both ij-cards vanish on this restriction because every nonzero one contains pq. Equation (F.46) then kills both lifts, contradicting $\mathbf{z} \in \operatorname{supp} \widetilde{f}$. Hence $z_p = z_q$ on the support. □

When $W_f \neq \varnothing$, Lemma F.7.7 supplies global equality constraints. The remaining issue is whether the coefficients separate across these pairs. The following lemma recovers that product structure from diagonal restrictions.

Lemma F.7.8 (Diagonal-slice integration)**.** *Let $r \geq 3$, and let $h : \mathbb{F}_2^r \to \mathbb{C}^\times$ satisfy $h(\mathbf{0}) = 1$. Suppose that, for every $i \neq j$, the restriction $h|_{t_i = t_j}$, viewed as a signature on the common bit and*

the other $r-2$ bits, is a product of unary signatures (equivalently, is a simple tensor). If $r \geq 4$, then

$$h(\mathbf{t}) = \prod_{i=1}^{r} \rho_i^{t_i} \qquad (\rho_i \in \mathbb{C}^\times).$$

If $r = 3$, there are $\rho_1, \rho_2, \rho_3, \kappa \in \mathbb{C}^\times$ such that, for distinct i, j,

$$h(\mathbf{e}_i) = \rho_i, \qquad h(\mathbf{e}_i + \mathbf{e}_j) = \kappa\rho_i\rho_j, \qquad h(\mathbf{1}^3) = \kappa\rho_1\rho_2\rho_3. \tag{F.48}$$

Proof. Use the multiplicative subset expansion

$$h(\mathbf{t}) = \prod_{\varnothing \neq S \subseteq [r]} \theta_S^{\prod_{i \in S} t_i}, \qquad \theta_S \in \mathbb{C}^\times. \tag{F.49}$$

Multiplicative Möbius inversion determines its parameters:

$$\theta_S = \prod_{T \subseteq S} h(\mathbf{1}_T)^{(-1)^{|S|-|T|}}.$$

The parameter θ_S measures the interaction among the coordinates in S. Unary factors account exactly for the parameters with $|S| = 1$. Thus the coefficient tensor is simple if and only if $\theta_S = 1$ for every $|S| \geq 2$. On setting $t_i = t_j$, a parameter indexed by a nonempty $U \subseteq [r] \setminus \{i, j\}$ and the common bit is

$$\theta_{U \cup \{i\}}\theta_{U \cup \{j\}}\theta_{U \cup \{i,j\}} = 1. \tag{F.50}$$

Parameters θ_S with $S \cap \{i, j\} = \varnothing$ and $|S| \geq 2$ must also equal one.

Assume $r \geq 4$. Choosing i, j outside S first gives $\theta_S = 1$ for $2 \leq |S| \leq r-2$. For $|S| = r-1$, choose distinct $i, j \in S$ and define the index set $U := S \setminus \{i, j\}$ in Equation (F.50). The other two factors have index-set cardinality $r-2$, so $\theta_S = 1$. Taking $U = [r] \setminus \{i, j\}$ then gives $\theta_{[r]} = 1$. Only the unary parameters remain in Equation (F.49), proving the product formula.

Let $r = 3$. The three instances of Equation (F.50) imply

$$\theta_{12} = \theta_{13} = \theta_{23} =: \kappa, \qquad \theta_{123} = \kappa^{-2}.$$

Substitution in Equation (F.49), with $\rho_i = \theta_{\{i\}}$, gives Equation (F.48). □

Lemma F.7.9 (Integration of a nonempty weight-two graph)**.** *Let f be the fixed minimum-arity nonbinary irreducible signature of arity $n \geq 6$, with $E(Z_f) = \varnothing$. If $W_f \neq \varnothing$, then either $n = 6$ and a constant-size two-copy gadget realizes the even-parity signature of arity four, or f admits a nontrivial tensor factorization. In particular, a minimum-arity nonbinary irreducible signature of arity at least eight has $W_f = \varnothing$.*

Proof. Fix $E_1 = \{u, v\} \in W_f$ and define the residual input set $R := [n] \setminus E_1$. For $k \in R$, let $A_k := C_{uk}^+$. Since $\widetilde{A}_k(\mathbf{0}) = 2a_\varnothing \neq 0$, this is a nonzero matching card. The global equality on E_1 gives the unique lift

$$\begin{aligned}
\widetilde{A}_k(z_v, \mathbf{z}_{R\setminus\{k\}}) &= 2\widetilde{f}(z_u = z_v, z_k = z_v, \mathbf{z}_{R\setminus\{k\}}),\\
\widetilde{A}_k(\mathbf{0}) &= 2a_\varnothing \neq 0,\\
\widetilde{A}_k(\mathbf{e}_p + \mathbf{e}_q) &= 2a_{pq} \quad (p, q \in R \setminus \{k\}),\\
\widetilde{A}_k(\mathbf{e}_v + \mathbf{e}_q) &= 2a_{\{u,v,k,q\}} \quad (q \in R \setminus \{k\}).
\end{aligned}$$

Its matching therefore consists of $W_f[R] - k$ and at most one edge from v. To justify the last assertion, let U be the set of vertices not covered by W_f. It excludes u, v. For every k, all vertices of $U \setminus \{k\}$ remain on the card boundary and have no available weight-two edge, except that at most one can be paired through v. If $W_f = \{E_1\}$, then $|U| = n - 2 \geq 4$, and the preceding restriction already gives a contradiction. Otherwise a covered vertex of W_f lies in R, so choose $k \in R \setminus U$. Hence $|U| \leq 1$. Since $|U| = n - 2|W_f|$ is even, $U = \varnothing$. Thus W_f is perfect. Write its edges $E_1, \ldots, E_s$, where $s = n/2$.

By the global equality conclusion of Lemma F.7.7, for every string $\mathbf{z}$ constant on each E_r, let $t_r(\mathbf{z})$ denote that common edge bit. There is a logical coefficient tensor $h : \mathbb{F}_2^s \to \mathbb{C}$ such that

$$\widetilde{f}(\mathbf{z}) = \begin{cases} h(t_1(\mathbf{z}), \ldots, t_s(\mathbf{z})), & \mathbf{z}|_{E_r} \text{ is constant for every } r, \\ 0, & \text{otherwise.} \end{cases}$$

For two distinct W_f-edges E_i, E_j, close one endpoint of E_i to one endpoint of E_j, forming separately the two cards with the realizable $+$ and $-$ kernels. Since cross coefficients between different W_f-edges vanish and $a_\varnothing \neq 0$, both cards are nonzero. Both matching supports impose $t_i = t_j$. The $+/-$ choice changes only the sign character in that common logical bit. Since $s = n/2 \geq 3$, every $\mathbf{t} \in \mathbb{F}_2^s$ has a pair $i \neq j$ with $t_i = t_j$. Use the plus card for that pair. The boundary string obtained from $\mathbf{t}$ by retaining the mates of E_i, E_j and the two inputs of every other E_r lies in this card's matching support. The closure equation has a unique compatible lift in the global equality code, namely the string with edge bits $\mathbf{t}$, so the plus-card value is a fixed nonzero scalar times $h(\mathbf{t})$ (there is no second lift with which it could cancel). A nonzero matching product is nonzero at every supported boundary string. Hence $h(\mathbf{t}) \neq 0$ for every $\mathbf{t}$, and h has full support. Rescale it by $a_\varnothing^{-1}$ (absorbing this nonzero scalar into f), so that $h(\mathbf{0}) = 1$. Every cross-pair diagonal slice $t_i = t_j$ is nonzero and rank one. By Lemma F.7.8, $s \geq 4$ would make f a product of s nonsingular binaries, which is the factorization alternative of the lemma. Hence, on the remaining branch, $s = 3$. Use the parameters $\rho_1, \rho_2, \rho_3, \kappa \neq 0$ of Equation (F.48). Let $H_i^{\pm}$ be the 2×2 logical coefficient matrix obtained by closing E_i with the $\pm$ kernel. For $\{i, j, k\} = \{1, 2, 3\}$, one has

$$\det H_i^{\pm} = \rho_j \rho_k (\kappa - 1)(1 - \kappa \rho_i^2).$$

If $\kappa = 1$, h, and hence f, has the asserted product factorization. On the remaining branch $\kappa \neq 1$, and rank one of the nonzero cards forces

$$\kappa \rho_1^2 = \kappa \rho_2^2 = \kappa \rho_3^2 = 1.$$

In Walsh coordinates, write $E_i = \{i^0, i^1\}$ for the two inputs of each matching edge. The realizable binary closing chains for the I/X-contractions have Walsh kernels proportional to $\epsilon^u \mathbf{1}[u = v]$, with $\epsilon = 1$ for I and $\epsilon = -1$ for X. Two such chains joining matching edges contribute $(\epsilon_0 \epsilon_1)^u$. Choosing their signs gives either character on the common bit, using the original X-edges.

Let E_1', E_2', E_3' denote the corresponding matching edges in the second copy. Use two such binary closing chains for each displayed edge-to-edge join: join the two inputs of E_3 in the first copy to the two inputs of E_1' in the second copy, and join the two inputs of E_2' to those of E_3' inside the second copy. Leave both inputs of E_1 and E_2 in the first copy external. Evaluating these contractions by Equation (F.46) gives the logical characters below, and every internal matching assignment has a unique lift. Choose the realizable I/X-signs

$$\theta = \kappa \rho_2 \rho_3, \qquad \eta = -\kappa \rho_1 \rho_3,$$

which lie in $\{\pm 1\}$. Writing $x = t_1$ and $y = t_2$ for the two external logical bits, the logical coefficient matrix is

$$Q(x,y) = \sum_{a,c\in\mathbb{F}_2} \eta^a \theta^c h(x,y,a) h(a,c,c).$$

Using $\sum_c \theta^c h(a,c,c) = 2\rho_1^a$, this becomes

$$Q \doteq \frac{\kappa-1}{\kappa} \begin{pmatrix} 1 & 0 \\ 0 & \kappa\rho_1\rho_2 \end{pmatrix}.$$

Thus the realized quaternary signature has Walsh coordinates

$$\widetilde{F} = c\big(\delta_{0000} + \sigma\delta_{1111}\big), \qquad c \neq 0, \qquad \sigma = \kappa\rho_1\rho_2 \in \{\pm 1\},$$

where the two nonzero coefficients follow from $\kappa \neq 1$. Applying the inverse analytic change of coordinates, namely $F = P^{\otimes 4}\widetilde{F}$, gives

$$F(\mathbf{x}) = c\big(1 + \sigma(-1)^{x_1+x_2+x_3+x_4}\big).$$

Define the even-parity signature by

$$\mathrm{Even}_4(\mathbf{x}) := \mathbf{1}[x_1 + x_2 + x_3 + x_4 = 0].$$

Thus the realizable tensor is a nonzero scalar multiple of Even_4 when $\sigma = 1$, and of its odd-parity coset when $\sigma = -1$. A realizable X-dressing on one exposed input toggles the coset. For the equality-anchor condition, use the matrix $U := D(i)P^{-1}$. It satisfies

$$U^{\otimes 4}\mathrm{Even}_4 \doteq (=_4), \qquad (U^{-1})^{\mathsf{T}} X U^{-1} = D(i)^{-1} P^{\mathsf{T}} X P D(i)^{-1} = 2I = 2\,(=_2).$$

Consequently a realizable Even_4 equality anchor is obtained, and Lemma F.3.15 applies. □

A nonempty W_f has therefore already produced an equality anchor or a proper factorization. After those alternatives have been resolved, the remaining case has

$$W_f = \varnothing.$$

Scale $a_\varnothing = 1$. With all weight-two coefficients absent, the common card matchings determine the entire Walsh support.

Lemma F.7.10 (Reed–Muller support reconstruction)**.** *Let f be the fixed minimum-arity nonbinary irreducible signature of arity $n \geq 6$, normalized by $a_\varnothing = 1$, and assume $E(Z_f) = \varnothing$ and $W_f = \varnothing$. For $i \neq j$, let M_{ij} be the common support matching of the two ij-cards. Fix an arbitrary ordering of its $n/2-1$ edges and, for $\mathbf{t} \in \mathbb{F}_2^{n/2-1}$, write t_e for the coordinate indexed by $e \in M_{ij}$. Let $D := \operatorname{supp} \widetilde{f}$ be the support set. Its equal-ij slice is the $(n/2-1)$-dimensional code*

$$D_{ij} = \Big\{ \mathbf{1}_{T(\mathbf{t})} + (\mathrm{wt}(\mathbf{t}) \bmod 2)(\mathbf{e}_i + \mathbf{e}_j) : \mathbf{t} \in \mathbb{F}_2^{n/2-1} \Big\}, \tag{F.51}$$

where $T(\mathbf{t}) = \bigcup_{e\in M_{ij}:\, t_e=1} e$. Every codeword of D_{ij} has weight divisible by four. Moreover,

$$n = 8, \qquad D = RM(1,3),$$

after a bijective relabelling of the inputs by $\mathbb{F}_2^3$.

Proof. By Lemma F.7.7, the two ij-cards survive with a common support matching M_{ij}. Normalize them at the zero string $\mathbf{t} = \mathbf{0}^{n/2-1}$ in matching coordinates, and write $r^+_{ij,e}$ and $r^-_{ij,e}$ for the corresponding normalized edge ratios (the $t_e = 1$ factor entry divided by its $t_e = 0$ entry). For each $e \in M_{ij}$, set $\rho_{ij,e} := r^-_{ij,e}/r^+_{ij,e}$. Their quotient on the matching code—the repetition code consisting of assignments constant on every edge of M_{ij}, parametrized by $\mathbf{t}$—is

$$\frac{\widetilde{C}^-_{ij}(\mathbf{t})}{\widetilde{C}^+_{ij}(\mathbf{t})} = \prod_{e \in M_{ij}} \rho^{t_e}_{ij,e}.$$

Each single-edge quotient is -1 by $a_{pq} = 0$ and Equation (F.46). Hence the quotient is $(-1)^{\mathrm{wt}(\mathbf{t})}$. Inverting the pencil selects the $ij = 00$ lift for even $\mathrm{wt}(\mathbf{t})$ and $ij = 11$ for odd $\mathrm{wt}(\mathbf{t})$, proving Equation (F.51). The selected string has weight $2\,\mathrm{wt}(\mathbf{t})$ in the even case and $2\,\mathrm{wt}(\mathbf{t}) + 2$ in the odd case, so its weight is divisible by four in either case.

For $\mathbf{x}, \mathbf{y} \in D$, some two coordinates have the same pair (x_i, y_i). The linear slice D_{ij} then contains $\mathbf{x}, \mathbf{y}, \mathbf{x} + \mathbf{y}$. Thus D is linear. For the linear functional $\ell_{ij} : \mathbb{F}_2^n \to \mathbb{F}_2$ defined by $\ell_{ij}(\mathbf{z}) := z_i + z_j$, one has

$$D \cap \ker \ell_{ij} = D_{ij}.$$

This subspace has dimension $n/2 - 1$. If $\dim D = n/2 - 1$, then $D = D_{ij} \subseteq \ker \ell_{ij}$ for every $i \neq j$. Hence every codeword of D has all coordinates equal, so $\dim D \leq 1$, contradicting $n/2 - 1 \geq 2$. Therefore $\dim D = n/2$. Every codeword of D lies in one of the displayed slices and hence has weight divisible by four. For $\mathbf{x}, \mathbf{y} \in D$,

$$0 \equiv \mathrm{wt}(\mathbf{x}) + \mathrm{wt}(\mathbf{y}) - \mathrm{wt}(\mathbf{x} + \mathbf{y}) = 2\mathbf{x} \cdot \mathbf{y} \pmod 4.$$

Thus $D \subseteq D^\perp$, and equal dimensions give

$$D = D^\perp.$$

For distinct i, j, p, the mate q of p in M_{ij} gives the unique weight-four codeword $\{i, j, p, q\}$. Now Lemma F.6.24 yields $n = 8$ and $D = RM(1, 3)$. □

Coefficient rigidity and the order of the cyclic group. The combinatorial reconstruction is complete: the surviving core has arity eight and Reed–Muller support. It remains to control its nonzero coefficients. The next lemma shows that they carry only a linear sign character, so an available X-dressing removes the resulting affine shift.

Lemma F.7.11 (Coefficients on $\mathrm{RM}(1,3)$)**.** *Let f be an arity-eight signature with $\operatorname{supp} \widetilde{f} = RM(1,3)$ whose proper I/X-cards are zero or have the weighted-equality matching form of Equation* (F.47)*. There are $c \neq 0$ and $\mathbf{w} \in \mathbb{F}_2^8$ such that*

$$\widetilde{f}(\mathbf{z}) = c(-1)^{\mathbf{w}\cdot\mathbf{z}} \mathbf{1}[\mathbf{z} \in RM(1,3)]. \tag{F.52}$$

Consequently, back in the original K-coordinates,

$$f(\mathbf{x}) \doteq \mathbf{1}[\mathbf{x} \in \mathbf{w} + RM(1,3)]. \tag{F.53}$$

Proof. Let the support code be $D := RM(1,3)$, and define $b(\mathbf{z}) := \widetilde{f}(\mathbf{z})/\widetilde{f}(\mathbf{0}^8)$. For an affine plane $A \subseteq \mathbb{F}_2^3$, its incidence vector $\mathbf{1}_A$ is a codeword of D, with the fixed lexicographic input indexing.

Fix distinct points $p, q \in \mathbb{F}_2^3$, and let A_1, A_2, A_3 be the three affine planes containing them. Under the identification of the equal-pq plus card with its three logical matching coordinates, the

codewords $\mathbf{1}_{A_1}, \mathbf{1}_{A_2}, \mathbf{1}_{A_3}$ are the three singleton logical codewords. Its zero codeword lifts only to $\mathbf{0}^8 \in D$, because D contains no codeword of weight two. Normalizing the card at that codeword, the weighted-product form therefore gives

$$b\left(\sum_{r\in S} \mathbf{1}_{A_r}\right) = \prod_{r\in S} b(\mathbf{1}_{A_r}) \qquad (S \subseteq \{1,2,3\})\,. \tag{F.54}$$

In particular, $\mathbf{1}_{A_1} + \mathbf{1}_{A_2} + \mathbf{1}_{A_3} = \mathbf{1}^8$. The same argument shows that whenever two affine planes A, B meet in a two-point affine line,

$$b(\mathbf{1}_A + \mathbf{1}_B) = b(\mathbf{1}_A)b(\mathbf{1}_B). \tag{F.55}$$

For a plane A, write $A^c := \mathbb{F}_2^3 \setminus A$. Now $\mathbf{1}_{A_1} + \mathbf{1}_{A_2} = \mathbf{1}_{A_3^c}$ and A_3^c is an affine plane. Since $A_2 \cap A_3 = \{p, q\}$, it meets A_2 in the other two points of A_2, hence in a two-point line. Applying Equation (F.55) first to A_1, A_2 and then to A_3^c, A_2 gives

$$b(\mathbf{1}_{A_1} + \mathbf{1}_{A_2}) = b(\mathbf{1}_{A_1})b(\mathbf{1}_{A_2}), \qquad b(\mathbf{1}_{A_1}) = b(\mathbf{1}_{A_1} + \mathbf{1}_{A_2})b(\mathbf{1}_{A_2}).$$

Thus $b(\mathbf{1}_{A_2})^2 = 1$. Every affine plane can play the role of A_2, so

$$b(\mathbf{1}_A) \in \{\pm 1\} \qquad \text{for every affine plane } A. \tag{F.56}$$

For any affine plane A, choose distinct $p, q \in A$ and call the other two planes through them B, C. Then $\mathbf{1}_{A^c} = \mathbf{1}_B + \mathbf{1}_C$, and Equations (F.54) and (F.56) give

$$b(\mathbf{1}_{A^c}) = b(\mathbf{1}_B)b(\mathbf{1}_C) = b(\mathbf{1}^8)b(\mathbf{1}_A). \tag{F.57}$$

The code D consists of $\mathbf{0}^8, \mathbf{1}^8$, and the fourteen affine-plane incidence strings. We can now check multiplicativity on all of D. The zero case is immediate. Equal planes use Equation (F.56). Distinct noncomplementary planes meet in a two-point line and use Equation (F.55). Complementary planes and sums of $\mathbf{1}^8$ with a plane use Equation (F.57). Finally, Equations (F.54) and (F.56) imply $b(\mathbf{1}^8)^2 = 1$, which handles $\mathbf{1}^8 + \mathbf{1}^8$. Hence

$$b(\mathbf{x} + \mathbf{y}) = b(\mathbf{x})b(\mathbf{y}) \qquad (\mathbf{x}, \mathbf{y} \in D).$$

Thus b is a sign character of D. Its binary exponent extends to a linear functional on $\mathbb{F}_2^8$, so there is $\mathbf{w} \in \mathbb{F}_2^8$ with $b(\mathbf{z}) = (-1)^{\mathbf{w}\cdot\mathbf{z}}$ for $\mathbf{z} \in D$. This proves Equation (F.52). Since $D = D^\perp$, Fourier inversion gives

$$f(\mathbf{x}) \doteq \sum_{\mathbf{z}\in D} (-1)^{(\mathbf{x}+\mathbf{w})\cdot\mathbf{z}} = |D|\mathbf{1}[\mathbf{x} + \mathbf{w} \in D].$$

This is Equation (F.53). □

Available X-dressings translate the coset, so the standard core RM_8 of Equation (F.40) is available. In lexicographic order it is the Shao–Cai signature $f_8 = \widehat{f_8}$. See Remark F.7.12.

Define the matrix

$$\Theta = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_2), \qquad \Theta^3 = I.$$

In a tensor $A \otimes B \otimes C$ of three ordered binary signatures, A, B, and C act on input pairs $(1,2)$, $(3,4)$, and $(5,6)$, respectively. With this convention, the six-input Pauli core is

$$\begin{aligned} \mathrm{H}_6 &= \sum_{\gamma\in\mathbb{F}_2^2} B_\gamma \otimes B_{\Theta\gamma} \otimes B_{\Theta^2\gamma} \\ &= I\otimes I\otimes I + Z\otimes Y\otimes X + X\otimes Z\otimes Y + Y\otimes X\otimes Z. \end{aligned} \tag{F.58}$$

We use the ordered core definitions in Equations (F.40) and (F.58), with the lexicographic input order for RM_8 and the displayed three ordered pairs for H_6, to compare their Bell contractions.

A Bell product is a *same-$\boldsymbol{\gamma}$ Bell product* when every factor label $\boldsymbol{\gamma}_e$ equals one fixed $\boldsymbol{\gamma}$.

An even-arity irreducible signature f has the *Bell property* when, for every ordered pair of distinct inputs (i, j) and every $\boldsymbol{\gamma} \in \mathbb{F}_2^2$, the contraction $\partial_{(i,j)}^{B_\gamma} f$ is a Bell product on the remaining inputs. It has the *strong Bell property* when each such contraction is, more specifically, a same-$\boldsymbol{\gamma}$ Bell product. These properties concern the displayed algebraic contractions. They use the ordered-input, projective form of the definitions [SC20, full version, Definition 6.12] and [CFS20b, full version, Definition 1.10; conference version, Definition 12].

Remark F.7.12 (The Shao–Cai f_8 and $\widehat{f_6}$ cores). With the preceding input order, RM_8 is the signature $f_8 = \widehat{f_8}$ of [SC20, full version, Section 8]. It has the strong Bell property: contracting any ordered input pair by B_γ gives, up to a nonzero scalar, a product of three copies of B_γ on the remaining inputs [CFS20b, Theorems 13–14]. Let $\widehat{f_6}$ denote the six-ary signature of Shao–Cai given in [SC20, full version, Eq. (6.1)]. Direct comparison with Equation (F.58) gives the input-permutation identity

$$\mathrm{H}_6(x_1, \ldots, x_6) = \widehat{f_6}(x_1, x_3, x_5, x_4, x_6, x_2).$$

The same results show that H_6 has the Bell property: every ordered Bell contraction is, up to a nonzero scalar, a product of two ordered Bell binaries [CFS20b, Theorems 13–14]. See also [SC20, full version, Definition 6.12 and Lemma 6.13]. The arguments below reconstruct signatures from their contraction properties and obtain a common presentation for the whole retained set. The direct H_6-to-RM_8 construction is given in Section F.9.5.2.

Remark F.7.13 (Limited appearance and internal $C_2^{(X)}$ rigidity). Once RM_8 is available, the limited-appearance method of Shao and Cai [SC20, full version, Section 8.2, Lemmas 8.9–8.10] provides an alternative route through a larger binary group. After the binary and arity-four cases are resolved, its reduction permits adjoining the Bell signatures. In our transfer convention, Z and Y give $[ZX] = [Y]$ and $[YX] = [Z]$, whose matrix representatives lie outside the current transfer span $\mathrm{span}\{I, X\}$. The enlarged set therefore has a strictly larger complete transfer group and returns to Theorem F.5.16 through the Turing reduction.

The proof below instead stays in the $C_2^{(X)}$ case. Reed–Muller networks, Radon contractions, and affine recognition establish the stronger structural conclusion that the whole retained set and its edge have one common flat-Lagrangian K-presentation. This is the conclusion used by Theorem F.7.1 in the final synthesis.

Thus the core itself is now fixed, up to the recorded equivalences. The last initial-core issue is the complete transfer group. On the continuing matching-deck branch, the coefficient classification also rules out every additional transfer class beyond $[I]$ and $[X]$.

Lemma F.7.14 (The surviving Reed–Muller core forces $G = C_2^{(X)}$)**.** *Let $\mathcal{F}$ be a retained normalized signature set, let $G := G(\mathcal{F})$ be its complete transfer group, and assume $\mathscr{K}_G = \mathrm{span}\{I, X\}$. Let $f \in \mathcal{F}$ be an irreducible Reed–Muller core such that every single-input dressing of f by a safe binary remains in the standing matching-deck branch. Then the complete transfer group is*

$$\{[I], [X]\} \cong C_2.$$

Proof. Put $D := RM(1, 3)$. By Lemma F.7.11, choose $c \neq 0$ and $\mathbf{w} \in \mathbb{F}_2^8$ such that $\widetilde{f}(\mathbf{z}) = c(-1)^{\mathbf{w}\cdot\mathbf{z}}$ for $\mathbf{z} \in D$, and define $b(\mathbf{z}) := \widetilde{f}(\mathbf{z})/c$. Let T be the transfer matrix of a safe binary signature in the original basis, and write

$$T^{\mathrm{W}} := P^{-1} T P$$

for its Walsh-coordinate matrix. Since the transfer span is diagonal in these coordinates, $T^{\mathrm{W}} \doteq D(r)$ for some $r \neq 0$. Dress one input p by T. This constant-size gadget preserves irreducibility, support, and the matching-deck hypotheses. Its coefficient table, after division by its value at $\mathbf{z} = \mathbf{0}^8$, is

$$b_T(\mathbf{z}) = (-1)^{\mathbf{w}\cdot\mathbf{z}} r^{z_p} \qquad (\mathbf{z} \in D).$$

By Lemma F.7.11, b_T, hence also the quotient $r^{z_p} = b_T(\mathbf{z})/b(\mathbf{z})$, is a character of D and takes values ± 1. Evaluating at $\mathbf{1}^8 \in D$ gives $r = \pm 1$, precisely the transfer classes in the original basis $[I]$ and $[X]$. □

F.7.4 Flat-Lagrangian networks and Radon access

Retain the RM_8 to build contexts around an arbitrary signature. A flat-Lagrangian signature has the form $c\,\mathbf{1}[\mathbf{x} \in \mathbf{a} + L]$, where $c \neq 0$ and $L = L^{\perp}$ as in Section F.2.10.1. The next lemma shows both what these networks preserve and what they can construct. Self-duality tracks the boundary after internal variables are summed out. Weight-four transvections provide the reverse construction from a basis of L.

Symmetric binary spaces. A bilinear form $b : W \times W \to \mathbb{F}_2$ is linear in each argument and is symmetric when $b(\mathbf{x}, \mathbf{y}) = b(\mathbf{y}, \mathbf{x})$. A *symmetric binary space* is a finite-dimensional $\mathbb{F}_2$-space with a specified symmetric form, written $\mathbf{x} \cdot \mathbf{y}$. On $\mathbb{F}_2^m$, the standard dot product is $\mathbf{x} \cdot \mathbf{y} := \sum_j x_j y_j$. For $U \leq W$, define

$$U^{\perp_W} := \{\mathbf{y} \in W : \mathbf{x} \cdot \mathbf{y} = 0 \text{ for every } \mathbf{x} \in U\}, \qquad \mathrm{rad}_W(U) := U \cap U^{\perp_W}.$$

We omit W when it is clear. The restricted form on U is *nondegenerate* when $\mathrm{rad}_W(U) = \{0\}$. The notation $W = U \perp P$ means $W = U \oplus P$ and $U \cdot P = 0$. Then $\pi_U(\mathbf{u} + \mathbf{p}) = \mathbf{u}$ and $\pi_P(\mathbf{u} + \mathbf{p}) = \mathbf{p}$ are the orthogonal projections.

If $R \leq \mathrm{rad}_W(U)$, the quotient U/R has the induced form $(\mathbf{x} + R) \cdot (\mathbf{y} + R) := \mathbf{x} \cdot \mathbf{y}$. When R acts by translations on an affine space with translation space U, its orbit space has translation space U/R. A linear map $\phi : U \to U'$ is an *isometry* when it is bijective and preserves the form. The corresponding affine isometry is $\mathbf{a} + \mathbf{u} \mapsto \mathbf{a}' + \phi(\mathbf{u})$.

A vector $\mathbf{x}$ is *isotropic* when $\mathbf{x} \cdot \mathbf{x} = 0$ and anisotropic otherwise. A subspace is *totally isotropic* when $U \subseteq U^{\perp_W}$, and is *self-dual in* W when $U = U^{\perp_W}$. In a nondegenerate symmetric binary space, the map $\mathbf{x} \mapsto \mathbf{x} \cdot \mathbf{x}$ is linear, so nondegeneracy identifies it with a unique vector $\mathbf{c}$ satisfying $\mathbf{x} \cdot \mathbf{x} = \mathbf{c} \cdot \mathbf{x}$. This is the *characteristic vector*. The space is alternating when $\mathbf{c} = 0$, and nonalternating otherwise.

We use the following facts about symmetric binary spaces: nondegenerate subspaces split off orthogonally. Every even-dimensional nondegenerate symmetric binary space has a self-dual subspace. Every nondegenerate nonalternating m-space is isometric to the standard dot-product space $\mathbb{F}_2^m$. Every odd-dimensional nondegenerate symmetric binary space is nonalternating and has an anisotropic vector. Here are short proofs. For a nondegenerate subspace P, its pairing map identifies P with $P^{\vee}$, so each ambient vector differs from a unique vector of P by an element of $P^{\perp}$. This gives the orthogonal splitting. A nonzero nondegenerate alternating space contains vectors $\mathbf{u}, \mathbf{v}$ with $\mathbf{u} \cdot \mathbf{v} = 1$. Their Gram matrix is $H = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Splitting off these planes shows that an alternating space has even dimension, and one vector from each plane spans a self-dual subspace. In a nonalternating space, split off norm-one vectors until the remainder is alternating. A norm-one vector $\mathbf{e}$ orthogonal to such a plane can absorb it: $\mathbf{e} + \mathbf{u}, \mathbf{e} + \mathbf{v}, \mathbf{e} + \mathbf{u} + \mathbf{v}$ are an orthogonal norm-one basis of their three-dimensional span. Repeating this gives the standard dot-product basis. In even dimension $2k$, the vectors $\mathbf{e}_1 + \mathbf{e}_2, \ldots, \mathbf{e}_{2k-1} + \mathbf{e}_{2k}$ span a self-dual subspace of that space (the empty

span applies in dimension zero). Finally, odd dimension excludes the alternating case and hence supplies a norm-one vector.

Lemma F.7.15 (Exact flat-Lagrangian network calculus)**.** *1. Let Ω be a tensor network with ordered boundary B. Suppose that the signature at each tensor vertex u is proportional to $\mathbf{1}[\mathbf{x}_u \in \mathbf{a}_u + L_u]$, where either $L_u = RM(1,3) \leq \mathbb{F}_2^8$ or $L_u = \mathrm{span}_{\mathbb{F}_2}\{(1,1)\} \leq \mathbb{F}_2^2$, and $\mathbf{a}_u$ is a shift in the corresponding ambient space. The binary case includes ordinary I/X-vertices. Suppose every internal connection is labelled by I or X. Then the boundary signature is either zero or*

$$\mathbf{x}_B \longmapsto c_B \mathbf{1}[\mathbf{x}_B \in \mathbf{a}_B + L], \qquad c_B \neq 0, \quad \mathbf{a}_B \in \mathbb{F}_2^B, \quad L = L^\perp \leq \mathbb{F}_2^B. \tag{F.59}$$

2. Conversely, for every $\mathbf{a}_B \in \mathbb{F}_2^{2k}$ and every $L = L^\perp \leq \mathbb{F}_2^{2k}$, a polynomial-size $\mathrm{RM}_8/I/X$-network directly realizes a nonzero scalar multiple of $\mathbf{x}_B \longmapsto \mathbf{1}[\mathbf{x}_B \in \mathbf{a}_B + L]$.

In particular every nonzero boundary produced by a network in item 1 has even arity.

Proof. For item 1, apply the boundary normalization following Equation (F.154). The isolated I/X components are flat-Lagrangian and contribute direct summands to the boundary translation space. It therefore suffices to treat a boundary B consisting of unpaired vertex inputs. Let E_{int} be the set of internal half-edge pairs and put

$$H_{\mathrm{int}} := \bigcup E_{\mathrm{int}}, \qquad V_{\mathrm{he}} := \mathbb{F}_2^B \oplus \mathbb{F}_2^{H_{\mathrm{int}}}.$$

For each $e = \{h, h'\} \in E_{\mathrm{int}}$, let $\eta_e = 0$ when e is labelled by I and $\eta_e = 1$ when it is labelled by X. Define

$$E_\eta := \{\mathbf{y} \in \mathbb{F}_2^{H_{\mathrm{int}}} : y_h + y_{h'} = \eta_e \text{ for every } e = \{h, h'\} \in E_{\mathrm{int}}\},$$

and let

$$E_{\mathrm{eq}} := \{\mathbf{y} \in \mathbb{F}_2^{H_{\mathrm{int}}} : y_h = y_{h'} \text{ for every } \{h, h'\} \in E_{\mathrm{int}}\}, \qquad H_{\mathrm{con}} := \mathbb{F}_2^B \oplus E_{\mathrm{eq}}.$$

The translation space E_{eq} is self-dual in $\mathbb{F}_2^{H_{\mathrm{int}}}$, and hence

$$H_{\mathrm{con}}^\perp = \{\mathbf{0}_B\} \oplus E_{\mathrm{eq}},$$

where, for any finite index set R, $\mathbf{0}_R$ denotes the zero vector in $\mathbb{F}_2^R$.

Let $V(\Omega)$ denote the tensor-vertex set of Ω, and put

$$C_0 := \bigoplus_{u \in V(\Omega)} L_u = C_0^\perp, \qquad \mathcal{K} := C_0 \cap H_{\mathrm{con}}.$$

Let $\mathbf{a}_{\mathrm{vert}} := (\mathbf{a}_u)_{u \in V(\Omega)}$ be the concatenated vertex shift. The feasible assignments form

$$(\mathbf{a}_{\mathrm{vert}} + C_0) \cap (\mathbb{F}_2^B \oplus E_\eta).$$

This intersection is either empty or an affine coset $\mathbf{y}_0 + \mathcal{K}$. Define

$$\pi_B : V_{\mathrm{he}} \to \mathbb{F}_2^B, \qquad L := \pi_B(\mathcal{K}), \qquad \mathbf{a}_B := \pi_B(\mathbf{y}_0).$$

Every nonempty boundary fibre is a coset of $\ker(\pi_B|_{\mathcal{K}})$, so all fibres have size $2^{\dim \ker(\pi_B|_{\mathcal{K}})}$. Multiplying this number by the nonzero vertex scalars gives $c_B \neq 0$ in Equation (F.59). The support and scalar are therefore determined by affine Gaussian elimination.

To prove $L \subseteq L^\perp$, take $\mathbf{k}, \mathbf{k}' \in \mathcal{K}$. Since $\mathbf{k}, \mathbf{k}' \in C_0 = C_0^\perp$, we have $\mathbf{k} \cdot \mathbf{k}' = 0$. For an internal pair $e = \{h, h'\}$, put

$$t := k_h = k_{h'}, \qquad s := k'_h = k'_{h'}.$$

That pair contributes $ts + ts = 0$ to $\mathbf{k} \cdot \mathbf{k}'$. Consequently $\pi_B(\mathbf{k}) \cdot \pi_B(\mathbf{k}') = 0$, proving $L \subseteq L^\perp$.

Moreover,

$$\mathcal{K}^\perp = C_0 + H_{\mathrm{con}}^\perp.$$

If $\mathbf{b} \in L^\perp$, then $(\mathbf{b}, \mathbf{0}_{H_{\mathrm{int}}}) \in \mathcal{K}^\perp$, so write

$$(\mathbf{b}, \mathbf{0}_{H_{\mathrm{int}}}) = \mathbf{c}_0 + \mathbf{d}, \qquad \mathbf{c}_0 \in C_0, \quad \mathbf{d} \in H_{\mathrm{con}}^\perp.$$

Both $(\mathbf{b}, \mathbf{0}_{H_{\mathrm{int}}})$ and $\mathbf{d}$ belong to H_{con}, hence $\mathbf{c}_0 \in C_0 \cap H_{\mathrm{con}} = \mathcal{K}$. Therefore $\mathbf{b} = \pi_B(\mathbf{c}_0) \in L$, proving $L = L^\perp$.

For the converse, put inputs $000, 100, 010, 001$ of one RM_8 vertex on the input side and inputs $111, 011, 101, 110$ on the output side, in the displayed orders. Direct evaluation gives

$$\mathbf{y} = (I_4 + J_4)\mathbf{x} = \mathbf{x} + (x_1 + x_2 + x_3 + x_4)\mathbf{1}^4.$$

Here $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^4$ are respectively the ordered input and output strings, I_4 is the 4×4 identity matrix, and $J_4 := \mathbf{1}^4(\mathbf{1}^4)^{\mathsf{T}}$ is the 4×4 all-ones matrix over $\mathbb{F}_2$. Thus realizable RM_8 vertices and permutations of inputs realize every weight-four transvection $\tau_{\mathbf{s}}(\mathbf{x}) = \mathbf{x} + (\mathbf{s} \cdot \mathbf{x})\mathbf{s}$. I-vertices (or $X - I - X$ chains) compose them. We claim by induction on k that every $L = L^\perp \leq \mathbb{F}_2^{2k}$ is obtained from $L_0 = \mathrm{span}_{\mathbb{F}_2}\{\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_3 + \mathbf{e}_4, \ldots\}$. The cases $k = 0, 1$ are immediate. For $k \geq 2$, self-orthogonality gives $\mathbf{x} \cdot \mathbf{x} = 0$ for every $\mathbf{x} \in L$, so all strings are even and $\mathbf{1} \in L$. Choose $\mathbf{z} \in L \setminus \{\mathbf{0}, \mathbf{1}\}$. One of $\mathbf{z}, \mathbf{z} + \mathbf{1}$ has even weight in $[2, k]$. If its weight is larger than two, three support coordinates and one zero coordinate give a weight-four $\mathbf{s}$ with $\mathbf{z} \cdot \mathbf{s} = 1$ and $\mathrm{wt}(\mathbf{z} + \mathbf{s}) = \mathrm{wt}(\mathbf{z}) - 2$. Apply the corresponding transvection to the whole current code and repeat, then permute coordinates, to obtain $\mathbf{p} = \mathbf{e}_1 + \mathbf{e}_2 \in L$. Every codeword has equal first two coordinates. Projection deleting them has kernel $\mathrm{span}_{\mathbb{F}_2}\{\mathbf{p}\}$, image dimension $k - 1$, and preserves dot products (the two deleted contributions cancel), so its image is self-dual. Recurse and invert the accumulated realizable transvections and permutations. The resulting circuit has size $O(k^2)$. Closing the standard input pairs by I gives one internal lift per boundary string. For each $(\mathbf{a}_B)_j = 1$, attach a retained ordinary I-vertex to the corresponding circuit input through one X-edge and expose the other ordinary input of that vertex. Since $XI = X$, this attachment complements exactly that boundary bit. Leave the other circuit inputs unchanged. Applying these dressings independently at the exposed vertex inputs realizes every affine coset $\mathbf{a}_B + L$. □

For an orthogonal map A, define the self-dual graph subspace $\mathrm{Graph}(A) := \{(\mathbf{x}, A\mathbf{x}) : \mathbf{x} \in \mathbb{F}_2^n\}$. The converse of Lemma F.7.15 realizes its indicator signature. Attaching a signature to its input and closing one output pair then sums the signature over prescribed two-point orbits. This supplies the following Radon cards for the high-arity argument.

Realizable Radon access. We use the symmetric-space conventions, including their ambient-relative orthogonal notation.

Let g have even arity n. Call $\mathbf{v} \in (\mathbf{1}^n)^\perp \setminus \{\mathbf{0}^n, \mathbf{1}^n\}$ *accessible*. Such a $\mathbf{v}$ is isotropic. For $\epsilon \in \mathbb{F}_2$, define

$$H_{\mathbf{v},\epsilon} := \{\mathbf{x} \in \mathbb{F}_2^n : \mathbf{x} \cdot \mathbf{v} = \epsilon\}.$$

Translation by $\mathbf{v}$ preserves $H_{\mathbf{v},\epsilon}$. For $\mathbf{x} \in H_{\mathbf{v},\epsilon}$, write

$$[\mathbf{x}]_{\mathbf{v}} := \mathbf{x} + \mathrm{span}_{\mathbb{F}_2}\{\mathbf{v}\}$$

for its translation orbit, and let $H_{\mathbf{v},\epsilon}/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{v}\}$ denote the set of these orbits. Define

$$[\mathbf{u}]_{\mathbf{v}} \cdot [\mathbf{w}]_{\mathbf{v}} := \mathbf{u} \cdot \mathbf{w} \qquad (\mathbf{u}, \mathbf{w} \in \mathbf{v}^{\perp}).$$

This is a well-defined nondegenerate symmetric form on the translation space $\mathbf{v}^{\perp}/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{v}\}$. Each $H_{\mathbf{v},\epsilon}/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{v}\}$ is an affine space over that quotient. Define

$$\mathcal{R}_{\mathbf{v},\epsilon} g([\mathbf{x}]_{\mathbf{v}}) = g(\mathbf{x}) + g(\mathbf{x}+\mathbf{v}), \qquad [\mathbf{x}]_{\mathbf{v}} \in H_{\mathbf{v},\epsilon}/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{v}\}.$$

The right side is invariant under replacing $\mathbf{x}$ by $\mathbf{x}+\mathbf{v}$, so the quotient signature is well defined. The construction below identifies the quotient with $n-2$ ordinary boundary inputs. Thus the quotient sum is a realizable arity-reducing operation in the original network model. Write

$$O(n,2) := \{A \in \mathrm{GL}_n(\mathbb{F}_2) : A^{\mathsf{T}} A = I_n\}.$$

Lemma F.7.16 (Accessible Radon directions)**.** *Let g be a retained signature of even arity $n \geq 4$. For each $\mathbf{v} \in (\mathbf{1}^n)^{\perp} \setminus \{\mathbf{0}^n, \mathbf{1}^n\}$, one can construct and record distinct inputs $i_{\mathbf{v}}, j_{\mathbf{v}}$ and an orthogonal matrix $A_{\mathbf{v}} \in O(n,2)$ such that*

$$A_{\mathbf{v}}^{\mathsf{T}}(\mathbf{e}_{i_{\mathbf{v}}} + \mathbf{e}_{j_{\mathbf{v}}}) = \mathbf{v}.$$

Define

$$\mathbf{r}_{\mathbf{v}} := A_{\mathbf{v}}^{\mathsf{T}} \mathbf{e}_{i_{\mathbf{v}}}, \qquad \Pi_{\mathbf{v}} := \operatorname{span}\{\mathbf{r}_{\mathbf{v}}, \mathbf{v}\}, \qquad W_{\mathbf{v}} := \Pi_{\mathbf{v}}^{\perp},$$

and, for $\epsilon \in \mathbb{F}_2$,

$$\iota_{\mathbf{v},\epsilon} : W_{\mathbf{v}} \longrightarrow H_{\mathbf{v},\epsilon}/\operatorname{span}\{\mathbf{v}\}, \qquad \mathbf{w} \longmapsto [\epsilon \mathbf{r}_{\mathbf{v}} + \mathbf{w}]_{\mathbf{v}}.$$

Then $\mathbb{F}_2^n = \Pi_{\mathbf{v}} \perp W_{\mathbf{v}}$, and $\iota_{\mathbf{v},\epsilon}$ is an affine isometry from $W_{\mathbf{v}}$ to $H_{\mathbf{v},\epsilon}/\operatorname{span}\{\mathbf{v}\}$. An $O(n^2)$-size network using one copy of g, RM_8, and I/X realizes $\mathcal{R}_{\mathbf{v},\epsilon} g$ in the boundary coordinates supplied by $A_{\mathbf{v}}|_{W_{\mathbf{v}}}$, up to a nonzero scalar. The available directions are exactly the displayed vectors $\mathbf{v}$.

Proof. First take $A_{\mathbf{v}}, i_{\mathbf{v}}, j_{\mathbf{v}}$ as in the statement. Their construction is given at the end of the proof. The definitions imply $\Pi_{\mathbf{v}} = A_{\mathbf{v}}^{\mathsf{T}} \operatorname{span}\{\mathbf{e}_{i_{\mathbf{v}}}, \mathbf{e}_{j_{\mathbf{v}}}\}$. Then

$$\mathbf{r}_{\mathbf{v}} \cdot \mathbf{r}_{\mathbf{v}} = 1, \qquad \mathbf{r}_{\mathbf{v}} \cdot \mathbf{v} = 1, \qquad \mathbf{v} \cdot \mathbf{v} = 0.$$

Hence $\Pi_{\mathbf{v}}$ is a nondegenerate two-plane and

$$\mathbb{F}_2^n = \Pi_{\mathbf{v}} \perp W_{\mathbf{v}}, \qquad \mathbf{v}^{\perp} = \operatorname{span}_{\mathbb{F}_2}\{\mathbf{v}\} \oplus W_{\mathbf{v}}.$$

The isometry $A_{\mathbf{v}}$ carries $\Pi_{\mathbf{v}}$ to the output-coordinate plane $\operatorname{span}\{\mathbf{e}_{i_{\mathbf{v}}}, \mathbf{e}_{j_{\mathbf{v}}}\}$ and $W_{\mathbf{v}}$ to its ordinary-coordinate complement. We use $A_{\mathbf{v}}$ to identify these pairs of spaces and call the corresponding output positions the $\Pi_{\mathbf{v}}$-inputs and the $W_{\mathbf{v}}$-inputs. Consequently $\iota_{\mathbf{v},\epsilon}$ has the asserted domain, range, and isometry property.

For $A \in O(n,2)$, the graph subspace $\mathrm{Graph}(A)$ is self-dual in $\mathbb{F}_2^n \oplus \mathbb{F}_2^n$: it is isotropic because $\mathbf{x} \cdot \mathbf{y} + (A\mathbf{x}) \cdot (A\mathbf{y}) = 0$, and it has half the ambient dimension. Hence Lemma F.7.15 realizes a nonzero scalar multiple of

$$(\mathbf{x}, \mathbf{y}) \longmapsto \mathbf{1}[\mathbf{y} = A_{\mathbf{v}}\mathbf{x}]$$

by a realizable $\mathrm{RM}_8/I/X$-network. Attach one copy of g to the graph input using an effective I-kernel on each input connection: insert one retained ordinary I-vertex between two X-edges, so that $XIX = I$. The output before closing a pair is therefore $g(A_{\mathbf{v}}^{\mathsf{T}}\mathbf{y})$, with the graph's nonzero scalar. Close the ordered output pair $(i_{\mathbf{v}}, j_{\mathbf{v}})$ with either realizable kernel I or realizable kernel X. The two internal lifts differ by $\mathbf{v}$. The choice of I or X fixes $\mathbf{x} \cdot \mathbf{v} = \epsilon$. Under the isometry $A_{\mathbf{v}}|_{W_{\mathbf{v}}}$,

the remaining output coordinates are precisely the $W_{\mathbf{v}}$-coordinates in the inverse of $\iota_{\mathbf{v},\epsilon}$. Thus the resulting boundary is $\mathcal{R}_{\mathbf{v},\epsilon}g$, up to its nonzero scalar.

The weight-four transvections constructed in Lemma F.7.15, together with coordinate permutations, send every nonzero even $\mathbf{v} \neq \mathbf{1}^n$ to a weight-two vector. Write the resulting orthogonal map as B, with $B\mathbf{v} = \mathbf{e}_{i_\mathbf{v}} + \mathbf{e}_{j_\mathbf{v}}$, and record $A_\mathbf{v} = B$. Then $A_\mathbf{v}^\mathsf{T}(\mathbf{e}_{i_\mathbf{v}} + \mathbf{e}_{j_\mathbf{v}}) = B^{-1}B\mathbf{v} = \mathbf{v}$, as required. Conversely, orthogonal maps preserve parity and fix $\mathbf{1}^n$, so neither $\mathbf{0}^n$ nor $\mathbf{1}^n$ lies in the orbit of a weight-two vector. This proves necessity as well. The transvection construction and graph network use $O(n^2)$ vertices. The associated binary linear systems are solved by Gaussian elimination. □

F.7.5 A common flat-Lagrangian presentation

We now consider an arbitrary unresolved irreducible occurrence g. For arity at most eight, its smaller flat-Lagrangian cards are matching products, so the reconstruction above applies. At higher arity, Radon cards compare two quotient layers. A failure to combine them into an affine signature can be retained on at most eight inputs. After that obstruction is excluded, affine patching and dimension counts recover a flat-Lagrangian g.

Affine phase and translation tools. The following two facts test whether the quotient layers combine into an affine signature. They use the affine class defined in Section F.2.10.1.

For a nowhere-zero signature $f : \mathbb{F}_2^k \to \overline{\mathbb{Q}} \setminus \{0\}$, define its multiplicative third-difference ratio $\mathcal{T}_f : (\mathbb{F}_2^k)^4 \to \overline{\mathbb{Q}}^\times$ by

$$\mathcal{T}_f(\mathbf{z},\mathbf{u},\mathbf{v},\mathbf{w}) = \frac{f(\mathbf{z}+\mathbf{u}+\mathbf{v}+\mathbf{w})f(\mathbf{z}+\mathbf{u})f(\mathbf{z}+\mathbf{v})f(\mathbf{z}+\mathbf{w})}{f(\mathbf{z}+\mathbf{u}+\mathbf{v})f(\mathbf{z}+\mathbf{u}+\mathbf{w})f(\mathbf{z}+\mathbf{v}+\mathbf{w})f(\mathbf{z})}.$$

Here $\mathbf{z},\mathbf{u},\mathbf{v},\mathbf{w} \in \mathbb{F}_2^k$, and repeated directions are permitted. For fixed $\mathbf{u},\mathbf{v}$, let $D_{\mathbf{u},\mathbf{v}}$ be the multiplicative second-difference operator acting on the entire signature f:

$$(D_{\mathbf{u},\mathbf{v}}f)(\mathbf{z}) := \frac{f(\mathbf{z}+\mathbf{u}+\mathbf{v})f(\mathbf{z})}{f(\mathbf{z}+\mathbf{u})f(\mathbf{z}+\mathbf{v})}.$$

Lemma F.7.17 (Affine phase criterion)**.** *Let $f : \mathbb{F}_2^k \to \overline{\mathbb{Q}} \setminus \{0\}$. Then f belongs to $\mathscr{A}$ if and only if $\mathcal{T}_f(\mathbf{z},\mathbf{u},\mathbf{v},\mathbf{w}) = 1$ for all $\mathbf{z},\mathbf{u},\mathbf{v},\mathbf{w} \in \mathbb{F}_2^k$.*

Proof. Define the scalar $\lambda = f(\mathbf{0}^k)$, the scalars $A_j = f(\mathbf{e}_j)/\lambda$ for $j \in [k]$, and, for $1 \le j < \ell \le k$, the scalars

$$B_{j\ell} = \frac{\lambda f(\mathbf{e}_j + \mathbf{e}_\ell)}{f(\mathbf{e}_j)f(\mathbf{e}_\ell)}.$$

Substitution of $\mathbf{z} = \mathbf{0}^k$ and $\mathbf{u} = \mathbf{v} = \mathbf{w} = \mathbf{e}_j$ gives $A_j^4 = 1$. Taking $\mathbf{u} = \mathbf{v} = \mathbf{e}_j$ and $\mathbf{w} = \mathbf{e}_\ell$ gives $B_{j\ell}^2 = 1$. Moreover, the identity $\mathcal{T}_f(\mathbf{z},\mathbf{u},\mathbf{v},\mathbf{w}) = 1$ says exactly that

$$(D_{\mathbf{u},\mathbf{v}}f)(\mathbf{z}+\mathbf{w}) = (D_{\mathbf{u},\mathbf{v}}f)(\mathbf{z}).$$

Because $\mathbf{w}$ is arbitrary, $D_{\mathbf{u},\mathbf{v}}f$ is constant on $\mathbb{F}_2^k$. In particular,

$$(D_{\mathbf{e}_j,\mathbf{e}_\ell}f)(\mathbf{z}) = (D_{\mathbf{e}_j,\mathbf{e}_\ell}f)(\mathbf{0}^k) = B_{j\ell} \qquad (1 \le j < \ell \le k).$$

We now give the Hamming-weight induction explicitly. Extend the notation symmetrically by $B_{\ell j} = B_{j\ell}$. If $z_j = 0$, start with $f(\mathbf{e}_j)/f(\mathbf{0}^k) = A_j$ and add the basis vectors in $\mathrm{supp}(\mathbf{z})$ one at a time. The identity $D_{\mathbf{e}_j,\mathbf{e}_\ell}f = B_{j\ell}$, rearranged after the previously added vector $\mathbf{y}$, is

$$\frac{f(\mathbf{y}+\mathbf{e}_\ell+\mathbf{e}_j)}{f(\mathbf{y}+\mathbf{e}_\ell)} = B_{j\ell}\frac{f(\mathbf{y}+\mathbf{e}_j)}{f(\mathbf{y})}.$$

It follows by induction on $\mathrm{wt}(\mathbf{z})$ that

$$\frac{f(\mathbf{z}+\mathbf{e}_j)}{f(\mathbf{z})} = A_j \prod_{\ell \neq j} B_{j\ell}^{z_\ell} \qquad (z_j = 0). \tag{F.60}$$

We next prove the desired formula for $f(\mathbf{z})$, again by induction on $\mathrm{wt}(\mathbf{z})$. It is immediate for $\mathbf{z} = \mathbf{0}^k$. For $\mathbf{z} \neq \mathbf{0}^k$, choose $j \in \mathrm{supp}(\mathbf{z})$ and define the predecessor string $\mathbf{y} := \mathbf{z} + \mathbf{e}_j$, so $y_j = 0$ and $\mathrm{wt}(\mathbf{y}) = \mathrm{wt}(\mathbf{z}) - 1$. Applying the induction hypothesis to $f(\mathbf{y})$ and Equation (F.60) to $f(\mathbf{y}+\mathbf{e}_j)/f(\mathbf{y})$ yields

$$f(\mathbf{z}) = \lambda \prod_j A_j^{z_j} \prod_{j<\ell} B_{j\ell}^{z_j z_\ell}, \qquad A_j \in \mu_4, \quad B_{j\ell} \in \{1, -1\},$$

because the new factors are precisely A_j and $B_{j\ell}$ for $\ell \in \mathrm{supp}(\mathbf{y})$. This is the full-support normal form in Equation (F.5). Conversely, each linear factor and each even-quadratic factor cancels in the multiplicative third difference, so $\mathcal{T}_f = 1$. □

Lemma F.7.18 (Two-translate closure of the affine class)**.** *If $h \in \mathscr{A}$ and $\mathbf{w}$ is a Boolean translation of its input coordinates, then $h + T_{\mathbf{w}} h \in \mathscr{A}$ (with the zero signature allowed), where the translation operator $T_{\mathbf{w}}$ is defined by $(T_{\mathbf{w}} h)(\mathbf{x}) := h(\mathbf{x} + \mathbf{w})$.*

Proof. The zero case is immediate. Let $n := \mathrm{arity}(h)$. For $h \neq 0$, write

$$h(\mathbf{x}) = c\, \mathbf{1}[\mathbf{x} \in \mathbf{a} + L] i^{Q(\mathbf{x})}, \qquad c \in \overline{\mathbb{Q}}^\times, \quad \mathbf{a} \in \mathbb{F}_2^n, \quad L \leq \mathbb{F}_2^n,$$

where $Q : \mathbb{F}_2^n \to \mathbb{Z}_4$ is a quadratic phase polynomial with even mixed coefficients. If $\mathbf{w} \notin L$, every string in the union of the two supports has a unique form $\mathbf{a} + \mathbf{y} + t\mathbf{w}$, with $\mathbf{y} \in L$ and $t \in \mathbb{F}_2$. On this affine space,

$$(h + T_{\mathbf{w}} h)(\mathbf{a} + \mathbf{y} + t\mathbf{w}) = c\, i^{Q(\mathbf{a}+\mathbf{y})}.$$

The inverse coordinates $(\mathbf{y}, t)$ are affine, so the displayed exponent is a quadratic phase polynomial with even mixed coefficients. Thus the sum is affine. For $\mathbf{w} \in L$, expand $Q(\mathbf{x} + \mathbf{w}) - Q(\mathbf{x}) = c_{\mathbf{w}} + 2\ell_{\mathbf{w}}(\mathbf{x}) \pmod 4$, where $c_{\mathbf{w}} \in \mathbb{Z}_4$ is constant and $\ell_{\mathbf{w}} : \mathbb{F}_2^n \to \mathbb{F}_2$ is linear. This gives

$$h(\mathbf{x}) + h(\mathbf{x} + \mathbf{w}) = c\, i^{Q(\mathbf{x})} \big(1 + i^{c_{\mathbf{w}}} (-1)^{\ell_{\mathbf{w}}(\mathbf{x})}\big) \quad (\mathbf{x} \in \mathbf{a} + L).$$

For even $c_{\mathbf{w}}$ the bracket is a scalar times an affine hyperplane indicator. For odd $c_{\mathbf{w}}$ it is a nonzero scalar times a fourth-root linear phase. Restricting the phase polynomial to an affine subspace and multiplying by this linear phase preserve $\mathscr{A}$, proving the claim. The displayed difference formula follows by expanding the multilinear representative of Q. All nonconstant coefficients of that difference are even. □

For an n-ary signature h, use the translation operator just defined. Write Id for the identity on the signature space and define the orbit-sum operator for $\mathbf{s} \in \mathbb{F}_2^n$ by

$$\mathsf{P}_{\mathbf{s}}^+ := \mathrm{Id} + T_{\mathbf{s}}.$$

Thus $(\mathsf{P}_{\mathbf{s}}^+ h)(\mathbf{x}) = h(\mathbf{x}) + h(\mathbf{x}+\mathbf{s})$ is, for $\mathbf{s} \neq 0$, the sum over the two-point translation orbit $\{\mathbf{x}, \mathbf{x}+\mathbf{s}\}$. For accessible $\mathbf{s}$, the two quotient signatures $\mathcal{R}_{\mathbf{s},0} h$ and $\mathcal{R}_{\mathbf{s},1} h$ are called the 0- and 1-layer *plus cards* of h in direction $\mathbf{s}$. Together they are the two quotient layers of $\mathsf{P}_{\mathbf{s}}^+ h$.

Lemma F.7.19 (Low-arity flat-Lagrangian signatures are matching products)**.** *Let $L = L^\perp \leq \mathbb{F}_2^{2r}$, where $r \leq 3$. After a permutation of inputs, L is a direct sum of r two-coordinate repetition lines $\mathrm{span}_{\mathbb{F}_2}\{\mathbf{e}_{2j-1} + \mathbf{e}_{2j}\}$. Consequently every flat-Lagrangian signature of arity at most six is a product of projective I/X binaries on a perfect matching.*

Proof. Every vector of L has even weight, so $\mathbf{1} \in L^{\perp} = L$. If $r > 1$, choose $\mathbf{z} \in L \setminus \{\mathbf{0}, \mathbf{1}\}$. One of $\mathbf{z}, \mathbf{z}+\mathbf{1}$ has weight at most r, hence weight two when $2r \le 6$. Thus $\mathbf{e}_i + \mathbf{e}_j \in L$ for some pair i, j. Self-duality now gives $x_i = x_j$ for every $\mathbf{x} \in L$, so this repetition line splits orthogonally from L. Induction proves the assertion. Translating the affine support changes an equality factor only to its disequality coset. □

Consequently every flat-Lagrangian proper card of an occurrence of arity at most eight is already a matching product. A nonmatching flat-Lagrangian proper card can first occur at arity ten. The Radon localization argument handles this remaining case.

For linearly independent vectors $\mathbf{r}_1, \dots, \mathbf{r}_s$, a *dual family* is a family $\mathbf{r}_1^*, \dots, \mathbf{r}_s^*$ with $\mathbf{r}_i \cdot \mathbf{r}_j^* = \mathbf{1}[i = j]$. Pairing the radical means adding such a family, chosen orthogonal to a previously fixed nondegenerate complement. Indeed, when the ambient W is nondegenerate and $R = \operatorname{span}_{\mathbb{F}_2}\{\mathbf{r}_1, \dots, \mathbf{r}_s\}$, the pairing map

$$W \longrightarrow R^{\vee}, \qquad \mathbf{w} \longmapsto (\mathbf{r} \mapsto \mathbf{r} \cdot \mathbf{w}),$$

is surjective: otherwise a nonzero element of R would pair to zero with all of W. Choosing preimages gives a dual family. If $U = P \perp R$ with P nondegenerate and $R = \operatorname{rad}_W(U)$, the same argument inside $P^{\perp_W}$ makes the family orthogonal to P. The new vectors are independent modulo U, as pairing a linear combination with each $\mathbf{r}_i$ shows. On R and its dual family the Gram matrix is $\left(\begin{smallmatrix} 0 & I \\ I & A \end{smallmatrix}\right)$, which is nonsingular. Adding the family therefore gives a nondegenerate extension of dimension $\dim U + \dim R$.

Lemma F.7.20 (Localization from two flat-Lagrangian Radon cards)**.** *Let g have even arity $n \ge 10$, let $\mathbf{v}$ be accessible, and suppose that, for both $\epsilon \in \mathbb{F}_2$, the card $\mathcal{R}_{\mathbf{v},\epsilon} g$ is zero or flat-Lagrangian. Define, only for the analysis, the orbit-sum signature*

$$F_{\mathbf{v}}(\mathbf{x}) = g(\mathbf{x}) + g(\mathbf{x} + \mathbf{v}).$$

If $F_{\mathbf{v}} \notin \mathscr{A}$, an $O(n^2)$-size direct $\mathrm{RM}_8/I/X$-context using one copy of g produces a nonzero $H \notin \mathscr{A}$ of arity at most eight.

Proof. Fix the recorded data $(A_{\mathbf{v}}, i_{\mathbf{v}}, j_{\mathbf{v}}, \mathbf{r}_{\mathbf{v}}, \Pi_{\mathbf{v}}, W_{\mathbf{v}})$ from Lemma F.7.16, and write $W := W_{\mathbf{v}}$. Before the final output pair is closed, the graph realization gives the orthogonal decomposition

$$\mathbb{F}_2^n = \Pi_{\mathbf{v}} \perp W.$$

By the recorded identification, the two output positions $i_{\mathbf{v}}, j_{\mathbf{v}}$ are the $\Pi_{\mathbf{v}}$-inputs, while the remaining output positions are the W-inputs.

Under the isometry $W \simeq \mathbf{v}^{\perp}/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{v}\}$, the characteristic vector of W corresponds to the class of $\mathbf{1}^n$. This class is nonzero because accessibility excludes $\mathbf{v} = \mathbf{1}^n$. Thus W is even-dimensional and nonalternating.

For $\epsilon \in \mathbb{F}_2$, write the Radon card in the chosen quotient coordinates as

$$h_\epsilon(\mathbf{w}) := F_{\mathbf{v}}(\epsilon \mathbf{r}_{\mathbf{v}} + \mathbf{w}), \qquad \mathbf{w} \in W.$$

This is well defined because $F_{\mathbf{v}}$ is invariant under translation by $\mathbf{v}$. Each nonzero Radon card has the form

$$h_\epsilon(\mathbf{w}) = c_\epsilon \mathbf{1}[\mathbf{w} \in \mathbf{a}_\epsilon + L_\epsilon], \qquad c_\epsilon \neq 0, \quad \mathbf{a}_\epsilon \in W, \quad L_\epsilon = L_\epsilon^{\perp_W} \le W.$$

If $h_{1-\epsilon} = 0$ and $h_\epsilon \neq 0$, then, in the decomposition $\mathbb{F}_2^n = \Pi_{\mathbf{v}} \perp W$,

$$F_{\mathbf{v}}(\mathbf{x}) = c_\epsilon \mathbf{1}[\mathbf{x} \in \epsilon \mathbf{r}_{\mathbf{v}} + \mathbf{a}_\epsilon + (\operatorname{span}_{\mathbb{F}_2}\{\mathbf{v}\} \oplus L_\epsilon)].$$

Thus $F_{\mathbf{v}} \in \mathscr{A}$ whenever one Radon card is zero, so both cards are nonzero under the hypothesis. Restoring the layer coordinate gives an affine support exactly when $L_0 = L_1$. With that condition satisfied, the coefficients define a signature in $\mathscr{A}$ exactly when $c_1/c_0 \in \mu_4$.

If $L_0 \neq L_1$, take $\mathbf{s} \in L_0 \setminus L_1$. Since $\mathbf{s} \notin L_1^{\perp_W} = L_1$, choose $\mathbf{t} \in L_1$ with $\mathbf{s} \cdot \mathbf{t} = 1$. Elementary completion in a nondegenerate binary symmetric space gives an even-dimensional, nonalternating, nondegenerate subspace

$$U \supseteq \operatorname{span}\{\mathbf{s}, \mathbf{t}, \mathbf{a}_1 - \mathbf{a}_0\}, \qquad \dim U \leq 6.$$

To obtain this dimension bound, use the symmetric-space conventions. Put $S := \operatorname{span}\{\mathbf{s}, \mathbf{t}, \mathbf{a}_1 - \mathbf{a}_0\}$. Since $\mathbf{s} \cdot \mathbf{t} = 1$, the plane $\operatorname{span}\{\mathbf{s}, \mathbf{t}\}$ is nondegenerate, so $\dim \operatorname{rad}(S) \leq 1$. If $\operatorname{rad}(S) = 0$, put $N := S$. Otherwise write $\operatorname{rad}(S) = \operatorname{span}_{\mathbb{F}_2}\{\mathbf{r}\}$, choose a nondegenerate complement S_0 of $\operatorname{span}_{\mathbb{F}_2}\{\mathbf{r}\}$ in S, and choose $\mathbf{d} \in W$ satisfying

$$\mathbf{r} \cdot \mathbf{d} = 1, \qquad \mathbf{d} \in S_0^{\perp}.$$

Put $N := S + \operatorname{span}_{\mathbb{F}_2}\{\mathbf{d}\}$. In either case N is nondegenerate and $\dim N \leq 4$. If $\dim N$ is odd, its nondegenerate orthogonal complement has odd dimension and therefore contains an anisotropic vector. Adding that orthogonal line makes N even-dimensional without exceeding dimension four. If the resulting even-dimensional N is alternating, then $N^{\perp}$ is nonalternating because W is. Add a nonalternating nondegenerate plane from $N^{\perp}$. The resulting U has all the asserted properties and dimension at most six.

If instead $L_0 = L_1$ and $c_1/c_0 \notin \mu_4$, every vector in an even-dimensional nondegenerate nonalternating symmetric binary space lies in a nonalternating nondegenerate plane. For a nonzero isotropic vector $\mathbf{z}$, choose $\mathbf{d}$ with $\mathbf{z} \cdot \mathbf{d} = 1$ and $\mathbf{d} \cdot \mathbf{d} = 1$. For an anisotropic $\mathbf{z}$, choose $\mathbf{d}$ with $\mathbf{z} \cdot \mathbf{d} = 1$ and $\mathbf{d} \cdot \mathbf{d} = 0$. For $\mathbf{z} = 0$, choose any nonalternating nondegenerate plane. We may therefore choose an even-dimensional, nonalternating, nondegenerate

$$U \supseteq \operatorname{span}\{\mathbf{a}_1 - \mathbf{a}_0\}, \qquad \dim U \leq 2.$$

Put

$$P := U^{\perp_W}, \qquad W = U \perp P,$$

and let π_U, π_P be the corresponding orthogonal projections. The space P is even-dimensional and nondegenerate. By the symmetric-binary Witt decomposition recalled in the symmetric-space conventions, P contains a half-dimensional totally isotropic subspace J. Equivalently,

$$J = J^{\perp_P} \leq P.$$

Because $\mathbf{a}_1 - \mathbf{a}_0 \in U$, the two shifts have the same P-projection. Call it $\mathbf{p}$. Choose an isometry $\phi : U \to \mathbb{F}_2^{\dim U}$, and define the affine relation

$$\Gamma_{\mathbf{p}} := \{(\mathbf{u} + \mathbf{p} + \mathbf{j}, \phi(\mathbf{u})) : \mathbf{u} \in U,\ \mathbf{j} \in J\} \subseteq W \oplus \mathbb{F}_2^{\dim U}.$$

Its translation space is self-dual: it has half the ambient dimension, and

$$(\mathbf{u} + \mathbf{j}) \cdot (\mathbf{u}' + \mathbf{j}') + \phi(\mathbf{u}) \cdot \phi(\mathbf{u}') = \mathbf{j} \cdot \mathbf{j}' = 0$$

for $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{j}, \mathbf{j}' \in J$. Thus after transporting the W-coordinate through $A_{\mathbf{v}}|_{W_{\mathbf{v}}}$, Lemma F.7.15 realizes the corresponding signature

$$(\mathbf{w}, \mathbf{z}) \longmapsto \mathbf{1}[(\mathbf{w}, \mathbf{z}) \in \Gamma_{\mathbf{p}}]$$

by a realizable $O(n^2)$-size $\mathrm{RM}_8/I/X$-network. Apply this signature to the W-inputs and leave the two $\Pi_{\mathbf{v}}$-inputs external.

When $L_0 \neq L_1$, the boundary points induced by $\mathbf{a}_0, \mathbf{a}_0+\mathbf{s}, \mathbf{a}_1$ survive but the fourth parallelogram point induced by $\mathbf{a}_1 + \mathbf{s}$ does not. Otherwise there would be $\mathbf{z} \in L_1$ with $\pi_U \mathbf{z} = \mathbf{s}$, and

$$0 = \mathbf{z} \cdot \mathbf{t} = (\pi_U \mathbf{z}) \cdot \mathbf{t} = \mathbf{s} \cdot \mathbf{t} = 1.$$

The output therefore retains a nonaffine parallelogram on at most $\dim U + 2 \leq 8$ inputs. When $L_0 = L_1$, the two fibers have equal size because their support translation spaces agree. Thus neither cancellation nor projection changes the ratio c_1/c_0 outside μ_4, and the output again lies outside $\mathscr{A}$.

Let H be the contraction of this realizable context with g, and let H^+ be the contraction with $F_{\mathbf{v}} = \mathsf{P}^+_{\mathbf{v}} g$. Order the boundary with the two retained $\Pi_{\mathbf{v}}$-inputs first and put

$$\mathbf{w}_\partial := (1, 1, \mathbf{0}_{\dim U}).$$

Indeed, translation by $\mathbf{v}$ changes the graph output by $\mathbf{e}_{i_{\mathbf{v}}} + \mathbf{e}_{j_{\mathbf{v}}}$, so it flips exactly the two $\Pi_{\mathbf{v}}$-inputs and acts trivially on the remaining boundary inputs. Consequently, with the same scalar on both terms,

$$H^+ = \mathsf{P}^+_{\mathbf{w}_\partial} H.$$

The missing parallelogram or forbidden layer ratio proves $H^+ \notin \mathscr{A}$. By Lemma F.7.18, $H \in \mathscr{A}$ would imply $H^+ = \mathsf{P}^+_{\mathbf{w}_\partial} H \in \mathscr{A}$, a contradiction. Therefore $H \neq 0$, $H \notin \mathscr{A}$, and its arity is at most eight. □

In the relative induction, the localized nonaffine signature supplied by the preceding lemma is excluded by the already closed lower-arity cases. Hence every accessible orbit sum $F_{\mathbf{v}}$ is affine. The next lemma patches these hyperplane-wise affine conclusions into a global affine conclusion for g.

Lemma F.7.21 (Affine hyperplane patching)**.** *Let $n \geq 10$ be even, and let $g : \mathbb{F}_2^n \to \overline{\mathbb{Q}}$. For every accessible $\mathbf{v}$, define the orbit-sum signature $F_{\mathbf{v}} := \mathsf{P}^+_{\mathbf{v}} g$. If $F_{\mathbf{v}} \in \mathscr{A}$ for every such $\mathbf{v}$, then $g \in \mathscr{A}$.*

Proof. We first record the Walsh-invariance of $\mathscr{A}$ used below. The Walsh kernel $(-1)^{\boldsymbol{\xi}\cdot\mathbf{x}}$ has an allowed quadratic modulo-four phase, so its product with an affine signature is affine in the joint variables $(\boldsymbol{\xi}, \mathbf{x})$. Gaussian-eliminate the affine support constraints and sum the remaining internal variables one at a time. If the current phase depends on a free bit y as

$$i^{\alpha y}(-1)^{y\ell}, \qquad \alpha \in \mathbb{Z}_4,$$

where ℓ is an affine linear form in the remaining variables and is independent of y, then summing over y either imposes the affine equation $\ell = \alpha/2 \pmod 2$ when α is even, or contributes a nonzero scalar times a fourth-root phase of the affine linear form ℓ when α is odd. Thus every elimination remains in $\mathscr{A}$, proving that the Walsh transform preserves $\mathscr{A}$. The converse follows from Walsh inversion.

Use the Walsh coordinates $\widetilde{g} = (P^{-1})^{\otimes n} g$ from Equation (F.24). Since

$$(\widetilde{F}_{\mathbf{v}})(\boldsymbol{\xi}) = (1 + (-1)^{\boldsymbol{\xi}\cdot\mathbf{v}})(\widetilde{g})(\boldsymbol{\xi}),$$

each restriction $(\widetilde{g})|_{\mathbf{v}^\perp}$ is affine or zero. The claim is immediate when $\widetilde{g} = 0$, so assume otherwise and choose $\boldsymbol{\zeta} \in \operatorname{supp}(\widetilde{g})$.

Given $\mathbf{x}, \mathbf{y} \in \operatorname{supp}(\widetilde{g})$, the space

$$E \cap \operatorname{span}\{\boldsymbol{\zeta}, \mathbf{x}, \mathbf{y}\}^\perp, \qquad E = \mathbf{1}^\perp,$$

has dimension at least $n - 4$, hence contains accessible $\mathbf{v}$. Affinity on $\mathbf{v}^\perp$ puts $\mathbf{x} + \mathbf{y} + \boldsymbol{\zeta}$ in the support, so $\operatorname{supp}(\widetilde{g})$ is a coset. Write it as $\boldsymbol{\zeta} + U$.

Choose an isomorphism $\beta : \mathbb{F}_2^{\dim U} \to U$ and define the nowhere-zero logical coefficient tensor

$$\psi(\mathbf{u}) := (\widetilde{g})(\boldsymbol{\zeta} + \beta\mathbf{u}).$$

Fix arbitrary

$$\mathbf{u}_0, \mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3 \in \mathbb{F}_2^{\dim U},$$

where $\mathbf{u}_0$ is the base point and $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are the three directions of a multiplicative third-difference cube of ψ. The four vectors

$$\boldsymbol{\zeta} + \beta\mathbf{u}_0, \quad \beta\mathbf{u}_1, \quad \beta\mathbf{u}_2, \quad \beta\mathbf{u}_3$$

span at most four dimensions. Hence

$$E \cap \operatorname{span}\{\boldsymbol{\zeta} + \beta\mathbf{u}_0, \beta\mathbf{u}_1, \beta\mathbf{u}_2, \beta\mathbf{u}_3\}^{\perp}$$

has dimension at least $n - 5 \geq 5$ and contains an accessible direction $\mathbf{s}$. The entire cube lies in $\mathbf{s}^{\perp}$, where the nonzero restriction of $\widetilde{g}$ is affine. Hence

$$\mathcal{T}_{\psi}(\mathbf{u}_0, \mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3) = 1.$$

By Lemma F.7.17, ψ has an allowed quadratic modulo-four phase. Consequently $\widetilde{g} \in \mathscr{A}$, and the Walsh-invariance proved above gives $g \in \mathscr{A}$. □

Affine patching gives $g \in \mathscr{A}$, but does not yet give the required flat-Lagrangian form. The final recognition lemma uses the dimensions and phases of all realizable Radon cards to force a self-dual support translation space and constant coefficient on that support.

Lemma F.7.22 (Affine-to-flat-Lagrangian Radon recognition)**.** *Let $0 \neq h \in \mathscr{A}$ have even arity $2r \geq 8$. Suppose that for every accessible direction $\mathbf{v}$ and both $\epsilon \in \mathbb{F}_2$, the realizable card $\mathcal{R}_{\mathbf{v},\epsilon}h$ is zero or flat-Lagrangian. Then h is flat-Lagrangian.*

Proof. Write the affine signature h in standard normal form as

$$h(\mathbf{x}) = c\mathbf{1}[\mathbf{x} \in \mathbf{a} + U]i^{Q(\mathbf{x})}, \qquad d := \dim U.$$

Here $c \neq 0$ is the scale, $\mathbf{a} + U$ is the affine support, Q is its quadratic phase polynomial, and d is the support dimension. Define the even hyperplane $E := \mathbf{1}^{\perp}$. For $\mathbf{v} \notin U$ there is no cancellation: the nonempty card has support dimension d if $\mathbf{v} \in U^{\perp}$, and both have dimension $d - 1$ otherwise. A flat-Lagrangian card has support dimension $r - 1$.

Neither $E \subseteq U$ nor $E \subseteq U^{\perp}$. Suppose first that $E \subseteq U$. Choose $\mathbf{p}, \mathbf{q} \in E$ so that $\mathbf{p}, \mathbf{q}, \mathbf{p} + \mathbf{q}$ are accessible. They exist because $\dim E = 2r - 1 \geq 7$. If

$$\mathsf{P}^{+}_{\mathbf{p}}h = \mathsf{P}^{+}_{\mathbf{q}}h = 0,$$

then $T_{\mathbf{p}}h = T_{\mathbf{q}}h = -h$, so $T_{\mathbf{p}+\mathbf{q}}h = h$ and $\mathsf{P}^{+}_{\mathbf{p}+\mathbf{q}}h = 2h \neq 0$. Thus at least one Radon card arising from the three directions $\mathbf{p}, \mathbf{q}, \mathbf{p} + \mathbf{q}$ is nonzero. Let $\mathbf{s} \in \{\mathbf{p}, \mathbf{q}, \mathbf{p} + \mathbf{q}\}$ be its direction. Passing from h to such a card can lower support dimension by at most one when restricting to its layer hyperplane and by one when quotienting by $\operatorname{span}_{\mathbb{F}_2}\{\mathbf{s}\}$. On an overlapping support orbit, factoring out $h(\mathbf{x})$ leaves

$$1 + \zeta(-1)^{\ell(\mathbf{x})}, \qquad \zeta \in \mu_4.$$

Here ℓ is a binary affine-linear form on the current support. For $\zeta \in \{\pm i\}$ this factor is nowhere zero. For $\zeta \in \{\pm 1\}$ its nonzero set is empty, the whole current support, or one affine hyperplane.

Since the selected card is nonzero, cancellation therefore lowers support dimension by at most one more. Hence its support dimension is at least

$$d - 3 \geq 2r - 4 > r - 1,$$

contradicting the support dimension of a flat-Lagrangian card.

If instead $E \subseteq U^\perp$, then $d \leq 1$. Choose accessible $\mathbf{v} \in E \setminus U$. Here $\mathbf{v} \in U^\perp \setminus U$, so there is no cancellation and the nonzero card has support dimension d, whereas a flat-Lagrangian card has dimension $r - 1 \geq 3$, again a contradiction.

Thus both $E \cap U$ and $E \cap U^\perp$ are proper subspaces of E. Two proper subspaces cannot cover E. Since $\dim E \geq 7$, we may also avoid $\mathbf{0}, \mathbf{1}$ and choose accessible $\mathbf{v} \notin U \cup U^\perp$. Its two cards have dimension $d - 1 = r - 1$, so $d = r$.

No accessible vector lies in $U^\perp \setminus U$, since it would give a dimension-r card. Put $A_0 := E \cap U^\perp$. Then

$$A_0 \setminus \{\mathbf{0}, \mathbf{1}\} \subseteq U.$$

If $\mathbf{1} \notin A_0$, or if $\mathbf{1} \in A_0 \cap U$, this gives $A_0 \subseteq U$. If instead $\mathbf{1} \in A_0 \setminus U$, choose $\mathbf{z} \in A_0 \setminus \{\mathbf{0}, \mathbf{1}\}$, which is possible because $\dim A_0 \geq r - 1 \geq 3$. Then both $\mathbf{z}$ and $\mathbf{z} + \mathbf{1}$ are accessible and lie in U by the preceding display, forcing $\mathbf{1} \in U$, a contradiction. Hence $A_0 \subseteq U$ in every case. If $U^\perp \subseteq E$, dimensions give $U = U^\perp$. Otherwise $\dim A_0 = r - 1$. Since $A_0 \subseteq U$ and $\dim U = r$, choose $\mathbf{u} \in U \setminus A_0$. Then $U = A_0 \oplus \operatorname{span}_{\mathbb{F}_2}\{\mathbf{u}\}$, and $A_0 \subseteq U^\perp$ makes $\mathbf{u}$ orthogonal to A_0. If $\mathbf{u}$ were even, then $\mathbf{u} \cdot \mathbf{u} = 0$, so $\mathbf{u} \in U^\perp$ and $U \subseteq U^\perp$. Equal dimensions would give $U = U^\perp$, contrary to the present case. Thus $\mathbf{u}$ is odd and

$$U = A_0 \oplus \operatorname{span}_{\mathbb{F}_2}\{\mathbf{u}\}, \qquad \mathbf{u} \cdot \mathbf{u} = 1.$$

For accessible $\mathbf{s} \in A_0$, put

$$\epsilon_{\mathbf{s}} := \mathbf{a} \cdot \mathbf{s}.$$

Because $\mathbf{s} \in U^\perp$, the support $\mathbf{a} + U$ lies wholly in $H_{\mathbf{s},\epsilon_{\mathbf{s}}}$. The Radon card on the other layer is identically zero. The translation space of the supported card $\mathcal{R}_{\mathbf{s},\epsilon_{\mathbf{s}}} h$ is $U/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{s}\}$, whose orthogonal is $U^\perp/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{s}\}$. These differ because $U \neq U^\perp$. Write its phase difference as

$$Q(\mathbf{x} + \mathbf{s}) - Q(\mathbf{x}) = \alpha_{\mathbf{s}} + 2\ell_{\mathbf{s}}(\mathbf{x}) \pmod 4, \qquad \alpha_{\mathbf{s}} \in \mathbb{Z}_4,$$

where $\ell_{\mathbf{s}}$ is affine linear on $\mathbf{a} + U$. Thus

$$\frac{h(\mathbf{x} + \mathbf{s})}{h(\mathbf{x})} = i^{\alpha_{\mathbf{s}}}(-1)^{\ell_{\mathbf{s}}(\mathbf{x})}.$$

If $\alpha_{\mathbf{s}}$ is odd, the supported plus card is nonzero on the full quotient and has the non-self-dual translation space $U/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{s}\}$. If $\alpha_{\mathbf{s}}$ is even and $\ell_{\mathbf{s}}$ is nonconstant, its nonzero support is a codimension-one hyperplane of dimension $r - 2$. If $\ell_{\mathbf{s}}$ is constant, the card is either zero or again has the full non-self-dual translation space. The hypothesis therefore forces

$$\mathcal{R}_{\mathbf{s},\epsilon_{\mathbf{s}}} h = 0, \qquad \text{equivalently} \qquad \mathsf{P}^+_{\mathbf{s}} h = 0.$$

Because $\dim A_0 = r - 1 \geq 3$, choose a two-dimensional subspace $B_0 \leq A_0$ with $\mathbf{1} \notin B_0$, and let $\mathbf{p}, \mathbf{q}$ be a basis of B_0. Then $\mathbf{p}, \mathbf{q}, \mathbf{p} + \mathbf{q}$ are all accessible, and $\mathsf{P}^+_{\mathbf{p}} h = \mathsf{P}^+_{\mathbf{q}} h = 0$, whereas $T_{\mathbf{p}+\mathbf{q}} = T_{\mathbf{p}} T_{\mathbf{q}}$ gives $\mathsf{P}^+_{\mathbf{p}+\mathbf{q}} h = 2h \neq 0$, a contradiction. Therefore $U = U^\perp$.

For accessible $\mathbf{s} \in U = U^\perp$, put $\epsilon_{\mathbf{s}} := \mathbf{a} \cdot \mathbf{s}$. The support lies wholly in $H_{\mathbf{s},\epsilon_{\mathbf{s}}}$, so $\mathcal{R}_{\mathbf{s},\epsilon_{\mathbf{s}}} h$ is the unique potentially nonzero Radon card. Use the same phase-difference form as above. If $\alpha_{\mathbf{s}}$ is even

and $\ell_{\mathbf{s}}$ is nonconstant, the nonzero part of the plus card has quotient-support dimension $r-2$, so it cannot be flat-Lagrangian. In every remaining case the plus card is either zero or has the full self-dual quotient support $U/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{s}\}$. By the flat-Lagrangian hypothesis it is then constant on that support. Taking $c_{\mathbf{s}}=0$ in the zero case, we obtain

$$h(\mathbf{x})+h(\mathbf{x}+\mathbf{s})=c_{\mathbf{s}} \qquad (\mathbf{x}\in\mathbf{a}+U).$$

Since $\dim U=r\geq 4$ and $U=U^{\perp}$, choose a two-dimensional subspace $B_1\leq U$ with $\mathbf{1}\notin B_1$, and let $\mathbf{p},\mathbf{q}$ be a basis. Then $\mathbf{p},\mathbf{q},\mathbf{p}+\mathbf{q}$ are all accessible. Using $T_{\mathbf{p}+\mathbf{q}}=T_{\mathbf{p}}T_{\mathbf{q}}$ to compare their three equations gives

$$2h(\mathbf{x})=c_{\mathbf{p}+\mathbf{q}}-c_{\mathbf{q}}+c_{\mathbf{p}},$$

so h is constant on $\mathbf{a}+U$, hence flat-Lagrangian. □

F.7.6 Completion of Reed–Muller rigidity

Completion of Theorem F.7.1. We first close the initial minimum-arity nonbinary irreducible signature f. If it has odd arity, apply Theorem F.2.9 to $K\mathcal{F}$. Its hardness conclusion pulls back through the holographic K-coordinate equivalence, while its tractable conclusion is exactly $\mathsf{Tract}(K\mathcal{F})$. If $\operatorname{arity}(f)=4$, Lemma F.7.5 and the arity-four reduction give either a resolved leaf or a strict stable successor, and the complete batch is resolved by Theorem F.5.22, Corollary F.4.8, Lemmas F.5.4 and F.5.12, and Definition F.5.10. Hence, on the sole remaining initial branch, f has even arity $n\geq 6$, and every proper card is zero or a nonsingular current-group matching signature.

Apply Lemmas F.7.6 and F.7.7. If $W_f\neq\varnothing$, then Lemma F.7.9 gives either the realizable Even_4 equality anchor constructed in its proof or a proper tensor factorization. After factor closure and group recomputation, both alternatives are resolved by Lemmas F.3.4, F.3.15 and F.5.12 and Definition F.5.10. Thus the remaining branch has $W_f=\varnothing$. Now Lemmas F.7.10 and F.7.11 give

$$n=8, \qquad f(\mathbf{x})=c\mathbf{1}[\mathbf{x}\in\mathbf{a}+RM(1,3)]$$

after a permutation of inputs, for a recorded $c\neq 0$. Available realizable single-input X-dressings remove $\mathbf{a}$, so the retained set contains the standard core RM_8. Regenerate the complete deck of every single-input dressing of this core by a safe binary. If one leaves the matching branch, its complete batch is resolved by the same case analysis. Otherwise Lemma F.7.14 gives

$$G=\{[I],[X]\}\cong C_2.$$

This proves the asserted description of the remaining initial case.

Retain this RM_8 and normalized I/X while resolving the remaining signature set. After adding signatures, consider the complete transfer group for the resulting set and process the complete decks of single-input dressings of RM_8 by safe binaries before applying the argument again. A decrease of ν or a strict group enlargement is charged to the stable-state measure Ξ_{full} of Lemma F.5.12. On the unchanged remaining branch the preceding argument using Lemma F.7.14 reapplies and the complete transfer group remains $C_2^{(X)}$.

For an unchanged stable state, a second measure controls the remaining irreducible occurrences and their lower-arity outputs. It records the arity, the number of unprocessed network outputs, and the remaining factorization potential. Let $\mathcal{W}$ be the finite multiset of unresolved irreducible occurrences in the retained factorization trees. When an occurrence u, carrying h_u, is first added to $\mathcal{W}$, fix the following finite reduction family $\mathscr{U}_{\mathrm{RM}}(u)$:

1. the complete $C_2^{(X)}$ matching-deck universe of h_u, with every oriented proper contraction and input dressing by I/X binaries, together with every ordered two-copy I/X-integration network displayed in Lemma F.7.9, for all input, orientation, and closing-chain sign choices;

2. for every accessible $\mathbf{v}$ and ϵ, the fixed realizable Radon network of Lemma F.7.16, including its recorded data $(A_{\mathbf{v}}, i_{\mathbf{v}}, j_{\mathbf{v}}, \mathbf{r}_{\mathbf{v}}, \Pi_{\mathbf{v}}, W_{\mathbf{v}})$, and, for every finite parameter tuple $(\mathbf{v}, U, J, \mathbf{p}, \phi)$ allowed in Lemma F.7.20, its realizable localization network applied to h_u using that same recorded Radon data;

3. the complete factorization tree of every output in the preceding two items.

Each network-output index specifies the ordered network, its boundary order, scalar, and direct gadget construction. Each attached factor index records its canonical signature and Turing reduction. The family is finite because the input set and all of its binary subspaces, vectors, and isometries are finite. Every individual listed network has the size bounds proved in the cited lemmas. These choices depend on the fixed retained set, not on the input Holant graph. Once selected, each network replaces a vertex by a fixed-size gadget, as required for the stated reductions. Partition it as

$$\mathscr{U}_{\mathrm{RM}}(u) = Q_{u,\mathrm{done}} \sqcup Q_u.$$

For $q \in Q_u$, let $\mathcal{W}_q$ be the multiset of unresolved occurrences in its attached factorization tree, and set

$$\Phi_u := \sum_{q \in Q_u} \Phi(\mathcal{W}_q),$$

using Equation (F.9). Define

$$\Theta_{\mathrm{RM}}(\mathcal{W}) = \{\{(\mathrm{arity}(h_u), |Q_u|, \Phi_u) : u \in \mathcal{W}\}\},$$

using lexicographic order on each triple and the well-founded multiset extension of that order. Explicitly, replacing one triple by any finite multiset of strictly smaller triples is a strict decrease.

Proving that an occurrence is flat-Lagrangian removes its triple. If the selected occurrence factors, its triple is replaced by triples of strictly smaller arity. Every batch used below starts from a fixed index: the matching-deck and integration steps use item 1, the Radon steps use the first family in item 2, and the two-layer localization step uses the second family in item 2. Completing the output indexed by q moves q to $Q_{u,\mathrm{done}}$, while an intermediate genuine factor split strictly decreases its summand $\Phi(\mathcal{W}_q)$. Any newly opened child occurrence has smaller arity: proper and Radon cards have arity at most $\mathrm{arity}(h_u) - 2$, localization outputs have arity at most eight when they are invoked from arity at least ten, and every proper factor is smaller than its parent. Thus the old parent triple is replaced by a strictly smaller parent triple together with child triples whose first coordinate is smaller. This strictly decreases Θ_{RM}. No fixed-(H, ν) batch creates a fresh unresolved index, because all of its possible outputs and complete factorization trees were included in $\mathscr{U}_{\mathrm{RM}}(u)$ when the family was fixed. Consequently this outer induction, rather than a multiset of arities alone, also controls passage to another occurrence of the same arity.

Run induction on Θ_{RM}. Before processing an occurrence, recursively close all its lower-arity proper cards and factors. Unary, odd-arity, and quaternary outputs are handled by Theorems F.2.9 and F.5.22. Hence consider a remaining irreducible $g \notin \mathsf{FLag}$ of even arity at least six.

First suppose that every surviving proper card of g is zero or a current-$C_2^{(X)}$ matching signature. The reconstruction proofs Lemmas F.7.6, F.7.7 and F.7.9 to F.7.11 use minimum arity only to exclude a surviving proper nonmatching card or nonbinary factor. Here the regenerated deck and prior closure of smaller outputs give that exclusion, so the same networks and coefficient identities apply

to g. They yield a resolved Even_4 equality anchor, a proper factorization, or the flat-Lagrangian arity-eight indicator of $RM(1,3)$. Thus no occurrence outside FLag with an all-matching proper deck survives.

Otherwise some nonzero proper card is nonmatching. Factor closure and prior closure nevertheless make every proper card zero or a tensor product of already proved flat-Lagrangian factors, hence zero or flat-Lagrangian. If $6 \le \mathrm{arity}(g) \le 8$, every such card has arity at most six, and Lemma F.7.19 makes it an I/X-matching signature after a permutation of inputs, contradicting the present case. The appearance of a nonmatching flat-Lagrangian card identifies the remaining case for Radon contractions, where $\mathrm{arity}(g) = n \ge 10$.

By Lemma F.7.16, every Radon card is realizable, has its fixed index in item 2, and has arity $n-2$. Processing that output includes its complete factor closure. Every surviving factor has already been proved flat-Lagrangian. Tensor closure therefore makes each card zero or flat-Lagrangian. Fix an accessible $\mathbf{v}$. If the analytic orbit sum

$$F_{\mathbf{v}} = \mathsf{P}^{+}_{\mathbf{v}} g$$

were outside $\mathscr{A}$, Lemma F.7.20 would produce a realizable $H \notin \mathscr{A}$ of arity at most eight. Its parameter tuple and realizable context have a fixed item 2 index, whose attached factorization tree is now processed. Factor closure of H must expose a nonaffine irreducible factor, since a tensor product of affine factors is affine. That factor has smaller arity and is processed earlier in the induction, contradicting the remaining branch. Hence $F_{\mathbf{v}} \in \mathscr{A}$ for every accessible $\mathbf{v}$.

It follows from Lemma F.7.21 that $g \in \mathscr{A}$. All its realizable Radon cards are zero or flat-Lagrangian, so Lemma F.7.22 gives $g \in \mathsf{FLag}$, contradicting the choice of a surviving occurrence outside FLag. Therefore every retained irreducible occurrence is eventually proved flat-Lagrangian.

Every concrete terminal, anchor, factor, or group-change batch above is closed before the next fixed index is selected. A stable-tuple transition strictly decreases Ξ_{full}. A step on the unresolved occurrences at an unchanged tuple strictly decreases Θ_{RM}. The two well-founded measures control changes of stable state and progress at a fixed state, respectively.

All membership conclusions were proved in the same retained normalized coordinates. The X-edge is itself flat-Lagrangian, and tensor products of flat-Lagrangian signatures remain flat-Lagrangian by taking direct sums of their self-dual support translation spaces. Factor closure reconstructs every member of the signature set from its proved irreducible factors. Hence the entire retained set and its edge have one common flat-Lagrangian presentation, which is exactly $\mathsf{Close}_{\{\mathsf{FLag}\}}(\mathcal{F}, G, f)$. □

F.8 Klein-Four Matching Decks

The two marked Klein-four groups lead to different conclusions from the matching deck. In the standard Pauli case, pair contractions reduce a signature that is not a scalar multiple of a real signature to arity four. Factor closure then gives a resolved complexity leaf or an outer deck successor. If every retained signature is a scalar multiple of a real signature, the real-valued Holant dichotomy applies to the whole set.

For the exotic Klein-four group, the matching condition instead forces a proper tensor factor in every even arity at least six. This excludes a distinguished irreducible signature at those arities, leaving the arity-four reduction. The six-input step uses the finite certificate in Appendix F.A.4.

Theorem F.8.1 (Klein-four relative rigidity)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, G, f), \qquad \mathrm{arity}(f) = \nu(\mathcal{F}), \qquad G \in \{V_4^{(\mathrm{std})}, V_4^{(\mathrm{ex})}\}.$$

In the standard Pauli case $G = V_4^{(\mathrm{std})}$, assume additionally that $\nu(\mathcal{F}) \geq 6$. Then $\mathsf{Close}_{\{\mathscr{A}\}}(\mathcal{F}, G, f)$.

The two cases are proved in Theorems F.8.3 and F.8.10. A tractable case satisfying Tract completes the branch under Equation (F.21).

Proof strategy. In the standard Pauli case, we compare a signature with its complex conjugate at every input pair and each of the four Bell labels. Above arity four, linear dependence of each pair of corresponding cards implies linear dependence of the two signatures. Hence failure to be real up to a scalar passes to a pair card and, by iteration, to arity four.

For the exotic group, the six-input classification first forces all nonzero cards at a fixed pair to use one residual matching. Unless a proper tensor factor already exists, the four labels on each residual edge must then vary by a permutation. Contracting across two such edges produces a coefficient support incompatible with a matching product. Two choices of contraction are needed at arity eight, and one suffices at arity at least ten.

F.8.1 The $V_4^{(\mathrm{std})}$ form: realizable Pauli contractions

Let $\mathcal{F}$ be retained and factor-closed, with the supplied equality I and complete transfer group $V_4^{(\mathrm{std})}$. For a Bell label $\boldsymbol{\gamma} = (d, \ell)^{\mathsf{T}} \in \mathbb{F}_2^2$, let $B_{\boldsymbol{\gamma}} = B_{d,\ell}$ be the ordered Bell signature defined in Section F.2.9. Its chosen realizable representative is $\beta_{\boldsymbol{\gamma}} B_{\boldsymbol{\gamma}}$, as in Section F.2.6. We divide out the known nonzero scalar $\beta_{\boldsymbol{\gamma}} \in \overline{\mathbb{Q}}^{\times}$ in the formulas below and use bilinear contractions, without complex conjugation. For an ordered pair p, define

$$C_p^{\boldsymbol{\gamma}}(h) := \partial_p^{B_{\boldsymbol{\gamma}}} h, \qquad \boldsymbol{\gamma} \in \mathbb{F}_2^2,$$

and omit the argument only when the parent signature is unambiguous. For a list $\mathbf{p} = (p_1, \ldots, p_s)$ of disjoint ordered pairs and $\boldsymbol{\eta} = (\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_s)$, put

$$C_{\mathbf{p}}^{\boldsymbol{\eta}}(h) := C_{p_s}^{\boldsymbol{\gamma}_s} \cdots C_{p_1}^{\boldsymbol{\gamma}_1}(h).$$

The order is immaterial for disjoint pairs. In this section, a *proper Pauli card* means such a realizable iterated contraction leaving at least two inputs. A *Pauli pair card* has $s = 1$.

F.8.2 Relative closure in the standard Pauli case

When the kernels span $\mathrm{Mat}_2(\mathbb{C})$, Theorem F.5.20 reconstructs the slices at one input pair from the coefficients of four cards. Here we compare cards at every input pair, allowing zero cards and independent rescalings. To apply the real-valued dichotomy we compare a signature with its complex conjugate. The following lemma tests linear dependence through the vanishing of the antisymmetrized product $\mathbf{u} \otimes \mathbf{v} - \mathbf{v} \otimes \mathbf{u}$.

Lemma F.8.2 (Pauli contractions determine a tensor line)**.** *Let $n \geq 5$, and let $\mathbf{u}, \mathbf{v} \in (\mathbb{C}^2)^{\otimes n}$. If $C_{rs}^{\boldsymbol{\gamma}}(\mathbf{u})$ and $C_{rs}^{\boldsymbol{\gamma}}(\mathbf{v})$ are linearly dependent for every pair $r < s$ and every $\boldsymbol{\gamma} \in \mathbb{F}_2^2$, with zero cards allowed, then $\mathbf{u}, \mathbf{v}$ are linearly dependent. Consequently, if every Pauli pair card of $\mathbf{u}$ is zero or a nonzero complex scalar multiple of a real tensor, the same holds for $\mathbf{u}$.*

Proof. Put $W = \mathbf{u} \otimes \mathbf{v} - \mathbf{v} \otimes \mathbf{u}$ and group its two copies by input, so that $W \in (\mathbb{C}^4)^{\otimes n}$. The hypothesis says

$$(C_{rs}^{\boldsymbol{\gamma}} \otimes C_{rs}^{\boldsymbol{\gamma}})W = 0 \qquad (r < s,\ \boldsymbol{\gamma} \in \mathbb{F}_2^2).$$

On the regrouped four-dimensional spaces at r, s, the four contraction matrices are

$$I \otimes I, \quad Z \otimes Z, \quad X \otimes X, \quad (ZX) \otimes (ZX).$$

Use the same orthonormal Bell basis $2^{-1/2}(\operatorname{vec} I, \operatorname{vec} Z, \operatorname{vec} X, \operatorname{vec}(ZX))$ on each regrouped input. These matrices are diagonal in that basis, with eigenvalue rows

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

The rows are independent. Thus the four contraction equations, for any fixed values on the other inputs, force each coefficient with equal Bell basis indices at r, s to vanish. Applying this to every pair leaves only strings whose n indices are all distinct. There are just four indices, so $n \geq 5$ gives $W = 0$, proving the first assertion.

For the last assertion take $\mathbf{v} = \overline{\mathbf{u}}$. The Pauli kernels are real, and a vector and its conjugate are dependent precisely when it is zero or a scalar multiple of a real vector: divide by its first nonzero coordinate to see the reverse implication. □

Theorem F.8.3 (Standard Pauli $V_4^{(\mathrm{std})}$ relative closure)**.** *If*

$$\mathsf{Stable}(\mathcal{F}, V_4^{(\mathrm{std})}, f), \qquad \nu(\mathcal{F}) = \operatorname{arity}(f) \geq 6,$$

then $\mathsf{Close}_{\{\mathscr{A}\}}(\mathcal{F}, V_4^{(\mathrm{std})}, f)$.

Proof. Apply the established reductions for odd, unary, and unsafe binary factors in Theorem F.2.9 and Lemmas F.3.4, F.3.8, F.3.11 and F.5.4. On the continuing branch every nonbinary irreducible signature is even, and all binary irreducible factors are nonsingular Pauli binaries up to nonzero scalars.

Suppose some retained signature is not a scalar multiple of a real signature. Factor closure supplies such an irreducible signature g, necessarily of even arity $n \geq \nu(\mathcal{F}) \geq 6$. The contrapositive of Lemma F.8.2 selects a Pauli pair card with the same property. Repeat while its arity is at least six, reaching a nonzero quaternary signature q that is not a scalar multiple of a real signature. All selected contractions are realizable: the complete transfer group $V_4^{(\mathrm{std})}$ supplies the four Pauli kernels, including their reversal and disequality-edge scalars. Their composition uses one g-vertex and $O(n)$ binary vertices. The coefficient tables select the contractions in this finite gadget.

Add q and process all its irreducible factors by Definition F.5.10. An odd or unsafe factor is resolved. A new finite-order transfer class gives a successor with a strictly larger complete transfer group, after which the deck is regenerated. Otherwise, if all factors were binaries, they would all be Pauli and their product would be a scalar multiple of a real signature, contrary to the choice of q. Hence an even nonbinary irreducible signature remains. It has arity four, so the signature set strictly lowers $\nu(\mathcal{F}) \geq 6$. The supplied I excludes the one-sided exception in factor closure. Thus processing this family gives a resolved complexity leaf or an outer deck successor.

It remains to treat the case in which every retained signature is a scalar multiple of a real signature. Divide each by its first nonzero coefficient and denote the resulting set of real-valued algebraic signatures by $\mathcal{F}_0$. These known scalars are accounted for per vertex, and $I \in \mathcal{F}_0$. The equality-edge-model equivalence Lemma F.5.1 gives

$$\text{K-Holant}(\mathcal{F}) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F}_0) \equiv_{\mathrm{T}} \text{Holant}(\mathcal{F}_0 \cup \{X\}).$$

Apply the real-valued dichotomy of Shao and Cai [SC20, full version, Theorems 1.2 and 2.22] to this whole finite real set. If its condition (T) fails, hardness composes along the displayed reductions. If it holds, the following argument gives a tractable case satisfying $\mathsf{Tract}(K\mathcal{F})$ in our coordinates.

The case $\mathcal{F}_0 \cup \{X\} \subseteq \langle \mathscr{T} \rangle$ immediately implies $K\mathcal{F} \subseteq \langle \mathscr{T} \rangle$, since invertible single-input maps and nonzero scalars preserve tensor products of unary and binary signatures. In each other case there is one algebraic matrix $M \in \mathrm{GL}_2(\overline{\mathbb{Q}})$, by Lemma F.3.1, and one $\mathscr{C} \in \{\mathscr{P}, \mathscr{A}, \mathscr{L}\}$ with

$$E := M^{\mathsf{T}} M \in \mathscr{C}, \qquad M^{-1}(\mathcal{F}_0 \cup \{X\}) \subseteq \mathscr{C}.$$

In particular $B := M^{-1} X M^{-\mathsf{T}} \in \mathscr{C}$, and

$$M^{\mathsf{T}} X M = EBE.$$

Both E and B are symmetric nonsingular members of $\mathscr{C}$. Therefore Lemma F.6.14 gives $EBE \in \mathscr{C}$ for each of the three possible classes. In all cases the same $U = M$ satisfies

$$U^{\mathsf{T}} X U \in \mathscr{C}, \qquad U^{-1}\mathcal{F} \subseteq \mathscr{C}.$$

This is a common K-presentation, hence $\mathsf{Tract}(K\mathcal{F})$, by Definition F.2.3.

All factors of the new family are processed, and each outer deck successor reapplies Lemma F.5.9. Thus the preceding outcomes satisfy Definitions F.5.10 and F.5.11 and give the claimed closure in Equation (F.21). □

F.8.3 The exotic Klein-four deck

For the exotic Klein-four group, we will show that the matching-card condition forces a proper tensor factor in every even arity at least six. This form is not $GO(X)$-conjugate to the Pauli form, so we use its original matrix representatives

$$\begin{aligned} E_0 &= I, & E_1 &= X, \\ E_2 &= \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}, & E_3 &= \begin{pmatrix} -i & 1 \\ -1 & i \end{pmatrix}. \end{aligned} \tag{F.61}$$

Identify $0, 1, 2, 3$ with $00, 01, 10, 11 \in V = \mathbb{F}_2^2$, define the exotic Klein-four group $V_4^{(\mathrm{ex})} := \{[E_a] : a \in V\}$, write addition as $\oplus$, and define the label permutation $\vartheta := (0)(1)(2\ 3)$. A superscript that is an ordered input pair, as in $E_a^{(1,3)}$, specifies the tensor inputs. An upper label, as in E^a, denotes the Frobenius-dual basis element defined below.

For an oriented perfect matching M, an *exotic M-matching product* is a nonzero scalar multiple of

$$\bigotimes_{e \in M} E_{\tau_e}^{(e)}, \qquad \tau \in V^M.$$

These are the $V_4^{(\mathrm{ex})}$-matching signatures on M. The products of the fixed representatives form the *M-basis* of $(\mathbb{C}^2)^{\otimes 2|M|}$. The one-dimensional vector subspace spanned by one product is its *matching line*.

To compare different matchings, we express a product on one matching in the basis of another, an operation called *recoupling*. The next lemma counts the nonzero coefficients in this expansion. This count will constrain the label maps in the later factor argument.

Lemma F.8.4 (Exotic Klein-four calculus and recoupling)**.** *For the bilinear Frobenius pairing,*

$$\langle E_a, E_b \rangle_{\mathrm{Frob}} = d_a \mathbf{1}[b = \vartheta(a)], \qquad (d_0, d_1, d_2, d_3) = (2, 2, 4, 4). \tag{F.62}$$

Thus define

$$E^a := d_a^{-1} E_{\vartheta(a)}, \qquad \langle E^a, E_b \rangle_{\mathrm{Frob}} = \mathbf{1}[a = b].$$

These duals are nonzero scalar multiples of the original representatives. Transposition and multiplication satisfy

$$E_a^{\mathsf{T}} \doteq E_{\vartheta(a)}, \qquad E_a E_b \doteq E_{a\oplus b}, \tag{F.63}$$

with nonzero omitted scalars. Relative to the matching $12|34$*,*

$$\delta^{\times}_{a,b,\gamma} := \vartheta(\gamma) \oplus b \oplus \vartheta(a),$$
$$\delta^{\parallel}_{a,b,\gamma} := \gamma \oplus \vartheta\big(b \oplus \vartheta(a)\big),$$

and define the recoupling coefficients by

$$\rho^{\times}_{a,b}(\gamma) := \left\langle (E^{\gamma})^{(1,2)} (E^{\delta^{\times}_{a,b,\gamma}})^{(3,4)}, E_a^{(1,3)} E_b^{(2,4)} \right\rangle_{\mathrm{Frob}},$$
$$\rho^{\parallel}_{a,b}(\gamma) := \left\langle (E^{\gamma})^{(1,2)} (E^{\delta^{\parallel}_{a,b,\gamma}})^{(3,4)}, E_a^{(1,4)} E_b^{(2,3)} \right\rangle_{\mathrm{Frob}}.$$

Here the pairing is the bilinear Frobenius pairing on quaternary tensors. The recoupling identities are

$$E_a^{(1,3)} E_b^{(2,4)} = \sum_{\gamma\in V} \rho^{\times}_{a,b}(\gamma) E_{\gamma}^{(1,2)} E^{(3,4)}_{\delta^{\times}_{a,b,\gamma}},$$
$$E_a^{(1,4)} E_b^{(2,3)} = \sum_{\gamma\in V} \rho^{\parallel}_{a,b}(\gamma) E_{\gamma}^{(1,2)} E^{(3,4)}_{\delta^{\parallel}_{a,b,\gamma}},$$

Every displayed coefficient lies in $\mathbb{Q}(i)^{\times}$*. For matchings* M, N *of* $2r$ *inputs, let* $c(M,N)$ *count alternating cycles, including common edges. An* N*-matching product then has exactly*

$$4^{r-c(M,N)} \tag{F.64}$$

nonzero coefficients in the M*-basis. Thus coefficient support of size four means one alternating four-cycle and exactly two varying* M*-edge labels.*

Proof. The displayed matrices satisfy

$$E_1^2 = I, \qquad E_2^2 = E_3^2 = -2I, \qquad E_1E_2 = -E_2E_1 = -iE_3,$$
$$E_0^{\mathsf{T}} = E_0, \qquad E_1^{\mathsf{T}} = E_1, \qquad E_2^{\mathsf{T}} = -E_3, \qquad E_3^{\mathsf{T}} = -E_2,$$
$$\mathrm{tr}(E_a) = 2\mathbf{1}[a=0].$$

These identities give the fusion and Gram formulas, including all nonzero scalar factors. For example, the coefficient of $E_{\gamma}^{(1,2)} E_{\delta}^{(3,4)}$ in the first recoupling is

$$\mathrm{tr}\Big((E^{\delta})^{\mathsf{T}} E_a^{\mathsf{T}} E^{\gamma} E_b\Big).$$

Fusion makes its matrix product a nonzero multiple of $E_{\delta\oplus\vartheta(a)\oplus\vartheta(\gamma)\oplus b}$. The trace therefore vanishes except at $\delta = \delta^{\times}_{a,b,\gamma}$, and is nonzero there. For the second recoupling the coefficient is

$$\mathrm{tr}\Big((E^{\delta})^{\mathsf{T}} E_b^{\mathsf{T}} (E^{\gamma})^{\mathsf{T}} E_a\Big),$$

which is nonzero exactly at $\delta = \gamma \oplus \vartheta(b) \oplus a = \delta^{\parallel}_{a,b,\gamma}$. All factors have entries in $\mathbb{Q}(i)$, proving the coefficient assertion.

For general M, N, contracting an N-product against a dual M-basis element separates over the alternating cycles of $M \cup N$. On a cycle of length 2ℓ, the coefficient is one trace of a product of the ℓ fixed N-matrices and ℓ dual M-matrices, transposed according to their orientations. Each M-label enters its fused label once, through either the identity or the bijection ϑ. After $\ell - 1$ labels are chosen, the zero-label trace condition determines the last label uniquely. The resulting trace is a nonzero scalar times $\mathrm{tr}(I) = 2$. Thus the cycle contributes exactly $4^{\ell-1}$ nonzero coefficients, including one for a common edge when $\ell = 1$. Multiplying over cycles gives $4^{\sum(\ell-1)} = 4^{r-c(M,N)}$. If this equals four, exactly one cycle has length four and the others are common edges, proving the last assertion. □

Realizable exotic contractions. For an ordered pair p, define the exotic pair card

$$C_p^a(f) := \partial_p^{E^a} f.$$

In this section, for disjoint ordered pairs $p_1, \ldots, p_s$, an *exotic card* is the corresponding iterated contraction by fixed representatives of the duals $E^{a_1}, \ldots, E^{a_s}$. It is *proper* when at least two inputs remain. Disjoint contractions commute, and nonzero dual scalars are divided out in the displayed card notation. With the fixed representatives, write

$$E_a E_b = \omega(a, b) E_{a \oplus b}, \qquad \omega(a, b) \in \mathbb{Q}(i)^\times.$$

The factors $\omega(a, b)$ and the corresponding transpose scalars are retained in coefficient calculations. To cross-contract two residual matching edges $e = \{u, u'\}$ and $e' = \{v, v'\}$, choose one endpoint u of e, one endpoint v of e', and one fixed $a \in V$. Contract the single pair u, v by the chosen representative of E^a, and leave u', v' as the endpoints of the fused residual edge. The endpoint orientations are part of the choice, and the scalar factors are retained.

The duals are realizable up to the recorded scalars. Their coordinate reconstruction identity is

$$f = \sum_{a \in V} E_a^{(p)} \otimes C_p^a, \qquad C_p^a = \partial_p^{E^a} f. \tag{F.65}$$

Here the cards are realizable contractions, and the sum expresses f in the basis on the pair p.

Lemma F.8.5 (The exotic Klein-four arity-six case)**.** *If every exotic pair card of a nonzero arity-six tensor h is zero or a product of two exotic binaries, then h is a product of three exotic binaries. Thus there is no exotic analogue of* H_6.

Proof. Normalize a nonzero card to $C_{12}^0 = E_0^{(3,4)} E_0^{(5,6)}$ by realizable dressings and a permutation. The six-input certificate Theorem F.A.13 covers all choices of zero and nonzero cards and their scalar coefficients, including cancellations. Every case is a product of three exotic binaries. Undoing the normalization preserves this conclusion. □

Lemma F.8.6 (Fixed residual matching for exotic cards)**.** *Let $f \neq 0$ have even arity at least eight, and suppose every proper exotic card of f is zero or an exotic matching product. For every deleted pair p, all nonzero cards C_p^a in Equation* (F.65) *use one residual perfect matching.*

Proof. Fix a deleted pair p, and suppose that two nonzero cards C_p^a and C_p^b use distinct residual matchings $M \neq N$. We construct a sequence of realizable exotic contractions that preserves both cards as nonzero matching products and preserves one residual disagreement.

The only local operation needed is the following. If two contracted inputs belong to distinct matching edges in a given distinguished card, then, up to the fixed transpose convention, the new edge factor is

$$E_\alpha E_\kappa E_\beta \doteq E_{\alpha\oplus\kappa\oplus\beta}.$$

It is nonzero because every E_γ is nonsingular. Choose two vertices at alternating distance two on an alternating cycle of $M \cup N$. They belong to distinct edges in both distinguished cards, so contracting them shortens that cycle by two vertices while keeping both distinguished cards nonzero. Repeating reduces a marked nontrivial cycle to length four.

Reduce every other nontrivial alternating cycle and splice it into the marked cycle by one cross-contraction. The same three-matrix identity keeps both distinguished cards nonzero and merges the two cycles. A common edge carrying the same label in both cards may be closed with its realizable Frobenius dual. If a common edge carries different labels, splice it into the marked cycle instead of closing it, and reduce the enlarged cycle again. After all common edges and all other cycles have been removed, the resulting realizable arity-six tensor h has two nonzero p-cards with different residual two-edge matchings.

Every exotic pair card of h is zero or a matching product. Indeed, bilinear contractions commute, so any card of h may be reordered so that its final pair contraction is applied first to f. By hypothesis that first card is zero or a matching product, and every subsequent contraction preserves this property by $E_\alpha E_\kappa E_\beta \doteq E_{\alpha\oplus\kappa\oplus\beta}$. Thus Lemma F.8.5 applies and makes h a product of three exotic binaries.

For a product of three binaries, all nonzero cards at a fixed deleted pair have one residual matching: if the deleted pair is a factor edge, only its contraction in the dual direction is nonzero. Otherwise fusion of the two incident factor edges leaves the same residual matching for every kernel giving a nonzero card. This contradicts the two preserved residual matchings of h. Hence all nonzero p-cards of f use one residual perfect matching. □

With the residual matching fixed, only its binary labels can vary. The next lemma shows that, in the absence of a proper tensor factor, each edge uses all four labels exactly once.

Lemma F.8.7 (Permutation label maps for a fixed residual matching)**.** *Let f satisfy the hypotheses of Lemma F.8.6, fix p and its residual matching M, and suppose f has no nontrivial positive-arity tensor factor. Then all four cards C_p^a are nonzero and have the form*

$$C_p^a = \lambda_a \bigotimes_{e\in M} E^{(e)}_{\tau_e(a)},$$

where $\lambda_a \in \mathbb{C}^\times$ are nonzero scalar coefficients and $\tau_e : V \to V$ are label maps. Every τ_e is a permutation.

Proof. Let

$$\Gamma = \{a \in V : C_p^a \neq 0\}.$$

Reconstruction and the fixed residual matching give

$$f = \sum_{a\in\Gamma} \lambda_a E_a^{(p)} \bigotimes_{e\in M} E^{(e)}_{\tau_e(a)}, \qquad \lambda_a \neq 0.$$

For an edge $e \in M$, contraction with the realizable Frobenius dual of E_b selects the fibre

$$\Gamma_{e,b} = \{a \in \Gamma : \tau_e(a) = b\}.$$

In the displayed $p \cup (M \setminus \{e\})$-basis, the resulting card has exactly $|\Gamma_{e,b}|$ nonzero coefficients, since distinct a's remain distinct on the p-coordinate. By Equation (F.64), every nonempty fibre therefore has size one or four.

If $|\Gamma| = 1$, the factor $E_a^{(p)}$ is common to the whole tensor, contrary to irreducibility. If $|\Gamma| \in \{2, 3\}$, every τ_e is injective on Γ. Choose inputs from two distinct edges of M and cross-contract them by any realizable exotic kernel. Projective fusion is nonzero in every summand of the reconstruction, and the p-labels remain distinct, so the resulting proper card has exactly $|\Gamma|$ coefficients in the induced matching basis. This is impossible because Equation (F.64) permits only powers of four. Consequently $\Gamma = V$.

Now every nonempty τ_e-fibre has size one or four. A four-point fibre makes τ_e constant and exposes $E_b^{(e)}$ as a proper tensor factor. Since f is irreducible, all fibres are singletons. Hence every $\tau_e : V \to V$ is a permutation, proving the stated form. □

Lemma F.8.8 (The exotic Klein-four arity-eight case)**.** *Let $h \neq 0$ have arity eight and suppose every exotic pair card is zero or a product of three exotic binaries. Then h has a proper tensor factor.*

Proof. Contraction closure under Equation (F.63) makes every iterated proper exotic card zero or a matching product. Suppose that h is irreducible. Then Lemmas F.8.6 and F.8.7 gives four cards on one matching. Normalize one to $C_{12}^0 = E_0^{(3,4)} E_0^{(5,6)} E_0^{(7,8)}$. Each residual label map is affine because $\mathrm{AGL}(2, 2) = \mathrm{Sym}(V)$, as recalled in Section F.6.5.2. Realizable edge dressings remove the translations, and their nonzero scalar factors are absorbed into the coefficients. Writing the labels as column vectors gives

$$h = \sum_{\gamma \in V} \lambda_\gamma E_\gamma^{(1,2)} E_{A_1\gamma}^{(3,4)} E_{A_2\gamma}^{(5,6)} E_{A_3\gamma}^{(7,8)}, \qquad A_j \in \mathrm{GL}_2(\mathbb{F}_2), \quad \lambda_\gamma \neq 0. \tag{F.66}$$

To obtain a card whose labels vary on three edges, consider separately the contractions of $(3, 5)$ and $(4, 5)$ by $E^0 = I/2$. On the two remaining endpoints of the affected edges, the respective matrices are

$$\tfrac{1}{2} E_{A_1\gamma}^{\mathsf{T}} E_{A_2\gamma}, \qquad \tfrac{1}{2} E_{A_1\gamma} E_{A_2\gamma}.$$

By Equation (F.63), their labels are

$$D_1\boldsymbol{\gamma} = (\vartheta A_1 + A_2)\boldsymbol{\gamma}, \qquad D_2\boldsymbol{\gamma} = (A_1 + A_2)\boldsymbol{\gamma},$$

with all sums over $\mathbb{F}_2$ and all omitted scalars nonzero. The two maps cannot both vanish, since that would imply $\vartheta A_1 = A_1$, contrary to the invertibility of A_1 and $\vartheta \neq I$. Choose a nonzero one, denoted by D. In the resulting six-input matching basis the four label tuples are $(\boldsymbol{\gamma}, D\boldsymbol{\gamma}, A_3\boldsymbol{\gamma})$. They are distinct because their first coordinates are distinct, and every coefficient is nonzero. All three coordinates vary: the first and third maps are bijective, and $D \neq 0$. This contradicts Equation (F.64), which allows only two varying edges in a four-term matching product. Thus h has a proper tensor factor. □

Proposition F.8.9 (High-arity exotic factor)**.** *Let $f \neq 0$ have arity $2r \geq 10$, and suppose every proper exotic card is zero or an exotic matching product. Then f has a proper tensor factor.*

Proof. Suppose that f is irreducible. At a fixed pair, Lemmas F.8.6 and F.8.7 gives four nonzero cards on one matching with permutation label maps. Choose two residual edges and perform one cross-contraction by a Frobenius dual as defined above. Every summand of the reconstruction remains nonzero because the three matrices in its endpoint-to-endpoint fusion are nonsingular. The

four resulting terms still vary at the fixed pair and at all $r-3$ untouched residual edges, hence in at least $r-2 \geq 3$ coordinates. But Equation (F.64) allows only two varying coordinates in a four-term matching product. This contradiction proves that a proper tensor factor exists. □

Theorem F.8.10 (Exotic Klein-four relative rigidity)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, V_4^{(\mathrm{ex})}, f), \qquad \mathrm{arity}(f) = \nu(\mathcal{F}).$$

Then

$$\mathsf{Close}_{\{\mathscr{A}\}}(\mathcal{F}, V_4^{(\mathrm{ex})}, f).$$

More explicitly, every continuing branch reaches a resolved complexity leaf or an outer deck successor after processing the realizable batch from the arity-four reduction by Lemma F.5.12. No tractable case requiring a common presentation remains at the current node.

Proof. First consider odd-arity factors, including unaries, by Theorem F.2.9, and unsafe binaries by Lemmas F.3.8 and F.5.4. On a continuing branch all retained irreducible signatures have even arity. Deck stability makes every exotic pair card zero or an exotic matching product. Repeated transpose/fusion using Equation (F.63) therefore makes every iterated proper exotic card zero or an exotic matching product. At arity four, item 4 of Theorem F.5.22 invokes Corollary F.4.8. Process its complete factor batch by Definitions F.5.10 and F.5.13 and Lemma F.5.12, obtaining a resolved complexity leaf or an outer deck successor. At arity six, Lemma F.8.5 contradicts irreducibility. At arity eight and at least ten, the factor conclusions of Lemma F.8.8 and Proposition F.8.9 contradict irreducibility as well. These cases exhaust the possible arities of the distinguished nonbinary irreducible signature f.

Hence every branch is resolved before a common presentation is needed. By Equation (F.21), this gives $\mathsf{Close}_{\{\mathscr{A}\}}(\mathcal{F}, V_4^{(\mathrm{ex})}, f)$. □

F.8.4 Completion of the Klein-four classification

Proof of Theorem F.8.1. If $G = V_4^{(\mathrm{std})}$, apply Theorem F.8.3. If $G = V_4^{(\mathrm{ex})}$, apply Theorem F.8.10. These are exactly the two Klein-four forms in the statement. □

F.9 Platonic Matching Decks

For a fixed input pair p, contraction with a kernel C gives a card $\Phi_p(C)$ on the remaining inputs. The realizable kernels form a finite set of projective points. By linearity, any linear relation among chosen kernel representatives holds among their cards with the same coefficients. Suppose all nonzero cards at p share one matching. For these kernel configurations, Lemma F.9.6 fixes the binary factors, up to scalar, on all but at most one edge of that matching. Above arity four there are at least two residual edges, so at least one fixed binary lifts to a proper factor of f by the slice reconstruction in Section F.5.5. If the matchings disagree, the incidence argument produces three distinct collinear card points with different matchings. Their matchings differ on only four residual inputs. Closing the common factors leaves a realizable six-input signature.

The finite classifications in Appendix F.A.5 then give the octahedral and icosahedral reductions. In the external tetrahedral case, the absence of a quaternary line with three different matchings forces a common matching at higher arity and hence a proper factor (Theorem F.9.28). The internal tetrahedral case also permits the affine core H_6. Here stabilizer-code contractions can shorten a nonaffine irreducible signature while preserving its failure of affine membership in the current

basis. The six- and eight-input cases complete the induction to a common affine presentation in Theorem F.9.1.

Theorem F.9.1 (Platonic deck rigidity)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, G, f), \qquad G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{T}^{(\mathrm{ext})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}, \mathcal{I}^{(0)}, \mathcal{I}^{(1)}\}.$$

Then

$$\begin{cases} \mathsf{Close}_{\{\not\sim\}}(\mathcal{F}, G, f), & G = \mathcal{T}^{(\mathrm{int})}, \\ \mathsf{Close}_{\varnothing}(\mathcal{F}, G, f), & G \in \{\mathcal{T}^{(\mathrm{ext})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}, \mathcal{I}^{(0)}, \mathcal{I}^{(1)}\}. \end{cases}$$

F.9.1 Common coordinates and realizable kernels

We use the quaternion coordinates of Equation (F.16), the marked-group classification of Theorem F.5.16, and the projective and tensor conventions of Section F.2.1.3 and the projective tensor conventions. In particular, $[F]$ denotes the projective class of a nonzero matrix or tensor, including a singular matrix.

A finite projective point configuration is a finite subset P of a projective space. Points are collinear when they lie on one projective line, and a k-rich line contains at least k points of P. A point–line incidence structure is a pair $(P, \mathcal{L})$, where $\mathcal{L}$ is a family of subsets of P. Its automorphisms are the permutations of P preserving $\mathcal{L}$.

Throughout this section, for a finite projective group G put

$$G^{\#} := \{[T^{\#}] : [T] \in G\}, \qquad T^{\#} := XT^{\mathsf{T}}X,$$

and call G *reversal-stable* when $G^{\#} = G$. A *marked Platonic state* means a retained, factor-closed state containing the supplied I, whose complete transfer group is one of

$$\mathcal{T}^{(\mathrm{int})},\ \mathcal{T}^{(\mathrm{ext})},\ \mathcal{O}^{(\mathrm{dd})},\ \mathcal{O}^{(\mathrm{tt})},\ \mathcal{O}^{(\mathrm{dt})},\ \mathcal{O}^{(\mathrm{td})},\ \mathcal{I}^{(0)},\ \mathcal{I}^{(1)},$$

with one fixed chosen representative and realization record for every transfer class.

For a signature u and an ordered pair $p = (a, b)$ of its inputs, define the contraction map, using the pair contraction from Section F.2.6,

$$\Phi_p^u : \mathrm{Mat}_2(\mathbb{C}) \longrightarrow (\mathbb{C}^2)^{\otimes(\mathrm{arity}(u)-2)}, \qquad \Phi_p^u(C) := \partial_p^C u. \tag{F.67}$$

When the signature is the active f, write $\Phi_p = \Phi_p^f$, and write $\langle C^{(p)}, u\rangle := \Phi_p^u(C)$ for this bilinear (not Hermitian) contraction. Equivalently, $\Phi_p^u(C)(\mathbf{y}) = \mathrm{tr}((F_{p,\mathbf{y}}^u)^{\mathsf{T}}C)$, using the two-input slice of Section F.5.5.1. Input permutations supply the other orderings.

For every two-element input set $\{a, b\}$, use the canonical orientation $p = (a, b)$ with $a < b$, and order complementary inputs increasingly unless both orientations are explicitly requested. Thus a quaternary signature has six canonical pairs and an arity-six tensor has fifteen. All other orientations are obtained by realizable permutations of inputs.

At a continuing fixed-I deck-stable node, fix for each $g \in G$ a matrix K_g with $[K_g] = g$. Since $[X] \in G$, choose a fixed realizable binary representative C_g of transfer class $[XK_g]$. The convention $T_{C_g} = C_gX$ gives $C_g \doteq XK_gX$. Attach its two ordered inputs to the selected pair through two X-edges. The effective bilinear kernel is then $XC_gX \doteq K_g$. Store its nonzero realization scalar and ordered-input record, and define the chosen representative set

$$\mathcal{K}_G^{\mathrm{act}} := \{K_g : g \in G\}. \tag{F.68}$$

Realizable group contractions use these fixed representatives. A resulting binary B is tested through its transfer BX. A zero card contributes zero. A nonzero singular B is processed by factor closure. If B is nonsingular and $[BX] \notin G$, then Lemmas F.3.8 and F.5.4 gives an infinite-order terminal or a strict enlargement of the complete transfer group. The ordered inputs and nonzero realization scalar are retained in each case.

The three nonstandard octahedral displays are as follows. For the transposition–transposition form put

$$H_{\mathrm{tt}} := \begin{pmatrix} \sqrt{2} & -i\sqrt{2} \\ 1-i & 1+i \end{pmatrix}, \qquad \mathcal{O}^{(\mathrm{tt})} := H_{\mathrm{tt}}^{-1}\mathcal{O}^{(\mathrm{dd})}H_{\mathrm{tt}}. \tag{F.69}$$

This realizes the marked type $\mathcal{O}^{(\mathrm{tt})}$ of Theorem F.5.16 in coefficient coordinates.

For the two mixed fixed-equality forms, put

$$\rho := \sqrt{2}-1, \qquad H_{\mathrm{dt}} := Q_0 - \rho Q_1, \qquad H_{\mathrm{td}} := Q_0 + \rho Q_3, \tag{F.70}$$

and, for $\alpha \in \{\mathrm{dt}, \mathrm{td}\}$, define

$$\theta_\alpha(U) := H_\alpha^{-1} U H_\alpha, \qquad \mathcal{O}^{(\alpha)} := \theta_\alpha(\mathcal{O}^{(\mathrm{dd})}). \tag{F.71}$$

Direct quaternion multiplication gives

$$\begin{aligned} &H_{\mathrm{dt}}H_{\mathrm{dt}}^{\mathsf{T}} = 2\rho(Q_0 - Q_1), && \theta_{\mathrm{dt}}(Q_1) = Q_1, && \theta_{\mathrm{dt}}(Q_2 - Q_3) = -\sqrt{2}Q_3, \\ &H_{\mathrm{td}}H_{\mathrm{td}}^{\mathsf{T}} = 2\rho(Q_0 + Q_3), && \theta_{\mathrm{td}}(Q_3) = Q_3, && \theta_{\mathrm{td}}(Q_1 - Q_2) = \sqrt{2}Q_1. \end{aligned} \tag{F.72}$$

Thus $\mathcal{O}^{(\mathrm{dt})}$ has marked class pattern $(dt;t)$, whereas $\mathcal{O}^{(\mathrm{td})}$ has pattern $(td;t)$. All three matrices $H_{\mathrm{tt}}, H_{\mathrm{dt}}, H_{\mathrm{td}}$ are used as coefficient similarities. Realizable contractions and dressings use representatives of the current group. The retained set and its binary equality remain in the fixed coordinates.

Lemma F.9.2 (The two marked tetrahedral forms)**.** *Let $G \le \mathrm{PGL}_2(\mathbb{C})$ be a finite reversal-stable complete transfer group of a retained set containing the supplied I, in the fixed coordinates, and define the transfer mark x, the reversal mark j, and the extension group $\widehat{G}$ by*

$$x = [X] \in G, \qquad j = [Z], \qquad \widehat{G} = \langle G, j\rangle.$$

If $G \cong A_4$, then, up to one global fixed-I-preserving $GO(X)$-normalization, exactly one of the following holds.

1. *$j \in G$, $\widehat{G} = G$, and $G = \mathcal{T}^{(\mathrm{int})}$, where $\mathcal{T}^{(\mathrm{int})}$ is the standard group in Equation* (F.17)*.*
2. *$j \notin G$, $\widehat{G} \cong S_4$, and $G = \mathcal{T}^{(\mathrm{ext})}$, where*

$$\begin{aligned} \mathcal{T}^{(\mathrm{ext})} = {} & \{[Q_0], [Q_1], [Q_2+Q_3], [Q_2-Q_3]\} \\ & \cup \{[Q_0 + \epsilon Q_1 + \delta\sqrt{2}Q_2], [Q_0 + \epsilon Q_1 + \delta\sqrt{2}Q_3] : \epsilon, \delta \in \{1,-1\}\}. \end{aligned}$$

Here $x = [Q_1] \in \mathcal{T}^{(\mathrm{ext})}$, whereas $j = [Q_3] \notin \mathcal{T}^{(\mathrm{ext})}$ and $xj = [Q_2] \notin \mathcal{T}^{(\mathrm{ext})}$.

Proof. Reversal gives $G \triangleleft \widehat{G}$ and $[\widehat{G} : G] \le 2$. If $j \notin G$, then $\widehat{G} \cong S_4$, with the commuting pair (x, j) marked as a double transposition and a transposition. If $j \in G$, the three involutions form the unique normal Klein four of A_4 by Lemma F.5.15. After the corresponding finite-subgroup embedding the remaining ambiguity is

$$C_{\mathrm{PGL}_2(\mathbb{C})}(\langle x, j\rangle) = \langle x, j\rangle,$$

represented inside $GO(X)$. Direct multiplication gives the two stated domains. □

For coefficient comparison, define the matrix R_{ext} and the coefficient-similarity map θ_{ext} by

$$R_{\mathrm{ext}} := Q_0 + \rho Q_1, \qquad \theta_{\mathrm{ext}}(U) := R_{\mathrm{ext}}^{-1} U R_{\mathrm{ext}}.$$

On projective sets, apply θ_{ext} representative-wise. Then

$$\theta_{\mathrm{ext}}(\mathcal{T}^{(\mathrm{int})}) = \mathcal{T}^{(\mathrm{ext})}, \qquad [R_{\mathrm{ext}}^{\mathsf{T}} X R_{\mathrm{ext}}] = [Q_0 - Q_1] \neq [X].$$

Thus $R_{\mathrm{ext}} \notin GO(X)$. We use θ_{ext} to compare coefficient tables and use realizable external representatives for gadgets.

Write $\mathcal{I}^{(1)}$ for the opposite icosahedral coordinate display. The two displays have the same abstract marked classification, but both are retained because their representatives in the original basis are different. Put

$$D_{\mathcal{I}^{(0)}} := Q_0 + Q_3, \qquad D_{\mathcal{I}^{(0)}}^{\mathsf{T}} X D_{\mathcal{I}^{(0)}} = 2X, \qquad \mathcal{I}^{(1)} = D_{\mathcal{I}^{(0)}} \mathcal{I}^{(0)} D_{\mathcal{I}^{(0)}}^{-1}.$$

The signed coordinate permutations defining $\mathcal{T}^{(\mathrm{int})}$ also give

$$D_{\mathcal{I}^{(0)}} \mathcal{T}^{(\mathrm{int})} D_{\mathcal{I}^{(0)}}^{-1} = \mathcal{T}^{(\mathrm{int})}, \qquad \mathcal{T}^{(\mathrm{int})} \subseteq \mathcal{I}^{(0)} \cap \mathcal{I}^{(1)}.$$

Indeed, conjugation by $D_{\mathcal{I}^{(0)}}$ sends (Q_1, Q_2, Q_3) to $(-Q_2, Q_1, Q_3)$. This gives the coefficient correspondence below, with realizations carried out in the original coordinates containing the supplied I.

Lemma F.9.3 (Fixed-equality correspondence for the opposite icosahedral display)**.** *Set $D = D_{\mathcal{I}^{(0)}}$ and*

$$\kappa_D(C) := D^{-\mathsf{T}} C D^{-1}.$$

Let $(\mathcal{F}, \mathcal{I}^{(1)}, f)$ be a marked Platonic state in the fixed coordinates containing the supplied I, and introduce only for coefficient analysis the tensor $f^{(D)} := D^{-\otimes \mathrm{arity}(f)} f$. For every ordered pair p,

$$\Phi_p^f(\kappa_D(C)) = D^{\otimes(\mathrm{arity}(f)-2)} \Phi_p^{f^{(D)}}(C). \tag{F.73}$$

The map $[C] \mapsto [\kappa_D(C)]$ bijects the standard $\mathcal{I}^{(0)}$-kernel classes with the realizable $\mathcal{I}^{(1)}$-kernel classes. Consequently every finite card or network identity proved in the standard display has a direct identity in the opposite display, obtained by replacing each kernel by its κ_D-image and recording the nonzero representative and edge scalars. This correspondence preserves zero cards, tensor factors, all flattening ranks, matching labels, endpoint support, and the property that an output binary transfer lies outside the complete transfer group. It does not change $\mathcal{F}$, the active occurrence, or the supplied I.

Proof. Direct contraction gives Equation (F.73). Moreover $D^{-\mathsf{T}} = \frac{1}{2} X D X$, and hence

$$[\kappa_D(C)] = [X\, D(XC)D^{-1}] \in \mathcal{I}^{(1)} \qquad ([C] \in \mathcal{I}^{(0)}).$$

Here $[X], [XC] \in \mathcal{I}^{(0)}$, conjugation sends the latter class into $\mathcal{I}^{(1)}$, and $[X] \in \mathcal{T}^{(\mathrm{int})} \subseteq \mathcal{I}^{(1)}$. The inverse formula proves bijectivity. Since $D^{\mathsf{T}} X D = 2X$, applying the covariance successively at the vertices of a finite network cancels the internal D, D^{-1} factors across every disequality edge, leaving only a nonzero scalar. A residual binary B is carried to DBD^{T}, whose transfer satisfies

$$[DBD^{\mathsf{T}} X] = [D(BX)D^{-1}].$$

Thus nonsingularity and exclusion from the current group are preserved. Finally, D is diagonal and invertible, so it preserves support (hence endpoint nondegeneracy), tensor products, matching labels, and all flattening ranks. All resulting kernels are replaced by the already fixed realizable $\mathcal{I}^{(1)}$ representatives in their projective classes, with their nonzero realization scalars recorded. The resulting identities therefore hold for realizations over the original retained set. □

F.9.2 Card maps and the arity-four boundary

The rank of Φ_p^h is the flattening rank across the selected pair and the remaining inputs. Since its domain is $\mathrm{Mat}_2(\mathbb{C})$, it is at most four. We treat ranks below four and rank four separately. All signatures used as gadgets are directly realized over $\mathcal{F}$, with ordered inputs, nonzero scalar, and predecessor reduction recorded. The required condition is

$$\Phi_p^h(K_g) = 0 \quad \text{or} \quad \Phi_p^h(K_g) \text{ is } G\text{-matching}, \qquad \text{for every canonical } p \text{ and } K_g \in \mathcal{K}_G^{\mathrm{act}}. \tag{F.74}$$

Write $\mathcal{D}_G := \{[C] : C \in \mathcal{K}_G^{\mathrm{act}}\}$ for the domain configuration of kernel classes. A *rich domain line* is a projective line containing at least three points of $\mathcal{D}_G$. Let $\mathscr{G}_G^{\mathrm{dom}}$ be the graph on $\mathcal{D}_G$ joining two points when they lie on a common rich domain line. Such a line is *kernel-avoiding* for a linear map Φ when it is disjoint from $\mathbb{P}(\ker \Phi)$. Here $\ker \Phi$ is a subspace of the contraction-matrix space.

For an original signature of arity at least six, localization retains p and four residual inputs. The rank of the original card map is used only to choose a kernel-avoiding rich domain line and to bound the resulting six-input rank. After that signature is directly realized, its own card-map rank is recomputed before either the rank-deficient or full-rank finite reduction is invoked.

Quaternion coordinates and quaternary matching signatures. Recall the quaternion basis from Equation (F.16). Its Frobenius Gram matrix is

$$\left(\mathrm{tr}(Q_\mu^{\mathsf{T}} Q_\nu)\right)_{\mu,\nu} = 2\,\mathrm{diag}(1,-1,1,-1),$$

so the bilinear dual basis is

$$Q^0 = Q_0/2, \qquad Q^1 = -Q_1/2, \qquad Q^2 = Q_2/2, \qquad Q^3 = -Q_3/2. \tag{F.75}$$

These dual matrices recover the coefficients of the original tensor: their role is linear reconstruction. For $C \in \mathrm{Mat}_2(\mathbb{C})$, let

$$c_Q(C) = (c_0, c_1, c_2, c_3)^{\mathsf{T}} \quad \Longleftrightarrow \quad C = c_Q(C) \cdot Q := \sum_{\mu=0}^{3} c_\mu Q_\mu.$$

We use $U^{(a,b)}$ for placement on the ordered pair (a, b), as in Section F.2.9.

Frobenius-dual reconstruction gives, for every signature f and canonical pair p,

$$f = \sum_{\mu=0}^{3} (Q^\mu)^{(p)} \otimes \Phi_p^f(Q_\mu). \tag{F.76}$$

More generally, for an ordered pair p, a finite-dimensional tensor space W, and a linear map $\Phi : \mathrm{Mat}_2 \to W$, write $f_{\Phi,p} := \sum_{\mu=0}^{3} (Q^\mu)^{(p)} \otimes \Phi(Q_\mu)$. Every assertion below that an abstract card map factors refers to this reconstructed tensor.

On four ordered residual inputs define the three matching products

$$\begin{aligned} P_0(U,V) &= U(x_1,x_2)V(x_3,x_4), \\ P_1(U,V) &= U(x_1,x_3)V(x_2,x_4), \\ P_2(U,V) &= U(x_1,x_4)V(x_2,x_3). \end{aligned}$$

For a finite projective transfer group G, define its quaternary projective matching deck by

$$\mathfrak{D}_4(G) = \{[P_r(U,V)] : r \in \{0,1,2\},\ [UX], [VX] \in G\}.$$

Thus $\mathfrak{D}_4(G)$ is the projective configuration $\{[F] : F \in \mathcal{M}_4(G)\}$ of the arity-four G-matching class defined above. Define the nonzero scalar cone $\mathrm{cone}(\mathfrak{D}) := \{\lambda F : \lambda \neq 0, [F] \in \mathfrak{D}\}$.

A *rich deck line* is a projective line containing at least three points of $\mathfrak{D}_4(G)$. It is an s-line when it contains exactly s deck points. Its *matching-label type* is the partition of s formed by the multiplicities of its residual matching labels. Type (s) is *monochromatic*, type $(1,1,1)$ is a *rainbow three-line*, and every other nonmonochromatic type is *mixed*. A rich deck line is *maximal* under the analogous deck-point containment order. In the fixed order $(12 \mid 34, 13 \mid 24, 14 \mid 23)$, the flattening rank of a deck point is one at its matching split and four at each of the other two splits. For every matching product F with matching M, and every input subset S,

$$\operatorname{rank} M_{\mathbf{x}_S, \mathbf{x}_{\bar{S}}}(F) = 2^{c_M(S)}, \tag{F.77}$$

where $c_M(S)$ counts edges crossing the cut.

For $L \in \mathcal{K}_G^{\mathrm{act}}$, the binary $B_L = XL^{\mathsf{T}}$ has $[B_L X] = [XL^{\mathsf{T}}X] = [L^{\#}] \in G$, and one disequality edge satisfies $\sum_u X(x,u)B_L(u,y) = L(y,x)$. Hence, for $\mathbf{L} = (L_1, L_2, L_3, L_4) \in (\mathcal{K}_G^{\mathrm{act}})^4$,

$$(\mathscr{D}_{\mathbf{L}} q)(\mathbf{y}) = \sum_{\mathbf{x}} q(\mathbf{x}) \prod_{j=1}^{4} L_j(y_j, x_j)$$

is a constant-size realizable dressing with nonzero scalar.

Lemma F.9.4 (Eight-vertex signatures with rank triples $(2,2,2)$ or $(3,3,3)$)**.** *Let $\mathcal{F}$ be a retained, factor-closed finite algebraic signature set containing the supplied equality I. If $\mathcal{F}$ directly realizes an endpoint-nondegenerate eight-vertex signature M of flattening ranks $(2,2,2)$ or $(3,3,3)$, then*

$$\text{K-Holant}(\mathcal{F}) \text{ is } \#\mathsf{P}\text{-hard} \qquad \text{or} \qquad \mathsf{Tract}(K\mathcal{F}) \text{ holds.}$$

Proof. Apply Corollary F.4.8 to M. Its first two outcomes are the asserted conclusions. In its sole continuing outcome, M itself has, up to input order and a nonzero scalar, a factorization $E \otimes E'$ into two nonsingular binary signatures. Its matching flattening ranks are therefore $(1,4,4)$, up to permutation, by Equation (F.77), contradicting either assumed rank triple. □

Lemma F.9.5 (The Platonic quaternary boundary)**.** *Let $(\mathcal{F}, G, q)$ be a deck-stable retained, factor-closed fixed-I state with nonzero irreducible quaternary signature q. Let G be the internal A_4 form $\mathcal{T}^{(\mathrm{int})}$, any of the four marked S_4 forms $\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$, or a marked A_5 display, and define the transfer-valued pair map*

$$L_p(C) = \Phi_p^q(C)X \in \mathrm{Mat}_2(\mathbb{C}).$$

If every fixed chosen representative K_g satisfies

$$L_p(K_g) = 0 \quad \text{or} \quad [L_p(K_g)] \in G \qquad \text{for all six canonical pairs } p,$$

then internal A_4 and either A_5 display have no irreducible remaining case. In the first S_4 form, there are $\lambda \in \mathbb{C}^{\times}$ and matrices U_i with $[U_i] \in \mathcal{O}^{(\mathrm{dd})}$ such that

$$q = \lambda\,(U_1 \otimes U_2 \otimes U_3 \otimes U_4)(=_4).$$

In the transposition–transposition form, four realizable $\mathcal{O}^{(\mathrm{tt})}$-matrices dress q to

$$M_{\mathcal{O}^{(\mathrm{tt})}}(\mathbf{x}) = \mathbf{1}[\mathrm{wt}(\mathbf{x}) \equiv 0 \pmod 2](-1)^{\sum_{i<j} x_i x_j}, \tag{F.78}$$

of ranks $(2,2,2)$. *In the two mixed forms the normalized rank-two remaining case is already an endpoint-nondegenerate eight-vertex signature, with all odd entries zero and, in the even-string order* $(0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111)$, *coefficient vectors*

$$\mathbf{e}(M_{\mathrm{dt}}) = (1,1,1,1,1,1,1,1), \qquad \mathbf{e}(M_{\mathrm{td}}) = (1,i,i,i,1,1,1,i). \tag{F.79}$$

Both have ranks $(2,2,2)$. *Hence the arity-four analysis gives a tensor factorization, an unsafe binary, a generalized equality, or an endpoint-nondegenerate eight-vertex signature.*

Proof. Let $A_p \in \mathrm{Mat}_4(\mathbb{C})$ represent L_p in the quaternion coordinates above, so that $c_Q(L_p(C)) = A_p c_Q(C)$. Up to scale, the bilinear Frobenius form is $J_{\mathrm{Frob}} = \mathrm{diag}(1,-1,1,-1)$. Write $\bar{p}$ for the complementary pair and define the right-X multiplication map $\mathsf{R}_X : \mathrm{Mat}_2(\mathbb{C}) \to \mathrm{Mat}_2(\mathbb{C})$ and its coordinate matrix $\mathbf{R}_X$ by

$$\mathsf{R}_X(C) := CX, \qquad c_Q(\mathsf{R}_X(C)) = \mathbf{R}_X c_Q(C).$$

Complementary-pair contraction gives

$$L_{\bar{p}} = \mathsf{R}_X \circ L_p^* \circ \mathsf{R}_X, \tag{F.80}$$

where * is the Frobenius adjoint. Thus

$$A_{\bar{p}} = \mathbf{R}_X A_p^{\sharp} \mathbf{R}_X, \qquad A_p^{\sharp} = J_{\mathrm{Frob}} A_p^{\mathsf{T}} J_{\mathrm{Frob}}. \tag{F.81}$$

Realizable right X and reversals turn the six pair conditions into

$$\begin{aligned} &A_p\mathbf{r} = 0 \text{ or } [A_p\mathbf{r}] \in \mathcal{R}_G, \\ &A_p^{\sharp}\mathbf{r} = 0 \text{ or } [A_p^{\sharp}\mathbf{r}] \in \mathcal{R}_G \end{aligned} \quad \text{for every } [\mathbf{r}] \in \mathcal{R}_G, \qquad \mathcal{R}_G = \{[c_Q(K_g)] : g \in G\}.$$

By the classification for complementary pairs in Theorem F.A.28(i), rank three is impossible. Rank zero gives $q = 0$, ranks one and four factor, and rank two occurs only for S_4. In the first marked form, realizable inverse dressings expose $=_4$, so K-Holant$(\mathcal{F}, =_4) \le_{\mathrm{T}}$ K-Holant$(\mathcal{F})$ and Corollary F.3.13 and Lemma F.3.11 apply. In the second form, its non-$GO(X)$ similarity identifies coefficient tables only. Theorem F.A.28(i) checks the dressing to Equation (F.78), directly in realizable $\mathcal{O}^{(\mathrm{tt})}$-representatives, with ranks $(2,2,2)$. For $\mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$, the direct reconstructed tensors in Theorem F.A.28(i) are exactly Equation (F.79). These three cases invoke Lemma F.9.4. Finally, the direct correspondence Lemma F.9.3 proves the same no-remaining case conclusion for $\mathcal{I}^{(1)}$ while leaving the binary equality untouched. □

The binary reductions, equality-anchor reductions, and eight-vertex reduction invoked in this proof complete the arity-four case, so no recursive arity-four call is needed.

Lemma F.9.6 (Quadratic extension of a fixed matching)**.** *Let* Q_0, Q_1, Q_2, Q_3 *be the quaternion basis in Equation* (F.16), *and let* $\mathcal{D} \subset \mathbb{P}(\mathrm{Mat}_2)$ *contain the twelve tetrahedral points*

$$[Q_\mu] \ (0 \le \mu \le 3), \qquad [Q_0 + \varepsilon_1 Q_1 + \varepsilon_2 Q_2 + \varepsilon_3 Q_3] \quad (\varepsilon_j \in \{1,-1\}). \tag{F.82}$$

Let M *be a nonempty finite index set, let* W_e *be a nonzero finite-dimensional complex vector space for every* $e \in M$, *and let* $\Phi : \mathrm{Mat}_2 \to \bigotimes_{e\in M} W_e$ *be a nonzero linear map. Define*

$$\Sigma_M := \left\{ \left[\bigotimes_{e\in M} \mathbf{w}_e \right] : 0 \ne \mathbf{w}_e \in W_e \right\} \subseteq \mathbb{P}\left(\bigotimes_{e\in M} W_e \right), \quad \mathrm{Cone}_0(\Sigma_M) := \{0\} \cup \left\{ \bigotimes_{e\in M} \mathbf{w}_e : \mathbf{w}_e \in W_e \right\},$$

so that Σ_M *and* $\mathrm{Cone}_0(\Sigma_M)$ *are, respectively, the Segre variety and its affine cone in the terminology of the projective tensor conventions. Suppose* $\Phi(C) \in \mathrm{Cone}_0(\Sigma_M)$ *for every* $[C] \in \mathcal{D}$*. Then there is one index* $e_* \in M$ *and fixed nonzero tensors* $U_e \in W_e$*,* $e \neq e_*$*, such that*

$$\Phi(C) = \left(\bigotimes_{e \neq e_*} U_e \right) \otimes \psi(C) \qquad (C \in \mathrm{Mat}_2) \tag{F.83}$$

for a linear map $\psi : \mathrm{Mat}_2 \to W_{e_*}$*. For the reconstruction conclusion, suppose* M *partitions the residual inputs into nonempty blocks,* $W_e = (\mathbb{C}^2)^{\otimes e}$*, and* $\Phi = \Phi_p^f$ *under the corresponding input identification. If* $|M| \geq 2$*, then* $f = f_{\Phi,p}$ *has a proper tensor factor.*

Proof of Lemma F.9.6. Every defining Segre minor pulls back to a homogeneous quadratic in the quaternion coordinates. The four axes kill its square coefficients. At the eight sign points the remaining expression is

$$\sum_{0 \leq \mu < \nu \leq 3} c_{\mu\nu} \varepsilon_\mu \varepsilon_\nu = 0, \qquad \varepsilon_0 = 1.$$

The six displayed functions are distinct characters of $\{\pm 1\}^3$, so every $c_{\mu\nu}$ vanishes. All Segre minors therefore vanish identically, including at zero, and $\operatorname{im} \Phi \subseteq \mathrm{Cone}_0(\Sigma_M)$.

If a linear subspace lies in $\mathrm{Cone}_0(\Sigma_M)$, take two nonzero simple tensors $\mathbf{u} = \bigotimes \mathbf{u}_e$ and $\mathbf{v} = \bigotimes \mathbf{v}_e$ in it. If $\mathbf{u}_i, \mathbf{v}_i$ and $\mathbf{u}_j, \mathbf{v}_j$ are both independent for $i \neq j$, apply dual functionals on all other factors. A 2×2 flattening minor of $\mathbf{u}_i \otimes \mathbf{u}_j + t\, \mathbf{v}_i \otimes \mathbf{v}_j$ is then nonzero, contradicting that $\mathbf{u} + t\mathbf{v}$ is simple. Thus each pair varies in at most one factor. Fixing one nonzero $\mathbf{u}$, two vectors that vary from $\mathbf{u}$ in different factors would violate this pairwise conclusion, so all variation is in one factor. Hence the image lies in a one-factor Segre ruling, giving Equation (F.83).

For $\Phi = \Phi_p^f$, Frobenius reconstruction using Equation (F.75) gives

$$f = \left(\bigotimes_{e \neq e_*} U_e \right) \otimes \left(\sum_{\mu=0}^{3} (Q^\mu)^{(p)} \otimes \psi(Q_\mu) \right).$$

The fixed factor occupies the nonempty set $\bigcup_{e \neq e_*} e$, while the other factor contains p, so the factorization is proper. Lemma F.3.4 exposes the resulting factors under a Turing reduction. □

Lemma F.9.7 (Rank-deficient Platonic card maps)**.** *Let* $(\mathcal{F}, G, h)$ *be a marked Platonic state, where* $G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}, \mathcal{I}^{(0)}, \mathcal{I}^{(1)}\}$*. Assume* $h \neq 0$ *is a directly realized arity-six signature over* $\mathcal{F}$*, satisfies Equation* (F.74) *for* G*, and, for a specified canonical ordered pair* p*, satisfies* $\operatorname{rank} \Phi_p^h \leq 3$*. Then* h *factors, or a realizable binary gives a singular/infinite-order terminal or a successor with a strictly larger complete transfer group. No rank-deficient incidence branch remains.*

Proof. For $G = \mathcal{I}^{(1)}$, apply the card covariance and direct network correspondence of Lemma F.9.3. It identifies the coefficient calculation with the standard $\mathcal{I}^{(0)}$ calculation while replacing every realizable kernel by its realizable κ_D-image in $\mathcal{I}^{(1)}$. Thus its fixed-matching branch uses Lemma F.9.6, and its mixed branch uses Theorem F.A.28(ii)–(iii), directly in the original fixed-I state. Hence assume below that $G \neq \mathcal{I}^{(1)}$. Rank zero contradicts $h \neq 0$ by Equation (F.76), and rank one factors. If all nonzero cards use one matching, apply Lemma F.9.6. In any nonstandard S_4 form precompose with its fixed coefficient similarity and pull the varying factor back by the inverse similarity. Pulling back the coefficient identity yields $\Phi_p^h(C) = (\bigotimes_{e \neq e_*} U_e) \otimes \psi(C)$. Lemma F.3.4 exposes the proper factor. If two nonzero cards use different matchings, the local reduction Theorem F.A.28(ii) gives a

realizable rainbow localization and a quaternary whose three flattening ranks are $(3,3,3)$, while the same interface gives its realizable outside transfer. The direct-gadget direction Equation (F.3) and binary case distinction Lemmas F.3.8 and F.5.4 then apply. □

For the rank-four cases below, a *separator* is a fixed realizable contraction kernel that produces one of the reduction outputs in the next lemma.

Lemma F.9.8 (Finite Platonic quaternary reductions)**.** *Let $(\mathcal{F}, G, f)$ be a marked Platonic state, and let q be directly realized over $\mathcal{F}$, with its canonical input order and nonzero realization scalar recorded. Suppose that q is one of the explicitly enumerated representatives in Theorem F.A.28*(iii)*, including every normalized output later supplied by a marked rank-four separator. Then* K-Holant$(\mathcal{F}, q) \le_{\mathrm{T}}$ K-Holant$(\mathcal{F})$*, and q directly realizes one of the following: a rank-$(2,2,2)$ endpoint-nondegenerate eight-vertex signature, a quaternary generalized equality, or a nonsingular binary B with $[BX] \notin G$. These outputs are handled by Lemmas F.3.16, F.5.4 and F.9.4.*

Proof. This is exactly the representative-wise three-way direct-realization statement of Theorem F.A.28(iii). □

Consequently the reductions Lemmas F.9.5, F.9.7 and F.9.8 are bounded local reductions. After their displayed output signature is constructed and its finite factorization tree is processed, every branch is a resolved complexity leaf, a lower-arity factor route, or a successor with a strictly larger complete transfer group. An intermediate factor split remains inside the originating card's finite factorization tree and strictly decreases the factor potential of Equation (F.9). Thus each unresolved output decreases the stored factor potential or gives an outer deck successor.

F.9.3 Rainbow localization and matching-contraction closure

Lemma F.9.9 (Platonic matching-contraction closure)**.** *Let $G \le \mathrm{PGL}_2$ contain $[X]$ and be reversal-stable. Let F be zero or a product of nonsingular binary factors on a perfect matching, with every factor transfer in G. Contract any disjoint collection of its inputs by realizable nonsingular G-binaries (including the explicit I- or X-edge links). The result is zero or again a product of nonsingular binary factors whose transfers lie in G.*

Proof. Superpose the factor matching with the contraction matching. Components are cycles and paths. A cycle contributes a scalar (possibly zero). A path splices B_1, C, B_2 to $B_1^{\mathsf{T}} C B_2$. Using $T_B = BX$, its transfer class is

$$[T^{\#}_{B_1} X T_C X T_{B_2}], \qquad T^{\#} = X T^{\mathsf{T}} X.$$

Reversal stability, $[X] \in G$, and closure of G keep this class in G. The omitted orientation scalar is nonzero. Iterating over components proves the claim. □

Lemma F.9.6 handles cards with one matching. We now show that three collinear cards with different matchings differ on only four inputs. Closing their common factors then converts this geometric statement into a six-input gadget.

Lemma F.9.10 (Localization of a rainbow matching line)**.** *Let F_0, F_1, F_2 be three distinct nonsingular matching products on one projective line, with pairwise distinct perfect matchings M_0, M_1, M_2. Then there is a unique set S of four inputs such that the three matchings restrict on S to the three pairwise distinct perfect matchings and the three tensors have common projective binary factors on*

S^c. Moreover, this projective line contains exactly three matching-product points. After compatible projective rescaling one may write

$$F_i = R \otimes E_i \qquad (0 \le i \le 2),$$

where $R \neq 0$ is the fixed matching product on S^c, and the quaternary tensors E_0, E_1, E_2 span a two-dimensional subspace.

In the gadget setting, let $(\mathcal{F}, G, f)$ be a deck-stable marked Platonic state, where f has even arity at least six, and suppose these three tensors occur as the images of one rich domain line under the card map Equation (F.67)*. If every common binary factor U satisfies $[UX] \in G$, and the realizable kernel set $\mathcal{K}_G^{\mathrm{act}}$ spans* $\mathrm{Mat}_2(\mathbb{C})$*, those factors can be closed by realizable G-kernels without annihilating the three selected cards. This realizes an arity-six signature h with one f-vertex and exactly* $(\mathrm{arity}(f) - 6)/2$ *realizable binary kernels, and*

$$\text{K-Holant}(\mathcal{F}, h) \le_{\mathrm{T}} \text{K-Holant}(\mathcal{F}). \tag{F.84}$$

Every other realizable G-card of h remains zero or a quaternary G-matching product of nonsingular safe binaries.

Proof of Lemma F.9.10. For an input set S, let $c_i(S)$ count the edges of M_i crossing $S \mid \bar{S}$. Nonsingularity of every binary factor makes the matching recoverable from flattening ranks and gives

$$\operatorname{rank} M_{\mathbf{x}_S, \mathbf{x}_{\bar{S}}}(F_i) = 2^{c_i(S)}.$$

In particular, proportional nonsingular matching products have the same matching. Collinearity also gives, for every permutation (i, j, k),

$$2^{c_i(S)} \le 2^{c_j(S)} + 2^{c_k(S)}.$$

Decompose $M_0 \cup M_1$ into common edges and alternating cycles. For each component C, both F_0 and F_1 have flattening rank one across $C \mid C^c$, so $\operatorname{rank}_C(F_2) \le 2$. The number $c_2(C)$ is even. Hence $2^{c_2(C)} \le 2$ forces $c_2(C) = 0$. Regard the components as tensor factors. A nontrivial linear combination of two simple tensors is simple only when the two summands differ projectively in at most one factor. Since M_2 is distinct from M_0, M_1, it differs on at least one alternating cycle (a common-edge component has only one internal matching). The simple-sum criterion permits only one such varying component. Hence there is exactly one exceptional alternating cycle, and the common-edge factors outside it agree projectively in all three tensors.

The matching M_2 cannot repeat M_0 or M_1 on the exceptional component. Indeed, flatten across the $2 \mid (2\ell - 2)$ cut defined by the two endpoints of an edge of the repeated matching: the two same-matching tensors have rank one, so the third has rank at most two, whereas its different matching has two edges crossing that cut and hence rank four. Thus the three restrictions on the exceptional component are pairwise distinct.

Write the exceptional cycle as 2ℓ cyclic inputs. If $\ell \ge 3$, consider its cyclic consecutive triples. Each M_2-edge belongs to at most two of them. If every triple contained an M_2-edge, equality in this incidence count would force every such edge to be a cycle edge and hence $M_2 = M_0$ or M_1. Choose instead a triple T containing no M_2-edge. Then

$$(c_0(T), c_1(T), c_2(T)) = (1, 1, 3),$$

contradicting $8 \le 2 + 2$. Hence $\ell = 2$, so the exceptional component is the asserted set of four inputs and carries the three perfect matchings. It is unique by unique factorization. A fourth

matching-product point on the same projective line would repeat one of the three quaternary matchings. Applying the preceding repeated-matching rank argument to it and either of the other two points gives a contradiction. Thus the line has exactly three matching-product points, and the displayed decomposition $F_i = R \otimes E_i$ follows.

For each common factor U, spanning by $\mathcal{K}_G^{\mathrm{act}}$ supplies a realizable C with $\mathrm{tr}(U^{\mathsf{T}}C) \neq 0$. Closing these factors preserves all three selected cards and uses one f-vertex and $(\mathrm{arity}(f) - 6)/2$ binary kernels, proving Equation (F.84). Every other card is zero or a G-matching product by Lemma F.9.9, since the marked group contains $[X]$ and is reversal-stable. This proves Equation (F.74). □

For the second octahedral display, with H_{tt} already fixed in Equation (F.69), define the coefficient-similarity map

$$\theta_{\mathrm{tt}}(U) := H_{\mathrm{tt}}^{-1} U H_{\mathrm{tt}}, \tag{F.85}$$

the transformation tuple

$$\mathbf{H}_{\mathrm{tt}} := (H_{\mathrm{tt}}^{-1}, H_{\mathrm{tt}}^{\mathsf{T}}, H_{\mathrm{tt}}^{-1}, H_{\mathrm{tt}}^{\mathsf{T}}),$$

and its induced quaternary coefficient operator

$$\mathcal{L}_{\mathrm{tt}} := \mathscr{D}_{\mathbf{H}_{\mathrm{tt}}}, \qquad \mathcal{L}_{\mathrm{tt}}(F) = (H_{\mathrm{tt}}^{-1} \otimes H_{\mathrm{tt}}^{\mathsf{T}} \otimes H_{\mathrm{tt}}^{-1} \otimes H_{\mathrm{tt}}^{\mathsf{T}})F.$$

These maps compare coefficient tables under the fixed-coordinate convention at the start of the section.

For $\alpha \in \{\mathrm{dt}, \mathrm{td}\}$, put

$$S_\alpha := H_\alpha H_\alpha^{\mathsf{T}}, \qquad \mathcal{L}_\alpha := H_\alpha^{-1} \otimes H_\alpha^{\mathsf{T}} \otimes H_\alpha^{-1} \otimes H_\alpha^{\mathsf{T}}.$$

By Equation (F.72), $[S_\alpha] \in \mathcal{O}^{(\mathrm{dd})}$. Ordered contraction gives the three matching formulas

$$\begin{aligned}
\mathcal{L}_\alpha P_0(U, V) &= P_0(\theta_\alpha(U), \theta_\alpha(V)), \\
\mathcal{L}_\alpha P_1(U, V) &= P_1(\theta_\alpha(U S_\alpha^{-1}), \theta_\alpha(S_\alpha V)), \\
\mathcal{L}_\alpha P_2(U, V) &= P_2(\theta_\alpha(U), \theta_\alpha(S_\alpha V S_\alpha^{-1})).
\end{aligned} \tag{F.86}$$

Thus $\mathcal{L}_\alpha$ bijects the standard and mixed matching decks, preserves their matching labels, and intertwines their realizable local group actions after the displayed current-group orientation factors are absorbed. The corresponding gadget operations use realizable $\mathcal{O}^{(\alpha)}$-representatives in the original coordinates.

The incidence counts, realizable rainbow-line orbits, and deletion connectivity used below are supplied by Theorem F.A.20.

Lemma F.9.11 (High-arity Platonic localization)**.** *Let $(\mathcal{F}, G, f)$ be a deck-stable retained, factor-closed fixed-I state, where $f \neq 0$ is a minimum-arity nonbinary irreducible signature of even arity $n \geq 6$ and*

$$G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}, \mathcal{I}^{(0)}, \mathcal{I}^{(1)}\}.$$

Fix an ordered pair p, define the original card-map rank $r := \mathrm{rank}\, \Phi_p^f \in \{2, 3, 4\}$, and suppose that, for every ordered pair of inputs and every realizable G-kernel, the corresponding original card is zero or a G-matching product of nonsingular safe binaries. Then at least one of the following structural routes applies. When both are available, choose route 1.

1. All nonzero p-cards use one residual matching, and f has a proper tensor factor.

2. *One f-vertex and exactly $(n-6)/2$ realizable binary kernels directly realize a nonzero arity-six signature h, retaining p, such that*

$$2 \le \operatorname{rank} \Phi_p^h \le r \le 4, \tag{F.87}$$

and every realizable G-card of h is zero or a quaternary G-matching product of nonsingular safe binaries. In particular, $\text{K-Holant}(\mathcal{F}, h) \le_{\mathrm{T}} \text{K-Holant}(\mathcal{F})$.

The second route recomputes the rank of the directly realized six-input card map. The original rank is only the upper bound in Equation (F.87).

Proof. Use the domain graph $\mathscr{G}_G^{\mathrm{dom}}$ defined above. The kernel-deleted connectivity assertion of Theorem F.A.20(iii) says that

$$\mathcal{D}_G \setminus \mathbb{P}(\ker \Phi_p^f)$$

is nonempty and connected. If all its nonzero images have one matching, Lemma F.9.6 gives a proper factor. For any nonstandard octahedral form, precompose the card map with its fixed invertible coefficient similarity and pull the resulting tensor identity back. For $\mathcal{I}^{(1)}$, use Lemma F.9.3. These are analytic reconstructions, and factor access is supplied by Lemma F.3.4.

Otherwise choose the first matching-changing edge along a path in this deleted graph, and let ℓ be a rich domain line containing that edge. If $r \in \{2, 3\}$ and ℓ met $\mathbb{P}(\ker \Phi_p^f)$, the restriction of Φ_p^f to the two-dimensional vector span of ℓ would have rank at most one. All nonkernel images on ℓ would then be projectively proportional and have the same matching, a contradiction. If $r = 4$, the nullspace of the card map is zero. Thus ℓ is kernel-avoiding. Let V_ℓ be its two-dimensional vector span. The matching-changing edge gives $\operatorname{rank}(\Phi_p^f|_{V_\ell}) \ge 2$, while kernel-avoidance gives $\operatorname{rank}(\Phi_p^f|_{V_\ell}) \le 2$. Hence the restriction has rank two and induces a projective injection on ℓ. Let F_0, F_1 be the matching-changing images, and choose any third domain point on this rich line, with image F_2. Its matching differs from both of theirs. Indeed, if $M_2 = M_0$, choose an edge of $M_0 \setminus M_1$ and flatten across its two inputs. The ranks of F_0, F_2 are one, whereas the rank of F_1 is four, because two nonsingular factors cross this cut. This contradicts collinearity. The case $M_2 = M_1$ is identical. Thus the three images satisfy the general-arity hypotheses of Lemma F.9.10. That lemma says their line contains exactly three matching-product points, so injectivity also forces ℓ to have exactly three domain points. The quaternary incidence classification is used only after this localization.

The same lemma supplies a unique set S of four residual inputs and compatible rescalings

$$\Phi_p^f(D_i) = R \otimes E_i, \qquad R = \bigotimes_{j=1}^{(n-6)/2} U_j, \qquad 0 \le i \le 2,$$

where the E_i form the residual rainbow deck line and $D_i \in \mathcal{K}_G^{\mathrm{act}}$ are the selected domain representatives. Use the realizable closing kernels $(K_j)_j$ from the proof of Lemma F.9.10, leaving $p \cup S$. They give a nonzero one-vertex signature h with common closing scalar $\kappa = \prod_j \operatorname{tr}(U_j^{\mathsf{T}} K_j) \ne 0$. Record p, ℓ, the ordered kernels, the output order π, and κ. If C denotes the resulting contraction on the residual inputs and P_π the invertible output reindexing induced by the recorded permutation of inputs, then

$$\Phi_p^h = P_\pi \circ C \circ \Phi_p^f.$$

This proves the upper rank bound, while the surviving rainbow deck line proves $\operatorname{rank} \Phi_p^h \ge 2$.

Every other card is obtained from an original G-matching card by the remaining realizable closures. Thus Lemma F.9.9 gives zero or a G-matching product of nonsingular safe binaries. □

Whenever localization supplies a rainbow deck line, use realizable dressings on the domain pair and on the four residual inputs to send the two lines to their specified representatives. The two operations act on disjoint input sets and commute. Retain their scalars and input orders. In the standard forms $G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{I}^{(0)}\}$, let Φ be this realizable card map after one common nonzero scalar is removed, and put $G_0 = G$. For a nonstandard octahedral form, write $\theta, \mathcal{L}$ for its fixed coefficient maps and Φ_{phys} for the normalized realizable card map in the current coordinates. Its domain and image representatives are the θ- and $\mathcal{L}$-images of the standard ones. Only for coefficient calculations put

$$\Phi = \mathcal{L}^{-1} \circ \Phi_{\mathrm{phys}} \circ \theta, \qquad \Phi_{\mathrm{phys}}(\theta C) = \mathcal{L}\Phi(C), \qquad G_0 = \mathcal{O}^{(\mathrm{dd})}. \tag{F.88}$$

Thus a kernel C in a standard formula means the current-form kernel θC, and its output signature is $\mathcal{L}\Phi(C)$. The opposite icosahedral display uses its own κ_D-correspondence in Lemma F.9.3. In these standard coefficient coordinates define

$$A = P_0(Q_0, Q_0), \qquad B = P_1(Q_0, Q_0), \qquad C = P_2(Q_2, Q_2) = A - B. \tag{F.89}$$

The two input matrices in the next display sum to $2Q_0$. Together with $C = A - B$, this fixes the relative image scalars. Dividing by their one common nonzero scalar, the normalized map satisfies

$$\Phi(Q_0 + Q_1 + Q_2 + Q_3) = A, \qquad \Phi(Q_0 - Q_1 - Q_2 - Q_3) = -B. \tag{F.90}$$

Thus, defining the image tensors $W_j := \Phi(Q_j)$,

$$\Phi(Q_0) = C/2, \qquad W_1 + W_2 + W_3 = (A + B)/2, \tag{F.91}$$

and each remaining sign point imposes

$$W_j, \quad A - 2W_j, \quad 2W_j - B \in \{0\} \cup \mathrm{cone}(\mathfrak{D}_4(G_0)). \tag{F.92}$$

These are coefficient identities of realizable cards, read through Equation (F.88) in the nonstandard forms. The reductions below apply to the output signatures in the current basis.

F.9.4 The four fixed-equality marked S_4 forms

The projective octahedral group has four ordered marked-involution forms in the coordinates containing the supplied I. They are the standard $(dd; d)$ form $\mathcal{O}^{(\mathrm{dd})}$, the $(tt; d)$ form $\mathcal{O}^{(\mathrm{tt})}$, and the mixed forms $\mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$ fixed in Equations (F.69) and (F.71). They are exhaustive by Theorem F.5.16. For a normalized rainbow map, the separated-sign and sum equations have six rank-deficient solutions, handled on that directly realized six-input map by Lemma F.9.7, and twenty-four full-rank solutions, the orderings of the four frames listed in Theorem F.A.27(i). Here a frame is an unordered triple of image tensors $\{W_1, W_2, W_3\}$, whose six orderings give the corresponding card maps.

Retain the even-entry order

$$\mathbf{e}(M) = (M_{0000}, M_{0011}, M_{0101}, M_{0110}, M_{1001}, M_{1010}, M_{1100}, M_{1111}). \tag{F.93}$$

Lemma F.9.12 (Uniform octahedral full-rank exits)**.** *Let Φ be a normalized full-rank map for which $(\Phi(Q_1), \Phi(Q_2), \Phi(Q_3))$ is one of the twenty-four orderings of the four frames in Theorem F.A.27*(i)*. In a nonstandard form, Φ is the coefficient map associated to the realizable Φ_{phys} by Equation* (F.88)*. Define*

$$K_{\mathcal{O}^{(\mathrm{dd})}} := Q_0 + Q_1, \qquad K_{\mathcal{O}^{(\mathrm{tt})}} := \theta_{\mathrm{tt}}(K_{\mathcal{O}^{(\mathrm{dd})}}), \qquad K_{\mathcal{O}^{(\alpha)}} := \theta_\alpha(K_{\mathcal{O}^{(\mathrm{dd})}}) \quad (\alpha \in \{\mathrm{dt}, \mathrm{td}\}).$$

For each marked form, apply its displayed kernel to the realizable card map. At most four further realizable group binaries, used in the input dressings or pair contraction, then directly realize one of: an endpoint-nondegenerate eight-vertex signature of ranks $(2,2,2)$*, a quaternary generalized equality, or a nonsingular binary whose transfer lies outside the current complete transfer group. For* $\mathcal{O}^{(\mathrm{dt})}$*, the first-frame entries* R_0, R_1, R_3 *give the eight-vertex exits and the other six named entries give outside binaries. For* $\mathcal{O}^{(\mathrm{td})}$*,* R_0, R_3 *give the eight-vertex exits,* R_1 *gives the generalized equality, and the other six give outside binaries. Thus each full-rank map gives one of these three outputs.*

Proof. This is Theorem F.A.27(i) together with the direct current-coordinate tables in Proposition F.A.16. Every listed kernel and dressing is a chosen representative in its own marked form. Hence Equation (F.3) gives the stated directions, followed by Lemmas F.3.16, F.5.4 and F.9.4 according to the listed exit. □

For the full-rank six-input statements below, assume

$$\begin{aligned} &\mathsf{Stable}(\mathcal{F}, G, h), \qquad \operatorname{arity}(h) = \nu(\mathcal{F}) = 6, \\ &h \text{ is directly realized over } \mathcal{F}, \qquad \operatorname{rank} \Phi_p^h = 4. \end{aligned} \tag{F.94}$$

The rank in Equation (F.94) is computed from the coefficients of the directly realized h. Under this condition, put $\Phi := \Phi_p^h$. For the octahedral forms, use the auxiliary map $\Psi := \Phi$ in the first form, $\Psi := \Phi \circ \theta_{\mathrm{tt}}$ for $\mathcal{O}^{(\mathrm{tt})}$, and $\Psi := \Phi \circ \theta_\alpha$ for $\mathcal{O}^{(\alpha)}$, $\alpha \in \{\mathrm{dt}, \mathrm{td}\}$.

Theorem F.9.13 (Octahedral deck rigidity)**.** *If Equation* (F.94) *holds and*

$$G \in \{\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}\},$$

then $\mathsf{Close}_\varnothing(\mathcal{F}, G, h)$.

Proof. If card matchings vary, clauses (i)–(iii) of Theorem F.A.20 and Equation (F.77) give a rainbow deck line. Lemmas F.9.10 and F.9.12, with Equation (F.88) in a nonstandard form, then give a resolved eight-vertex, generalized-equality, or outside-binary construction.

If all nonzero cards use one matching M, the twelve test points under Ψ map to zero or an M-matching product. Lemma F.9.6 factors the tensor reconstructed from Ψ. In a nonstandard form, composing the varying factor with the inverse coefficient similarity pulls this identity back to Φ. Thus h has a proper factor, available by Lemma F.3.4.

Close that finite factorization tree directly. A nonbinary irreducible factor has arity below six and therefore lowers ν. A singular or unsafe binary is terminal, and a safe binary whose transfer class lies outside G strictly enlarges the complete transfer group. If every factor were a safe binary whose transfer class already lies in G, the original signature would be a G-matching product, contrary to the retained irreducible branch. Hence every branch is a resolved complexity leaf or an outer deck successor, proving $\mathsf{Close}_\varnothing(\mathcal{F}, G, h)$. The rank and factorization used here belong to the realized tensor h. □

F.9.5 The internal A_4 form

The internal tetrahedral case uses the standard twelve-point model $\mathcal{T}^{(\mathrm{int})}$ in Equation (F.17). It is closed under multiplication, inverse, and transpose. All normalizations below use realizable left and right $\mathcal{T}^{(\mathrm{int})}$-multiplication and permutations of inputs. The exceptional six-input signatures in this case supply H_6, from which we construct RM_8. These cores provide gadgets for testing the other retained signatures. After retaining the cores and their factors, we check the complete group and stabilize again. If the internal tetrahedral case survives, the task is to show that every remaining irreducible signature is affine in the same basis.

Theorem F.9.14 (Tetrahedral deck rigidity)**.** *If Equation* (F.94) *holds with* $G = \mathcal{T}^{(\mathrm{int})}$*, then* $\mathsf{Close}_{\{\mathscr{A}\}}(\mathcal{F}, \mathcal{T}^{(\mathrm{int})}, h)$*.*

F.9.5.1 The Pauli cores

The supplied equality and the Pauli subgroup $V_4^{(\mathrm{std})} \leq \mathcal{T}^{(\mathrm{int})}$ provide the resources for the following construction. Use the Pauli contraction notation of Section F.8.1. Recall the cores H_6 and RM_8 from Equations (F.40) and (F.58). For H_6, use the ordered pairs $A = (1,2), B = (3,4), C = (5,6)$ and the same order-three Bell-label map Θ. Direct expansion gives

$$\begin{aligned} \mathrm{supp}(\mathrm{H}_6) &= \left\{ \mathbf{x} \in \mathbb{F}_2^6 : \sum_{j=1}^{6} x_j = 0 \right\}, \\ \mathrm{H}_6(\mathbf{x}) &= (-1)^{x_1+x_1x_2+x_1x_6+x_2x_5+x_3x_5+x_3x_6} \quad (\mathbf{x} \in \mathrm{supp}(\mathrm{H}_6)). \end{aligned}$$

Hence $\mathrm{H}_6 \in \mathscr{A}$. Its full even-parity support makes H_6 irreducible. By Remark F.7.12, H_6 and RM_8 are the Shao–Cai $\widehat{f_6}$ and $f_8 = \widehat{f_8}$, up to the stated permutation of inputs. Consequently every Pauli pair card of H_6 is a nonzero product of two Pauli binaries, while every Pauli pair card of RM_8 is, up to a nonzero scalar, a product of three copies of the contracted Pauli binary. See [SC20, full version, Definition 6.12 and Lemma 6.13] and [CFS20b, Theorems 13–14].

F.9.5.2 The H_6-to-RM_8 gadget

A *complete Bell block* is an ordered Bell signature $B_{\boldsymbol{\gamma}}^{(p)}$ with both inputs of p available for attachment. An *I-path* between two equally oriented complete blocks is one copy of the retained equality I joining corresponding endpoints. Thus joining two blocks uses two I-paths.

We realize RM_8 from four copies of H_6 joined by equality signatures. In copy $\mathrm{H}_6^{(v)}$, write the ordered blocks $(\mathsf{A}_v, \mathsf{B}_v, \mathsf{C}_v) = (B_{\boldsymbol{\gamma}_v}, B_{\Theta\boldsymbol{\gamma}_v}, B_{\Theta^2\boldsymbol{\gamma}_v})$ and leave the A_v external. Connect equally oriented complete blocks by

$$\mathsf{B}_1 - \mathsf{B}_2, \qquad \mathsf{C}_2 - \mathsf{C}_3, \qquad \mathsf{B}_3 - \mathsf{B}_4, \qquad \mathsf{C}_4 - \mathsf{C}_1.$$

Each connection uses two I-paths and contributes $2\delta_{\boldsymbol{\gamma}_u, \boldsymbol{\gamma}_v}$, so all labels agree without cancellation:

$$2^4 \sum_{\boldsymbol{\gamma} \in \mathbb{F}_2^2} B_{\boldsymbol{\gamma}}^{\otimes 4} = 32\mathrm{RM}_8. \tag{F.95}$$

Restoring the known nonzero realization and path scalars gives

$$\text{K-Holant}(\mathcal{F}, \mathrm{RM}_8) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}, \mathrm{H}_6).$$

If H_6 is already realizable or available by reduction over $\mathcal{F}$, then $\text{K-Holant}(\mathcal{F}, \mathrm{H}_6) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F})$ as well. If the starting complete transfer group is $V_4^{(\mathrm{std})}$, adding the realizable gadget output leaves that complete group unchanged. The gadget requires only H_6 and realizable I-paths, so it also applies in a signature set with a larger complete transfer group. A reverse gadget construction using only RM_8 and Pauli resources is excluded by Corollary F.D.2.

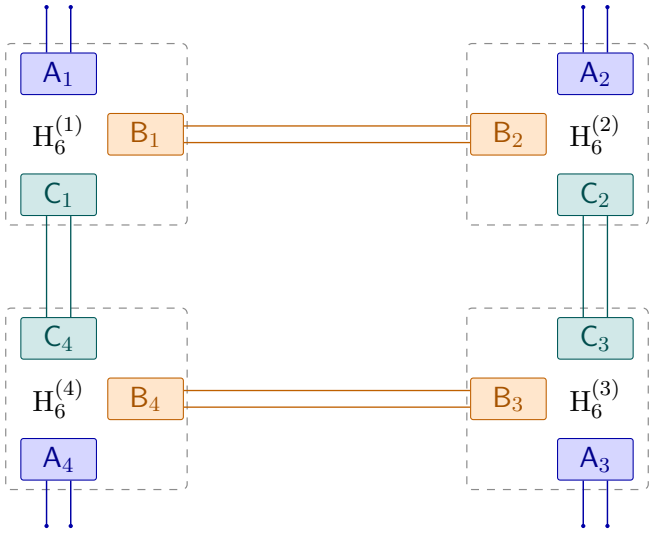


Figure F.2: The four-copy H_6-to-RM_8 gadget. In copy v, $\mathsf{A}_v, \mathsf{B}_v, \mathsf{C}_v$ denote the three ordered input pairs. Each doubled internal connection represents two I-paths joining corresponding inputs. The four A_v pairs remain external and give the eight inputs of RM_8.

F.9.5.3 Six-input tetrahedral signatures

Let $\Phi : \mathrm{Mat}_2 \to (\mathbb{C}^2)^{\otimes 4}$ be the full-rank map localized from a tetrahedral rainbow deck line. We use the common normalization Equations (F.89) to (F.92) with $G = \mathcal{T}^{(\mathrm{int})}$.

Lemma F.9.15 (Solutions of the tetrahedral separated-sign equations)**.** *The simultaneous solution set of Equation* (F.92) *is*

$$\mathcal{B}_{\mathcal{T}^{(\mathrm{int})}} = \left\{0, \frac{A}{2}, \frac{B}{2}, R_0, R_1, R_3\right\},$$

where

$$R_0 = \frac{1}{2}P_2(Q_0, Q_0), \qquad R_1 = -\frac{1}{2}P_2(Q_1, Q_1), \qquad R_3 = -\frac{1}{2}P_2(Q_3, Q_3).$$

The sum equation leaves exactly twelve ordered maps: the six rank-two permutations of $(0, A/2, B/2)$ *and the six rank-four permutations of* (R_0, R_1, R_3).

Proof. The exact separated-sign and sum calculation, including dependent lines, is Theorem F.A.27(ii). □

For the rank-four maps write $\sigma = (a, b, c)$ for a permutation of $(0, 1, 3)$. The corresponding six-input signature is uniquely reconstructed as

$$f_\sigma = \sum_{\mu=0}^{3} (Q^\mu)^{(1,2)} \otimes F^\sigma_\mu, \qquad (F^\sigma_0, F^\sigma_1, F^\sigma_2, F^\sigma_3) = (C/2, R_a, R_b, R_c).$$

Lemma F.9.16 (The two permutation classes)**.** *The six rank-four maps split into two sets:*

$$\Sigma_+ = \{(0,1,3), (1,3,0), (3,0,1)\},$$
$$\Sigma_- = \{(0,3,1), (1,0,3), (3,1,0)\}.$$

For $\sigma \in \Sigma_+$*, all fifteen deleted-pair contractions by all twelve realizable tetrahedral kernels are nonzero matching products. For* $\sigma \in \Sigma_-$*, contracting inputs* $1, 3$ *by the realizable kernel* $Q_0 = I$

gives a directly realizable quaternary q_σ whose three flattening ranks are $(2,2,2)$. For each such σ, four realizable tetrahedral input dressings transform q_σ into an endpoint-nondegenerate eight-vertex signature of the same ranks. The contraction and dressings are direct gadgets from f_σ, so Lemma F.9.4 applies in the required direction.

Proof. Theorem F.A.27(ii) gives the split, common Q_0-card, and realizable endpoint-nondegenerate eight-vertex signature dressings. Thus the two direct-gadget reductions point back to f_σ, and Lemma F.9.4 applies. □

Lemma F.9.17 (The Σ_+ signatures are affine)**.** *For every $\sigma \in \Sigma_+$, the tensor f_σ belongs to the affine class $\mathscr{A}$ in the same computational basis as all twelve tetrahedral binaries.*

Proof. Theorem F.A.27(ii) gives affine support and even mixed quadratic coefficients. Equation (F.17) puts all twelve binaries in that same affine basis. □

To lift this statement to the signature set, define the tetrahedral matrices

$$R_+ = Q_0 + Q_1 + Q_2 + Q_3, \qquad R_- = Q_0 - Q_1 - Q_2 - Q_3.$$

Then

$$R_+R_- = 4I, \qquad R_+Q_1 = -Q_0 + Q_1 - Q_2 + Q_3,$$

so $[R_+], [R_-], [Q_1], [R_+Q_1] \in \mathcal{T}^{(\mathrm{int})}$, and all four matrices have realizable inverses up to nonzero scalar. For a matrix M, let $L_j(M)$ denote its action $M(y_j, x_j)$ on input j, and let Π_{ab} exchange output inputs a, b. Products of permutations below act from right to left. Entrywise expansion in the fixed input convention gives the identities

$$\begin{aligned}
\Pi_{45}\Pi_{23}\Pi_{34}\big[I \otimes R_+ \otimes I^{\otimes 3} \otimes R_+\big]\mathrm{H}_6 &= -16 f_{(0,1,3)},\\
\Pi_{34}\big[I \otimes R_- \otimes (R_+Q_1) \otimes R_+ \otimes I^{\otimes 2}\big]\mathrm{H}_6 &= -32 f_{(1,3,0)},\\
\Pi_{23}\big[I \otimes R_+ \otimes R_+ \otimes I \otimes R_+ \otimes Q_1\big]\mathrm{H}_6 &= 32 f_{(3,0,1)}.
\end{aligned} \tag{F.96}$$

The bracketed tensor products act on the six inputs by the L_j-actions. Because $R_+^{-1} = R_-/4$ and $Q_1^{-1} = -Q_1$, every local action and permutation is projectively realizable and invertible. Thus all three equivalences in Equation (F.96) are reversible identities. Every f_σ, $\sigma \in \Sigma_+$, therefore directly gives H_6. Four copies then directly give RM_8 by Equation (F.95). Lemma F.9.26 retains both cores and regenerates the deck if the group stays $\mathcal{T}^{(\mathrm{int})}$. The remaining deck-stable set then satisfies the hypotheses of Theorem F.9.18. For any retained set $\mathcal{F}$ directly realizing f_σ, the direct-gadget reductions point from K-Holant$(\mathcal{F}, \mathrm{H}_6, \mathrm{RM}_8)$ through K-Holant$(\mathcal{F}, f_\sigma)$ to K-Holant$(\mathcal{F})$.

After the arity-four case, let a signature f have even arity at least six. Rank zero is impossible and rank one factors. In ranks two through four, Lemma F.9.11 either gives the proper fixed-matching factor or directly realizes an arity-six signature satisfying Equation (F.74). The six-input card-map rank is then recomputed from its own coefficients before Lemmas F.9.15 and F.9.16 is invoked.

F.9.5.4 Relative rigidity of the retained set

Theorem F.9.18 (Tetrahedral relative rigidity)**.** *Assume*

$$\mathsf{Stable}(\mathcal{F}, \mathcal{T}^{(\mathrm{int})}, f), \qquad \{\alpha \mathrm{H}_6, \beta \mathrm{RM}_8\} \subseteq \mathcal{F}, \qquad \alpha\beta \neq 0.$$

Then $\mathsf{Close}_{\{\mathscr{A}\}}(\mathcal{F}, \mathcal{T}^{(\mathrm{int})}, f)$.

To obtain a common affine basis, every retained occurrence g must be classified with all twelve tetrahedral kernels. A binary factor may have transfer in $\mathcal{T}^{(\mathrm{int})} \setminus V_4^{(\mathrm{std})}$, so the complete family is $\{\Phi_p^g(C) : C \in \mathcal{K}^{\mathrm{act}}_{\mathcal{T}^{(\mathrm{int})}}\}$ for each pair p. The six-input classification supplies the two sets below. An eight-input base and a stabilizer-code shortening argument then handle larger arities.

Define the projective six-input matching set, H_6-orbit, and their union by

$$\mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})}) = \left\{ \left[\prod_{\{a,b\}\in M} (U_{ab})^{(a,b)} \right] : M \text{ is a perfect matching of } [6],\ [U_{ab}X] \in \mathcal{T}^{(\mathrm{int})} \right\},$$
$$\mathscr{H}_6 = \mathrm{Orb}_{(\mathcal{T}^{(\mathrm{int})})^6 \rtimes S_6}([\mathrm{H}_6]), \qquad \mathscr{U}_6 = \mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})}) \sqcup \mathscr{H}_6.$$

Here U_{ab} is the binary matrix assigned to $\{a, b\}$, ordered with $a < b$. Define their affine cone by $\mathscr{C}_6 := \{0\} \cup \mathrm{cone}(\mathscr{U}_6)$. Here the local action uses the fixed representatives of $\mathcal{T}^{(\mathrm{int})}$, so every orbit equivalence is a realizable reversible input dressing. Their disjointness and exact cardinalities are given by Theorem F.A.23.

Proposition F.9.19 (The tetrahedral arity-eight case)**.** *Let $g \neq 0$ be an arity-eight signature in the normalized tetrahedral branch whose every tetrahedral pair card lies in $\mathscr{C}_6$. Then g factors across a $2|6$ cut or $g \in \mathscr{A}$ in the normalized basis. Thus every irreducible eight-input remaining case is affine. Factors are exposed only through factor closure.*

Proof. For a pair map $\Phi = \Phi_p^g$, choose a kernel with nonzero image. By a realizable tetrahedral multiplication on the deleted pair, take its domain point to be $[Q_0]$. Dress the residual inputs and permute them to normalize $\Phi(Q_0)$ projectively to a three-tetrahedral-binary product or H_6. By Equation (F.17), the four domain lines through $[Q_0]$ have opposite endpoints represented by $Q_0 \pm S$, where $S = Q_1 \pm Q_2 \pm Q_3$. Their images sum to $2\Phi(Q_0)$. If the two images on each line are dependent, their difference $2\Phi(S)$ lies in $\mathrm{span}\{\Phi(Q_0)\}$. The four directions S span $\mathrm{span}\{Q_1, Q_2, Q_3\}$, so $\mathrm{rank}\,\Phi = 1$. Dual-basis reconstruction then gives a tensor factor across the $2|6$ cut. Otherwise the exhaustive classification with arbitrary scalar coefficients Theorem F.A.23 says that every nonaffine parent has a realizable card outside $\mathscr{C}_6$, contrary to the hypothesis. The remaining signatures are affine, and any factors are available by Lemma F.3.4. □

To compare the twelve cards, we describe their coefficient space by a stabilizer encoder. Its input tensor factors are called logical qubits, and its output factors are the physical qubits corresponding to the original signature inputs. The next two lemmas bound the varying part by four logical qubits. The erasure criterion then identifies an output qubit that can be separated, allowing the shortening argument to select a realizable tetrahedral kernel.

F.9.5.5 Hilbert-space conventions

The stabilizer argument uses the Hermitian inner product $\langle \mathbf{u}, \mathbf{v}\rangle = \mathbf{u}^\dagger \mathbf{v}$, conjugate-linear in its first argument, with $A^\dagger$ denoting conjugate transpose. Put $\mathcal{H}_n := (\mathbb{C}^2)^{\otimes n}$, and write

$$|\mathbf{x}\rangle := |x_1\rangle \otimes \cdots \otimes |x_n\rangle, \qquad \langle \mathbf{x}| := |\mathbf{x}\rangle^\dagger \quad (\mathbf{x} \in \mathbb{F}_2^n).$$

A unitary on $\mathcal{H}_n$ satisfies $U^\dagger U = UU^\dagger = I_{\mathcal{H}_n}$, while an isometry $V : \mathcal{H}_k \to \mathcal{H}_n$ satisfies $V^\dagger V = I_{\mathcal{H}_k}$. These Hermitian conventions are distinct from the bilinear Frobenius pairing used for Holant contractions and from isometries of symmetric binary forms.

A density operator ρ has trace one and satisfies $\mathbf{u}^\dagger \rho \mathbf{u} \geq 0$ for every $\mathbf{u}$. For a nonzero vector $|\psi\rangle$, its pure-state density operator is

$$\rho_\psi = \frac{|\psi\rangle\langle\psi|}{\langle\psi, \psi\rangle}.$$

It is unchanged by nonzero scalar multiplication of $|\psi\rangle$. For a density operator ρ on $\mathcal{H}_A \otimes \mathcal{H}_B$, fix an orthonormal basis $\{|b_j\rangle\}_j$ of $\mathcal{H}_B$. The partial trace is specified, for every $|a\rangle, |a'\rangle \in \mathcal{H}_A$, by

$$\langle a| \operatorname{Tr}_B(\rho)|a'\rangle = \sum_j \langle a, b_j|\rho|a', b_j\rangle.$$

The result is independent of the chosen orthonormal basis of $\mathcal{H}_B$. The operator $\operatorname{Tr}_B(\rho)$ is the *marginal* on $\mathcal{H}_A$. A one-qubit marginal is maximally mixed when it is $I_{\mathbb{C}^2}/2$. Every bipartite pure state has a Schmidt decomposition

$$|\psi\rangle = \sum_{j=1}^{s} \sigma_j |\mathbf{u}_j\rangle \otimes |\mathbf{v}_j\rangle, \qquad \sigma_j > 0,$$

with both displayed families orthonormal. s is the Schmidt rank [NC10, Section 2.5]. For $n \geq 1$, $r \in [n]$, and a one-qubit linear operator $R : \mathbb{C}^2 \to \mathbb{C}^2$, define

$$R_r := I_{\mathbb{C}^2}^{\otimes(r-1)} \otimes R \otimes I_{\mathbb{C}^2}^{\otimes(n-r)} : \mathcal{H}_n \to \mathcal{H}_n.$$

F.9.5.6 Pauli and stabilizer conventions

The n-qubit Pauli group is

$$\mathcal{P}_n^{(4)} := \{i^c \bigotimes_{j=1}^n X^{a_j} Z^{b_j} : c \in \mathbb{Z}_4,\ \mathbf{a}, \mathbf{b} \in \mathbb{F}_2^n\}.$$

Its projectivization by the scalar subgroup $\{\pm I_{\mathcal{H}_n}, \pm i I_{\mathcal{H}_n}\}$ is identified with $\mathsf{P}([n]) = \mathbb{F}_2^n \oplus \mathbb{F}_2^n$. The label $\mathbf{a} \mid \mathbf{b}$ is the concatenation of the two binary strings. This label space has the symplectic form

$$\omega((\mathbf{a} \mid \mathbf{b}), (\mathbf{a}' \mid \mathbf{b}')) = \mathbf{a} \cdot \mathbf{b}' + \mathbf{b} \cdot \mathbf{a}'.$$

Here symplectic means alternating $\omega(\mathbf{v}, \mathbf{v}) = 0$ and nondegenerate $\{\mathbf{v} : \omega(\mathbf{v}, \mathbf{w}) = 0 \text{ for all } \mathbf{w}\} = \{0\}$. For a subspace S, put $S^{\perp\omega} := \{\mathbf{v} : \omega(\mathbf{v}, \mathbf{s}) = 0 \text{ for every } \mathbf{s} \in S\}$. S is *symplectic Lagrangian* when $S = S^{\perp\omega}$. If S is Lagrangian, a *Lagrangian complement* of S is a Lagrangian subspace T satisfying $\mathsf{P}([n]) = S \oplus T$.

A *projective Clifford* is the class, modulo scalars in S^1, of a unitary U satisfying $U\mathcal{P}_n^{(4)}U^\dagger = \mathcal{P}_n^{(4)}$. Clifford conjugation induces a symplectic linear map on Pauli labels [NC10, Chapter 10]. A stabilizer code is the common eigenspace

$$\{\mathbf{v} : P\mathbf{v} = \chi(P)\mathbf{v} \text{ for every } P \in \widetilde{S}\}$$

of a commuting Pauli subgroup $\widetilde{S}$, where $\chi : \widetilde{S} \to \mu_4$ is an eigenvalue character agreeing with the scalar by which every scalar Pauli acts. A one-dimensional stabilizer code is a stabilizer state, and a nonzero tensor spanning it is a stabilizer tensor. Up to nonzero scalar normalization, the nonzero affine signatures are exactly the stabilizer states: their amplitudes have affine support and a quadratic phase polynomial with even mixed coefficients [DD03, Theorem 5]. Clifford changes preserve this class.

A stabilizer isometry is an isometry whose image is a stabilizer code and whose logical Paulis have physical Pauli representatives. Equivalently it is a Clifford encoder with fixed stabilizer-state ancillas. Here an *encoder* is the code isometry, a *physical operator* acts on its ambient output space, the corresponding *logical operator* is the induced operator on the encoded input space, and an *ancilla* is a fixed initialized tensor factor. If S is the projective stabilizer label space, the logical

Pauli label space is $S^{\perp_\omega}/S$. For a 2^k-dimensional code it has binary dimension $2k$. A Clifford decoder is a Clifford unitary D satisfying

$$DV|\psi\rangle = |\psi\rangle \otimes |\eta\rangle \qquad (|\psi\rangle \in \mathcal{H}_k)$$

for a fixed stabilizer ancilla $|\eta\rangle$. These standard encoder–decoder and logical-Pauli facts follow from the stabilizer normal form [Got97, Chapters 3–4]. In particular, decoding a one-dimensional stabilizer code on m qubits sends its spanning tensor to a nonzero multiple of $\delta_{\mathbf{0}^m}$, as used below.

Lemma F.9.20 (Sharp stabilizer-line bound)**.** *Let u, v be nonproportional nonzero affine signatures of common arity m, and let $c \neq 0$. If $u + cv$ is also nonzero and affine, then, after applying the same analytic Clifford change to both tensors, followed by a common nonzero scalar normalization sending u to $\delta_{\mathbf{0}^m}$, the support of the transformed v is either a singleton disjoint from zero or a linear subspace of dimension at most two. In the already normalized case $u = \delta_{\mathbf{0}^m}$, the same support alternative holds for v in the current coordinates, without a further Clifford change.*

Proof. Choose a unitary Clifford C with $Cu = \alpha\delta_{\mathbf{0}^m}$, where $\alpha \neq 0$, and apply $\alpha^{-1}C$ to both tensors. Relabel the resulting second tensor as v. Write v in affine normal form as $v = \lambda\mathbf{1}[\mathbf{a} + L]i^Q$, where $\lambda \in \mathbb{C}^\times$ is the scale, $\mathbf{a} + L$ is the affine support, Q is the quadratic phase polynomial, and $d := \dim L$. If $0 \notin \mathbf{a} + L$, the support size $1 + 2^d$ forces $d = 0$. If $0 \in \mathbf{a} + L = L$ and the zero coefficient cancels, $2^d - 1$ forces $d = 1$. Otherwise both v and $w := \delta_{\mathbf{0}^m} + cv$ have support L. Their quotient $q(\mathbf{x}) := w(\mathbf{x})/(cv(\mathbf{x}))$ equals one on $L \setminus \{0\}$, whereas $q(0) = 1 + 1/(cv(0)) \neq 1$. If $d \geq 3$, choose independent $\mathbf{a}, \mathbf{b}, \mathbf{e} \in L$. Since both phases are quadratic, their quotient satisfies

$$q(0)q(\mathbf{a}+\mathbf{b})q(\mathbf{a}+\mathbf{e})q(\mathbf{b}+\mathbf{e}) = q(\mathbf{a})q(\mathbf{b})q(\mathbf{e})q(\mathbf{a}+\mathbf{b}+\mathbf{e}).$$

Every displayed argument except zero is nonzero, so the identity forces $q(0) = 1$, a contradiction. Hence $d \leq 2$. The support argument starts only from $u = \delta_{\mathbf{0}^m}$. It therefore also proves the already normalized clause in the statement. □

Lemma F.9.21 (Tetrahedral card maps have at most four logical qubits)**.** *Let $\Phi : \mathrm{Mat}_2(\mathbb{C}) \to (\mathbb{C}^2)^{\otimes m}$ be linear. Suppose that $\Phi(C_{\mathrm{ker}})$ is zero or affine for every binary contraction kernel C_{ker} with $[C_{\mathrm{ker}}] \in \mathcal{T}^{(\mathrm{int})}$. Then there are an analytic Clifford operator C on the m output qubits and a subspace $R \leq \mathbb{F}_2^m$, $\dim R \leq 4$, such that*

$$\mathrm{supp}(C\Phi(C_{\mathrm{ker}})) \subseteq R \qquad (C_{\mathrm{ker}} \in \mathrm{Mat}_2(\mathbb{C})). \tag{F.97}$$

Equivalently, the image of Φ is contained in one stabilizer code with at most four logical qubits.

Proof. If $\Phi = 0$, take $R = \{0\}$. Otherwise at least one tetrahedral test point has nonzero image because those points span Mat_2. Precompose Φ by a realizable tetrahedral multiplication carrying that point to $A := Q_0 - Q_1 - Q_2 - Q_3$, relabel the resulting map as Φ, and hence assume $\Phi(A) \neq 0$. Define

$$B_0 := 2Q_0 - A, \qquad B_j := A + 2Q_j \quad (1 \leq j \leq 3).$$

These are the four opposite-parity tetrahedral sign points paired with A. They form a basis and satisfy

$$A + B_0 = 2Q_0, \qquad B_j - A = 2Q_j \ (1 \leq j \leq 3), \qquad 2A = -B_0 + B_1 + B_2 + B_3.$$

Choose an analytic Clifford C with $C\Phi(A) = \alpha\delta_{\mathbf{0}^m}$, where $\alpha \neq 0$. For the following analysis put $\Psi(C_{\mathrm{ker}}) := \alpha^{-1}C\Phi(C_{\mathrm{ker}})$. The common scalar leaves every support unchanged. The already

normalized clause of Lemma F.9.20 applies in these fixed coordinates: $\Psi(A) = \delta_{\mathbf{0}^m}$, and the displayed identities give the required affine sums for the B_j. Thus each nonzero $\Psi(B_j)$ not proportional to $\delta_{\mathbf{0}^m}$ has, off zero, either one point or the three nonzero points of a two-dimensional $\mathbb{F}_2$-subspace. The last displayed identity makes every point in their union occur at least twice. If its span had dimension at least five, total incidence at most twelve would force sizes $(3,3,3,1)$ or $(3,3,3,3)$. In type $(3,3,3,1)$, double incidence leaves at most five distinct nonzero points, and any three-point support is the nonzero part $\{\mathbf{x}, \mathbf{y}, \mathbf{x}+\mathbf{y}\}$ of a two-dimensional $\mathbb{F}_2$-subspace. Its linear dependence is nontrivial, so those at most five points have span at most four.

In type $(3,3,3,3)$, double incidence leaves at most six points. The only remaining extremal case has six points, each occurring in exactly two of the four supports. Regard a point shared by supports i, j as an edge between vertices i, j. The four supports form a loopless three-regular multigraph on four vertices, and the linear dependence at each vertex is the sum of its three incident edge labels. The binary vertex–edge incidence matrix has rank at least two: its rank is $4 - c$, where the graph has $c \leq 2$ connected components. Hence the six edge labels satisfy at least two independent linear dependencies and again span at most four. Thus their union spans some R with $\dim R \leq 4$. Since the B_j's form a basis and the common nonzero scalar leaves supports unchanged, Equation (F.97) follows. □

For shortening, use the identification of the octahedral group $\mathcal{O}^{(\mathrm{dd})}$ with the projective one-qubit Clifford group in the quaternion coordinates of Equation (F.16). The normal subgroup $\mathcal{T}^{(\mathrm{int})} \triangleleft \mathcal{O}^{(\mathrm{dd})}$ has index two. Each of its two cosets contains the following eight full-support representatives:

$$\begin{aligned} \mathcal{C}_+ &= \{Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3 : \epsilon_j \in \{\pm 1\}\}, \\ \mathcal{C}_- &= \{Q_d + \epsilon Q_o : d \in \{0,3\},\ o \in \{1,2\},\ \epsilon \in \{\pm 1\}\}. \end{aligned} \tag{F.98}$$

On the open set $C_{01}C_{10} \neq 0$, the coefficient cross ratio is $\mathrm{cr}(C) = C_{00}C_{11}/(C_{01}C_{10})$. Direct use of Equation (F.16) gives

$$C_{xy} \neq 0, \qquad \mathrm{cr}(C) = -1 \qquad (C \in \mathcal{C}_+ \cup \mathcal{C}_-). \tag{F.99}$$

Indeed, in $\mathcal{C}_+$ the diagonal and off-diagonal products are 2 and -2, while in $\mathcal{C}_-$ they are 1 and -1. After division by one entry, every such coefficient signature and its pointwise inverse have entries in μ_4, so both are nowhere-zero affine phase tables.

F.9.5.7 Stabilizer error correction

A stabilizer tableau is a binary symplectic generator matrix whose rows label Pauli generators, together with their phases. Multiplying generators is an elementary row operation, while a Clifford coordinate change acts by a symplectic change of columns [Got97, Chapter 4]. A hyperbolic pair is a pair $\mathbf{p}, \mathbf{q}$ with $\omega(\mathbf{p}, \mathbf{q}) = 1$. Symplectic Gaussian elimination is row reduction by changes of symplectic basis. A nowhere-zero stabilizer quadratic phase is

$$\zeta(\mathbf{z}) = \zeta_0 i^{Q(\mathbf{z})}, \qquad \zeta_0 \in \mathbb{C}^\times,$$

where $Q : \mathbb{F}_2^k \to \mathbb{Z}_4$ is multilinear of degree at most two with even mixed coefficients. After fixed physical and logical Clifford coordinate changes, a stabilizer isometry has amplitude normal form

$$V|\mathbf{z}\rangle = \zeta(\mathbf{z})|\mathbf{c} + M^{\mathsf{T}}\mathbf{z}\rangle, \qquad \mathbf{c} \in \mathbb{F}_2^n, \quad M \in \mathrm{Mat}_{k\times n}(\mathbb{F}_2), \quad \mathrm{rank}\, M = k,$$

a standard consequence of stabilizer-tableau reduction [Got97, Chapter 4].

Erasure of physical output qubit r of an isometry V is correctable exactly when the single-site Knill–Laflamme condition holds:

$$V^\dagger R_r V \in \mathbb{C} I_{\mathcal{H}_k} \qquad \text{for every one-qubit operator } R.$$

For a specified erased qubit, the error-operator products are the matrix units $(|a\rangle\langle b|)_r$, $a, b \in \{0, 1\}$. They span its operator space, giving the displayed criterion [KL97]. Equivalently, one may test the Pauli basis $I, X, -iY, Z$. The erased qubit then carries no information about the logical input. In the stabilizer setting, an encoder tableau is the tableau of a Clifford encoder. Its symplectic row reduction records physical representatives of logical Paulis and the residual stabilizer generators.

Lemma F.9.22 (Stabilizer erasure and column normal form)**.** *Let $V : (\mathbb{C}^2)^{\otimes k} \to (\mathbb{C}^2)^{\otimes N}$ be a stabilizer isometry and let r be a physical output qubit. If erasure of r is correctable, meaning $V^\dagger R_r V$ is scalar for every one-qubit operator R, exactly one of the following normal forms applies after a fixed one-qubit Clifford on r:*

1. there are a one-qubit stabilizer state $|\chi\rangle_r$ and a stabilizer isometry W such that

$$V|\psi\rangle = |\chi\rangle_r \otimes W|\psi\rangle; \tag{F.100}$$

2. there are an auxiliary qubit a and a stabilizer isometry $W : \mathbb{C}^2_a \otimes (\mathbb{C}^2)^{\otimes k} \to (\mathbb{C}^2)^{\otimes(N-1)}$ such that

$$V|\psi\rangle = 2^{-1/2} \sum_{c \in \mathbb{F}_2} |c\rangle_r W(|c\rangle_a \otimes |\psi\rangle). \tag{F.101}$$

If no single-site erasure is correctable, then, after postcomposing the outputs by a tensor product of one-qubit Cliffords and precomposing the input by one logical Clifford, there are a vector $\mathbf{c} \in \mathbb{F}_2^N$, and a matrix $M = (\mathbf{m}_1 \ \cdots \ \mathbf{m}_N) \in \mathbb{F}_2^{k \times N}$ such that

$$V|\mathbf{z}\rangle = \zeta(\mathbf{z})|\mathbf{c} + M^{\mathsf{T}}\mathbf{z}\rangle, \qquad \operatorname{rank} M = k, \qquad \mathbf{m}_r \neq 0 \quad (1 \le r \le N), \tag{F.102}$$

where $\zeta(\mathbf{z}) = \zeta_0 i^{Q(\mathbf{z})}$ is a nowhere-zero stabilizer quadratic phase in the precise sense of Section F.9.5.7.

Proof. By the Hilbert-space and error-correction conventions of Sections F.9.5.5 and F.9.5.7, correctability of erasure at r is the single-site Knill–Laflamme condition

$$V^\dagger P_r V = \alpha_P I \qquad (P \in \{I, X, -iY, Z\}). \tag{F.103}$$

Here the Hermitian Pauli matrix is $\sigma_y = -iY$, with $Y = ZX$. Conjugate P_r through a Clifford encoder and then evaluate its fixed ancillas. The three nonidentity expectations in Equation (F.103) are either all zero or exactly one is ± 1. Thus the input-independent marginal at r is respectively maximally mixed or a pure one-qubit stabilizer state. In the pure case its Schmidt rank is one, which gives Equation (F.100). A one-qubit Clifford sends the pure factor to a computational-basis state. Its fixed Pauli generator can then be removed from the encoder tableau. The remaining stabilizer generators and logical representatives are those of W, so W is a stabilizer isometry as asserted.

In the maximally mixed case a Schmidt decomposition gives

$$V|\psi\rangle = 2^{-1/2}\big(|0\rangle_r W_0|\psi\rangle + |1\rangle_r W_1|\psi\rangle\big).$$

Applying Equation (F.103) to the four matrix units on r gives $W_c^\dagger W_d = \delta_{cd} I$. Hence $W(|c\rangle_a \otimes |\psi\rangle) = W_c|\psi\rangle$ is an isometry. To show that W is a stabilizer isometry, we eliminate the erased qubit from

the encoder tableau as follows. First let U be any stabilizer isometry and let P be a Hermitian Pauli with $U^\dagger PU = 0$. Some Hermitian code stabilizer S, with $SU = U$, anticommutes with P. Otherwise P would preserve the code, and its induced unitary on the code would be zero. Since $(PS)^2 = -I$, the operator

$$C_P := 2^{-1/2}(I + PS), \qquad C_P U = 2^{-1/2}(I + P)U$$

is a Clifford quarter turn: conjugating a Pauli that commutes with PS leaves it fixed, and conjugating one that anticommutes with PS multiplies it by PS. Thus the normalized positive Pauli projection on the right preserves the stabilizer-isometry property, including the physical representatives of logical Paulis.

Apply this observation to $U_0 = I_a \otimes V$, with output order $(a, r, \text{remaining qubits})$, and to the commuting Hermitian Paulis $P_Z = Z_a Z_r$ and $P_X = X_a X_r$. The maximally mixed case gives

$$U_0^\dagger P_Z U_0 = U_0^\dagger P_X U_0 = U_0^\dagger P_Z P_X U_0 = 0.$$

For $U_1 = 2^{-1/2}(I + P_Z)U_0$, commutation and $P_Z^2 = I$ give $U_1^\dagger P_X U_1 = U_0^\dagger (P_X + P_Z P_X) U_0 = 0$. Consequently both normalized projections are stabilizer isometries. Writing $|\beta\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, their exact coefficient identity is

$$U_2 := \tfrac{1}{2}(I + P_X)(I + P_Z)(I_a \otimes V) = |\beta\rangle_{ar} \otimes W.$$

Indeed, the coefficient at $|cc\rangle_{ar}$ is $\sum_d \langle d|_r V \psi_d = 2^{-1/2} W(\sum_d |d\rangle_a \psi_d)$, and the coefficients at $|01\rangle, |10\rangle$ vanish. A controlled addition from a to r, followed by a Hadamard on a, sends $|\beta\rangle$ to $|00\rangle$. The resulting tableau has the fixed generators Z_a, Z_r. Multiplying its other stabilizer rows by these generators removes their entries on a, r, and the same operation removes those entries from the logical representatives. Deleting the two fixed-zero coordinates therefore leaves the code and logical-Pauli data of W. This proves Equation (F.101). These are analytic encoder operations. The realizable Holant kernel is selected in the shortening argument that follows.

Suppose now that no single-site erasure is correctable. For each r, failure of the displayed Knill–Laflamme condition gives a one-site Pauli whose pullback through the stabilizer encoder is nonscalar. By the logical Pauli convention in Section F.9.5.6, this is a nontrivial logical Pauli with a physical representative supported on r alone. Representatives on different inputs commute, so their logical Paulis commute. Independent physical one-qubit Cliffords send them to Z_r, and one logical Clifford sends the generated commuting logical subgroup to Z-type. Consequently each $V|\mathbf{z}\rangle$ is a simultaneous eigenvector of every physical Z_r, and hence is one computational-basis ket times a phase. Its r-th bit is $c_r + \mathbf{m}_r \cdot \mathbf{z}$. Nontriviality of the logical Pauli gives $\mathbf{m}_r \neq 0$. If $\operatorname{rank} M < k$, some $0 \neq \mathbf{d} \in \ker M^{\mathsf{T}}$ makes $V|\mathbf{z}\rangle$ and $V|\mathbf{z} + \mathbf{d}\rangle$ proportional, contrary to isometry. Thus $\operatorname{rank} M = k$, and the stabilizer amplitude normal form in Section F.9.5.7 makes the remaining nowhere-zero phase quadratic modulo four, proving Equation (F.102). □

The next lemma applies the code structure to a signature with some inputs left unencoded. Its physical inputs include both those untouched inputs and the output qubits of the encoder.

Lemma F.9.23 (Shortening a nonaffine encoded signature by a tetrahedral contraction)**.** *Let $t \geq 1$, let $V : (\mathbb{C}^2)^{\otimes k} \to (\mathbb{C}^2)^{\otimes N}$ be a stabilizer isometry with $k \leq 4$ and $N \geq k+2$, and let $\xi \in (\mathbb{C}^2)^{\otimes(t+k)} \setminus \mathscr{A}$ be the logical coefficient tensor, including the t untouched inputs. Put $F := (I^{\otimes t} \otimes V)\xi$. Then F factors or one realizable tetrahedral contraction of two physical qubits is nonzero and nonaffine.*

Proof. Apply Lemma F.9.22. If erasure of a physical output qubit r of the encoder is correctable, its pure-marginal case Equation (F.100) gives a unary tensor factor. In the maximally-mixed case Equation (F.101) isolates an auxiliary qubit a. Contract r with one of the t untouched inputs using the realizable kernel Q_0, whose class lies in $\mathcal{T}^{(\mathrm{int})}$. In the analytic local frame of r, this acts on the untouched-input/a coefficient array by an invertible one-qubit Clifford. The residual encoder is a nonzero stabilizer isometry, and a Clifford and its inverse preserve $\mathscr{A}$. Consequently the shortened card is nonaffine.

Assume now that no one-site erasure is correctable. Write the normal form of Equation (F.102) explicitly as

$$\widetilde{V} = \left(\bigotimes_{j=1}^{N} C_j^{-1}\right) V L, \qquad \widetilde{\xi} = (I^{\otimes t} \otimes L^{-1})\xi,$$

where L is the logical Clifford and the C_j's map normal-form output coordinates back to original coordinates. Then $\widetilde{\xi} \notin \mathscr{A}$, since Clifford maps and their inverses preserve $\mathscr{A}$, and $F = (I^{\otimes t} \otimes (\bigotimes_j C_j)\widetilde{V})\widetilde{\xi}$. Let $\mathbf{c}$ and $M = (\mathbf{m}_1 \ \cdots \ \mathbf{m}_N)$ be the data for $\widetilde{V}$. Choose a k-column basis of M. Since $N \geq k+2$, choose two positions r, s outside it. Deleting their columns leaves rank k. Choose matrix representatives C_r, C_s of the two analytic local Clifford frames used in the normal form, with $[C_r], [C_s] \in \mathcal{O}^{(\mathrm{dd})}$. Normality and index two imply that

$$\{[C_r^{\mathsf{T}} C_{\mathrm{ker}} C_s] : [C_{\mathrm{ker}}] \in \mathcal{T}^{(\mathrm{int})}\}$$

is one of the two projective $\mathcal{T}^{(\mathrm{int})}$-cosets in $\mathcal{O}^{(\mathrm{dd})}$ (the X-twist only relabels that coset). By Equations (F.98) and (F.99), choose a realizable C_{ker} with $[C_{\mathrm{ker}}] \in \mathcal{T}^{(\mathrm{int})}$ whose effective contraction matrix $C_{\mathrm{eff}} = C_r^{\mathsf{T}} C_{\mathrm{ker}} C_s$ has full support and cross ratio -1. Thus the analytic frames determine a realizable kernel C_{ker} in the original coordinates.

For a logical string $\mathbf{z}$, the two contracted bits are

$$\ell_r(\mathbf{z}) = c_r + \mathbf{m}_r \cdot \mathbf{z}, \qquad \ell_s(\mathbf{z}) = c_s + \mathbf{m}_s \cdot \mathbf{z}.$$

Each logical string $\mathbf{z}$ determines a unique physical codeword, so the contraction multiplies its coefficient by

$$\omega(\mathbf{z}) = C_{\mathrm{eff}}(\ell_r(\mathbf{z}), \ell_s(\mathbf{z})).$$

The full-support property makes ω nowhere zero. By Equation (F.99), after multiplication by a common nonzero scalar, $\omega = i^{q(\mathbf{z})}$ for a modulo-four quadratic q, and ω^{-1} has the same property. The undeleted columns retain rank k, so the residual map to computational-basis strings is injective. Multiplication by ω, its inverse, and this injective stabilizer encoding preserve affine membership in both directions, as does the normal-form phase $\zeta(\mathbf{z})$. Since $\widetilde{\xi} \notin \mathscr{A}$, the realizable two-input C_{ker}-card is nonzero and nonaffine. □

Proposition F.9.24 (Gadget tetrahedral localization)**.** *Let g be an even-arity irreducible signature in a fixed-I standard marked A_4 state whose complete transfer group remains $\mathcal{T}^{(\mathrm{int})}$ after factor closure. Assume the established terminal exits do not apply. If $g \notin \mathscr{A}$ and* $\mathrm{arity}(g) \geq 10$*, some realizable tetrahedral pair card of g is nonzero and nonaffine. Iteration therefore reaches a realized nonaffine signature on at most eight inputs, unless a factor, terminal, or restart with a strictly larger complete transfer group occurs, using one g-vertex and $O(\mathrm{arity}(g))$ realizable binaries.*

Proof. If every realizable pair card were zero or affine, reconstruction and Lemma F.9.21 would place each card-map image in a stabilizer code with $k \leq 4$ logical qubits. Frobenius reconstruction would then write $g = (I^{\otimes 2} \otimes V)\xi$, with V encoding those k qubits into $N = \mathrm{arity}(g) - 2 \geq 8$

sites and $\xi \notin \mathscr{A}$. Since $N \geq k+2$, Lemma F.9.23 gives a factor or a nonaffine realizable card, a contradiction. Iterating this two-input descent reaches arity at most eight unless a terminal, factor, or recomputed-group restart occurs. The direct-gadget and factor-closure directions are Equation (F.3) and Lemma F.3.4. □

F.9.5.8 Induction over the retained signatures

We close the relative tetrahedral argument by a finite induction on unresolved occurrences and their remaining contractions. On entering the argument, consider nonzero odd-arity signatures by Theorem F.2.9, unaries by Lemma F.3.11, nullaries by preprocessing, and newly exposed binaries by Lemmas F.3.8 and F.5.4. Thus, on a continuing branch, every unresolved nonbinary irreducible occurrence has even arity and every binary transfer belongs to the recomputed group $\mathcal{T}^{(\mathrm{int})}$.

Let $\mathcal{P}$ be the finite multiset of unresolved nonbinary irreducible occurrences. When a token u, carrying g_u, is selected, fix the finite current-coordinate universe consisting of all iterated proper tetrahedral pair cards of g_u, all realizable dressings and permutations of inputs used by the finite reductions below, and the complete factorization tree of every output. This universe is finite: every proper contraction or factor lowers arity, $\mathcal{T}^{(\mathrm{int})}$ and the reduction list are finite, each arity has finitely many input choices and dressings, and each tensor has a finite irreducible-factorization tree. Write its remaining indices as Q_u and let $\varphi_u \in \mathbb{N}$ be the sum of their factor potentials from Equation (F.9). Define

$$\Theta_{\mathrm{Pl}}(\mathcal{P}) = \{\{(\mathrm{arity}(g_u), |Q_u|, \varphi_u) : u \in \mathcal{P}\}\},$$

using lexicographic order on triples and its well-founded multiset extension.

Before applying the next induction step to u, regenerate its complete oriented current-group deck. Process each proper card and factor, either resolving it or entering it as a lower-arity occurrence with smaller Θ_{Pl}. The queue Q_u then contains only unconsumed finite reduction or certificate indices, not unresolved proper signatures. Before a proper card is returned, consume its unique originating index in Q_u and attach that index, marked completed, to the child's reduction data. It is not copied into the child's unfinished index set. If the card factors, its finite factorization tree is processed under that same originating index, and every intermediate split strictly lowers the stored factor potential.

Processing u therefore replaces its triple by a finite multiset of child triples. Every proper child has smaller first coordinate. If a residual copy of u remains, it has the same arity but one fewer fixed card/factor index or smaller factor potential. Thus every inserted triple is strictly smaller than the removed parent triple in lexicographic order. Certification removes the triple altogether. The standard multiset extension therefore makes every same-coordinate local step strictly decrease Θ_{Pl}. A terminal, lower ν, or strict enlargement of the complete transfer group is instead an outer exit.

Lemma F.9.25 (The arity-four case for Platonic groups)**.** *Let $\mathcal{F}$ be the current retained factor-closed fixed-I signature set with marked group of the resulting set*

$$G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}, \mathcal{I}^{(0)}, \mathcal{I}^{(1)}\}.$$

Let $q \in \mathcal{F}$ be a nonzero irreducible quaternary occurrence whose complete oriented deck and all proper cards and factors have been processed as above. For each canonical pair p, put $L_p := \mathsf{R}_X \circ \Phi_p^q$. Suppose

$$L_p(K_g) = 0 \quad \text{or} \quad [L_p(K_g)] \in G \qquad (K_g \in \mathcal{K}_G^{\mathrm{act}})$$

for all six canonical pairs. Then q has exactly the factor, generalized equality, endpoint-nondegenerate eight-vertex, or outside-binary outcomes of Lemma F.9.5. In particular, no irreducible quaternary occurrence continues in the internal tetrahedral or either icosahedral form.

Proof. The proof of Lemma F.9.5 uses deck stability only to supply the displayed pair conditions and to exclude an unresolved proper card or factor. Here the pair conditions are assumed explicitly, and the preceding steps have already resolved or entered every such proper object at a smaller value of Θ_{Pl}. The finite preserver classification for complementary pairs and all of its realizable terminal gadgets therefore apply. □

Lemma F.9.26 (Adding the tetrahedral core signatures)**.** *Assume Equation* (F.94) *with* $G = \mathcal{T}^{(\mathrm{int})}$*, and suppose the full-rank analysis reaches a signature* f_σ*,* $\sigma \in \Sigma_+$*, by reversible realizable tetrahedral dressings and a permutation of inputs. Then nonzero scalars* α, β *are recorded such that the signature set*

$$\mathcal{F}_{\mathrm{HR}} := \mathcal{F} \cup \{\alpha \mathrm{H}_6, \beta \mathrm{RM}_8\}$$

satisfies K-Holant$(\mathcal{F}_{\mathrm{HR}}) \equiv_{\mathrm{T}}$ K-Holant$(\mathcal{F})$*. Close the finite core batch and consider its group. On an unchanged* $\mathcal{T}^{(\mathrm{int})}$*-branch, regenerate the complete deck of* $\alpha \mathrm{H}_6$*, retain every card and its factorization tree, close the retained set under irreducible factors, consider all outputs, and form the group and deck for this set. Every branch is then an outer exit or has a retained residue* $\mathcal{F}^\star_{\mathrm{HR}}$ *with*

$$\mathsf{Stable}(\mathcal{F}^\star_{\mathrm{HR}}, \mathcal{T}^{(\mathrm{int})}, \alpha \mathrm{H}_6), \qquad \{\alpha \mathrm{H}_6, \beta \mathrm{RM}_8\} \subseteq \mathcal{F}^\star_{\mathrm{HR}}.$$

Only these deck-stable reduction data are passed to Theorem F.9.18.

Proof. The reversible identities Equation (F.96) produce $\alpha \mathrm{H}_6$, and four copies followed by the forward gadget Equation (F.95) produce $\beta \mathrm{RM}_8$. Direct realization and set inclusion give the displayed Turing equivalence. First preprocess and jointly close the two cores under irreducible factors, consider every terminal output, and consider the complete transfer group. A terminal, lower minimum arity, or strict enlargement of the complete transfer group is an outer exit. Otherwise the group remains $\mathcal{T}^{(\mathrm{int})}$ and the minimum nonbinary arity remains six.

Now regenerate the complete oriented deck of $\alpha \mathrm{H}_6$. Retain every card with its ordered kernel and scalar factor, process its entire finite factorization tree, consider all resulting binaries and terminals, and consider the complete transfer group. The core H_6 is irreducible by Section F.9.5.1. Its pair cards are matching products because this holds for f_σ by Lemma F.9.16 and the displayed equivalence is a realizable reversible $(\mathcal{T}^{(\mathrm{int})})^6 \rtimes S_6$-dressing. Lemma F.9.9 then makes every iterated proper realizable card zero or a matching product of safe tetrahedral binaries. Thus any new singular or outside binary, lower-arity nonbinary factor, or group change is an outer exit. On the unchanged branch, regenerate the deck once more. It is complete, all of its cards and factors are retained, and every card is zero or a $\mathcal{T}^{(\mathrm{int})}$-matching product of safe binaries. Hence Definition F.5.8 gives the asserted stable tuple. The two cores are certified affine signatures and are excluded from every subsequent unresolved-occurrence set. Adding these signatures does not create a fresh fixed-(H, ν) token. □

Proof of Theorem F.9.18. Let $\mathcal{P}$ contain every unresolved nonbinary irreducible occurrence of $\mathcal{F}$. The two displayed cores are already known to be affine and are excluded from this multiset. Prepare a selected occurrence and induct on $\Theta_{\mathrm{Pl}}(\mathcal{P})$.

Arity four is Lemma F.9.25. For the next step, assume the complete oriented deck and all proper cards and factors have been processed as above, and consider a six-input occurrence g. Its quaternary cards have already been closed, so every continuing one is a product of two safe tetrahedral binaries. For a canonical pair p, compute $\rho = \operatorname{rank} \Phi^g_p$. Rank zero gives $g = 0$, rank one factors, and ranks two or three are closed by Lemma F.9.7. At rank four, fixed residual matchings factor by Lemma F.9.6. Otherwise clauses (i)–(iii) of Theorem F.A.20, together with Lemmas F.9.10, F.9.15 and F.9.16, give either the resolved endpoint-nondegenerate eight-vertex leaf or a signature f_σ with $\sigma \in \Sigma_+$.

Its projective class lies in $\mathscr{H}_6$ by Equation (F.96), and f_σ is affine by Lemma F.9.17. Because g already has six inputs, this rainbow localization closes no common-factor inputs: f_σ differs from g only by a nonzero scalar, a permutation of inputs, and reversible realizable $(\mathcal{T}^{(\mathrm{int})})^6$-dressings. Affine membership and the $\mathscr{H}_6$-orbit conclusion therefore pull back to g. Thus the projective class of every continuing six-input occurrence lies in $\mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})}) \cup \mathscr{H}_6$.

Let g have arity eight. Prepare every tetrahedral pair card. For a factored six-input card, process its factorization tree under the originating g-index. A continuing result consists only of safe tetrahedral binaries and hence has projective class in $\mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})})$. Before an irreducible six-input card is entered as a child occurrence, consume its originating g-index. Replacing the parent triple by the residual eight-input triple with one fewer index and the six-input child triple is a strict multiset decrease of Θ_{Pl}. The preceding six-input base then places the child's projective class in $\mathscr{H}_6$. Therefore all pair cards lie in $\mathscr{C}_6$, and Proposition F.9.19 says that g factors or is affine.

Finally let $\mathrm{arity}(g) \geq 10$. If $g \notin \mathscr{A}$, Proposition F.9.24 gives a terminal output, a factor, a successor with a strictly larger complete transfer group, or a nonzero nonaffine realizable card of smaller arity. The first three are outer exits or decrease the current token. In the last case the parent card index is consumed before its factor-closed child is entered, so Θ_{Pl} strictly decreases. Induction makes every child factor affine. Their tensor product would then be affine, contradicting the nonaffine card. Hence g is affine.

After adding signatures, consider the complete transfer group. A strict change is an outer exit. Otherwise prepare the next occurrence from its regenerated current-group deck. Every continuing step removes or strictly lowers a triple of Θ_{Pl}, so every irreducible occurrence is eventually affine. The disequality edge and all twelve tetrahedral binaries are affine in this basis by Equation (F.17). H_6 is affine by Section F.9.5.1, and RM_8 is affine because $RM(1,3)$ in Equation (F.40) is a linear code. Tensor closure, the normalization data, and Definition F.2.3 give $\mathsf{Pres}_K(K\mathcal{F}; \mathscr{A})$, proving the claimed relative closure. □

F.9.5.9 Completion of the internal tetrahedral case

Proof of Theorem F.9.14. Varying card matchings give a realizable rainbow localization by Lemma F.9.10. The separated-sign classification and permutation split then give either the resolved endpoint-nondegenerate eight-vertex leaf or a signature f_σ, $\sigma \in \Sigma_+$. Fixed matchings expose a proper factor by Lemmas F.3.4 and F.9.6. Closing its finite factorization tree gives a resolved complexity leaf, lower ν, or a successor with a strictly larger complete transfer group, as in the octahedral proof.

In the Σ_+ case, apply Lemma F.9.26. Every outer-exit branch is already closed. On its sole deck-stable $\mathcal{T}^{(\mathrm{int})}$ branch, Theorem F.9.18 gives the common affine K-presentation, and the recorded Turing equivalence transports that conclusion back to $\mathcal{F}$. This proves $\mathsf{Close}_{\{\mathscr{A}\}}(\mathcal{F}, \mathcal{T}^{(\mathrm{int})}, h)$. □

F.9.6 The external A_4 form

The external form has the same domain incidence as $\mathcal{T}^{(\mathrm{int})}$, but its matching deck has no rainbow line. Its arity-four case follows from the conditions for complementary pairs below. At higher arity, connected domain incidence forces all cards to share one matching and hence a proper factor.

Proposition F.9.27 (External tetrahedral preservers for complementary pairs)**.** *For a projective matrix set $\mathcal{S}$, define its projective coefficient set by*

$$\mathcal{R}_{\mathcal{S}} := \{[c_Q(C)] : [C] \in \mathcal{S}\}.$$

Coordinate matrices act on these projective sets representative-wise. Let $\Theta_{\mathrm{ext}} := c_Q \circ \theta_{\mathrm{ext}} \circ c_Q^{-1}$*, and use the Frobenius matrix* $J_{\mathrm{Frob}} = \mathrm{diag}(1,-1,1,-1)$ *and adjoint* $A^\sharp = J_{\mathrm{Frob}} A^{\mathsf{T}} J_{\mathrm{Frob}}$ *from Equation* (F.81)*. Consider the coordinate maps* A *that satisfy the conditions for both complementary pairs. For every* $[\mathbf{r}] \in \mathcal{R}_{\mathcal{T}^{(\mathrm{ext})}}$*, we require*

$$\begin{aligned} A\mathbf{r} = 0 \quad &or \quad [A\mathbf{r}] \in \mathcal{R}_{\mathcal{T}^{(\mathrm{ext})}}, \\ A^\sharp \mathbf{r} = 0 \quad &or \quad [A^\sharp \mathbf{r}] \in \mathcal{R}_{\mathcal{T}^{(\mathrm{ext})}}. \end{aligned} \tag{F.104}$$

This is representative-independent. There is one zero map. Up to nonzero scalar, the nonzero maps have rank distribution $(144, 0, 0, 576)$ *in ranks* $(1,2,3,4)$*, respectively. Under the Frobenius-dual reconstruction Equation* (F.76)*, every rank-one or rank-four map reconstructs a tensor with a binary-pair factorization.*

Proof. The adjoint is taken for the bilinear Frobenius form, without complex conjugation. Since $R_{\mathrm{ext}}^{\mathsf{T}} = R_{\mathrm{ext}}$, conjugation by R_{ext} is self-adjoint, so $\Theta_{\mathrm{ext}}^\sharp = \Theta_{\mathrm{ext}}$. Moreover,

$$[R_{\mathrm{ext}}^2] = [Q_0 + Q_1] \in N_{\mathrm{PGL}_2}(\mathcal{T}^{(\mathrm{int})}), \qquad \Theta_{\mathrm{ext}}^2(\mathcal{R}_{\mathcal{T}^{(\mathrm{int})}}) = \mathcal{R}_{\mathcal{T}^{(\mathrm{int})}}.$$

Put $A' = \Theta_{\mathrm{ext}}^{-1} A \Theta_{\mathrm{ext}}$. If $[\mathbf{r}] \in \mathcal{R}_{\mathcal{T}^{(\mathrm{int})}}$, then $[\Theta_{\mathrm{ext}}\mathbf{r}] \in \mathcal{R}_{\mathcal{T}^{(\mathrm{ext})}}$, so the first condition for A implies the standard first condition for A'. Also

$$(A')^\sharp = \Theta_{\mathrm{ext}} A^\sharp \Theta_{\mathrm{ext}}^{-1}.$$

Because $\Theta_{\mathrm{ext}}^2(\mathcal{R}_{\mathcal{T}^{(\mathrm{int})}}) = \mathcal{R}_{\mathcal{T}^{(\mathrm{int})}}$, one has

$$\Theta_{\mathrm{ext}}^{-1}(\mathcal{R}_{\mathcal{T}^{(\mathrm{int})}}) = \Theta_{\mathrm{ext}}(\mathcal{R}_{\mathcal{T}^{(\mathrm{int})}}) = \mathcal{R}_{\mathcal{T}^{(\mathrm{ext})}};$$

the adjoint condition transports in the same way. Reversing the calculation gives the converse. Thus the external conditions for both complementary pairs correspond to the standard tetrahedral conditions in the A_4 row of Theorem F.A.28(i). Rank-one reconstruction gives a binary-pair factorization. In rank four, for some $U, V \in \mathrm{GL}_2(\mathbb{C})$, the standard map is $C \mapsto UCV$ or $C \mapsto UC^{\mathsf{T}}V$. Since $A = \Theta_{\mathrm{ext}} A' \Theta_{\mathrm{ext}}^{-1}$, transport gives, respectively,

$$C \mapsto \theta_{\mathrm{ext}}(U) C \theta_{\mathrm{ext}}(V), \qquad C \mapsto \theta_{\mathrm{ext}}(U) R_{\mathrm{ext}}^{-2} C^{\mathsf{T}} R_{\mathrm{ext}}^2 \theta_{\mathrm{ext}}(V).$$

Either display reconstructs two crossing nonsingular binaries. □

Theorem F.9.28 (External tetrahedral deck rigidity)**.** *Let* $(\mathcal{F}, \mathcal{T}^{(\mathrm{ext})}, f)$ *be a deck-stable retained, factor-closed state containing the supplied* I *and* $\mathrm{arity}(f) = \nu(\mathcal{F})$*. Then an established terminal, a restart with a strictly larger complete transfer group, or a proper tensor factor of* f *occurs. Thus this case has no irreducible remaining case or tractable case with a common affine presentation.*

Proof. At arity four, use the transfer-valued pair maps $L_p = \mathsf{R}_X \circ \Phi_p^f$. The exact complementary-pair adjoint identity Equations (F.80) and (F.81) gives the conditions on the map and its adjoint in Equation (F.104). By Proposition F.9.27, rank zero makes $f = 0$, ranks one and four factor, and ranks two and three do not occur.

For $\mathrm{arity}(f) \geq 6$, fix a pair and let Φ be its card map. The realizable kernel set $\mathcal{K}_{\mathcal{T}^{(\mathrm{ext})}}^{\mathrm{act}}$ spans Mat_2, so rank zero is impossible and rank one factors. In ranks two through four, the kernel-deleted rich-line graph is connected by Theorem F.A.20(iii). A change of matching along an edge would give a kernel-avoiding rich domain line with three distinct images. Their matchings are pairwise distinct: if two repeated a matching, flattening across one of its edges absent from the third would give ranks $1, 1, 4$, contradicting collinearity. Apply Lemma F.9.10 before using the quaternary incidence

classification. Each common factor has transfer in $\mathcal{T}^{(\mathrm{ext})}$, and the realizable kernels span Mat_2, so realizable closures preserve all three images and produce a quaternary $\mathcal{T}^{(\mathrm{ext})}$-rainbow line. This contradicts Theorem F.A.20(i). Thus all nonzero cards use one matching M.

Now $\Psi = \Phi \circ \theta_{\mathrm{ext}}$ maps the twelve standard test points to zero or an M-matching product. Lemma F.9.6 factors the tensor reconstructed from Ψ, and composing its varying factor with $\theta_{\mathrm{ext}}^{-1}$ pulls the tensor identity back to Φ. Since $|M| \geq 2$, dual reconstruction and Lemma F.3.4 expose a proper factor. Its binaries give the stated terminal or restart. □

F.9.7 The two icosahedral displays

Recall $\phi = (1+\sqrt{5})/2$, $\tau = \phi^{-1} = \phi - 1$, and the standard sixty-point model $\mathcal{I}^{(0)}$ from Equation (F.19). Its axes, tetrahedral sign points, and additional even-permutation points contribute $4, 8, 48$ classes. Theorem F.A.27(iii) proves directly that these points are closed under multiplication, inverse, and transpose.

For the opposite display, use the fixed-equality card and network correspondence Lemma F.9.3. It replaces every standard kernel by its realizable κ_D-image while leaving the retained state and the supplied I unchanged.

Use the common rainbow normalization and separated-sign system Equations (F.89), (F.91) and (F.92) with $G = \mathcal{I}^{(0)}$.

Lemma F.9.29 (Icosahedral separated-sign solutions and uniform separator)**.** *The separated-sign system for normalized maps $\Phi : \mathrm{Mat}_2(\mathbb{C}) \to (\mathbb{C}^2)^{\otimes 4}$ with $G = \mathcal{I}^{(0)}$ has exactly six rank-deficient solutions and thirty full-rank solutions. For each full-rank solution Φ, the realizable kernel*

$$K_{\mathcal{I}^{(0)}} = \phi Q_1 - \tau Q_2 + Q_3, \qquad [K_{\mathcal{I}^{(0)}}] \in \mathcal{I}^{(0)}$$

realizes the quaternary $q_{\mathcal{I}^{(0)}} := \Phi(K_{\mathcal{I}^{(0)}})$, whose three flattening ranks are $(3,3,3)$, and the realizable kernel on the canonical pair $(1,2)$

$$L_{\mathcal{I}^{(0)}} = Q_0 - \tau^2 Q_1 + \tau Q_3, \qquad [L_{\mathcal{I}^{(0)}}] \in \mathcal{I}^{(0)}$$

gives the nonsingular sequential contraction $B_{\mathrm{out}} := \Phi_{(1,2)}^{q_{\mathcal{I}^{(0)}}}(L_{\mathcal{I}^{(0)}})$, with $[B_{\mathrm{out}}X] \notin \mathcal{I}^{(0)}$ in every case. For $\mathcal{I}^{(1)}$, the corresponding realizable kernels are

$$K_{\mathcal{I}^{(1)}} := \kappa_D(K_{\mathcal{I}^{(0)}}), \qquad L_{\mathcal{I}^{(1)}} := \kappa_D(L_{\mathcal{I}^{(0)}}).$$

They give the covariant quaternary and a nonsingular binary whose transfer lies outside $\mathcal{I}^{(1)}$, directly in the fixed-I coordinates.

Proof. The line classification, separated-sign solutions, five frames, and all thirty full-rank cases, including arbitrary scales and cancellations, are covered by Theorem F.A.27(iii). The same interface supplies the realizable binary with transfer outside the complete transfer group. Thus Equation (F.3) gives the required direction. Lemmas F.3.8 and F.5.4 resolves infinite order, while finite order contradicts completeness. Lemma F.9.3 gives the stated direct gadget certificate for the other display. □

Theorem F.9.30 (Icosahedral deck rigidity)**.** *If Equation* (F.94) *holds and*

$$G \in \{\mathcal{I}^{(0)}, \mathcal{I}^{(1)}\},$$

then $\mathsf{Close}_{\varnothing}(\mathcal{F}, G, h)$.

Proof. For $G = \mathcal{I}^{(0)}$, varying card matchings and connected incidence supply a mixed line. By Theorem F.A.20(i) it is not a mixed five-line. Its rainbow three-line and Lemmas F.9.10 and F.9.29 therefore give a nonsingular binary with transfer outside the complete transfer group. For $G = \mathcal{I}^{(1)}$, apply the same construction through Lemma F.9.3, which preserves the binary equality.

If all nonzero cards use one matching M, the twelve test points map to zero or an M-matching product. Since $|M| = 2$, Lemmas F.3.4 and F.9.6 exposes a proper factor. Close its finite factorization tree: a nonbinary factor lowers arity, a singular or unsafe binary is terminal, and a safe binary whose transfer class lies outside the current group strictly enlarges that group. All-safe current-group binary factors would contradict the irreducible branch. Thus every branch is a resolved complexity leaf or outer deck successor, proving the conclusion. The separator uses only the rank recomputed on the realizable six-input coefficients. □

F.9.8 Completion of the Platonic classification

Proof of Theorem F.9.1. Work at the stable tuple in the theorem. First apply the established reductions to a nonzero odd signature by Theorem F.2.9, a unary by Lemma F.3.11, a nullary factor by preprocessing, and every new binary by Lemmas F.3.8 and F.5.4. Thus a continuing branch is all-even and every binary transfer lies in the complete transfer group.

If $G = \mathcal{T}^{(\mathrm{ext})}$, apply Theorem F.9.28. Its factor branch is closed by factor closure. For every other Platonic form, including $\mathcal{I}^{(1)}$, use the direct coefficient interfaces already proved above. Hence assume

$$G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}, \mathcal{I}^{(0)}, \mathcal{I}^{(1)}\}.$$

At arity four, Lemmas F.3.11, F.9.4 and F.9.5 and Corollary F.3.13 and the local-reduction conclusion following Lemma F.9.8 close every output. Suppose $n := \mathrm{arity}(f) = \nu(\mathcal{F}) \geq 6$.

Fix a canonical pair p and compute $r = \operatorname{rank} \Phi_p^f$ from the coefficients. Rank zero gives $f = 0$ by Equation (F.76). At rank one there are nonzero tensors U, F_{res} with

$$\Phi_p^f(C) = \operatorname{tr}(U^{\mathsf{T}} C) F_{\mathrm{res}}, \qquad f = U^{(p)} \otimes F_{\mathrm{res}},$$

so Lemma F.3.4 gives a proper factor route. Closing that finite factorization tree either lowers ν, reaches a binary terminal or successor with a strictly larger complete transfer group, or expresses f entirely as safe current-group binaries, contradicting its irreducible choice.

Let $r \in \{2, 3, 4\}$. Deck stability gives Equation (F.74), so Lemma F.9.11 applies. The fixed-matching route again gives a proper factor. Close its finite factorization tree by the same alternatives. An all-safe current-group binary product again contradicts irreducibility. The mixed-matching case realizes a nonzero six-input h, using one f-vertex and exactly $(n-6)/2$ realizable kernels, with

$$2 \leq \operatorname{rank} \Phi_p^h \leq r$$

and Equation (F.74). Form the signature set $\mathcal{F}_h := \mathcal{F} \cup \{h\}$. Direct realization and set inclusion give

$$\text{K-Holant}(\mathcal{F}_h) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}), \qquad \text{K-Holant}(\mathcal{F}) \leq_{\mathrm{T}} \text{K-Holant}(\mathcal{F}_h).$$

Preprocess and close this entire finite batch under irreducible factors, apply the established reductions to its newly exposed outputs, and consider the complete transfer group. If h factors, its selected varying matchings exclude a product entirely of safe current-group binaries, so a lower-arity nonbinary factor or an earlier terminal occurs. If $n > 6$, a retained six-input irreducible signature lowers ν. A strict enlargement of the complete transfer group is likewise an outer deck successor. If $n = 6$, the localization has no closing kernels: the contraction C in the proof of Lemma F.9.11

is the identity, and h is only the recorded permutation of inputs and nonzero scalar multiple of f. We identify it with the same occurrence and reuse its fixed card/factor index. No fresh fixed-(H,ν) occurrence is inserted.

On the only remaining irreducible six-input branch with unchanged marked group, regenerate the complete oriented deck of h. Retain every card with its ordered kernel, scalar, and reduction to the preceding set. Process its complete finite factorization tree under factor closure. Consider every resulting binary and terminal. Recompute the complete transfer group. A lower-arity factor or group change is an outer deck successor. On the unchanged branch regenerate the deck again. Condition Equation (F.74) and Lemma F.9.9 make every card zero or a matching product of safe binaries with transfers in the current group, and all card factors are retained. Consequently Definition F.5.8 gives the stable tuple

$$\mathsf{Stable}(\mathcal{F}_h^\star, G, h), \qquad \operatorname{arity}(h) = \nu(\mathcal{F}_h^\star) = 6.$$

Only for these deck-stable reduction data compute $\rho_h := \operatorname{rank} \Phi_p^h$. If $\rho_h \le 3$, apply Lemma F.9.7. If $\rho_h = 4$, apply, according to the marked group of the resulting set, Theorems F.9.13, F.9.14 and F.9.30. The only remaining tractable case is the internal tetrahedral one, whose theorem supplies the stated common affine presentation for the whole retained set. The other outcomes are resolved complexity leaves or outer deck successors.

These reductions exhaust the stable six-input residue. Each construction records its input order and nonzero scalar, and its reduction points to the preceding retained problem. The finite separated-sign classifications have constant size. Localization uses $(n-6)/2$ kernels. These bounds and the local occurrence induction complete the proof. □

F.10 Synthesis of the Binary-Disequality Dichotomy

The four group analyses give either a complexity conclusion for the retained set or a construction that makes strict progress. We now combine them using the descent measure from Lemma F.5.12. The resulting proof of Theorem F.5.14 classifies deck-stable states. Corollary F.10.2 then uses stabilization to classify every retained factor-closed set containing I. Finally, factor closure and the equality-edge-model equivalence give Theorems F.1.1 and F.5.2 for the original input sets. At each step, hardness follows the reductions, while a tractable case supplies the criterion for the whole preceding set.

Every continuing signature set $\mathcal{F}$ in this section contains I, so $Z = K^{\otimes 2} I \in K\mathcal{F}$. Hence Lemma F.2.7 gives

$$\mathsf{Tract}(K\mathcal{F}) \quad \Longleftrightarrow \quad \mathsf{ReTract}(K\mathcal{F}).$$

This equivalence identifies the branchwise criterion Tract with the equality-accessible criterion ReTract.

F.10.1 Synthesis of the Equality-Accessible Matching-Deck Classification

Lemma F.10.1 (Descent in the four group cases)**.** *For every deck-stable tuple $(\mathcal{F}, G, f)$, work in the fixed coordinate matrix $A = I$, put*

$$H = \overline{G}(I, \mathcal{F}) = G(\mathcal{F}), \qquad \mathscr{U}^{\mathrm{fac}}(I, f, H) = Q_{\mathrm{done}} \sqcup Q,$$

and fix the displayed partition. Then the unfinished-token batches and the further reductions invoked by the active-signature completion proofs in Sections F.6 to F.9 satisfy Definition F.5.11.

Proof. Fix the displayed partition for the deck-stable tuple $(\mathcal{F}, G, f)$ in the fixed-I coordinates. Each nonzero oriented one-copy proper card determines a member of $\mathscr{U}(I, f, H)$. All constructions with that normalized output share a token in the one-copy queue. A continuing one-copy step with unchanged (H, ν) therefore works on a single unfinished token. Each genuine split in its stored tree decreases the factor potential by Lemma F.3.4 and Equation (F.9). Completing the card and its tree moves the token to Q_{done}.

It remains to check reductions beyond the fixed one-copy deck. In the standard marked-dihedral branch, Theorem F.6.12 resolves the whole retained set at arity four through Theorem F.1.2. Its higher-arity arguments exclude the distinguished irreducible signature: the arity-six base is Lemma F.6.11, and localization followed by the common-factor consequence gives a proper factor at larger arity. Thus these local reductions end in a factorization or a resolved complexity leaf.

For $D_{2m}^{(Y)}$, the open-network correspondence in the proof of Theorem F.6.15 transports arity, factorization, ordered cards, and their factorization trees to the $D_{2m}^{(Z)}$ deck. It therefore transports the fixed partition bijectively. For $D_{2m}^{(X,0)}$, the fixed-I identities Equations (F.28) and (F.29) and the deck relabelling in the proof of Theorem F.6.16 give the same bijective transport to the $D_{2m}^{(Z)}$ reductions, including their fixed card and factor indices. For $D_{2m}^{(X,1)}$, Lemma F.6.33 transports the weighted local comparisons used in Theorem F.6.17. Together with the arity-four reduction Lemma F.6.32, these resolve the complexity, factor the selected occurrence, or give an outer deck successor. The factorization completes the selected card and factor indices.

In the cyclic branch, let $N = |G|$. The $N = 2$ boundary uses Theorem F.5.22 and Lemma F.5.21. For $N \geq 4$, Lemma F.7.5 gives either a realizable irreducible endpoint-nondegenerate eight-vertex signature, resolved by Theorem F.1.2, or a nonsingular anti-diagonal factor of a two-copy output. Factor closure and Lemma F.5.4 make the latter available and resolve its complexity or strictly enlarge the complete transfer group. In the Reed–Muller argument, Lemma F.7.9 gives a proper factor or a realizable Even_4, resolved by Lemma F.3.15. These cyclic multi-copy reductions therefore give a resolved complexity leaf or an outer deck successor.

In the standard and exotic Klein-four branches, arity-four batches are processed through Theorem F.5.22, Corollary F.4.8, and Lemma F.5.21. At higher arity, Theorem F.8.3 either resolves the whole set through the real-valued dichotomy or supplies a realizable quaternary batch. Its complete factor batch gives a resolved complexity leaf, a strict group enlargement, or a decrease of ν. Theorem F.8.10 excludes the remaining exotic higher-arity cases by proper factorization.

For the external tetrahedral form, Theorem F.9.28 gives a resolved complexity leaf, strict group enlargement, or a proper tensor factor. For the other Platonic forms at arity at least six, Lemma F.9.11 gives a proper tensor factor or a directly realized six-input signature, with its kernel tuple, input order, and scalar. Process this output and all its factors before continuing. A nonbinary irreducible factor of smaller arity lowers ν, and a new transfer class triggers the complete-group comparison. On the remaining six-input deck-stable state, recompute the card-map rank. Lemma F.9.7 handles rank at most three. At rank four, the hypotheses of Equation (F.94) permit Theorems F.9.13, F.9.14 and F.9.30, according to the current marked form. In the internal tetrahedral case, the core retention and regenerated decks in Lemma F.9.26 justify the finite occurrence induction of Theorem F.9.18, giving a common affine presentation for the entire retained set. This is the assembly in Theorem F.9.1.

The multi-copy constructions above are resolved by their local contraction, factorization, or finite-induction arguments. After those arguments are completed, their outputs have outcomes 1–3 of Definition F.5.11. Any fixed-(H, ν) proper-factor continuation arising from a one-copy reduction remains within its originating unfinished token and satisfies outcome 4: completion decreases $|Q|$, while an intermediate split decreases its stored factor potential. The fixed-I reduction data preserve

these indices under the coordinate correspondences above. Thus outcome 4 has the required strict progress, and all conditions of Definition F.5.11 hold. □

Proof of Theorem F.5.14. Run the procedure of Definition F.5.10 and Lemma F.5.12 in the fixed-I coordinates, with $A = I$. Fix the coefficient field L_0 of Lemma F.3.7. Gadget evaluation and pivot-normalized factorization keep the pulled-back data over this field. The same lemma restores the original coordinates after any continuing normalization in Section F.4. Consequently Equation (F.20) gives the uniform bound $|G(\mathcal{F})| \le M_{L_0}$ at every continuing state. Keep the finite worklist of irreducible occurrences. For the selected minimum-arity root f, put $H = G(\mathcal{F})$ and use the canonical card and factor indices of $\mathscr{U}^{\mathrm{fac}}(I, f, H)$ from Definition F.5.10. While (H, ν) is unchanged, carry its fixed partition into completed and unfinished indices. A token is completed only after its stored card and entire factorization tree have been processed. At each outer deck successor, apply Theorem F.2.9 and Lemmas F.3.8 and F.3.11 to any odd-arity member, unary factor, or unsafe binary. If only unary and binary irreducible factors remain, the arity-two case applies. Otherwise choose a minimum-arity nonbinary irreducible signature and reapply Lemma F.5.9. An interrupted token is carried to the successor until its processing is complete. Thus every continuing node consists of a deck-stable all-even tuple $(\mathcal{F}, H, f)$, together with its existing queue Q, where $H = \overline{G}(I, \mathcal{F}) = G(\mathcal{F})$ is finite by Lemma F.5.3. Use the previously defined measures $\Xi(I, \mathcal{F})$ and $\Xi_{\mathrm{full}}(I, \mathcal{F}, f, Q)$ from Lemma F.5.12 and Equation (F.23). By Lemma F.10.1, all four group cases satisfy Definition F.5.11. Thus Lemma F.5.12 permits well-founded induction on Ξ_{full}, including every restart. Strict group enlargement lowers the first coordinate, a lower minimum arity lowers the second, fixed-(H, ν) closure decreases $|Q|$ in the third, and a genuine factor split lowers Φ_Q in the fourth. A newly exposed safe binary with transfer class outside H gives an outer deck successor. If the resulting complete transfer group remains H, the binary is processed within the current node. An unsafe binary gives a resolved complexity leaf. Reductions compose toward the predecessor. A tractability criterion or common presentation passes back by Lemmas F.3.5 and F.3.7. In the group cases below, G denotes the current complete transfer group $H = G(\mathcal{F})$.

First suppose $\mathrm{arity}(f) = 4$. Apply Theorem F.5.16. The following list is exhaustive.

1. If the transfer span is proper and $G = C_2^{(X)}$, use item 1 of Theorem F.5.22. If $G = C_N^{(X)}$, with even $N \ge 4$, use Lemma F.7.5 and Theorem F.1.2.

2. If $G = D_{2m}^{(Z)}$, $m \ge 3$, use item 2 of Theorem F.5.22.

3. If $G = D_{2m}^{(Y)}$, with $m \ge 3$, use Theorem F.6.15. If $G = D_{2m}^{(X,0)}$, with even $m \ge 4$, use Theorem F.6.16. If $G = D_{2m}^{(X,1)}$, with even $m \ge 4$, use Lemma F.6.32 and Corollary F.4.8.

4. For the standard Pauli or exotic Klein group, use items 3 and 4, respectively, of Theorem F.5.22.

5. For $G = \mathcal{T}^{(\mathrm{int})}$, any of the four octahedral forms $\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$, or either icosahedral display $\mathcal{I}^{(0)}, \mathcal{I}^{(1)}$, use Lemma F.9.5. For $G = \mathcal{T}^{(\mathrm{ext})}$, use Theorem F.9.28.

Each cited construction is applied to the retained set and its complete oriented deck, with the factor processing specified after Theorem F.5.22. For endpoint-nondegenerate irreducible quaternary outputs, Lemma F.5.21 supplies the binary case distinction. The sector matrices $U_Y, P, U_{X,1}$ are used through their stated equivalences or coefficient identities, with continuing outputs expressed in the fixed-I coordinates. A directly realized endpoint-nondegenerate irreducible quaternary is added to the set and handled by Theorem F.1.2. For a possibly reducible eight-vertex output, Corollary F.4.8 instead gives a complexity conclusion or its full factor batch. The latter is processed by Lemma F.5.4 as described after Theorem F.5.22. A directly realized generalized equality is handled by Lemma F.3.16. Thus every arity-four case yields a resolved complexity leaf or a successor

with smaller Ξ_{full}. In a tractable case, Lemma F.2.7 gives $\mathsf{ReTract}(K\mathcal{F})$. The retained irreducible quaternary excludes the arity-two case, leaving one common $\mathscr{A}$-, $\mathscr{P}$-, or $\mathscr{L}$-presentation, with the class supplied by the cited result.

Assume henceforth that $\text{arity}(f) \geq 6$. Apply Theorem F.5.20. If the transfer span is $\text{span}\{I, X\}$, then Theorem F.7.1 applies to the current minimum-arity deck-stable signature. Its arity-four reductions are Theorem F.5.22 and Lemma F.7.5. It gives a resolved complexity leaf, a strict stable successor, or one common flat-Lagrangian K-presentation. The last is an affine tractable case.

It remains to consider the full transfer span. Apply Theorem F.5.16. Its marked list is exhaustive because

$$G \lhd \langle G, [Z] \rangle, \qquad [\langle G, [Z] \rangle : G] \leq 2, \qquad \langle [X], [Z] \rangle \cong V_4.$$

The continuing state is all-even and has minimum nonbinary arity at least six, so it satisfies the arity and parity hypotheses of the following results:

marked form of the complete transfer group	relative theorem
$D_{2m}^{(Z)}, D_{2m}^{(Y)}$ $(m \geq 3)$, $D_{2m}^{(X,0)}, D_{2m}^{(X,1)}$ (even $m \geq 4$)	Theorem F.6.1
standard Pauli $V_4^{(\text{std})}$ and $V_4^{(\text{ex})}$	Theorem F.8.1
all marked Platonic forms	Theorem F.9.1

Within the marked-dihedral row, the $D_{2m}^{(Y)}$ branch is a whole-problem equality-edge-model equivalence. The $D_{2m}^{(X,0)}$ branch uses the fixed-I whole-problem/open-network equivalence Equations (F.28) and (F.30). Its two edge-replacement constructions use only I, Y, and its final common witness is pulled back inside Theorem F.6.16. For $D_{2m}^{(X,1)}$, Theorem F.6.17 resolves arity four by its reduction for the whole signature set and excludes every higher-arity distinguished irreducible signature. The sector matrix $U_{X,1}$ expresses the weighted pencils and fusion identities used in that proof. The realized contractions remain in the current fixed-I coordinates. Each relative theorem gives the alternatives specified in Definition F.5.13: a resolved complexity leaf or a strict stable successor. Group enlargement and arity descent lower Ξ. At fixed group and arity, the card and factor processing verified in Lemma F.10.1 lowers the remaining coordinates of Ξ_{full}. The induction therefore handles every continuing successor.

The two transfer spans and the marked group list exhaust the possible states. A surviving common product, affine, or local-affine presentation satisfies $\mathsf{ReTract}(K\mathcal{F})$ by Definitions F.2.3 and F.2.6. The flat-Lagrangian case is contained in the affine case. Each common presentation uses one matrix for the transformed edge and every retained signature. It pulls back through the stated whole-problem equivalences and Lemma F.3.7. Other tractable cases already carry the criterion by Definition F.5.13. The anchor equivalence at the start of this section identifies it with $\mathsf{Tract}(K\mathcal{F})$.

If $\mathsf{Tract}(K\mathcal{F})$ holds, Theorem F.3.3 and Equation (F.2) gives polynomial-time evaluation. If it fails, preservation under taking subsets and the preceding pullbacks exclude every tractable conclusion. The induction leaves a hard problem reducing to the original retained problem. This proves both cases of Theorem F.5.14. □

Corollary F.10.2 (Classification of equality-accessible signature sets)**.** *Let $\mathcal{F}_0$ be a retained, factor-closed finite algebraic signature set containing the supplied equality I. If $\mathsf{ReTract}(K\mathcal{F}_0)$ holds, then* K-Holant$(\mathcal{F}_0) \in \mathsf{FP}$. *Otherwise* K-Holant$(\mathcal{F}_0)$ *is* #P*-hard. Here $\mathsf{ReTract}(K\mathcal{F}_0)$ is equivalent to $\mathsf{Tract}(K\mathcal{F}_0)$, since $Z = K^{\otimes 2} I \in K\mathcal{F}_0$. The factor access used in reaching this conclusion is a Turing reduction. Every deck is regenerated over the current signature set and its complete transfer group.*

Proof. Because $Z = K^{\otimes 2} I \in K\mathcal{F}_0$, Lemma F.2.7 identifies $\mathsf{ReTract}(K\mathcal{F}_0)$ with $\mathsf{Tract}(K\mathcal{F}_0)$. If this criterion holds, Theorem F.3.3 and Equation (F.2) gives K-Holant$(\mathcal{F}_0) \in \mathsf{FP}$.

Suppose the criterion fails. Apply Theorem F.2.9 in K-coordinates to any odd-arity member, and Lemmas F.3.8 and F.3.11 to any unsafe binary. Their tractable cases are excluded by the failed criterion. A state whose irreducible factors all have arity at most two would satisfy the arity-two case. Otherwise a minimum-arity nonbinary irreducible signature exists and has even arity.

Apply Lemma F.5.9 to that signature. Its lower-arity outcome is handled by strong induction on ν. Its stable outcome satisfies the hypotheses of Theorem F.5.14, whose Ξ_{full}-induction handles all subsequent group changes and restarts. Every continuing extension is Turing-equivalent to its predecessor and retains the preceding set. A tractable criterion on an extension would restrict to $\mathcal{F}_0$ by Lemma F.3.5, and Lemma F.3.7 pulls back any intervening normalization and common presentation. This would contradict the assumed failure. Thus the remaining outcome is hardness, which transfers along the reductions to K-Holant($\mathcal{F}_0$). □

F.10.2 From the equality-accessible dichotomy to the binary-disequality dichotomy

Corollary F.10.2 gives polynomial-time evaluation when ReTract holds and hardness otherwise. Its tractable cases are the arity-two tensor closure and the common affine, product, or local-affine classes. The flat-Lagrangian case lies in the affine class. We first apply this criterion to the input set in Theorem F.5.2, then use the edge-model equivalence to deduce Theorem F.1.1.

Proof of Theorem F.5.2. Apply Section F.2.4, retaining the name $\mathcal{F}$. The supplied I survives. Since $Z = K^{\otimes 2} I \in K\mathcal{F}$, Lemma F.2.7 identifies $\mathsf{ReTract}(K\mathcal{F})$ with $\mathsf{Tract}(K\mathcal{F})$. If this criterion holds, Theorem F.3.3 and Equation (F.2) gives K-Holant($\mathcal{F}$) $\in \mathsf{FP}$, with the deleted nullary scalars restored.

Suppose the criterion fails. The supplied I excludes the common-one-sided exception. By Lemma F.3.4, we may add $\Pi(\mathcal{F})$ by a Turing equivalence. A tractable presentation of the extension would restrict to the original set by Lemmas F.2.7 and F.3.5, so its criterion also fails. Each original signature is reconstructed from its irreducible factors by tensor product, an input permutation, and a nonzero scalar, as described after Lemma F.3.4. All four cases of ReTract in Definition F.2.6 are preserved by these operations with the same common basis. Thus a tractable presentation of $K\Pi(\mathcal{F})$ would extend to $K(\mathcal{F} \cup \Pi(\mathcal{F}))$, a contradiction. The reconstruction and Equation (F.8) give K-Holant($\Pi(\mathcal{F})$) $\equiv_{\mathrm{T}}$ K-Holant($\mathcal{F}$), so we may replace the initial set by $\Pi(\mathcal{F})$ with the criterion still false. Retain the name $\mathcal{F}$ for this irreducible set. This finite set is factor-closed, and the irreducible binary equality I remains present.

The retained set now satisfies Corollary F.10.2, which gives hardness because the criterion fails. The factor-closure equivalence transfers it to the original input problem. For the original input set, decidability follows directly from Theorem F.3.3 and Lemma F.2.7, before factor replacement. □

Proof of Theorem F.1.1. Apply Section F.2.4, retaining the name $\mathcal{F}$. By Lemma F.5.1,

$$\text{Holant}(\mathcal{F}, X) \equiv_{\mathrm{T}} \text{K-Holant}(\mathcal{F} \cup \{I\}).$$

Apply Theorem F.5.2 directly to $\mathcal{F} \cup \{I\}$. Its criterion is $\mathsf{ReTract}(K(\mathcal{F} \cup \{I\}))$, which is equivalent to $\mathsf{ReTract}(\mathcal{F} \cup \{X\})$ by the criterion correspondence following Lemma F.5.1. Both complexity conclusions transfer through the displayed equivalence, and the criterion is decidable by Theorem F.5.2. The preprocessing convention gives the same criterion for the original input set. The edge replacements are linear-size, and the deleted nullary scalars are restored exactly. □

F.11 Conclusion

The supplied binary disequality signature yields a decidable FP-versus-#P-hard classification for every fixed finite set of algebraic complex-valued Boolean signatures. Theorems F.1.1 and F.5.2 express this classification using, respectively, ordinary edges with an X-vertex and X-edges with an I-vertex. The binary anchor leaves four tractable cases: the arity-two tensor closure and common affine, product, or local-affine transformability.

The proof combines arity reduction with the complete transfer group. A new contraction or factor can lower the minimum nonbinary arity or enlarge that group, changing the applicable structural analysis. The progress mechanism in Lemma F.5.12 controls both changes and completes the finite card and factor processing between them. It thereby assembles the four group analyses into a classification of the entire signature set.

The eight-vertex reductions provide the common arity-four step. Theorem F.1.2 extends the single-signature classification of Cai and Fu [CF23, arXiv version 2, Theorem 3.1] to a signature set containing an endpoint-nondegenerate irreducible quaternary, with the same reduced predicate ReTract and without requiring binary equality. The finite classifications in Appendix F.A supply the stated low-arity calculations used by the structural branches.

Code Availability

The accompanying Lean formalization (fixed source revision `ec3e598`) gives the public statements in `Holant/Statements.lean` and their proofs in `Holant/Proofs.lean`. In the `Holant.Theorems` namespace, `binaryDisequality` proves the complexity dichotomy in Theorem F.1.1, `quaternary` the one in Theorem F.1.2, and `literalEquality` the one in Theorem F.5.2. For each of these names, `Holant/Decidability.lean` exports the corresponding exact-input recognition theorem with suffix `_uniform`. Supplementary lemmas and exact finite certificates are included through `Holant/Library.lean`. The statement-only reference `Verification/Challenge.lean` and `Verification/comparator.json` support independent comparison using Lean's Comparator. Build and verification instructions are given in the repository README.

The original Python companion repository is *Holant-X-Dichotomy*. References below to `certificates/` and its Python scripts refer to the separately distributed source and instructions in that repository. The Lean release supplies its own exact certificates under `Holant/Certificates/` and does not require the Python companion.

F.A Finite Classifications and Explicit Contractions

The finite classifications below supply specific inputs to the group analyses. The main text verifies their hypotheses, realizes the required contractions, and applies the resulting reductions to the retained signature set.

Results	Input to the main proof	Main use
Theorems F.A.4 to F.A.6	Affine endpoint normal forms and contractions of quaternary signatures with full odd-parity support and fourth-root phases	Proposition F.4.6 and Corollary F.4.7
Theorem F.A.8	Equality-accessible arity-four constructions	Theorem F.5.22; also Section F.6
Theorem F.A.11	Six-to-four support projection and contraction	Lemma F.6.21
Theorem F.A.13	Exotic Klein six-input classification	Lemma F.8.5
Theorem F.A.14	Normalized eight-input factor argument	Records the proof of Lemma F.8.8
Theorem F.A.20	Rich-line incidence and deletion connectivity	Lemma F.9.11; also Section F.9
Theorem F.A.23	Eight-input tetrahedral contraction	Proposition F.9.19
Theorem F.A.27	Separated-sign solutions and realizable bridge outputs	(i) Lemma F.9.12, (ii) Lemma F.9.15, (iii) Lemma F.9.29
Theorem F.A.28	Arity-four ranks, reconstructions, and realizable outputs	(i) Lemma F.9.5, (ii) Lemma F.9.7, (iii) Lemma F.9.8

Theorem F.A.5 is also used in Section F.6.

Each calculation states its parameter space, tests, and exact characteristic-zero field. Where completeness depends on exhaustive computation, the proof identifies that dependency and the relevant script in `certificates/` of the Python companion repository. These programs emit assertions rather than independent proof objects. For results proved structurally, the scripts provide supplementary checks. The Lean repository has its own finite certificates, as described in Code Availability.

In a histogram, s^m means m objects of size s. Projective representatives are normalized by their first nonzero coordinate. We use the conventions in Sections F.2.1.3 and F.2.8 and Appendix F.C, together with the projective tensor conventions and the symmetric-space conventions.

F.A.1 Endpoint-Nondegenerate Eight-Vertex Signatures

We first classify the loop and two-copy contractions of quaternary signatures with full odd-parity support whose nonzero entries, after normalization, are fourth roots of unity. We then determine the affine endpoint normal forms. Theorems F.A.4 to F.A.6 state the consequences used in Section F.4.

F.A.1.1 Fourth-root phases on full odd-parity support

In this subsubsection, label the four inputs by $0, 1, 2, 3$, with local label j corresponding to input $j + 1$ in the main text. Here $\mathbf{e}_j$ denotes the string with a single 1 at the locally labelled input.

Normalize a quaternary signature with full odd-parity support by $a_j = f(\mathbf{e}_j)$, $b_j = f(\mathbf{1} - \mathbf{e}_j)$, and $a_0 = 1$. Its fourth-root phase tuple is

$$a_j = i^{\alpha_j}, \qquad b_j = i^{\beta_j}, \qquad (\alpha_1, \alpha_2, \alpha_3, \beta_0, \ldots, \beta_3) \in \mathbb{Z}_4^7. \tag{F.105}$$

There are $4^7 = 16\,384$ phase tuples. Write $N(z) = z\bar{z}$ for the Gaussian norm. For $i < j$, with $k < l$ the complementary pair, the six disequality self-loops and six equality-mediated loops are

$$L_{ij}^X = \operatorname{diag}(a_i + a_j, b_i + b_j), \qquad L_{ij}^I = \begin{pmatrix} 0 & a_l + b_k \\ a_k + b_l & 0 \end{pmatrix}. \tag{F.106}$$

The latter inserts one equality vertex using two X-edges. A series equality support-flips the former:

$$\operatorname{diag}(P, Q)XI = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix}, \qquad T_{\operatorname{diag}(P,Q)XI} = \operatorname{diag}(P, Q). \tag{F.107}$$

In either construction, when $P, Q \neq 0$, the resulting transfer has eigenvalue ratio P/Q.

Proposition F.A.1 (Deterministic twelve-loop test)**.** *For each phase tuple in Equation* (F.105)*, inspect the twelve ordered pairs in Equation* (F.106)*:*

(i) *If exactly one entry of some pair is zero, declare a rank-one exit.*

(ii) *Otherwise, for every pair with two nonzero coordinates P, Q, compare their Gaussian norms. If $N(P) \neq N(Q)$, declare an infinite-projective-order exit.*

(iii) *Otherwise retain the phase tuple and all zero–zero pairs.*

The resulting disjoint partition is

$$16\,384 = 15\,792 + 224 + 368, \tag{F.108}$$

where the summands are respectively rank-one, infinite-order, and remaining phase tuples.

Proof. The entries lie in $\mathbb{Z}[i]$. Up to a fourth root of unity, every nonzero sum of two fourth roots is either 2 or $1 + i$. Consequently two such sums have the same Gaussian norm precisely when their quotient lies in μ_4. If their norms differ, the quotient has modulus different from one and therefore cannot have finite multiplicative order. Thus, for exactly the nonzero sums occurring here, P/Q is torsion if and only if $N(P) = N(Q)$. Hence Equation (F.107) makes the three tests exact and disjoint. Exhaustive evaluation gives Equation (F.108). □

Exactly 64 remaining phase tuples satisfy

$$b_j = \varepsilon a_j \quad (0 \leq j \leq 3), \qquad \varepsilon \in \{1, -1\}, \qquad \prod_{j=0}^{3} a_j \in \{i, -i\}. \tag{F.109}$$

Indeed, after the normalization $a_0 = 1$, exactly half of the 4^3 choices of (a_1, a_2, a_3) have product in $\{i, -i\}$, and ε has two choices. Hence the count is $32 \cdot 2 = 64$. Join inputs $0, 1$ of two copies A, B by X-edges. Denote the resulting quaternary signature by Q. In the order (A_2, A_3, B_2, B_3), its coefficients are

$\mathbf{x}$	0000	1111	0011, 1100	0101	1010	0110, 1001
$Q(\mathbf{x})$	$2a_0a_1$	$2b_0b_1$	$a_0b_0 + a_1b_1$	$2a_3b_2$	$2a_2b_3$	$a_2b_2 + a_3b_3$.

The contraction of two odd-parity signatures along two X-edges has even external parity, and the two endpoint values $2a_0a_1$ and $2b_0b_1$ are nonzero. Under Equation (F.109), exactly one of $a_0^2+a_1^2, a_2^2+a_3^2$ vanishes: indeed, each a_j^2 is a sign and $\prod_j a_j^2=-1$, so exactly one of the two displayed pairs consists of opposite signs. Hence Q is a support-six eight-vertex signature.

For each other remaining phase tuple and $\delta\in\mathbb{Z}_4$, define the phase-shifted signature f_δ by

$$f_\delta(\mathbf{e}_j)=i^{\alpha_j},\qquad f_\delta(\mathbf{1}-\mathbf{e}_j)=i^{\beta_j+\delta}$$

and keep $f_\delta(\mathbf{x})=0$ on every even-weight input. Parameterize the odd coset by

$$\mathbf{x}(\mathbf{y})=(y_0,y_1,y_2,1\oplus y_0\oplus y_1\oplus y_2),\qquad \mathbf{y}\in\mathbb{F}_2^3.$$

Define the phase-exponent function $\varphi_\delta:\mathbb{F}_2^3\to\mathbb{Z}_4$ by $f_\delta(\mathbf{x}(\mathbf{y}))/f_\delta(\mathbf{x}(000))=i^{\varphi_\delta(\mathbf{y})}$, and define its linear coefficients ℓ_j and mixed-quadratic coefficients q_{jk} by

$$\ell_j=\varphi_\delta(\mathbf{e}_j),\qquad q_{jk}=\varphi_\delta(\mathbf{e}_j+\mathbf{e}_k)-\ell_j-\ell_k.$$

Then f_δ is affine if and only if

$$q_{jk}\in\{0,2\}\ (j<k),\qquad \varphi_\delta(111)=\ell_0+\ell_1+\ell_2+q_{01}+q_{02}+q_{12}\pmod 4. \tag{F.110}$$

Indeed, in the Boolean Möbius convention of Section F.2.8, the expansion of φ_δ over $\mathbb{Z}_4$ has the three q_{jk} as its quadratic coefficients, while the difference between the two sides of the second congruence is its cubic coefficient. A modulo-four phase is affine exactly when the quadratic coefficients are even and that cubic coefficient vanishes. Exactly two δ's pass: $\{0,2\}$ for 176 phase tuples and $\{1,3\}$ for 128 phase tuples.

The 512 normalized affine signatures with full odd-parity support are given by

$$(a_0,a_1,a_2,a_3)=(1,A,B,C),\qquad (b_0,b_1,b_2,b_3)=(xyzABC,zBC,yAC,xAB), \tag{F.111}$$

where $A,B,C\in\mu_4$ and $x,y,z\in\{1,-1\}$. The $4^3\cdot 2^3$ choices give 512 distinct normalized phase tuples. The locus having no rank-one X-loop is the disjoint union

$$x=y=z \qquad \text{(128 phase tuples)}, \tag{F.112}$$

$$x,y,z \text{ not all equal},\quad (A^2,B^2,C^2)=(xy,xz,yz) \qquad \text{(48 phase tuples)}. \tag{F.113}$$

The other 336 phase tuples have a rank-one loop. For the remaining 176, let G_{0j} join inputs $0,j$ of two copies by corresponding X-edges and retain the other inputs in copy order. Call a resulting quaternary *dense product-transformable* when it has the full even-parity form in Equation (F.11), namely

$$g(\mathbf{x})=\rho r^{\mathrm{wt}(\mathbf{x})}(-1)^{\mathbf{s}\cdot\mathbf{x}}\mathbf{1}[\mathrm{wt}(\mathbf{x})\text{ is even}]\quad\text{for some }\rho,r\neq 0,\ \mathbf{s}\in\mathbb{F}_2^4.$$

Then

$$\begin{aligned}
&G_{01}\text{ is dense product-transformable}\iff \mathbf{x}=\mathbf{y}\text{ and }B^2=C^2,\\
&G_{02}\text{ is dense product-transformable}\iff \mathbf{x}=z\text{ and }A^2=C^2,\\
&G_{03}\text{ is dense product-transformable}\iff \mathbf{y}=z\text{ and }A^2=B^2.
\end{aligned} \tag{F.114}$$

In Equation (F.112) two squares agree. In Equation (F.113) the unique equal pair of x,y,z has the corresponding equal squares. Hence one test in Equation (F.114) always succeeds. These formulas are a self-contained mathematical certificate.

F.A.1.2 Affine endpoint normal forms

The affine classification used in Section F.4 reduces to eighteen coefficient transforms and three endpoint-support types. For a nonzero affine signature with support $\mathbf{s}+L$, choose the lexicographically first string in the support as origin and the lexicographically first spanning basis of L as intrinsic coordinates, and divide by the value at that origin. Define the eighteen representative matrices

$$\begin{aligned}\mathscr{K}_{\mathrm{aff}} = \{I, X\} \\ \cup \left\{ \begin{pmatrix} 1 & r \\ 1 & -r \end{pmatrix} : r \in \mu_8 \right\} \\ \cup \left\{ \begin{pmatrix} 1 & u(1+\varepsilon\sqrt{2}) \\ 1 & u(1-\varepsilon\sqrt{2}) \end{pmatrix} : u \in \mu_4,\ \varepsilon \in \{1,-1\} \right\}.\end{aligned} \tag{F.115}$$

Proposition F.A.2 (Exact affine endpoint-nondegenerate eight-vertex certificate)**.** *For every canonically projectively normalized nonzero arity-four affine signature $f \in \mathscr{A}$ and every $K_0 \in \mathscr{K}_{\mathrm{aff}}$, define the transformed signature q by*

$$q := K_0^{\otimes 4} f.$$

Retain (K_0, f) precisely when

$$q(\mathbf{x}) = 0 \quad \text{for odd } \mathrm{wt}(\mathbf{x}), \qquad q(\mathbf{0}^4)q(\mathbf{1}^4) \neq 0. \tag{F.116}$$

Then the following assertions hold.

(i) *Writing each affine support as $\mathbf{s} + L$, there are 36 720 normalized projective affine signatures:*

$\dim L$	0	1	2	3	4
# *affine supports*	16	120	140	30	1
# *normalized signatures*	16	480	4480	15360	16384.

Among the $36\,720 \cdot 18$ pairs (K_0, f), exactly 4200 satisfy Equation (F.116)*. Six representative matrices retain 612 inputs each and the other twelve retain 44 inputs each.*

(ii) *Every retained q has the form*

$$q = \gamma\, D(t)^{\otimes 4} h, \qquad \gamma, t \in \overline{\mathbb{Q}}^{\times}, \tag{F.117}$$

where $h \in \mathscr{A}$, $h(\mathbf{0}^4) = 1$, and

$$\mathrm{supp}(h) \in \left\{ \{\mathbf{0}^4, \mathbf{1}^4\},\ \{\mathbf{0}^4, \mathbf{1}^4, \mathbf{1}_I, \mathbf{1}_{I^c}\},\ \{\mathbf{x} : \mathrm{wt}(\mathbf{x}) \equiv 0 \pmod 2\} \right\},$$

with $|I| = 2$. Let $\mathscr{R}_{\mathrm{aff}}$ be the finite set of all such normalized affine tensors h. Their support-size histogram is

$$2^4, \qquad 4^{96}, \qquad 8^{512}.$$

The three support cases form, respectively, $1, 24, 256$ diagonal orbits. Here

$$g \sim_{\mathrm{diag}} g' \quad \Longleftrightarrow \quad g' = \gamma\, D(t)^{\otimes 4} g \quad \text{for some } \gamma, t \in \overline{\mathbb{Q}}^{\times}.$$

More explicitly, fix the lexicographic order on the six weight-two strings. If $v_1, \ldots, v_6$ are the corresponding central coefficients of an endpoint-nondegenerate eight-vertex signature g, define its diagonal-orbit invariant vector by

$$\mathcal{J}(g) := \left(\frac{v_r v_s}{g(\mathbf{0}^4) g(\mathbf{1}^4)} \right)_{1 \le r \le s \le 6} \tag{F.118}$$

Then $\mathcal{J}(g)$ is a complete invariant of these diagonal orbits.

(iii) *For a support-four reference, let $I \sqcup I^c = [4]$ be its two pairs of inputs and write its logical coefficient matrix as*

$$F = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad ABCD \neq 0.$$

Here the row bit is the common value on the I-block, the column bit is the common value on the I^c-block, and

$$A = h(\mathbf{0}^4), \qquad B = h(\mathbf{1}_{I^c}), \qquad C = h(\mathbf{1}_I), \qquad D = h(\mathbf{1}^4).$$

Then $AD/BC \in \{1, -1\}$. If $AD = BC$, the signature is a tensor product of two nonsingular weighted equalities. If $AD = -BC$, take two labelled copies, join each of the two inputs in the I^c-block of the first copy to the corresponding input of the second copy by a X-edge, and retain the two I-blocks as the four external inputs. The resulting logical coefficient matrix is

$$FXF^{\mathsf{T}} = \begin{pmatrix} 2AB & AD + BC \\ AD + BC & 2CD \end{pmatrix} = \begin{pmatrix} 2AB & 0 \\ 0 & 2CD \end{pmatrix}, \tag{F.119}$$

and hence the output signature is a nonzero generalized equality on its four external inputs.

(iv) *The 512 reference signatures with full even-parity support have the disjoint first-witness partition*

$$512 = 16 + 336 + 96 + 48 + 16. \tag{F.120}$$

The first 16 have the dense product-transformable form of Equation (F.11). The next 336 have a rank-one X-self-loop. For this construction, label the four inputs by $0, 1, 2, 3$ (local label j corresponds to input $j + 1$ in the main text). For each remaining reference signature, take two labelled copies and, successively for $j = 1, 2, 3$, join inputs $0, j$ of the first copy by crossed X-edges to inputs $j, 0$ of the second. Order the external inputs as the uncontracted inputs of the first copy in increasing order, followed by the corresponding inputs of the second copy. The first support-four product occurs for, respectively, $96, 48, 16$ new reference signatures. It has the form

$$cD(r) \otimes D(r), \qquad c \in \mathbb{C}^\times, \qquad r \in \mu_4,$$

with respect to the induced perfect matching and displayed external-input order. Here c is a nonzero scalar and $D(r)$ is the diagonal binary matrix. The histogram in the order $r = 1, -1, i, -i$ is

$$(56, 56, 24, 24).$$

(v) *Every normalized fourth-root signature p with full odd-parity support, as in Equation (F.105), is classified as follows. The twelve loops in Equation (F.106) give the partition*

$$16\,384 = 15\,792 + 224 + 368$$

of Proposition F.A.1. Among the 368 remaining phase tuples, exactly 64 satisfy Equation (F.109). The displayed two-copy gadget there is a support-six endpoint-nondegenerate eight-vertex signature.

For each of the other 304 phase tuples, let p_δ multiply the four weight-three values of p by i^δ while leaving its weight-one values fixed. Exactly two $\delta \in \mathbb{Z}_4$ make p_δ affine. The valid sets are $\{0, 2\}$ for 176 phase tuples and $\{1, 3\}$ for 128 phase tuples. Every such p_δ lies in the 512-member family Equation (F.111). Of that family, 336 phase tuples have a rank-one X-loop. Every one of the remaining 176 phase tuples has at least one two-copy output of the dense product-transformable form among the three gadgets in Equation (F.114).

Proof. For a support $\mathbf{s}+L$, put $d=\dim L$. In the canonical intrinsic coordinates $\mathbf{x}=(x_1,\dots,x_d)$, the normalized phases are exactly

$$i^{\sum_{j=1}^{d}\ell_j x_j+2\sum_{1\le j<k\le d}q_{jk}x_jx_k},\qquad \ell_j\in\mathbb{Z}_4,\qquad q_{jk}\in\mathbb{F}_2.$$

There are

$$2^{4-d}\begin{bmatrix}4\\ d\end{bmatrix}_2 4^d 2^{\binom{d}{2}}$$

such signatures of support dimension d: the Gaussian binomial counts the linear spaces, where $\begin{bmatrix}4\\ d\end{bmatrix}_2=\prod_{j=0}^{d-1}(2^4-2^j)/(2^d-2^j)$ and the empty product is one. The factor 2^{4-d} counts their cosets, and the last two factors the linear and mixed phase coefficients. This proves the first table in *(i)*. Applying the eighteen matrices in Equation (F.115) and testing Equation (F.116) gives the 4200 retained pairs.

For completeness of the orbit test, normalize the first endpoint to one, write the central layer as $\mathbf{v}$, and write the second endpoint as e. If $\mathbf{v}=0$, two signatures are diagonally equivalent after choosing $t^4=e'/e$. Otherwise equality of Equation (F.118), using one nonzero coordinate of $\mathbf{v}$, gives

$$\mathbf{v}'=\lambda\mathbf{v},\qquad e'=\lambda^2 e.$$

Choosing $t^2=\lambda$ proves Equation (F.117). The support and orbit histograms then follow from the exhaustive retained list.

On a support-four affine plane, choose the logical coordinates u,v used in the displayed table. After projective normalization its phase is

$$h(u,v)=i^{\ell_1 u+\ell_2 v+2quv},\qquad \ell_1,\ell_2\in\mathbb{Z}_4,\quad q\in\mathbb{F}_2.$$

Therefore $AD/(BC)=(-1)^q\in\{1,-1\}$, proving the asserted cross-ratio. The gadget construction in Equation (F.119) is then a direct coefficient identity. For full even-parity support, the exact replay tests the six disequality self-loops and the three crossed two-copy contractions in the stated first-witness order. This gives Equation (F.120) and the r-histogram.

Finally it evaluates all 4^7 phase tuples of Equation (F.105), the twelve ordered pairs in Equation (F.106), the exceptional two-copy output, all four tests in Equation (F.110), and the three gadgets G_{0j}. This gives the last assertion.

The verification script `certificates/verify_affine_q4.py` regenerates all the finite objects above from their definitions and evaluates the identities over $\mathbb{Z}[i,\sqrt{2}]$ and $\mathbb{Q}(i,\sqrt{2})$. □

Remark F.A.3 (Use in the main proof). Proposition F.A.2 classifies coefficients for the eighteen displayed representative matrices. The main proof supplies the transformability witness, reduces it to those representative matrices, and implements the required constructions in the retained Holant problem. In particular, the support-four gadget is the two-copy construction in Equation (F.119). The retained-list partition and the partitions for full even-parity support use exhaustive exact computation in `certificates/verify_affine_q4.py`.

Theorem F.A.4 (Affine endpoint normal forms)**.** *The matrix set used by the affine endpoint certificate is exactly the eighteen-element set $\mathscr{K}_{\mathrm{aff}}$ displayed in Equation* (F.115)*. Let $f\in\mathscr{A}$ be a nonzero canonically projectively normalized arity-four signature, let $K_0\in\mathscr{K}_{\mathrm{aff}}$, and put $q=K_0^{\otimes 4}f$. If*

$$q(\mathbf{x})=0\quad(\mathrm{wt}(\mathbf{x})\ \textit{odd}),\qquad q(\mathbf{0}^4)q(\mathbf{1}^4)\neq 0,$$

then there are $\gamma, t \in \overline{\mathbb{Q}}^{\times}$ *and a normalized* $h \in \mathscr{A}$*, with* $h(\mathbf{0}^4) = 1$*, such that*

$$q = \gamma\, D(t)^{\otimes 4} h.$$

Moreover, $\operatorname{supp}(h)$ *is exactly one of the following:*

(i) *the complementary endpoint pair* $\{\mathbf{0}^4, \mathbf{1}^4\}$*;*

(ii) *a four-point plane* $\{\mathbf{0}^4, \mathbf{1}^4, \mathbf{1}_I, \mathbf{1}_{I^c}\}$ *with* $|I| = 2$*;*

(iii) *the full even-parity support.*

Proof. This is the direct-use form of Proposition F.A.2(i)–(ii). The three supports have distinct cardinalities, so the alternatives are disjoint. □

Theorem F.A.5 (Gadgets from affine reference signatures)**.** *Let* h *be supplied by Theorem F.A.4.*

(i) *If* $|\operatorname{supp}(h)| = 2$*, then* h *is a generalized equality.*

(ii) *If* $|\operatorname{supp}(h)| = 4$*, its logical coefficient matrix* $F = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ *satisfies* $AD/BC \in \{1, -1\}$*. If* $AD = BC$*, then* h *factors across its two pairs of inputs into nonsingular weighted equalities. If* $AD = -BC$*, the two-copy* X*-edge construction directly realizes a nonzero generalized equality.*

(iii) *If* h *has full even-parity support, then either it has the dense product-transformable form of Equation* (F.11)*, an* X*-self-loop is nonzero of rank one, or one of the three crossed two-copy contractions specified in Proposition F.A.2*(iv) *directly realizes* $cD(r) \otimes D(r)$*, where* $c \neq 0$ *and* $r \in \mu_4$*.*

Proof. The support-two assertion is immediate, and the other two assertions are exactly Proposition F.A.2(iii)–(iv). □

Theorem F.A.6 (Contractions of signatures with full odd-parity support and fourth-root phases)**.** *Let* p *be a normalized quaternary signature with full odd-parity support and fourth-root phases. With* X*-edges and binary equality* I*, the twelve-loop test of Proposition F.A.1 selects a nonzero rank-one binary if one exists. Failing that, it selects a nonsingular binary of infinite projective transfer order if available. If neither occurs, retain* p *and all its zero–zero loops. Then either two copies directly realize a support-six endpoint-nondegenerate eight-vertex signature, or a certified phase shift* $p_\delta \in \mathscr{A}$ *has a rank-one disequality self-loop or a dense product-transformable two-copy output. These conclusions hold at every nonzero scale.*

Proof. Combine the deterministic test in Proposition F.A.1 with Proposition F.A.2(v). □

F.A.2 Equality-Accessible Matching Decks

This subsection gives the shared finite quaternary certificate statement used in Section F.5 and by the later finite-group branches.

Proposition F.A.7 (Gadget equality-accessible quaternary certificate)**.** *Let* $(\mathcal{F}, G, q)$ *be a deck-stable retained state, with* $\mathcal{F}$ *factor-closed and containing the supplied equality* I*, and* $q \neq 0$ *a quaternary irreducible signature. Suppose its complete deck consists only of zero cards and nonzero scalar multiples of the fixed realizable binary representatives of* G*, and suppose* G *is* $C_2^{(X)} = \{1, [X]\}$*, a monomial dihedral group* $D_{2m}^{(Z)} = \{[D(t)], [XD(t)] : t \in \mu_m\}$ *with* $m \geq 3$*, the standard Pauli* $V_4^{(\mathrm{std})}$*, or the exotic group*

$$V_4^{(\mathrm{ex})} = \left\{ [I], [X], \left[\begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix} \right], \left[\begin{pmatrix} -i & 1 \\ -1 & i \end{pmatrix} \right] \right\}.$$

Then a gadget using at most two copies of q produces an endpoint-nondegenerate eight-vertex signature. The assertion covers all zero cards, both input orientations, all six unordered contracted pairs, and the fixed nonzero scalars of the realizable group representatives. In every case, the endpoint-nondegenerate eight-vertex signature can moreover be chosen not to be a G-matching signature.

Proof. Input dressings and reversal only permute the complete group, multiply cards by invertible factors, and contribute known nonzero scalars. It therefore suffices to check every standard group kernel on the six unordered pairs.

The $C_2^{(X)}$ case. Use Walsh coordinates only for this calculation, and define the Walsh coefficients

$$a_S = \widetilde{q}(\mathbf{1}_S), \qquad S \subseteq [4].$$

For any Walsh coefficient tensor indexed by subsets of $[4]$, call it *Walsh-complement-symmetric* when $a_{\bar{S}} = a_S$ for every S, where $\bar{S} = [4] \setminus S$. For a deleted pair ij, the two realizable I/X-cards satisfy

$$\widetilde{C}_{ij}^{\pm}(\mathbf{1}_T) = 2(a_T \pm a_{Tij}).$$

Here $Tij := T \cup \{i, j\}$, with T a subset of the two residual inputs. The allowed nonzero Walsh binary cards are scalar multiples of I or Z. Singleton residual coordinates force $a_S = 0$ for odd $|S|$. For $\{i, j\} \sqcup \{k, l\} = [4]$, define the four Walsh coefficient scalars

$$u = a_{\varnothing}, \quad v = a_{1234}, \quad x = a_{ij}, \quad y = a_{kl}.$$

Safety for the ij-cards gives

$$(y + v)^2 = (u + x)^2, \qquad (y - v)^2 = (u - x)^2,$$

and safety for the complementary kl-cards gives

$$(x + v)^2 = (u + y)^2, \qquad (x - v)^2 = (u - y)^2.$$

Adding and subtracting these four equations gives

$$v^2 = u^2, \qquad y^2 = x^2, \qquad yv = ux, \qquad xv = uy.$$

If $u \neq 0$, one $\epsilon \in \{1, -1\}$ satisfies

$$v = \epsilon u, \qquad a_{\bar{S}} = \epsilon a_S.$$

When $u = v = 0$, call a complementary weight-two block $S, \bar{S}$ *active* if $a_S \neq 0$ (equivalently $a_{\bar{S}} \neq 0$), and call $a_{\bar{S}}/a_S \in \{1, -1\}$ its sign. If all active blocks have one common sign, the same conclusion holds. In both cases Fourier inversion gives one parity in the original basis and complement closure: support only on even-weight Walsh indices gives $q(\mathbf{x}) = q(\bar{\mathbf{x}})$, while $a_{\bar{S}} = \epsilon a_S$ makes the inverse transform vanish unless $(-1)^{\mathrm{wt}(\mathbf{x})} = \epsilon$. A realizable X-dressing then moves a nonzero complementary pair in the original variables to $\mathbf{0}^4, \mathbf{1}^4$. The result is an endpoint-nondegenerate eight-vertex signature and remains irreducible, whereas every quaternary $C_2^{(X)}$-matching signature is a two-binary product.

It remains to treat mixed signs, which necessarily require at least two active complementary blocks. Input permutations permute the three block signs, and a realizable X-dressing reverses all of them. We may therefore choose a positive active block and a negative active block, make any third active block positive, and write the Walsh coefficient scalars as x, y, z, with $xz \neq 0$, such that one X-dressing and a permutation give

$$a_{12} = a_{34} = x, \qquad a_{13} = a_{24} = y, \qquad a_{14} = z, \qquad a_{23} = -z, \qquad xz \neq 0,$$

where y may vanish. For $p = 12, 13, 14$, join the p-inputs of two copies by the realizable path $X - I - X$, and call the output r_p. Its realizable kernel is I, its Walsh kernel is $2I$, and direct contraction gives

	common value on $0000, 1111$	values on the three complementary weight-two blocks
$\widetilde{r}_{12}/4$	x^2	$(y^2 + z^2, 0, 0)$
$\widetilde{r}_{13}/4$	y^2	$(x^2 + z^2, 0, 0)$
$\widetilde{r}_{14}/4$	z^2	$(x^2 + y^2, 2xy, 0)$.

Every nonzero row is supported only on even-weight Walsh indices and is Walsh-complement-symmetric. The same inverse-transform identities just used therefore make it a complement-symmetric, one-parity signature in the original basis. A realizable X-dressing makes its two endpoints nonzero. Hence it becomes an endpoint-nondegenerate eight-vertex signature.

A $C_2^{(X)}$-matching tensor whose Walsh coefficients are complement-symmetric has Walsh support on the endpoints and exactly one weight-two block, with the two common values differing by a sign. Indeed, in Walsh coordinates the group is C_2^{std}, and each realizable binary factor is projectively I or Z. Their tensor product on a perfect matching has precisely that support, and Walsh-complement symmetry forces the two factor signs to agree.

If $y = 0$ and $x^2 + z^2 \neq 0$, r_{13} has zero Walsh endpoints and is not matching. If $x^2 + z^2 = 0$, the nonzero part of the $13 \mid 24$ flattening is

$$\begin{pmatrix} x & -z \\ z & x \end{pmatrix}.$$

All other rows and columns vanish, so this matrix is the complete nonzero block and has rank one, contradicting irreducibility. The local Walsh transforms are invertible and therefore preserve flattening rank and irreducibility.

If $y \neq 0$ and all three outputs were matching, the first two rows would give $\sigma, \tau \in \{1, -1\}$ with

$$y^2 + z^2 = \sigma x^2, \qquad x^2 + z^2 = \tau y^2.$$

Since $2xy \neq 0$, the last row forces $x^2 + y^2 = 0$. Then the first equation gives $\sigma = 1, z^2 = 2x^2$, while the second gives $3x^2 = -\tau x^2$, impossible. Thus a realizable endpoint-nondegenerate eight-vertex signature output is outside the $C_2^{(X)}$-matching products.

The monomial dihedral case. For one pair, write its four residual slices as A, B, C, D_{11}. Define the even and odd residual-card pencils produced by the standard realizable kernels by

$$E(t) := A + tD_{11}, \qquad O(t) := C + tB, \qquad t \in \mu_m.$$

The pencil normal form Lemma F.6.10 applies verbatim to each of these pencils. It says that the two endpoints are proportional on one monomial line, or, in one fixed parity plane with axes E_0, E_1,

$$U + tV = \lambda(E_0 + \alpha t E_1) \quad \text{or} \quad U + tV = \lambda(tE_0 + \alpha E_1), \qquad \alpha \in \mu_m.$$

Its proof uses the exact Möbius classification Lemma F.6.9. In particular, a zero sampled value belongs only to the proportional branch.

Apply this fact to both pencils for all six pairs. The resulting support conditions have exactly the following solutions: choose one total parity and a nonempty subset of its four complementary string pairs. Their size histogram is

$\lvert \operatorname{supp} q \rvert$	2	4	6	8
#	8	12	8	2.

(F.121)

Hence the support is already one parity and complement-closed. A realizable X-dressing makes the same copy an endpoint-nondegenerate eight-vertex signature.

The two Klein cases. In the Pauli Bell basis, one flattening writes

$$q = \sum_{\boldsymbol{\alpha},\boldsymbol{\beta}\in\mathbb{F}_2^2} t_{\boldsymbol{\alpha}\boldsymbol{\beta}} B_{\boldsymbol{\alpha}} \otimes B_{\boldsymbol{\beta}}.$$

Its row and column cards first make $T = (t_{\boldsymbol{\alpha}\boldsymbol{\beta}})$ a partial monomial matrix. Intersecting the four allowed Bell-line preimages for every kernel on the other four pairs leaves exactly the four maximal four-planes

$$V_{\pi,\epsilon} = \{q : q(\mathbf{x}) = 0 \text{ if } \mathrm{wt}(\mathbf{x}) \not\equiv \pi \pmod 2,\ q(\bar{\mathbf{x}}) = \epsilon q(\mathbf{x})\}, \quad \pi \in \mathbb{F}_2, \quad \epsilon \in \{1,-1\}. \tag{F.122}$$

Thus a realizable Pauli X-dressing makes every nonzero member an endpoint-nondegenerate eight-vertex signature.

For $V_4^{(\mathrm{ex})}$, define the representative matrices

$$E_0 = I, \quad E_1 = X, \quad E_2 = \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}, \quad E_3 = \begin{pmatrix} -i & 1 \\ -1 & i \end{pmatrix}.$$

Their Frobenius duals are $E_0/2, E_1/2, E_3/4, E_2/4$, so every dual contraction is again a realizable group direction. The locus satisfying all six pair-card conditions is the union of twenty-four maximal two-dimensional linear subspaces. The support-stratified calculation also produces sixteen one-dimensional solution subspaces, but every one is contained in one of those twenty-four planes and hence contributes no additional component to this locus. Realizable local exotic dressings and permutations of inputs act transitively on the twenty-four planes. A representative is

$$q_{a,b} = a(E_0 \otimes E_0 + E_1 \otimes E_1) + b(E_2 \otimes E_2 + E_3 \otimes E_3),$$

whose coefficient array is

$$q_{a,b}(\mathbf{x}) = \begin{cases} a - 2b, & \mathbf{x} \in \{0000, 0110, 1001, 1111\}, \\ a + 2b, & \mathbf{x} \in \{0011, 0101, 1010, 1100\}, \\ 0, & \mathrm{wt}(\mathbf{x}) \text{ odd}. \end{cases} \tag{F.123}$$

If $a - 2b \neq 0$, this is an endpoint-nondegenerate eight-vertex signature. If $a = 2b \neq 0$, realizable X-dressings on the last two inputs make both endpoints $4b$. This proves the exotic case.

In each of the latter three cases, the displayed endpoint-nondegenerate eight-vertex signature is obtained from the original q only by invertible group dressings and a permutation of inputs. It is therefore irreducible. Since every quaternary $G(\mathcal{F})$-matching signature is a tensor product of two nonsingular group binaries, these endpoint-nondegenerate eight-vertex signatures are not $G(\mathcal{F})$-matching. The strengthened conclusion in the $C_2^{(X)}$ case was proved separately above.

The support and subspace completeness claims in Equations (F.121) to (F.123) use the exact enumeration in `certificates/verify_equality_q4.py`. It tests all $2^{16} - 1$ nonempty computational supports in the monomial case and all 209 partial-injection supports in each Bell basis. For the latter, it intersects the allowed preimages for four cross-pairs and four kernels, retaining zero cards and reducing the equations over $\mathbb{Q}$ or $\mathbb{Q}(i)$. The preceding pencil proof supplies the coefficient constraint before this finite support classification. □

Theorem F.A.8 (Classification of equality-accessible quaternary signatures)**.** *Let $(\mathcal{F}, G, q)$ be a deck-stable retained, factor-closed state containing the supplied equality I, and let $q \neq 0$ be an irreducible quaternary signature. Suppose every realizable card is zero or a nonzero scalar multiple of one of the fixed representatives of G, where G is $C_2^{(X)}$, $D_{2m}^{(Z)}$ with $m \geq 3$, $V_4^{(\mathrm{std})}$, or $V_4^{(\mathrm{ex})}$. Then a gadget using at most two copies of q realizes an endpoint-nondegenerate eight-vertex signature that is not G-matching. More precisely:*

(i) *In the $C_2^{(X)}$ row, put $a_S := \widetilde{q}(\mathbf{1}_S)$. A complementary weight-two block $S, \bar{S}$ is* active *when $a_S \neq 0$, and its sign is $a_{\bar{S}}/a_S \in \{1, -1\}$. If the two Walsh endpoints $a_\varnothing, a_{[4]}$ are nonzero or all active blocks have one common sign, one dressed copy suffices. Otherwise at least two active blocks have different signs. This is the* mixed-sign *case. For $p \in \{12, 13, 14\}$, let r_p join the p-inputs of two copies by the realizable path $X - I - X$. One of these three contractions is endpoint-nondegenerate and nonmatching.*

(ii) *In the monomial-dihedral row, the support is exactly one total parity together with a nonempty subset of its four complementary string pairs. One realizable X-dressing makes the same copy endpoint-nondegenerate.*

(iii) *In either Klein row, realizable dressings and permutations of inputs give the asserted irreducible endpoint-nondegenerate nonmatching copy.*

All clauses include zero cards, both orientations, all six contracted pairs, and the fixed nonzero scalars of the chosen representatives.

Proof. The overall conclusion is Proposition F.A.7. Its $C_2^{(X)}$, monomial-dihedral, and two Klein proof cases give clauses (i), (ii), and (iii), respectively. The containment sentence in the exotic-Klein case makes its representative-plane argument exhaustive. □

F.A.3 Marked Dihedral Matching Decks

The six-input support classification below is used by the standard $D_{2m}^{(Z)}$ branch. The whole-network reductions in Theorems F.6.15 and F.6.16 carry the $D_{2m}^{(Y)}$ and $D_{2m}^{(X,0)}$ cases to that branch. For $D_{2m}^{(X,1)}$, Lemmas F.6.33 and F.6.34 preserve this support argument with weighted kernels. We reduce its 2304 labelled parameter choices to the explicit 48-case table below.

Fix the distinguished pair $p = \{1, 2\}$ and residual inputs $R = \{3, 4, 5, 6\}$. For each sector $\epsilon \in \mathbb{F}_2$, choose a perfect matching

$$M_\epsilon = \{e_\epsilon, e'_\epsilon\} \quad \text{of } R,$$

distinguish the fixed edge $e_\epsilon = \{u_\epsilon, v_\epsilon\}$, and order the other edge as $e'_\epsilon = (r_\epsilon, s_\epsilon)$ with $r_\epsilon < s_\epsilon$. Choose $a_\epsilon, b_\epsilon, o_\epsilon \in \mathbb{F}_2$. Define the eight-string support $S \subseteq \mathbb{F}_2^6$ by taking, for every $\epsilon, z \in \mathbb{F}_2$, the two strings satisfying

$$\begin{aligned} (x_1, x_2) &= (z, z + \epsilon), \\ x_{u_\epsilon} + x_{v_\epsilon} &= a_\epsilon, \qquad \text{over } \mathbb{F}_2. \\ (x_{r_\epsilon}, x_{s_\epsilon}) &= (o_\epsilon + z, o_\epsilon + z + b_\epsilon), \end{aligned} \tag{F.124}$$

This is exactly the support form obtained from four complementary flat endpoints: the fixed monomial factor supplies the two strings in a slice, and the two endpoints of the variable factor are complementary.

Proposition F.A.9 (Six-to-four support lowering in the standard $D_{2m}^{(Z)}$ case)**.** *Among the $3^2 \cdot 2^8 = 2304$ labelled choices in Equation* (F.124)*, exactly 96 give a matching coset: for some perfect*

matching M and constants $c_e \in \mathbb{F}_2$, the support is

$$\{\mathbf{x} \in \mathbb{F}_2^6 : x_u + x_v = c_{\{u,v\}} \text{ for every } \{u,v\} \in M\}.$$

For every other choice there are an input pair $e = \{e_1, e_2\}$ and $\gamma \in \mathbb{F}_2$ such that projection away from e is injective on

$$S \cap \{\mathbf{x} : x_{e_1} + x_{e_2} = \gamma\},$$

and the projected quaternary support has size six or eight. Choosing a largest such projection gives 1152 *support-six cases and* 1056 *support-eight cases.*

Proof. Permute the four residual inputs so that the fixed edge in sector zero is 34, and its ordered varying edge is $(5, 6)$. Independently flipping bits on inputs $3, 5, 6$ removes a_0, o_0, b_0, respectively: use the flips $a_0, o_0, o_0 + b_0$. Thus sector zero is

$$S_0 = \{(z, z, w, w, z, z) : z, w \in \mathbb{F}_2\}.$$

These permutations and bit flips preserve matching cosets, projection injectivity, and projected support sizes. For each of the $3 \cdot 2 \cdot 2^3 = 48$ choices of sector-zero data, fix one such normalization. It bijects the 48 choices of sector-one data with the following canonical choices: a fixed edge $e \subseteq \{3, 4, 5, 6\}$, the increasingly ordered complementary edge $e' = (r, s)$, and $(a, b, o) \in \mathbb{F}_2^3$. Their four strings satisfy

$$(x_1, x_2) = (z, z+1), \qquad x_u = w, \quad x_v = w+a \quad (e = \{u, v\},\ u < v), \qquad (x_r, x_s) = (o+z, o+z+b).$$

The table gives a projection witness for every canonical choice. An entry ij^s_γ means: retain strings with $x_i + x_j = \gamma$, delete inputs i, j, and obtain s distinct projected strings from exactly s retained strings. Thus the projection is injective. M denotes a matching coset.

$e \backslash (a,b,o)$	000	001	010	011	100	101	110	111
34	15^8_0	25^8_0	M	M	15^8_0	25^8_0	15^8_0	16^8_0
35	16^8_0	26^8_0	24^6_1	14^6_1	14^6_0	24^6_0	26^8_0	16^8_0
36	15^8_0	25^8_0	24^6_1	14^6_1	14^6_0	24^6_0	25^8_0	15^8_0
45	16^8_0	26^8_0	23^6_1	13^6_1	13^6_0	23^6_0	26^8_0	16^8_0
46	15^8_0	25^8_0	23^6_1	13^6_1	13^6_0	23^6_0	25^8_0	15^8_0
56	13^6_0	13^6_1	15^6_0	15^6_0	13^6_0	13^6_1	13^6_0	13^6_1

Each entry follows by substituting $(z, w) \in \mathbb{F}_2^2$ in the two displayed sectors. For example, at $e = 35$, $(a, b, o) = 010$, the 24^6_1 projection leaves

$$\{0001, 0100, 0111, 1000, 1011, 1110\}$$

on inputs $(1, 3, 5, 6)$, with one preimage per string.

For maximality of the entries marked 6, an eight-string projection would require a pair with constant XOR on both sectors. The pairs with constant XOR on S_0 are $12, 15, 16, 25, 26, 34, 56$. Substitution of the sector-one equations in each 6-entry eliminates these pairs, or leaves a projection that identifies two strings. A seven-string parity slice is impossible: on each four-string affine sector its size is $0, 2$, or 4. Thus the displayed six-string images are maximal. At the two M-entries the matching equations are, respectively,

$$x_3 = x_4,\ x_1 = x_5,\ x_2 = x_6; \qquad x_3 = x_4,\ x_1 = x_6,\ x_2 = x_5.$$

A parity contraction of a three-edge matching coset has at most four projected strings, so none of the other entries is a matching coset. The canonical table therefore has $2, 24, 22$ cases of the three stated types. Multiplying by the 48 sector-zero choices gives

$$48(2, 24, 22) = (96, 1152, 1056),$$

as required. □

Remark F.A.10 (From support projection to contraction). In the standard $D_{2m}^{(Z)}$ setting, the supplied equality I gives the contraction kernel $XIX = I$, and a direct edge gives the kernel X. They implement parity $\gamma = 0$ and $\gamma = 1$, respectively. Injectivity leaves one nonzero summand for each projected string, so the conclusion holds for arbitrary nonzero coefficients on the support. The main proof supplies the sector-ruling hypothesis and exposes any resulting factors by Lemma F.3.4. The realizable contractions in the following theorem are taken in this setting.

Theorem F.A.11 (Six-to-four contractions in the standard $D_{2m}^{(Z)}$ case)**.** *Let h be an arity-six signature whose support is a set S of the form Equation* (F.124)*, and suppose every coefficient of h on S is nonzero. Then either S is a three-edge matching coset, or there are an input pair $e = \{e_1, e_2\}$ and $\gamma \in \mathbb{F}_2$ for which projection away from e is injective on*

$$S \cap \{\mathbf{x} : x_{e_1} + x_{e_2} = \gamma\}$$

and has image of size six or eight. In the latter case, contraction by the realizable kernel I for $\gamma = 0$, or X for $\gamma = 1$, directly realizes a cancellation-free quaternary of support size six or eight. That output is not a product of two nonsingular monomial binaries.

Proof. Proposition F.A.9 gives the exhaustive support alternative and injectivity. The selected parity kernel leaves exactly one nonzero summand above each projected string, proving noncancellation. A product of two nonsingular monomial binaries has support size four. □

F.A.4 Klein-Four Matching Decks

The following certificate supplies the exotic Klein finite cases used in Section F.8.

F.A.4.1 The exotic Klein form $V_4^{(\mathrm{ex})}$

Proposition F.A.12 (Exotic-Klein six- and eight-input certificate)**.** *Use the four exotic binary directions E_a, their Frobenius duals, and the label set $V = \mathbb{F}_2^2$ from Equations* (F.61) *and* (F.62)*. The following finite statements hold.*

(i) *Let $h \neq 0$ have arity six, suppose every exotic pair card is zero or a product of two exotic binaries, and normalize one nonzero card by*

$$C_{12}^0 = E_0^{(3,4)} E_0^{(5,6)}.$$

The other three card directions range over the complete set of normalized card-direction assignments

$$\left(\{0\} \cup \mathfrak{D}_4(V_4^{(\mathrm{ex})})\right)^3, \qquad (1 + 48)^3 = 117{,}649.$$

Over arbitrary algebraic nonzero-card scales, including every cancellation locus in every other card context, exactly seventeen projective scale lines survive. Every one reconstructs a product of three exotic binaries. No irreducible arity-six line survives.

(ii) *Let h have arity eight and suppose it has the normal form Equation* (F.66)*, with one matching and all four cards nonzero. If every one of its $28 \cdot 4 = 112$ exotic pair cards is zero or a product of three exotic binaries, then h has a proper tensor factor.*

Proof. All matching lines and card maps below are generated directly from the four matrices in Equation (F.61). The calculation first verifies the Gram matrix, the dual basis

$$(E_0/2, E_1/2, E_3/4, E_2/4),$$

and projective fusion closure, so its contractions use the same normalized directions as the main argument.

For part (i), there are $3 \cdot 4^2 = 48$ projective quaternary matching lines. Encode zero by -1, and encode the line with matching $m \in \{0, 1, 2\}$ and labels $a, b \in V$ by $16m + 4a + b$. Thus a nonnegative code denotes the canonically normalized line $P_m(E_a, E_b)$. On each base residual edge 34 and 56, the stabilizer under realizable operations contains the paired endpoint dressings with label pairs

$$(1, 1), \qquad (2, 3), \qquad (3, 2),$$

which fix the base E_0-edge projectively. Together with the permutations of inputs

$$(3\,4), \qquad (5\,6), \qquad (3\,5)(4\,6),$$

these generate the stabilizer under realizable operations used here. Its induced action on the 48 residual matching lines has order 64. It partitions the full 49^3 card-direction assignments into 4,045 orbits, and the generated orbit-size histogram is

$$1^{27}, \quad 2^{49}, \quad 4^{351}, \quad 8^{521}, \quad 16^{577}, \quad 32^{1830}, \quad 64^{690}.$$

The orbit sizes weighted by these multiplicities sum to 117 649, so the generated orbits cover every card-direction assignment. On one representative of every orbit, exact $\mathbb{Q}(i)$-row reduction computes the union of the preimages of the forty-eight matching lines within each nonbase pair/kernel context, then intersects these unions over all 56 contexts. A zero output lies in every matching-line preimage, and nonzero-card scales remain independent linear variables. In particular, no cancellation locus is divided away.

The only surviving direction triples are

$$(-1, -1, -1), \quad (1, 2, 3), \quad (1, 3, 2), \quad (4, 8, 12), \quad (4, 12, 8).$$

The first has the single scale line (1). Each of the other four has exactly the four scale lines

$$(1, \epsilon, \delta/2, \epsilon\delta/2), \qquad \epsilon, \delta \in \{1, -1\}.$$

Thus the nonzero-card histogram consists of one line with a single nonzero card and sixteen lines with every card nonzero. An independent enumeration of all $15 \cdot 4^3 = 960$ three-binary product lines reconstructs exactly these seventeen tensors, proving part (i).

For part (ii), the prescribed form is Equation (F.66), with arbitrary $\lambda_\gamma \neq 0$ and $A_j \in \mathrm{GL}_2(\mathbb{F}_2)$. Apply the two-contraction argument in the proof of Lemma F.8.8, on $(3, 5)$ and $(4, 5)$ with $E^0 = I/2$. It uses exactly these conditions to contradict the matching-product overlap formula Equation (F.64), giving the asserted factor conclusion.

Every scale condition in part (i) is a finite union of homogeneous linear systems over $\mathbb{Q}(i)$. Prescribed liveness is also exact: over an infinite field, a linear space contains a vector avoiding all coordinate hyperplanes exactly when no coordinate vanishes identically on that space. Base change to $\overline{\mathbb{Q}}$, or to $\mathbb{C}$, therefore creates no additional component, and zero target cards are kept as zero-card subspaces. The exact calculation for part (i) regenerates the states, orbits, scale intersections, and factor tensors. The Python verification script is `certificates/verify_exotic_v4_q6_q8.py`. □

Theorem F.A.13 (Six-input tensors for the exotic Klein group)**.** *Let $h \neq 0$ have arity six. If every exotic pair card of h is zero or a product of two exotic binaries, then h is a product of three exotic binaries. Consequently no irreducible arity-six remaining case exists. This includes arbitrary algebraic scales and every cancellation locus.*

Proof. This is Proposition F.A.12(i) after undoing the normalization of one nonzero card. □

Theorem F.A.14 (Eight-input tensors for the exotic Klein group)**.** *Let h have arity eight and satisfy the displayed normal form Equation* (F.66)*, with one matching and all four cards nonzero. If all* 112 *exotic pair cards of h are zero or products of three exotic binaries, then h has a proper tensor factor.*

Proof. This is Proposition F.A.12(ii). □

F.A.5 Platonic Matching Decks

The Platonic analysis first solves the separated-sign equations for localized card maps. The incidence classification then determines how these local configurations fit together. The final calculations handle the tetrahedral arity-eight case and the arity-four contractions used by the main proof. All configurations use the quaternion basis in Equation (F.16).

Whenever a mixed three-point incidence is localized, realizable dressings put its rainbow image in the common normalization

$$A = P_0(Q_0, Q_0), \qquad B = P_1(Q_0, Q_0), \qquad C = P_2(Q_2, Q_2) = A - B. \tag{F.125}$$

Recall that matching-label type $(1,1,1)$ is rainbow, and (s) is monochromatic. Set

$$\begin{gathered} R_0 = \tfrac{1}{2}P_2(Q_0, Q_0), \qquad R_1 = -\tfrac{1}{2}P_2(Q_1, Q_1), \qquad R_3 = -\tfrac{1}{2}P_2(Q_3, Q_3), \\ R_+ = Q_0 + Q_1 + Q_2 + Q_3, \qquad R_- = Q_0 - Q_1 - Q_2 - Q_3. \end{gathered} \tag{F.126}$$

Here R_+, R_- are the matrices already used in Equation (F.96).

Lemma F.A.15 (Matching-product solutions of the separated-sign equations)**.** *Let S be the swap matrix, defined by $S(\mathbf{u} \otimes \mathbf{v}) = \mathbf{v} \otimes \mathbf{u}$. For $U, V \in \mathrm{GL}_2(\mathbb{C})$ and $\lambda \neq 0$,*

$$\operatorname{rank}\big((U \otimes V^{\mathsf{T}})S - \lambda I_4\big) = 1 \quad \Longleftrightarrow \quad V = \lambda U^{-\mathsf{T}}. \tag{F.127}$$

For $G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{I}^{(0)}\}$, put $\mathscr{C}_G = \{0\} \cup \operatorname{cone}(\mathfrak{D}_4(G))$. Then

$$\{W : W, A - 2W, 2W - B \in \mathscr{C}_G\} = \{0, A/2, B/2\} \cup \left\{\tfrac{1}{2}P_2(J, J^{-\mathsf{T}}) : [J] \in G,\ J^{\mathsf{T}} = J\right\}. \tag{F.128}$$

Write a symmetric group matrix as $J_{\mathbf{u}} = aQ_0 + bQ_1 + dQ_3$, choosing a nonzero real coefficient vector $\mathbf{u} = (a, b, d)$, and put $R_{\mathbf{u}} = \frac{1}{2}P_2(J_{\mathbf{u}}, J_{\mathbf{u}}^{-\mathsf{T}})$. A triple of solutions to the separated-sign equations sums to $(A + B)/2$ precisely when it is a permutation of $(0, A/2, B/2)$, or is $(R_{\mathbf{u}}, R_{\mathbf{v}}, R_{\mathbf{w}})$ with $\mathbf{u}, \mathbf{v}, \mathbf{w}$ pairwise orthogonal in $\mathbb{R}^3$.

Proof. Conjugation by $U^{-1} \otimes I$ reduces the rank condition to $(I \otimes W^{\mathsf{T}})S - \lambda I_4$, with $W = U^{\mathsf{T}}V = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Its matrix is

$$\begin{pmatrix} a - \lambda & 0 & c & 0 \\ b & -\lambda & d & 0 \\ 0 & a & -\lambda & c \\ 0 & b & 0 & d - \lambda \end{pmatrix}.$$

The second row has nonzero second entry. Rank one forces the first row to vanish, hence $a = \lambda, c = 0$. Comparing the second and third rows then gives $b = 0, d = \lambda$. Conversely, $S - I_4$ has rank one. This proves Equation (F.127).

The three exceptional separated-sign solutions $0, A/2, B/2$ satisfy the conditions, since $C = A - B = P_2(Y, Y)$ and $[Y] \in G$. For any other solution, all three tested vectors are nonzero. A line with two points of one matching and a point of another would give flattening ranks $1, 1, 4$, which is impossible. If $W = P_0(U, V)$, the condition on $2W - B$ therefore gives matching 2. Its rank-one flattening and Equation (F.127) imply $W = \frac{1}{2}P_0(U, U^{-\mathsf{T}})$. The condition on $A - 2W$ has matching 0, so $I \otimes I - U \otimes U^{-\mathsf{T}}$ is simple. The simple-tensor sum criterion forces U to be scalar, giving $W = A/2$, a contradiction. The matching-1 case similarly gives $W = B/2$.

Thus $W = P_2(U, V)$, and $A - 2W$ has matching 1. Its $13 \mid 24$ flattening is $I_4 - 2(U \otimes V^{\mathsf{T}})S$. Hence $W = \frac{1}{2}P_2(U, U^{-\mathsf{T}})$. Apply the invertible local matrices $(U^{-1}, I, U^{\mathsf{T}}, I)$, which fix B. They send $2W$ to $P_2(I, U^{-\mathsf{T}}U)$. Since $2W - B$ has matching 0, Equation (F.127) with $\lambda = 1$ now gives $U^{-\mathsf{T}}U = I$, or $U^{\mathsf{T}} = U$. Conversely, for symmetric U, the two differences are local images of $P_1(Y, Y)$ and $-P_0(Y, Y)$. All leg matrices belong projectively to G, so both differences are in its matching cone. This proves Equation (F.128), including arbitrary scales.

For the sum equation, polarization of $\frac{1}{2}P_2(J_{\mathbf{u}}, \mathrm{adj}(J_{\mathbf{u}}))$, where $\mathrm{adj}(J_{\mathbf{u}}) = aQ_0 - bQ_1 - dQ_3$, defines an injective linear map T from the symmetric 3×3 matrices to quaternary tensors with

$$T(\mathbf{u}\mathbf{u}^{\mathsf{T}}) = \tfrac{1}{2}P_2(J_{\mathbf{u}}, \mathrm{adj}(J_{\mathbf{u}})), \qquad T(I_3) = (A + B)/2, \qquad R_{\mathbf{u}} = T\left(\frac{\mathbf{u}\mathbf{u}^{\mathsf{T}}}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}\right).$$

The six coefficient tensors are independent and $C \notin \operatorname{im} T$: in the $14 \mid 23$ binary basis they use only $Q_\mu \otimes Q_\nu$ with $\mu, \nu \in \{0, 1, 3\}$, whereas $C = Q_2 \otimes Q_2$. Also

$$A/2 = \tfrac{1}{2}T(I_3) + C/4, \qquad B/2 = \tfrac{1}{2}T(I_3) - C/4.$$

Comparison of the C-coefficient forces equal numbers of $A/2$ and $B/2$. If they occur, the third summand is zero. Otherwise the trace of $T^{-1}R_{\mathbf{u}}$ is one, so all three summands are nonzero R-vectors. Their sum is $T(I_3)$ exactly when their normalized real coefficient columns form a matrix E with $EE^{\mathsf{T}} = I_3$. This is equivalent to $E^{\mathsf{T}}E = I_3$, proving the orthogonality criterion. The three orthogonal rank-one projectors are independent. Together with $C \notin \operatorname{im} T$, they give card-map rank four. The exceptional triple spans A, B and gives rank two. □

F.A.5.1 The four fixed-equality octahedral forms $\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$

For the octahedral certificate, use the standard 24-point set $\mathcal{O}^{(\mathrm{dd})}$ in Equations (F.16) and (F.18), ordered as the four axes, the eight sign points, and the twelve pair points $Q_\mu \pm Q_\nu$, with pairs lexicographic. The second unmixed distinguished-involution normal form is

$$\mathcal{O}^{(\mathrm{tt})} = H_{\mathrm{tt}}^{-1}\mathcal{O}^{(\mathrm{dd})}H_{\mathrm{tt}}. \tag{F.129}$$

Use the coefficient-similarity map θ_{tt}, transformation tuple $\mathbf{H}_{\mathrm{tt}}$, and induced coefficient operator $\mathcal{L}_{\mathrm{tt}}$ fixed in Equations (F.69) and (F.85). The similarity induces the corresponding map on projective group classes. Here P'_m denotes the same ordered matching placement as P_m, but in the $\mathcal{O}^{(\mathrm{tt})}$ coefficient coordinates. Writing $X = Q_1$ projectively, direct ordered contraction gives

$$\begin{aligned}
\mathcal{L}_{\mathrm{tt}}P_0(U, V) &\doteq P'_0(\theta_{\mathrm{tt}}(U), \theta_{\mathrm{tt}}(V)),\\
\mathcal{L}_{\mathrm{tt}}P_1(U, V) &\doteq P'_1(\theta_{\mathrm{tt}}(UX), \theta_{\mathrm{tt}}(XV)),\\
\mathcal{L}_{\mathrm{tt}}P_2(U, V) &\doteq P'_2(\theta_{\mathrm{tt}}(U), \theta_{\mathrm{tt}}(XVX)).
\end{aligned} \tag{F.130}$$

The operator $\mathcal{L}_{\mathrm{tt}}$ compares coefficient arrays. Realizable dressings use the corresponding $\theta_{\mathrm{tt}}(U)$ representatives with their disequality-edge orientations and nonzero scalars, as in the fixed-coordinate convention of Section F.9.

The coset classification in Proposition F.A.19 gives

domain points	three-point lines	four-point lines
24	32	18

Every point lies on seven rich lines. The three-lines and four-lines are single realizable left/right orbits, represented by

$$\{[Q_0],[R_+],[R_-]\}$$

and

$$\{[Q_0],[Q_1],[Q_0+Q_1],[Q_0-Q_1]\},$$

respectively.

The quaternary deck has $3\cdot 24^2 = 1728$ projective points. Its complete maximal-rich-line histogram is

line size	matching-label type	number
3	$(1,1,1)$	13 824
3	(3)	4 608
4	(4)	2 592
		21 024

There is no $(2,1)$ line and no mixed four-line. Representatives of the three rows are

$$\{P_0(Q_0,Q_0),P_1(Q_0,Q_0),P_2(Q_2,Q_2)\},$$
$$\{P_0(Q_0,Q_0),P_0(Q_0,R_+),P_0(Q_0,R_-)\},$$
$$\{P_0(Q_0,Q_0),P_0(Q_0,Q_1),P_0(Q_0,Q_0+Q_1),P_0(Q_0,Q_0-Q_1)\},$$

where R_+ and R_- are the two explicitly fixed representatives in Equation (F.126).

For a mixed three-point incidence, use the normalization (F.125). For a selected ordered frame, let Φ be the linear card map determined by $\Phi(Q_0)=C/2$ and by $(\Phi(Q_1),\Phi(Q_2),\Phi(Q_3))$ equal to that ordered frame. Define the image tensors $W_j := \Phi(Q_j)$, and write $\operatorname{rank}\Phi := \dim\operatorname{im}\Phi$. Before imposing the twelve pair points of (F.18), the bridge is

$$\{0, A/2, B/2\}\cup\mathcal{R},$$

where the subscripts in $R^{\epsilon\delta}_{\mu\nu}$ specify the two quaternion axes and the superscripts specify the displayed signs:

$$\begin{aligned}\mathcal{R} = \{&R_0,R_1,R_3,\\ &R^{+-}_{01}=\tfrac14 P_2(Q_0+Q_1,Q_0-Q_1),\quad R^{-+}_{01}=\tfrac14 P_2(Q_0-Q_1,Q_0+Q_1),\\ &R^{+-}_{03}=\tfrac14 P_2(Q_0+Q_3,Q_0-Q_3),\quad R^{-+}_{03}=\tfrac14 P_2(Q_0-Q_3,Q_0+Q_3),\\ &R^{++}_{13}=-\tfrac14 P_2(Q_1+Q_3,Q_1+Q_3),\quad R^{--}_{13}=-\tfrac14 P_2(Q_1-Q_3,Q_1-Q_3)\}.\end{aligned}$$

By Lemma F.A.15, these are all bridge vectors: the symmetric octahedral directions are the three axes and the six $Q_\mu\pm Q_\nu$ directions with $\mu,\nu\in\{0,1,3\}$. An orthogonal triple is either the three

axes or a pair of opposite diagonal directions in one coordinate plane together with the remaining axis. There are therefore four unordered frames. The sum equation $W_1 + W_2 + W_3 = (A + B)/2$ leaves 30 ordered maps: the six rank $\Phi = 2$ maps permute $(0, A/2, B/2)$, and the other twenty-four are the six orderings of each of the four rank $\Phi = 4$ frames

$$\begin{aligned} &\{R_0, R_1, R_3\}, \qquad \{R_0, R_{13}^{++}, R_{13}^{--}\}, \\ &\{R_1, R_{03}^{+-}, R_{03}^{-+}\}, \qquad \{R_3, R_{01}^{+-}, R_{01}^{-+}\}. \end{aligned} \tag{F.131}$$

The 1,727 deck points other than $[A]$ lie on 1,683 projective lines through $[A]$. The numbers of other deck points on those lines are

$$1^{1645}, \qquad 2^{32}, \qquad 3^{6}.$$

The single realizable kernel

$$K_* = Q_0 + Q_1, \qquad [K_*] \in \mathcal{O}^{(\mathrm{dd})}$$

has the following property: for every ordered rank-four frame, $q_R = \Phi(K_*) = C/2 + R$ is not a matching-deck point. The 24 supports have sizes

$$2^{4}, \qquad 8^{12}, \qquad 16^{8}.$$

The coefficient correspondence in Equation (F.130) identifies the second normal form. Its realizable constructions use the $\mathcal{O}^{(\mathrm{tt})}$-kernels and dressings in Proposition F.A.16. Invertible local coefficient maps preserve the three ranks.

For the two mixed forms, use the coefficient similarities $H_{\mathrm{dt}}, H_{\mathrm{td}}$, groups $\mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$, and operators $\mathcal{L}_{\mathrm{dt}}, \mathcal{L}_{\mathrm{td}}$ fixed in Equations (F.70), (F.71) and (F.86). The three exact transport identities there identify every matching label and absorb their displayed orientation factors into matrices of the current mixed group. They compare coefficient tables, while the following proposition supplies constructions in the current marked group.

Proposition F.A.16 (Gadget octahedral signature-separator certificate)**.** *For every ordering of every frame in Equation* (F.131)*, use the realizable kernel $K_* = Q_0 + Q_1$ in the $\mathcal{O}^{(\mathrm{dd})}$ form and the realizable kernel $K'_* := \theta_{\mathrm{tt}}(K_*)$ in the $\mathcal{O}^{(\mathrm{tt})}$ form. The resulting direct quaternary can be dressed, using only realizable local matrices in the current marked octahedral group, to an endpoint-nondegenerate eight-vertex signature of flattening ranks $(2, 2, 2)$. Similarity by H_{tt} identifies the coefficient tables.*

For $\alpha \in \{\mathrm{dt}, \mathrm{td}\}$, put $K_{*,\alpha} := \theta_\alpha(K_*)$, so $[K_{*,\alpha}] \in \mathcal{O}^{(\alpha)}$, and use the fixed realizable kernel in that projective class, with its realization scalar recorded. The resulting direct quaternary has, using only current-group contractions and dressings, one of three certified exits: an endpoint-nondegenerate eight-vertex signature of ranks $(2, 2, 2)$, a generalized equality, or a nonsingular binary B with $[BX] \notin \mathcal{O}^{(\alpha)}$. The same three-way conclusion holds for each of the six rank-deficient maps in each mixed form. All representatives, determinants, and outside-transfer directions are given in the proof below.

Proof. Define the tetrahedral sign matrices by

$$G_{\epsilon_1\epsilon_2\epsilon_3} = Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3.$$

For the first marked form the separator is $q_R = C/2 + R$, where R is the first member of the ordered frame. In the even-string order Equation (F.93), exact substitution gives the following realizable dressings. Every omitted odd-weight coefficient is zero, and the displayed vectors are projective.

R	$\mathbf{L}$	$\mathbf{e}(\mathscr{D}_{\mathbf{L}} q_R)$
R_0	$I^{\otimes 4}$	$(1,1,-1,1,1,-1,1,1)$
R_1	(I,I,Q_1,Q_1)	$(1,0,0,0,0,0,0,1)$
R_3	$I^{\otimes 4}$	$(1,1,-1,-1,-1,-1,1,1)$
R_{01}^{+-}	$G_{+++}^{\otimes 4}$	$(1,-1,1,-i,i,1,-1,1)$
R_{01}^{-+}	$G_{+++}^{\otimes 4}$	$(1,-1,1,i,-i,1,-1,1)$
R_{03}^{+-}	$I^{\otimes 4}$	$(1,1,-1,i,-i,-1,1,1)$
R_{03}^{-+}	$I^{\otimes 4}$	$(1,1,-1,-i,i,-1,1,1)$
R_{13}^{++}	$G_{++-}^{\otimes 4}$	$(1,i,-i,-i,-i,-i,i,-1)$
R_{13}^{--}	$G_{++-}^{\otimes 4}$	$(1,-i,i,i,i,i,-i,-1)$

Both endpoints are nonzero in every row. Since the separator depends only on the first frame member, these nine rows cover all twenty-four ordered maps. Direct row reduction of the three flattenings of every displayed q_R gives $(2,2,2)$.

For the second marked form, use the coefficient-transport operator $\mathcal{L}_{\mathrm{tt}}$ above, and define the transported card map Φ' and transported direct card q'_R by

$$\Phi'(\theta_{\mathrm{tt}} C_{\ker}) = \mathcal{L}_{\mathrm{tt}} \Phi(C_{\ker}), \qquad q'_R := \Phi'(K'_*) = \mathcal{L}_{\mathrm{tt}} q_R.$$

Here $[K'_*] \in \mathcal{O}^{(\mathrm{tt})}$, so q'_R is a direct card in the second marked form. With the nonzero comparison scalar $\lambda := 2\sqrt{2}(1-i)$, the identities

$$H_{\mathrm{tt}} H_{\mathrm{tt}}^{\mathsf{T}} = \lambda X, \qquad H_{\mathrm{tt}}^{-\mathsf{T}} H_{\mathrm{tt}}^{-1} = \lambda^{-1} X$$

show that $\mathcal{L}_{\mathrm{tt}}$ bijects the two coefficient decks. This is only an analytic correspondence. The following input matrices are separately verified to belong projectively to the realizable group $\mathcal{O}^{(\mathrm{tt})}$. In the table, θ_{tt} acts componentwise on every displayed tuple:

R	$\mathbf{L}'$	$\mathbf{e}(\mathscr{D}_{\mathbf{L}'} q'_R)$
R_0	$\theta_{\mathrm{tt}}(G_{++-}, G_{++-}, G_{++-}, G_{+-+})$	$(1,-1,-1,-1,-1,-1,-1,1)$
R_1	$\theta_{\mathrm{tt}}(I,I,I,I)$	$(1,1,1,-1,-1,1,1,1)$
R_3	$\theta_{\mathrm{tt}}(G_{+++})^{\otimes 4}$	$(1,-1,1,1,1,1,-1,1)$
R_{01}^{+-}	$\theta_{\mathrm{tt}}(G_{+++}, G_{--+}, G_{--+}, G_{+++})$	$(1,0,0,0,0,0,0,1)$
R_{01}^{-+}	$\theta_{\mathrm{tt}}(G_{+++}, G_{+--}, G_{+--}, G_{+++})$	$(1,-1,-1,1,1,-1,-1,1)$
R_{03}^{+-}	$\theta_{\mathrm{tt}}(I,I,Q_1,Q_1)$	$(1,1,-1,-1,-1,-1,1,1)$
R_{03}^{-+}	$\theta_{\mathrm{tt}}(I,I,Q_2,Q_2)$	$(1,0,0,0,0,0,0,1)$
R_{13}^{++}	$\theta_{\mathrm{tt}}(G_{++-}, G_{-++}, G_{++-}, G_{-++})$	$(1,-1,-1,1,1,-1,-1,1)$
R_{13}^{--}	$\theta_{\mathrm{tt}}(G_{++-}, G_{---}, G_{+-+}, G_{-++})$	$(1,0,0,0,0,0,0,1)$

Again both endpoints are nonzero. All dressings are invertible, so the three certified ranks remain $(2,2,2)$.

It remains to verify the mixed rows directly in their fixed- I coordinates. For $\alpha \in \{\mathrm{dt}, \mathrm{td}\}$, absorb the three matching-dependent orientation factors in Equation (F.86) into the current- group representatives. The kernel $K_{*,\alpha}$ first gives the realizable card $\mathcal{L}_\alpha q_R$. On inputs $2,4$, apply the current-group dressing $\theta_\alpha(S_\alpha^{-1})$, where $S_\alpha = H_\alpha H_\alpha^{\mathsf{T}}$. Its product with the H_α^{T} leg of $\mathcal{L}_\alpha$ is H_α^{-1}, so the resulting direct card $q^\alpha_{R,\mathrm{act}}$ has the normalized representative

$$q_R^\alpha := (H_\alpha^{-1})^{\otimes 4} q_R, \qquad q^\alpha_{R,\mathrm{act}} \doteq q_R^\alpha. \tag{F.132}$$

These realizable dressings preserve the retained reduction data. When another dressing on inputs $2, 4$ is listed below, compose the two group elements into one chosen representative on that input. The total is at most four dressings, or two dressings and one pair contraction for an outside-binary row. All exact determinants below refer to this normalized representative. The corresponding realizable output differs by a nonzero scalar, so only determinant nonvanishing is used. Put

$$\delta_\alpha(B) := [c_Q(H_\alpha(BX)H_\alpha^{-1})].$$

Thus $\delta_\alpha(B) \notin \mathcal{R}_{\mathcal{O}^{(\mathrm{dd})}}$ is exactly the assertion $[BX] \notin \mathcal{O}^{(\alpha)}$.

For $\mathcal{O}^{(\mathrm{dt})}$, the three first-frame entries in the following table give endpoint-nondegenerate eight-vertex signatures. All odd coefficients vanish, and the last column is in the even-string order fixed above.

R	realizable dressing	$\mathbf{e}(\mathscr{D}q_R^{dt})$
R_0	$\theta_{\mathrm{dt}}(G_{--+})^{\otimes 4}$	$(1,-1,1,-1,-1,1,-1,1)$
R_1	$\theta_{\mathrm{dt}}(G_{---})^{\otimes 4}$	$(1,1,-1,-1,-1,-1,1,1)$
R_3	identity	$(1,1,-1,-1,-1,-1,1,1)$

The other six possible first entries have the uniform outside-binary exits listed below. The kernel acts on inputs $1, 2$.

R	realizable kernel	$\delta_{\mathrm{dt}}(B)$	$\det B$
R_{01}^{+-}	$\theta_{\mathrm{dt}}(G_{---})$	$[1,-1,1,0]$	$(51+36\sqrt{2})/64$
R_{01}^{-+}	$\theta_{\mathrm{dt}}(G_{---})$	$[1,-1,0,-1]$	$(51+36\sqrt{2})/64$
R_{03}^{+-}	$\theta_{\mathrm{dt}}(Q_0)$	$[1,-3,1,-1]$	$(51+36\sqrt{2})/256$
R_{03}^{-+}	$\theta_{\mathrm{dt}}(Q_0)$	$[1,-3,-1,1]$	$(51+36\sqrt{2})/256$
R_{13}^{++}	$\theta_{\mathrm{dt}}(Q_0)$	$[1,-3,1,1]$	$(51+36\sqrt{2})/256$
R_{13}^{--}	$\theta_{\mathrm{dt}}(Q_0)$	$[1,-3,-1,-1]$	$(51+36\sqrt{2})/256$

For $\mathcal{O}^{(\mathrm{td})}$, no dressing is needed for the two endpoint rows:

$$\mathbf{e}(q_{R_0}^{td}) = (1,i,-i,i,i,-i,i,-1),$$
$$\mathbf{e}(q_{R_3}^{td}) = (1,i,-i,-i,-i,-i,i,-1).$$

Both have zero odd coefficients and ranks $(2,2,2)$. The R_1 row is

$$q_{R_1}^{td} = \frac{3+2\sqrt{2}}{8}(\mathbf{1}[0011] + \mathbf{1}[1100]),$$

so realizable X-dressings on inputs $3, 4$ give $\frac{3+2\sqrt{2}}{8}(=_4)$, a generalized equality. Compose the dressing on input 4 with its preceding correction. Together with the corrections on inputs $2, 4$, this uses only three dressed inputs. The remaining rows have the following outside-binary exits on inputs $1, 2$:

R	realizable kernel	$\delta_{\mathrm{td}}(B)$	$\det B$
R_{01}^{+-}	$\theta_{\mathrm{td}}(Q_0)$	$[0,2,-1,1]$	$(51+36\sqrt{2})/256$
R_{01}^{-+}	$\theta_{\mathrm{td}}(Q_0)$	$[0,2,-1,-1]$	$(51+36\sqrt{2})/256$
R_{03}^{+-}	$\theta_{\mathrm{td}}(G_{--+})$	$[1,2,0,1]$	$(51+36\sqrt{2})/64$
R_{03}^{-+}	$\theta_{\mathrm{td}}(G_{--+})$	$[1,-2,0,-1]$	$(51+36\sqrt{2})/64$
R_{13}^{++}	$\theta_{\mathrm{td}}(Q_0)$	$[1,2,-1,0]$	$(51+36\sqrt{2})/256$
R_{13}^{--}	$\theta_{\mathrm{td}}(Q_0)$	$[1,-2,1,0]$	$(51+36\sqrt{2})/256$

Finally let q_π^α denote the six normalized mixed reconstructed rank-deficient outputs in the same input convention as Equation (F.132). Put

$$K_{\mathrm{dt}}^{\mathrm{def}} := \theta_{\mathrm{dt}}(Q_0 - Q_2), \qquad K_{\mathrm{td}}^{\mathrm{def}} := \theta_{\mathrm{td}}(Q_2 - Q_3),$$

and let $B_{\pi,\alpha}^{\mathrm{def}} := \partial_{12}^{K_\alpha^{\mathrm{def}}} q_\pi^\alpha$. The fixed chosen representative of $[K_\alpha^{\mathrm{def}}]$ on inputs $1, 2$ works uniformly for all six outputs in each row:

α	K_α^{def}	$\{\delta_\alpha(B_{\pi,\alpha}^{\mathrm{def}}) : \pi \in S_3\}$	$\det B_{\pi,\alpha}^{\mathrm{def}}$
dt	$\theta_{\mathrm{dt}}(Q_0 - Q_2)$	$\{[1,-1,0,-1]^2, [1,-3,1,-1]^2, [1,-3,-1,-1]^2\}$	$(51+36\sqrt{2})/256$
td	$\theta_{\mathrm{td}}(Q_2 - Q_3)$	$\{[0,2,1,1]^2, [1,2,0,1]^2, [1,2,1,0]^2\}$	$(51+36\sqrt{2})/256.$

Here $[a_0, a_1, a_2, a_3]$ denotes the projective quaternion-coordinate point $[(a_0, a_1, a_2, a_3)^{\mathsf{T}}]$, equivalently the matrix class $[a_0Q_0 + a_1Q_1 + a_2Q_2 + a_3Q_3]$ under c_Q, and an exponent records multiplicity. Every determinant displayed in the mixed tables is nonzero. Every pulled-back direction has support three, or support four with unequal nonzero magnitudes, whereas the quaternion-coordinate points of $\mathcal{R}_{\mathcal{O}^{(\mathrm{dd})}}$ have support one, two, or four with equal nonzero magnitudes. Hence all the listed transfer classes are outside their realizable mixed group. Direct row reduction gives rank pattern $(2, 2, 2)$ in every endpoint row. This proves all full-rank and rank-deficient mixed exits using only current-form kernels and dressings, while the similarities serve only to write the coefficient arrays. □

For the fixed-matching residue in the standard form, the four axes and eight tetrahedral sign points give the quadratic test set of Lemma F.9.6. In any nonstandard form, apply the same test after precomposing Φ with the corresponding one of $\theta_{\mathrm{tt}}, \theta_{\mathrm{dt}}, \theta_{\mathrm{td}}$, and then pull the resulting tensor identity back by its inverse. In all four forms, the evaluation matrix of the ten homogeneous quadratic monomials in the four quaternion coordinates has full column rank. Hence every pulled-back Segre minor vanishes identically, giving a genuine common factor rather than a new octahedral remaining case.

Proposition F.A.19 proves the domain and deck incidences by cyclic cosets, tensor factorization, and realizable group actions. Lemma F.A.15 proves the bridge intersections and reduces their sum equations to the displayed orthogonal frames.

F.A.5.2 The internal tetrahedral permutation classes

For the tetrahedral certificate, order the standard twelve points $\mathcal{T}^{(\mathrm{int})}$ in Equation (F.17) as the four axes followed by the eight sign points. The coset classification in Proposition F.A.19 gives 16 three-point lines in one orbit under realizable left and right multiplication by $\mathcal{T}^{(\mathrm{int})}$. The quaternary deck has $3 \cdot 12^2 = 432$ distinct projective points. Exactly 1728 of its three-point lines meet all three matching classes, again in one orbit under realizable local dressings and permutations of inputs. Thus the common normalization above applies.

Scale the normalized card map $\Phi : \mathrm{Mat}_2(\mathbb{C}) \to (\mathbb{C}^2)^{\otimes 4}$ so that

$$\Phi(R_+) = A, \qquad \Phi(R_-) = -B.$$

Then $\Phi(Q_0) = C/2$. Define the image tensors $W_j := \Phi(Q_j)$. Thus $W_1 + W_2 + W_3 = (A+B)/2$. For each $W \in \{W_1, W_2, W_3\}$, require

$$W,\ A - 2W,\ 2W - B \in \{0\} \cup \mathrm{cone}(\mathfrak{D}_4(\mathcal{T}^{(\mathrm{int})})).$$

The resulting exact bridge is

$$\mathcal{B}_{\mathcal{T}^{(\mathrm{int})}} = \{0, A/2, B/2, R_0, R_1, R_3\}, \tag{F.133}$$

where R_0, R_1, R_3 are defined in Equation (F.126). The remaining bridge vectors obey

W	$A-2W$	$2W-B$
R_0	$P_1(Q_2,Q_2)$	$-P_0(Q_2,Q_2)$
R_1	$-P_1(Q_3,Q_3)$	$P_0(Q_3,Q_3)$
R_3	$-P_1(Q_1,Q_1)$	$P_0(Q_1,Q_1)$

Lemma F.A.15 proves completeness of this list. The symmetric tetrahedral directions are exactly Q_0, Q_1, Q_3, so their three coordinate axes form the unique orthogonal frame.

The sum equation on $\mathcal{B}^3_{\mathcal{T}^{(\mathrm{int})}}$ leaves precisely twelve maps:

$$\begin{aligned} &6 \text{ maps with } \operatorname{rank}\Phi = 2: \quad (W_1,W_2,W_3) \text{ permutes } (0,A/2,B/2),\\ &6 \text{ maps with } \operatorname{rank}\Phi = 4: \quad (W_1,W_2,W_3) \text{ permutes } (R_0,R_1,R_3). \end{aligned} \tag{F.134}$$

For a permutation $\sigma=(a,b,c)$ of $(0,1,3)$, the Frobenius-dual basis is

$$Q^0 = Q_0/2, \quad Q^1 = -Q_1/2, \quad Q^2 = Q_2/2, \quad Q^3 = -Q_3/2,$$

and the uniquely reconstructed six-input ancestor is

$$f_\sigma = \sum_{\mu=0}^{3} (Q^\mu)^{(1,2)} \otimes F^\sigma_\mu, \qquad (F^\sigma_0, F^\sigma_1, F^\sigma_2, F^\sigma_3) = (C/2, R_a, R_b, R_c).$$

The duals in this formula are coefficient-reconstruction devices, not available gadgets.

Contract all fifteen input pairs by all twelve realizable tetrahedral kernels. The exact 1080-card histogram is

σ	zero or matching-deck cards	nonmatching cards
$(0,1,3),(1,3,0),(3,0,1)$, each	180	0
$(0,3,1),(1,0,3),(3,1,0)$, each	36	144

All 180 cards for each of the three orientations in the first data row are nonzero. For the three rejected orientations, the single contraction of inputs 1 and 3 by $Q_0 = I$ gives the transparent certificate

σ	support size	three flattening ranks
$(0,3,1)$	8	$(2,2,2)$
$(1,0,3)$	16	$(2,2,2)$
$(3,1,0)$	16	$(2,2,2)$

Normalize each card by its first nonzero entry, as stipulated for the projective enumerations above. In the three σ-rows of the preceding table, respectively, one nonzero 2×2 minor in each of the flattening orders $(12 \mid 34, 13 \mid 24, 14 \mid 23)$ is

$$(-1,-1,-2), \qquad (-2,2i,2i), \qquad (-2i,2,-2i).$$

Thus none of the three outputs is a hidden two-binary matching product.

Proposition F.A.17 (Tetrahedral permutation classes and eight-vertex signatures)**.** *Let the exceptional orientation set be*

$$\Sigma_- := \{(0,3,1),(1,0,3),(3,1,0)\}.$$

For each $\sigma \in \Sigma_-$, let q_σ be the quaternary obtained by bilinearly contracting inputs $1,3$ of f_σ with the realizable kernel $Q_0 = I$. Then q_σ admits realizable tetrahedral input dressings to an endpoint-nondegenerate eight-vertex signature. The three flattening ranks remain $(2,2,2)$.

Proof. Define the tetrahedral representative matrix $G_{\epsilon_1\epsilon_2\epsilon_3} := Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3$, and use the even-string order in Equation (F.93) on the residual inputs $(2, 4, 5, 6)$. Direct expansion gives

σ	$\mathbf{L}$	$\mathbf{e}(\mathscr{D}_{\mathbf{L}} q_\sigma)$
$(0,3,1)$	$(Q_0, G_{++-}, G_{+++}, Q_1)$	$(1,0,0,0,0,0,0,-i)$
$(1,0,3)$	(Q_0, Q_0, G_{+++}, Q_0)	$(1,1,-i,i,1,-1,i,i)$
$(3,1,0)$	$(Q_0, G_{+++}, Q_0, G_{++-})$	$(1,-1,1,1,-i,-i,i,-i)$

Every omitted odd entry is zero and both endpoints are nonzero. Each displayed dressing is realizable and invertible, so it preserves the certified flattening ranks. □

For the three surviving orientations, the following coefficient formulas prove membership in the same affine class. Let $f_{013}, f_{130}, f_{301}$ be the three surviving arity-six signatures and $q_{013}, q_{130}, q_{301}$ their quadratic phase polynomials, and put

$$\mathbf{x} = (x_1, \ldots, x_6) \in \mathbb{F}_2^6.$$

In the display below, the indicator constraints and their additions are over $\mathbb{F}_2$, the exponents q_σ are read modulo four, and the displayed prefactors are complex scalars:

$$\begin{aligned}
f_{013}(\mathbf{x}) &= -\frac{i}{4}\,\mathbf{1}[x_3 + x_4 + x_5 + x_6 = 0]\, i^{q_{013}(\mathbf{x})},\\
q_{013} &= 2x_3 + 3x_4 + x_5 + 2(x_1x_2 + x_1x_3 + x_1x_5 + x_2x_3 + x_2x_4 + x_4x_5),\\
f_{130}(\mathbf{x}) &= -\frac{i}{4}\,\mathbf{1}[x_3 + x_4 + x_5 + x_6 = 0]\, i^{q_{130}(\mathbf{x})},\\
q_{130} &= 3x_1 + x_2 + 3x_4 + x_5 + 2(x_1x_2 + x_1x_3 + x_1x_4 + x_2x_3 + x_2x_5 + x_3x_4 + x_3x_5),\\
f_{301}(\mathbf{x}) &= \frac{1-i}{4}\,\mathbf{1}_{\left[\substack{x_3+x_4+x_5+x_6=0,\\ x_1+x_2+x_4+x_5=1}\right]}\, i^{q_{301}(\mathbf{x})},\\
q_{301} &= 2 + x_1 + 2x_2 + x_5 + x_6 + 2(x_1x_2 + x_1x_3 + x_1x_4 + x_3x_4).
\end{aligned}$$

The displayed supports are affine and every mixed phase coefficient is even, so the three ancestors are affine. Separately, direct expansion of the twelve realizable binary representatives $C_{\ker}X$, $[C_{\ker}] \in \mathcal{T}^{(\mathrm{int})}$, verifies that they are affine in this same normalized basis.

F.A.5.3 The icosahedral forms $\mathcal{I}^{(0)}$ and $\mathcal{I}^{(1)}$

For the icosahedral certificate, use the standard set $\mathcal{I}^{(0)}$ in Equations (F.16) and (F.19), ordered as the four axes, the eight tetrahedral sign points, and the 48 distinct even-permutation points. Exact arithmetic over $\mathbb{Q}(\phi, i)$ gives

$$|\mathcal{I}^{(0)}| = 60, \qquad \#\{g : \mathrm{ord}(g) = 1, 2, 3, 5\} = (1, 15, 20, 24).$$

The set is closed under multiplication, inverse, and transpose. It is the projectivization of the binary icosahedral group, so $\mathcal{I}^{(0)} \cong A_5$. The apparent opposite chirality gives no additional distinguished-involution case. If $\mathcal{I}^{(1)}$ is obtained by using odd coordinate permutations in the additional part of (F.19), then, for $D = Q_0 + Q_3$,

$$D^{\mathsf{T}} X D = 2X, \qquad \mathcal{I}^{(1)} = D\mathcal{I}^{(0)}D^{-1}, \qquad (DQ_1D^{-1}, DQ_2D^{-1}, DQ_3D^{-1}) = (-Q_2, Q_1, Q_3).$$

Lemma F.9.3 uses this identity to translate kernel and card formulas while preserving the retained fixed-I problem.

The maximal-cyclic-coset classification in Proposition F.A.19 gives 200 three-point C_3-coset lines and 72 five-point C_5-coset lines. Every point is on ten three-lines and six five-lines. The quaternary matching-product deck has $3 \cdot 60^2 = 10\,800$ distinct projective points, and its complete maximal-rich-line histogram is

line size	matching-label type	number
3	$(1,1,1)$	216 000
3	(3)	72 000
5	(5)	25 920

There is no $(2,1)$ line and no mixed five-line.

Normalize a mixed three-point incidence by (F.125), and scale its card map so that $\Phi(R_+) = A$ and $\Phi(R_-) = -B$. Define the image tensors $W_j := \Phi(Q_j)$. Then

$$\Phi(Q_0) = C/2, \qquad W_1 + W_2 + W_3 = (A+B)/2.$$

The other six tetrahedral sign points require each of $W_j, A-2W_j, 2W_j-B$ to lie in the zero/matching-product cone

$$\mathscr{C}_{\mathcal{I}^{(0)}} := \{0\} \cup \{\lambda P_m(U,V) : \lambda \in \mathbb{C}^\times,\ m \in \{0,1,2\},\ [U],[V] \in \mathcal{I}^{(0)}\}.$$

Define the following 15-point slice in coefficient projective space:

$$\begin{aligned} \mathcal{H} &= \{[a:b:d] \in \mathbb{P}^2 : [aQ_0 + bQ_1 + dQ_3] \in \mathcal{I}^{(0)}\}, & |\mathcal{H}| &= 15, \\ J_{\mathbf{u}} &= aQ_0 + bQ_1 + dQ_3, & \operatorname{adj}(J_{\mathbf{u}}) &= aQ_0 - bQ_1 - dQ_3, \\ R_{\mathbf{u}} &= \frac{P_2(J_{\mathbf{u}}, \operatorname{adj}(J_{\mathbf{u}}))}{2(a^2+b^2+d^2)}. \end{aligned}$$

Here $\mathbf{u} = (a,b,d)$ is any nonzero representative of $[a:b:d] \in \mathcal{H}$. The displayed quotient is unchanged when $\mathbf{u}$ is rescaled, so $R_{\mathbf{u}}$ is well defined by the projective coefficient point. Moreover, $a^2+b^2+d^2 = \det J_{\mathbf{u}} \neq 0$: the equality follows directly from the quaternion matrices in Equation (F.16), and nonvanishing follows from $[J_{\mathbf{u}}] \in \mathcal{I}^{(0)} \subset \mathrm{PGL}_2(\mathbb{C})$. The exact bridge intersection is the eighteen-point set

$$\mathcal{B}_{\mathcal{I}^{(0)}} := \{W : W, A-2W, 2W-B \in \mathscr{C}_{\mathcal{I}^{(0)}}\} = \{0, A/2, B/2\} \cup \{R_{\mathbf{u}} : [\mathbf{u}] \in \mathcal{H}\}.$$

Lemma F.A.15 proves this equality. The symmetric icosahedral directions are precisely the three axes with no Q_2 component and the twelve additional points in that coordinate plane. For such a direction, $J_{\mathbf{u}}^{-1} = \operatorname{adj}(J_{\mathbf{u}})/(a^2+b^2+d^2)$, which gives the displayed formula for $R_{\mathbf{u}}$. Thus the list includes every complex scale and every cancellation case. As a finite completeness check, group the other 10,799 deck points by their line through $[A]$. Counting points other than $[A]$ on each line gives the histogram

$$1^{10\,591}, \qquad 2^{80}, \qquad 4^{12}.$$

Here a^b means that b lines each contain a other deck points.

The sum equation on $\mathcal{B}^3_{\mathcal{I}^{(0)}}$ has exactly 36 ordered solutions. Six permute $(0, A/2, B/2)$. The other thirty are the six orderings of the following five unordered frames, where a vector $\mathbf{u}$ represents

the tensor $R_{\mathbf{u}}$:

$$\begin{aligned}
\mathrm{Fr}_0 &= \{(0,0,1),(0,1,0),(1,0,0)\},\\
\mathrm{Fr}_1 &= \{(1,-\tau^2,\tau),(1,-\phi,-\phi^2),(1,\phi,-\tau)\},\\
\mathrm{Fr}_2 &= \{(1,-\tau^2,-\tau),(1,-\phi,\phi^2),(1,\phi,\tau)\},\\
\mathrm{Fr}_3 &= \{(1,-\phi,\tau),(1,\phi,\phi^2),(1,\tau^2,-\tau)\},\\
\mathrm{Fr}_4 &= \{(1,-\phi,-\tau),(1,\phi,-\phi^2),(1,\tau^2,\tau)\}.
\end{aligned} \tag{F.135}$$

Lemma F.A.15 reduces the nondegenerate sum solutions to orthogonal triples in $\mathcal{H}$. The five displayed frames contain the fifteen points of $\mathcal{H}$ once each. Using $\phi^2 = \phi + 1$, their dot products show that each point is orthogonal to precisely the other two points in its displayed frame. These are therefore all orthogonal triples. This proves the $6 + 5 \cdot 6 = 36$ solutions without a search through $\mathcal{B}^3_{\mathcal{I}^{(0)}}$.

One legal group element suffices for all thirty orderings:

$$K_* = \phi Q_1 - \tau Q_2 + Q_3, \qquad [K_*] \in \mathcal{I}^{(0)}.$$

For an ordering $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$ of any frame, its direct card is

$$q_{\mathbf{u}_1,\mathbf{u}_2,\mathbf{u}_3} = \phi R_{\mathbf{u}_1} - \tau R_{\mathbf{u}_2} + R_{\mathbf{u}_3}. \tag{F.136}$$

Exact row reduction of the thirty explicit tensors gives, uniformly,

$$(\mathrm{rank}_{12|34}, \mathrm{rank}_{13|24}, \mathrm{rank}_{14|23}) = (3,3,3), \tag{F.137}$$

with support-size histogram $8^6, 12^4, 16^{20}$. Each output is therefore a direct nonmatching quaternary card.

For the no-mixed-line branch, first discard rank at most one, which is already rank lowering. On the remaining branch $\dim \ker \Phi \le 2$. Along every rich line avoiding $\mathbb{P}(\ker \Phi)$, the absence of a mixed line forces all images to use one matching. The kernel-deleted connectedness assertion Equation (F.139) propagates this matching to all nonzero card images among the sixty domain points, placing them in the fixed-matching Segre variety: the projective simple-tensor locus Σ_M of Lemma F.9.6, with the two matching edges as its factors. Because the icosahedral domain contains the test set in Equation (F.82), Lemma F.9.6 then gives, after a permutation of inputs,

$$\Phi(C_{\ker}) = U_* \otimes \psi(C_{\ker}) \quad \text{for all } C_{\ker}, \qquad \text{or} \qquad \Phi(C_{\ker}) = \psi(C_{\ker}) \otimes U_*. \tag{F.138}$$

Here U_* is a fixed nonzero binary factor and $\psi : \mathrm{Mat}_2(\mathbb{C}) \to \mathrm{Mat}_2(\mathbb{C})$ is a binary-valued linear map. Combining the rank-two solutions, (F.137), and (F.138) gives the bounded-interface trichotomy: rank lowering, a realizable nonmatching quaternary, or a genuine common factor.

The rejecting kernel and rank assertions are evaluated on the thirty displayed outputs over $\mathbb{Q}(\phi, i)$.

F.A.5.4 The complete Platonic incidence atlas

Lemma F.A.18 (Realizable rainbow-line orbit certificate)**.** *For a marked group G, let Γ_G be the finite permutation group of the projective deck induced by the realizable local operations*

$$\mathscr{L}_{j,g} : [F] \longmapsto [(I^{\otimes(j-1)} \otimes K_g \otimes I^{\otimes(4-j)})F] \quad (g \in G,\ 1 \le j \le 4), \qquad (12), (23), (34),$$

where the first operations mean the attachment of a realizable binary $D_g \doteq XK_g^{\mathsf{T}}$ in the orientation with effective leg map $D_g^{\mathsf{T}}X \doteq K_g$, which induces the displayed projective action on that leg and records its nonzero representative scalar. The last three are permutations of inputs. Let the standard representative rainbow line be

$$\ell_0^{\mathrm{std}} = \{[P_0(Q_0, Q_0)], [P_1(Q_0, Q_0)], [P_2(Q_2, Q_2)]\}.$$

Write $\mathscr{S}_G$ for the displayed finite generating set (all $\mathscr{L}_{j,g}$ together with the three adjacent swaps), so that $\Gamma_G = \langle \mathscr{S}_G \rangle$. Here K_g is the fixed chosen representative of g. For the second octahedral form, put $\theta := \theta_{\mathrm{tt}}$. The coefficient operator sends the standard representative to the line

$$\ell_0^{\mathcal{O}^{(\mathrm{tt})}} := \mathcal{L}_{\mathrm{tt}}(\ell_0^{\mathrm{std}}) = \{[P_0'(\theta(Q_0), \theta(Q_0))], [P_1'(\theta(Q_1), \theta(Q_1))], [P_2'(\theta(Q_2), \theta(Q_2))]\}.$$

Set $\ell_0^G := \ell_0^{\mathrm{std}}$ for $G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{I}^{(0)}\}$. The orbit of ℓ_0^G under these realizable operations has the following size:

marked form	*deck points*	*all rainbow three-lines*	*orbit size*
$\mathcal{T}^{(\mathrm{int})}$	*432*	*1 728*	*1 728*
$\mathcal{O}^{(\mathrm{dd})}$	*1 728*	*13 824*	*13 824*
$\mathcal{O}^{(\mathrm{tt})}$	*1 728*	*13 824*	*13 824*
$\mathcal{I}^{(0)}$	*10 800*	*216 000*	*216 000*
$\mathcal{T}^{(\mathrm{ext})}$	*432*	*0*	*0*

For $\mathcal{O}^{(\mathrm{tt})}$, the representative in the orbit column is $\ell_0^{\mathcal{O}^{(\mathrm{tt})}}$. For the other rows with a nonzero rainbow count it is ℓ_0^{std}. Consequently every rainbow line in the first four rows is sent to that row's representative by a sequence of realizable local dressings and permutations of inputs. The corresponding scalar and ordered permutation of inputs are part of the finite witness. For $\alpha \in \{\mathrm{dt}, \mathrm{td}\}$, define the initial mixed line as the $\mathcal{L}_\alpha$-image of ℓ_0^{std}. The three identities in Equation (F.86) *intertwine every standard orbit generator with the corresponding realizable $\mathcal{O}^{(\alpha)}$-representative and preserve its nonzero scalar and input order. Hence each mixed form has the same* 1 728 *deck points and* 13 824 *rainbow three-lines, in one realizable local-dressing orbit. These two rows are omitted from the table only because their three numerical entries coincide with the two displayed octahedral rows.*

Proof. First let G contain $[X]$ and be reversal-stable, so it is closed under inverse transpose. Every binary factor of a G-matching tensor has its matrix class in G, because its transfer does and $[X] \in G$. Consider any rainbow line. Realizable local dressings send its matching-0 point to $A = P_0(I, I)$. Write its matching-1 point as $P_1(U, V)$. The further dressing $(U^{-1}, U^{\mathsf{T}}, I, I)$ fixes A and sends that point to $P_1(I, W)$, where $W = U^{\mathsf{T}}V$.

The third point has rank one across $14 \mid 23$. Since the three points are distinct, both coefficients of its linear combination of A and $P_1(I, W)$ are nonzero. Equation (F.127) therefore forces W to be scalar, and the line is $\{[A], [B], [C]\}$, with $C = A - B = P_2(Y, Y)$. Thus every rainbow line has a realizable local normalization to this line. It exists exactly when $[YX] = [Z] \in G$. In particular, the external form has none by Lemma F.9.2.

For $G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{I}^{(0)}\}$, the local action of G^4 preserves the three matching labels. A stabilizer of the normalized line therefore fixes its three points, and the first two give

$$T_2 = T_1^{-\mathsf{T}}, \qquad T_3 = T_1^{-\mathsf{T}}, \qquad T_4 = T_1$$

projectively. Conversely, $(T, T^{-\mathsf{T}}, T^{-\mathsf{T}}, T)$ fixes all three, using $TYT^{\mathsf{T}} = \det(T)Y$. The stabilizer has $|G|$ elements, so the complete rainbow orbit has $|G|^3 = 1728, 13824, 216000$ elements, respectively.

This establishes both transitivity and the counts directly. For $\mathcal{O}^{(\mathrm{tt})}$, the ordered transport formulas Equation (F.130) give a bijection under $\mathcal{L}_{\mathrm{tt}} = \mathscr{D}_{\mathbf{H}_{\mathrm{tt}}}$ that preserves matching labels and rich-line incidence. This map also conjugates the local group $(\mathcal{O}^{(\mathrm{dd})})^4$ to $(\mathcal{O}^{(\mathrm{tt})})^4$: on an H^{-1} leg this is $U \mapsto H^{-1}UH$, while on an H^{T} leg it is

$$H^{\mathsf{T}}UH^{-\mathsf{T}} = H^{-1}(SUS^{-1})H, \qquad S = HH^{\mathsf{T}}, \quad [S] \in \mathcal{O}^{(\mathrm{dd})}.$$

The same argument applies to $H_{\mathrm{dt}}, H_{\mathrm{td}}$, with their already verified S-matrices. Hence all three nonstandard octahedral forms have the same transitive local orbit, starting at their displayed $\mathcal{L}$-image of the standard line. The operator $\mathcal{L}$ only proves this correspondence. Each conjugated leg matrix belongs to the current group. For $\mathcal{T}^{(\mathrm{ext})}$, the preceding rank-one argument and $[Z] \notin \mathcal{T}^{(\mathrm{ext})}$ exclude every rainbow line.

To obtain a normalization witness, choose coset representatives as follows. In the standard forms, right multiplication by a stabilizing tuple gives each local coset a unique representative (I, T_2, T_3, T_4). Enumerate these $|G|^3$ tuples and their images. Each rainbow line occurs once. The nonstandard correspondence gives the same enumeration in its current-group representatives. Applying the inverse recorded tuple sends any selected line to the prescribed one. The input order and all nonzero representative scalars are obtained by these exact matrix products. □

Proposition F.A.19 (Exact Platonic incidence atlas)**.** *For the standard tetrahedral, first octahedral, external tetrahedral, and icosahedral marked configurations, the domain and quaternary matching decks have the following complete maximal-rich-line data.*

configuration	*domain points and lines*	*deck points*	*deck lines by matching-label type*
$\mathcal{T}^{(\mathrm{int})}$	$12;\ 16_{[3]}$	432	$1728(1,1,1),\ 1152(3)$
$\mathcal{O}^{(\mathrm{dd})}$	$24;\ 32_{[3]}, 18_{[4]}$	1728	$13824(1,1,1),\ 4608(3),\ 2592(4)$
$\mathcal{T}^{(\mathrm{ext})}$	$12;\ 16_{[3]}$	432	$1152(3)$
$\mathcal{I}^{(0)}$	$60;\ 200_{[3]}, 72_{[5]}$	10800	$216000(1,1,1),\ 72000(3),\ 25920(5)$

In the second column the entry before the semicolon is the number of domain points, and $n_{[s]}$ denotes n rich lines of size s. A parenthesized partition in the last column is the matching-label type. Thus the standard tetrahedral deck has only rainbow or monochromatic three-lines. The external tetrahedral deck has only monochromatic three-lines. The octahedral and icosahedral decks have no type $(2,1)$ line and no mixed four- or five-line, respectively. The octahedral row also gives the abstract incidence of $\mathcal{O}^{(\mathrm{tt})} = H_{\mathrm{tt}}^{-1}\mathcal{O}^{(\mathrm{dd})}H_{\mathrm{tt}}$ under the coefficient transport in Equation (F.129)*, where $H_{\mathrm{tt}} \notin GO(X)$. The bridge kernels and input dressings in the second marked form use the realizable $\mathcal{O}^{(\mathrm{tt})}$-representatives specified below. Explicitly, the coefficient operator $\mathcal{L}_{\mathrm{tt}} = \mathscr{D}_{\mathbf{H}_{\mathrm{tt}}}$ satisfies $\mathcal{L}_{\mathrm{tt}}P_M(U,V) \doteq P'_M(U'_M, V'_M)$. Writing $\theta = \theta_{\mathrm{tt}}$ and $X = Q_1$ projectively, the three explicit instances are*

$$\begin{aligned}
\mathcal{L}_{\mathrm{tt}}P_0(U,V) &\doteq P'_0(\theta(U), \theta(V)),\\
\mathcal{L}_{\mathrm{tt}}P_1(U,V) &\doteq P'_1(\theta(UX), \theta(XV)),\\
\mathcal{L}_{\mathrm{tt}}P_2(U,V) &\doteq P'_2(\theta(U), \theta(XVX)).
\end{aligned}$$

The X's are the fixed orientation factors from the ordered $H_{\mathrm{tt}}^{-1}/H_{\mathrm{tt}}^{\mathsf{T}}$ pair contractions, and each displayed matrix lies in the current $\mathcal{O}^{(\mathrm{tt})}$-representatives. Thus $\mathcal{L}_{\mathrm{tt}}$ preserves matching labels and rich-line incidence. The realizable $\mathcal{O}^{(\mathrm{tt})}$ input witnesses are supplied by Lemma F.A.18 and the separate bridge tables.

For $\mathcal{O}^{(\mathrm{dt})}$ and $\mathcal{O}^{(\mathrm{td})}$, the direct transport Equation (F.86) *likewise preserves every matching label, rich-line incidence, and kernel-deleted adjacency, while intertwining the generators with chosen*

representatives of the current mixed form. Thus both mixed rows have the octahedral counts and connectivity in the table. The coefficient correspondence transports the realizable kernels and dressings for the gadget constructions.

For $\mathcal{T}^{(\mathrm{int})}$, each domain point lies on four three-lines. For $\mathcal{O}^{(\mathrm{dd})}$, each lies on seven rich lines. For $\mathcal{I}^{(0)}$, each lies on ten three-lines and six five-lines. In each standard marked form, every displayed domain-line family is a realizable left/right group orbit. The intersection graphs of rich domain lines—vertices are rich lines and adjacency means nonempty intersection—are connected. Their edge counts for $\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{I}^{(0)}$ are respectively

$$72, \qquad 504, \qquad 7200.$$

For the localization interface, define the projective domain set

$$\mathcal{D}_G = \{[K_g] : g \in G\}$$

and join two distinct domain points when they lie on a common rich line. For every complex subspace $W \leq \mathrm{Mat}_2$ with $\dim W \leq 2$, the induced graph on

$$\mathcal{D}_G \setminus \mathbb{P}(W) \tag{F.139}$$

is nonempty and connected. The distinct deletion sets tested for $\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{I}^{(0)}$ number, respectively,

$$\begin{aligned} 47 &= 1 + 12 + (18 + 16), \\ 147 &= 1 + 24 + (72 + 32 + 18), \\ 783 &= 1 + 60 + (450 + 200 + 72). \end{aligned} \tag{F.140}$$

Here the parenthesized summands count the distinct two-point spans by their numbers of domain points: 18 of size two and 16 of size three for $\mathcal{T}^{(\mathrm{int})}$; $72, 32, 18$ of sizes two, three, four for $\mathcal{O}^{(\mathrm{dd})}$; and $450, 200, 72$ of sizes two, three, five for $\mathcal{I}^{(0)}$. The same assertion for $\mathcal{O}^{(\mathrm{tt})}$ and $\mathcal{T}^{(\mathrm{ext})}$ is transported by their respective invertible coefficient similarities. Each $\mathcal{O}^{(\mathrm{tt})}$ rainbow line used below has a separately checked normalization witness from realizable $(\mathcal{O}^{(\mathrm{tt})})^4 \rtimes S_4$ input operations.

Finally, the realizable kernel representatives span Mat_2. One may take

$$\begin{aligned} \mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{I}^{(0)} &: Q_0, Q_1, Q_2, Q_3, \\ \mathcal{O}^{(\mathrm{tt})} &: \theta_{\mathrm{tt}}(Q_0), \theta_{\mathrm{tt}}(Q_1), \theta_{\mathrm{tt}}(Q_2), \theta_{\mathrm{tt}}(Q_3), \\ \mathcal{T}^{(\mathrm{ext})} &: Q_0, Q_1, Q_2 + Q_3, Q_2 - Q_3. \end{aligned} \tag{F.141}$$

The tetrahedral incidence counts, and the rainbow deck-line normalizations used in the octahedral and icosahedral bridges, are realized by independent input dressings and permutations of inputs. For any finite configuration in $\mathbb{P}(\mathscr{V})$, and distinct points $[F_0], [F]$, define the quotient direction of $[F]$ at $[F_0]$ *to be*

$$[F + \mathrm{span}_{\mathbb{C}}\{F_0\}] \in \mathbb{P}(\mathscr{V} / \mathrm{span}_{\mathbb{C}}\{F_0\}),$$

using arbitrary nonzero representatives. This class is independent of those representatives, and two points have the same quotient direction exactly when they lie with $[F_0]$ on one projective line. For $\mathcal{T}^{(\mathrm{ext})}$, realizable external-group input actions are transitive on the 432 deck points and permute the three matching labels. At $P_0(Q_0, Q_0)$ the quotient-direction histogram is

$$1^{415}, \qquad 2^8,$$

and all eight rich lines there are monochromatic. No use of the analytic twist R is made in this deck transitivity assertion.

For completeness, at the normalized octahedral and icosahedral deck points the exact quotient-direction histograms are, respectively,

$$1^{1645},\ 2^{32},\ 3^{6}, \qquad 1^{10591},\ 2^{80},\ 4^{12}.$$

The first gives all octahedral rich-line types through the base point. The second gives all icosahedral rich-line types through it. Hence the global counts in the table follow by realizable transitivity and incidence double counting, not by assuming an abstract group action.

Proof. A secant line of a point configuration contains at least two of its points.

We first classify every domain secant. Let ℓ contain distinct $[g], [h] \in G$. Left multiplication by g^{-1} sends it to $\mathbb{P}(\mathrm{span}\{I, T\})$, where $[T] = [g^{-1}h] \neq 1$. A nonidentity finite-order projectivity has two distinct eigenlines, so T is diagonal in their basis. Every matrix in this two-dimensional span is diagonal there. The nonsingular group points on the line form $C = G \cap \mathbb{T}$, where $\mathbb{T}$ is that diagonal torus. A finite subgroup of $\mathbb{C}^\times$ is cyclic, so C is cyclic. It is maximal cyclic in G: any cyclic extension containing its nonidentity element $[T]$ has the same eigenlines. Conversely, every maximal cyclic subgroup lies on its eigenline torus's projective line. Thus the domain secants are exactly the left cosets of maximal cyclic subgroups, with line size equal to the subgroup order.

The permutation cycle types of A_4, S_4, A_5 give the following numbers of maximal cyclic subgroups:

	$\lvert C\rvert = 2$	$\lvert C\rvert = 3$	$\lvert C\rvert = 4$	$\lvert C\rvert = 5$
A_4	3	4	0	0
S_4	6	4	3	0
A_5	15	10	0	6.

For example, the three double transpositions of S_4 are squares of its three cyclic order-four subgroups. Only the six transpositions contribute maximal order-two subgroups. The order-three and order-five counts divide their numbers of generators by two and four, respectively. A subgroup of order d has $|G|/d$ left cosets, and distinct maximal cyclic subgroups give distinct coset sets. Hence the complete secant counts are

$$A_4 : (18, 16), \qquad S_4 : (72, 32, 18), \qquad A_5 : (450, 200, 72),$$

ordered by the nonzero subgroup-order columns above. Each point belongs to one coset of each subgroup. This also proves every stated number of rich lines through a point. Conjugation in G is transitive on the maximal cyclic subgroups of each displayed order, and left multiplication moves between their cosets, proving the asserted realizable left/right line orbits.

The deck-line classification then follows structurally. A line containing three matching products either has one matching or has all three: repetition of one matching together with another would give flattening ranks $1, 1, 4$, contradicting collinearity. In the latter case, Lemmas F.9.10 and F.A.18 gives exactly three points and the stated $|G|^3$ count in each standard form. For a line with one matching, the simple-tensor sum criterion in Lemma F.9.6 implies that exactly one binary factor varies. For each domain line there are three matchings, two choices of varying factor, and $|G|$ fixed factors. Thus each monochromatic line count is $6|G|$ times its domain-line count. This proves the complete deck table from the domain table.

We next prove kernel-deleted connectivity. The domain-point graph has

$$(n, \delta, s) = (12, 8, 3),\ (24, 17, 4),\ (60, 44, 5),$$

where n is its number of vertices, δ its degree, and s the maximum line size. The degree is the sum of $t-1$ over the rich lines of size t through a point: respectively $4\cdot 2$, $4\cdot 2+3\cdot 3$, and $10\cdot 2+6\cdot 4$. A subspace of dimension at most two removes $k\le s$ points. The remaining graph has $n-k>0$ vertices and minimum degree at least $\delta-k$. If it were disconnected, two components would each have at least $\delta-k+1$ vertices. But every displayed triple satisfies

$$2(\delta-k+1)>n-k \qquad (0\le k\le s),$$

a contradiction. This proves Equation (F.139) without graph traversal. The deletion counts follow from the domain secant counts by adding the empty and singleton deletions, as displayed in Equation (F.140). Every complex subspace is covered: an intersection with at least two domain points spans its two-dimensional vector space. Smaller intersections are empty or singletons. The rich-line intersection graphs are also connected: a path in the connected domain-point graph gives a sequence of rich lines with consecutive nonempty intersections. Their edge counts $72, 504, 7200$ follow by summing $\binom{d_v}{2}$ over domain points, where $d_v=4,7,16$, respectively. Distinct projective lines meet in at most one point, so this counts every edge once.

The coefficient transports preserve these domain and deck incidences. Lemma F.A.18 supplies realizable local normalization matrices. The spanning assertions in Equation (F.141) follow from the displayed four independent matrices in each row.

For the external form, the coefficient similarity θ_{ext} transports the twelve domain points and their sixteen three-lines. Its deck has no rainbow line by Lemma F.A.18. The monochromatic count is therefore $6\cdot 12\cdot 16=1152$. Realizable independent input actions are transitive on each matching class, and permutations of inputs exchange the three classes, so the full action is transitive on the 432 deck points. At each point, the two factor choices and four domain lines through each factor give eight rich three-lines. These account for 16 of the other 431 points, proving the quotient-direction histogram $1^{415},2^8$.

The octahedral and icosahedral quotient-direction histograms follow in the same way. Rainbow transitivity gives $|G|$ rainbow lines through each deck point, while domain incidences give the monochromatic lines. The remaining points each determine a separate two-point line. The programs `certificates/verify_a4_s4_certificates.py`, `certificates/verify_a4_external_form.py`, `certificates/verify_a5_certificate.py`, and `certificates/verify_platonic_extended.py` also reproduce the complete finite data. The coset, rank-one, and degree arguments above prove this proposition independently of those computations. □

Theorem F.A.20 (Incidence of Platonic kernel lines)**.** *For the marked Platonic domain and quaternary matching decks, the following statements hold.*

(i) *The four entries in each row below count, in order, rainbow three-lines and monochromatic lines of sizes three, four, and five:*

$\mathcal{T}^{(\mathrm{int})}$	$(1728,1152,0,0)$
$\mathcal{O}^{(\mathrm{dd})},\mathcal{O}^{(\mathrm{tt})},\mathcal{O}^{(\mathrm{dt})},\mathcal{O}^{(\mathrm{td})}$	$(13824,4608,2592,0)$
$\mathcal{T}^{(\mathrm{ext})}$	$(0,1152,0,0)$
$\mathcal{I}^{(0)},\mathcal{I}^{(1)}$	$(216000,72000,0,25920)$.

In particular, no deck has a line of type $(2,1)$, no octahedral deck has a mixed four-line, neither icosahedral deck has a mixed five-line, and the external tetrahedral deck has no rainbow line.

(ii) *Every rainbow line for $G\in\{\mathcal{T}^{(\mathrm{int})},\mathcal{O}^{(\mathrm{dd})},\mathcal{O}^{(\mathrm{tt})},\mathcal{O}^{(\mathrm{dt})},\mathcal{O}^{(\mathrm{td})},\mathcal{I}^{(0)},\mathcal{I}^{(1)}\}$ lies in one orbit under realizable local G-dressings and permutations of inputs. A witness includes the ordered permutation of inputs and its nonzero realization scalar.*

(iii) *Let $\mathcal{D}_G$ be the projective domain set for any of the marked rows above. For every complex subspace $W \le \mathrm{Mat}_2(\mathbb{C})$ with $\dim W \le 2$, the rich-line domain graph induced on $\mathcal{D}_G \setminus \mathbb{P}(W)$ is nonempty and connected.*

Proof. The complete line counts and kernel-deleted connectivity are Proposition F.A.19. Realizable rainbow transitivity, including the separate $\mathcal{O}^{(\mathrm{tt})}$ witnesses, is Lemma F.A.18. For the mixed rows, Equation (F.86) gives the same counts and intertwines the standard orbit generator-by-generator with current-form matrices. The fixed-equality coefficient correspondence Lemma F.9.3 gives both icosahedral statements in $\mathcal{I}^{(1)}$, with realizable κ_D-kernels and nonzero scalars, without changing the I. □

Remark F.A.21 (Computational completeness and gadget interpretation). The finite preserver and hereditary eight-input completeness claims retain an exact-computation component over the stated number fields. The bridge classifications have the structural proof in Lemma F.A.15. The incidence classification and deletion connectivity have the structural proof in Proposition F.A.19. The Python verification programs regenerate the configurations and test the stated predicates. The main lemmas apply these classifications using realizable kernels and dressings, factor exposure, and reductions of the retained Holant problem. The mixed-octahedral exits are given by the explicit coefficient and contraction tables in Proposition F.A.16.

F.A.5.5 Tetrahedral arity-eight signatures with pair cards in $\mathscr{C}_6$

Proposition F.A.22 (Tetrahedral eight-input classification)**.** *Let $\mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})}), \mathscr{H}_6, \mathscr{U}_6$, and $\mathscr{C}_6$ be the six-input sets defined before Proposition F.9.19. Let the base matching-product signature be M_0 and the base signature be H_0:*

$$M_0 = (Q_0)^{(1,2)}(Q_0)^{(3,4)}(Q_0)^{(5,6)}, \qquad H_0 = \mathrm{H}_6.$$

Then

$$|\mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})})| = 25920, \qquad |\mathscr{H}_6| = 31104, \qquad |\mathscr{U}_6| = 57024, \tag{F.142}$$

and the union defining $\mathscr{U}_6$ is disjoint. Through either $[M_0]$ or $[H_0]$, the quotient-direction histogram of the other points of $\mathscr{U}_6$ is

$$1^{56855}, \qquad 2^{84}. \tag{F.143}$$

Here two points $[F], [F'] \in \mathscr{U}_6 \setminus \{[F_0]\}$ have the same quotient direction at $[F_0]$ when $[F + \mathrm{span}_{\mathbb{C}}\{F_0\}] = [F' + \mathrm{span}_{\mathbb{C}}\{F_0\}]$ in the projectivized quotient. Thus the histogram means that 56,855 directions contain one other point of $\mathscr{U}_6$ and 84 directions contain two.

Of the 84 rich lines through $[M_0]$, 48 have both other points in $\mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})})$ and 36 have both in $\mathscr{H}_6$. Their orbit sizes are $36, 36, 12$ under the realizable stabilizer subgroup specified below. Of those through $[H_0]$, 60 have one other point in each set and 24 have both in $\mathscr{H}_6$. The orbit sizes are $24, 12, 8, 12, 12, 4, 12$ under its specified realizable stabilizer subgroup.

For each of the ten resulting line-orbit representatives and either endpoint orientation, denote the oriented endpoint tensors by A, D and the base signature by F_0, scaled so that $A + D = 2F_0$. Define the bridge set

$$\mathcal{B}(A, D) = \{W : W,\ A - 2W,\ D + 2W \in \mathscr{C}_6\}. \tag{F.144}$$

Then $|\mathcal{B}(A, D)| = 6$, and exactly twelve ordered triples in $\mathcal{B}(A, D)^3$ sum to $(A - D)/2$. The resulting $10 \cdot 2 \cdot 12 = 240$ reconstructed eight-input parents are projectively distinct. For a reconstructed parent g, let

$$\Phi_g : \mathrm{Mat}_2(\mathbb{C}) \longrightarrow (\mathbb{C}^2)^{\otimes 6}, \qquad \Phi_g(C) := \Phi_{12}^g(C),$$

be its distinguished-pair card map. Testing the $\binom{8}{2} \cdot 12 = 336$ *realizable tetrahedral cards in each of the reconstructed parents gives*

$\operatorname{rank} \Phi_g$	*outcome*	*number*
2	*a realizable card outside* $\mathscr{C}_6$	120
3	*no parent*	0
4	*a realizable card outside* $\mathscr{C}_6$	60
4	*all* 336 *cards in* $\mathscr{C}_6$	60.

Every one of the final sixty parents is affine in the normalized computational basis. Twenty-four also have an exact nontrivial rank-one flattening. For each of the other 180 *parents there is a specified input pair and realizable tetrahedral kernel producing a card outside* $\mathscr{C}_6$*.*

Proof. The sets $\mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})})$ and $\mathscr{H}_6$ are generated over $\mathbb{Q}(i)$: matching products are canonicalized directly, while the H_6-orbit is generated to closure under realizable local tetrahedral matrices and permutations of inputs. Exact projective comparison gives Equations (F.142) and (F.143). The subgroups used for this audit are generated by the following realizable operations. At M_0, use the pair stabilizers $(U, U^{-\mathsf{T}})$, $U \in \mathcal{T}^{(\mathrm{int})}$, and the permutations preserving its three-pair matching. At H_0, first enumerate the local two-input tetrahedral actions that permute the four projective Bell signatures. Retain the triples of such pair-block actions that fix $[H_0]$, and add all residual-permutations of inputs that directly fix $[H_0]$. Every selected generator fixes its base tensor projectively. Close their induced permutations on the complete set of 84 rich lines. These explicitly generated stabilizer subgroups give the three plus seven orbits and sizes stated above. In a fixed lexicographic ordering of normalized coefficient vectors, the first line of each orbit is the deterministic line-orbit representative used below. Applying the inverse stabilizer word transports every bridge solution, reconstructed parent, and recorded outside-card witness from that representative to every other line in its orbit, with the input pair, kernel, input order, and nonzero scalar transported along the same realizable operation. Thus the ten representative calculations cover every rich line through either base point.

Fix representatives over $\mathbb{Q}(i)$ on one such line. Since its endpoints are independent, $A + D = 2F_0$ uniquely fixes both endpoint scales. To solve Equation (F.144), the case $W = 0$ is tested separately. If W and $A - 2W$ are nonzero and nonproportional, their projective classes lie on one of the exhaustively enumerated rich lines through $[A]$, and the equation fixes both scales. In the remaining nonzero case write $W = tA$ for a nonzero scalar t. Then $D + 2tA \neq 0$, and the third direction in $\mathscr{U}_6$ on $\langle A, D\rangle$ fixes t. Thus the six bridge points and twelve sum triples are a complete symbolic solve over $\mathbb{Q}(i)$, including all zero and complex-scale cases.

For every ordered bridge triple (W_1, W_2, W_3), put $F_j = W_j$ for $1 \leq j \leq 3$, retain the displayed base tensor as F_0, and use the Frobenius-dual quaternion basis to reconstruct

$$g = \sum_{\mu=0}^{3} (Q^{\mu})^{(1,2)} \otimes F_{\mu}.$$

Exact projective comparison gives 240 distinct parents. The Python verifier replay `certificates/verify_a4_b8.py`, invoked with `-fresh` by the suite runner, then scans input pairs in lexicographic order and the twelve realizable kernels in the point order of Equation (F.17). For each card it tests zero first and otherwise tests projective membership in the definition-level regenerated set $\mathscr{U}_6$. For a parent with a card outside $\mathscr{C}_6$, it records and rechecks the first such card. Otherwise it checks all 336 cards. Affineness of each remaining case is tested from the definition: affine-coset support, fourth-root ratios, linear and even mixed quadratic phase coefficients, and the recovered phase on

every support point. Factor cuts are tested by exact flattening minors. This exhaustive calculation regenerates the two orbits, their rich lines, bridges, parents, and outside witnesses over $\mathbb{Q}(i)$. This classification of the parents whose cards all lie in $\mathscr{C}_6$ depends on exhaustive exact computation. □

Theorem F.A.23 (Tetrahedral eight-input contractions)**.** *Let $\mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})})$ and $\mathscr{H}_6$ be the realizable tetrahedral orbits of the base matching product and H_6, and put $\mathscr{U}_6 = \mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})}) \sqcup \mathscr{H}_6$ and $\mathscr{C}_6 = \{0\} \cup \mathrm{cone}(\mathscr{U}_6)$. Then*

$$|\mathfrak{D}_6(\mathcal{T}^{(\mathrm{int})})| = 25920, \qquad |\mathscr{H}_6| = 31104, \qquad |\mathscr{U}_6| = 57024.$$

Every normalized eight-input parent reconstructed from a rich line through either base point is realizable-equivalent to one of the 240 audited parents. No audited parent has distinguished card-map rank three. 180 have a recorded realizable tetrahedral card outside $\mathscr{C}_6$, and the remaining 60 have all 336 realizable cards in $\mathscr{C}_6$ and belong to the affine class $\mathscr{A}$. These assertions include arbitrary nonzero complex scales and both endpoint orientations.

Proof. This is the finite-level conclusion of Proposition F.A.22. Its realizable-stabilizer covariance transports the ten representative line calculations, their reconstructed parents, and their recorded card witnesses to every rich line through either base point. The realizable tetrahedral input dressings and permutations of inputs preserve $\mathscr{A}$, because the corresponding realizable tetrahedral binaries are affine, so the affine classification transports as well. □

F.A.5.6 Deficient mixed-card localization

Proposition F.A.24 (Platonic contractions of rank less than four)**.** *Let $\mathcal{F}$ be a signature set, let G be the standard marked tetrahedral form, either unmixed octahedral form, or the marked icosahedral form, and let h be a localized arity-six signature directly realizable from $\mathcal{F}$, whose card map Φ has rank two or three. Suppose every realizable G-card is zero or a nonsingular G-matching product and that two nonzero cards use different matchings. Then the nonkernel domain points contain a rich line disjoint from the projectivized kernel of Φ whose three images are nonzero and use the three distinct residual matchings. For the standard tetrahedral, first octahedral, and icosahedral forms, contractions produce one of six support-ten quaternaries q_π, indexed by $\pi \in S_3$, each satisfying*

$$\left(\mathrm{rank}_{12|34}\, q_\pi, \mathrm{rank}_{13|24}\, q_\pi, \mathrm{rank}_{14|23}\, q_\pi\right) = (3,3,3). \tag{F.145}$$

For the second octahedral form they instead produce the corresponding transported quaternaries q'_π. These need not have support size ten, but invertible local transport preserves the same three ranks. Writing $q^\star_\pi = q_\pi$ in the standard forms and $q^\star_\pi = q'_\pi$ in the second octahedral form, one has

$$\text{K-Holant}(\mathcal{F}, q^\star_\pi) \le_{\mathrm{T}} \text{K-Holant}(\mathcal{F}, h) \le_{\mathrm{T}} \text{K-Holant}(\mathcal{F}).$$

The statement includes the second octahedral form in its own chosen representatives.

Proof. Apply the kernel-deleted connectivity assertion Equation (F.139) to $W = \ker \Phi$, whose dimension is two or one. Along a path between two nonzero cards with different matchings, choose the first matching-changing edge and a rich line containing it. That line cannot meet $\mathbb{P}(\ker \Phi)$: its two-dimensional vector span would otherwise map with rank at most one, making all nonkernel images projectively proportional. The exact deck-line types in Proposition F.A.19 exclude type $(2,1)$ and mixed four- or five-lines. The selected line is therefore a kernel-avoiding rainbow three-line.

Apply Lemma F.9.10. It gives a realizable six-input map Ψ with the normalized rainbow line and $\operatorname{rank}\Psi \le 3$. The three exact bridge classifications have only six rank-two maps and their listed rank-four frames, so Ψ has rank two. In the common coefficient normalization,

$$(F_0^\pi, F_1^\pi, F_2^\pi, F_3^\pi) = (C/2, \pi(0, A/2, B/2)), \qquad \pi \in S_3.$$

Using the Frobenius-dual quaternion basis, reconstruct the arity-six signature

$$h_\pi := \sum_{\mu=0}^{3} (Q^\mu)^{(1,2)} \otimes F_\mu^\pi.$$

Contract inputs $1, 3$ by the realizable kernel $Q_0 = I$, and order the surviving inputs $(2, 4, 5, 6)$ as $(1, 2, 3, 4)$. Direct expansion gives support size ten for every q_π, and exact row reduction gives Equation (F.145). All preceding normalizations and the final contraction are realizable. Undoing them gives the displayed reduction in the three standard forms. In the second octahedral form the coefficient transport identifies $q'_\pi = (H_{\mathrm{tt}}^{-1})^{\otimes 4} q_\pi$. The corresponding localizations and contraction kernels are checked separately in realizable $\mathcal{O}^{(\mathrm{tt})}$-representatives. Local invertibility preserves the three flattening ranks, but no support-size claim is transported. □

F.A.5.7 Root geometry for complementary input pairs at arity four

For a quaternary tensor, the contraction maps for complementary input pairs are related by the Frobenius adjoint as in Equation (F.80). In quaternion coordinates, both the map and its adjoint must send each specified line to a line in the same configuration or to zero. We classify the maps by rank and reconstruct the quaternary tensor in each case. This is the finite geometric input to Lemma F.9.5.

Root configurations and cross-ratios. Let E be a real vector space with a positive-definite symmetric bilinear form $\langle\ ,\ \rangle$. This is a real inner-product space. A *reduced finite root system* in E is a finite spanning set Φ of nonzero vectors such that each reflection

$$\mathbf{v} \longmapsto \mathbf{v} - 2\frac{\langle \mathbf{v}, \boldsymbol{\alpha}\rangle}{\langle \boldsymbol{\alpha}, \boldsymbol{\alpha}\rangle}\boldsymbol{\alpha} \qquad (\boldsymbol{\alpha} \in \Phi)$$

permutes Φ, and $\Phi \cap \mathbb{R}\boldsymbol{\alpha} = \{\boldsymbol{\alpha}, -\boldsymbol{\alpha}\}$. Its root-line configuration is

$$\mathcal{R}(\Phi) := \{[\boldsymbol{\alpha}] : \boldsymbol{\alpha} \in \Phi\} \subseteq \mathbb{P}(E \otimes_{\mathbb{R}} \mathbb{C}).$$

In root-system notation, $\Phi(D_4), \Phi(F_4), \Phi(H_4)$ denote the standard systems of those types, and

$$\mathcal{R}(D_4) := \mathcal{R}(\Phi(D_4)), \quad \mathcal{R}(F_4) := \mathcal{R}(\Phi(F_4)), \quad \mathcal{R}(H_4) := \mathcal{R}(\Phi(H_4)).$$

These are root-system types [Hum90]. In particular, the type D_4 differs from the order-four dihedral group $D_4 \cong V_4$.

For four distinct points $p_i = [\mathbf{v}_i]$ on a projective line $\ell = \mathbb{P}(U)$, choose an ordered basis of the two-dimensional vector space U. Let $\det_U(\mathbf{v}, \mathbf{w})$ be the 2-by-2 determinant of the coordinate columns of $\mathbf{v}, \mathbf{w}$ in that basis, and define

$$[p_1, p_2; p_3, p_4] := \frac{\det_U(\mathbf{v}_1, \mathbf{v}_3)\det_U(\mathbf{v}_2, \mathbf{v}_4)}{\det_U(\mathbf{v}_1, \mathbf{v}_4)\det_U(\mathbf{v}_2, \mathbf{v}_3)}.$$

Changing a representative scales one numerator and one denominator factor, and changing the ordered basis scales every determinant by the same nonzero factor. Hence the scalar is well defined. The ordered quadruple is *harmonic* when this value is -1. A harmonic projectivity between two four-point lines is a projective isomorphism carrying one set to the other and preserving exactly the ordered quadruples of cross-ratio -1. This projective cross-ratio is distinct from a cross-ratio of signature coefficients.

The fixed representatives from Equation (F.68), written in quaternion coordinates, give the following root-point configurations:

$$\begin{aligned}\mathcal{R}_{\mathcal{T}^{(\mathrm{int})}} &:= \{[c_Q(C)] : [C] \in \mathcal{T}^{(\mathrm{int})}\} \cong \mathcal{R}(D_4),\\ \mathcal{R}_{\mathcal{O}^{(\mathrm{dd})}} &:= \{[c_Q(C)] : [C] \in \mathcal{O}^{(\mathrm{dd})}\} \cong \mathcal{R}(F_4),\\ \mathcal{R}_{\mathcal{I}^{(0)}} &:= \{[c_Q(C)] : [C] \in \mathcal{I}^{(0)}\} \cong \mathcal{R}(H_4).\end{aligned}$$

where the displayed coordinate models are those of Equations (F.17) to (F.19) and the displayed isomorphisms mean projective identification in the sense of Section F.2.1.3 and Appendix F.A.5.7. Their sizes are $12, 24, 60$. Recall the Frobenius-form matrix and define the associated adjoint map by

$$J_{\mathrm{Frob}} := \mathrm{diag}(1, -1, 1, -1), \qquad A^{\sharp} := J_{\mathrm{Frob}} A^{\mathsf{T}} J_{\mathrm{Frob}}.$$

All calculations are exact over $\mathbb{Q}(i)$ for D_4, F_4 and over $\mathbb{Q}(\phi, i)$ for H_4.

Proposition F.A.25 (Platonic preserver classification for complementary pairs)**.** *For $G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{I}^{(0)}\}$, let $\mathcal{R}_G$ be its projective coefficient set and define*

$$\mathscr{E}_G = \{A \in \mathrm{End}(\mathbb{C}^4) : \ \forall [\mathbf{r}] \in \mathcal{R}_G,\ (A\mathbf{r} = 0 \text{ or } [A\mathbf{r}] \in \mathcal{R}_G),\ (A^{\sharp}\mathbf{r} = 0 \text{ or } [A^{\sharp}\mathbf{r}] \in \mathcal{R}_G)\}. \quad \text{(F.146)}$$

Here $\mathbf{r}$ denotes any nonzero representative of $[\mathbf{r}]$. The two conditions are independent of that choice. Up to nonzero scalar, the nonzero members of $\mathscr{E}_G$, together with the single zero map recorded in the rank-zero column, have the exact rank distribution

type of G	rank 0	rank 1	rank 2	rank 3	rank 4
A_4	1	144	0	0	576
S_4	1	576	2592	0	1152
A_5	1	3600	0	0	7200.

(F.147)

Rank one reconstructs a binary-pair tensor factor. Rank two occurs only for S_4. All 2592 maps form one orbit under realizable local octahedral actions in each current marked form. In the first normal form the reconstructed tensors form the realizable local-$\mathcal{O}^{(\mathrm{dd})}$ orbit of $=_4$. In the second normal form they admit realizable local-$\mathcal{O}^{(\mathrm{tt})}$ dressings to the endpoint-nondegenerate eight-vertex signature

$$M_{\mathcal{O}^{(\mathrm{tt})}}(\mathbf{x}) = \mathbf{1}[\mathrm{wt}(\mathbf{x}) \equiv 0 \pmod 2](-1)^{\sum_{i<j} x_i x_j}.$$

In the mixed forms the normalized directly reconstructed remaining cases have the even-entry vectors in Equation (F.79), all odd entries zero, and ranks $(2, 2, 2)$. Every rank-four map has one of the forms

$$C \longmapsto UCV \qquad \text{or} \qquad C \longmapsto UC^{\mathsf{T}}V, \quad \text{(F.148)}$$

and reconstructs a product of two nonsingular binaries on a crossing matching. The rank counts hold for all four fixed-equality octahedral forms $\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}$, while their realizable rank-two outputs are the ones just stated.

Proof. We first exclude rank three without enumerating arbitrary linear maps. A plane section of a point configuration P is $P \cap \Pi$, where Π is a projective plane.

In the following table, the secants are the distinct projective lines spanned by pairs of points of $\mathcal{R}_G$, while the plane sections are the distinct intersections with planes spanned by noncollinear triples of those points. Exponents record the number of distinct lines or sections of each size. Exact row reduction gives the complete line and plane-section data

type of G	secant-line sizes	plane sections
A_4	$2^{18}, 3^{16}$	$3^{12}, 6^{12}$
S_4	$2^{72}, 3^{32}, 4^{18}$	$4^{96}, 9^{24}$
A_5	$2^{450}, 3^{200}, 5^{72}$	$4^{600}, 6^{660}, 15^{60}$.

(F.149)

For a projective point $k = [\mathbf{k}]$, let $d(k)$ be the number of distinct nonzero projective classes $[\mathbf{r} + \operatorname{span}_{\mathbb{C}}\{\mathbf{k}\}]$ in $\mathbb{P}(\mathbb{C}^4/\operatorname{span}_{\mathbb{C}}\{\mathbf{k}\})$, as $r = [\mathbf{r}]$ ranges over the root points other than k. If k lies on at most one secant containing two root points, the maximum line sizes $3, 4, 5$ give

$$d(k) \geq 10, \qquad d(k) \geq 21, \qquad d(k) \geq 56,$$

respectively. The centers lying on at least two secants are the finite set of pairwise secant intersections. Exact quotient row reduction gives

A_4	$7^{12}, 9^{12}$
S_4	$13^{24}, 19^{96}$
A_5	$31^{60}, 49^{300}, 51^{360}, 55^{600}$.

(F.150)

A rank-three map is projection from the point representing its nullspace to its image plane, and the induced map from the quotient is injective. Hence its quotient directions must fit into a plane section of the target configuration. The bounds and every count in Equation (F.150) exceed the plane maxima $6, 9, 15$ in Equation (F.149). This proves the rank-three column of Equation (F.147).

We next classify rank two. For $A \in \mathscr{E}_G$, define the projective image line $W := \mathbb{P}(\operatorname{im} A^\sharp)$. Because the root points span $\mathbb{C}^4$, W is a secant line. Moreover $\ker A = (\operatorname{im} A^\sharp)^{\perp_{J_{\mathrm{Frob}}}}$. Projecting the root points modulo this kernel gives the following complete table:

type of G	$(\lvert W \cap \mathcal{R}_G\rvert, \#$ quotient directions, $\#$ killed points$)$
A_4	$(2,4,2)^{18},\ (3,6,0)^{16}$
S_4	$(2,6,2)^{72},\ (3,6,3)^{32},\ (4,4,4)^{18}$
A_5	$(2,12,2)^{450},\ (3,12,3)^{200},\ (5,10,5)^{72}$.

The image of A is also a projective line, containing at most $3, 4, 5$ root points. The middle entry in every row is larger than that bound except for $(4,4,4)^{18}$ in the octahedral row. The eighteen surviving four-lines form one orbit under realizable left and right $\mathcal{O}^{(\mathrm{dd})}$-actions. There are eighteen choices for $\mathbb{P}(\operatorname{im} A^\sharp)$, eighteen for $\mathbb{P}(\operatorname{im} A)$. Each surviving four-point set is harmonic in the cross-ratio convention of Appendix F.A.5.7, and exactly eight projective isomorphisms between any two such lines preserve its harmonic pairings. Thus there are exactly eight harmonic projectivities, and consequently

$$18 \cdot 18 \cdot 8 = 2592$$

projective rank-two maps survive. They form one realizable local-octahedral orbit of the rank-two endomorphism

$$A_0 := \operatorname{diag}(1, 1, 0, 0).$$

Bilinear-dual reconstruction, including the fixed output X in the transfer convention $L_p(C) = \Phi_p(C)X$, gives a realizable local-octahedral equivalent of the quaternary signature

$$q_0(\mathbf{x}) := \frac{1}{2}\mathbf{1}[x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 2].$$

The realizable octahedral matrix $W_0 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, with $[W_0] = [Q_1 + Q_3] \in \mathcal{O}^{(\mathrm{dd})}$, satisfies $W_0^{\otimes 4}(=_4) = 2\mathbf{1}[x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 2]$ and $W_0^{\otimes 4} q_0 = 4(=_4)$. It follows that every rank-two remaining case has, for some matrices U_j with $[U_j] \in \mathcal{O}^{(\mathrm{dd})}$,

$$q = \lambda(U_1 \otimes U_2 \otimes U_3 \otimes U_4)(=_4), \qquad \lambda \neq 0.$$

Conversely, direct substitution verifies the conditions for both complementary pairs on this orbit. The complete four-line orbit above leaves no additional projective rank-two degeneration. Every surviving tensor has flattening ranks $(2, 2, 2)$.

For the second octahedral form, analytic conjugation by H_{tt} transports the equations for both complementary pairs, preserving the count 2592 and the rank pattern. The realizable local gadgets require a separate calculation. Indeed, the realizable $(\mathcal{O}^{(\mathrm{tt})})^{\otimes 4}$-orbit of the second-form remaining case has 2592 projective tensors with support histogram

$$8^{16}, \qquad 10^{128}, \qquad 12^{384}, \qquad 16^{2064},$$

and hence does not meet the realizable $(\mathcal{O}^{(\mathrm{tt})})^{\otimes 4}$-orbit of the support-two signature $=_4$. For the normalized second-form rank-two remaining case, use the four matrices

$$\begin{aligned} L_1 &= \theta_{\mathrm{tt}}(Q_0 + Q_2), \\ L_2 = L_3 &= \theta_{\mathrm{tt}}(Q_0 - Q_1 + Q_2 - Q_3), \\ L_4 &= X\theta_{\mathrm{tt}}(Q_2 - Q_3). \end{aligned} \tag{F.151}$$

Exact substitution gives the endpoint-nondegenerate eight-vertex signature even-entry vector

$$(1, -1, -1, -1, -1, -1, -1, 1), \tag{F.152}$$

with every odd entry zero and ranks $(2, 2, 2)$. Acting before and after this certificate by the realizable group proves the stated eight-vertex realization for the entire second-form orbit.

For $\alpha \in \{\mathrm{dt}, \mathrm{td}\}$, complementary-pair reconstruction of the same normalized coefficient map uses the invertible coefficient operator

$$\mathscr{T}_\alpha = H_\alpha^{\mathsf{T}} \otimes H_\alpha^{-1} \otimes H_\alpha^{-1} \otimes X H_\alpha^{\mathsf{T}} X.$$

Direct exact substitution in q_0 gives, projectively,

$$\mathbf{e}(\mathscr{T}_{\mathrm{dt}} q_0) = (1, 1, 1, 1, 1, 1, 1, 1), \qquad \mathbf{e}(\mathscr{T}_{\mathrm{td}} q_0) = (1, i, i, i, 1, 1, 1, i),$$

with all odd entries zero. These are the direct current-coordinate reconstructions of the remaining cases. Exact row reduction gives ranks $(2, 2, 2)$. Acting by realizable $\mathcal{O}^{(\alpha)}$-matrices covers the entire rank-two orbit.

The rank-zero member is the zero map. Because the quaternion basis spans $\mathrm{Mat}_2(\mathbb{C})$, its reconstructed quaternary is zero. For rank one, choose nonzero column vectors $\mathbf{u}, \mathbf{v} \in \mathbb{C}^4$ and write $A = \mathbf{u}\mathbf{v}^{\mathsf{T}}$. The conditions on A and $A^\sharp$ force $[\mathbf{u}], [J_{\mathrm{Frob}}\mathbf{v}] \in \mathcal{R}_G$. Conversely every such pair works. This gives $|G|^2 = 144,576,3600$ projective maps and a rank-one $p \mid \bar{p}$ flattening of the reconstructed tensor.

Finally, a rank-four member of $\mathscr{E}_G$ permutes $\mathcal{R}_G$. Exact root-incidence automorphism classification gives precisely Equation (F.148), with

$$[V] \in N_{\mathrm{PGL}_2}(G), \qquad [UV] \in G.$$

The projective counts of the two displayed families are

$$2|G|\,|N_{\mathrm{PGL}_2}(G)| = 2 \cdot 12 \cdot 24, \quad 2 \cdot 24^2, \quad 2 \cdot 60^2,$$

namely $576, 1152, 7200$. If the contraction map is $C \mapsto UCV$, undoing the output X gives $q_{uvyz} = U_{yu}(VX)_{vz}$. The transpose case gives $q_{uvyz} = U_{yv}(VX)_{uz}$. In either case the quaternary is a product of two nonsingular binaries on a crossing matching.

For $\mathcal{T}^{(\mathrm{int})}$, imposing the condition of Equation (F.146) on A alone gives 1536 projective rank-two solutions. Each fails the condition on $A^\sharp$ from the complementary pair, giving the zero in the A_4 rank-two column.

The coefficient similarities identify the four rank tables. The formulas Equations (F.151) and (F.152) and the two direct mixed reconstructions supply the required gadget reductions in the respective current forms.

Here the finite incidence hypergraph has the root points as vertices and the rich lines and plane sections as its distinguished hyperedge families. Its automorphisms preserve each of these families.

For all three rows, exact enumeration generates the root configurations and then recovers every secant, plane section, multiple-secant center, and quotient direction by characteristic-zero row reduction: over $\mathbb{Q}(i)$ for D_4, F_4, and over $\mathbb{Q}(\phi, i)$ for H_4. It enumerates all projectivities between the surviving octahedral four-lines, giving $18^2 \cdot 8 = 2592$. For the full-rank case it constructs the complete finite incidence hypergraph of root points, rich lines, and plane sections, enumerates its automorphisms, solves for the inducing projective linear map, and retains the map only after checking both A and $A^\sharp$ on every root point. The resulting automorphism counts equal the cardinalities of the explicit left–right/transpose families for D_4, F_4, H_4, proving completeness of all three full-rank lists rather than merely verifying their displayed members. □

F.A.5.8 Contractions of the Platonic quaternary signatures

This subsection gives the contractions used in Lemma F.9.8. Inputs of every quaternary are numbered $1, 2, 3, 4$ in their displayed order. If inputs r, s are contracted by a realizable kernel C, the surviving ordered binary is denoted $B = \partial^C_{rs} q$. The pairing is bilinear, with no conjugation. Each group-membership test checks whether the transfer class $[T_B] = [BX]$ belongs to G.

Proposition F.A.26 (Contractions of Platonic quaternary signatures)**.** *For each quaternary in the following list, let G denote the marked Platonic group from whose row the quaternary is obtained. Every one of the six rank-deficient quaternaries q_π constructed in Proposition F.A.24 and satisfying Equation* (F.145)*, one of the three orientation-reversing tetrahedral outputs q_σ specified in Proposition F.A.17, one of the twenty-four octahedral separator outputs (in any of the four marked forms) specified in Proposition F.A.16, or one of the thirty icosahedral frame outputs q_{u_1,u_2,u_3} in Equations* (F.135) *and* (F.136) *has one of the following directly realized signatures:*

1. *an endpoint-nondegenerate eight-vertex signature with flattening ranks $(2, 2, 2)$;*

2. *a generalized equality of arity four;*

3. *a nonsingular ordered binary B whose transfer class $[BX]$ is outside G.*

For either mixed octahedral form the list also includes the six direct current-coordinate counterparts of the rank-deficient outputs. This holds for all four octahedral distinguished-involution normal forms and for both icosahedral coordinate displays. Every such signature uses one quaternary vertex and at most four realizable group binaries.

Proof. First consider the six support-ten rank-deficient outputs in Proposition F.A.24. For each fixed output, exact enumeration of all six input pairs and all realizable kernels gives the following table. "Inside" means $[BX]$ is in the displayed group, and "singular" counts nonzero singular cards. Zero cards are kept separate.

current group	cards per q	zero	inside	outside	singular	one uniform outsider
$\mathcal{T}^{(\mathrm{int})}$	72	3	9	60	0	$\partial_{12}^{Q_0} q$
$\mathcal{O}^{(\mathrm{dd})}$	144	6	108	30	0	$\partial_{24}^{Q_0} q$
$\mathcal{I}^{(0)}$	360	3	36	321	0	$\partial_{12}^{Q_0} q$

Thus the same displayed pair and kernel work for all six outputs in each row. The first two rows are computed over $\mathbb{Q}(i)$, and the $\mathcal{I}^{(0)}$ row is computed over $\mathbb{Q}(\phi, i)$.

For the second octahedral form, define $\sigma_{\mathrm{tt}} := \sqrt{2}$ and the transport matrix $S_{\mathrm{tt}} := H_{\mathrm{tt}}^{-1}$. The transported rank-deficient outputs are $q'_\pi = S_{\mathrm{tt}}^{\otimes 4} q_\pi$. The single realizable kernel

$$K'_{\mathrm{def}} = H_{\mathrm{tt}}^{-1}(Q_0 - Q_1 - Q_2 - Q_3)H_{\mathrm{tt}}, \qquad [K'_{\mathrm{def}}] \in \mathcal{O}^{(\mathrm{tt})}$$

on inputs $1, 2$ works for all six. Direct expansion gives $\det(\partial_{12}^{K'_{\mathrm{def}}} q'_\pi) = -3/512$. As π ranges over S_3, the pulled-back transfer directions $H_{\mathrm{tt}}(B'X)H_{\mathrm{tt}}^{-1}$, in quaternion coordinates, form the multiset

$$[3,-1,1,1],\ [1,-1,0,1],\ [3,-1,1,1],\ [1,-1,0,1],\ [3,-1,-1,1],\ [3,-1,-1,1].$$

None is in $\mathcal{O}^{(\mathrm{dd})}$: its support has size three, or size four with unequal nonzero magnitudes. Hence every corresponding $[B'X]$ is outside $\mathcal{O}^{(\mathrm{tt})} = H_{\mathrm{tt}}^{-1}\mathcal{O}^{(\mathrm{dd})}H_{\mathrm{tt}}$.

For $\mathcal{O}^{(\mathrm{dt})}$ and $\mathcal{O}^{(\mathrm{td})}$, the final exact table in Proposition F.A.16 gives one current-group kernel on inputs $1, 2$, a common nonzero determinant, and all six pulled-back outside-transfer directions. This proves the rank-deficient rows directly in the retained fixed-I coordinates.

The notation here is compatible with the earlier separator transport $q'_R = \mathcal{L}_{\mathrm{tt}} q_R$, but the two formulas use different displayed input-orientation conventions. The matching-dependent X-factors in Equations (F.130) and (F.153) convert the alternating $H_{\mathrm{tt}}^{-1}/H_{\mathrm{tt}}^{\mathsf{T}}$ convention of $\mathcal{L}_{\mathrm{tt}}$ into the uniform $S_{\mathrm{tt}}^{\otimes 4}$ convention after those fixed orientation factors are absorbed into the current $\mathcal{O}^{(\mathrm{tt})}$-representatives. Thus both notations parametrize the same marked second-form coefficient families in their stated input orders. Neither equality is being used as a gadget.

For the three tetrahedral outputs q_σ specified in Proposition F.A.17, the one realizable kernel

$$K^{\mathrm{out}}_{\mathcal{T}^{(\mathrm{int})}} = Q_0 + Q_1 - Q_2 + Q_3, \qquad [K^{\mathrm{out}}_{\mathcal{T}^{(\mathrm{int})}}] \in \mathcal{T}^{(\mathrm{int})}$$

on inputs $1, 2$ gives, projectively,

σ	$B = \partial_{12}^{K^{\mathrm{out}}_{\mathcal{T}^{(\mathrm{int})}}} q_\sigma$	BX
$(0,3,1)$	$\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$
$(1,0,3)$	$\left(\begin{smallmatrix} 1 & -i \\ -i & 1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} -i & 1 \\ 1 & -i \end{smallmatrix}\right)$
$(3,1,0)$	$\mathrm{diag}(1, i)$	$\left(\begin{smallmatrix} 0 & 1 \\ i & 0 \end{smallmatrix}\right)$

All three determinants are nonzero, and exact projective comparison with the twelve points in Equation (F.17) excludes all three transfer classes.

We next treat the first octahedral form. Define the octahedral sign matrices by

$$S_{\epsilon_1\epsilon_2\epsilon_3} = Q_0 + \epsilon_1 Q_1 + \epsilon_2 Q_2 + \epsilon_3 Q_3.$$

For an ordered frame in Equation (F.131), its separator is $q = C/2 + W_1$, where W_1 is the first entry. The twelve ordered outputs with

$$W_1 \in \{R_0, R_3, R_{03}^{+-}, R_{03}^{-+}\}$$

are endpoint-nondegenerate eight-vertex signatures on all eight even-parity strings and have ranks $(2,2,2)$. The four with $W_1 = R_1$ have support $\{0011, 1100\}$. Realizable X-dressings on the last two inputs turn them into generalized equalities. The remaining eight outputs occur twice for each row of the following table:

W_1	kernel on 12	B	$[BX]$
R_{13}^{++}	S_{++-}	$\left(\begin{smallmatrix} 1+i & -i \\ -i & 1-i \end{smallmatrix}\right)$	$[Q_0 + Q_1 - Q_2]$
R_{13}^{--}	S_{+++}	$\left(\begin{smallmatrix} 1-i & -i \\ -i & 1+i \end{smallmatrix}\right)$	$[Q_0 + Q_1 + Q_2]$
R_{01}^{+-}	S_{+++}	$\left(\begin{smallmatrix} 1-i & -1 \\ 1 & 1+i \end{smallmatrix}\right)$	$[Q_1 + Q_2 - Q_3]$
R_{01}^{-+}	S_{++-}	$\left(\begin{smallmatrix} 1+i & -1 \\ 1 & 1-i \end{smallmatrix}\right)$	$[Q_1 - Q_2 - Q_3]$

Every determinant equals 3. The points of $\mathcal{O}^{(\mathrm{dd})}$ in Equation (F.18) have quaternion-coordinate support size one, two, or four, never three, so all four displayed transfer directions are outside $\mathcal{O}^{(\mathrm{dd})}$.

For the second octahedral form, we compute the realizable kernels in its fixed coordinates. Define the nonzero scalar $c_{\mathrm{tt}} := 2\sigma_{\mathrm{tt}}(1-i)$. Direct multiplication gives

$$H_{\mathrm{tt}}^{\mathsf{T}} X H_{\mathrm{tt}} = c_{\mathrm{tt}} I, \qquad S_{\mathrm{tt}} U S_{\mathrm{tt}}^{\mathsf{T}} = c_{\mathrm{tt}}^{-1} H_{\mathrm{tt}}^{-1} (UX) H_{\mathrm{tt}}. \tag{F.153}$$

Thus $S_{\mathrm{tt}}^{\otimes 4}$ transports every $\mathcal{O}^{(\mathrm{dd})}$-matching factor to an $\mathcal{O}^{(\mathrm{tt})}$-matching factor. More precisely, if h is the standard-form six-input ancestor with card map Φ, define, in its displayed input order,

$$h' := (H_{\mathrm{tt}}^{\mathsf{T}} \otimes H_{\mathrm{tt}}^{-1} \otimes S_{\mathrm{tt}}^{\otimes 4}) h.$$

The card map Φ' of h' is parametrized by

$$\Phi'(H_{\mathrm{tt}}^{-1} C_{\mathrm{ker}} H_{\mathrm{tt}}) = S_{\mathrm{tt}}^{\otimes 4} \Phi(C_{\mathrm{ker}}).$$

This invertible coefficient correspondence exhausts the second-form tensors. The contractions below use the listed current-group kernels, with their ordered inputs and nonzero realization scalars.

Define the conjugated octahedral sign matrices by $S'_{\epsilon_1\epsilon_2\epsilon_3} := H_{\mathrm{tt}}^{-1} S_{\epsilon_1\epsilon_2\epsilon_3} H_{\mathrm{tt}}$. For the twelve ordered outputs in the next table, direct contraction of $q' = S_{\mathrm{tt}}^{\otimes 4} q$ on inputs $1, 2$ gives $\det B' = -3/256$:

W_1	realizable kernel	$H_{\mathrm{tt}}(B'X)H_{\mathrm{tt}}^{-1}$
R_{13}^{++}	S'_{+++}	$[2Q_0 + Q_1 + Q_2]$
R_{13}^{--}	S'_{+-+}	$[Q_1 + Q_2 + 2Q_3]$
R_{03}^{+-}	S'_{++-}	$[Q_0 + 2Q_1 + Q_3]$
R_{03}^{-+}	S'_{+++}	$[Q_0 + 2Q_2 + Q_3]$
R_{01}^{+-}	S'_{++-}	$[2Q_0 + Q_1 - Q_2]$
R_{01}^{-+}	S'_{+++}	$[2Q_0 - Q_1 + Q_2]$

Each row occurs for two orderings. Every pulled-back direction has three nonzero quaternion coordinates of magnitudes $2, 1, 1$, so it is outside $\mathcal{O}^{(\mathrm{dd})}$, proving $[B'X] \notin \mathcal{O}^{(\mathrm{tt})}$.

The remaining twelve ordered $\mathcal{O}^{(\mathrm{tt})}$-outputs have $W_1 = R_0, R_1, R_3$, four of each. Realizable local dressings turn them into endpoint-nondegenerate eight-vertex signatures. In the even-string order

$$(0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111),$$

the exact outputs are

W_1	dressing	even coefficients, up to scale
R_0	$(S'_{++-})^{\otimes 4}$	$(1, 1, -1, 1, 1, -1, 1, 1)$
R_1	identity	$(1, -1, 1, 1, 1, 1, -1, 1)$
R_3	$(S'_{+++})^{\otimes 4}$	$(1, -1, 1, 1, 1, 1, -1, 1)$

All odd-weight coefficients vanish, both endpoints are nonzero, and local invertible dressings preserve ranks $(2, 2, 2)$.

The two mixed full-rank rows are the direct tables in Proposition F.A.16. For $\mathcal{O}^{(\mathrm{dt})}$ they partition into endpoint and outside-binary exits. For $\mathcal{O}^{(\mathrm{td})}$ they partition into endpoint, generalized equality, and outside-binary exits. The tables cover every possible first frame entry, hence all twenty-four orderings, and use only matrices of the current mixed group.

Finally, for each of the thirty icosahedral outputs in Equation (F.136), use the one chosen representative

$$L_{\mathcal{I}^{(0)}} = Q_0 - \tau^2 Q_1 + \tau Q_3 \qquad \text{with} \qquad [L_{\mathcal{I}^{(0)}}] \in \mathcal{I}^{(0)}$$

on inputs $1, 2$. Exact arithmetic over $\mathbb{Q}(\phi, i)$ proves that all thirty resulting binaries are nonsingular, their thirty projective transfers $[BX]$ are distinct, and none belongs to the complete sixty-point set $\mathcal{I}^{(0)}$. Enumerating all $30 \cdot 6 \cdot 60 = 10800$ realizable cards gives

$$\text{zero } 180, \qquad \text{inside } 1260, \qquad \text{outside } 9360, \qquad \text{nonzero singular } 0.$$

For the opposite display, use the fixed-equality coefficient correspondence Lemma F.9.3: replace the two standard kernels by their realizable κ_D-images. Card covariance conjugates the complete transfer group and preserves nonsingularity and group exclusion, while the I is untouched. This exhausts every family in the proposition.

The displayed kernels and dressing tables prove the listed bounded exits by exact matrix multiplication. The complete output histograms and the completeness claims used earlier are also reproduced by `certificates/verify_a4_s4_certificates.py`, `certificates/verify_s4_second_form.py`, `certificates/verify_a5_certificate.py`, and `certificates/verify_platonic_extended.py` over the number fields specified above. In particular, the full preserver and hereditary eight-input classifications retain the exhaustive computations described in their own proofs. □

F.A.5.9 Consequences of the Platonic classifications

Theorem F.A.27 (Platonic separated-sign classifications)**.** *Under the normalized rainbow and separated-sign hypotheses of Equations* (F.89)*,* (F.91) *and* (F.92)*, the following separated-sign classifications hold.*

(i) *For each marked octahedral form, there are exactly six rank-deficient maps and* 24 *full-rank maps, the latter being the six orderings of each of*

$$\{R_0, R_1, R_3\}, \quad \{R_0, R_{13}^{++}, R_{13}^{--}\}, \quad \{R_1, R_{03}^{+-}, R_{03}^{-+}\}, \quad \{R_3, R_{01}^{+-}, R_{01}^{-+}\}.$$

The fixed realizable kernel is $K_{\mathcal{O}^{(\mathrm{dd})}} = Q_0 + Q_1$ in the first form and $K_{\mathcal{O}^{(\mathrm{tt})}} = \theta_{\mathrm{tt}}(K_{\mathcal{O}^{(\mathrm{dd})}})$ in the second, while $K_{\mathcal{O}^{(\mathrm{dt})}} = \theta_{\mathrm{dt}}(K_{\mathcal{O}^{(\mathrm{dd})}})$ and $K_{\mathcal{O}^{(\mathrm{td})}} = \theta_{\mathrm{td}}(K_{\mathcal{O}^{(\mathrm{dd})}})$ are the two mixed-form kernels. In the two unmixed forms, current-group dressings send every full-rank map to an endpoint-nondegenerate eight-vertex signature of ranks $(2,2,2)$. In either mixed form, the fixed kernel and the direct current-coordinate tables give one of the three exits: such an eight-vertex signature, a generalized equality, or a nonsingular binary B with $[BX] \notin G$, where G is the current marked group.

(ii) *For the internal tetrahedral form, the bridge set is $\{0, A/2, B/2, R_0, R_1, R_3\}$, and the sum equation has exactly six rank-deficient and six full-rank solutions. Write*

$$\Sigma_+ = \{(0,1,3),(1,3,0),(3,0,1)\}, \qquad \Sigma_- = \{(0,3,1),(1,0,3),(3,1,0)\}.$$

For every $\sigma \in \Sigma_+$, all 180 realizable pair cards of the reconstructed parent are zero or $\mathcal{T}^{(\mathrm{int})}$-matching signatures, and the parent lies in $\mathscr{A}$. For every $\sigma \in \Sigma_-$, contraction of inputs $1,3$ by the fixed realizable kernel $Q_0 = I$, followed by four realizable dressings, gives an endpoint-nondegenerate eight-vertex signature of ranks $(2,2,2)$.

(iii) *The sixty displayed projective classes form a projective group isomorphic to A_5 and are closed under induced projective multiplication, inverse, and transpose. Its bridge has exactly six rank-deficient and 30 full-rank solutions, the latter being the six orderings of the five displayed frames $\mathrm{Fr}_0, \dots, \mathrm{Fr}_4$. The fixed realizable kernel $K_{\mathcal{I}^{(0)}} := \phi Q_1 - \tau Q_2 + Q_3$ gives a quaternary of ranks $(3,3,3)$, and the fixed subsequent realizable kernel $L_{\mathcal{I}^{(0)}} := Q_0 - \tau^2 Q_1 + \tau Q_3$ gives a nonsingular binary B with $[BX] \notin \mathcal{I}^{(0)}$, for all 30 solutions. The statement includes arbitrary complex scales and cancellations. For $\mathcal{I}^{(1)}$, the realizable kernels $K_{\mathcal{I}^{(1)}} = \kappa_D(K_{\mathcal{I}^{(0)}})$ and $L_{\mathcal{I}^{(1)}} = \kappa_D(L_{\mathcal{I}^{(0)}})$ give the corresponding direct fixed-I identities.*

Proof. Clause (i) is the exact octahedral bridge Equation (F.131) together with the separate gadget certificate Proposition F.A.16. Clause (ii) is Equations (F.133) and (F.134) and the all-card, affine-phase, and realizable-orientation checks in Appendix F.A.5.2, including Proposition F.A.17. Clause (iii) is the exact group, bridge, frame, and rank calculation in Appendix F.A.5.3, followed by the uniform explicit gadget reduction in Proposition F.A.26. Lemma F.9.3 gives the opposite display with realizable κ_D-kernels and leaves the supplied equality unchanged. □

Theorem F.A.28 (Classification of Platonic quaternaries)**.** *Let*

$$G \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}, \mathcal{I}^{(0)}, \mathcal{I}^{(1)}\}.$$

The finite quaternary certificates have the following directly usable form.

(i) *Let q be quaternary and define $L_p(C) = \Phi_p^q(C)X$. If, for every one of the six pairs p and every fixed chosen representative K_g,*

$$L_p(K_g) = 0 \qquad \text{or} \qquad [L_p(K_g)] \in G,$$

then, for each pair p, put $r_p := \operatorname{rank} L_p$. One has $r_p \neq 3$. If $r_p = 0$, then $q = 0$. If $r_p \in \{1,4\}$, then q has a binary-pair factorization. $r_p = 2$ can occur only for the octahedral forms: it is actually dressable to $=_4$ for $\mathcal{O}^{(\mathrm{dd})}$, and to an endpoint-nondegenerate eight-vertex signature of ranks $(2,2,2)$ for $\mathcal{O}^{(\mathrm{tt})}$, $\mathcal{O}^{(\mathrm{dt})}$, and $\mathcal{O}^{(\mathrm{td})}$, with the mixed vectors in Equation (F.79).

(ii) *Let $h \neq 0$ have arity six and let Φ_p^h have rank two or three. Suppose, for every $g \in G$, the card $\Phi_p^h(K_g)$ formed with its fixed chosen representative is zero or a nonsingular G-matching*

product, with two nonzero cards in this p-family using different residual matchings. Realizable localization produces a quaternary of ranks $(3,3,3)$*, and a contraction of that quaternary gives a nonsingular binary* B *with* $[BX] \notin G$*. This includes the separately verified* $\mathcal{O}^{(\mathrm{tt})}$ *and mixed representatives and both icosahedral displays.*

(iii) *For each fixed* G*, every corresponding enumerated rank-deficient quaternary has the following conclusion. The same conclusion holds for the tetrahedral-orientation outputs when* $G = \mathcal{T}^{(\mathrm{int})}$*, the octahedral-frame outputs when* $G \in \{\mathcal{O}^{(\mathrm{dd})}, \mathcal{O}^{(\mathrm{tt})}, \mathcal{O}^{(\mathrm{dt})}, \mathcal{O}^{(\mathrm{td})}\}$*, and the icosahedral-frame outputs when* $G \in \{\mathcal{I}^{(0)}, \mathcal{I}^{(1)}\}$*. Each such quaternary directly realizes one of: an endpoint-nondegenerate eight-vertex signature of ranks* $(2,2,2)$*, a generalized equality of arity four, or a nonsingular binary* B *with* $[BX] \notin G$*. Each realization uses the quaternary and at most four realizable* G*-binaries.*

Proof. For clause (i), use Lemma F.9.3 for the $\mathcal{I}^{(1)}$ coefficient calculation, replacing every realizable kernel by its realizable κ_D-image and leaving the I fixed. Fix a pair p, and let A_p be the matrix of L_p in the corresponding quaternion root coordinates. The card conditions at p give the first condition in Equation (F.146). For the complementary pair $\bar{p}$, let $\mathbf{R}_X$ be the coordinate matrix of right multiplication $C \mapsto CX$. Direct complementary-pair reshaping gives $A_{\bar{p}} = \mathbf{R}_X A_p^{\sharp} \mathbf{R}_X$. The fixed X-reversals permute the root configuration, so the card conditions at $\bar{p}$ give the second condition. In the normalized coordinates this places A_p in $\mathscr{E}_{G_0}$, where $G_0 \in \{\mathcal{T}^{(\mathrm{int})}, \mathcal{O}^{(\mathrm{dd})}, \mathcal{I}^{(0)}\}$ is the standard representative for G. Now Proposition F.A.25 gives all stated rank and reconstruction alternatives, including the separately verified $\mathcal{O}^{(\mathrm{tt})}$ eight-vertex output and the direct mixed reconstructions above.

Clause (ii) combines the realizable localization and rank calculation in Proposition F.A.24 with the corresponding uniform outside-binary row of Proposition F.A.26. Clause (iii) is the representative-wise realization statement of the latter proposition. The card and network covariance in Lemma F.9.3 supplies all three clauses for $\mathcal{I}^{(1)}$ directly with realizable κ_D-kernels. □

F.B General Boundary Representations

The gadget convention of Section F.2.3 uses exposed vertex inputs and isolated edges. We now describe the general representation with free edge endpoints and prove that it gives the same replacement signature, gluing operations, and charge identities.

We also allow an edge to end at a free stub, a half-edge with no tensor vertex. In this extension, $H(\Omega) = (\bigsqcup_v H_v) \sqcup H_{\mathrm{free}}(\Omega)$. Every free stub is paired and belongs to $\partial\Omega$. A vertex half-edge is unpaired exactly when it belongs to $\partial\Omega$. Thus $E_{\mathrm{int}}(\Omega)$, the matching of pairs of half-edges, contains each nonboundary vertex half-edge and each free stub exactly once. An edge box E_W denotes one edge with default kernel W, contributing one factor $W(a,b)$. It has no vertex or separate H_v. By contrast, an I or X supplied as an input signature labels an ordinary vertex, counted and transformed with all other tensor vertices.

Write $\mathbf{W} = (W_e)_{e \in E_{\mathrm{int}}(\Omega)}$, with $W_e \in \{I, X\}$. For a string $\mathbf{z}_{\partial\Omega}$ in boundary order, define the raw boundary signature by

$$R_\Omega(\mathbf{z}_{\partial\Omega}) = \sum_{\substack{\tau : H(\Omega) \to \{0,1\} \\ \tau|_{\partial\Omega} = \mathbf{z}_{\partial\Omega}}} \prod_v f_v(\tau|_{H_v}) \prod_{\{h,h'\} \in E_{\mathrm{int}}(\Omega)} W_{\{h,h'\}}(\tau_h, \tau_{h'}).$$

For the j-th boundary half-edge, put $A_j = I$ at a vertex half-edge and $A_j = W_e$ at a free endpoint

of the edge e. The boundary tensor used for vertex replacement is

$$T_\Omega = \left(\bigotimes_{j=1}^{|\partial\Omega|} A_j \right) R_\Omega, \tag{F.154}$$

$$Z_{\mathbf{W}}(\Omega) := T_\Omega(\varnothing) \qquad (\partial\Omega = \varnothing). \tag{F.155}$$

For the ordinary boundary convention of Section F.2.3, $T_\Omega = R_\Omega$. At a free endpoint, $W_e^2 = I$ cancels the incident edge: deleting that endpoint and edge and exposing the ordinary mate preserves T_Ω and the boundary order. An edge with two free endpoints is an isolated E_W, with boundary tensor $WWW = W$. These cancellations reduce every open network to unpaired vertex half-edges and isolated edge boxes. Gluing and vertex replacement are specified in Section F.2.6.

For homogeneous edges, abbreviate $Z_{\mathbf{W}}$ by Z_W. The empty network is the nullary unit 1.

F.B.1 Gluing and coordinate transport

In either homogeneous model, joining two boundary half-edges contracts their boundary-tensor coordinates against the default kernel W. The remaining boundary order is inherited. The ordinary-boundary case is given in Section F.2.6. For one paired free endpoint and one unpaired vertex half-edge, splice the latter into the former's incidence, retaining its existing edge. For two free endpoints on distinct edges, delete those endpoints and their two edges and join their mates by one W-edge. If both endpoints belong to the same isolated E_W, remove that component and record the scalar $\sum_{a,b} W(a,b)^2 = 2$. These rules follow from $W^2 = I$ and $W^3 = W$. In particular, vertex replacement first cancels the one-free-end edges as above and then inserts the resulting ordinary-boundary gadget. Each remaining isolated E_W splices into a single edge. Closed components containing only such edge boxes contribute known factors of 2. This proves Equation (F.3), including loops, without leaving free stubs in the oracle instance.

When transporting an open gadget with disequality edges by $D \in GO(X)$, first cancel its one-free-end edges in this way. Every remaining internal edge between vertex half-edges contributes the scalar from $D^{\mathsf{T}} X D = \lambda X$. An isolated edge remains available because $DXD^{\mathsf{T}} = \lambda X$. Thus its boundary tensor is transported by D up to a nonzero scalar. The closed-network edge count in Equation (F.1) is used only for closed instances.

F.B.2 Boundary charge

For a homogeneous X-edge network (all structural pairs of half-edges have kernel X, any explicitly supplied I or X is an ordinary tensor vertex), let $U(\Omega) = H(\Omega) \setminus \bigcup_{e \in E_{\mathrm{int}}(\Omega)} e$ be the unpaired boundary half-edges (by definition, $U(\Omega) \subseteq \partial\Omega$). Call an assignment *edge-supported* when every structural edge-kernel factor is nonzero on the two bits assigned to its endpoints. Every edge-supported assignment has one 0-bit and one 1-bit on each X-pair. Hence, for every edge-supported assignment τ,

$$\sum_v q_{\mathrm{arity}(f_v)}(\tau|_{H_v}) + \sum_{h \in H_{\mathrm{free}}(\Omega)} (2\tau_h - 1) = \sum_{h \in U(\Omega)} (2\tau_h - 1).$$

For a contribution to the boundary tensor $T_\Omega(\mathbf{z})$, an ordinary boundary bit equals τ_h, while a free boundary bit equals $1 - \tau_h$ by Equation (F.154). Moving the free-stub sum to the right therefore gives $q_{|\partial\Omega|}(\mathbf{z}) = \sum_v q_{\mathrm{arity}(f_v)}(\tau|_{H_v})$. In standard closed instances there are no free stubs and $U(\Omega) = \varnothing$, so $\sum_v q_{\mathrm{arity}(f_v)}(\tau|_{H_v}) = 0$. Thus X-contraction preserves the total charge in every closed contributing assignment.

F.C Exact Algebraic Tools

This appendix fixes the standard conventions used to decide common transformability and extract the algebraic witnesses in Lemma F.3.1.

Work over the exact coefficient field L. For a finite polynomial system $f_1, \dots, f_s \in R = L[x_1, \dots, x_n]$, let $I = (f_1, \dots, f_s)$ be the ideal they generate and write

$$V_E(I) = \{\mathbf{a} \in E^n : f(\mathbf{a}) = 0 \text{ for every } f \in I\}$$

for its zero locus over an algebraically closed extension $E \supseteq L$. A projective zero locus is defined by homogeneous equations in the specified projective space. Zero loci are Zariski closed, their complements are Zariski open, and finite unions of intersections of open and closed sets are constructible. Nonvanishing conditions $p_1 \cdots p_t \neq 0$ can be encoded by an auxiliary variable z and the equation $zp_1 \cdots p_t = 1$, as in Lemma F.3.1.

For a fixed monomial order, a Gröbner basis of I has leading monomials generating the leading-monomial ideal of I. Buchberger's algorithm computes such a basis and terminates by the ascending-chain condition on monomial ideals. Division by the basis decides ideal membership. In particular, the weak Nullstellensatz gives

$$V_{\overline{L}}(I) = \varnothing \quad \Longleftrightarrow \quad 1 \in I,$$

equivalently, the reduced Gröbner basis is $\{1\}$ [CLO15, Chapters 2–4].

Exact algebraic points can be obtained by elimination and triangular decomposition. For an ordered variable split, the elimination ideal is $I \cap L[x_{k+1}, \dots, x_n]$. In a triangular set, successive equations have distinct largest variables; the leading coefficient in that variable is the initial. For $\Delta \in R$, saturation is

$$I : \Delta^\infty = \{f : \Delta^m f \in I \text{ for some } m \geq 0\}.$$

A regular chain is a triangular set whose successive initials are non-zero-divisors modulo the ideal obtained by saturating the preceding equations by their initials. Its constructible branch requires those initials to be nonzero. Regular-chain decomposition produces finitely many such branches. Generic algebraic specialization assigns algebraic values to free coordinates outside the exceptional zero loci of the required initials and denominators. Minimal polynomials with isolating data then represent exact algebraic sample points [ALMM99, Coh93].

These tools supply the emptiness test and exact witness construction in Lemma F.3.1.

For a field extension L'/L, base change extends the coefficients and sends I to $I_{L'} = I\,L'[x_1, \dots, x_n]$. When $L \subseteq \mathbb{C}$, complexification is base change to $L' = \mathbb{C}$.

F.D Character-Flat Extensions of the Reed–Muller Calculus

Allowing all four Bell signatures extends the flat-Lagrangian network construction to signatures with linear signs on their support. This appendix characterizes that class and proves signed Radon localization and recognition results. These structural extensions complement the $C_2^{(X)}$ result of Theorem F.7.1. Closure for $V_4^{(\mathrm{std})}$ is proved by the tensor-line argument in Theorem F.8.3. We retain the symmetric-space conventions.

We call an arity-$2k$ signature *character-flat* when a linear sign character is also allowed:

$$\mathsf{CFlat}_{2k} = \left\{c(-1)^{\ell(\mathbf{x})}\mathbf{1}[\mathbf{x} \in \mathbf{a} + L] : c \in \overline{\mathbb{Q}}^\times,\ \mathbf{a} \in \mathbb{F}_2^{2k},\ \ell \in (\mathbb{F}_2^{2k})^\vee,\ L = L^\perp\right\} \cup \{0\}. \tag{F.156}$$

Define the signature class $\mathsf{CFlat} = \bigcup_{k \geq 0} \mathsf{CFlat}_{2k}$. This is a subclass of $\mathscr{A}$. The following network-calculus lemma describes its realization by RM_8 and all four Bell binaries.

A common character-flat K-presentation is defined by Definition F.2.3 with $\mathscr{C} = \mathsf{CFlat}$. It implies a common affine K-presentation because $\mathsf{CFlat} \subseteq \mathscr{A}$. Here an RM_8*–Pauli network* means a tensor network whose nonbinary vertices are retained copies of RM_8, whose binary vertices are realizable ordered Bell binaries, and whose internal connections are X-edges. The next lemma gives the exact Pauli network calculus for this class. The remaining lemmas extend Radon localization and recognition from FLag to CFlat.

Lemma F.D.1 (Character-flat Pauli network calculus)**.** *1. Every network whose vertices are* RM_8 *and realizable ordered Bell binaries $B_{d,\ell}$, and whose internal edges are X-edges, has boundary signature in* CFlat.

2. If RM_8 *and realizable nonzero representatives of all four ordered Bell binaries $B_{d,\ell}$ are retained, then, for every $0 \neq f \in \mathsf{CFlat}_{2k}$, an $O(k^2)$-size realizable* RM_8*–Pauli network has boundary equal to a nonzero scalar multiple of f.*

3. Every nonzero character-flat signature of arity at most six is, after a permutation of inputs, a nonzero scalar multiple of a product of ordered Bell binaries on a perfect matching.

4. CFlat *is closed under nonzero scalars, tensor products, permutations of inputs, single-input ordered-Bell attachments through X-edges, and nonzero tensor factors.*

Proof. Let B be the displayed boundary-input set and let V_{vert} be the tensor-vertex set. Use the boundary normalization of Lemma F.7.15, so B consists of unpaired vertex inputs. Let H_{int} be the set of internal half-edges. For every vertex let L_v be the translation space of its affine support. The Reed–Muller code and the two-coordinate repetition space of every Bell binary are self-dual. Define the internal equality subspace E_{eq}, the ambient constraint subspace H_{con}, and the vertex-code sum C_0 by

$$\begin{aligned} E_{\mathrm{eq}} &= \{\mathbf{y} \in \mathbb{F}_2^{H_{\mathrm{int}}} : y_h = y_{h'} \text{ on every internal pair } \{h, h'\}\}, \\ E_{\mathrm{eq}} &= E_{\mathrm{eq}}^{\perp}, \qquad\qquad H_{\mathrm{con}} = \mathbb{F}_2^B \oplus E_{\mathrm{eq}}, \\ C_0 &= \bigoplus_{v \in V_{\mathrm{vert}}} L_v = C_0^{\perp}. \end{aligned}$$

Exactly as in the proof of Lemma F.7.15, the feasible boundary translation space is $L = \pi_B(C_0 \cap H_{\mathrm{con}}) = L^{\perp}$.

Define the feasible half-edge code $\mathcal{K} := C_0 \cap H_{\mathrm{con}}$ and its internal-fiber subspace

$$J := \mathcal{K} \cap (\{\mathbf{0}_B\} \oplus \mathbb{F}_2^{H_{\mathrm{int}}}),$$

where $\mathbf{0}_B$ is the zero boundary string. The product of the ordered-Bell signs is a linear character $(-1)^{\lambda(\mathbf{z})}$ on the half-edge variables. For a fixed feasible boundary string the internal fiber is $\mathbf{z}_0 + J$, and

$$\sum_{\mathbf{j} \in J} (-1)^{\lambda(\mathbf{z}_0 + \mathbf{j})} = (-1)^{\lambda(\mathbf{z}_0)} \sum_{\mathbf{j} \in J} (-1)^{\lambda(\mathbf{j})}.$$

If $\lambda|_J \neq 0$, every nonempty boundary fiber cancels. Otherwise the same nonzero scalar remains on every fiber and the character descends to the boundary, proving item 1.

Conversely, the network in Lemma F.7.15 realizes the indicator of $\mathbf{a} + L$. Attaching a Bell binary through X induces, up to a nonzero sign, one of I, X, Z, Y, so realizable ordered-Bell attachments implement the shift and linear character, proving item 2. More explicitly, $B_{d,\ell}X$ is respectively X, Y, I, Z for $(d, \ell) = (0,0), (0,1), (1,0), (1,1)$, up to the fixed orientation signs. Arity zero is the

empty matching. At positive arity at most six, Lemma F.7.19 splits the self-dual translation space into two-coordinate repetition lines. Support shifts and characters turn their factors into ordered Bell signatures, proving item 3.

For the factor claim in item 4, let $f = f_1 \otimes f_2 \neq 0$. The two projections of its affine support are affine. Write its Cartesian support as $(\mathbf{a}_1 + L_1) \times (\mathbf{a}_2 + L_2)$, with $L = L_1 \oplus L_2$. Since $L = L^\perp$, each L_j is self-dual in its own coordinate space. Fixing a supported string of the other factor restricts the character of f to a character on $\mathbf{a}_j + L_j$, up to a nonzero scalar. Hence f_j is character-flat. The other closure properties follow by the same direct-sum and character identities. □

Corollary F.D.2 (No Pauli-network inverse from RM_8 to H_6)**.** *No network using only* RM_8*, realizable ordered Bell binaries, and X-edges has boundary proportional to* H_6*.*

Proof. Every nonzero such boundary lies in CFlat by Lemma F.D.1. A character-flat six-input support is a coset of a self-dual subspace of $\mathbb{F}_2^6$, hence has dimension three. By direct expansion of Equation (F.58), the support of H_6 is the six-bit even-parity space, of dimension five. Thus $\mathrm{H}_6 \notin \mathsf{CFlat}$, a contradiction. □

The support-coset and character data in Equation (F.156) can be evaluated by affine Gaussian elimination.

For every even n, define the accessible-direction set

$$\mathrm{Acc}_n = (\mathbf{1}^n)^\perp \setminus \{\mathbf{0}^n, \mathbf{1}^n\}.$$

Equivalently, Acc_n consists of the nonzero even-weight vectors other than $\mathbf{1}^n$. In the two smallest even arities,

$$\mathrm{Acc}_2 = \varnothing, \qquad \mathrm{Acc}_4 = \{\mathbf{e}_i + \mathbf{e}_j : 1 \leq i < j \leq 4\}.$$

Recall

$$O(n,2) := \{A \in \mathrm{GL}_n(\mathbb{F}_2) : A^\mathsf{T} A = I_n\}.$$

For $A \in O(n,2)$, the graph subspace $\mathrm{Graph}(A)$ is self-dual, as in Lemma F.7.16. Let Γ_A denote a fixed realizable $\mathrm{RM}_8/I/X$-network supplied by Lemma F.7.15 whose boundary is a nonzero scalar multiple of $\mathbf{1}[\mathrm{Graph}(A)]$.

Lemma F.D.3 (Realizable signed Radon cards)**.** *Assume* RM_8 *and chosen representatives of all four ordered Bell binaries are retained, and let g have even arity n. For each* $\mathbf{v} \in \mathrm{Acc}_n$*, use the recorded data* $(A_\mathbf{v}, i_\mathbf{v}, j_\mathbf{v}, \mathbf{r}_\mathbf{v}, \Pi_\mathbf{v}, W_\mathbf{v})$ *fixed in Lemma F.7.16, and fix the graph network* $\Gamma_{A_\mathbf{v}}$*. For* $d, \ell \in \mathbb{F}_2$*, using one copy of g, this graph network, X-edges, and the realizable ordered Bell binary* $B_{d,\ell}$*, one directly realizes, up to a nonzero scalar, the signed card*

$$\mathcal{S}_{\mathbf{v},d,\ell} g([\mathbf{x}]_\mathbf{v}) = (-1)^{\ell (A_\mathbf{v}\mathbf{x})_{i_\mathbf{v}}} (g(\mathbf{x}) + (-1)^\ell g(\mathbf{x}+\mathbf{v})), \qquad [\mathbf{x}]_\mathbf{v} \in H_{\mathbf{v},d}/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{v}\}. \tag{F.157}$$

The cases $\ell = 0$ *and* $\ell = 1$ *are called respectively the* plus *and* minus *signed cards on layer d. The prefactor* $(-1)^{\ell (A_\mathbf{v}\mathbf{x})_{i_\mathbf{v}}}$ *is the exterior Bell sign in the chosen quotient coordinates. After stripping this sign and the nonzero scalar, the two orbit numerators are* $g(\mathbf{x}) + g(\mathbf{x}+\mathbf{v})$ *and* $g(\mathbf{x}) - g(\mathbf{x}+\mathbf{v})$*. The remaining* $n-2$ *graph-output coordinates give an explicit affine isometry from this quotient to the boundary coordinate space. Replacing* $(A_\mathbf{v}, i_\mathbf{v}, j_\mathbf{v})$ *by another such choice changes the displayed analytic card only by a quotient affine isometry, a nonzero scalar, and a linear sign character.*

Proof. Connect g to the input side of $\Gamma_{A_\mathbf{v}}$ through X-edges, and let $\mathbf{x}$ denote the variables at g. The graph input is therefore $\mathbf{x} + \mathbf{1}$. Every $A \in O(n,2)$ fixes $\mathbf{1}$: invariance of $\mathbf{z} \cdot \mathbf{z} = \mathbf{z} \cdot \mathbf{1}$ gives

$A^{\mathsf{T}}\mathbf{1} = \mathbf{1}$, and hence $A\mathbf{1} = \mathbf{1}$. Thus the graph output is $\mathbf{z} = A_{\mathbf{v}}(\mathbf{x} + \mathbf{1}) = A_{\mathbf{v}}\mathbf{x} + \mathbf{1}$. Connect output positions $i_{\mathbf{v}}, j_{\mathbf{v}}$ to $B_{d,\ell}$ through their two X-edges. The ordered Bell arguments are then

$$u = z_{i_{\mathbf{v}}} + 1 = (A_{\mathbf{v}}\mathbf{x})_{i_{\mathbf{v}}}, \qquad w = z_{j_{\mathbf{v}}} + 1 = (A_{\mathbf{v}}\mathbf{x})_{j_{\mathbf{v}}}.$$

Consequently its support equation is

$$u + w = \mathbf{x} \cdot A_{\mathbf{v}}^{\mathsf{T}}(\mathbf{e}_{i_{\mathbf{v}}} + \mathbf{e}_{j_{\mathbf{v}}}) = \mathbf{x} \cdot \mathbf{v} = d.$$

For fixed remaining graph-output coordinates there are two lifts, differing by $\mathbf{e}_{i_{\mathbf{v}}} + \mathbf{e}_{j_{\mathbf{v}}}$ in $\mathbf{z}$ and hence by $\mathbf{v}$ in $\mathbf{x}$. The Bell sign is $(-1)^{\ell u} = (-1)^{\ell (A_{\mathbf{v}}\mathbf{x})_{i_{\mathbf{v}}}}$ and changes between those lifts by $(-1)^{\ell}$, giving Equation (F.157). The graph relation is single-valued before the pair contraction, so the realization scalar is nonzero.

Each choice identifies the quotient affinely and isometrically with its remaining graph-output coordinates. The transition between two choices is therefore an affine isometry. Their selected-coordinate functionals differ on the quotient by a linear form and a constant, which gives the last assertion. □

Lemma F.D.4 (Localization from two character-flat Radon cards)**.** *Assume the resources of Lemma F.D.3 are retained. Let g have even arity $n \geq 10$, fix $\mathbf{v} \in \mathrm{Acc}_n$, and suppose that the two plus cards $\mathcal{S}_{\mathbf{v},0,0}g$ and $\mathcal{S}_{\mathbf{v},1,0}g$ are zero or character-flat. Define, only for the analysis, the orbit-sum signature*

$$F_{\mathbf{v}} = \mathsf{P}_{\mathbf{v}}^{+} g.$$

If $F_{\mathbf{v}} \notin \mathscr{A}$, connecting one copy of g to a RM_8–Pauli network realizes a nonzero $H \notin \mathscr{A}$ of arity at most eight.

Proof. Use the recorded data $(A_{\mathbf{v}}, i_{\mathbf{v}}, j_{\mathbf{v}}, \mathbf{r}_{\mathbf{v}}, \Pi_{\mathbf{v}}, W_{\mathbf{v}})$ from Lemma F.7.16, and put $W := W_{\mathbf{v}}$. The plane $\Pi_{\mathbf{v}}$ is nondegenerate, and the map $\iota_{\mathbf{v},\epsilon}$ identifies W isometrically with $H_{\mathbf{v},\epsilon}/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{v}\}$. Since $F_{\mathbf{v}}$ is invariant under $T_{\mathbf{v}}$, its two plus Radon cards, in these quotient coordinates, are

$$f_\epsilon(\mathbf{x}) := F_{\mathbf{v}}(\epsilon \mathbf{r}_{\mathbf{v}} + \mathbf{x}), \qquad \mathbf{x} \in W.$$

For each nonzero Radon card choose a presentation

$$f_\epsilon(\mathbf{x}) = c_\epsilon (-1)^{\lambda_\epsilon(\mathbf{x})} \mathbf{1}[\mathbf{x} \in \mathbf{a}_\epsilon + M_\epsilon], \qquad M_\epsilon = M_\epsilon^{\perp} \leq W.$$

Here $\mathbf{a}_\epsilon$ is chosen modulo M_ϵ, and λ_ϵ is an ambient extension of the character on the support. Changing either choice changes c_ϵ only by a sign, so membership of c_1/c_0 in μ_4 is independent of these choices.

If one Radon card is zero, the other gains only one affine layer equation and $F_{\mathbf{v}} \in \mathscr{A}$. Suppose both are nonzero. Their union has affine support exactly when $M_0 = M_1$. When the translation spaces agree, the difference of the two characters is absorbed by an allowed quadratic mixed term with the layer bit, and the layer scales extend to an affine fourth-root phase exactly when $c_1/c_0 \in \mu_4$. Thus $F_{\mathbf{v}} \notin \mathscr{A}$ gives

$$M_0 \neq M_1, \qquad \text{or} \qquad M_0 = M_1 \ \text{ and } \ c_1/c_0 \notin \mu_4.$$

Represent $\lambda_0 + \lambda_1$ by $\boldsymbol{\rho} \in W$, so that $(\lambda_0 + \lambda_1)(\mathbf{x}) = \boldsymbol{\rho} \cdot \mathbf{x}$. When $M_0 \neq M_1$, choose $\mathbf{s} \in M_0 \setminus M_1$ and $\mathbf{t} \in M_1$ with $\mathbf{s} \cdot \mathbf{t} = 1$, and put $\boldsymbol{\Delta} = \mathbf{a}_1 - \mathbf{a}_0$. The plane $\operatorname{span}_{\mathbb{F}_2}\{\mathbf{s}, \mathbf{t}\}$ is nondegenerate. Put $S_0 = \operatorname{span}\{\mathbf{s}, \mathbf{t}, \boldsymbol{\Delta}, \boldsymbol{\rho}\}$. Its radical has dimension at most two. Pairing a basis of that radical with dual vectors gives a nondegenerate extension $N \supseteq S_0$ of dimension at most six. If N has odd dimension, then its nondegenerate orthogonal complement is odd-dimensional and contains an

anisotropic vector. Adding that orthogonal line preserves the bound and makes the extension even-dimensional. Denote the result by

$$U \supseteq \operatorname{span}\{\mathbf{s}, \mathbf{t}, \boldsymbol{\Delta}, \boldsymbol{\rho}\}, \qquad \dim U \le 6.$$

When $M_0 = M_1$, start from $S_0 = \operatorname{span}\{\boldsymbol{\Delta}, \boldsymbol{\rho}\}$. Pairing its radical needs at most two dual vectors and gives a nondegenerate extension of dimension at most four. If its dimension is odd, add an anisotropic vector from its odd-dimensional nondegenerate complement. This gives an even-dimensional nondegenerate $U \supseteq \operatorname{span}\{\boldsymbol{\Delta}, \boldsymbol{\rho}\}$ with $\dim U \le 4$. Put

$$P := U^{\perp_W}, \qquad W = U \perp P,$$

and let π_U, π_P be the corresponding orthogonal projections. A Witt decomposition of the even-dimensional nondegenerate space P supplies

$$J \le P, \qquad J = J^{\perp_P}.$$

Because $\boldsymbol{\Delta}, \boldsymbol{\rho} \in U$, the two shifts have the same P-projection

$$\mathbf{p}_0 := \pi_P(\mathbf{a}_0) = \pi_P(\mathbf{a}_1),$$

and the two characters have the same restriction $\lambda_P := \lambda_0|_P = \lambda_1|_P$.

Let $S := \Pi_{\mathbf{v}} \perp U$, put $m := \dim S$, and choose an isometry $\phi : S \to \mathbb{F}_2^m$. Such an isometry exists because S contains the nonalternating plane $\Pi_{\mathbf{v}}$. We have $m \le 8$ when $M_0 \ne M_1$ and $m \le 6$ when $M_0 = M_1$. Define

$$R_0 := \{(\mathbf{s}_0 + \mathbf{j}, \phi(\mathbf{s}_0)) : \mathbf{s}_0 \in S,\ \mathbf{j} \in J\} \le \mathbb{F}_2^n \oplus \mathbb{F}_2^m.$$

For two elements of R_0, their inner product is

$$(\mathbf{s}_0 + \mathbf{j}) \cdot (\mathbf{s}_0' + \mathbf{j}') + \phi(\mathbf{s}_0) \cdot \phi(\mathbf{s}_0') = \mathbf{j} \cdot \mathbf{j}' = 0.$$

Moreover $\dim R_0 = m + (n-m)/2 = (n+m)/2$, so $R_0 = R_0^{\perp}$. On R_0, define the linear sign character

$$\chi(\mathbf{s}_0 + \mathbf{j}, \phi(\mathbf{s}_0)) := (-1)^{\lambda_P(\mathbf{j})}.$$

Choose an ambient linear-character extension and denote it again by χ. Hence, for $(\mathbf{x}, \mathbf{y}) \in \mathbb{F}_2^n \oplus \mathbb{F}_2^m$,

$$C(\mathbf{x}, \mathbf{y}) := \chi(\mathbf{x} - \mathbf{p}_0, \mathbf{y})\, \mathbf{1}[(\mathbf{x}, \mathbf{y}) \in (\mathbf{p}_0, \mathbf{0}) + R_0] \tag{F.158}$$

is character-flat. By Lemma F.D.1, choose an $O(n^2)$-size RM_8–Pauli network whose boundary is κC, where the realization scalar $\kappa \in \overline{\mathbb{Q}}^{\times}$ is recorded. Any fixed coordinate complements contributed by its X-attachments are absorbed into the displayed affine shift.

Contract this context with $F_{\mathbf{v}}$. Its boundary is

$$H^+(\phi(\mathbf{s}_0)) = \kappa \sum_{\mathbf{j} \in J} (-1)^{\lambda_P(\mathbf{j})} F_{\mathbf{v}}(\mathbf{s}_0 + \mathbf{p}_0 + \mathbf{j}), \qquad \mathbf{s}_0 \in S.$$

When $M_0 \ne M_1$, write $\mathbf{a}_\epsilon = \mathbf{u}_\epsilon + \mathbf{p}_0$ with $\mathbf{u}_\epsilon \in U$. The three boundary points $\phi(\mathbf{u}_0)$, $\phi(\mathbf{u}_0 + \mathbf{s})$, and $\phi(\mathbf{r}_{\mathbf{v}} + \mathbf{u}_1)$ give nonzero sums. On each support fiber the context character cancels the common P-restriction of the character of the Radon card, so there is no cancellation. The fourth point $\phi(\mathbf{r}_{\mathbf{v}} + \mathbf{u}_1 + \mathbf{s})$ has value zero: otherwise some $\mathbf{j} \in J \subseteq P$ would satisfy $\mathbf{s} + \mathbf{j} \in M_1$, and then

$$0 = (\mathbf{s} + \mathbf{j}) \cdot \mathbf{t} = \mathbf{s} \cdot \mathbf{t} = 1.$$

Thus the support of H^+ is not affine. When $M_0 = M_1$, the two corresponding fibers have the same nonzero cardinality $|J \cap M_0|$. Their value ratio differs from c_1/c_0 only by a sign, and therefore remains outside μ_4. In both cases $H^+ \neq 0$ and $H^+ \notin \mathscr{A}$.

Finally, contract the same realizable context with g, and call its boundary H. Put $\mathbf{w} := \phi(\mathbf{v})$. Translation by $\mathbf{v}$ on the input of Equation (F.158) is translation by $\mathbf{w}$ on its boundary, and the context character is trivial on $(\mathbf{v}, \mathbf{w})$. Therefore

$$H^+ = \mathsf{P}^+_{\mathbf{w}} H.$$

If $H \in \mathscr{A}$, then Lemma F.7.18 would give $H^+ \in \mathscr{A}$, a contradiction. Thus $H \neq 0$, $H \notin \mathscr{A}$, and its arity m is at most eight. □

Lemma F.D.5 (Affine-to-character-flat recognition)**.** *Let $0 \neq g \in \mathscr{A}$ have even arity $2r \geq 10$. If every signed card $\mathcal{S}_{\mathbf{v},d,\ell} g$, for $\mathbf{v} \in \mathrm{Acc}_{2r}$ and $d, \ell \in \mathbb{F}_2$, is zero or character-flat, then $g \in \mathsf{CFlat}$.*

Proof. Input translation permutes the two values of d and reindexes the corresponding quotient by an affine isometry. Under this reindexing, it changes a signed card only by a nonzero scalar and a linear sign character. We may therefore translate the affine support to a linear subspace and write

$$g(\mathbf{x}) = c\, i^{Q(\mathbf{x})} \mathbf{1}[\mathbf{x} \in U], \qquad c \neq 0, \qquad d_U := \dim U.$$

Put $E := \mathbf{1}^\perp$. Every nonzero character-flat card has arity $2r - 2$ and hence support dimension $r - 1$.

We first prove $U = U^\perp$. If $E \subseteq U$, then $d_U \geq 2r - 1$. For an accessible $\mathbf{v} \in E$, at least one signed card on a nonempty layer is nonzero. Restriction to the layer and quotienting by $\mathrm{span}_{\mathbb{F}_2}\{\mathbf{v}\}$ can each remove at most one dimension. More explicitly, on an overlapping support orbit, factoring out $g(\mathbf{x})$ leaves

$$1 \pm \zeta(-1)^{\theta(\mathbf{x})}, \qquad \zeta \in \mu_4,$$

where θ is a binary linear form. If $\zeta \in \{\pm i\}$, this factor is nowhere zero. If $\zeta \in \{\pm 1\}$, its nonzero set is either empty, the whole current support, or one affine hyperplane. Thus a nonzero signed card loses at most one additional support dimension through cancellation and has support dimension at least

$$d_U - 3 \geq 2r - 4 > r - 1,$$

a contradiction. If $E \subseteq U^\perp$, then $d_U \leq 1$. Choosing accessible $\mathbf{v} \in E \setminus U$ gives a nonzero card of support dimension d_U, again a contradiction. Hence neither containment holds.

The two proper subspaces $E \cap U$ and $E \cap U^\perp$ do not cover E. Since $\dim E = 2r - 1 \geq 9$, choose

$$\mathbf{v} \in \mathrm{Acc}_{2r} \setminus (U \cup U^\perp).$$

There is no cancellation because $\mathbf{v} \notin U$, and each nonempty card has support dimension $d_U - 1$. Thus $d_U = r$.

No accessible vector can lie in $U^\perp \setminus U$, because such a vector would give a nonzero card of support dimension r. Put $A := E \cap U^\perp$. It follows that

$$A \setminus \{\mathbf{0}, \mathbf{1}\} \subseteq U.$$

If $\mathbf{1} \notin A$, or if $\mathbf{1} \in A \cap U$, this gives $A \subseteq U$. If instead $\mathbf{1} \in A \setminus U$, then $\dim A \geq r - 1 \geq 4$. Choose $\mathbf{z} \in A \setminus \{\mathbf{0}, \mathbf{1}\}$. Both $\mathbf{z}$ and $\mathbf{z} + \mathbf{1}$ are accessible and lie in U by the preceding display, forcing $\mathbf{1} \in U$, a contradiction. Thus $A \subseteq U$ in every case.

If $U^\perp \subseteq E$, equal dimensions give $U = U^\perp$. Otherwise $\dim A = r - 1$. Since $A \subseteq U$ and $\dim U = r$, choose $\mathbf{u} \in U \setminus A$, so $U = A \oplus \mathrm{span}_{\mathbb{F}_2}\{\mathbf{u}\}$. Because $A \subseteq U^\perp$, the vector $\mathbf{u}$ is orthogonal

to A. If $\mathbf{u}$ were even, then $\mathbf{u}\cdot\mathbf{u}=0$, hence $\mathbf{u}\in U^{\perp}$ and $U\subseteq U^{\perp}$. Equal dimensions would again give $U=U^{\perp}$. Therefore $\mathbf{u}$ is odd and

$$U = A\oplus \operatorname{span}_{\mathbb{F}_2}\{\mathbf{u}\},\qquad \mathbf{u}\cdot\mathbf{u}=1.$$

Since $\mathbf{u}$ is odd, $\mathbf{1}\cdot\mathbf{u}=1$, whereas every vector of A is orthogonal to $\mathbf{u}$. Hence $\mathbf{1}\notin A$. Fix $\mathbf{0}\neq\mathbf{s}\in A$. Here $\dim A = r-1\geq 4$, so $\mathbf{s}$ is accessible. For this $\mathbf{s}$, a nonzero signed card has either the full quotient translation space $U/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{s}\}$, of dimension $r-1$, or a codimension-one subspace of dimension $r-2$. The first is not self-dual because its orthogonal complement is $U^{\perp}/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{s}\}\neq U/\operatorname{span}_{\mathbb{F}_2}\{\mathbf{s}\}$. The second has the wrong dimension. On the supported layer $d=0$, at least one of the plus and minus cards is nonzero: after stripping their respective exterior Bell signs and nonzero scalars, their orbit brackets are

$$g(\mathbf{x})+g(\mathbf{x}+\mathbf{s})\quad\text{and}\quad g(\mathbf{x})-g(\mathbf{x}+\mathbf{s}),$$

whose sum is $2g(\mathbf{x})$. This contradicts the hypothesis. Therefore

$$U=U^{\perp}.$$

The orthogonal reduction in the proof of Lemma F.7.15 sends U to the code

$$\mathsf{Pair}_r := \{\mathbf{x}(\mathbf{t}) : \mathbf{t}\in\mathbb{F}_2^r\},\qquad \mathbf{x}(\mathbf{t}) := (t_1,t_1,\dots,t_r,t_r).$$

Orthogonal maps preserve $\mathbf{1}$ and hence accessible directions. Their induced quotient affine isometries, together with nonzero scalar and linear-character changes, preserve character-flatness, so the signed-card hypothesis is invariant under this coordinate reduction. Put

$$g(\mathbf{x}(\mathbf{t})) = c\,h(\mathbf{t}),$$

where

$$h(\mathbf{t}) = i^{q_0+\sum_j \alpha_j t_j+2\sum_{j<k}\beta_{jk}t_jt_k},\qquad q_0,\alpha_j\in\mathbb{Z}_4,\quad \beta_{jk}\in\mathbb{F}_2. \tag{F.159}$$

Extend the quadratic coefficients symmetrically by setting $\beta_{jk}:=\beta_{kj}$ whenever $k<j$, and put $\beta_{jj}:=0$. Let $\bar{\mathbf{e}}_j$ be the j-th basis vector of $\mathbb{F}_2^r$, and define the j-th logical pair and its accessible direction in the original variables by

$$P_j := \{2j-1,2j\},\qquad \mathbf{v}_j := \mathbf{e}_{2j-1}+\mathbf{e}_{2j}.$$

Let

$$\mathcal{T}_j := \{\mathbf{t}\in\mathbb{F}_2^r : t_j=0\}.$$

Using the representatives $\mathbf{t}\in\mathcal{T}_j$, and stripping the exterior Bell signs and nonzero scalars, the $d=0$ plus and minus cards in the $\mathbf{v}_j$-direction have logical coefficient tensors

$$A_j(\mathbf{t}) = h(\mathbf{t})+h(\mathbf{t}+\bar{\mathbf{e}}_j),\qquad B_j(\mathbf{t}) = h(\mathbf{t})-h(\mathbf{t}+\bar{\mathbf{e}}_j),\qquad \mathbf{t}\in\mathcal{T}_j. \tag{F.160}$$

Whenever one of these cards has support $\mathcal{T}_j$, character-flatness makes its logical coefficient tensor a nonzero scalar times a sign character on $\mathcal{T}_j$.

Define

$$L_j(\mathbf{t}) := \sum_{k\neq j}\beta_{jk}t_k\in\mathbb{F}_2,\qquad \frac{h(\mathbf{t}+\bar{\mathbf{e}}_j)}{h(\mathbf{t})} = i^{\alpha_j}(-1)^{L_j(\mathbf{t})}.$$

Suppose first that some α_j is even. If $L_j\neq 0$, the two logical coefficient tensors in Equation (F.160) are supported on complementary proper hyperplanes, so neither nonzero card is character-flat. Thus

$L_j = 0$, which gives $\beta_{jk} = 0$ for every $k \neq j$. One of A_j, B_j is then zero and the other is a nonzero scalar multiple of $h|_{t_j=0}$. Its being a sign character forces

$$\alpha_k \equiv 0 \pmod 2 \quad (k \neq j), \qquad \beta_{km} = 0 \quad (k, m \neq j).$$

Together with the evenness of α_j, this proves that all α_k are even and all β_{km} vanish.

It remains to exclude the alternative in which every α_j is odd. Write $i^{\alpha_j} = \delta_j i$, where $\delta_j \in \{\pm 1\}$. The plus bracket in A_j satisfies

$$1 + \delta_j i(-1)^{L_j} = (1 + \delta_j i)(-\delta_j i)^{L_j}.$$

The multilinear modulo-four expansion of the Boolean parity L_j is

$$L_j(\mathbf{t}) \equiv \sum_{k \neq j} \beta_{jk} t_k + 2 \sum_{\substack{k<m \\ k,m \neq j}} \beta_{jk}\beta_{jm} t_k t_m \pmod 4.$$

Since the nonzero full-support card A_j is a sign character, all mixed quadratic coefficients in its phase vanish. Hence, for all distinct j, k, m,

$$\beta_{km} = \beta_{jk}\beta_{jm}. \tag{F.161}$$

View β as the adjacency matrix of a simple graph. These identities make the graph either empty or complete. Indeed, if $\beta_{ab} = 1$, then for every $c \notin \{a, b\}$,

$$1 = \beta_{ab} = \beta_{ca}\beta_{cb},$$

so c is adjacent to both a and b. Applying Equation (F.161) with $j = a$ joins every remaining pair.

The empty case is impossible: when $L_j = 0$, either nonzero card in Equation (F.160) is a scalar multiple of $h|_{t_j=0}$, whose ratio in any remaining coordinate k is $i^{\alpha_k} = \pm i$, not the ± 1 ratio of a sign character.

In the complete case, choose distinct logical pairs a, b, c and the weight-two direction in the original variables

$$\mathbf{w} := \mathbf{e}_{2a-1} + \mathbf{e}_{2b-1} \in \mathrm{Acc}_{2r}\,.$$

This vector is not in Pair_r. The support of the plus card $\mathcal{S}_{\mathbf{w},0,0}g$ is the image of $\{\mathbf{t} : t_a + t_b = 0\}$, and on the two-point $\mathbf{w}$-translation orbit represented by $\mathbf{x}(\mathbf{t})$ its coefficient is $c\,h(\mathbf{t})$, because $\mathbf{x}(\mathbf{t}) + \mathbf{w} \notin U$. Toggling the third logical pair c preserves this support, but its coefficient ratio is

$$\frac{h(\mathbf{t} + \bar{\mathbf{e}}_c)}{h(\mathbf{t})} = i^{\alpha_c}(-1)^{t_c + \sum_{k \neq c} t_k} \in \{\pm i\}.$$

A character-flat signature has ratio ± 1 under translation by any vector in its support translation space. This contradiction eliminates the complete case.

Thus the all-odd alternative is impossible. The earlier even-α_j case applies, and Equation (F.159) reduces to a nonzero scalar times a linear sign character on the self-dual support U. This is exactly Equation (F.156). □